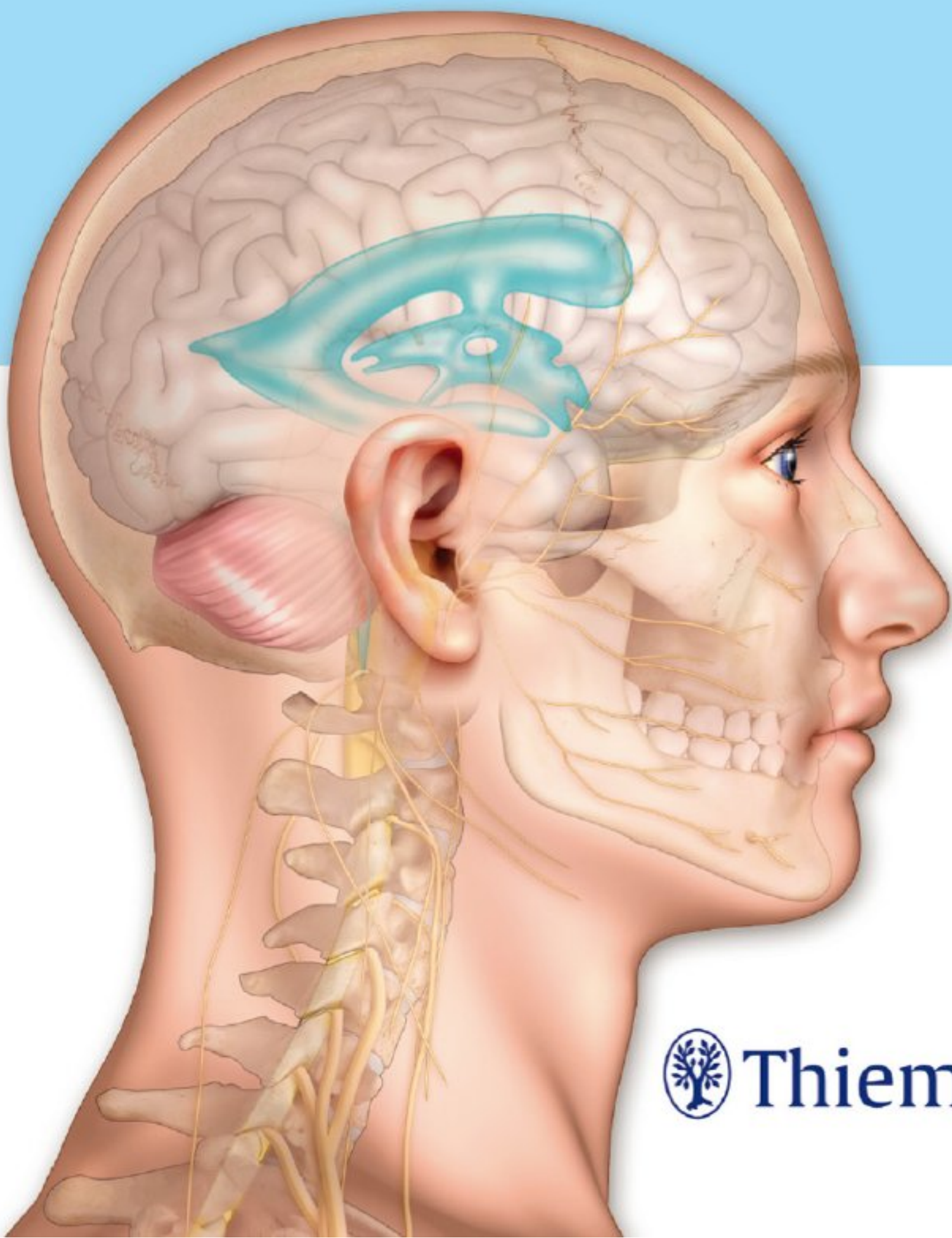


Handbook of Neurosurgery

Mark S. Greenberg

Eighth Edition



Thieme

Handbook of Neurosurgery

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Eighth Edition

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Dedication

The eighth edition of the Handbook of Neurosurgery is dedicated to my wonderful wife, Debbie. She inspires me to be the best I can be.

This book is also dedicated to “Duke” (August Henry Wagner III), beloved uncle of contributor Jayson Sack, who suffered a fatal aneurysmal subarachnoid hemorrhage as this book was going to press. His selfless life was an inspiration to Jayson and his family. He will be missed and never forgotten.

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Preface

Shortly after finishing my residency training in 1989, in a time long before the internet, I wrote the first edition of the Handbook of Neurosurgery with the intent to make practical, actionable neurosurgical information accessible at the fingertips of the practitioner.

Over the years, I used desktop publishing technology to expand the scope of the book, and to incorporate diagrams and literature references. I always wanted to make the book “academic” by presenting data that backed up the assertions. My goal was to never publish a cookbook.

Without advertising, I relied solely on word-of-mouth to promote the book. I believed in the same thing that Edwin Land must have when he said, “Marketing is what you do when your product is no good.” I had a test run of 600 copies of the book printed at a small vanity press in Ann Arbor, and sold copies to individuals by mail. My big break came when the medical bookstore at the State University of New York in Syracuse purchased 6 copies to put on their shelves. After that, I regularly packaged 5 or 6 books in a box destined for various bookstores and dropped them off at the loading dock on my way to the O.R. in the morning. After a time, the major medical book distributors in the country began ordering cases of books and my garage became a shipping department. (I still have the pallet jack that I used to unload the delivery trucks.) Then, somehow, international orders began to arrive.

I was finally able to get out of the packing and transporting business when I partnered with Thieme Medical Publishers in 2001. At that time the internet was just becoming “a thing,” and while it was a great way to make connections, the information explosion was still to come. The goal of the book regardless was still the same—to make useful information readily available.

By the time the 6th edition was published in 2006, the process I used to produce the book was becoming more difficult. The page layout software that had been my workhorse for over 10 years was no longer supported, and the developers of the reference manager I utilized had changed their focus and the product was now compatible primarily with word processors (which were poorly suited for complex multi-chapter books like this one). As a result, I was forced to use an obsolete PowerPC computer and had to hack the chapter files in order to trick the reference manager to work with them! After the 7th edition was completed, I could see no way forward with this approach. I never imagined that desktop publishing was going to atrophy and fall victim to the same fate that would later come to claim print newspapers. With the availability of the world wide web, the internet was quickly becoming the chief means of accessing information on demand.

The birth of this 8th edition of the handbook that you are now reading was a particularly difficult one. Thanks to the people at Thieme, the material was painstakingly converted from the defunct software platform to a contemporary format that will facilitate continued updates and availability in digital media. This time-consuming and labor-intensive process included porting thousands of cross references, index entries, and literature citations.

Interestingly, the objectives of the handbook have also gone through a transformation. It is challenging to distill what all of those objectives are, but I believe that it is important to present material in a framework that can serve as a foundation for studying the field of neurosurgery. This book is intended to be a place that brings together the important information that is increasingly scattered across the literature and the web (for instance, practice guidelines for disparate subjects like spinal cord injury, stroke, aneurysms...) that might not necessarily be encountered unless one is actively seeking it out.

My goal with the book has always been to present information succinctly and clearly. To that end, this edition is a completely restructured version of the Handbook of Neurosurgery, presenting the entire content in a collection of more than 100 well-ordered chapters of comparable length and format while retaining the wealth of crosslinks and references the book is known for. This new organization was designed to make the content more easily digestible and accessible – for both browsing the print book and navigating any electronic version.

With its new structure and format, as well as revised and updated content, I hope I have provided readers with an even more valuable resource in this 8th edition of the Handbook of Neurosurgery.

Acknowledgments

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Thanks also to my colleagues and the residents in the neurosurgery program at the University of South Florida from whom I learn every day. Special thanks to our chairman, Dr. Harry van Loveren, for his advice, calm leadership and for inspiring excellence in neurosurgery.

Abbreviations and Symbols

Abbreviations used only locally are defined in that section using boldface type. Numbers following entries below indicate the page number for the relevant section.

Abbreviations	
a.	artery (aa. =arteries)
AA	anaplastic astrocytoma (p.616)
ABC	aneurysmal bone cyst (p.784)
Abx.	antibiotics
AC	arachnoid cyst (p.248)
ACA	anterior cerebral artery
ACAS	asymptomatic carotid artery stenosis (p.1275) or Asymptomatic Carotid Atherosclerosis Study (p.1276)
ACDF	anterior cervical discectomy & fusion (p.1072)
ACE	angiotensin-converting enzyme
ACh	acetylcholine (neurotransmitter)
AChA	anterior choroidal artery
ACoA	anterior communicating artery
ACTH	adrenocorticotropic hormone (corticotropin) (p.151)
AD	autosomal dominant
ADH	antidiuretic hormone (p.151)
ADI	atlantodental interval (p.213)
ADPKD	autosomal dominant polycystic kidney disease (p.1193)
ADQ	abductor digiti quinti (or minimi)
AED	anti-epileptic drug (anticonvulsant) (p.443)
AFP	alpha-fetoprotein (p.600)
Ag	antigen
AHCPR	Agency for Health Care Policy and Research (of the U. S. Public Health Service)
AICA	anterior inferior cerebellar artery (p.83)
AIDP	acute inflammatory demyelinating polyradiculoneuropathy (p.185)
AIDS	acquired immunodeficiency syndrome (p.329)
AIN	anterior interosseous neuropathy (p.518)
AD	autosomal dominant (inheritance)
AFO	ankle-foot-orthosis (p.537)
AKA	also known as
ALIF	anterior lumbar interbody fusion (p.1493)
ALARA	As Low As Reasonably Achievable (p.224)

A-line	arterial line
ALL	anterior longitudinal ligament
ALS	amyotrophic lateral sclerosis (p.1086)
AMS	acute mountain sickness (p.848)
AN	acoustic neuroma (p.670)
ANA	antinuclear antibodies
AOD	atlantooccipital dislocation (p.963)
AOI	atlantooccipital interval (p.964)
AP	antero-posterior
APAG	antipseudomonal aminoglycoside
APAP	acetaminophen (p.137)
APD	afferent pupillary defect (p.562)
APTT	(or PTT) activated partial thromboplastin time
ARDS	adult respiratory distress syndrome
ASA	American Society of Anesthesiologists or aspirin (acetylsalicylic acid)
ASAP	as soon as possible
ASD	antisiphon device
AT	anterior tibialis (tibialis anterior)
AT/RT	atypical teratoid/rhabdoid tumor (p.666)
ASHD	atherosclerotic heart disease
AVM	arteriovenous malformation (p.1238)
AVP	arginine vasopressin (p.151)
β-hCG	beta-human chorionic gonadotropin (p.600)
BA	basilar artery
BBB	blood-brain barrier (p.90)
BC	basal cisterns (p.921)
BCP	birth control pills (oral contraceptives)
BCVI	blunt cerebrovascular injury (p.849)
BG	basal ganglia
BI	basilar impression/invagination (p.217)
BMD	bone mineral density (p.1009)
BMP	bone morphogenic protein (p.1439)
BOB	benign osteoblastoma (p.792)
BP	blood pressure
BR	bed rest (activity restriction)
BSF	basal skull fracture (p.884)

BSG	brainstem glioma (p. 633)
Ca	cancer
CA	cavernous angioma (p. 1247)
CAA	cerebral amyloid angiopathy (p. 1334)
CABG	coronary artery bypass graft
CAD	coronary artery disease
CAT	(or CT) computerized (axial) tomography
CBF	cerebral blood flow (p. 1264)
CBV	cerebral blood volume
CBZ	carbamazepine (p. 449)
CCB	calcium-channel blocker
CCF	carotid-cavernous (sinus) fistula (p. 1256)
CCHD	congenital cyanotic heart disease
CD	Cushing's disease (p. 723)
CEA	carotid endarterectomy (p. 1290) or carcinoembryonic antigen (p. 601)
CECT	contrast enhanced CT
cf	(Latin: confer) compare
cGy	centi-Gray (1 cGy = 1 rad)
CHF	congestive heart failure
CI	confidence interval (statistics)
CIDP	chronic inflammatory demyelinating polyradiculoneuropathy (p. 186)
CIP	critical illness polyneuropathy (p. 542)
CJD	Creutzfeldt-Jakob disease (p. 367)
CM	cavernous malformation (p. 1247)
CMAP	compound motor action potential (EMG)
CMRO ₂	cerebral metabolic rate of oxygen consumption (p. 1265)
CMT	Charcot-Marie-Tooth (p. 541)
CMV	cytomegalovirus
CNL	chemonucleolysis
CNS	central nervous system
cCO	continuous cardiac output
CO	cardiac output or carbon monoxide (p. 208)
CPA	cerebellopontine angle
CPM	central pontine myelinolysis (p. 115)
CPN	common peroneal nerve (p. 535)
CPP	cerebral perfusion pressure (p. 856)

Cr. N.	cranial nerve(s)
CRH	corticotropin-releasing hormone (p.151)
CRP	C-reactive protein
CRPS	complex regional pain syndrome (p.497)
CSM	cervical spondylotic myelopathy (p.1084)
CSO	craniosynostosis (p.252)
CSW	cerebral salt wasting (p.118)
CTA	CT angiogram (p.227)
CTP	CT perfusion (p.228)
CTS	carpal tunnel syndrome (p.519)
CVA	cerebrovascular accident (stroke) (p.1264)
CVP	central venous pressure
CVVT	cerebrovascular venous thrombosis (p.1308)
CVR	cerebrovascular resistance (p.1264)
CVS	cerebral vasospasm (p.1178)
CXR	chest x-ray
DACA	distal anterior cerebral artery
DAI	diffuse axonal injury (p.848)
DBM	demineralized bone matrix (p.1439)
D/C	discontinue
DDAVP	1-deamino-8-D-arginine vasopressin (desmopressin) (p.125)
DDx	differential diagnosis (p.1395)
DBS	deep brain stimulation (p.1524)
DI	diabetes insipidus (p.120)
DIND	delayed ischemic neurologic deficit (p.1179)
DIG	desmoplastic infantile astrocytoma and ganglioglioma (p.645)
DISH	diffuse idiopathic skeletal hyperostosis (p.1129)
DKA	diabetic keto-acidosis
DLC	disco-ligamentous complex (p.986)
DLIF	direct lateral lumbar interbody fusion (p.1498)
DOC	drug of choice
DM	diabetes mellitus
DMZ	dexamethasone
DNT	(or DNET) dysembryoplastic neuroepithelial tumors (p.646)
DOE	dyspnea on exertion
DOMS	delayed onset muscle soreness (p.1101)

DPL	diagnostic peritoneal lavage
DREZ	dorsal root entry zone lesion (p.1550)
DSA	digital subtraction angiogram
DSD	degenerative spine disease (p.1096)
DST	dural sinus thrombosis (p.1308)
DTs	delirium tremens (p.206)
DTT	diffusion tensor tractography MRI (p.234)
DVT	deep-vein thrombosis (p.167)
DWI	(or DWMRI) diffusion-weighted imaging (MRI) (p.232)
EAC	external auditory canal
EAM	external auditory meatus
EAST	Eastern Association for the Surgery of Trauma
EBRT	external beam radiation therapy
EBV	Epstein-Barr Virus
ECM	erythema chronicum migrans (p.334)
EDC	electrolytically detachable coils
EDH	epidural hematoma (p.892)
EHL	extensor hallicus longus
ELISA	enzyme-linked immunosorbent assay
ELST	endolymphatic sac tumors (p.705)
EM	electron microscope (microscopy)
ENG	electronystagmography (p.674)
ENT	ear, nose and throat (otolaryngology)
EOM	extra-ocular muscles (p.565)
EOO	external oculomotor ophthalmoplegia
ESR	erythrocyte sedimentation rate
EST	endodermal sinus tumor (p.660)
EtOH	ethyl alcohol (ethanol)
ET tube	endotracheal tube
ETV	endoscopic third ventriculostomy (p.415)
EVD	external ventricular drain (ventriculostomy)
FCU	flexor carpi ulnaris
FDP	flexor digitorum profundus
FIM	Functional Independence Measure (p.1362)
FLAIR	fluid-attenuated inversion recovery (on MRI) (p.229)
FM	face mask

FMD	fibromuscular dysplasia (p.200)
FSH	follicle stimulating hormone (p.151)
F/U	follow-up
FUO	fever of unknown origin
GABA	gamma-aminobutyric acid
GBM	glioblastoma (multiforme) (p.616)
GBS	Guillain-Barré syndrome (p.184)
GCA	giant cell arteritis (p.195)
GCS	Glasgow coma scale (p.296)
GCT	granular cell tumor (p.727) or germ cell tumor (p.659)
GD	Graves’ disease
GFAP	glial fibrillary acidic protein (p.598)
GGT	gamma glutamyl transpeptidase
GH	growth hormone (p.151)
GH-RH	growth hormone releasing hormone (p.151)
GMH	germinal matrix hemorrhage (p.1346)
GNR	gram negative rods
GnRH	gonadotropin-releasing hormone (p.151)
GSW	gunshot wound
GTC	generalized tonic-clonic (seizure)
H/A	headache (p.174)
H&H	Hunt and Hess (SAH grade) (p.1162)
H&P	history and physical exam
HBsAg	hepatitis B surface antigen
HCD	herniated cervical disc (p.1069)
hCG	human chorionic gonadotropin (p.600)
HCP	hydrocephalus (p.394)
HDT	hyperdynamic therapy (p.1186)
HGB	hemangioblastoma (p.701)
Hgb-A1C	hemoglobin A1C
hGH	human growth hormone
HH	hypothalamic hamartomas (p.261) or homonymous hemianopsia
HHT	hereditary hemorrhagic telangiectasia (p.1246)
HIV	human immunodeficiency virus
HLD	herniated lumbar disc (p.1046)
HLA	human leukocyte antigen

H.O.	house officer
HNP	herniated nucleus pulposus (herniated disc) (p. 1046)
HNPP	hereditary neuropathy with liability to pressure palsies (p. 541)
HOB	head of bed
HPA	hypothalamic-pituitary-adrenal axis
HSE	herpes simplex encephalitis (p. 364)
HTN	hypertension
IAC	internal auditory canal
IASDH	infantile acute subdural hematoma (p. 898)
ICA	internal carotid artery
ICG	indocyanine green
ICH	intracerebral hemorrhage (p. 1330)
IC-HTN	intracranial hypertension (increased ICP)
ICP	intracranial pressure (p. 856)
ICU	intensive care unit
IDDM	insulin-dependent diabetes mellitus
IDET	intradiscal endothermal therapy (p. 1053)
IEP	immune electrophoresis
IG	image guidance (intra-operative)
IGF-1	insulin-like growth factor-1 (AKA somatomedin-C) (p. 151)
IIH	idiopathic intracranial hypertension (pseudotumor cerebri) (p. 766)
IIHWOP	idiopathic intracranial hypertension without papilledema (p. 768)
IJV	internal jugular vein
IMRT	intensity modulated radiation therapy
INO	internuclear ophthalmoplegia (p. 565)
INR	international normalized ratio (p. 164)
IPS	inferior petrosal sinus
IPA	idiopathic paralysis agitans (Parkinson’s disease) (p. 176)
ISAT	International Subarachnoid Hemorrhage Aneurysm Trial (p. 1195)
IT	intrathecal
ITB	intrathecal baclofen (p. 1531)
IVC	intraventricular catheter or inferior vena cava
IVH	intraventricular hemorrhage (p. 1386)
IVP	intravenous push (medication route) or intravenous pyelogram (x-ray study)
JPS	joint position sense
LBP	low back pain (p. 1024)

IDD	Ihermitte-Duclos disease (p.647)
IE	lower extremity
LFTs	liver function tests
IGG	low-grade glioma
IH	luteinizing hormone (p.151)
IH-RH	luteinizing hormone releasing hormone (p.151)
IMD	low molecular weight dextran
IMN	lower motor neuron (p.504)
LMW	low-molecular-weight (e.g. heparins)
IOC	loss of consciousness
IOH	loss of heterozygosity
LP	lumbar puncture (p.1504)
ISO	lumbo-sacral orthosis
MAC	mycobacterium avian complex (p.354)
MAOI	monoamine oxidase inhibitor
MAP	mean arterial pressure
MAST®	military anti-shock trousers
MB	medulloblastoma (p.664)
MBEN	medulloblastoma with extensive nodularity (p.665)
MBI	modified Barthel index (□ Table 88.6)
MBS	medulloblastoma (p.664)
MCA	middle cerebral artery
mcg	(or µg) microgram
MCP	mean carotid pressure or metacarpal phalangeal
MDCTA	multidetector CT angiography
MDMA	methylenedioxymethamphetamine (p.177)
mg	milligram
MI	myocardial infarction
MIB-1	monoclonal anti-Ki-67 antibody (p.599)
MIC	minimum inhibitory concentration (for antibiotics)
MID	multi-infarct dementia
MISS	minimally invasive spine surgery
mJOA	modified Japanese Orthopedic Association scale (p.1086)
MLF	medial longitudinal fasciculus
MLS	midline shift (p.921)
MM	myelomeningocele (p.265) or multiple myeloma (p.714)

MMD	moyamoya disease (p.1313)
MMN	multifocal motor neuropathy (p.1410)
MMPI	Minnesota Multiphasic Personality Inventory
mos	months
MPTP	1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (p.177)
MRA	MRI angiogram (p.232)
MRS	MRI spectroscopy (p.233)
MRSA	methicillin resistant staphylococcus aureus
MS	microsurgery or multiple sclerosis (p.179)
MSO ₄	morphine sulfate
MTP	metatarsal phalangeal
MTT	meant transit time (on CT perfusion) (p.228)
MUAP	motor unit action potential (p.243)
MVA	motor vehicle accident
MVD	microvascular decompression (p.488)
MW	molecular weight
n.	nerve (nn.=nerves)
Na	(or Na ⁺) sodium
N ₂ O	nitrous oxide (p.105)
NAA	N-acetyl aspartate (p.233)
NAP	nerve action potential (p.509)
NASCET	North American Symptomatic Carotid Endarterectomy Trial (p.1290)
NB	(Latin: nota bene) note well
NC	nasal cannula
NCCN	National Comprehensive Cancer Network
NCD	neurocutaneous disorders (p.603)
NCV	nerve conduction velocity
NEC	neurenteric cyst (p.290) or necrotizing enterocolitis
NEXUS	National Emergency X-Radiography Utilization Study (p.953)
NF	(or NFT) neurofibromatosis (p.603)
NF1	neurofibromatosis type 1 (p.604)
NF2	neurofibromatosis type 2 (p.605)
NG tube	nasogastric tube
NGGCT	non-germinomatous germ cell tumors (p.659)
NIHSS	NIH Stroke Scale (p.1282)
NMBA	neuromuscular blocking agent (p.134)

NMO	neuromyelitis optica (Devic disease) (p.1409)
NPH	normal pressure hydrocephalus (p.403)
NPS	neuropathic pain syndrome (p.476)
NS	normal saline
NSAID	non-steroidal anti-inflammatory drug (p.137)
NSCLC	non-small-cell cancer of the lung (p.803)
NSF	nephrogenic systemic fibrosis (p.231)
NSM	neurogenic stunned myocardium (p.1177)
N/V	nausea and vomiting
NVB	neurovascular bundle
OAD	occipital atlantal dislocation, see atlantooccipital dislocation (p.963)
OALL	ossification of the anterior longitudinal ligament (p.1129)
OC	occipital condyle
OCB	oligoclonal bands (in CSF) (p.181)
OCF	occipital condyle fracture (p.885)
ODG	oligodendroglioma (p.638)
OEF	oxygen extraction fraction
OFC	occipital-frontal (head) circumference
OGST	oral glucose suppression test (for growth hormone) (p.736)
OMO	open-mouth odontoid (C-spine x-ray view)
OMP	oculomotor (third nerve) palsy
ONSF	optic nerve sheath fenestration (p.772)
OP	opening pressure (on LP) (p.1505)
OPLL	ossification of the posterior longitudinal ligament (p.1127)
ORIF	open reduction/internal fixation
OS	overall survival
OTC	over the counter (i.e. without prescription)
PACU	post-anesthesia care unit (AKA recovery room, PAR)
PADI	posterior atlantodental interval (p.213)
PAN	poly- (or peri-) arteritis nodosa (p.199)
PBPP	perinatal brachial plexus palsy (p.552)
p _B O ₂	brain tissue oxygen tension (p.865)
PC	pineal cyst (p.658)
PCA	pilocytic astrocytoma (p.629) or posterior cerebral artery
PCB	pneumatic compression boot
PCC	prothrombin complex concentrate (p.166)

PCI	prophylactic cranial irradiation
PCN	penicillin
PCNSL	primary CNS lymphoma (p. 710)
P-comm	posterior communicating artery
PCV	procarbazine, CCNU, & vincristine (chemotherapy)
PCWP	pulmonary capillary wedge pressure
PDA	patent ductus arteriosus
PDN	painful diabetic neuropathy (p. 476)
PDR	Physicians Desk Reference®
peds	pediatrics (infants & children)
PEEK	poly-ether-ether-ketone (graft material)
PET	positron emission tomography (scan)
p-fossa	posterior fossa
PFS	progression-free survival
PFT	pulmonary function test
PHN	postherpetic neuralgia (p. 493)
PHT	phenytoin (Dilantin®) (p. 446)
PICA	posterior inferior cerebellar artery (p. 82)
PIF	prolactin release inhibitory factor (p. 151)
PIN	posterior interosseous neuropathy (p. 532)
PION	posterior ischemic optic neuropathy (p. 1056)
PIVH	periventricular-intraventricular hemorrhage (p. 1346)
PLAP	placental alkaline phosphatase (p. 660)
PLEDs	periodic lateralizing epileptiform discharges
PLIF	posterior lumbar interbody fusion
PM	pars marginalis (p. 60)
PMA	progressive muscular atrophy (p. 183) or pilomyxoid astrocytoma (p. 632)
PMH	pure motor hemiparesis
PML	progressive multifocal leukoencephalopathy (p. 329)
PMMA	polymethylmethacrylate (methylmethacrylate)
PMR	polymyalgia rheumatica (p. 198)
PMV	pontomesencephalic vein
PNET	primitive neuroectodermal tumor (p. 663)
POD	post-operative day
PPV	positive predictive value: in unselected patients who test positive, PPV is the probability that the patient has the disease
PR	per rectum

PRES	posterior reversible encephalopathy syndrome (p. 194)
PRF	prolactin releasing factor (p. 151)
PRIF	prolactin (releasing) inhibitory factor (p. 151)
PRN	as needed
PRSP	penicillinase resistant synthetic PCN
PSNP	progressive supra-nuclear palsy (p. 178)
PSR	percutaneous stereotactic rhizotomy (for trigeminal neuralgia) (p. 483)
PSW	positive sharp waves (on EMG) (p. 242)
pt	patient
PT	physical therapy or prothrombin time
PTC	pituicytoma (p. 728)
PTR	percutaneous trigeminal rhizotomy
PTT	(or APTT) partial thromboplastin time
PUD	peptic ulcer disease
PVP	percutaneous vertebroplasty (p. 1011)
PWI	perfusion-weighted imaging (MRI) (p. 233)
PXA	pleomorphic xanthoastrocytoma (p. 635)
q	(Latin: quaque) every (medication dosing)
RA	rheumatoid arthritis
RAPD	relative afferent pupillary defect (p. 562)
RASS	Richmond agitation-sedation scale (p. 132)
RCVS	reversible cerebral vasoconstrictive syndrome (p. 1158)
rem	roentgen-equivalent man
REZ	root entry zone
RFR	radiofrequency rhizotomy (p. 483)
rFVIIa	recombinant (activated) factor VII
RH	recurrent artery of Heubner
rhBMP	recombinant human BMP (p. 1439)
R/O	rule out
ROM	range of motion
RPA	recursive partitioning analysis
RPDB	randomized prospective double-blind
RPLS	reversible posterior leukoencephalopathy syndrome; see posterior reversible encephalopathy syndrome (p. 194)
RPNB	randomized prospective non-blinded
RTOG	Radiation Therapy Oncology Group
RTP	return to play (sports)

rt-PA	recombinant tissue plasminogen activator (AKA tissue plasminogen activator)
RTX	(or XRT) radiation therapy (p.1560)
S/S	signs and symptoms
SAH	subarachnoid hemorrhage (p.1191)
SBE	subacute bacterial endocarditis
SBO	spina bifida occulta (p.265)
SBP	systolic blood pressure
SCA	superior cerebellar artery
SCLC	small-cell lung cancer (p.802)
SCD	sequential compression device
SCI	spinal cord injury (p.943)
SCM	sternocleidomastoid (muscle)
SD	standard deviation
SDE	subdural empyema (p.327)
SDH	subdural hematoma (p.895)
SE	status epilepticus (for seizures) (p.468)
SEA	spinal epidural abscess (p.349)
SEP	(or SSEP) somatosensory evoked potential
SG	specific gravity
SIAD	syndrome of inappropriate antidiuresis (p.112)
SIADH	syndrome of inappropriate antidiuretic hormone (ADH) secretion (p.114)
SIDS	sudden infant death syndrome
SIH	spontaneous intracranial hypotension (p.389)
SIRS	septic inflammatory response syndrome
SjVO ₂	jugular venous oxygen saturation (p.865)
SLAD	surgical laser aiming device
SLE	systemic lupus erythematosus
SLIC	subaxial injury classification (p.986)
SMC	spinal meningeal cyst (p.1142)
SMT	spinal manipulation therapy (p.1034)
SNAP	sensory nerve action potential (EMG) (p.243)
SNUC	sinonasal undifferentiated carcinoma (p.1387)
SOMI	sternal-occipital-mandibular immobilizer (p.935)
SON	supraorbital neuralgia (p.491)
S/P	status-post
SPAM	subacute progressive ascending myelopathy (p.1019)

SPECT	single positron emission computed tomography (scan)
SPEP	serum protein electrophoresis
sPNET	supratentorial primitive neuroectodermal tumor (p. 666)
SQ	subcutaneous injection
SRS	stereotactic radiosurgery (p. 1564)
SRT	stereotactic radiotherapy (p. 1564)
SSEP	(or SEP) somatosensory evoked potential
SSPE	subacute sclerosing panencephalitis (p. 238)
SSRI	selective serotonin reuptake inhibitors
SSS	superior sagittal sinus
STA	superficial temporal artery
STICH	Surgical Trial in Intracerebral Haemorrhage (p. 1343)
STIR	short tau inversion recover (MRI image)
STN	subthalamic nucleus
STSG	Spine Trauma Study Group
SUNCT	short-lasting unilateral neuralgiform H/A with conjunctival injection and tearing (p. 478)
SVC	superior vena cava
SVM	spinal vascular malformations (p. 1140)
SVR	systemic venous resistance
SVT	supraventricular tachycardia
Sz.	seizure (p. 440)
T1 WI	T1 weighted image (on MRI) (p. 228)
T2 WI	T2 weighted image (on MRI) (p. 229)
TAL	transverse atlantal ligament (p. 70)
TBA	total bilateral adrenalectomy (p. 743)
TBI	traumatic brain injury
TCA	tricyclic antidepressants
TCD	transcranial doppler (p. 1182)
TDL	tumefactive demyelinating lesions (p. 181)
TE	time to echo (on MRI) (p. 228)
TEE	transesophageal echocardiogram
TEN	toxic epidermal necrolysis
TENS	transcutaneous electrical nerve stimulation
TGN	trigeminal neuralgia (p. 479)
T-H lines	Taylor-Haughton lines (p. 61)
TIA	transient ischemic attack (p. 1264)

TICH	traumatic intracerebral hemorrhage (hemorrhagic contusion) (p. 891)
TIVA	total intravenous anesthesia
TLIF	transforaminal lumbar interbody fusion (p. 1497)
TLISS	thoracolumbar injury severity score (p. 1006)
TLJ	thoracolumbar junction
TLSO	thoracolumbar-sacral orthosis
TM	tympanic membrane
TMB	transient monocular blindness (amaurosis fugax) (p. 1271)
t-PA	tissue plasminogen activator
TR	time to repetition (on MRI) (p. 228)
TRH	thyrotropin releasing hormone; AKA TSH-RH (p. 151)
TS	transverse sinus
TSC	tuberous sclerosis complex (p. 606)
TSH	thyroid-stimulating hormone (thyrotropin) (p. 151)
TSV	thalamostriate vein
TTP	thrombotic thrombocytopenic purpura
TVO	transient visual obscurations (p. 768)
Tx.	treatment
UBOs	unidentified bright objects (on MRI)
UE	upper extremity
UMN	upper motor neuron (p. 504)
UTI	urinary tract infection
URI	upper respiratory tract infection
U/S	ultrasound
VA	vertebral artery or ventriculoatrial
VB	vertebral body
VBI	vertebrobasilar insufficiency (p. 1305)
VEMP	vestibular evoked myogenic potential (p. 675)
VHL	von Hippel-Lindau (disease) (p. 703)
VMA	vanillylmandelic acid
VP	ventriculoperitoneal
VS	vestibular schwannoma (p. 670)
VZV	(herpes) varicella zoster virus
WBC	white blood cell (count)
WBXRT	whole brain radiation therapy (p. 810)
WFNS	World Federation of Neurosurgical Societies (grading SAH) (p. 1163)

WHO	World Health Organization. For tumor grading, e.g. WHO II indicates WHO grade II
wks	weeks
WNL	within normal limits
w/o	without
WRS	word recognition score (p.673)
W/U	work-up (evaluation)
XLIF	extreme lateral lumbar interbody fusion (p. 1498)
XRT	(or RTX) radiation therapy (p.1560)
Symbols	
□	prescribing information
→	causes or leads to
Δ	change
□	check (e.g. lab or exam item to check)
↑	increased
↓	decreased
≈	approximately
□	innervates (nerve distribution)
□	vascular supply
□	a branch of the preceding nerve
□	crucial point
□	caution; possible danger; negative factor...
Σ	summary
□	therefore
Instrumentation: the following shorthand allows rapid identification of metrics for spinal instrumentation:	
ENTRY	screw entry site
TRAJ	screw trajectory
TARGET	object to aim for
SCREWS	typical screw specifications

Conventions

□ Box types. The Handbook of Neurosurgery uses the following seven box types:

Drug info

Drug description & dosage.

Key concepts

Foundational knowledge in brief.

Practice guideline

Evidence-based guidelines. See below (in this section) for definitions. For a listing of evidence-based guidelines contained in this book, see the index under “Practice guideline.”

Booking the case

These sections appear under certain specific operations to help when scheduling that surgery. Default information appears below (in this section), for example, a specific type of anesthesia will only be mentioned if something other than general anesthesia is typically used. A list of operations addressed by this means can be found in the index under “Booking the case.”

Σ

Summarizing or synthesizing information from the associated text.

Side information

E.g., Greenberg IMHO.

Signs / symptoms

A description of signs and symptoms.

- Cross references. Cross references: the terms “see below” and “see above” are normally used when the referenced item is on the same page, or at most on the following (or preceding) page. When further excursions are needed, the page number will usually be included.
- Default values. These details are not repeated in each section or “Booking the case” box.
1. position: (depends on the operation)
 2. pre-op:
 - a) NPO after midnight the night before except meds with sips of water
 - b) antithrombotics: discontinue Coumadin® ≥ 3 days prior to surgery, Plavix® 5–7 d pre-op, aspirin 7–10 d pre-op, other NSAIDs 5 d pre-op

- 3. cardiology/medical clearance as needed
- 4. anesthesia: default =general anesthesia, unless otherwise specified
- 5. equipment: special devices such as ultrasonic aspirator, image guidance...
- 6. instrumentation: standard surgical instrument trays for a specific operation are assumed. Special instrumentation resident in the hospital will be listed
- 7. implants: this usually requires scheduling with a vendor (manufacturers representative/distributor) to provide
- 8. neuromonitoring will be listed if typically used
- 9. post-op: default care is on the ward (ICU is typically needed after craniotomy)
- 10. blood availability: specified if recommended
- 11. consent (these items use lay terms for the patient – not all-inclusive):
 - Disclaimers: informed consent for surgery requires disclosure of risks and benefits that would substantively affect a normal person’s decision to have the operation. It cannot and should not attempt to include every possibility. The items listed in this section are included as memory joggers for some items for various procedures, but are not meant to be all inclusive. The omission of information from this memory aid is not to be construed as implying that the omitted item is not important or should not be mentioned.
 - a) procedure: the typical operation and some possible common contingencies
 - b) alternatives: non-surgical (AKA “conservative”) treatment is almost always an option
 - c) complications:
 - risks of general anesthesia include: heart attack, stroke, pneumonia
 - infection: a risk with any invasive procedure
 - usual craniotomy complications include: bleeding intra-op and postop, seizure, stroke, coma, death, hydrocephalus, meningitis, and neurologic deficit related to the area of surgery including (for applicable locations): paralysis, language or sensory disturbances, coordination impairment...
 - usual spine surgery complications include: injury to nerve or spinal cord with possible numbness, weakness or paralysis, failure of the operation to achieve the desired result, dural opening which may cause a CSF leak which occasionally needs to be surgical repair. Hardware complications (when used) include: breakage, pull-out, malposition. Although a rare complication, it is serious enough that it bears mentioning in cases positioned prone with possible significant blood loss (>2 L): blindness (due to PION (p.1056))

□ Evidence-Based Medicine: Definitions. These definitions are referred to in the “Practice guideline” boxes.

Strength of recommendation		Description
Level I, II, III ^a	Level A, B, C, D ^b	
Level I High degree of clinical certainty	Level A	Based on consistent Class I evidence (well-designed, prospective randomized controlled studies)
	Level B	Single Class I study or consistent Class II evidence or strong Class II evidence especially when circumstances preclude randomized clinical trials
Level II Moderate degree of clinical certainty	Level C	Usually derived from Class II evidence (one or more well-designed comparative clinical studies or less well-designed randomized studies) or a preponderance of Class III evidence
Level III Unclear clinical certainty	Level D	Generally based on Class III evidence (case series, historical controls, case reports and expert opinion). Useful for educational purposes and to guide future research

^a as used in the Guidelines for the Management of Severe Traumatic Brain Injury, 3rd edition (Brain Trauma Foundation: Introduction. J Neurotrauma 24, Suppl 1: S1–2, 2007).
^b as used in the Guidelines for the Surgical Management of Cervical Degenerative Disease (Matz P G, et al.: Introduction and methodology. J Neurosurg: Spine 11 (2): 101–3, 2009).

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Part I

Anatomy and Physiology

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1 Gross Anatomy, Cranial and Spine

1.1 Cortical surface anatomy

1.1.1 Lateral cortical surface

□ Fig. 1.1. For abbreviations, see □ Table 1.1 and □ Table 1.2. The middle frontal gyrus (MFG) is usually more sinuous than the IFG or SFG, and it often connects to the pre-central gyrus via a thin isthmus.¹ The central sulcus joins the Sylvian fissure in only 2% of cases (i.e. in 98% of cases there is a “subcentral” gyrus). The intraparietal sulcus (ips) separates the superior and inferior parietal lobules. The IPL is composed primarily of the AG and SMG. The Sylvian fissure terminates in the SMG (Brodmann’s area 40). The superior temporal sulcus terminates in the AG.

1.1.2 Brodmann’s areas

□ Fig. 1.1 also identifies the clinically significant areas of Brodmann’s (Br.) map of the cytoarchitectonic fields of the human brain. Functional significance of these areas is as follows:

- 1. Br. areas 3, 1, 2: primary somatosensory cortex
- 2. Br. areas 41 & 42: primary auditory areas (transverse gyri of Heschl)
- 3. Br. area 4: precentral gyrus, primary motor cortex (AKA “motor strip”). Large concentration of giant pyramidal cells of Betz
- 4. Br. area 6: premotor area or supplemental motor area. Immediately anterior to motor strip, it plays a role in contralateral motor programming
- 5. Br. area 44: (dominant hemisphere) Broca’s area (motor speech)
- 6. Br. area 17: primary visual cortex

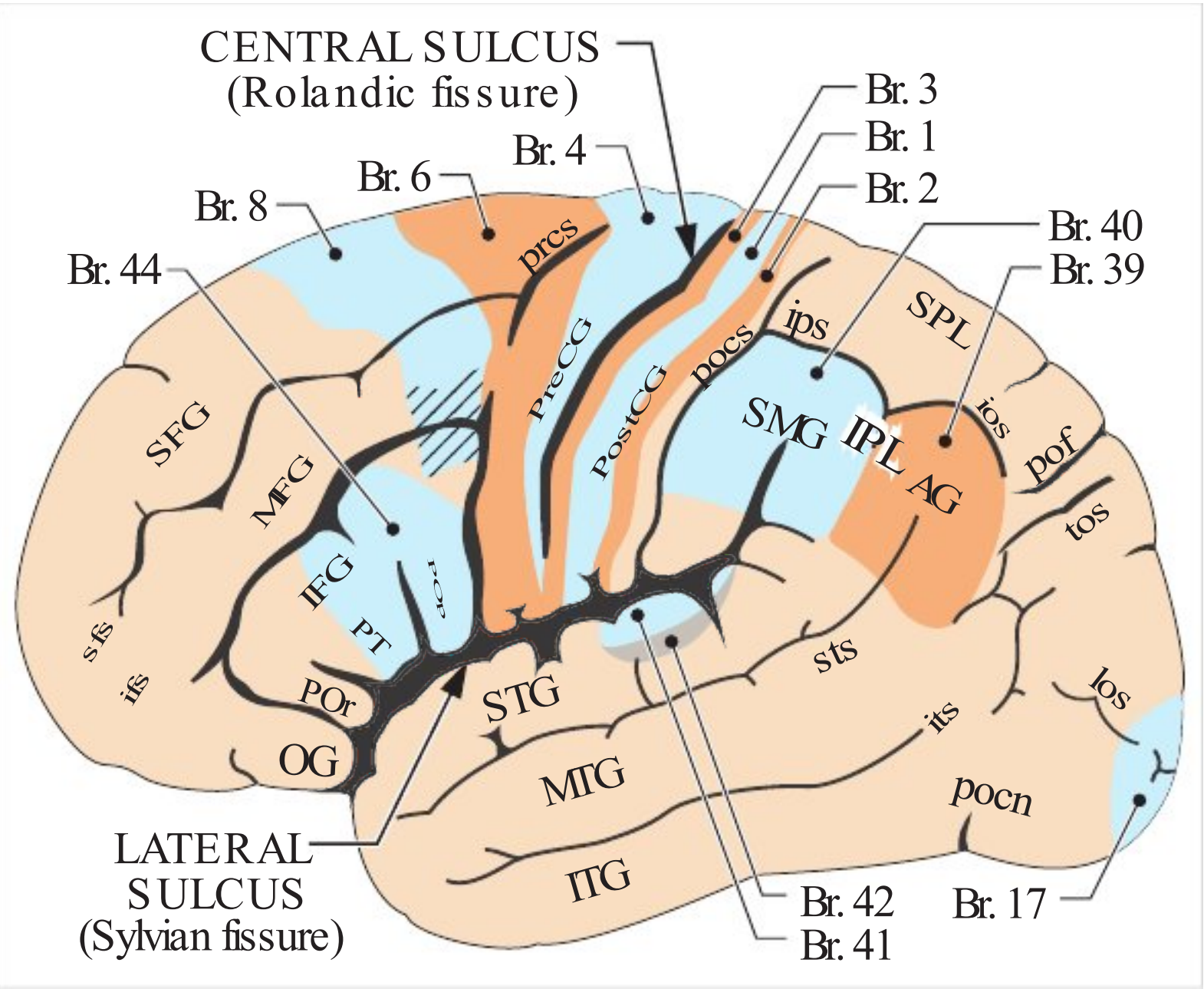


Fig. 1.1 Left lateral cerebral cortical surface anatomy. Br. = Brodmann’s area (shaded). See □ Table 1.1 and □ Table 1.2 for abbreviations (lowercase = sulci, UPPERCASE = gyri).

Table 1.1 Cerebral sulci (abbreviations)	
Abbreviation	Sulcus
cins	cingulate sulcus
cs	central sulcus
ips-ios	intraparietal-intraoccipital sulcus
los	lateral occipital sulcus
pM	pars marginalis
pocn	pre-occipital notch
pocs	post-central sulcus
pof	parieto-occipital fissure
pos	parieto-occipital sulcus
prcs	pre-central sulcus
sfs, ifs	superior, inferior frontal sulcus
sps	superior parietal sulcus
sts, its	superior, inferior temporal sulcus
tos	trans occipital sulcus

Table 1.2 Cerebral gyri and lobules (abbreviations)	
Abbreviation	Gyrus / lobule
AG	angular gyrus
CinG	cingulate gyrus
Cu	cuneus
LG	lingual gyrus
MFG, SFG	middle & superior frontal gyrus
OG	orbital gyrus
PCu	precuneous
PreCG, PostCG	pre- and post-central gyrus
PL	paracentral lobule (upper SFG and PreCG and PostCG)
IFG • POp • PT • POr	inferior frontal gyrus • pars opercularis • pars triangularis • pars orbitalis
STG, MTG, ITG	superior, middle & inferior temporal gyrus
SPL, IPL	superior & inferior parietal lobule
SMG	supramarginal gyrus

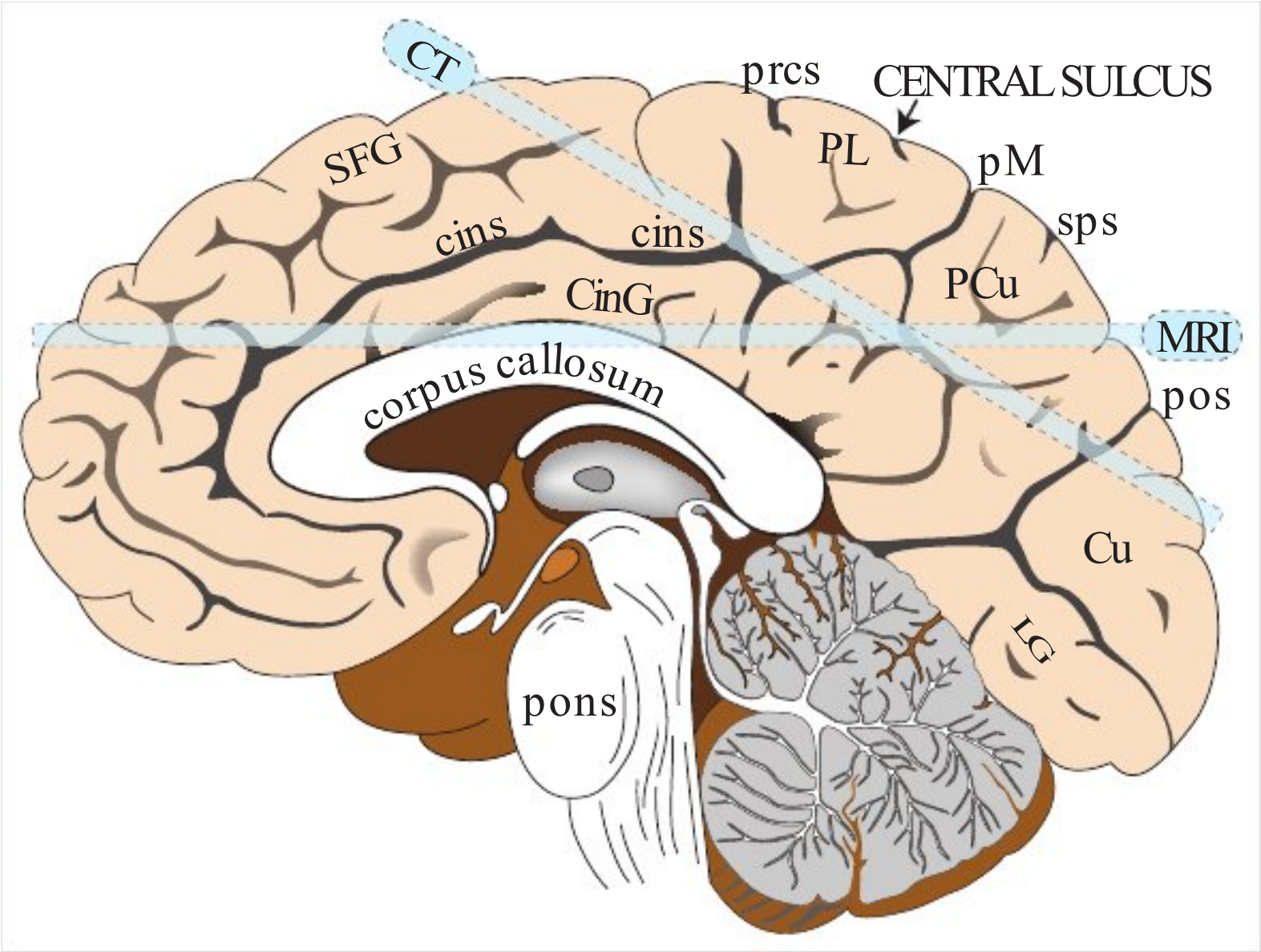


Fig. 1.2 Medial aspect of the right hemisphere.
“CT” & “MRI” bars depict typical axial slice orientation for CT & MRI scans. See □ Table 1.1 and □ Table 1.2 for abbreviations.

7. Wernicke’s area (language): in the dominant hemisphere, most of Br. area 40 and a portion of Br. area 39 (may also include $\approx$ posterior third of STG)
8. the striped portion of Br. area 8 in □ Fig. 1.1 (frontal eye field) initiates voluntary eye movements to the opposite direction

Brodmann’s area 44, Wernicke’s area: Language function cannot be reliably localized on anatomic grounds due to individual variability in its exact location; in order to perform maximal brain resections with minimal risk of aphasia, techniques such as intra-operative brain mapping² or looking for phase reversal on intraoperative cortical SSEP³ should be employed.

1.1.3 Medial surface

□ Fig. 1.2. The cingulate sulcus terminates posteriorly in the pars marginalis (pM) (plural: partes marginales). On axial imaging, the pMs: are visible on 95%of CTs and 91%of MRIs,⁴ are usually the most prominent of the paired grooves straddling the midline, and they extend a greater distance into the hemispheres.⁴ On axial CT, the pM is located slightly posterior to the widest biparietal diameter⁴; on the typically more horizontally oriented MRI slices the pM assumes a more posterior position. The pMs curve posteriorly in lower slices and anteriorly in higher slices (here, the paired pMs form the “pars bracket” – a characteristic “handlebar” configuration straddling the midline).

1.2 Central sulcus on axial imaging

See □ Fig. 1.3. Identification of the central sulcus is important to localize the motor strip (contained in the PreCG). The central sulcus (CS) is visible on 93%of CTs and 100%of MRIs.⁴ It curves posteriorly as it approaches the interhemispheric fissure (IHF), and often terminates in the paracentral lobule, just anterior to the pars marginalis (pM) within the pars bracket (see above)⁴ (i.e. the CS often does not reach the midline).

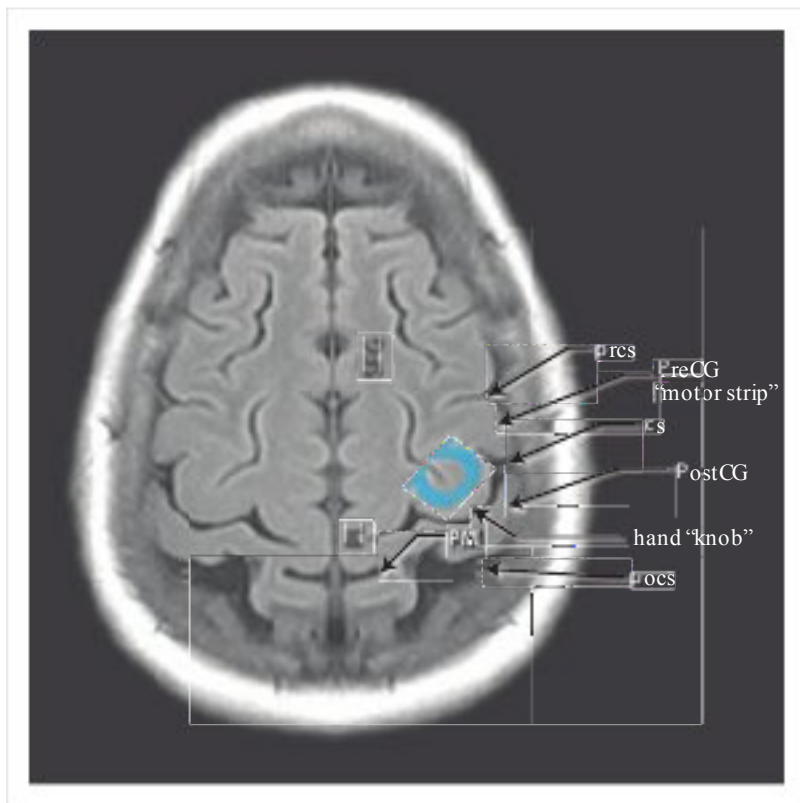


Fig. 1.3 Retouched axial FLAIR MRI with labels for gyri/sulci shown in the left hemisphere, and an unlabeled mirror image shown as the right hemisphere for reference. The inverted Ω illustrates the hand “knob” (see text).

See □ Table 1.1 and □ Table 1.2 for abbreviations.

Pointers:

- parieto-occipital sulcus (pos) (or fissure): more prominent over the medial surface, and on axial imaging is longer, more complex, and more posterior than the pars marginalis⁵
- post-central sulcus (pocs): usually bifurcates and forms an arc or parenthesis (“lazy-Y”) cupping the pM. The anterior limb does not enter the pM-bracket and the posterior limb curves behind the pM to enter the IHF

Hand “Knob”: The alpha motor neurons for hand motor function are located in the superior aspect of the prefrontal gyrus.⁶ On axial imaging, this appears as a knob-like protrusion (shaped like an inverted greek letter omega Ω) of the precentral gyrus projecting posterolaterally into the central sulcus⁷ □ Fig. 1.3. On sagittal imaging it has a posteriorly projecting hook-like appearance and is even with the posterior limit of the Sylvian fissure.⁷

1.3 Surface anatomy of the cranium

1.3.1 Craniometric points

See □ Fig. 1.4.

Pterion: region where the following bones are approximated: frontal, parietal, temporal and sphenoid (greater wing). Estimated as 2 finger-breadths above the zygomatic arch, and a thumb’s breadth behind the frontal process of the zygomatic bone (blue circle in □ Fig. 1.4).

Asterion: junction of lambdoid, occipitomastoid and parietomastoid sutures. Usually lies within a few millimeters of the posterior-inferior edge of the junction of the transverse and sigmoid sinuses (not always reliable⁸ – may overlie either sinus).

Vertex: the topmost point of the skull.

Lambda: junction of the lambdoid and sagittal sutures.

Stephanion: junction of coronal suture and superior temporal line.

Glabella: the most forward projecting point of the forehead at the level of the supraorbital ridge in the midline.

Opisthion: the posterior margin of the foramen magnum in the midline.

Bregma: the junction of the coronal and sagittal sutures.

Sagittal suture: midline suture from coronal suture to lambdoid suture. Although often assumed to overlie the superior sagittal sinus (SSS), the SSS lies to the right of the sagittal suture in the majority of specimens⁹ (but never by >11 mm).

The most anterior mastoid point lies just in front of the sigmoid sinus.¹⁰

1.3.2 Relation of skull markings to cerebral anatomy – Taylor-Haughton lines

Taylor-Haughton (T-H) lines can be constructed on an angiogram, CT scout film, or skull x-ray, and can then be reconstructed on the patient in the O.R. based on visible external landmarks.¹¹ T-H lines are shown as dashed lines in □ Fig. 1.5.

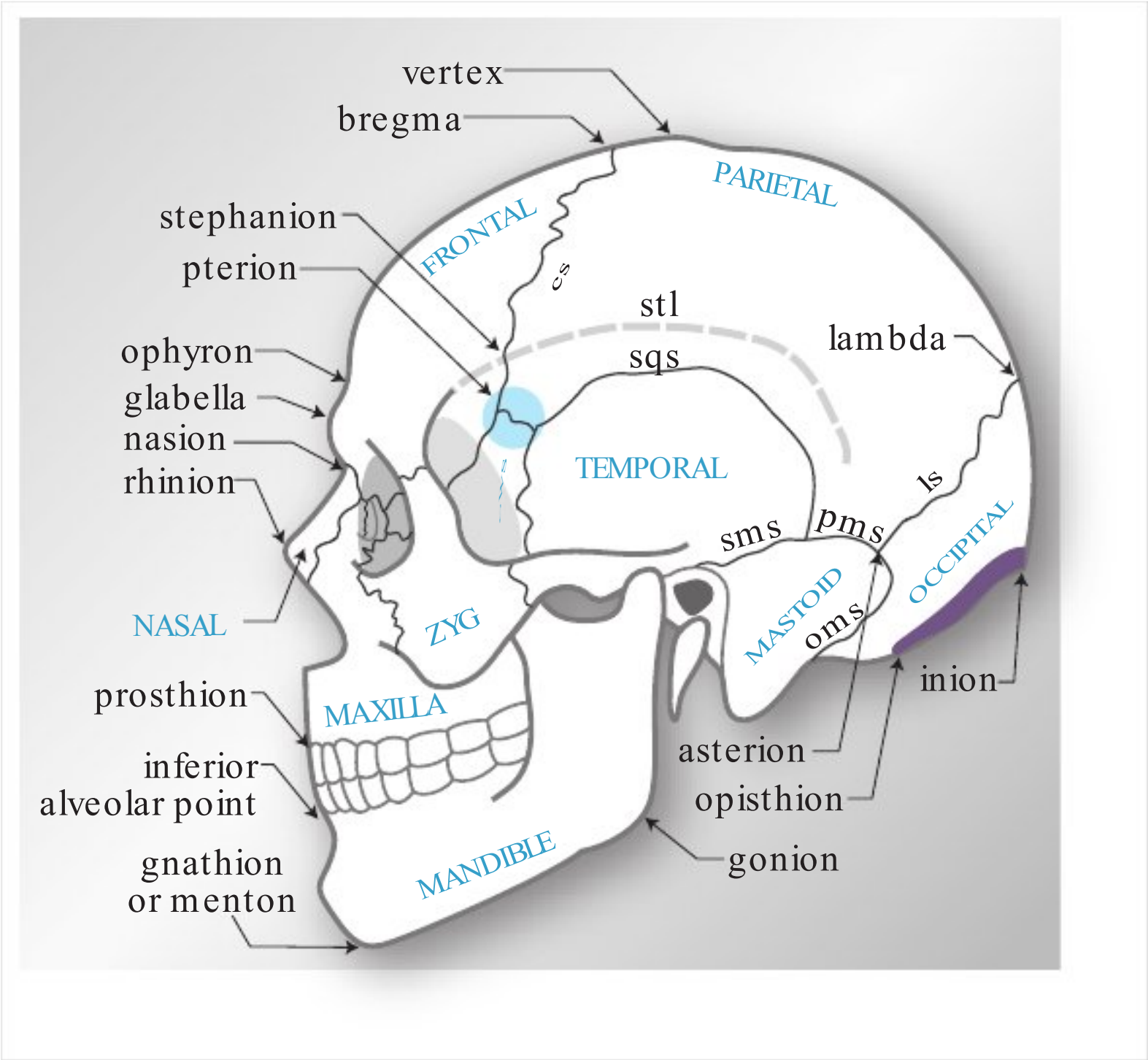


Fig. 1.4 Craniometric points & cranial sutures.
Named bones appear in all upper case letters.
Abbreviations: GWS=greater wing of sphenoid bone, NAS=nasal bone, stl=superior temporal line, ZYG=zygomatic.
Sutures: cs=coronal, ls=lambdoid, oms=occipitomastoid, pms=parietomastoid, sms=squamomastoid, sqs=squamosal

1. Frankfurt plane, AKA baseline: line from inferior margin of orbit through the upper margin of the external auditory meatus (EAM) (as distinguished from Reid's base line: from inferior orbital margin through the center of the EAM)^{12 (p 313)}
2. the distance from the nasion to the inion is measured across the top of the calvaria and is divided into quarters (can be done simply with a piece of tape which is then folded in half twice)
3. posterior ear line: perpendicular to the baseline through the mastoid process
4. condylar line: perpendicular to the baseline through the mandibular condyle
5. T-H lines can then be used to approximate the sylvian fissure (see below) and the motor cortex (also see below)

Sylvian fissure AKA lateral fissure

Approximated by a line connecting the lateral canthus to the point 3/4 of the way posterior along the arc running over convexity from nasion to inion (T-H lines).

Angular gyrus

Located just above the pinna, important on the dominant hemisphere as part of Wernicke's area. Note: there is significant individual variability in the location.²

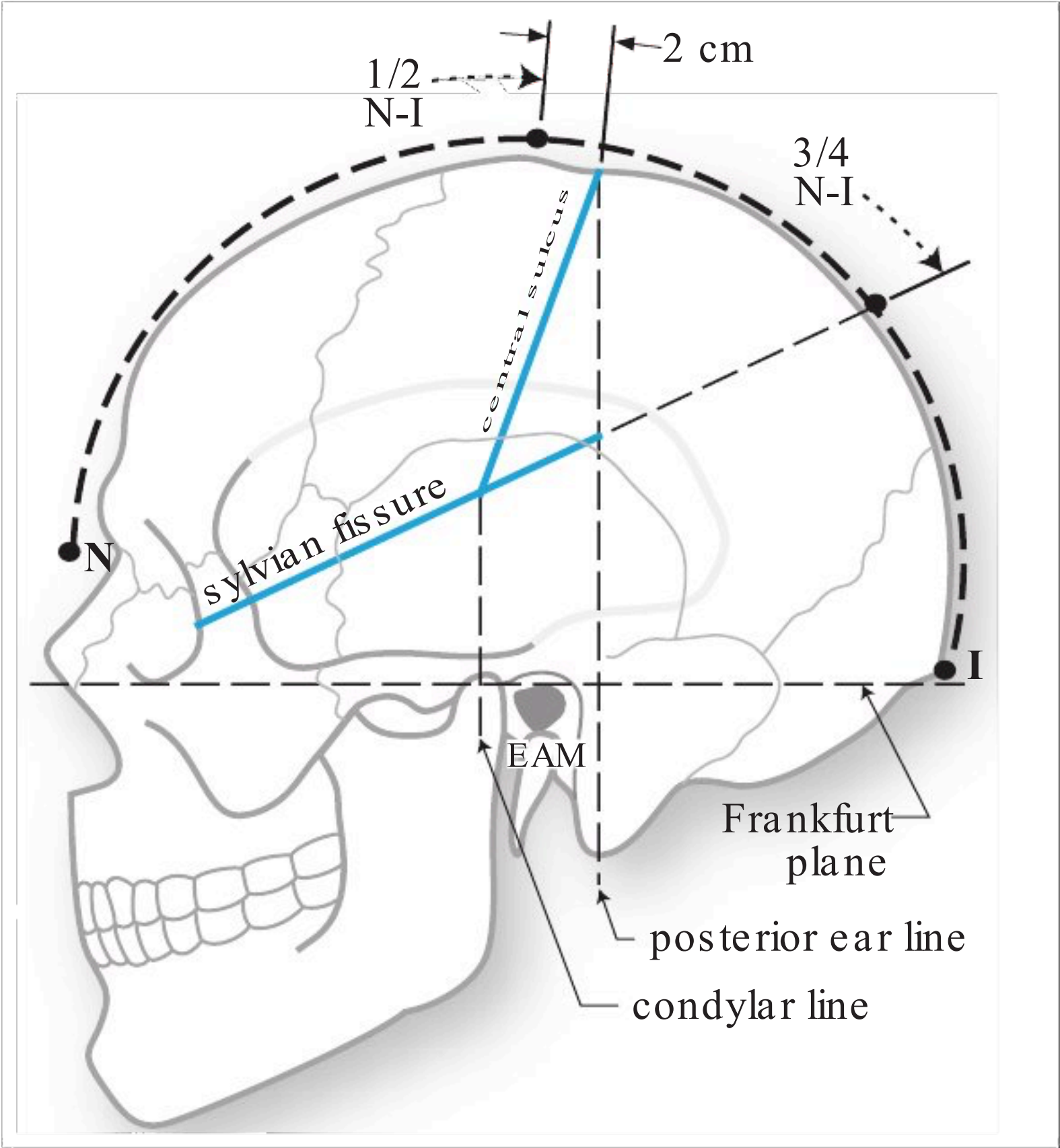


Fig. 1.5 Taylor-Haughton lines and other localizing methods

Angular artery

Located 6 cm above the EAM.

Motor cortex

Numerous methods utilize external landmarks to locate the motor strip (pre-central gyrus) or the central sulcus (Rolandic fissure) which separates motor strip anteriorly from primary sensory cortex posteriorly. These are just approximations since individual variability causes the motor strip to lie anywhere from 4 to 5.4 cm behind the coronal suture.¹³ The central sulcus cannot even be reliably identified visually at surgery.¹⁴

- 1. method 1: the superior aspect of the motor cortex is almost straight up from the EAM near the midline
- 2. method 2¹⁵: the central sulcus is approximated by connecting:
 - a) the point 2 cm posterior to the midposition of the arc extending from nasion to inion (illustrated in □ Fig. 1.5), to
 - b) the point 5 cm straight up from the EAM

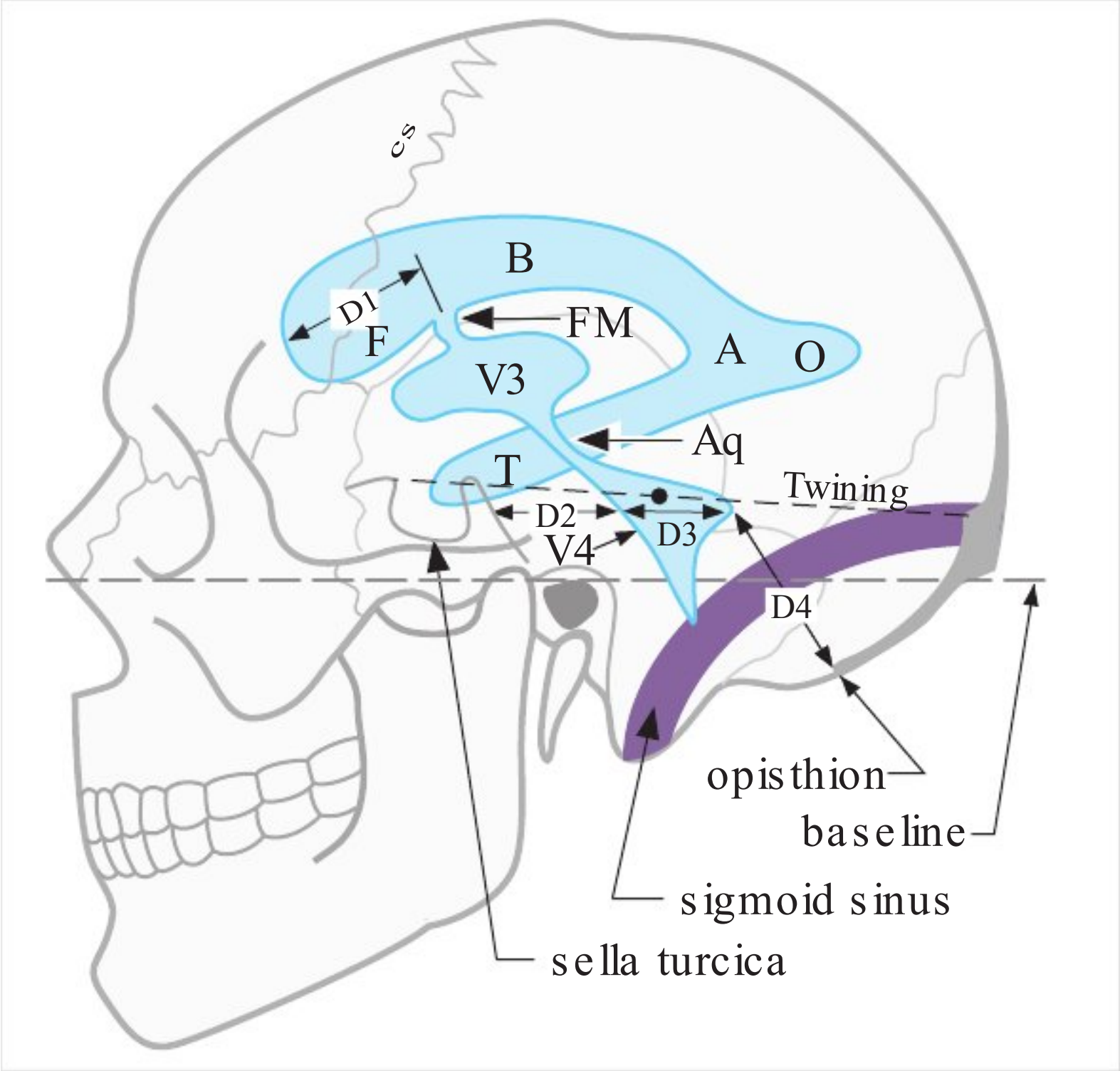


Fig. 1.6 Relationship of ventricles to skull landmarks
Abbreviations: (F=frontal horn, B=body, A=atrium, O=occipital horn, T=temporal horn) of lateral ventricle. FM=foramen of Monro. Aq=sylvian aqueduct. V3=third ventricle. V4=fourth ventricle. cs=coronal suture. Dimensions D1–4 see □ Table 1.3

- 3. method 3: using T-H lines, the central sulcus is approximated by connecting:
 - a) the point where the “posterior ear line” intersects the circumference of the skull (□ Fig. 1.5; usually about 1 cm behind the vertex, and 3–4 cm behind the coronal suture), to
 - b) the point where the “condylar line” intersects the line representing the sylvian fissure
- 4. method 4: a line drawn 45° to Reid’s base line starting at the pterion points in the direction of the motor strip^{16 (p 584–5)}

1.3.3 Relationship of ventricles to skull

□ Fig. 1.6 shows the relationship of non-hydrocephalic ventricles to the skull in the lateral view. Some dimensions of interest are shown in □ Table 1.3.¹⁷

In the non-hydrocephalic adult, the lateral ventricles lie 4–5 cm below the outer skull surface. The center of the body of the lateral ventricle sits in the midpupillary line, and the frontal horn is intersected by a line passing perpendicular to the calvaria along this line.¹⁸ The anterior horns extend 1–2 cm anterior to the coronal suture.

Average length of third ventricle≈ 2.8 cm.
The midpoint of Twining’s line (• in □ Fig. 1.6) should lie within the 4th ventricle.

Table 1.3 Dimensions from □ Fig. 1.6				
Dimension (□ Fig. 1.6)	Description	Lower limit (mm)	Average (mm)	Upper limit (mm)
D1	length of frontal horn anterior to FM		25	
D2	distance from clivus to floor of 4th ventricle at level of fastigium ^a	33.3	36.1	40.0
D3	length of 4th ventricle at level of fastigium ^a	10.0	14.6	19.0
D4	distance from fastigium ^a to opisthion	30.0	32.6	40.0
^a the fastigium is the apex of the 4th ventricle within the cerebellum				

Table 1.4 Cervical levels ¹⁹	
Level	Landmark
C1–2	angle of mandible
C3–4	1 cm above thyroid cartilage (≈ hyoid bone)
C4–5	level of thyroid cartilage
C5–6	crico-thyroid membrane
C6	carotid tubercle
C6–7	cricoid cartilage

1.4 Surface landmarks of spine levels

Estimates of cervical levels for anterior cervical spine surgery may be made using the landmarks shown in □ Table 1.4. Intra-operative C-spine x-rays are essential to verify these estimates.

The scapular spine is located at about T2–3.

The inferior scapular pole is ≈ T6 posteriorly.

Intercristal line: a line drawn between the highest point of the iliac crests across the back will cross the midline either at the interspace between the L4 and L5 spinous processes, or at the L4 spinous process itself.

1.5 Cranial foramina and their contents

1.5.1 Summary

Table 1.5 Cranial foramina and their contents ^a	
Foramen	Contents
nasal slits	anterior ethmoidal nn., a. & v
superior orbital fissure	Cr. Nn. III, IV, VI, all 3 branches of VI (ophthalmic division divides into nasociliary, frontal, and lacrimal nerves); superior ophthalmic vv.; recurrent meningeal br. from lacrimal a.; orbital branch of middle meningeal a.; sympathetic filaments from ICA plexus
inferior orbital fissure	Cr. N. V-2 (maxillary div.), zygomatic n.; filaments from pterygopalatine branch of maxillary n.; infraorbital a. & v.; v. between inferior ophthalmic v. & pterygoid venous plexus
foramen lacerum	usually nothing (ICA traverses the upper portion but doesn't enter, 30% have vidian a.)
carotid canal	internal carotid a., ascending sympathetic nerves

Table 1.5 continued	
Foramen	Contents
incisive foramen	descending septal a.; nasopalatine nn.
greater palatine foramen	greater palatine n., a., & v.
lesser palatine foramen	lesser palatine nn.
internal acoustic meatus	Cr. N. VII (facial); Cr. N. VIII (stato-acoustic) – see text & □ Fig. 1.7
hypoglossal canal	Cr. N. XII (hypoglossal); a meningeal branch of the ascending pharyngeal a.
foramen magnum	spinal cord (medulla oblongata); Cr. N. XI (spinal accessory nn.) entering the skull; vertebral aa.; anterior & posterior spinal arteries
foramen cecum	occasional small vein
cribriform plate	olfactory nn.
optic canal	Cr. N. II (optic); ophthalmic a.
foramen rotundum	Cr. N. V2 (maxillary div.), a. of foramen rotundum
foramen ovale	Cr. N. V3 (mandibular div.)+portio minor (motor for CrN V)
foramen spinosum	middle meningeal a. & v.
jugular foramen	internal jugular v. (beginning); Cr. Nn. IX, X, XI
stylomastoid foramen	Cr. N. VII (facial); stylomastoid a.
condyloid foramen	v. from transverse sinus
mastoid foramen	v. to mastoid sinus; branch of occipital a. to dura mater
^a Abbreviations: a. = artery, aa. = arteries, v. = vein, vv. = veins, n. = nerve, nn. = nerves, br. = branch, Cr. N. = cranial nerve, fmn. = foramen, div. = division	

1.5.2 Porus acusticus

AKA internal auditory canal (□ Fig. 1.7)

The filaments of the acoustic portion of VIII penetrate tiny openings of the lamina cribrosa of the cochlear area.²⁰

Transverse crest: separates superior vestibular area and facial canal (above) from the inferior vestibular area and cochlear area (below).²⁰

Vertical crest (AKA Bill’s bar – named after Dr. William House): separates the meatus to the facial canal anteriorly (containing VII and nervus intermedius) from the vestibular area posteriorly (containing the superior division of vestibular nerve). Bill’s bar is deeper in the IAC than the transverse crest.

The “5 nerves” of the IAC:

1. facial nerve (VII) (mnemonic: “7-up” as VII is in superior portion)
2. nervus intermedius: the somatic sensory branch of the facial nerve primarily innervating mechanoreceptors of the hair follicles on the inner surface of the pinna and deep mechanoreceptors of nasal and buccal cavities and chemoreceptors in the taste buds on the anterior 2/3 of the tongue
3. acoustic portion of the VIII nerve (mnemonic: “Coke down” for cochlear portion)
4. superior branch of vestibular nerve: passes through the superior vestibular area to terminate in the utricle and in the ampullæ of the superior and lateral semicircular canals (mnemonic superior = LSU (Lateral & Superior semicircular canals and the Utricle))
5. inferior branch of vestibular nerve: passes through inferior vestibular area to terminate in the saccule

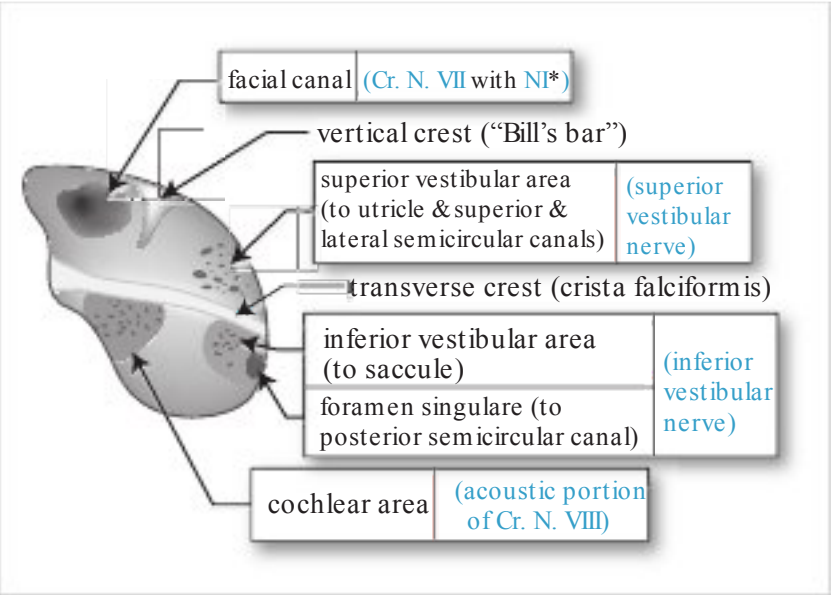


Fig. 1.7 Right internal auditory canal (porus acusticus) & nerves
* NI=nervus intermedius

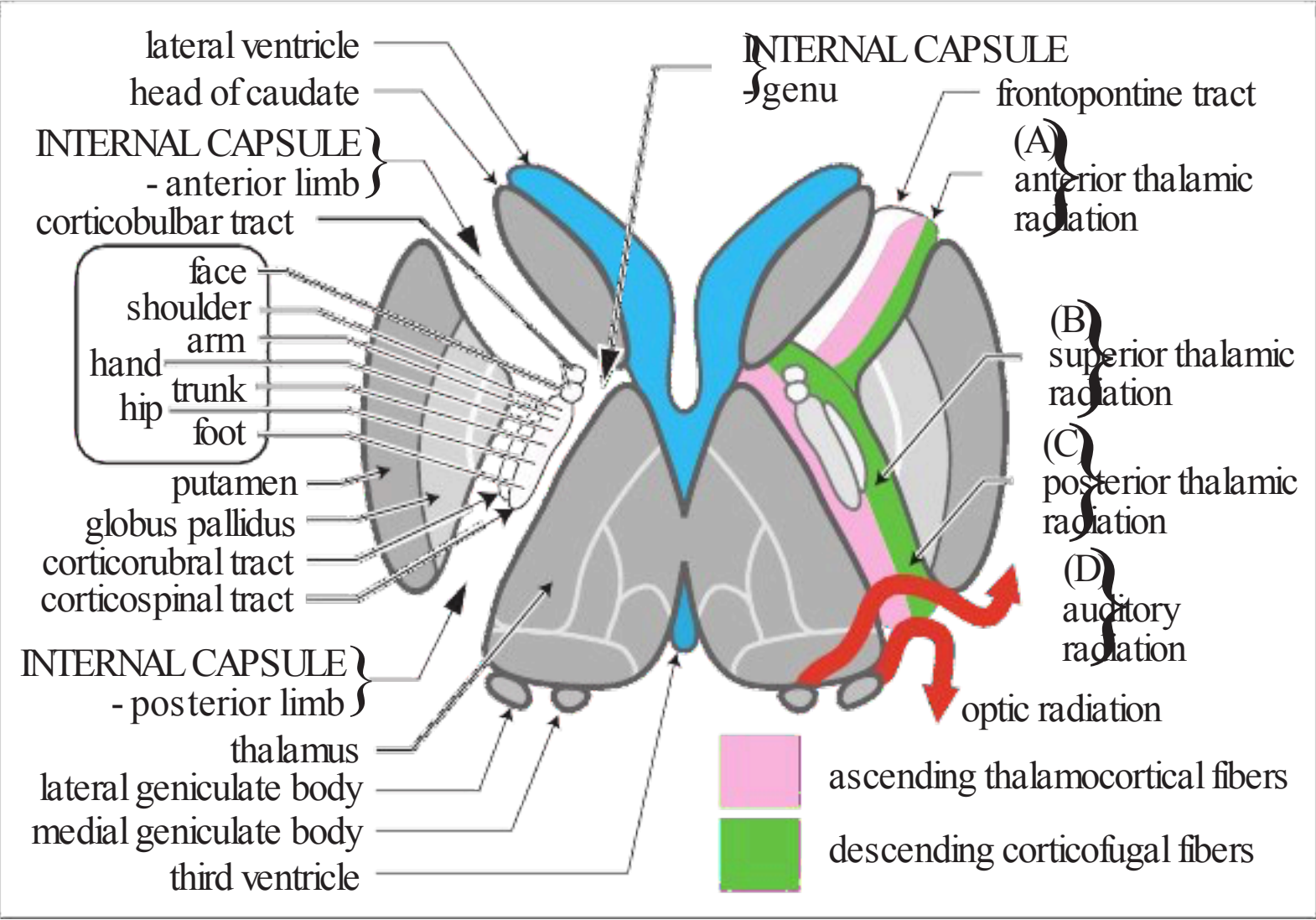


Fig. 1.8 Internal capsule schematic diagram (left side shows tracts, right side shows radiations)

1.6 Internal capsule

1.6.1 Architectural anatomy

For a schematic diagram, □ Fig. 1.8; □ Table 1.6 delineates the thalamic subradiations.
Most IC lesions are caused by vascular accidents (thrombosis or hemorrhage).

1.6.2 Vascular supply of the internal capsule (IC)

1. anterior choroidal: □ all of retrolenticular part (includes optic radiation) and ventral part of posterior limb of IC
2. lateral striate branches (AKA capsular branches) of middle cerebral artery: □ most of anterior AND posterior limbs of IC
3. genu usually receives some direct branches of the internal carotid artery

1.7 Cerebellopontine angle anatomy

For normal anatomy of right cerebellopontine angle, see □ Fig. 1.9.

Table 1.6 Four Thalamic “subradiations” (AKA thalamic peduncles) , labeled A-D in □ Fig. 1.8				
Radiation		Connection		Comments
anterior (A)	medial & anterior thalamic nucleus	↔	frontal lobe	
superior (B)	rolandic areas	↔	ventral thalamic nuclei	general sensory fibers from body & head to terminate in postcentral gyrus (areas 3,1,2)
posterior (C)	occipital & posterior parietal	↔	caudal thalamus	
inferior (D)	transverse temporal gyrus of Heschl	↔	MGB	(small) includes auditory radiation

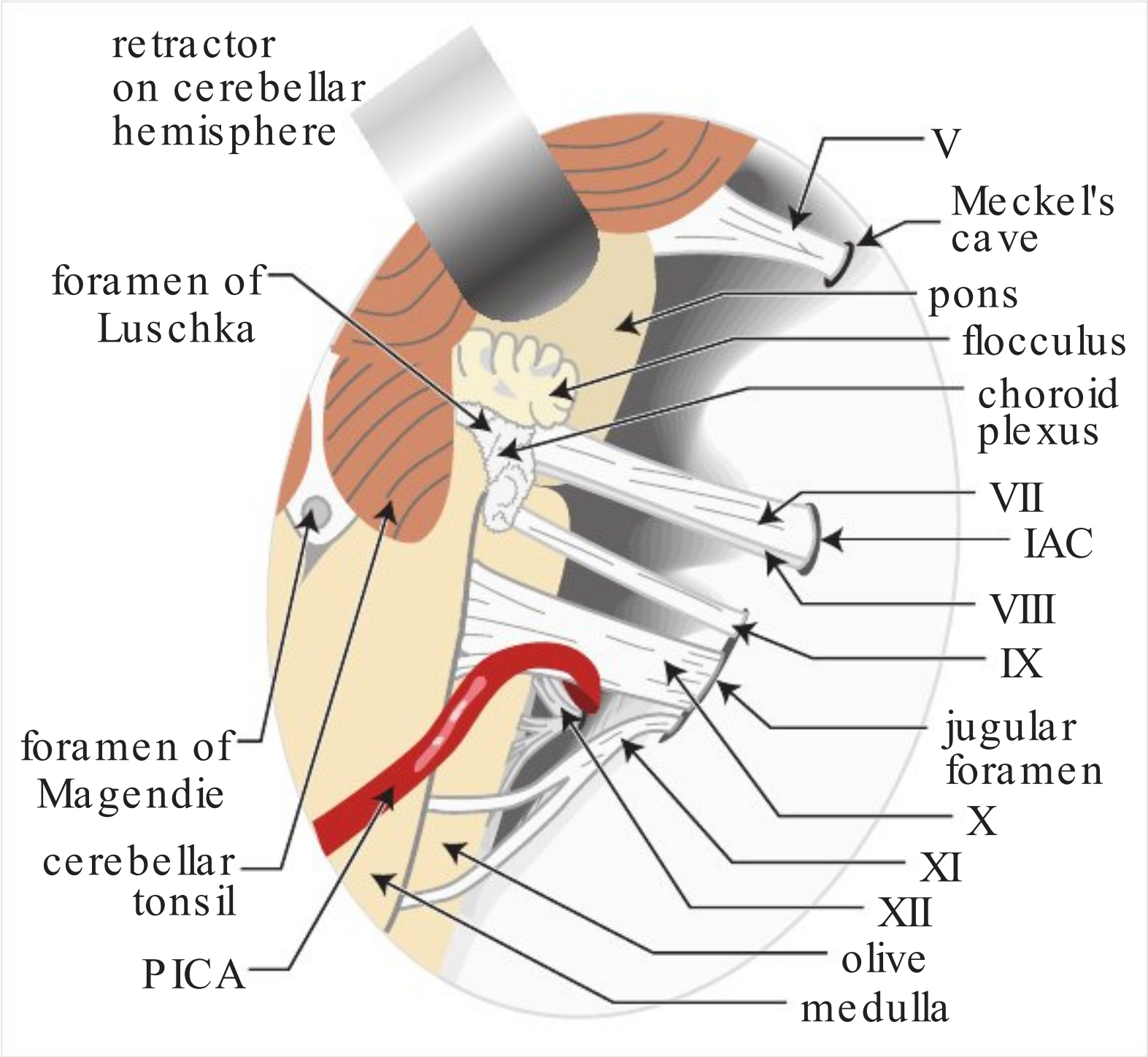


Fig. 1.9 Normal anatomy of right cerebellopontine angle viewed from behind (as in a suboccipital approach)²⁰

1.8 Occipitoatlantoaxial-complex anatomy

□ Ligaments of the occipitoatlantoaxial complex. Stability of the occipitoatlantal joint is primarily due to ligaments, with little contribution from bony articulations and joint capsules (see □ Fig. 1.10, □ Fig. 1.11, □ Fig. 1.12):

- ligaments that connect the atlas to the occiput:
 - anterior atlanto-occipital membrane: cephalad extension of the anterior longitudinal ligament. Extends from anterior margin of foramen magnum (FM) to anterior arch of C1

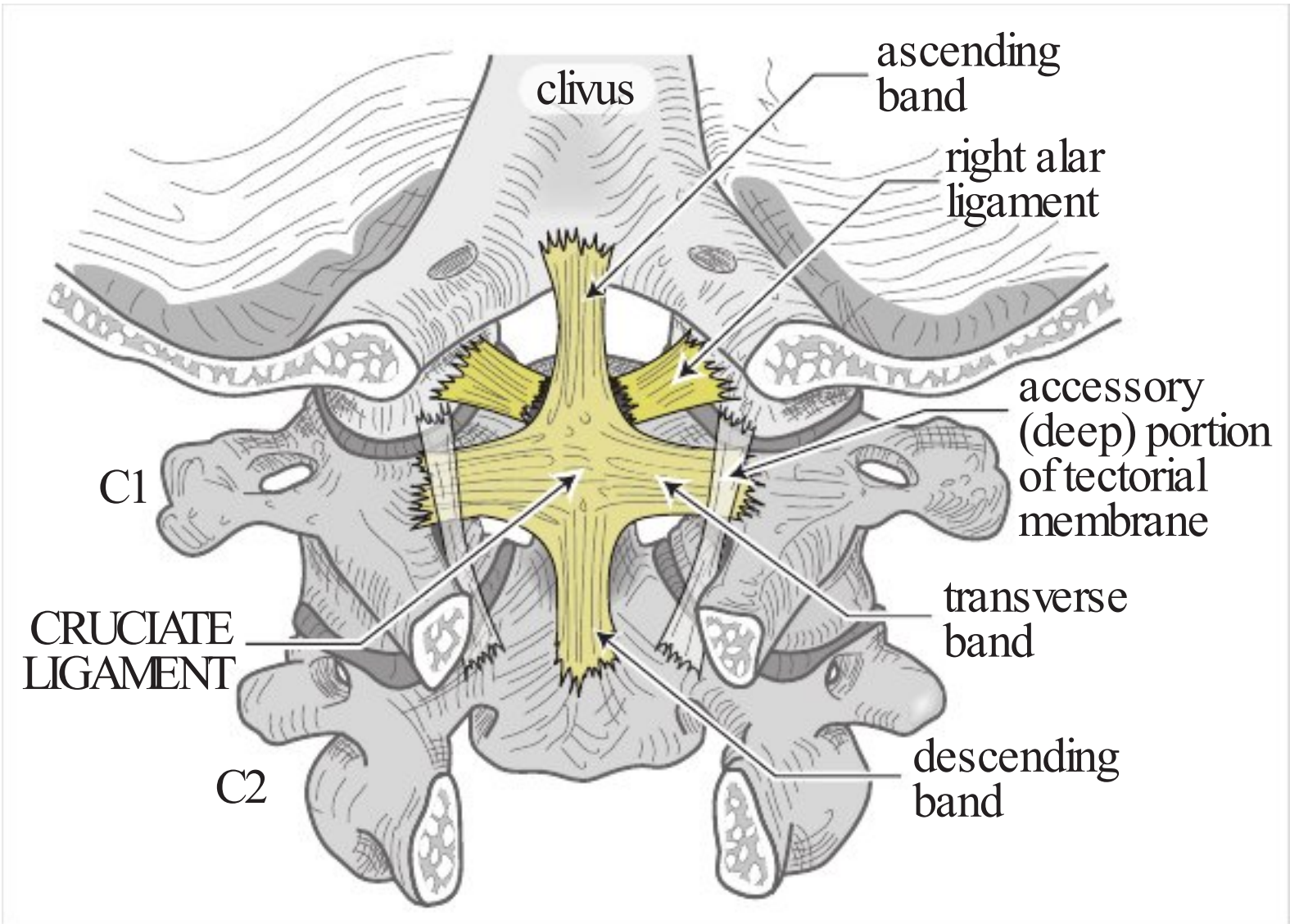


Fig. 1.10 Sagittal view of the ligaments of the craniocervical junction (Modified with permission from “In Vitro Cervical Spine Biomechanical Testing” BNI Quarterly, Vol.9, No. 4, 1993)

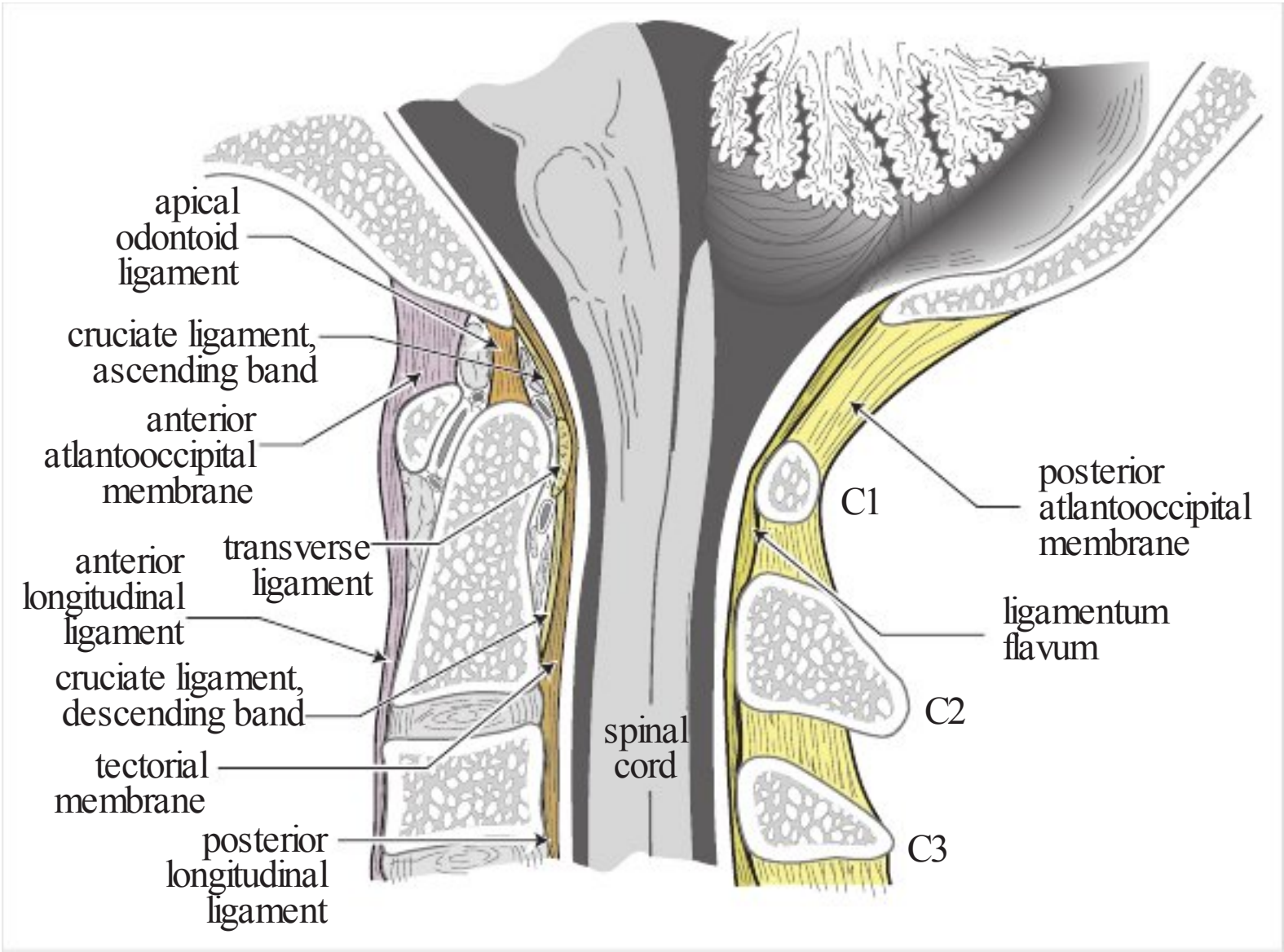


Fig. 1.11 Dorsal view of the cruciate and alar ligaments
Viewed with tectorial membrane removed. (Modified with permission from “In Vitro Cervical Spine Biomechanical Testing” BNI Quarterly, Vol.9, No. 4, 1993)

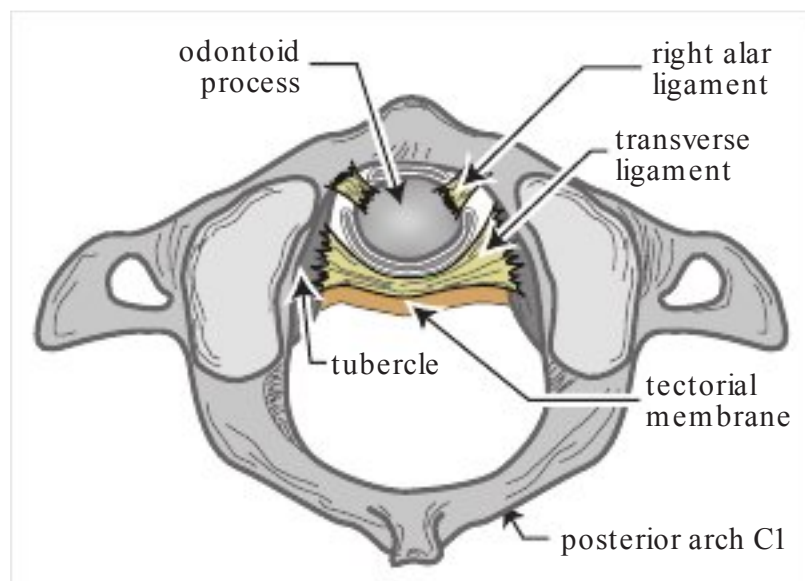


Fig. 1.12 C1 viewed from above, showing the transverse and alar ligaments (Modified with permission from “In Vitro Cervical Spine Biomechanical Testing” BNI Quarterly, Vol.9, No. 4, 199)

- b) posterior atlanto-occipital membrane: connects the posterior margin of the FM to posterior arch of C1
- c) the ascending band of the cruciate ligament
- 2. ligaments that connect the axis (viz. the odontoid) to the occiput:
 - a) tectorial membrane: some authors distinguish 2 components
 - superficial component: cephalad continuation of the posterior longitudinal ligament. A strong band connecting the dorsal surface of the dens to the ventral surface of the FM above, and dorsal surface of C2 & C3 bodies below
 - accessory (deep) portion: located laterally, connects C2 to occipital condyles
 - b) alar (“check”) ligaments²¹
 - occipito-alar portion: connects side of the dens to occipital condyle
 - atlanto-alar portion: connects side of the dens to the lateral mass of C1
 - c) apical odontoid ligament: connects tip of dens to the FM. Little mechanical strength
- 3. ligaments that connect the axis to the atlas:
 - a) transverse (atlantoaxial) ligament: the horizontal component of the cruciate ligament. Traps the dens against the anterior atlas via a strap-like mechanism (□ Fig. 1.12). Provides the majority of the strength (“the strongest ligament of the spine”²²)
 - b) atlanto-alar portion of the alar ligaments (see above)
 - c) descending band of the cruciate ligament

The most important structures in maintaining atlanto-occipital stability are the tectorial membrane and the alar ligaments. Without these, the remaining cruciate ligament and apical dentate ligament are insufficient.

1.9 Spinal cord anatomy

1.9.1 Dentate Ligament

The dentate ligament separates dorsal from ventral nerve roots in the spinal nerves. The spinal accessory nerve (Cr. N. XI) is dorsal to the dentate ligament.

1.9.2 Spinal cord tracts

Anatomy

□ Fig. 1.13 depicts a cross-section of a typical spinal cord segment, combining some elements from different levels (e.g. the intermediolateral grey nucleus is only present from T1 to $\approx$ L1 or L2 where there are sympathetic (thoracolumbar outflow) nuclei). It is schematically divided into ascending and descending halves, however, in actuality, ascending and descending paths coexist on both sides.

□ Fig. 1.13 also depicts some of the laminae according to the scheme of Rexed. Lamina II is equivalent to the substantia gelatinosa. Laminae III and IV are the nucleus proprius. Lamina VI is located in the base of the posterior horn.

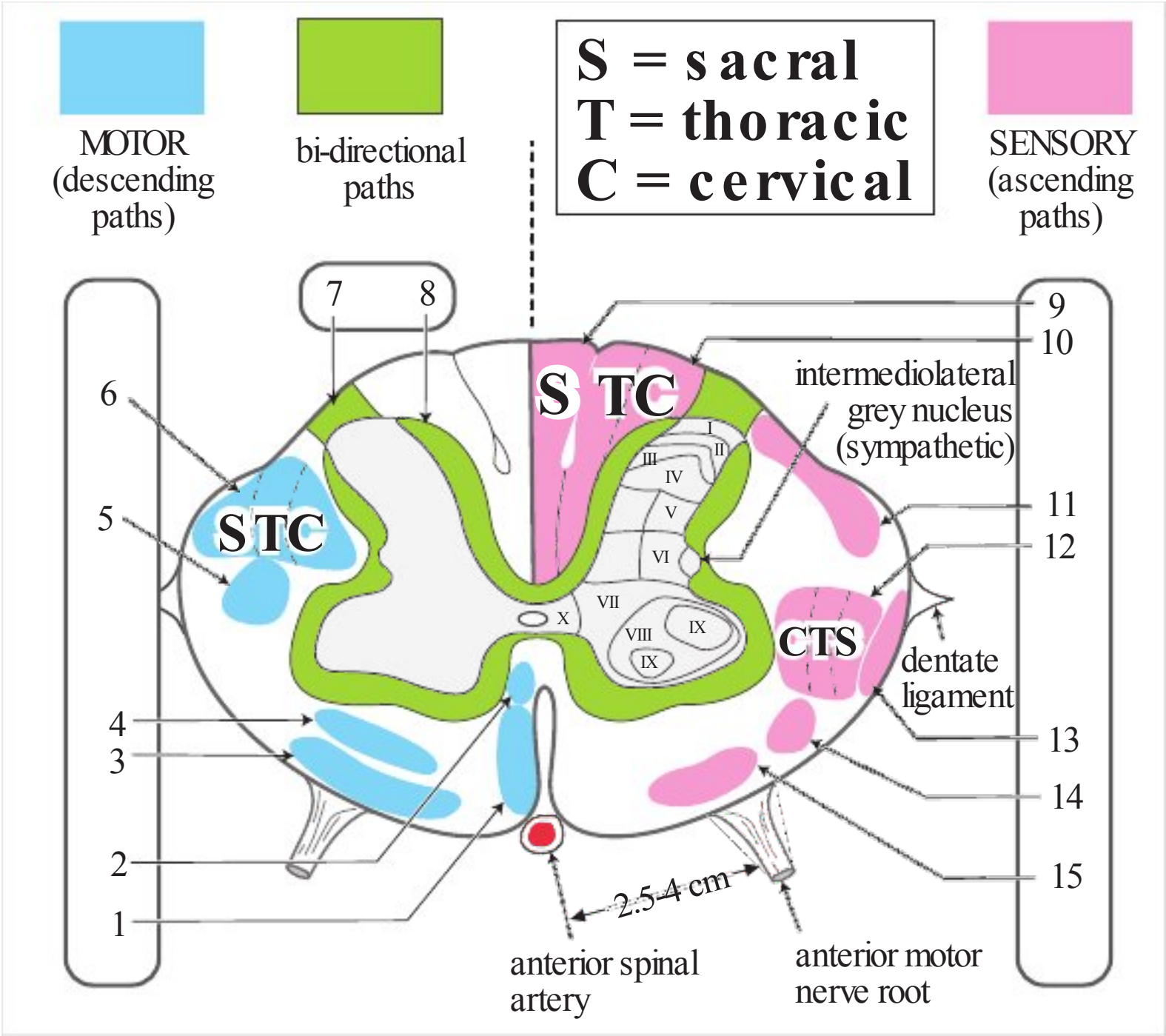


Fig. 1.13 Schematic cross-section of cervical spinal cord. See Table 1.7, Table 1.8 and Table 1.9 for path names.

Table 1.7 Descending (motor) tracts (↓) in Fig. 1.13			
Number (Fig. 1.13)	Path	Function	Side of body
1	anterior corticospinal tract	skilled movement ^a	opposite
2	medial longitudinal fasciculus	?	same
3	vestibulospinal tract	facilitates extensor muscle tone	same
4	medullary (ventrolateral) reticulospinal tract	automatic respirations?	same
5	rubrospinal tract	flexor muscle tone	same
6	lateral corticospinal (pyramidal) tract	skilled movement	same

^aThe terminal fibers of this uncrossed tract usually cross in the anterior white commissure to synapse on alpha motor neurons or on internuncial neurons. It is true that some of these fibers do stay on the same side, but it is felt to be a minority. Also, the anterior corticospinal tract is easily identified only in the cervical and upper thoracic regions.

Table 1.8 Bi-directional tracts in □ Fig. 1.13		
Number (□ Fig. 1.13)	Path	Function
7	dorsolateral fasciculus (of Lissauer)	
8	fasciculus proprius	short spinospinal connections

Table 1.9 Ascending (sensory) tracts (↑) in □ Fig. 1.13			
Number (□ Fig. 1.13)	Path	Function	Side of body
9	fasciculus gracilis	joint position, fine touch, vibration	same
10	fasciculus cuneatus		
11	posterior spinocerebellar tract	stretch receptors	same
12	lateral spinothalamic tract	pain & temperature	opposite
13	anterior spinocerebellar tract	whole limb position	opposite
14	spinotectal tract	unknown, ? nociceptive	opposite
15	anterior spinothalamic tract	light touch	opposite

Sensation

Pain and temperature: body

Receptors: free nerve endings (probable).

1st order neuron: small, finely myelinated afferents; soma in dorsal root ganglion (no synapse). Enter cord at dorsolateral tract (zone of Lissauer). Synapse: substantia gelatinosa (Rexed II).

2nd order neuron axon cross obliquely in the anterior white commissure ascending ≈ 1–3 segments while crossing to enter the lateral spinothalamic tract.

Synapse: VPL thalamus. 3rd order neurons pass through IC to postcentral gyrus (Brodmann’s areas 3, 1, 2).

Fine touch, deep pressure and proprioception: body

Fine touch AKA discriminative touch. Receptors: Meissner’s & pacinian corpuscles, Merkel’s disks, free nerve endings.

1st order neuron: heavily myelinated afferents; soma in dorsal root ganglion (no synapse). Short branches synapse in nucleus proprius (Rexed III & IV) of posterior gray; long fibers enter the ipsilateral posterior columns without synapsing (below T6: fasciculus gracilis; above T6: fasciculus cuneatus).

Synapse: nucleus gracilis/cuneatus (respectively), just above pyramidal decussation. 2nd order neuron axons form internal arcuate fibers, decussate in lower medulla as medial lemniscus.

Synapse: VPL thalamus. 3rd order neurons pass through IC primarily to postcentral gyrus.

Light (crude) touch: body

Receptors: as fine touch (see above), also peritrichial arborizations.

1st order neuron: large, heavily myelinated afferents (Type II); soma in dorsal root ganglion (no synapse). Some ascend uncrossed in post. columns (with fine touch); most synapse in Rexed VI & VII.

2nd order neuron axons cross in anterior white commissure (a few don’t cross); enter anterior spinothalamic tract.

Synapse: VPL thalamus. 3rd order neurons pass through IC primarily to postcentral gyrus.

1.9.3 Dermatomes and sensory nerves

Dermatomes are ares of the body where sensation is suberved by a single nerve root.

Peripheral nerves generally receive contributions from more than one dermatome.

Lesions in peripheral nerves and lesions in nerve roots may sometimes be distinguished in part by the pattern of sensory loss. A classic example is splitting of the ring finger in median nerve or ulnar nerve lesions, which does not occur in C8 nerve root injuries.

□ Fig. 1.14 shows anterior and posterior view, each schematically separated into sensory dermatomes (segmental) and peripheral sensory nerve distribution.

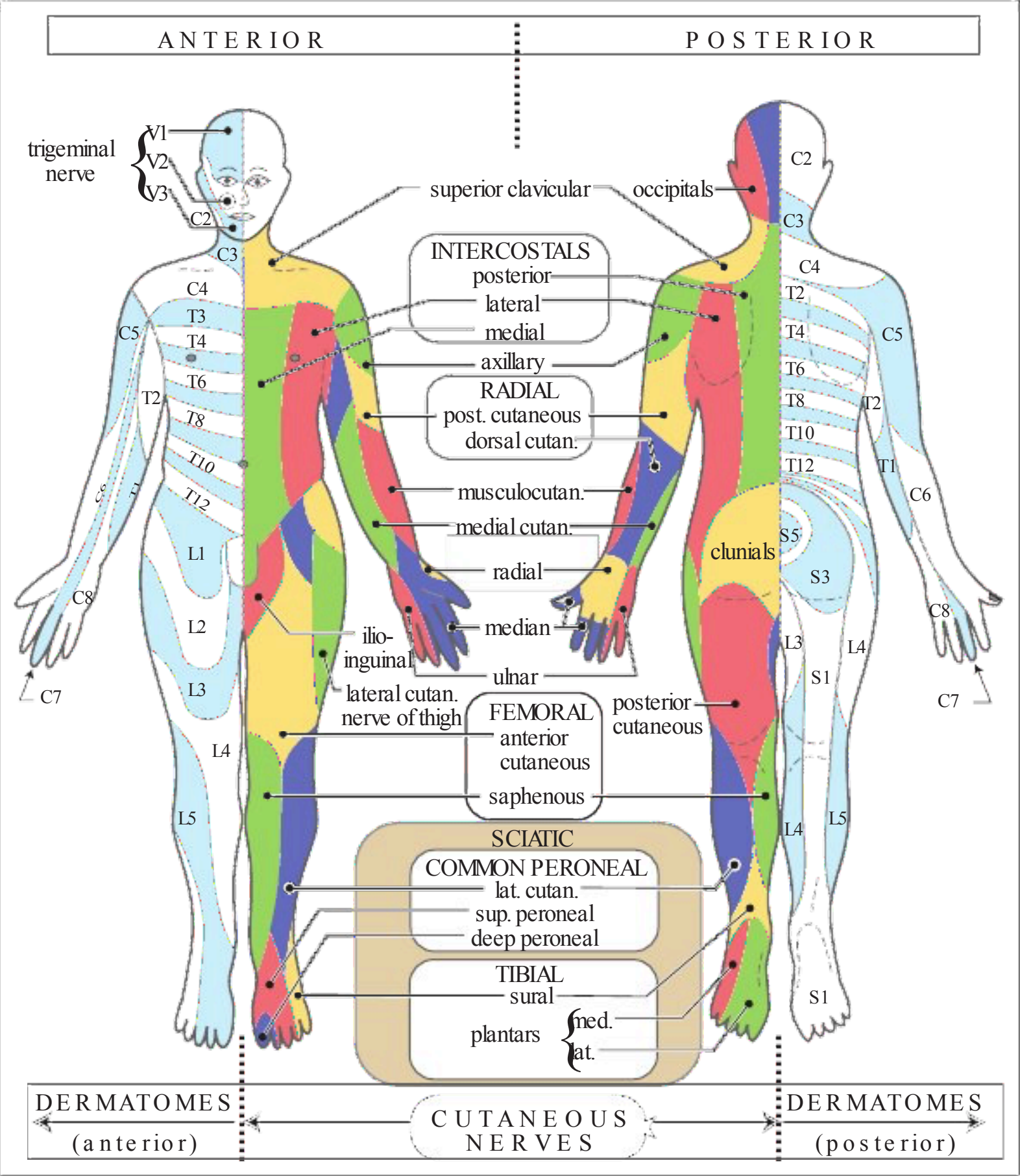


Fig. 1.14 Dermatomal and sensory nerve distribution (Redrawn from "Introduction to Basic Neurology", by Harry D. Patton, John W. Sundsten, Wayne E. Crill and Phillip D. Swanson, © 1976, pp 173, W. B. Saunders Co., Philadelphia, PA, with permission)

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2 Vascular Anatomy

2.1 Cerebral vascular territories

□ Fig. 2.1 depicts approximate vascular distributions of the major cerebral arteries. There is considerable variability of the major arteries¹ as well as the central distribution. The lenticulostriates may have origins off of different segments of the middle or anterior cerebral artery). Recurrent artery of Heubner (RAH) (AKA medial striate artery) origin: junction of the ACA and a-comm in 62.3% proximal A2 in 23.3% A1 in 14.3%²

2.2 Cerebral arterial anatomy

2.2.1 General information

The symbol “□” is used to denote a region supplied by the indicated artery. See Angiography (cerebral) (p.236) for angiographic diagrams of the following anatomy.

2.2.2 Circle of Willis

See □ Fig. 2.2. A balanced configuration of the Circle of Willis is present in only 18% of the population. Hypoplasia of 1 or both p-comms occurs in 22–32% absent or hypoplastic A1 segments occurs in 25%

Key point: the anterior cerebral arteries pass over the superior surface of the optic chiasm.

2.2.3 Anatomical segments of intracranial cerebral arteries

1. carotid artery: the traditional numbering system³ was from rostral to caudal (counter to the direction of flow, and to the numbering scheme of the other arteries). A number of systems have been described to address this inconsistency and also to identify anatomically important segments of the ICA that were not originally delineated (e.g. see □ Table 2.1⁴). Also see below for more detail

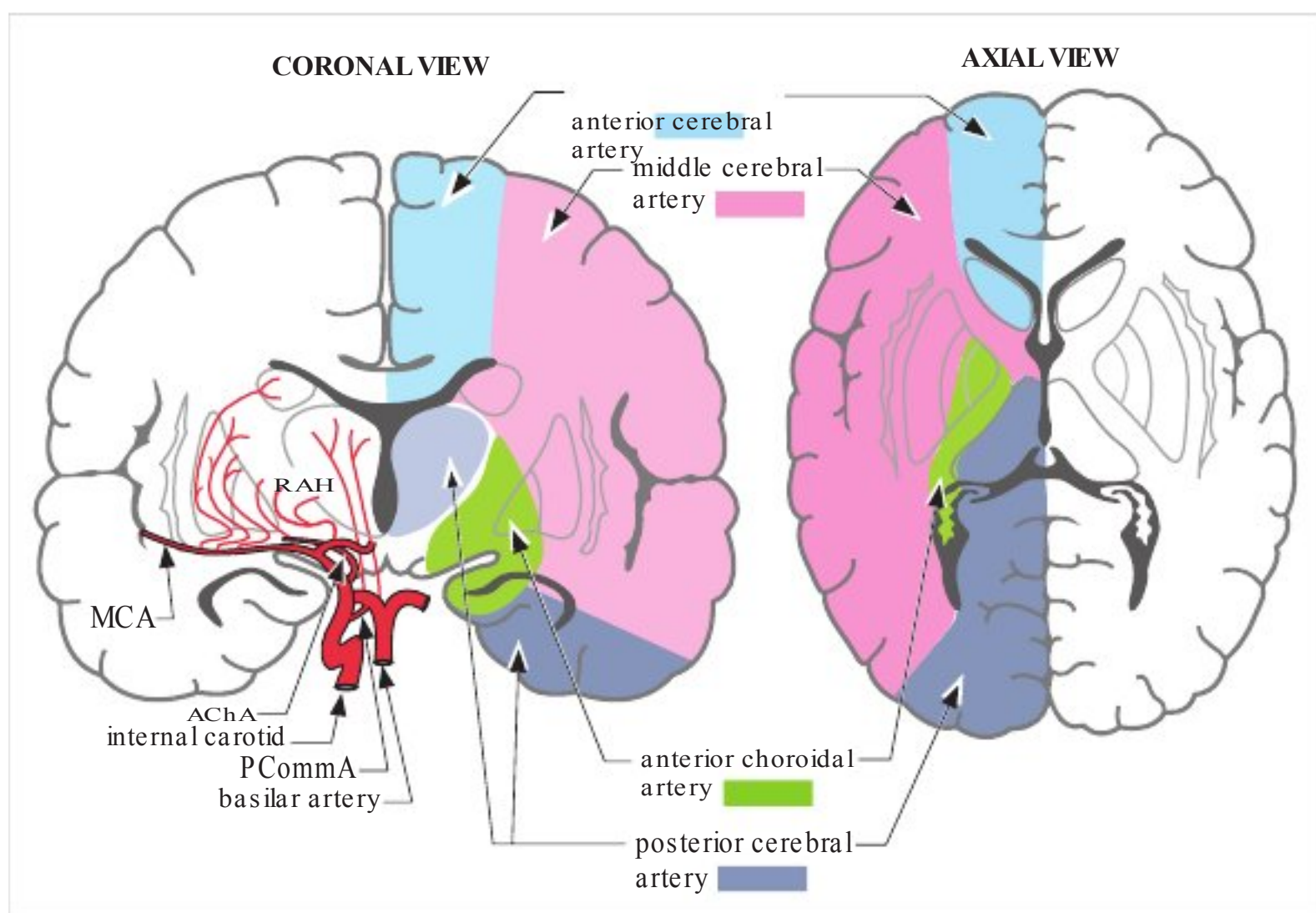


Fig. 2.1 Vascular territories of the cerebral hemispheres. RAH=recurrent artery of Heubner.

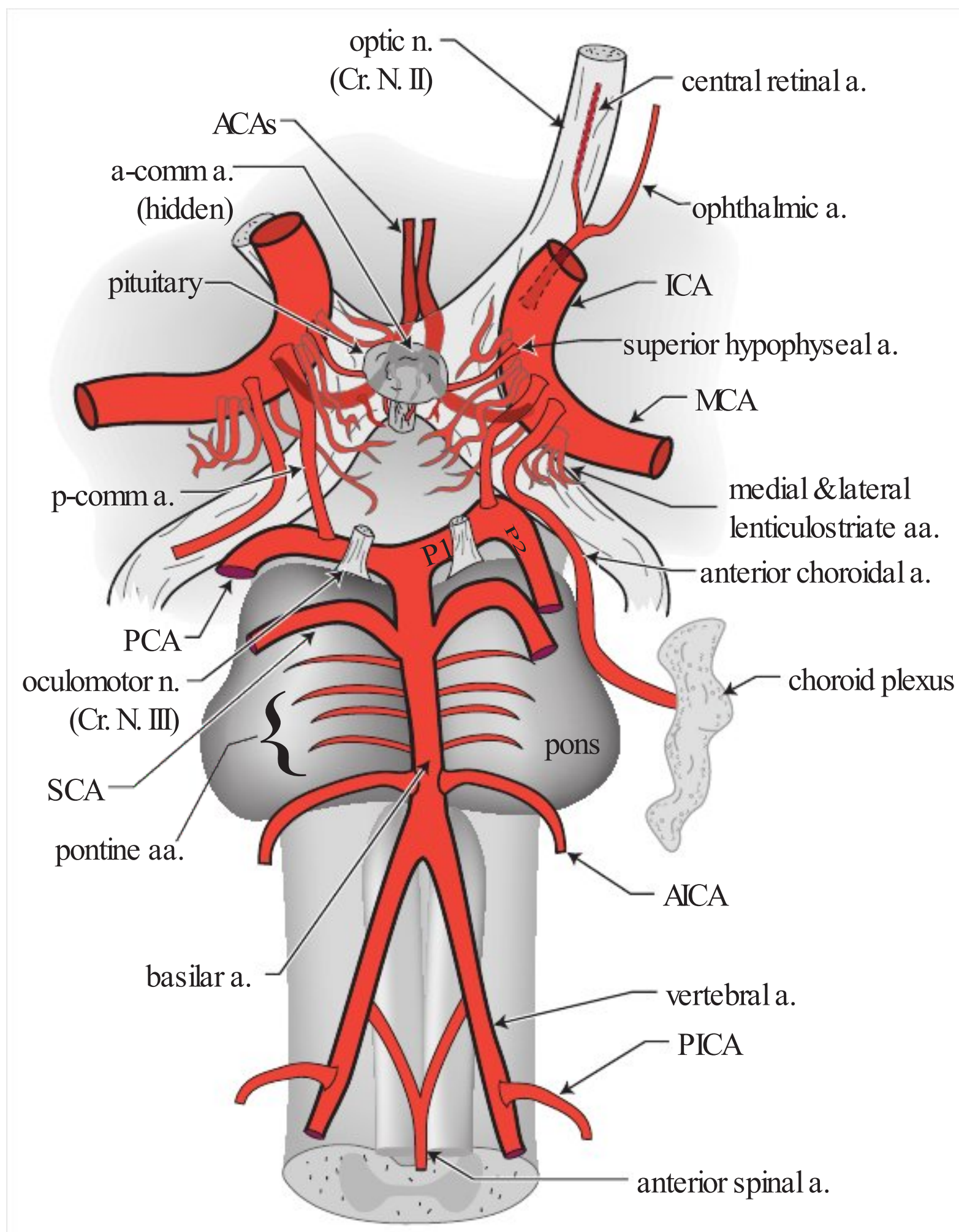


Fig. 2.2 Circle of Willis viewed from in front of and below the brain

2. anterior cerebral⁵:

- A1 (precommunicating): ACA from origin to ACoA
- A2 (postcommunicating): ACA from ACoA to branch-point of callosomarginal
- A3 (precallosal): from branch-point of callosomarginal curving around the genu of the corpus callosum to superior surface of corpus callosum 3 cm posterior to the genu
- A4: (supracallosal)
- A5: terminal branch (postcallosal)

3. middle cerebral⁶:

- M1: MCA from origin to bifurcation (horizontal segment on AP angiogram). A classical bifurcation into relatively symmetrical superior and inferior trunks is seen in 50%, no bifurcation occurs in 2%, 25% have a very proximal branch (middle trunk) arising from the superior (15%) or the inferior (10%) trunk creating a "pseudo-trifurcation", a pseudo-tetrafurcation occurs in 5%

Table 2.1 Segments of the ICA	
Cincinnati system	System of Fischer
C1 (cervical)	Not described
C2 (petrous)	
C3 (lacerum)	C5
C4 (cavernous)	C4 + part of C5
C5 (clinoid)	C3
C6 (ophthalmic)	C2
C7 (communicating)	C1

- lateral fronto-orbital and prefrontal branches arise from M1 or superior M2 trunk
 - precentral, central, anterior and posterior parietal arteries arise from a superior (60%) or middle (25%) or inferior (15%) trunk
 - the superior M2 trunk does not give any branches to the temporal lobe
- b) M2: MCA trunks from bifurcation to emergence from Sylvian fissure
- c) M3–4: distal branches
- d) M5: terminal branch
4. posterior cerebral (PCA) (several nomenclature schemes exist^{5,7}):
- a) P1: PCA from the origin to posterior communicating artery (AKA mesencephalic, precommunicating, circular, peduncular, basilar...). The long and short circumflex and thalamoperforating arteries arise from P1
- b) P2: PCA from origin of p-comm to the origin of inferior temporal arteries (AKA ambient, postcommunicating, perimesencephalic), P2 traverses the ambient cistern, Hippocampal, anterior temporal, peduncular perforating and medial posterior choroidal arteries arise from P2
- c) P3: PCA from the origin of the inferior temporal branches to the origin of the terminal branches (AKA quadrigeminal segment). P3 traverses the quadrigeminal cistern
- d) P4: segment after the origin of the parieto-occipital and calcarine arteries, includes the cortical branches of the PCA

2.2.4 Anterior circulation

Anatomic variants

Bovine circulation: the common carotids arise from a common trunk off the aorta.

External carotid

1. superior thyroid a.: 1st anterior branch
2. ascending pharyngeal a.
 - a) neuromeningeal trunk of the ascending pharyngeal a.: supplies IX, X & XI (important when embolizing glomus tumors, 20% of lower cranial nerve palsy if this branch is occluded)
 - b) pharyngeal branch: usually the primary feeder for jugular foramen tumors (essentially the only cause of hypertrophy of the ascending pharyngeal a.)
3. lingual a.
4. facial a.: branches anastomose with ophthalmic a.; important in collateral flow with ICA occlusion (p. 1265)
5. occipital a. □ posterior scalp
6. posterior auricular
7. superficial temporal
 - a) frontal branch
 - b) parietal branch
8. (internal) maxillary a. – initially within parotid gland
 - a) middle meningeal a.
 - anterior branch
 - posterior branch

- b) accessory meningeal
- c) inferior alveolar
- d) infra-orbital
- e) others: distal branches of which may anastomose with branches of ophthalmic artery in the orbit

Internal carotid artery (ICA)

Lies posterior & medial to the external carotid (ECA).

Segments of the ICA and its branches

See □ Fig. 2.3 for branches, and reference.⁴

1. C1 (cervical): begins in neck at carotid bifurcation where the common carotid artery divides into internal and external carotids. Travels in carotid sheath with IJV and vagal nerve, encircled with postganglionic sympathetic nerves (PGSN). C1 ends where the ICA enters carotid canal of petrous bone. No branches
2. C2 (petrous): still surrounded by PGSNs. Ends at the posterior edge of the foramen lacerum (f-Lac) (inferomedial to the edge of the Gasserian ganglion in Meckel's cave). Three divisions:

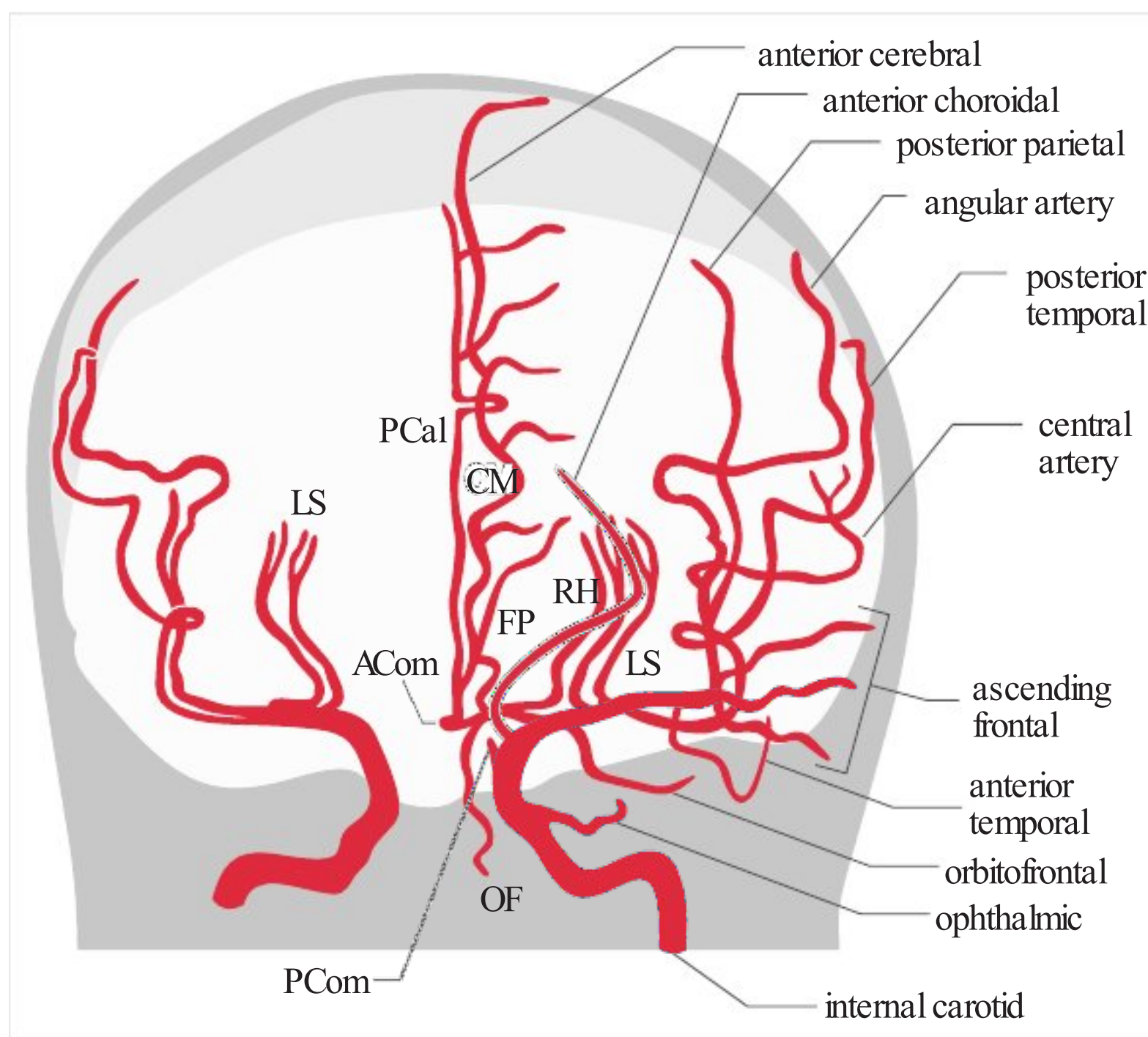


Fig. 2.3 Internal carotid arteriogram (AP view).

ACom: anterior communicating artery

CM: callosomarginal artery

FP: frontopolar artery

LS: lenticulostriate arteries

OF: orbitofrontal artery

PCal: pericallosal artery

PCom: posterior communicating artery

RH: recurrent artery of Heubner (Reprinted courtesy of Eastman Kodak Company)

- a) vertical segment: ICA ascends then bends as the...
- b) posterior loop: anterior to cochlea, bends antero-medially becoming the...
- c) horizontal segment: deep and medial to greater and lesser superficial petrosal nerves, anterior to tympanic membrane (TM)
3. C3 (lacerum): the ICA passes over (but not through) the f-Lac forming the lateral loop. Ascends in the canalicular portion of the f-Lac to juxtaseal position, piercing the dura as it passes the petrolingual ligament to become the cavernous segment. Branches (usually not visible angiographically):
 - a) caroticotympanic (inconsistent) □ tympanic cavity
 - b) pterygoid (vidian) branch: passes through foramen lacerum, present in only 30% may continue as artery of pterygoid canal
4. C4 (cavernous): covered by vascular membrane lining sinus, still surrounded by PGSNs. Passes anteriorly then supero-medially, bends posteriorly (medial loop of ICA), travels horizontally, and bends anteriorly (part of anterior loop of ICA) to anterior clinoid process. Ends at the proximal dural ring (incompletely encircles ICA). Many branches, main ones include:
 - a) meningohypophyseal trunk (MHT) (largest & most proximal). 2 causes of a prominent MHT:
 - 1) tumor (usually petroclival meningioma – see below), 2) dural AVM (p.1251)
 - a. of tentorium (AKA artery of Bernasconi & Cassinari): the blood supply of petroclival meningiomas
 - dorsal meningeal a. (AKA dorsal clival a.)
 - inferior hypophyseal a. (□ posterior lobe of pituitary): post-partum occlusion causes pituitary infarcts (Sheehan's necrosis), however, DI is rare because the stalk is spared
 - b) anterior meningeal a.
 - c) a. to inferior portion of cavernous sinus (present in 80%)
 - d) capsular aa. of McConnell (in 30%): supply the capsule of the pituitary⁸
5. C5 (clinoid): begins at proximal dural ring, ends at distal dural ring (which completely encircles ICA) where the ICA becomes intradural
6. C6 (ophthalmic): begins at distal dural ring, ends just proximal to p-comm. Branches:
 - a) ophthalmic a.: the origin from the ICA is distal to the cavernous sinus in 89%(intracavernous in 8% the ophthalmic artery is absent in 3%) and can vary from 5 mm anterior to 7 mm posterior to the anterior clinoid.⁸ Passes through the optic canal into the orbit (the intracranial course is very short, usually 1–2 mm⁸). Has a characteristic bayonet-like “kink” on lateral angiogram
 - b) superior hypophyseal a. branches □ anterior lobe of pituitary & stalk (1st branch of supraclinoid ICA)
7. C7 (communicating): begins just proximal to p-comm origin, travels between Cr. N. II & III, terminates just below anterior perforated substance where it bifurcates into the ACA & MCA
 - a) posterior communicating a. (p-comm)
 - few anterior thalamoperforators (□ optic tract, chiasm & posterior hypothalamus): below
 - plexal segment: enters supracornual recess of temporal horn, □ only this portion of choroid plexus
 - cisternal segment: passes through crural cistern
 - b) anterior choroidal artery¹⁰: takeoff 2–4 mm distal to p-comm □ (variable) portion of optic tract, medial globus pallidus, genu of internal capsule (IC) (in 50%), inferior half of posterior limb of IC, uncus, retrolenticular fibers (optic radiation), lateral geniculate body; for occlusion syndromes (p.1265)
8. “Carotid siphon”: not a segment, but a region incorporating the cavernous, ophthalmic and communicating segments. Begins at the posterior bend of the cavernous ICA, and ends at the ICA bifurcation

Differentiating p-comm from ACh on arteriogram

1. p-comm origin is proximal to that of the anterior choroidal artery (ACh)
2. p-comm is usually larger than ACh
3. p-comm usually goes up or down a little, then straight back & usually bifurcates
4. ACh usually has a superior “hump” (plexal point) where it pass through the choroidal fissure to enter the ventricle

Anterior cerebral artery (ACA)

Passes between Cr. N. II and anterior perforated substance. See □ Fig. 2.4. Branches:

1. recurrent artery (of Heubner): typically arises from the area of the A1/A2 junction. Various statistics can be found in the literature regarding the percentage that arise from distal A1 vs. proximal A2.¹¹ It is most important to be mindful that the takeoff is variable, e.g. when treating aneurysms (one of the larger medial lenticulostriates, remainder of lenticulostriates may arise from this artery) □ head of caudate, putamen, and anterior internal capsule
2. medial orbitofrontal artery
3. frontopolar artery
4. callosomarginal
 - a) internal frontal branches
 - anterior
 - middle
 - posterior
 - b) paracentral artery
5. pericallosal artery (continuation of ACA)
 - a) superior internal parietal (precuneate) artery
 - b) inferior internal parietal artery

Anatomic variants

Hypoid: having only one anterior cerebral artery (as in a horse).

Middle cerebral artery (MCA)

See □ Fig. 2.5 and anatomy (p. 76). Branches vary widely, 10 common ones:

1. medial (3–6 per side) and lateral lenticulostriate arteries
2. anterior temporal
3. posterior temporal
4. lateral orbitofrontal
5. ascending frontal (candelabra)
6. precentral (prerolandic)
7. central (rolandic)
8. anterior parietal (postrolandic)
9. posterior parietal
10. angular

Posterior circulation

Anatomic variants

Fetal circulation: 15–35% of patients supply their posterior cerebral artery on one or both sides primarily from the carotid (via p-comm) instead of via the vertebrobasilar system.

Vertebral artery (VA)

The VA is the first and usually the largest branch of the subclavian artery. Variant: the left VA arises off the aortic arch in ≈ 4%. Diameter ≈ 3 mm. Mean blood flow ≈ 150 ml/min. The left VA is dominant in 60%. The right VA will be hypoplastic in 10% and the left will be hypoplastic in 5%. The VA is atretic and does not communicate with the BA on the left in 3% and on the right in 2% (the VA may terminate in PICA).

Four segments:

- V1 prevertebral: from subclavian artery, courses superiorly and posteriorly and enters the foramen transversarium, usually of the 6th vertebral body
- V2 ascends vertically within the transverse foramina of the cervical vertebrae surrounded by sympathetic fibers (from the stellate ganglion) and a venous plexus. It is situated anterior to the cervical roots. It turns laterally to enter the foramen within the transverse process of the axis
- V3 exits the foramen of the axis and curves posteriorly and medially in a groove on the upper surface of the atlas and enters the foramen magnum
- V4 pierces the dura (location somewhat variable) and immediately enters the subarachnoid space. Joins the contralateral VA at the vertebral confluens located at the lower pontine border to form the basilar artery (BA)

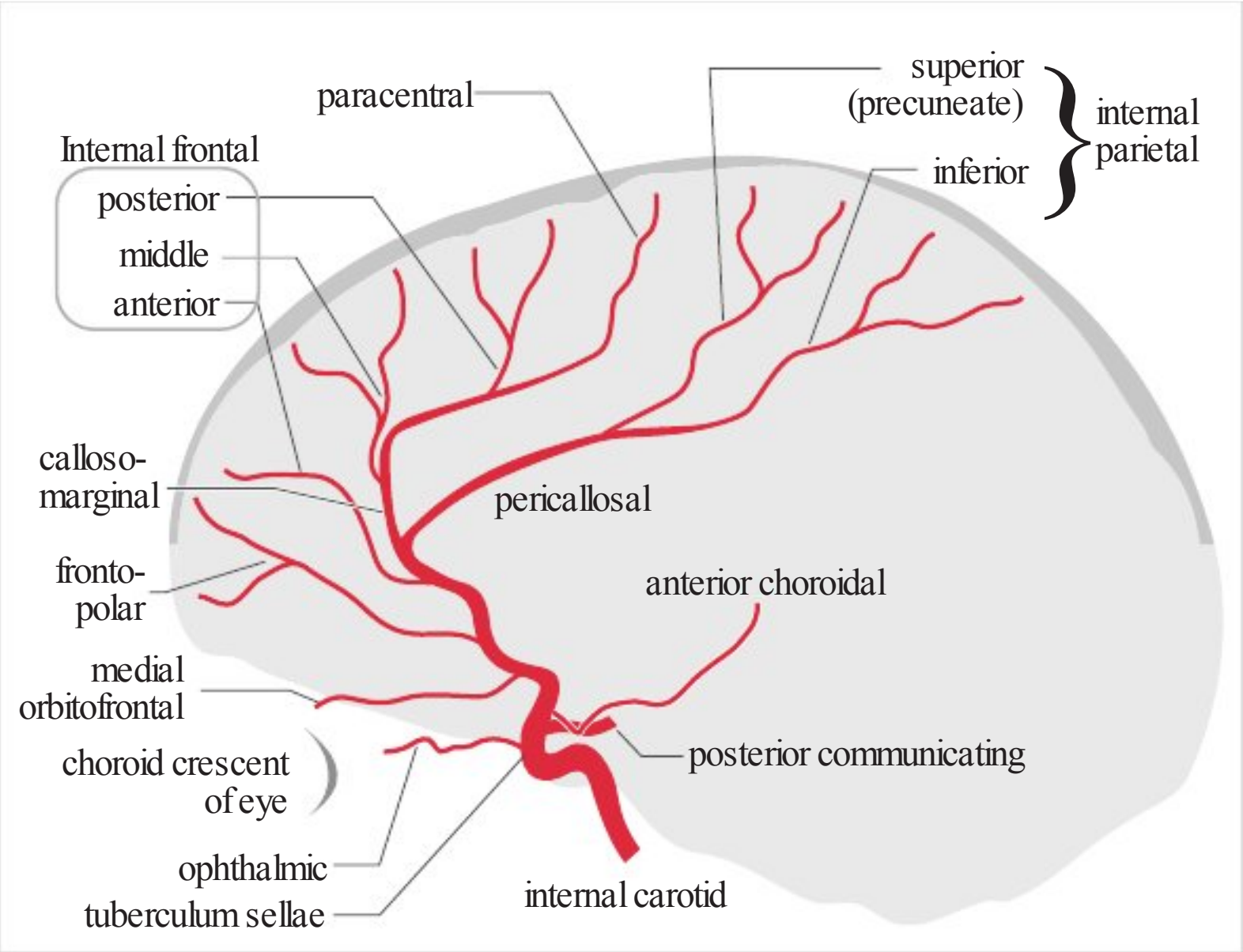


Fig. 2.4 Anterior cerebral arteriogram (lateral view) (Reprinted courtesy of Eastman Kodak Company)

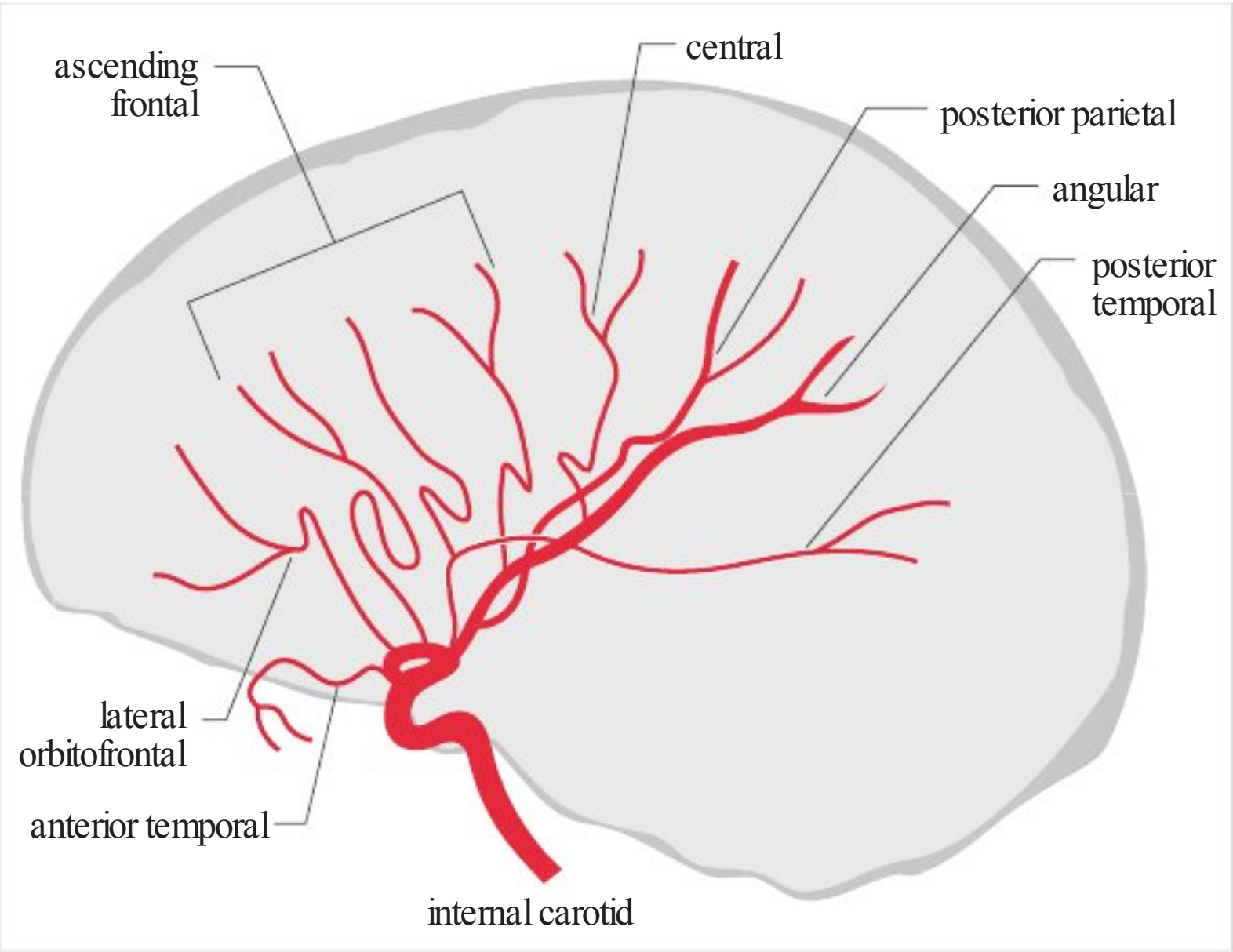


Fig. 2.5 Middle cerebral arteriogram (lateral view) (Reprinted courtesy of Eastman Kodak Company)

Branches

- Anterior meningeal. Arises at body of C2 (axis), may feed chordomas or foramen magnum meningiomas, may also act as collateral in vascular occlusion
- Posterior meningeal. May be a source of blood for some dural AVMs (p.1251)
- Medullary (bulbar) aa
- Posterior spinal
- Posterior inferior cerebellar artery (PICA) (largest branch). Usually arises ≈ 10 mm distal to point where VA becomes intradural, ≈ 15 mm proximal to the vertebrobasilar junction (□ Fig. 2.6)

1. anatomic variants:

- a) in 5–8% the PICA has an extradural origin
- b) “AICA-PICA”: origin is off basilar trunk (where AICA would usually originate)

2. 5 segments¹² (some systems some describe only 4). During surgery, the first three must be preserved, but the last 2 may usually be sacrificed with minimal deficit¹³:

- a) anterior medullary: from PICA origin to inferior olivary prominence. 1 or 2 short medullary short circumflex branches □ ventral medulla
- b) lateral medullary: to origin of nerves IX, X & XI. Up to 5 branches that supply brainstem
- c) tonsillomedullary: to tonsillar midportion (contains caudal loop on angio)

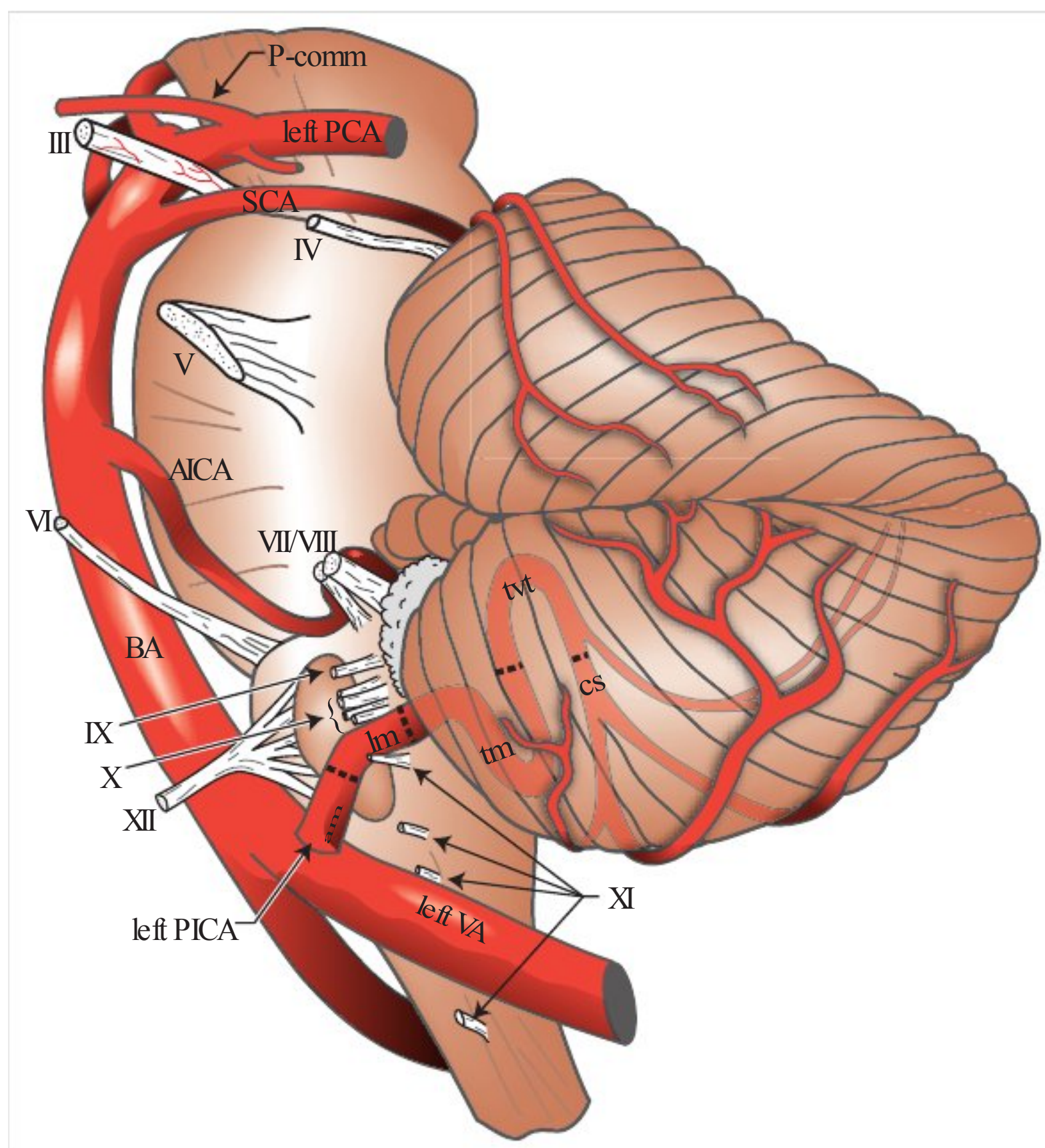


Fig. 2.6 Intradural VA and PICA segments (lateral view) (Modified with permission from: Lewis SB, Chang DJ, Peace DA, Lafrentz PJ, Day AL. Distal posterior inferior cerebellar artery aneurysms: clinical features and management. J Neurosurg 2002;97(4):756-66)

- d) telovelotonsillar (supratonsillar): ascends in tonsillomedullary fissure (contains cranial loop on angio)
- e) cortical segments
- 3. 3 branches
 - a) choroidal a. (BRANCH 1) arises from cranial loop (choroidal point), □ choroid plexus of 4th ventricle
 - b) terminal branches:
 - tonsillohemispheric (BRANCH 2)
 - inferior vermian (BRANCH 3) inferior inflection = copular point on angio
- Anterior spinal

Basilar artery (BA)

Formed by the junction of the 2 vertebral arteries. Branches:

1. anterior inferior cerebellar artery (AICA): from lower part of BA, runs posterolaterally anterior to VI, VII & VIII. Often gives off a loop that runs into the IAC and gives off the labyrinthine artery and then emerges to supply the anterolateral inferior cerebellum and then anastomoses with PICA
2. internal auditory (labyrinthine)
3. pontine branches
4. superior cerebellar a. (SCA)
 - a) sup. vermian
5. posterior cerebral: joined by p-comms $\approx$ 1 cm from origin (the p-comm is the major origin of the PCA in 15% and is termed “fetal” circulation, bilateral in 2%).
 - 3 segments (named for surrounding cistern) and their branches:
 - a) peduncular segment (P1)
 - mesencephalic perforating aa. (□ tectum, cerebral peduncles, and these nuclei: Edinger-Westphal, oculomotor and trochlear)
 - interpeduncular thalamoperforators (1st of 2 groups of posterior thalamoperforating aa.)
 - medial post. choroidal (most from P1 or P2)
 - “artery of Percheron”: a rare anatomic variant¹⁴ in which a solitary arterial trunk arising from the proximal segment of one PCA supplies the paramedian thalami and rostral mid-brain bilaterally
 - b) ambient segment (P2)
 - lateral post. choroidal (most from P2)
 - thalamogeniculate thalamoperforators (2nd of 2 groups of posterior thalamoperforating aa.) □ geniculate bodies + pulvinar
 - anterior temporal (anastomoses with anterior temporal br. of MCA)
 - posterior temporal
 - parieto-occipital
 - calcarine
 - c) quadrigeminal segment (P3)
 - quadrigeminal & geniculate branches □ quadrigeminal plate
 - post. pericallosal (splenial) (anastomoses with pericallosal of ACA)

Posterior cerebral artery (PCA)

See □ Fig. 2.7.

Carotid-vertebrobasilar anastomoses

P-comm artery: the “normal” (most common) anastomosis.

Persistent fetal anastomoses¹⁵ (□ Fig. 2.8) result from failure to involute as the VAs and p-comms develop (order of involution: otic, hypoglossal, primitive trigeminal, proatlantal). Most are asymptomatic. However, some may be associated with vascular anomalies such as aneurysms or AVMs, and occasionally cranial nerve symptoms (e.g. trigeminal neuralgia with PPTA) can occur.

Four types (from cranial to caudal – the 1st 3 are named for the associated cranial nerve):

1. persistent primitive trigeminal artery (PPTA): seen in $\approx$ 0.6% of cerebral angiograms. The most common of the persistent fetal anastomoses (83%). May be associated with trigeminal neuralgia (p.479). Connects the cavernous carotid to the basilar artery. Arises from the ICA proximal to the origin of the meningo-hypophyseal trunk (50% go through sella, 50% exit the cavernous sinus & course with the trigeminal nerve) and connects to the upper basilar artery between AICA & SCA. The VAs may be small. Saltzman type 1 variant: the p-comms are hypoplastic and the PPTA provides significant blood supply to the distributions of the distal BA, PCA and the SCAs (the basilar

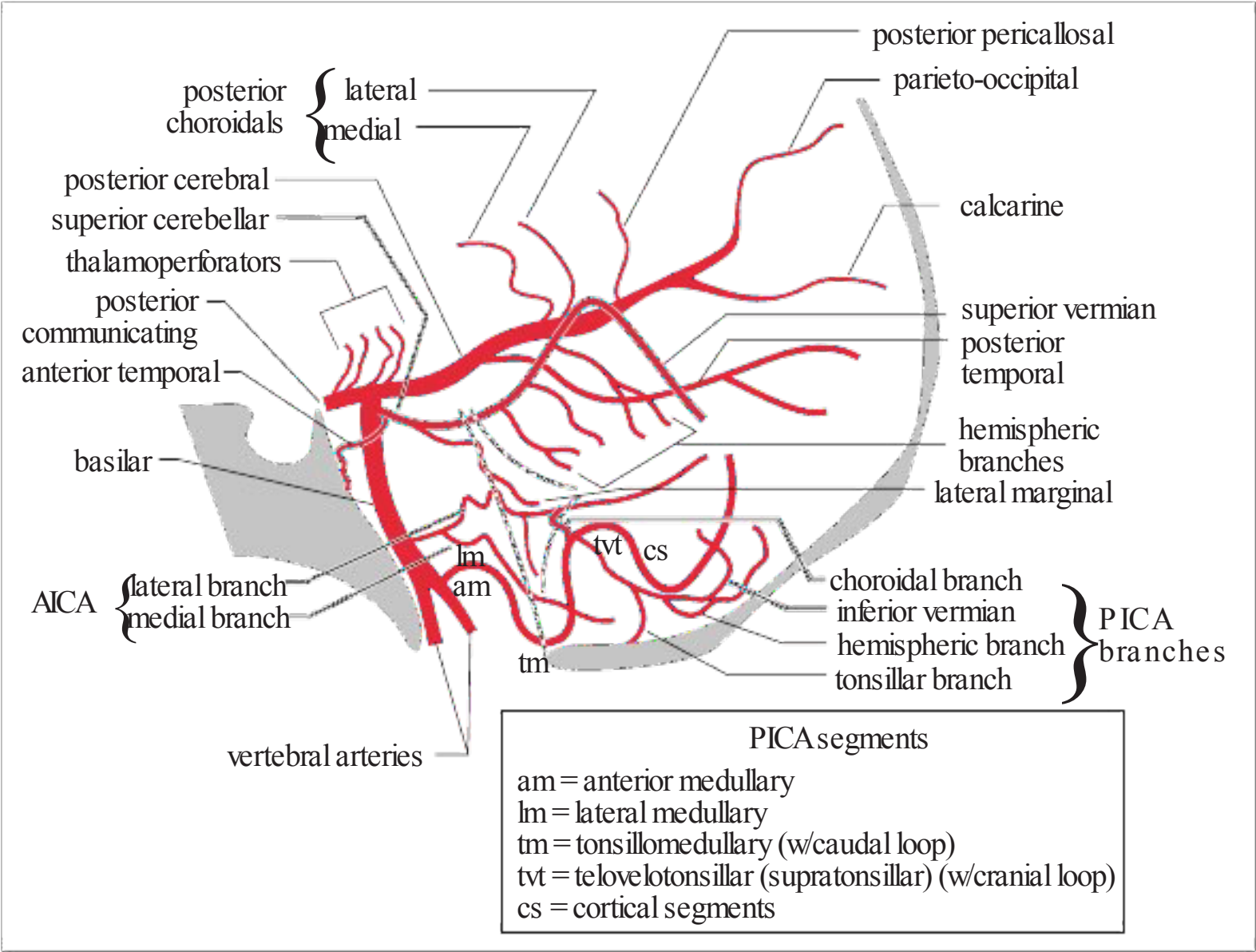


Fig. 2.7 Vertebrobasilar arteriogram (lateral view) (Reprinted courtesy of Eastman Kodak Company)

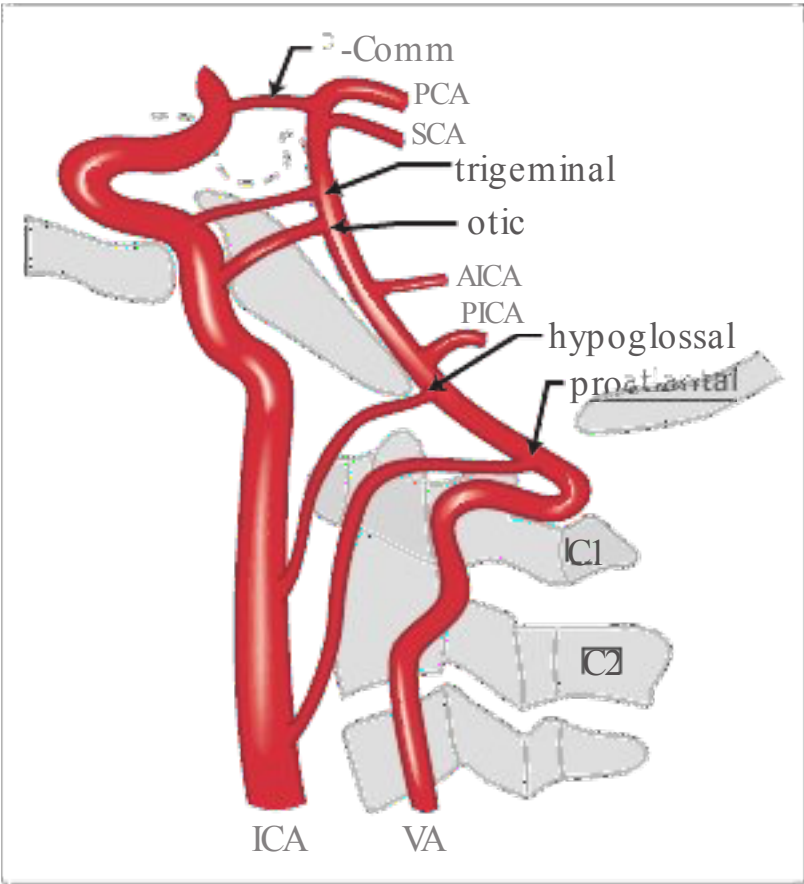


Fig. 2.8 Carotid-vertebrobasilar anastomoses

- artery is often hypoplastic). Saltzman type 2: p-comm supplies PCA. Saltzman type 3: PPTA joins the SCA (instead of the BA). It is critical to recognize a PPTA before doing a Wada test (p.1553) because of the risk anesthetizing the brainstem, and in doing transsphenoidal surgery because of risk of arterial injury. May rarely be an explanation of posterior fossa symptoms in a patient with carotid disease
2. otic: the first to involute, and the rarest to persist (8 cases reported). Passes through IAC to connect petrous carotid to basilar artery
 3. hypoglossal: connects petrous or distal cervical ICA (origin usually between C1-C3) to VA. Traverses the hypoglossal canal. Does not cross foramen magnum

4. proatlantal intersegmental: connects cervical ICA to VA. May arise from: bifurcation of common carotid, ECA, or ICA from C2-C4. Anastomosis with VA in suboccipital region. 50% have hypoplastic proximal VA. 40 cases reported

2.3 Cerebral venous anatomy

2.3.1 Supratentorial venous system

Major veins and tributaries

See □ Fig. 2.9 for angiogram and branches.

The left and right internal jugular veins (IJVs) are the major source of outflow of blood from the intracranial compartment. The right IJV is usually dominant. Other sources of outflow include orbital veins and the venous plexuses around the vertebral arteries. Diploic and scalp veins may act as collateral pathways, e.g. with superior sagittal sinus obstruction.¹⁶ The following outline traces the venous drainage back from the IJVs.

Inferior petrosal sinus

Terminates (i.e. drains to) ≤ 1 cm of junction of sigmoid and transverse sinuses.

Sigmoid sinus

superior petrosal sinus

Drains to IJV near junction with sigmoid sinus

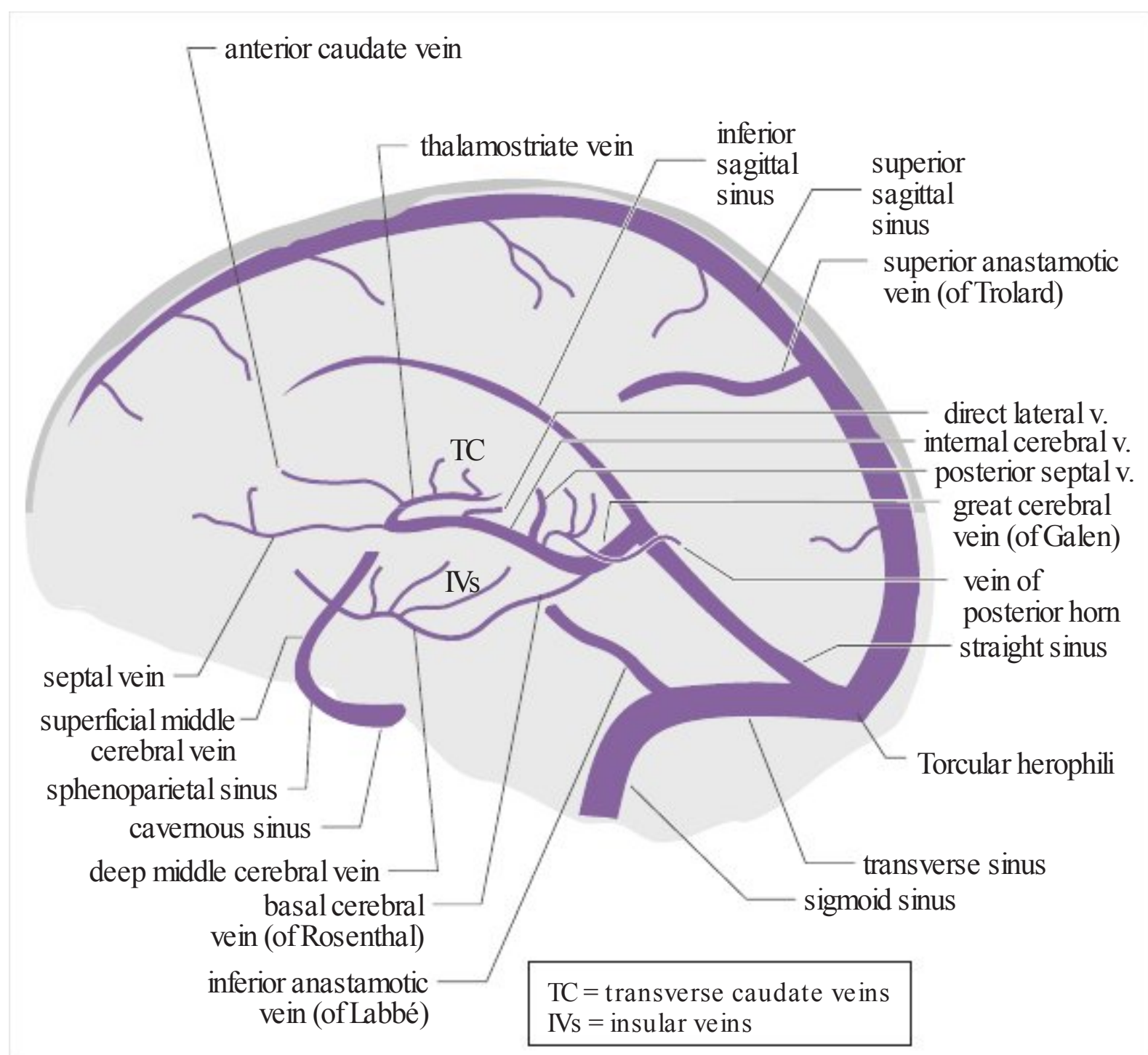


Fig. 2.9 Internal carotid venogram (lateral view) (Reprinted courtesy of Eastman Kodak Company)

Transverse sinus

R>L in 65%

- V. of Labbe. (Inferior anastomotic v.)
- Confluens of sinuses. (Torcular herophili)

1. occipital sinus
2. superior sagittal sinus
 - a) v. of Trolard (superior anastomotic v.): the prominent superficial vein on the non-dominant side (Labbe is more prominent on the dominant side)
 - b) cortical veins
3. straight sinus
 - a) inferior sagittal sinus
 - b) great cerebral v. (of Galen)
 - pre-central cerebellar v.
 - basal vein of Rosenthal
 - internal cerebral v.: joined at the foramen of Monro (venous angle) by:
 - anterior septal v.
 - thalamostriate v.

Cavernous sinus

Originally named for its superficial resemblance to the corpora cavernosa. Although classical teaching depicts the cavernous sinus as a large venous space with multiple trabeculations, injection studies¹⁷ and surgical experience¹⁸ instead supports the concept of the cavernous sinus as a plexus of veins. It is highly variable between individuals and from side-to-side. □ Fig. 2.10 is an oversimplified schematic of one section through the right cavernous sinus.

1. inflowing veins:
 - a) superior & inferior ophthalmic veins
 - b) superficial middle cerebral veins
 - c) sphenoparietal sinus
 - d) superior & inferior petrosal sinus
2. outflow:
 - a) sphenoparietal sinus
 - b) superior petrosal sinus
 - c) basilar plexus (which drains to the inferior petrosal sinus)
 - d) pterygoid plexus
 - e) the right and left cavernous sinuses communicate anteriorly and posteriorly via the circular sinus
3. contents¹⁹
 - a) Oculomotor n. (III)
 - b) Trochlear n. (IV)

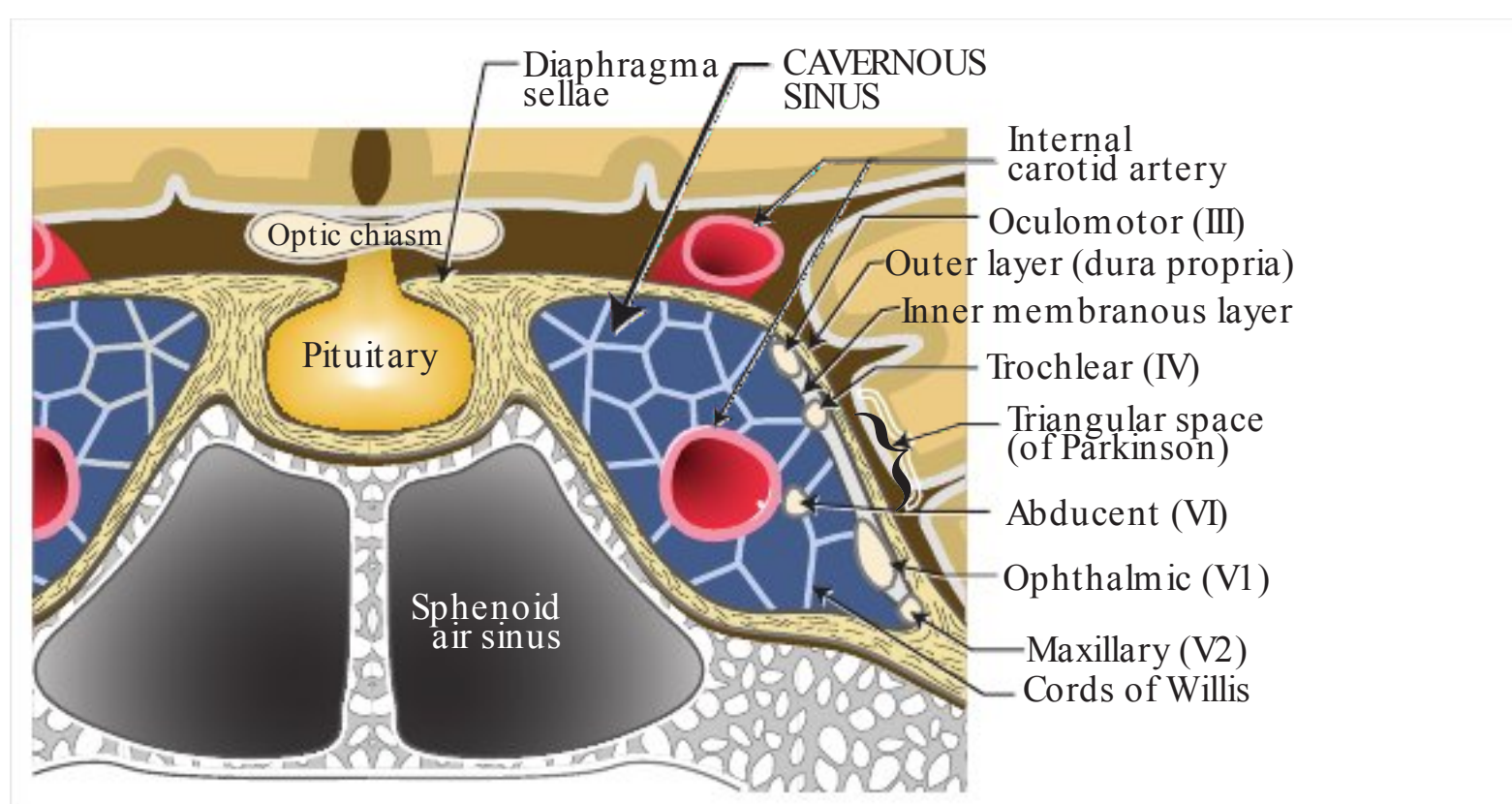


Fig. 2.10 Right cavernous sinus (coronal section)

- c) Ophthalmic division of trigeminal (V1)
 - d) Maxillary division of trigeminal (V2): the only nerve of the cavernous sinus that doesn't exit the skull through the superior orbital fissure (it exits through foramen rotundum)
 - e) Carotid artery (ICA). 3 segments within the cavernous sinus
 - posterior ascending segment: immediately after ICA enters the sinus
 - horizontal segment: after ICA turns anteriorly (the longest segment of the intracavernous ICA)
 - anterior ascending segment: ICA turns superiorly
 - f) Abducens n. (VI): the only nerve NOT attached to lateral dural wall, sometimes referred to as the only cranial nerve inside the cavernous sinus
4. triangular space (of Parkinson): superior border formed by Cr. N. III & IV, and the lower margin formed by V1 & VI (a landmark for surgical entrance to the cavernous sinus)^{20,21 (p 3007)}

2.3.2 Posterior fossa venous anatomy

See □ Fig. 2.11.

2.4 Spinal cord vasculature

See □ Fig. 2.12.

Although a radicular artery from the aorta accompanies the nerve root at many levels, most of these contribute little flow to the spinal cord itself. The anterior spinal artery is formed from the junction of two branches, each from one of the vertebral arteries. Major contributors of blood supply to the anterior spinal cord is from 6–8 radicular arteries at the following levels (“radiculomedullary arteries”, the levels listed are fairly consistent, but the side varies^{22 (p 1180–1)}):

1. C3 – arises from vertebral artery
2. C6 and C8 ($\approx 10\%$ of population lack an anterior radicular artery in lower cervical spine²³)
 - a) C6 – usually arises from deep cervical artery
 - b) C8 – usually from costocervical trunk
3. T4 or T5
4. artery of Adamkiewicz AKA arteria radicularis anterior magna
 - a) the main arterial supply for the spinal cord from $\approx$ T8 to the conus
 - b) located on the left in 80% ⁴
 - c) situated between T9 & L2 in 85% (between T9 & T12 in 75% ; in remaining 15% between T5 & T8 (in these latter cases, there may be a supplemental radicular artery further down)

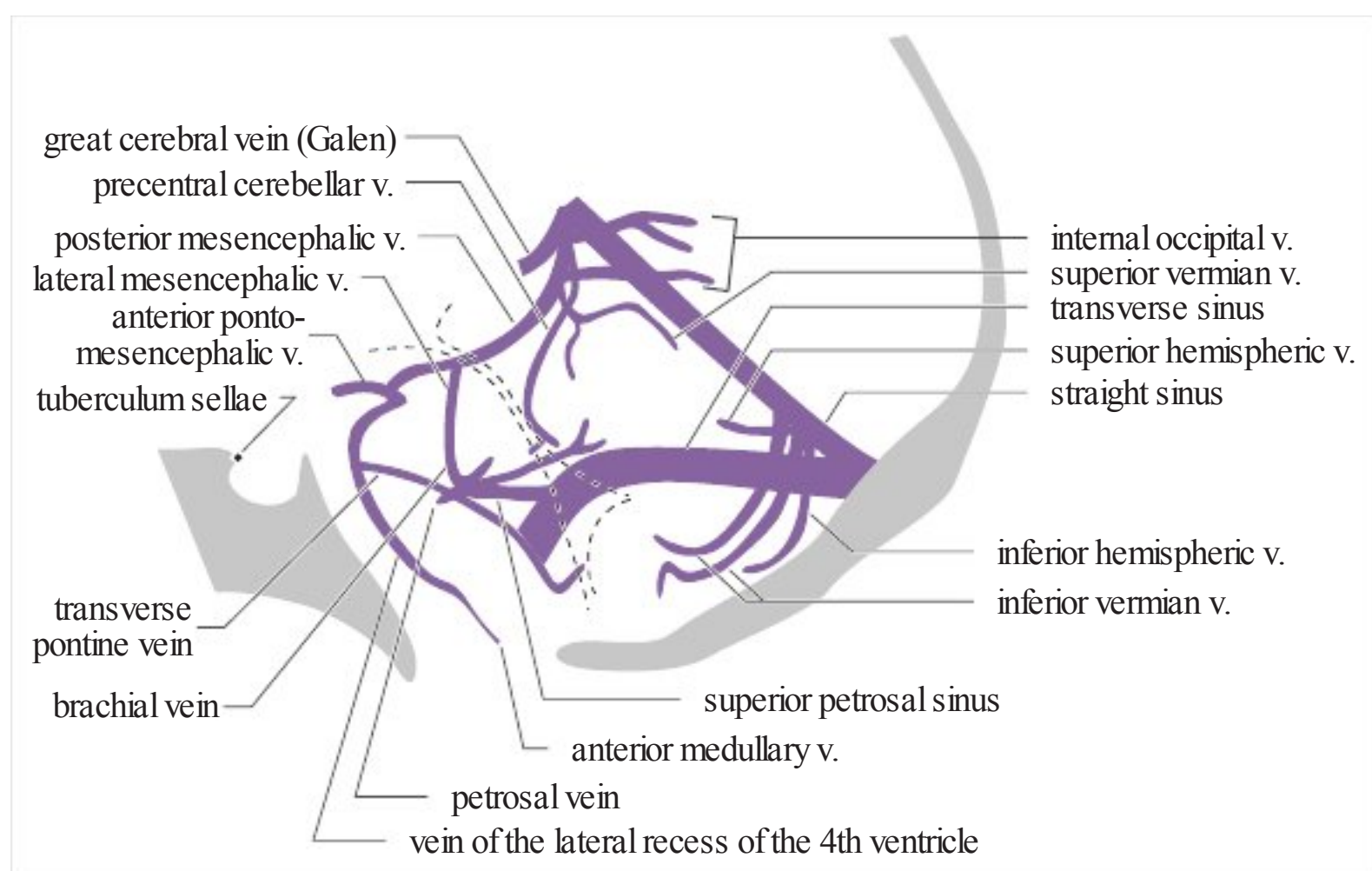


Fig. 2.11 Vertebrobasilar venogram (lateral view) (Reprinted courtesy of Eastman Kodak Company)

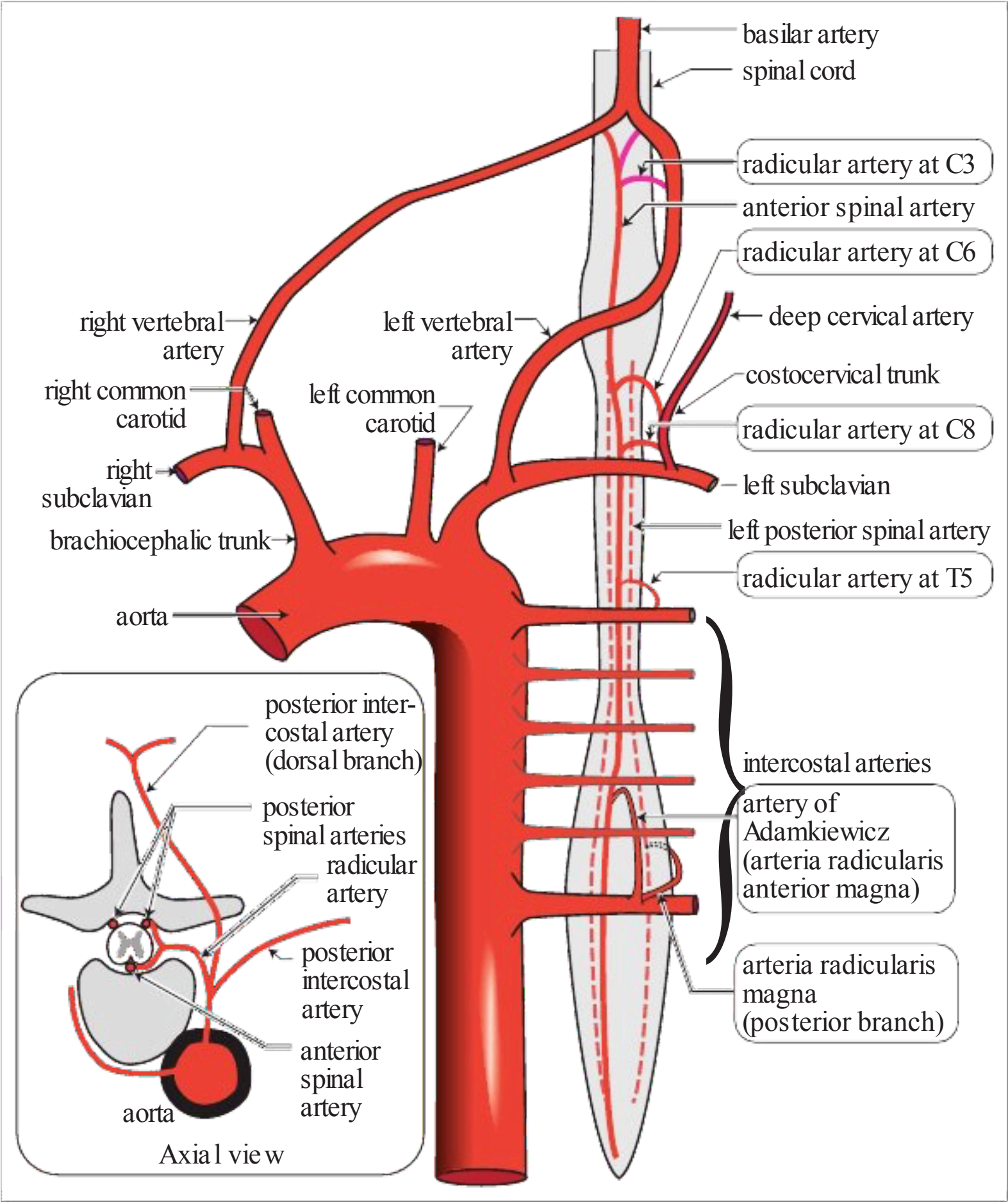


Fig. 2.12 Schematic diagram of spinal cord arterial supply (Modified from Diagnostic Neuroradiology, 2nd ed., Volume II, pp. 1181, Taveras JM, Woods EH, editors, © 1976, the Williams and Wilkins Co., Baltimore, with permission))

- d) usually fairly large, gives off cephalic and caudal branch (latter is usually larger) giving a characteristic hair-pin appearance on angiography

The paired posterior spinal arteries are less well defined than the anterior spinal artery, and are fed by 10–23 radicular branches.

The midthoracic region has a tenuous vascular supply (“watershed zone”), possessing only the above noted artery at T4 or T5. It is thus more susceptible to vascular insults.

□ Anatomic variants. Arcade of Lazorthes: normal variant where the anterior spinal artery joins with the paired posterior spinal arteries at the conus medullaris.

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3 Neurophysiology and Regional Brain Syndromes

3.1 Neurophysiology

3.1.1 Blood-brain barrier

General information

The passage of water-soluble substances from the blood to the CNS is limited by tight junctions (zonulae occludentes) which are found between cerebral capillary endothelial cells, limiting penetration of the cerebral parenchyma (blood-brain barrier, BBB), as well as between choroid plexus epithelial cells (blood-CSF barrier).¹ A number of specialized mediated transport systems allow transmission of, among other things, glucose and certain amino acids (especially precursors to neurotransmitters).

The efficacy of the BBB is compromised in certain pathological states (e.g. tumor, infection, trauma, stroke, hepatic encephalopathy...), and can also be manipulated pharmacologically (e.g. hypertonic mannitol increases the permeability, whereas steroids reduce the penetration of small hydrophilic molecules).

The BBB is absent in the following areas: choroid plexus, hypophysis, tuber cinereum, area postrema, pineal and preoptic recess.

Means of assessing the integrity of the BBB:

- visible dyes: Evan's blue, fluorescein
- radioopaque dyes (imaged with CTscan²): iodine (protein-bound contrast agent)
- paramagnetic (imaged on MRI): gadolinium (protein-bound contrast agent)
- microscopic: horseradish peroxidase
- radiolabeled: albumin, sucrose

Cerebral edema and the blood brain barrier

Three basic types of cerebral edema; diffusion-weighted MRI (p.232) may be able to differentiate:

1. cytotoxic: BBB is closed, therefore no protein extravasation, therefore no enhancement on CT or MRI. Cells swell then shrink. Seen e.g. in head injury
2. vasogenic: BBB disrupted. Protein (serum) leaks out of vascular system, and therefore may enhance on imaging. Extracellular space (ECS) expands. Cells are stable. Responds to corticosteroids (e.g. dexamethasone). Seen e.g. surrounding metastatic brain tumor
3. ischemic: a combination of the above. BBB closed initially, but then may open. ECS shrinks then expands. Fluid extravasates late. May cause delayed deterioration following intracerebral hemorrhage (p.1337).

3.1.2 Babinski sign and Hoffmann sign

Introduction

Although the Babinski sign is regarded as the most famous sign in neurology, there is still disagreement over what constitutes a normal response and when abnormal responses should occur.³ The following represents one interpretation.

The plantar reflex (PR) (AKA Babinski sign after Joseph François Félix Babinski (1857–1932) a French neurologist of Polish descent) is a primitive reflex, present in infancy, consisting of extension of the great toe in response to a noxious stimulus applied to the foot. The small toes may fan, but this is not a consistent nor clinically important component. The PR disappears usually at ≈ 10 months age (range: 6 mos to 12 yrs), presumably under inhibitory control as myelination of the CNS occurs, and the normal response then converts to plantarflexion of the great toe. An upper motor neuron (UMN) lesion anywhere along the pyramidal (corticospinal) tract from the motor strip down to $\approx L4$ will result in a loss of inhibition, and the PR will be “unmasked” producing extension of the great toe. With such an UMN lesion, there may also be exaggeration of flexor synergy resulting in dorsiflexion of the ankle, and flexion of the knee and hip (AKA triple flexor response) in addition to extension of the great toe.

Neuroanatomy

The afferent limb of the reflex originates in cutaneous receptors restricted to the first sacral dermatome (S1) and travels proximally via the tibial nerve. The spinal cord segments involved in the reflex-arc lie within L4-S2. The efferent limb to the toe extensors travels via the peroneal nerve.

Differential diagnosis

Etiologies

Lesions producing a PR need not be structural, but may be functional and reversible. The roster of possible etiologies is extensive, some are listed in □ Table 3.1.

Eliciting the PR, and variations

The optimal stimulus consists of stimulation of the lateral plantar surface and transverse arch in a single movement lasting 5–6 seconds.⁴ Other means for applying noxious stimuli may also elicit the plantar reflex (even outside the S1 dermatome, although these do not produce toe flexion in normals). Described maneuvers include: Chaddock (scratch the lateral foot; positive in 3% where plantar stimulation was negative), Schaeffer (pinch the Achilles tendon), Oppenheim (slide knuckles down shin), Gordon (momentarily squeeze lower gastrocnemius), Bing (light pinpricks on dorsolateral foot), Gonda or Stronsky (pull the 4th or 5th toe down and out and allow it to snap back).

Hoffman’s (or Hoffmann’s or Hoffmann) sign

Attributed to Johann Hoffmann, a German neurologist practicing in the late 1800. May signify a similar UMN interruption to the upper extremities. Elicited by flicking downward on the nail of the middle or ring finger: a positive (pathologic) response consists of involuntary flexion of the adjacent fingers and/or thumb (may be weakly present in normals).⁵ Differs from the plantar reflex since it is monosynaptic (synapse in Rexed lamina IX).

Can sometimes be seen as normal in young individual with diffusely brisk reflexes & positive jaw jerk, usually symmetric. When present pathologically, represents disinhibition of a C8 reflex, □ indicates lesion above C8.

Hoffmann sign was observed in 68% of patients operated for cervical spondylotic myelopathy.⁵ In 11 patients presenting with lumbar symptoms but no myelopathy, a bilateral Hoffman sign was associated with occult cervical spinal cord compression in 10 (91%).⁵ The Hoffmann test has a sensitivity of 33-68% specificity of 59-78% a positive predictive value of 26-62% and negative predictive value of 67-75%⁶

3.1.3 Bladder neurophysiology

Central pathways

The primary coordinating center for bladder function resides within the nucleus locus coeruleus of the pons. This center synchronizes bladder contraction with relaxation of the urethral sphincter during voiding.⁷

Voluntary cortical control primarily involves inhibition of the pontine reflex, and originates in the anteromedial portion of the frontal lobes and in the genu of the corpus callosum. In an uninhibited bladder (e.g. infancy) the pontine voiding center functions without cortical inhibition and the detrusor muscle contracts when the bladder reaches a critical capacity. Voluntary suppression from the cortex via the pyramidal tract may contract the external sphincter and may also inhibit detrusor contraction. Cortical lesions in this location → urgency incontinence with inability to suppress the micturition reflex.^{8 (p 1031)}

Efferents to the bladder travel in the dorsal portion of the lateral columns of the spinal cord (shaded areas in □ Fig. 3.1).

Table 3.1 Differential diagnosis of the plantar reflex (PR)
Etiologies
• spinal cord injuries^a • cervical spinal myelopathy • lesions in motor strip or internal capsule (stroke, tumor, contusion...) • subdural or epidural hematoma • hydranencephaly • toxic-metabolic coma • seizures • trauma • TIAs • hemiplegic migraine • motor neuron disease (ALS)
^a in spinal cord injuries, the PR may initially be absent during the period of spinal “shock” (p.931)

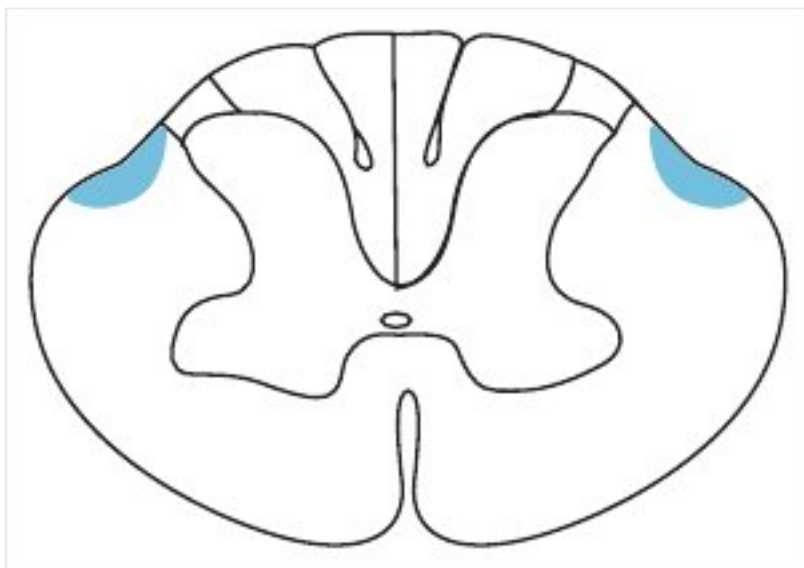


Fig. 3.1 Location of spinal cord bladder efferents (shaded)

Motor

There are two sphincters that prevent the flow of urine from the bladder: internal (autonomic, involuntary control), and external (striated muscle, voluntary control).

Parasympathetics (PSN)

The detrusor muscle of the bladder contracts and the internal sphincter relaxes under PSN stimulation. PSN preganglionic cell bodies reside in the intermediolateral grey of spinal cord segments S2–4. Fibers exit as ventral nerve roots and travel via pelvic splanchnic nerves (nervi erigentes) to terminate on ganglia within the wall of the detrusor muscle in the body and dome of the bladder.

Somatic nerves

Somatic voluntary control descends in the pyramidal tract to synapse on motor nerves in S2–4, and then travels via the pudendal nerve to the external sphincter. This sphincter may be voluntarily contracted, but relaxes reflexly with opening of the internal sphincter at the initiation of micturition. Primarily maintains continence during ↑ vesical pressure (e.g. valsalva).

Sympathetics

Sympathetic cell bodies lie within the intermediolateral gray column of lumbar spinal cord from segments T12 —L2. Preganglionic axons pass through the sympathetic chain (without synapsing) to the inferior mesenteric ganglion. Postganglionic fibers pass through the inferior hypogastric plexus to the bladder wall and internal sphincter. Sympathetics heavily innervate the bladder neck and trigone. Sympathetics have little effect on bladder motor activity, but alpha adrenergic stimulation results in bladder neck closure which is necessary for bladder filling.

Pelvic nerve stimulation → increased sympathetic tone → detrusor relaxation & increased bladder neck tone (allowing larger volume to be accommodated).

Sensory

Less well understood than motor innervation. Bladder wall stretch receptors sense bladder filling and send afferent signals via pelvic, pudendal and hypogastric nerves to spinal cord segments T10–L2 & S2–4. Fibers ascend primarily in the spinothalamic tract.

Urinary bladder dysfunction

The term neurogenic bladder describes bladder dysfunction due to lesions within the central or peripheral nervous systems. Some use the term synonymously with detrusor areflexia.

Dorsal (sensory) roots lesions interrupt the afferent limb, producing an atonic bladder that fills until dribbling and overflow incontinence occur. No sensation of bladder fullness is appreciated. Voluntary voiding is still possible, but is usually incomplete.

Detrusor hyperreflexia

Can result from interruption of efferents anywhere from cortex to sacral cord. When a critical volume is attained, reflex bladder emptying occurs. Clinically associated with frequent, uncontrollable,

precipitous voiding. Cerebral lesions include: stroke, head injury, brain tumors, hydrocephalus, Parkinson's disease, various dementias, and MS. Cord lesions include anything that causes myelopathy (p.1407).

Detrusor areflexia

Clinically correlates with difficulty initiating micturition, interrupted flow, and significant residual urine. Incontinence may result from over-distention of the bladder (overflow incontinence), or may be associated with absence of sphincter tone. Etiologies include: chronic infection, long-term bladder catheterization, certain drugs (especially phenothiazines), injury or tumor of the cauda equina or conus medullaris, myelomeningocele, and diabetes mellitus (autonomic neuropathy).

3

Specific injuries affecting the bladder

In general, regarding discrete neurologic lesions affecting the bladder⁹:

1. supraspinal (lesions above the brain stem): loss of centrally mediated inhibition of the pontine voiding reflex. Usually produces involuntary bladder contractions with smooth and striated sphincter synergy, often with preserved sensation and voluntary striated sphincter function. Symptoms: urinary frequency or urgency, urgency incontinence, and nocturia.⁷ If sensory pathways are interrupted, unconscious incontinence occurs (incontinence of the unawares type). Since muscles are coordinated, normal bladder pressures are maintained and there is low risk of high-pressure related renal dysfunction. Voluntary bladder emptying is usually maintained, and timed voiding together with anticholinergic medications (see below) are used in management. Areflexia may sometimes occur
2. complete (or near complete) spinal cord lesions:
 - a) suprasacral (lesion above the S2 spinal cord level, which is $\approx$ T12/L1 vertebral body level in an adult): the sacral voiding center is located in the conus medullaris. Etiologies: spinal cord injuries, tumors, transverse myelitis.
 - Initially following spinal cord injury, there may be spinal shock. During spinal shock (p.931), the bladder is acontractile and areflexic (detrusor areflexia); sphincter tone usually persists and urinary retention is the rule (urinary incontinence generally does not occur except with overdistention)
 - After spinal shock subsides, most develop detrusor hyperreflexia $\rightarrow$ involuntary bladder contractions without sensation (automatic bladder), smooth sphincter synergy, but striated dyssynergy (involuntary contraction of the external sphincter during voiding which produces a functional outlet obstruction with poor emptying and high vesical pressures). Bladder fills and empties spontaneously (or in response to lower extremity cutaneous stimulation). Bladder compliance is often reduced. Managed by intermittent catheterizations + anticholinergics
 - b) infrasacral lesions (lesion below the S2 spinal cord level): includes injury to conus medullaris, cauda equina or peripheral nerves (formerly referred to as lower motor neuron lesions). Etiologies: large HLD, trauma with compromise of spinal canal. Usually develop detrusor areflexia, and do not have involuntary bladder contractions. Reduced urinary flow rate or retention results, and voluntary voiding may be lost. Overflow incontinence develops. There may be reduced compliance during filling, and paralysis of the smooth sphincter (the neurologic basis of this has not been settled, and may be due to sympathetic or PSN involvement). Usually associated with loss of bulbocavernosus and anal wink reflex (preserved in suprasacral lesions, except when spinal shock is present (p.931)) and perineal sensory loss
3. interruption of the peripheral reflex arc: may produce disturbances similar to low spinal cord injury with detrusor areflexia, low compliance and inability to relax the striated sphincter
4. herniated lumbar disc (p.1046): most consist initially of difficulty voiding, straining, or urinary retention. Later, irritative symptoms may develop
5. spinal stenosis (lumbar or cervical): urologic symptoms vary, and depend on the spinal level(s) involved and the type of involvement (e.g. in cervical spinal stenosis, detrusor hyperactivity or underactivity may occur depending on whether the involvement of the micturition neural axis is compression of the inhibitory reticulospinal tracts or myelopathy involving the posterior funiculus)
6. cauda equina syndrome (p.1050): usually produces urinary retention, although sometimes incontinence may occur (some cases are overflow incontinence)
7. peripheral neuropathies: such as with diabetes, usually produce impaired detrusor activity
8. neurospinal dysraphism: most myelodysplastic patients have an areflexic bladder with an open bladder neck. The bladder usually fills until the resting residual fixed external sphincter pressure is exceeded and the leakage occurs

9. multiple sclerosis: 50–90% of patients develop voiding symptoms at some time. The demyelination primarily involves the posterior and lateral columns of the cervical spinal cord. Detrusor hyperreflexia is the most common urodynamic abnormality (in 50–99% of cases), with bladder areflexia being less common (5–20%)

3

Urinary retention

Etiologies of urinary retention:

1. bladder outlet obstruction (a brief differential diagnosis list is presented here)
 - a) urethral stricture: retention tends to be progressive over time
 - b) prostatic enlargement in males:
 - benign prostatic hypertrophy (BPH) & prostate cancer: retention tends to be progressive over time
 - acute prostatitis: onset of retention may be sudden
 - rare: extruded prostatic stone
 - c) women may develop a cystocele which can produce a urethral kink
 - d) rare: urethral cancer
2. detrusor areflexia (p. 93) or hypotonia
 - a) spinal cord injury
 - b) cauda equina syndrome (p. 1050)
 - c) chronic infection
 - d) long-term bladder catheterization
 - e) certain drugs (narcotics, phenothiazines)
 - f) injury of the cauda equina or conus medullaris, or of the spinal cord at or below the sacrum
 - trauma
 - tumor
 - myelomeningocele
 - g) diabetes mellitus (autonomic neuropathy)
 - h) herpes zoster at the level of the sacral dorsal root ganglia⁹ (p. 967)
 - i) incomplete opening of the bladder neck during voiding: occurs almost exclusively in young and middle-aged males with longstanding obstructive and irritative symptoms⁹ (p. 968)
 - j) initially following severe bladder over distention or with chronic distention and decompression from any of the above
3. postoperative retention: well-recognized but poorly understood. More common after lower urinary tract, perineal, gynecologic and anorectal operations. Anesthesia and analgesia may contribute to a number of factors⁹ (p. 969)
4. psychogenic

Evaluation of bladder function

Urodynamics

Usually combined with x-ray (cystometrogram (CMG)) or fluoro (videourodynamics). Measures intravesicular pressures during retrograde bladder filling through a urethral catheter, usually combined with sphincter electromyography. Presence or absence (detrusor areflexia, see below) of detrusor reflex is detected. If present, procedure is repeated, asking patient to suppress the urge to void. Inability to suppress is called an uninhibited detrusor reflex (AKA detrusor hyperreflexia, see above).

Sphincter electromyography (EMG)

Either via needle electrodes, or with externally mounted surface electrodes. Voluntary sphincter contraction tests intactness of supraspinal innervation. When combined with CMG, detects electrical activity in sphincters during associated phases of detrusor contraction.

Voiding cystourethrogram and intravenous pyelography (IVP)

Voiding cystourethrogram (VCUG) detects urethral pathology (diverticula, strictures...), abnormalities of bladder (diverticula, detrusor trabeculations associated with longstanding contractions against high resistance...), and vesical-ureteral reflux.

Pharmacologic Treatment

Goals are to preserve renal function (which usually involves prevention of UTIs, renal calculi, and ureteral reflux due to high intravesicular pressures) and optimization of urinary continence. Patients with inadequate emptying or increased bladder pressure are often managed by intermittent catheterizations and anticholinergics (see below). Anticholinergics and behavioral therapy are used for patients with maintained voluntary bladder emptying with urinary frequency or urgency incontinence.

The majority of the neurologic involvement in bladder contraction is ACh-mediated stimulation of postganglionic parasympathetic muscarinic cholinergic receptors on bladder smooth muscle.

Detrusor hyperreflexia

The following are all synthetic anticholinergics that block postganglionic synapses (muscarinic action) without blocking skeletal neuromuscular or autonomic ganglia (nicotinic junctions). This increases the volume at which automatic (reflex) contraction occurs in the neurogenic (uninhibited) bladder, effectively increasing bladder capacity. These agents increase the threshold at which involuntary bladder contraction occurs, but they do not increase the warning time and they do not increase the ability to suppress contraction, therefore, urgency and incontinence will still occur unless treatment is combined with a regimen of timed voiding.^{9 (p 972)}

All are contraindicated in glaucoma as the anticholinergic effects include mydriasis. Overdosage results in the classic anticholinergic symptoms (“red as a beet, hot as a stove, dry as a rock, mad as a hatter”). Use is often limited by side effects such as dry mouth.

Drug info: Oxybutynin (Ditropan®)

Probably the most widely prescribed agent. Combines anticholinergic activity with independent musculotropic relaxant effect and local anesthetic activity.

- Adults: usual dose is 5 mg BID-TID (maximum 4 times daily).
- Peds: not recommended for age <5 years; usual dose is 5 mg BID (maximum 5 mg TID). Supplied: 5 mg tablets, 5 mg/5 ml syrup.

Drug info: Tolterodine (Detrol®)

Milder side effects than oxybutynin, but may also be less effective.¹⁰

- 2 mg PO BID. Can be lowered to 1 mg PO BID in some patients. Supplied: 1 & 2 mg tablets. Detrol® LA 2 & 4 mg capsule

Drug info: Flavoxate HCl (Urispas®)

Weak anticholinergic. Direct smooth muscle inhibitor. Few reported side effects. Some studies have shown no benefit in the elderly population.^{9 (p 974)}

- Adult: 100–200 mg PO TID-QID.

Drug info: Imipramine HCl (Tofranil®)

A tricyclic antidepressant. Mechanism of beneficial effect is controversial. Does possess some anticholinergic activity as well as other properties.^{9 (p 977)} Appears to decrease bladder contractility and increase outlet resistance.

Detrusor areflexia

Drug info: Bethanechol (Urecholine®)

Aparasympathomimetic agent, primarily muscarinic with little nicotinic activity; related to acetylcholine but not destroyed by cholinesterase. Increases the tone of the detrusor muscle, aiding bladder emptying. Also increases gastric motility. Sub-Q administration produces a more intense effect on the bladder than PO. Always have atropine available when giving by sub-Q route. Effects occur within 30–90 minutes after PO dose, and within 15 minutes of sub-Q dose.

Indicated for acute post-op non-obstructive urinary retention and for neurogenic atony due to spinal cord injury or dysfunction.

Side effects: sweating and diarrhea are not uncommon but of little danger. Can precipitate severe bronchospasm in asthmatics. Nausea may be reduced by giving with an empty stomach. Atropine is a specific antidote for overdosage (atropine sub-Q: 0.6 mg in adults, or 0.01 mg/kg in children <12 yrs).

□ Start with 5–10 mg PO, and increase hourly until desired effect obtained or 50 mg given. Then, continue minimum effective dosage TID-QID (usual: 10–50 mg PO TID-QID). Sub-Q (have atropine available): 0.5–1 ml, repeat q 15 mins until desired response or 4 doses given; continue minimal effective dose TID-QID. Supplied: 5, 10, 25 & 50 mg tabs. Injection: 5.15 mg/ml (for sub-Q use only).

Bladder management after cauda equina compression

In situations where there is urinary retention with some prospect of return of function (e.g. following surgery for acute cauda equina compression) the following bladder management regimen may be employed:

- teach the patient or a family member to perform clean intermittent catheterizations (CIC), if CICs can be done:
 - have them monitor post-void residuals (PVR)
 - start tamsulosin (Flomax®) 0.4 mg PO q d (see below)
 - if PVRs ever fall to <75 cc, discontinue CICs
- if CICs cannot be performed, manage with an indwelling Foley catheter for a week, and check the PVR after that time
- if after 1 week the PVR is ≥75 cc, D/C tamsulosin if used, and refer the patient to a urologist for urodynamics (urodynamics earlier than this time will usually not result in a change in management)

Drug info: Tamsulosin (Flomax®)

Aprostate alpha_{1A}adrenoreceptor antagonist. Used to treat voiding difficulties resulting from outlet obstruction due to benign prostatic hypertrophy (BPH). Has been shown to have some effectiveness in women via other mechanisms. Similar to terazosin (Hytrin®) and doxazosin (Cardura®), but has an advantage for acute relief because the dose of tamsulosin does not need to be gradually ramped up (it can be started at the therapeutic dose). It takes at least 5–7 days to work.

Side effects : very few. Rhinitis, retrograde or diminished ejaculation, or postural hypotension may occur.¹¹

□: 0.4 mg PO q d (usually given 30 minutes after the same meal each day). If there is no response by 2–4 weeks, a dose of 0.8 mg PO q d can be tried.¹¹

3.2 Regional brain syndromes

This section serves to briefly describe typical syndrome associated with lesions in various areas of the brain. Unless otherwise noted, lesions considered are destructive.

3.2.1 Overview

1. frontal lobe
 - a) unilateral injury:
 - may produce few clinical findings except with very large lesions
 - bilateral or large unilateral lesions: apathy, abulia

- the frontal eye field (for contralateral gaze) is located in the posterior frontal lobe (Br. area 8, shown as the striped area in □ Fig. 1.1). Destructive lesions impair gaze to the contralateral side (patient looks towards the side of the lesion), whereas irritative lesions (i.e. seizures) cause the center to activate, producing contralateral gaze (patient looks away from the side of the lesion). See also Extraocular muscle (EOM) system (p.565) for more details.
- b) bilateral injury: may produce apathy, abulia
- c) olfactory groove region: may produce Foster Kennedy syndrome (see below)
- d) prefrontal lobes control “executive function”: planning, prioritizing, organizing thoughts, suppressing impulses, understanding the consequences of decisions
- 2. parietal lobe: major features (see below for details)
 - a) either side: cortical sensory syndrome, sensory extinction, contralateral homonymous hemianopia, contralateral neglect
 - b) dominant parietal lobe lesion (left in most): language disorders (aphasias), Gerstmann’s syndrome (p.98), bilateral astereognosis
 - c) non-dominant parietal lobe lesions: topographic memory loss, anosognosia and dressing apraxia
- 3. occipital lobe: homonymous hemianopsia
- 4. cerebellum
 - a) lesions of the cerebellar hemisphere cause ataxia in the ipsilateral limbs
 - b) lesions of the cerebellar vermis cause truncal ataxia
- 5. brainstem: usually produces a mixture of cranial nerve deficits and long tract findings (see below for some specific brainstem syndromes)
- 6. pineal region
 - a) Parinaud’s syndrome (p.99)

3.2.2 Parietal lobe syndromes

See reference.¹² (p 308–12)

Parietal lobe anatomy

The parietal lobe is located behind the central sulcus, above the Sylvian fissure, merging posteriorly into the occipital lobe (the border on the medial surface of brain is defined by a line connecting the parieto-occipital sulcus to the pre-occipital notch).

Parietal lobe neurophysiology

- either side: anterior parietal cortex organizes tactile precepts (probably contralateral) and integrates with visual and auditory sensation to build awareness of body and its spatial relations
- dominant side (on left in 97% of adults): understanding language, includes “cross-modal matching” (auditory-visual, visual-tactile, etc.). Dysphasia present with dominant lobe lesions often impedes assessment
- non-dominant side (right in most): integrates visual and proprioceptive sensation to allow manipulation of body and objects, and for certain constructional activities

Clinical syndromes of parietal lobe disease

Overview

1. unilateral parietal lobe disease (dominant or non-dominant):
 - a) cortical sensory syndrome (see below) and sensory extinction (neglecting 1 of 2 simultaneously presented stimuli). Large lesion → hemianesthesia
 - b) congenital injury → mild hemiparesis & contralateral muscle atrophy
 - c) homonymous hemianopia or visual inattentiveness
 - d) occasionally: anosognosia
 - e) neglect of contralateral half of body and visual space (more common with right side lesions)
 - f) abolition of optokinetic nystagmus to one side
2. additional effects of dominant parietal lobe lesion (left in most):
 - a) language disorders (aphasias)
 - b) speech-related or verbally mediated functions, e.g. cross-modal matching (e.g. patient understands spoken words and can read, but cannot understand sentences with elements of relationships)

- c) Gerstmann's syndrome, classically:
 - agraphia without alexia (patients can read but cannot write)
 - left-right confusion
 - digit agnosia: inability to identify finger by name
 - acalculia
 - d) tactile agnosia (bilateral astereognosis)
 - e) bilateral ideomotor apraxia (inability to carry out verbal commands for activities that can otherwise be performed spontaneously with ease)
3. additional effects of non-dominant parietal lobe lesions (usually right):
- a) topographic memory loss
 - b) anosognosia and dressing apraxia

Cortical sensory syndrome

Lesion of postcentral gyrus, especially area that maps to hand.

- sensory deficits:
 - a) loss of position sense and of passive movement sense
 - b) inability to localize tactile, thermal, and noxious stimuli
 - c) astereognosis (inability to judge object size, shape, and identity by feel)
 - d) agraphesthesia (cannot interpret numbers written on hand)
 - e) loss of two point discrimination
- preserved sensations: pain, touch, pressure, vibration, temperature
- other features
 - a) easy fatigability of sensory perceptions
 - b) difficulty distinguishing simultaneous stimulations
 - c) prolongation of superficial pain with hyperpathia
 - d) touch hallucinations

Anton-Babinski syndrome

A unilateral asomatagnosia. May seem more common with non-dominant (usually right) parietal lesions because it may be obscured by the aphasia that occurs with dominant (left) sided lesions.

1. anosognosia (indifference or unawareness of deficits, patient may deny that paralyzed extremity is theirs)
2. apathy (indifference to failure)
3. allocheiria (one-sided stimuli perceived contralaterally)
4. dressing apraxia: neglect of one side of body in dressing and grooming
5. extinction: patient is unaware of contralateral stimulus when presented with double-sided simultaneous stimulation
6. inattention to an entire visual field (with or without homonymous hemianopia), with deviation of head, eyes, and torsion of body to unaffected side

Parietal lobe aphasias

1. Wernicke's aphasia: lesion of auditory association areas or their separation from angular gyrus and primary auditory cortex. A fluent aphasia (normal sentence length & intonation, devoid of meaning). May include paraphasias. Lesion in region of Wernicke's area (Brodmann areas 40 & 39, □ Fig. 1.1)
2. Broca's (motor) aphasia: in reality, "apraxia" of motor sequencing for speech (speech and phonation muscles aren't paralyzed, and function for other activities), producing faltering, dysarthric speech. Lesion in region of Broca's area (Brodmann area 44, □ Fig. 1.1)
3. global aphasia: usually due to lesion that destroys large portion of language center; all aspects of speech and language affected
 - a) unable to speak except for some clichés, habitual phrases, or expletives
 - b) anomia (inability to name objects or parts of objects)
 - c) verbal and motor perseveration
 - d) unable to understand all except for a few words
 - e) inability to read or write
4. conduction aphasia: due to disruption of connections between frontal and temporal speech areas, usually involving supramarginal gyrus. Similar to Wernicke's (fluent spontaneous speech and paraphasias), but patients understand spoken or written words, and are aware of their deficit. Repetition is severely affected

5. pure word blindness: AKA alexia without agraphia (rare) due to lesion in parieto-occipital lobe that interrupts connections between left angular gyrus and both occipital lobes. Patients can write, but are unable to read what they've written, and frequently seem unconcerned about this. Often accompanied by loss of ability to name colors. Reading and naming numbers usually preserved

3.2.3 Foster Kennedy syndrome

Named after neurologist Robert Foster Kennedy. Usually from olfactory groove or medial third sphenoid wing tumor (usually meningioma). Now rare due to early detection by CT or MRI. Classic triad:

1. ipsilateral anosmia
2. ipsilateral central scotoma (with optic atrophy due pressure on optic nerve)
3. contralateral papilledema (from elevated ICP)

Occasionally ipsilateral proptosis will also occur due to orbital invasion by tumor.

3.2.4 Brain stem and related syndromes

Weber's syndrome

Cr. N. III palsy with contralateral hemiparesis; also see Lacunar strokes (p.1267). Third nerve palsies from parenchymal lesions may be relatively pupil sparing.

Benedikt's syndrome

Similar to Weber's, plus red nucleus lesion. Cr. N. III palsy with contralateral hemiparesis except arm which has hyperkinesia, ataxia, and a coarse intention tremor. Lesion: midbrain tegmentum involving red nucleus, brachium conjunctivum, and fascicles of III.

Millard-Gubler syndrome

Facial (VII) & abducens (VI) palsy + contralateral hemiplegia (corticospinal tract) from lesion in base of pons (usually ischemic infarct, occasionally tumor).

3.2.5 Parinaud's syndrome

Definition

AKA dorsal midbrain syndrome, AKA pretectal syndrome. As originally described, a supranuclear paralysis of vertical gaze resulting from damage to the mesencephalon.¹³

There are a number of somewhat varying descriptions, however most include:

- supranuclear upward gaze palsy (i.e. upgaze palsy affecting both voluntary saccadic and pursuit movements, with preservation of vestibulo-ocular or oculocephalic (doll's eyes) reflexes in most cases). Horizontal eye movements are spared
- lid retraction (Collier's sign): NB: upgaze palsy + lid retraction produces the "setting sun sign"
- convergence palsy
- accommodation palsy
- less common associations: pseudoabducens palsy (AKA thalamic esotropia), see-saw nystagmus, fixed pupils, dissociated light-near response (pseudo-Argyll Robertson), convergence spasm, nystagmus retractorius, internuclear ophthalmoplegia (INO)

Skew deviation may be a unilateral variant of Parinaud's syndrome.

Syndrome of the Sylvian aqueduct: Parinaud's syndrome (PS) combined with downgaze palsy.

Differential diagnosis

Etiologies

1. masses pressing directly on quadrigeminal plate (e.g. pineal region tumors)
2. elevated ICP: secondary to compression of mesencephalic tectum by dilated suprapineal recess, e.g. in hydrocephalus

3. stroke or hemorrhage in upper brainstem
4. multiple sclerosis (MS)
5. occasionally seen with toxoplasmosis

Conditions affecting ocular motility that could mimic the upgaze palsy of PS:

1. Guillain-Barré syndrome
2. myasthenia gravis
3. botulism
4. hypothyroidism
5. there may be a gradual benign loss of upgaze with senescence

3.3 Jugular foramen syndromes

3.3.1 Applied anatomy

The jugular foramen (JF) is one of a pair of openings between the lateral part of the occipital bone and the petrous part of the temporal bone. The foramen is usually divided in 2 by a bony spine from the petrous temporal bone that attaches via a fibrous bridge (which is bony in 26%) to the jugular process of the occipital bone.¹⁴ The right JF is usually larger than the left.^{14,15} The carotid ridge separates the JF from the nearby carotid canal. Contents of jugular foramen (JF): Cr. N. IX, X, XI, petrosal sinus, sigmoid sinus, some meningeal branches from the ascending pharyngeal and occipital arteries.¹⁶

Nearby: Cr. N. XII passes through the hypoglossal canal just above the occipital condyle. The carotid artery with the sympathetic plexus enters the carotid canal.

Compartmentalization of the jugular foramen remains controversial. As many as 4 foramina have been described over the years. Although it had been recognized previously, an early 2-compartment description was published in 1967 by Hovelacque.¹⁷ In this, the bony spine ($\pm$ its fibrous septum) divide the foramen into:

- pars vascularis: the larger posterolateral compartment containing the vagal nerve (and branching Arnold's nerve), spinal accessory nerve and the internal jugular vein
- pars nervosa: the smaller anteromedial compartment containing the glossopharyngeal nerve (and branching Jacobson's nerve), inferior petrosal sinus and meningeal branch of the ascending pharyngeal artery

A publication in 1997 described these 3 compartments¹⁸:

- Sigmoid: large posterolateral compartment containing sigmoid sinus
- Petrosal: smaller anteromedial compartment containing petrosal sinus
- Intrajugular or neural: CN 9, 10 and 11

3.3.2 Clinical syndromes

General information

A number of eponymous syndromes with some conflicting findings in the literature have been described. See □ Table 3.2 for a summary and □ Fig. 3.2 for a schematic diagram of deficits in various JF syndromes.

Vernet's syndrome: CN IX, X & XI palsy

AKA syndrome of the jugular foramen. Usually due to intracranial lesion.

Etiologies include: jugular foramen tumors, ICA dissections, mycotic aneurysms of the external carotid, thrombosis of the jugular vein, following carotid endarterectomy.

Symptoms: unilateral paralysis of the palate, vocal cords, sternocleidomastoid, trapezius, with loss of taste in the posterior 1/3 tongue, anesthesia of the soft palate, larynx and pharynx.

Collet-Sicard syndrome

Palsies of CN IX, X, XI & XII without sympathetic involvement. More likely with lesion outside skull. If caused by an intracranial lesion, it would have to be of such a large size that it would usually produce brain stem compression → long tract findings.

Table 3.2 Cranial nerve dysfunction in jugular foramen syndromes							
Nerve	Result of lesion	Syndrome					
		Vernet	Collet Sicard	Villaret	Tapia	Jackson	Schmidt
IX	loss of taste and sensation in posterior third of tongue	X	X	X			
X	paralysis of vocal cords & palate, anesthesia of pharynx & larynx	X	X	X	X	X	X
XI	weak trapezius & SCM	X	X	X	±	X	X
XII	tongue paralysis & atrophy		X	X	X	X	
sympathetics	Horner's syndrome			X	±		

Key: X indicates dysfunction / deficit (lesion) of that nerve; ± indicates involvement may or may not occur

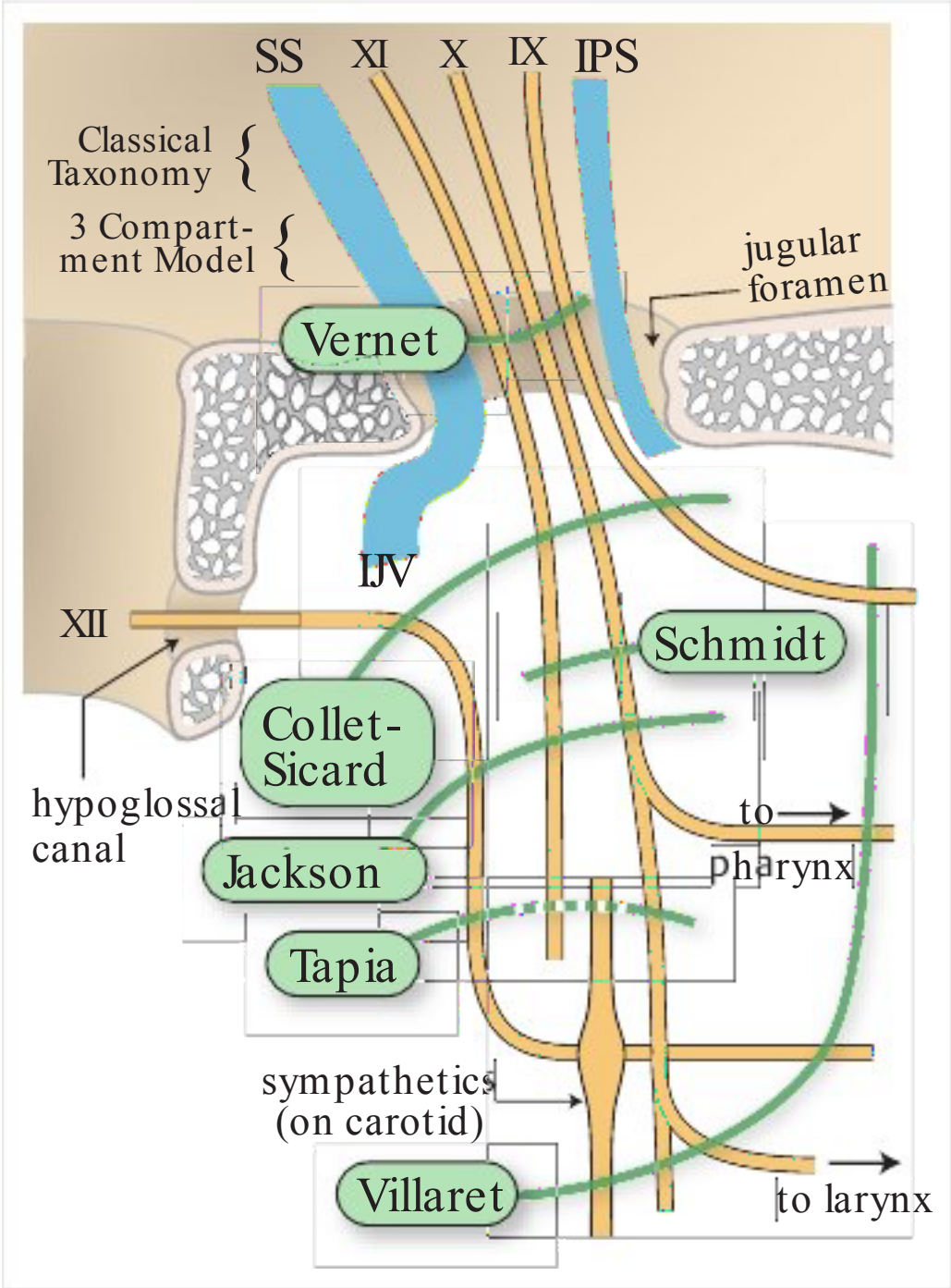


Fig. 3.2 Schematic diagram of the jugular foramen (coronal section through left jugular foramen viewed from the front). Includes the classic 2 compartment model and the 3 compartment classification of Katsuta et al.¹⁸ Jugular foramen syndromes are illustrated: a solid line through a nerve indicates a deficit, dashed line indicates ± involvement.

Etiologies include: condylar and Jefferson’s fractures, internal carotid dissection, primary and metastatic tumors, Lyme disease and fibromuscular dysplasia.

Symptoms: unilateral paralysis of the palate, vocal cords, sternocleidomastoid, trapezius, tongue, loss of taste in posterior 1/3 tongue, anesthesia of soft palate, larynx and pharynx.

Villaret's syndrome: CN IX, X, XI & XII palsy + sympathetic dysfunction

AKA posterior retropharyngeal syndrome, AKA the nervous syndrome of the posterior retroparotid space). Collet-Sicard syndrome with sympathetic involvement. Usually due to retropharyngeal lesions.

Etiologies include: parotid tumors, metastases, external carotid aneurysm and osteomyelitis of the skull base.

Symptoms: as with Collet-Sicard + Horner's syndrome.

Tapia syndrome: CN X & XII palsy (± XI)

AKA Matador's disease (first described in a bullfighter by Antonio Garcia Tapia). Some authors describe an intracranial and extracranial form.¹⁹

Etiologies include: oral intubation (majority of cases prior to 2013), metastases, rarely associated with carotid or vertebral artery dissections.

Symptoms: hoarseness of voice, dysphagia secondary to incoordination of tongue and food bolus propulsion, unilateral atrophy and paralysis of the tongue, ± paralysis of sternocleidomastoid & trapezius, sparing the soft palate.

(Hughlings) Jackson's syndrome: CN X, XI & XII palsy

First described in 1864 with unilateral paralysis of the soft palate, larynx, sternocleidomastoid, trapezius and tongue.

Schmidt syndrome: CN X & XI

AKA vago-spinal syndrome. Schmidt first described this in 1892. Unilateral vocal cord and paralysis of sternocleidomastoid, soft palate, larynx and trapezius.

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Part II

General and Neurology

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III

4 Neuroanesthesia

4.1 General information

□ Table 4.1 shows the American Society of Anesthesiologists (ASA) grading system to estimate anesthetic risk for various conditions.

For issues related to intracranial pressure (ICP), cerebral perfusion pressure (CPP), intracranial constituents, etc., see ICP principles (p.856). For cerebral blood flow (CBF) and cerebral metabolic rate of oxygen consumption (CMRO₂), see CBF and oxygen utilization (p.1264).

Parameters of primary relevance to neurological surgery that can be modulated by the anesthesiologist:

1. blood pressure: one of the factors that determines CPP as well as spinal cord perfusion. May need to be manipulated (e.g. reduced when working on an aneurysm, or increased to enhance collateral circulation during cross clamping). Measurement by arterial line is most accurate and depending on the patient's presentation and the planned procedure, often should be placed prior to induction of anesthesia. For intracranial procedures, the arterial line should be calibrated at the external auditory meatus to most closely reflect intracranial blood pressure
2. jugular venous pressure: one of the factors that influences ICP
3. arterial CO₂ tension (PaCO₂): CO₂ is the most potent cerebral vasodilator. Hyperventilation reduces PaCO₂ (hypocapnea) which decreases CBV but also CBF. Goal is generally end tidal CO₂ (ETCO₂) of 25–30 mm Hg with a correlating PaCO₂ of 30–35. Use with care for stereotactic procedures to minimize shift of intracranial contents when using this method to control ICP⁴
4. arterial O₂ tension
5. hematocrit: in neurosurgery it is critical to balance oxygen carrying capacity (decreased by anemia) against improved blood rheology (impaired by elevated Hct)
6. patient temperature: mild hypothermia provides some protection against ischemia by reducing the cerebral metabolic rate of oxygen (CMRO₂) by $\approx 7\%$ for each 1° C drop
7. blood glucose level: hyperglycemia exacerbates ischemic deficits⁵
8. CMRO₂: reduced with certain neuro-protective agents and by hypothermia which helps protect against ischemic injury
9. in cases where a lumbar drain or a ventricular drain has been placed: CSF output
10. elevation of the head of the patient: lowering the head increases arterial blood flow, but also increases ICP by impairing venous outflow
11. intravascular volume: hypovolemia can impair blood flow in neurovascular cases. In surgery in the prone position, excessive fluids may contribute to facial edema which is one of the risk factors for PION (p.1056)
12. positioning injuries: during the procedure, the patient's position may change and be unnoticed due to draping. Careful and frequent examination of the patient's position may prevent injuries associated with prolonged malpositioning
13. post operative nausea and vomiting (PONV): may adversely affect ICP and may negatively impact recent cervical surgical procedures. Avoidance of anesthetic agents known to cause PONV or pretreatment to prevent PONV may be prudent

4.2 Drugs used in neuroanesthesia

4.2.1 Inhalational agents

General information

Most reduce cerebral metabolism (except nitrous oxide) by suppressing neuronal activity. These agents disturb cerebral autoregulation and cause cerebral vasodilatation which increases cerebral blood volume (CBV) and can increase ICP. With administration >2 hrs they increase CSF volume which can also potentially contribute to increased ICP. Most agents increase the CO₂ reactivity of cerebral blood vessels. These agents affect intra-operative EP monitoring (p.107).

Drug info: Nitrous oxide

A potent vasodilator that markedly increases CBF and minimally increases cerebral metabolism. Contributes to post-op N/V.

Nitrous oxide, pneumocephalus and air embolism: The solubility of nitrous oxide (N₂O) is ≈ 34 times that of nitrogen.⁶ When N₂O comes out of solution in an airtight space it can increase the pressure which may convert pneumocephalus to “tension pneumocephalus.” It may also aggravate air embolism. Thus caution must be used especially in the sitting position where significant post-op pneumocephalus and air embolism are common. The risk of tension pneumocephalus may be reduced by filling the cavity with fluid in conjunction with turning off N₂O about 10 minutes prior to completion of dural closure. See Pneumocephalus (p. 887).

Halogenated agents

Agents in primary usage today are shown below. All suppress EEG activity and may provide some degree of cerebral protection.

Drug info: Isoflurane (Forane®)

Can produce isoelectric EEG without metabolic toxicity. Improves neurologic outcome in cases of incomplete global ischemia (although in experimental studies on rats, the amount of tissue injury was greater than with thiopental⁷).

Drug info: Desflurane (Suprane®)

A cerebral vasodilator, increases CBF and ICP. Decreases CMRO₂ which tends to cause a compensatory vasoconstriction.

Drug info: Sevoflurane (Ultane®)

Mildly increases CBP and ICP, and reduces CMRO₂. Mild negative inotrope, cardiac output not as well maintained as with isoflurane or desflurane.

4.2.2 Intravenous anesthetic agents

Agents generally used for induction

1. propofol: exact mechanism of action unknown. Short half-life with no active metabolites. May be used for induction and as a continuous infusion during total intravenous anesthesia (TIVA). Causes dose dependent decrease in mean arterial blood pressure (MAP) and ICP. See also information other than use in induction (p. 106). Is more rapidly cleared than, and has largely replaced, thiopental
2. barbiturates: produce significant reduction in CMRO₂ and scavenge free radicals among other effects (p. 1202). Produce dose-dependent EEG suppression which can be taken all the way to isoelectric. Minimally affect EPs. Most are anticonvulsant, but methohexital (Brevital®) (p. 132) can lower the seizure threshold. Myocardial suppression and peripheral vasodilatation from barbiturates may cause hypotension and compromise CPP, especially in hypovolemic patients
 - sodium thiopental (Pentothal®): the most common agent. Rapid onset, short acting. Minimal effect on ICP, CBF and CMRO₂
3. etomidate (Amidate®): a carboxylated imidazole derivative. Anesthetic and amnestic, but no analgesic properties. Sometimes produces myoclonic activity which may be confused with seizures. Impairs renal function and should be avoided in patients with known renal disease. May

produce adrenal insufficiency. See Miscellaneous drugs in neuroanesthesia (p.106) for information other than use in induction.

4. ketamine: NMDA receptor antagonist. Produces a dissociative anesthesia. Maintains cardiac output. May slightly increase both heart rate and blood pressure. ICP increases in parallel with increased cardiac output.

Narcotics in anesthesia

Nonsynthetic narcotics

Narcotics increase CSF absorption and minimally reduce cerebral metabolism. They slow the EEG but will not produce an isoelectric tracing. □ All narcotics cause dose-dependent respiratory depression which can result in hypercarbia and concomitant increased ICP in non-ventilated patients. Often also contribute to post-op N/V

Morphine: does not significantly cross the BBB.

□ Disadvantages in neuro patients:

1. causes histamine release which
 - a) may produce hypotension
 - b) may cause cerebrovascular vasodilation → increased ICP⁸ (p.1593)
 - c) the above together may compromise CPP
2. in renal or hepatic insufficiency, the metabolite morphine-6-glucuronide can accumulate which may cause confusion

Synthetic narcotics

These do not cause histamine release, unlike morphine and meperidine.

□ Remifentanyl (Ultiva®); see also detailed information (p.133): reduces CMRO₂, CBV and ICP. Large doses may be neurotoxic to limbic system and associated areas. May be used for awake craniotomy (p.1432).

Fentanyl: crosses the BBB. Reduces CMRO₂, CBV and ICP. May be given as bolus and/or as a continuous infusion.

Sufentanyl: more potent than fentanyl. Does not increase CBF. □ Raises ICP (may be due to hypoventilation – which can occur with any narcotic) and is thus often not appropriate for neurosurgical cases. Expensive.

4.2.3 Miscellaneous drugs in neuroanesthesia

□ Benzodiazepines. These drugs are GABA agonists and decrease CMRO₂. They also provide anti-convulsant action and produce amnesia. See also agents and reversal (p.205).

□ Etomidate (p.105) . Used primarily for induction (p.105).

- a cerebrovasoconstrictor which therefore: reduces CBF and ICP; reduces CMRO₂ but no longer promoted as a cerebral protectant based on experimental studies⁹ and a drop in pBtO₂ with temporary MCA clipping¹⁰
- does not suppress brainstem activity
- suppresses adrenocortical function cortisol production. This usually occurs with prolonged administration, but can occur even after single dose for induction and may persist up to 8 hrs (no adverse outcomes from short-term suppression have been reported)
- increases activity of seizure foci which may be used for mapping foci during seizure surgery but may also induce seizures

□ Propofol. A sedative hypnotic. Useful for induction (p.105). Reduces cerebral metabolism, CBF and ICP. Has been described for cerebral protection (p.1203) and for sedation (p.133). Short half-life permits rapid awakening which may be useful for awake craniotomy (p.1434). Not analgesic.

□ Lidocaine. Given IV, suppresses laryngeal reflexes which may help blunt ICP elevations that normally follow endotracheal intubation or suctioning. Anticonvulsant at low doses, may provoke seizures at high concentrations.

□ Esmolol. Selective beta-1 adrenergic antagonist, blunts the sympathetic response to laryngoscopy and intubation. Less sedating than equipotent doses of lidocaine or fentanyl used for the same purpose. Half life: 9 minutes. See also dosing, etc. (p.127)

□ Dexmedetomidine (Precedex®). Alpha 2 adrenergic receptor agonist, used for control of hypertension post operatively, as well as for its sedating qualities during awake craniotomy either alone or in conjunction with propofol (p.105). Also used to help patients tolerate endotracheal tube without sedatives/narcotics to facilitate extubation.

4.2.4 Paralytics for intubation

Paralytics (neuromuscular blocking agents (NMBA)): administered to facilitate tracheal intubation and to improve surgical conditions when indicated. Administration of paralytics ideally should always be guided by neuromuscular twitch monitoring. Also see Sedatives & paralytics (p.132). In addition to paralytics, all conscious patients should also receive a sedative to blunt awareness.

Paralytics should not be given until it has been determined that patient can be ventilated manually, unless treating laryngospasm (may be tested with thiopental). Use with caution in non-fixated patients with unstable C-spine.

Due to long action, pancuronium (Pavulon®) is not indicated as the primary paralytic for intubation, but may be useful once patient is intubated or in low dose as an adjunct to succinylcholine.

Drug info: Succinylcholine (Anectine®)

The only depolarizing agent. May be used to secure airway for emergency intubation, but due to possible side effects (p. 135), should not be used acutely following injury or in adolescents or children (a short acting nondepolarizing blocker is preferred). May transiently increase ICP. Prior dosing with 10% of the ED95 dose of a non-depolarizing muscle relaxant reduces muscle fasciculations.

□ Intubating dose: 1–1.5 mg/kg (supplied as 20 mg/ml → 3.5–5 cc for a 70 kg patient), onset 60–90 sec, duration 3–10 min, may repeat same dose × 1.

Drug info: Rocuronium (Zemuron®)

Intermediate acting, aminosteroid, non-depolarizing muscle relaxant. The only nondepolarizing neuromuscular blocking agent approved for rapid sequence intubation. Duration of action and onset are dose dependent. □ (p.135).

Drug info: Vecuronium (Norcuron®)

See details (p. 136).

Aminosteroid with activity similar to that of rocuronium, however, does not cause histamine release and is not approved for rapid sequence intubation. □.

Drug info: Cisatracurium (Nimbex®)

See details (p. 136).

Metabolized by Hoffman degradation (temperature dependent), intermediate acting, no significant increases in histamine. □

4.3 Anesthetic requirements for intra-operative evoked potential monitoring

For details of intra-operative evoked potential (EP) monitoring itself, see Intra-operative evoked potentials (p.239).

All volatile anesthetics produce dose-dependent reduction in SSEP peak amplitude and an increase in peak latency. Adding nitrous oxide increases this sensitivity to anesthetic agents.

Anesthesia issues related to intra-operative evoked potential (EPs) monitoring:

1. induction: minimize pentothal dose (produces ≈ 30 minutes of suppression of EPs), or use etomidate (which increases both SSEP amplitude and latency¹¹)
2. total intravenous anesthesia (TIVA) is ideal (i.e. no inhalational agents)
3. nitrous/narcotic technique is a distant second choice
4. if inhalational anesthetic agents are required:
 - a) use <1 MAC (maximal alveolar concentration), ideally <0.5 MAC
 - b) avoid older agents such as Halothane
5. nondepolarizing muscle relaxants have little effect on EP (in monkeys¹²)
6. propofol has a mild effect on EP: total anesthesia with propofol causes less EP depression than inhalational agents at the same depth of anesthesia¹³
7. benzodiazepines have a mild-to-moderate depressant effect on EPs
8. continuous infusion of anesthetic drugs is preferred over intermittent boluses
9. SSEPs can be affected by hyper- or hypo-thermia, and changes in BP
10. hypocapnia (down to end tidal $\text{CO}_2 = 21$) causes minimal reduction in peak latencies¹⁴
11. antiepileptic drugs: phenytoin, carbamazepine and phenobarbital do not affect SSEP¹⁵

4.4 Malignant hyperthermia

4.4.1 General information

Malignant hyperthermia (MH) is a hypermetabolic state of skeletal muscle due to idiopathic block of Ca^{++} re-entry into sarcoplasmic reticulum. Transmitted by a multifactorial genetic predisposition. Total body O_2 consumption increases $\times 2-3$.

Incidence: 1 in 15,000 anesthetic administrations in peds. 1 in 40,000 adults. 50% had previous anesthesia without MH. Frequently associated with administration of halogenated inhalational agents and the use of succinylcholine (fulminant form: muscle rigidity almost immediately after succinylcholine, may involve masseters $\rightarrow$ difficulty intubating). Initial attack and recrudescence may also occur post-op. 30% mortality.¹⁶

4.4.2 Presentation

1. earliest possible sign: increase in end-tidal pCO_2
2. tachycardia (early) and other arrhythmias
3. with progression:
 - a) coagulation disorder (DIC) (bleeding from surgical wound and body orifices)
 - b) ABG: increasing metabolic acidosis & decreasing pO_2
 - c) pulmonary edema
 - d) elevated body temperature (may reach $\geq 44^\circ \text{C}$ (113°F) at rate of $1^\circ \text{C}/5\text{-min}$) (normal patients become hypothermic with general anesthesia)
 - e) limb muscle rigidity (common, but late)
 - f) rhabdomyolysis $\rightarrow$ elevated CPK & myoglobin (late)
4. terminal:
 - a) hypotension
 - b) bradycardia
 - c) cardiac arrest

4.4.3 Treatment

1. eliminate offending agents (stop the operation, D/C inhalation anesthesia and change tubing on anesthesia machine)
2. dantrolene sodium (Dantrium®) 2.5 mg/kg IV usually effective, infuse until symptoms subside, up to 10 mg/kg
3. hyperventilation with 100% O_2
4. surface and cavity cooling: IV, in wound, per NG, PR
5. bicarbonate 1–2 mEq/kg for acidosis
6. IV insulin and glucose (lowers K^+ , glucose acts as energy substrate)
7. procainamide for arrhythmias
8. diuresis: volume loading + osmotic diuretics

4.4.4 Prevention

1. identification of patients at risk:
 - a) only reliable test: 4 cm viable muscle biopsy for in-vitro tests at a few regional test centers (abnormal contracture to caffeine or halothane)
 - b) family history: any relative with syndrome puts patient at risk
 - c) related traits: 50% of MH patients have heavy musculature, Duchenne type muscular dystrophy, or scoliosis
 - d) patients who exhibit masseter spasm in response to succinylcholine
2. in patients at risk: avoid succinylcholine (nondepolarizing blockers preferred if paralysis essential), may safely have non-halogenated anesthetics (narcotics, barbiturates, benzodiazepines, droperidol, nitrous...)
3. prophylactic oral dantrolene: 4–8 mg/kg/day for 1–2 days (last dose given 2 hrs before anesthesia) is usually effective

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5 Sodium Homeostasis and Osmolality

5.1 Serum osmolality and sodium concentration

Clinical significance of various serum osmolality values is shown in □ Table 5.1.

□ Serum osmolality. May be estimated using Eq (5.1).

Osmolarity mOsm/L

$$= 2 \times [\text{Na}^+] + \frac{[\text{BUN}]}{2.8} + \frac{[\text{glucose}]}{18}$$

5:15

(with [Na⁺] in mEq/L or mmol/L, and glucose and BUN in mg/dl).
NB: terms in square brackets [] represent the serum concentrations (in mEq/L for electrolytes).

□ Sodium content. In the diet: usually expressed in grams Na⁺ (not NaCl), a low sodium diet is considered 2 gm of Na⁺ per day or less.
1 teaspoon of table salt (NaCl) has 2.3 gm of Na⁺.
1 mg NaCl has 17 mEq Na⁺. 1 mg Na⁺ has 43 mEq Na⁺.
Normal saline has 0.9 gm of NaCl/100 ml. 3%NaCl has 3 gm NaCl/100 ml.

5.2 Hyponatremia

5.2.1 General information

Key concepts

- definition: serum [Na⁺] <135 mEq/L. Common etiologies:
 - SIADH: hypotonic hyponatremia (effective serum osmol<275 mOsm/L) with inappropriately high urinary concentration (urine osmol>100 mOsm/L) and euolemia or hypervolemia
 - cerebral salt wasting (CSW): similar to SIADH but with extracellular fluid volume depletion due to renal sodium loss (urinary [Na}>20 mEq/L)
- minimum W/U: □ serum [Na⁺], □ serum osmolality, □ urine osmolality, □ clinical assessment of volume status. If volume status is high or low: □ urinary [Na⁺] □ TSH (to R/O hypothyroidism)
- treatment: based on acuity, severity, symptoms & etiology; see SIADH (p. 115) or CSW (p. 118) as appropriate
- risk of overly rapid correction: osmotic demyelination (including central pontine myelinolysis)

□ Classification. [Na⁺] <135 mEq/L=mild, <130 =moderate, <125 =severe hyponatremia.

□ Hyponatremia in neurosurgical patients. Chiefly seen in:

- syndrome of inappropriate antidiuretic hormone secretion (p.114), SIADH (p.114): dilutional hyponatremia with normal or elevated intravascular volume. The most common type of

Table 5.1 Clinical correlates of serum osmolality	
Value (mOsm/L)	Comment
282–295	normal
<240 or>321	panic values
>320	risk of renal failure
>384	produces stupor
>400	risk of generalized seizures
>420	usually fatal

- hyponatremia.¹ Usually treated with fluid restriction. May be associated with numerous intracranial abnormalities (□ Table 5.2) and following transsphenoidal surgery
- cerebral salt wasting (CSW): inappropriate natriuresis with volume depletion. Treated with volume replacement (opposite to SIADH) and sodium; symptoms from derangements due to CSW may be exacerbated by fluid restriction (p.118).² Etiology of 6%of cases of hyponatremia following aneurysmal SAH³

Table 5.2 Etiologies of SIAD ^a
Malignant tumors
1. especially bronchogenic small-cell Ca 2. tumors of GI or GU tract 3. lymphomas 4. Ewing’s sarcoma
CNS disorders
1. infection: a) encephalitis b) meningitis: especially in peds c) TB meningitis d) AIDS e) brain abscess 2. head trauma: 4.6% prevalence 3. increased ICP: hydrocephalus, SDH... 4. SAH 5. brain tumors 6. cavernous sinus thrombosis 7. □ post craniotomy, especially following surgery for pituitary tumors, craniopharyngiomas, hypothalamic tumors 8. MS 9. Guillain-Barré 10. Shy-Drager 11. delirium tremens (DTs)
Pulmonary disorders
1. infection: pneumonia (bacterial & viral), abscess, TB, aspergillosis 2. asthma 3. respiratory failure associated with positive pressure respiration
Drugs
1. drugs that release ADH or potentiate it a) chlorpropamide (Diabinese®): increases renal sensitivity to ADH b) carbamazepine (Tegretol®), even more common with oxcarbazepine c) HCTZ d) SSRIs, TCAs e) clofibrate f) vincristine g) antipsychotics h) NSAIDs i) MDMA (“ecstasy”) 2. ADH analogues a) DDAVP b) oxytocin: ADH cross activity, may also be contaminated with ADH
Endocrine disturbances
1. adrenal insufficiency 2. hypothyroidism
Miscellaneous
1. anemia 2. stress, severe pain, nausea or hypotension (all can stimulate ADH release), postoperative state 3. acute intermittent porphyria (AIP)
^a excerpted and modified ^{9,1}

□ Other etiologies of hyponatremia:

- renal failure
- volume overload (e.g. as in congestive heart failure)
- pseudohyponatremia: osmotically active solutes (e.g. glucose, mannitol, marked hyperlipidemia, or hyperproteinemia (which can occur in multiple myeloma)⁴) draw water from cells and also reduce the water fraction of plasma and produce artifactually low sodium values (an artifact of indirect lab techniques). For every 100 mg/dl increase of glucose, serum [Na] decreases by 1.6–2.4 mEq/L. It is necessary to measure serum osmolality to rule-out pseudohyponatremia
- postoperative hyponatremia: a rare condition usually described in young, otherwise healthy women undergoing elective surgery⁵ and may be related to administration of even only mildly hypotonic fluids (sometimes in modest amounts)⁶ and the actions of ADH (which may be increased due to stress, pain or medications)

5.2.2 Evaluation of hyponatremia

□ Fig. 5.1 shows an algorithm for evaluating the etiology of hyponatremia⁷ which drives treatment decisions. Work-up requires assessment of:

1. serum sodium: must be <135 mEq/L to qualify as hyponatremia
2. the *effective* serum osmolality (AKA tonicity) is shown in Eq (5.2)

$$\text{effective serum osmolality} = \text{measured osmolality} - \frac{\text{BUN (mg/dl)}}{2} \quad (5.2)$$

and should be used when the blood urea nitrogen (BUN) level is elevated (for a normal [BUN] of 7–18 mg/dl, just subtract 5 from the measured osmolality). Values <275 mOsm/kg indicate hypotonic hyponatremia

1. urine osmolality: values >100 mOsm/kg are inappropriately high if serum tonicity is <275 mOsm/kg
2. volume status: differentiates SIADH from CSW
 - a) clinical assessment: better for hypervolemia (edema, upward trend in patient weights) but is insensitive in identifying extracellular fluid depletion as an etiology of hyponatremia⁸ (look for dry mucous membranes, loss of skin turgor, orthostatic hypotension)
 - b) normal saline infusion test used in uncertain cases. If baseline urine osmolality is <500 mOsm/kg, it is usually safe to infuse 2 L of 0.9% saline over 24–48 hours. Correction of the hyponatremia suggests extracellular fluid volume depletion was the cause
 - c) central venous pressure (CVP) may be used: CVP <5–6 cm H₂O suggests hypovolemia in patients with normal cardiac function^{3,7}
3. check urinary [Na⁺] if volume status is high or low
4. determine duration of hyponatremia:
 - a) duration documented as <48 hours is considered acute
 - b) hyponatremia of >48 hours duration or of unknown duration is chronic
 - c) hyponatremia that occurs outside the hospital is usually chronic and asymptomatic except in marathoners and MDMA (“ecstasy”) drug users

5.2.3 Symptoms

Due to slow compensatory mechanisms in the brain, a gradual decline in serum sodium is better tolerated than a rapid drop. Symptoms of mild ([Na] <130 mEq/L) or gradual hyponatremia include: anorexia, headache, difficulty concentrating, irritability, dysgeusia and muscle weakness. Severe hyponatremia (<125 mEq/L) or a rapid drop (>0.5 mEq/hr) can cause neuromuscular excitability, cerebral edema, muscle twitching and cramps, nausea/vomiting, confusion, seizures, respiratory arrest and possibly permanent neurologic injury, coma or death.

5.2.4 Syndrome of inappropriate antidiuresis (SIAD)

This term covers excess water retention in the face of hyponatremia, including cases due to inappropriate ADH secretion (SIADH) as well as others without increased circulating levels of ADH (e.g.

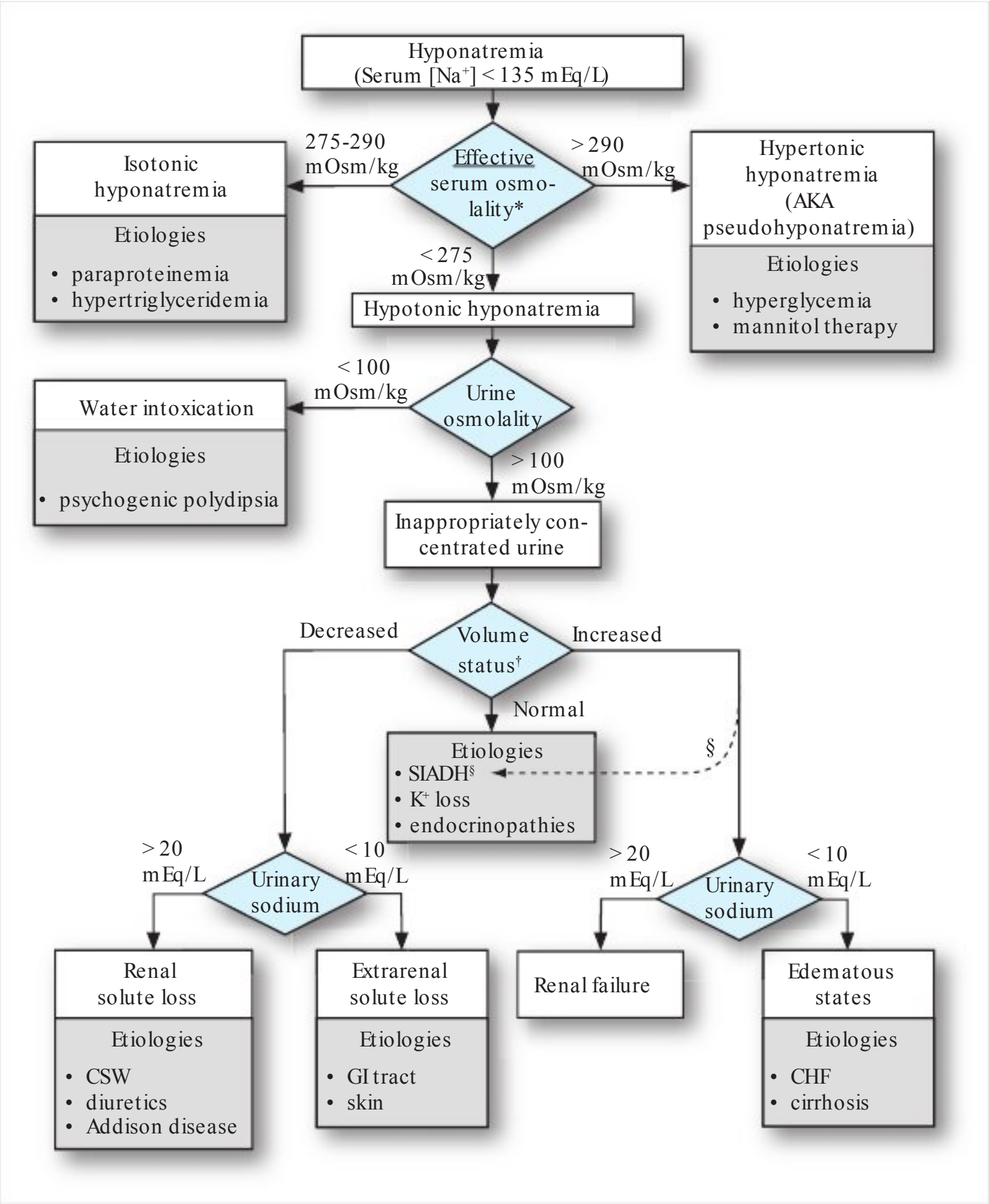


Fig. 5.1 Evaluation of the etiology of hyponatremia (adapted⁷)
* effective serum osmolality=measured osmolality – [BUN]/2.8 (Eq (5.2))
† volume status is usually assessed clinically, but this may be insensitive to volume depletion
§ SIADH may be associated with euolemia or hypervolemia

heightened response to ADH, certain drugs...). A partial list of etiologies is shown in □ Table 5.2 (see references^{1,9} for details).
The diagnostic criteria of SIAD is shown in □ Table 5.3. It is critical to measure serum osmolality to rule-out pseudohyponatremia (p.112), an artifact of indirect lab techniques.

Table 5.3 Diagnostic criteria for SIAD ¹	
Essential features	
• decreased effective serum osmolality^a (<275 mOsm/kg of water) • simultaneous urine osmolality>100 mOsm/kg of water • clinical euvoemia a) no clinical signs of extracellular (EC) volume orthostatic hypotension (orthostasis, tachycardia, decreased skin turgor, dry mucous membranes...) b) no clinical signs of excess EC volume (edema, ascites...) • urinary [Na]>40 mEq/L with normal dietary Na intake • normal thyroid and adrenal function • no recent diuretic use	
Supplemental features	
• plasma [uric acid]<4 mg/dl • [BUN]<10 mg/dl • fractional Na excretion>1%[‡] fractional urea excretion>55% • NS infusion test: failure to correct hyponatremia with IV infusion of 2 L 0.9% saline over 24–48 hrs • ^bcorrection of hyponatremia with fluid restriction • abnormal result on water load test^c: a) <80% excretion of 20 ml of water/kg body weight over 5 hours, or b) inadequate urinary dilution (<100 mOsm/kg of water) • elevated plasma [ADH] with hyponatremia and euvoemia	
^a effective osmolality (AKA tonicity)=(measured osmolality) – [BUN]/2.8 with [BUN] measured in mg/dl	
^b this test is used in uncertain cases (corrects volume depletion), and is usually safe when baseline urine osmolality is<500 mOsm/L	
^c water load test & [ADH] levels are rarely recommended; see text for details (p.115)	

5.2.5 Syndrome of inappropriate antidiuretic hormone secretion (SIADH)

General information

Key concepts
• definition: release of ADH in the absence of physiologic (osmotic) stimuli • results in hyponatremia with hypervolemia (occasionally with euvoemia) with inappropriately high urine osmolality (>100 mOsm/L) • may be seen with certain malignancies and many intracranial abnormalities • critical to distinguish from cerebral salt wasting which produces hypovolemia • treatment: initial guidelines in brief, see details (p.115) ◦ avoid rapid correction or overcorrection to reduce risk of osmotic demyelination (p.115). Check serum [Na⁺] q 2–4 hours and do not exceed 1 mEq/L per hour, or 8 mEq/L in 24 hrs or 18 mEq/L in 48 hrs ◦ severe ([Na⁺]⁺<125 mEq/L of<48 hrs duration or with severe symptoms (coma, Sz): start 3% saline at 1–2 ml/kg body weight/hr + furosemide 20 mg IV qd ◦ severe ([Na⁺]⁺<125 mEq/L of duration>48 hours or unknown without severe symptoms: normal saline infusion @ 100 ml/hr + furosemide 20 mg IV qd ◦ chronic or unknown duration and asymptomatic: fluid restriction (□ Table 5.4) with dietary salt and protein, and, if necessary, adjuvant drugs (demeclocycline, conivaptan...)

SIADH, AKA Schwartz-Bartter syndrome, was first described with bronchogenic cancer which is one cause of SIAD. SIADH is the release of antidiuretic hormone (ADH), AKA arginine vasopressin (AVP) (p.151), in the absence of physiologic (osmotic) stimuli. Result: elevated urine osmolality, and expansion of the extracellular fluid volume leading to a dilutional hyponatremia which can produce fluid overload (hypervolemia), but SIADH may also occur with euvoemia. For unclear reasons, edema does not occur.

The hyponatremia of SIADH must be differentiated from that due to cerebral salt wasting (CSW) due to differences in treatment recommendations (p.118).

Etiologies: □ Table 5.2.

Table 5.4 Fluid restriction recommendations ¹	
Solute ratio ^a	Recommended fluid intake
> 1	<500 ml/d
1	500–700 ml/d
< 1	< 1 L/d
^a solute ratio defined as: $\frac{\text{urinary } \text{Na}^+/\text{urinary } \text{K}^+}{\text{plasma } \text{Na}^+/\text{plasma } \text{K}^+}$	

Diagnosis of SIADH

In general, 3 diagnostic criteria are: hyponatremia, inappropriately concentrated urine, and no evidence of renal or adrenal dysfunction. In more detail:

1. low serum sodium (hyponatremia): usually <134 mEq/L
2. low effective serum osmolality: <275 mOsm/L
3. high urinary sodium (salt wasting): at least >18 mEq/L, often 50–150. Note: there has not been an adequate explanation of the high urinary sodium in SIADH
4. high ratio of urine:serum osmolality: often 1.5–2.5:1, but may be 1:1
5. normal renal function (check BUN & creatinine): BUN commonly <10
6. normal adrenal function (no hypotension, no hyperkalemia)
7. no hypothyroidism
8. no signs of dehydration or overhydration (in many patients with acute brain disease, there is significant hypovolemia often due to CSW (p.118) and as this is a stimulus for ADH secretion, the ADH release may be “appropriate”¹⁰). In uncertain cases, the normal saline infusion test (p.112) may be used.

If further testing is required, the following are options, but are rarely recommended:

1. measure serum or urinary levels of ADH. Rarely indicated since urine osmolality >100 mOsm/kg is usually sufficient to indicate excessive ADH.¹ ADH is normally undetectable in etiologies of hyponatremia other than SIADH
2. water-load test: considered to be the definitive test.¹¹ The patient is asked to consume a water load of 20 ml/kg up to 1500 ml. In the absence of adrenal or renal insufficiency, the failure to excrete 65%of the water load in 4 hrs or 80%in 5 hrs indicates SIAD.
□ CONTRAINDICATIONS: this test is dangerous if the starting serum [Na⁺] is ≤124 mEq/L or if the patient has symptoms of hyponatremia

Symptoms of SIADH

Symptoms of SIADH are those of hyponatremia (p.112) and possibly fluid overload. If mild, or if descent of [Na⁺] is gradual, it may be tolerated. [Na⁺] <120–125 mEq/L is almost always symptomatic. These patients often have a paradoxical (inappropriate) thirst.

Treatment of hyponatremia with SIADH

Management is based on the severity and duration of hyponatremia, and the presence of symptoms. Two caveats:

1. □ be sure that hyponatremia is not due to CSW (p.118) before restricting fluids
2. avoid too rapid correction and avoid correcting to normal or supranormal (overcorrection) sodium to reduce the risk of osmotic demyelination syndrome

Osmotic demyelination syndrome

A complication associated with some cases of treatment for hyponatremia. While excessively slow correction of acute hyponatremia is associated with increased morbidity and mortality,¹² some cases of inordinately rapid treatment have been associated with osmotic demyelination syndrome (which includes central pontine myelinolysis (CPM) a rare disorder of pontine white matter¹³ (□ Fig. 5.2) and extrapontine myelinolysis (□ Fig. 5.3), as well as other areas of cerebral white matter). First described in alcoholics,¹⁴ producing insidious flaccid quadriplegia, mental status changes, and cranial nerve abnormalities with a pseudobulbar palsy appearance. In one review,¹⁵ no patient developed CPM when treated slowly as

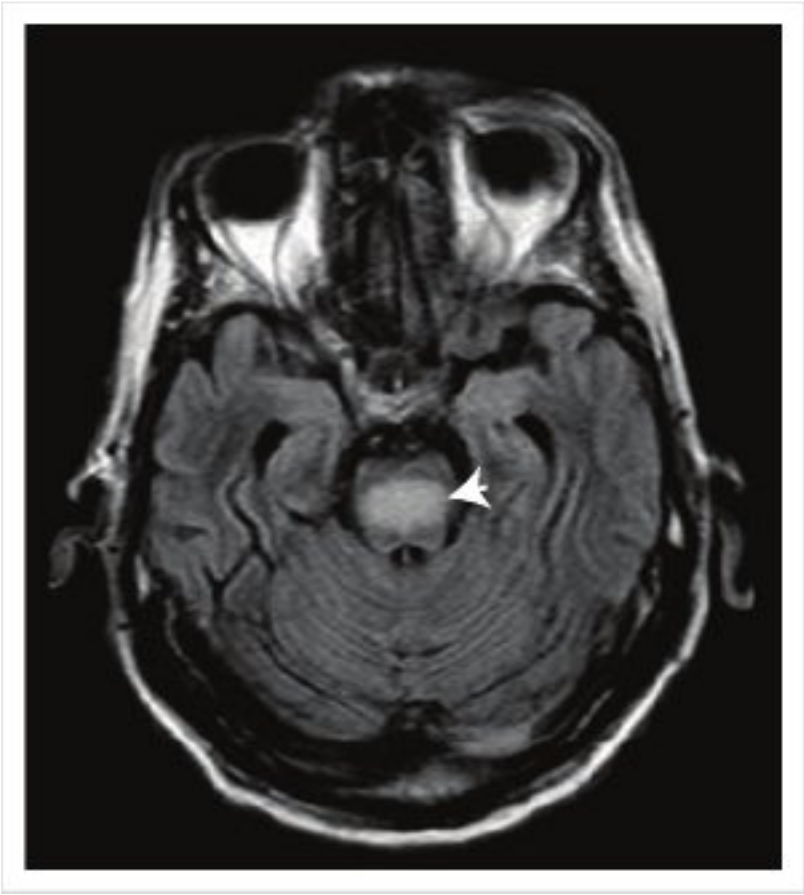


Fig. 5.2 Central pontine myelinolysis (arrowhead). Axial FLAIR MRI

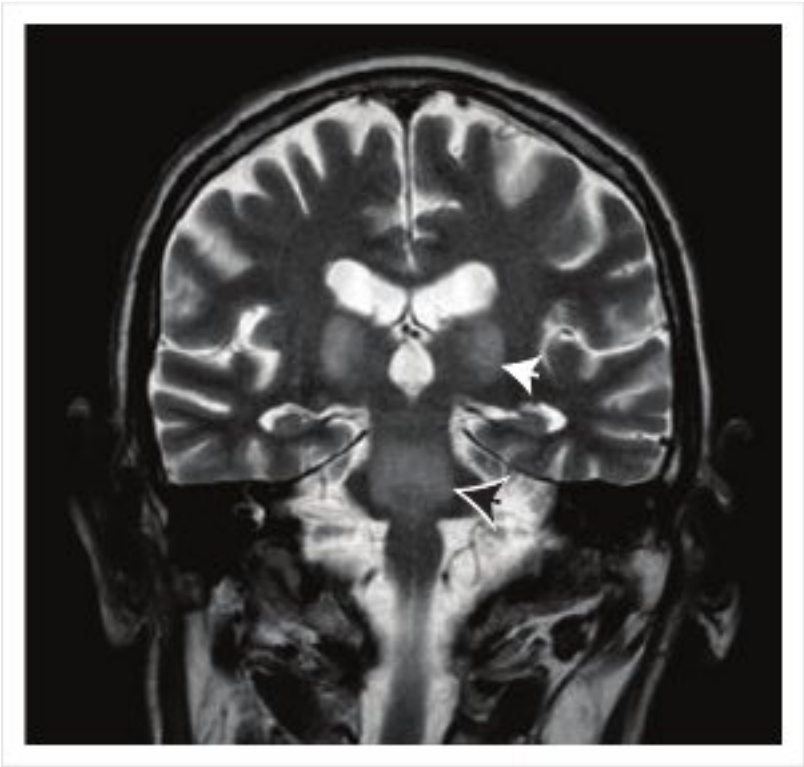


Fig. 5.3 Osmotic demyelination of pons (black arrowhead) & thalamus (white arrowhead). Coronal T2WI MRI

outlined below. And yet, the rate of correction correlates poorly with CPM; it may be that the magnitude is another critical variable.¹⁶ Features common to patients who develop CPM are¹⁵:

- delay in the diagnosis of hyponatremia with resultant respiratory arrest or seizure with probable hypoxemic event
- rapid correction to normo- or hyper-natremia (>135 mEq/L) within 48 hours of initiating therapy
- increase of serum sodium by >25 mEq/L within 48 hours of initiation of therapy
- over-correcting serum sodium in patients with hepatic encephalopathy
- NB: many patients developing CPM were victims of chronic debilitating disease, malnourishment, or alcoholism and never had hyponatremia. Many had an episode of hypoxia/anoxia¹⁷
- presence of hyponatremia >24 hrs prior to treatment¹⁶

The only definitive treatment is treatment of the underlying cause

- if caused by anemia: usually responds to transfusion
- if caused by malignancy, may respond to antineoplastic therapy
- most drug related cases respond rapidly to discontinuation of the offending drug

Treatment algorithms

□ Fig. 5.4 depicts an algorithm for selecting the correct SIADH treatment protocol.

□ Aggressive treatment protocol. Indications (also refer to □ Fig. 5.4):

1. severe hyponatremia (serum [Na⁺] <125 mEq/L)

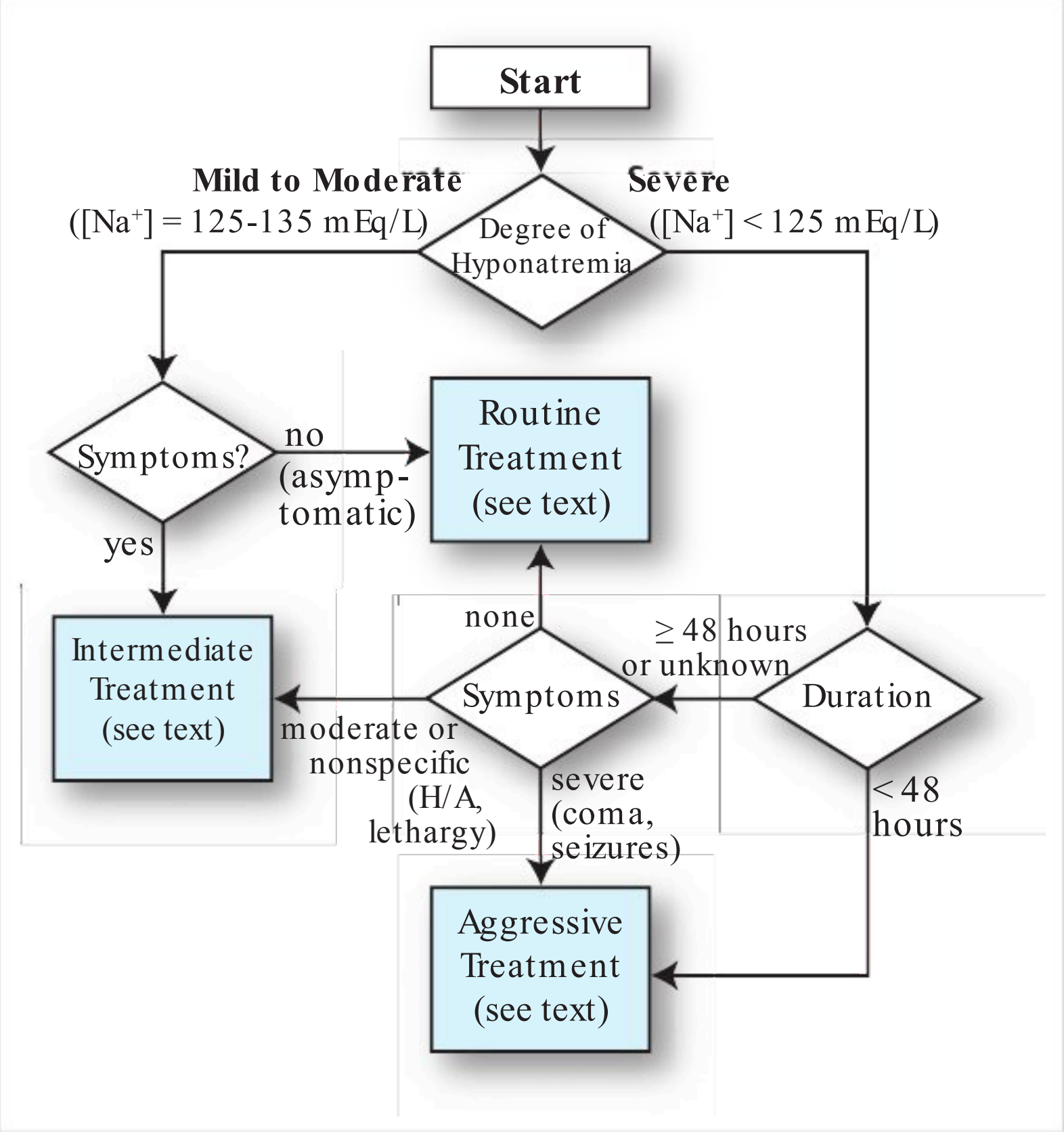


Fig. 5.4 Treatment protocol selection for hyponatremia in SIADH

2. AND either
- a) duration known to be <48 hours
 - b) or severe symptoms (coma, seizures)

Treatment

- transfer patient to ICU
- interventions
 - 3%saline: start infusion 1–2 ml/kg body weight per hour (infusion rate may be doubled to 2–4 ml/kg/hr for limited periods in patients with coma or seizures¹)
 - and furosemide (Lasix®) 20 mg IV q d (furosemide accelerates the increase in [Na⁺] and prevents volume overload with subsequent increase in atrial natriuretic factor and resultant urinary dumping of the extra Na⁺ being administered)
- monitoring and adjustments
 - check serum [Na⁺] every 2–3 hours and adjust infusion rate of 3%saline
 - goal: raise serum sodium by 1–2 mEq/L/hr¹⁸ (use lower end of range for hyponatremia >48 hours duration or unknown duration)
 - limits: do not exceed 8–10 mEq/L in 24 hrs and 18–25 mEq/L in 48 hrs¹ (use lower end of these ranges for hyponatremia >48 hours duration or unknown duration)
 - measure K⁺ lost in urine and replace accordingly

- if symptoms of osmotic demyelination occur (early symptoms are lethargy and affective changes, usually after initial improvement): deficits may improve by stopping treatment and modestly relowering the serum sodium e.g. with DDAVP^{19,20}

□ Intermediate treatment protocol. Indications (also refer to □ Fig. 5.4):

1. symptomatic nonsevere hyponatremia (serum $[Na^+] = 125-135$ mEq/L), or
2. severe hyponatremia (serum $[Na^+] < 125$ mEq/L), AND
 - a) duration > 48 hours or unknown AND
 - b) only moderate symptoms or nonspecific symptoms (e.g. H/A, or lethargy)

Treatment

1. interventions
 - a) 0.9% saline (normal saline) infusion
 - b) and furosemide (Lasix®) 20 mg IV q d
 - c) consider conivaptan for refractory cases
2. monitoring: check serum $[Na^+]$ every 4 hours and adjust infusion rate of normal saline
3. goals: $[Na^+]$ increase of 0.5–2 mEq/L/hr
4. limits: do not exceed 8–10 mEq/L in 24 hrs and 18–25 mEq/L in 48 hrs¹

□ Routine treatment protocol and maintenance therapy. Indications (also refer to □ Fig. 5.4):

1. asymptomatic nonsevere hyponatremia (serum $[Na^+] = 125-135$ mEq/L), or
2. severe hyponatremia (serum $[Na^+] < 125$ mEq/L) AND
 - a) duration > 48 hours or unknown AND
 - b) asymptomatic

Treatment

1. interventions
 - a) fluid restriction □ Table 5.4 for adults, for peds: 1 L/m²/day) while encouraging use of dietary salt and protein. Caution restricting fluids in hyponatremia following SAH (p. 1166).
 - b) for refractory cases, consider
 - demeclocycline: a tetracycline antibiotic that partially antagonizes the effects of ADH on the renal tubules.^{21,22,23} Effects are variable, and nephrotoxicity may occur. □ 300–600 mg PO BID
 - conivaptan (Vaprisol®): a nonpeptide antagonist of V1A & V2 vasopressin receptors. FDA approved for euvolemic and hypervolemic moderate-to-severe hyponatremia in hospitalized patients (NB: severe symptoms of seizures, coma, delirium... warrants aggressive treatment with hypertonic saline¹). Use in the neuro-ICU has been described for treating elevated ICP when serum $[Na]$ is not responding to traditional methods²⁴ (off-label use – use with caution). □ loading dose 20 mg IV over 30 minutes, followed by infusions of 20 mg over 24 hours $\times 3$ days. If serum $[Na^+]$ are not rising as desired, the infusion may be increased to the maximal dose of 40 mg over 24 hours. Use is approved for up to 4 days total. Caution re drug interactions
 - lithium: not very effective and many side effects. Not recommended

5.2.6 Cerebral salt wasting

Cerebral salt wasting (CSW): renal loss of sodium as a result of intracranial disease, producing hyponatremia and a decrease in extracellular fluid volume.¹¹ CAUTION: patients with aneurysmal SAH may have CSW with hyponatremia which mimics SIADH, however there is usually also hypovolemia in CSW. In this setting, fluid restriction may exacerbate vasospasm induced ischemia.^{11,25,26,27}

The mechanism whereby the kidneys fail to conserve sodium in CSW is not known, and may be either a result of an as yet unidentified natriuretic factor or direct neural control mechanisms (see Hyponatremia following SAH (p. 1166)).

Laboratory tests (serum and urinary electrolytes and osmolalities) may be identical with SIADH and CSW.²⁸ Furthermore, hypovolemia in CSW may stimulate ADH release. To differentiate: CVP, PCWP, and plasma volume (a nuclear medicine study) are low in hypovolemia (i.e. CSW). □ Table 5.5 compares some features of CSW and SIADH, the two most important differences being extracellular volume and salt balance. An elevated serum $[K^+]$ with hyponatremia is incompatible with the diagnosis of SIADH.

Treatment of CSW

- Goals:
 - volume replacement
 - positive salt balance

Table 5.5 Comparison of CSW and SIADH ¹¹		
Parameter	CSW	SIADH
□ Plasma volume	↓ (<35 ml/kg)	↑ or WNL
□ Salt balance	negative	variable
Signs & symptoms of dehydration	present	absent
Weight	↓	↑ or no Δ
PCWP	↓ (<8 mm Hg)	↑ or WNL
CVP	↓ (<6 mm Hg)	↑ or WNL
Orthostatic hypotension	+	±
Hematocrit	↑	↓ or no Δ
Serum osmolality	↑ or WNL ^a	↓
Ratio of serum [BUN]:[creatinine]	↑	WNL
Serum [protein]	↑	WNL
Urinary [Na ⁺]	↑ ↑	↑
Serum [K ⁺]	↑ or no Δ	↓ or no Δ
Serum [uric acid]	WNL	↓
Abbreviations: ↓ =decreased, ↑ =increased, ↑ ↑ = significantly increased, WNL=within normal limits, no Δ=no change, [] =concentration, += present, ±=may or may not be present		
^a In reality, serum osmolality is usually ↓ in CSW		

- avoid excessively rapid correction of hyponatremia or overcorrection which may be associated with osmotic demyelination (p.115) as with SIADH (p.115).
- Interventions
 - Hydrate patient with 0.9%NS at 100–125 ml/hr. For severe cases, 3%saline at 25–50 cc/hr is occasionally required.
 - Do not give furosemide.
 - Salt may also be simultaneously replaced orally.
 - Blood products may be needed if anemia is present.
 - Medications
 - a) Fludrocortisone acetate acts directly on the renal tubule to increase sodium absorption. Benefits of giving 0.2 mg IV or PO q d in CSW have been reported,²⁹ but significant complications of pulmonary edema, hypokalemia and HTN may occur.
 - b) Urea: an alternative treatment using urea may be applicable to the hyponatremia of either SIADH or CSW, and therefore may be used before the cause has been ascertained: urea (Ureaphil®) 0.5 grams/kg (dissolve 40 gm in 100–150 ml NS) IV over 30–60 mins q 8 hrs.³⁰ Use NS+20 mEq KCl/L at 2 ml/kg/hr as the main IV until the hyponatremia is corrected (unlike mannitol, urea does not increase ADH secretion). They supplemented with colloids (viz. 250 ml of 5%albumin IV q 8–12 hrs x 72 hrs).

5.3 Hyponatremia

5.3.1 General information

Definition: serum sodium >150 mEq/L. In neurosurgical patients, this is most often seen in the setting of diabetes insipidus (DI).

Since normal total body water (TBW) is ≈ 60%of the patient’s normal body weight, the patient’s current TBW may be estimated by Eq (5.3).

$$TBW_{current} = \frac{1}{4} \left(\frac{Na^b_{normal}}{Na^b_{current}} \right) TBW_{normal} = \frac{1}{4} \left(\frac{140mEq/L}{Na^b_{current}} \right) \times 0.6 \times \text{usual body wt} \times kg$$

05:30

The free water deficit to be replaced is given by Eq (5.4). Correction must be made slowly to avoid exacerbating cerebral edema. One half the water deficit is replaced over 24 hours, and the remainder is given over 1–2 additional days. Judicious replacement of deficient ADH in cases of true DI must also be made.

$$\text{free water deficit} = \frac{1}{4} \left(\frac{0.6 \times \text{usual body wt} \times kg}{Na^b_{current}} \right) - TBW_{current}$$

05:40

5

5.3.2 Diabetes insipidus

General information

Key concepts

- due to low levels of ADH (or, rarely, renal insensitivity to ADH)
- high output of dilute urine (<200 mOsmol/L or SG < 1.003) with normal or high serum osmolality and high serum sodium
- often accompanied by craving for water, especially ice-water
- danger of severe dehydration if not managed carefully

Diabetes insipidus (DI) is due to insufficient ADH activity at the kidneys, and results in the excessive renal loss of water and electrolytes. DI may be produced by two different etiologies:

- central or neurogenic DI: subnormal levels of ADH caused by hypothalamic-pituitary axis dysfunction. This is the type most often seen by neurosurgeons
- “nephrogenic DI”: due to relative resistance of the kidney to normal or supra-normal levels of ADH. Seen with some drugs (drugs: 1)

Etiologies of DI³¹:

1. (neurogenic AKA central) diabetes insipidus
 - a) familial (autosomal dominant)
 - b) idiopathic
 - c) posttraumatic (brain injury, including surgery)
 - d) tumor: craniopharyngioma, metastasis, lymphoma...
 - e) granuloma: neurosarcoidosis, histiocytosis
 - f) infectious: meningitis, encephalitis
 - g) autoimmune
 - h) vascular: aneurysm, Sheehan’s syndrome (rarely causes DI)
2. nephrogenic diabetes insipidus
 - a) familial (X-linked recessive)
 - b) hypokalemia
 - c) hypercalcemia
 - d) Sjögren’s syndrome
 - e) drugs: lithium, demeclocycline, colchicine...
 - f) chronic renal disease: pyelonephritis, amyloidosis, sickle cell disease, polycystic kidney disease, sarcoidosis

Central DI

85% of ADH secretory capacity must be lost before clinical DI ensues. Characteristic features: high urine output (polyuria) with low urine osmolality, and (in the conscious patient) craving for water (polydipsia), especially ice-water.

Differential diagnosis of DI:

1. (neurogenic) diabetes insipidus (true DI)
2. nephrogenic diabetes insipidus
3. psychogenic
 - a) idiopathic: from resetting of the osmostat
 - b) psychogenic polydipsia (excess free water intake)
4. osmotic diuresis: e.g. following mannitol, or with renal glucose spilling
5. diuretic use: furosemide, hydrochlorothiazide...

Central DI may be seen in the following situations:

1. following transsphenoidal surgery or removal of craniopharyngioma: (usually transient, therefore avoid long-acting agents until it can be determined if long-term replacement is required). Injury to the posterior pituitary or stalk usually causes one of three patterns of DI³²:
 - a) transient DI: supra-normal urine output (UO) and polydipsia which typically normalizes $\approx$ 12–36 hrs post-op
 - b) “prolonged” DI: UO stays supra-normal for prolonged period (may be months) or even permanently: only about one-third of these patients will not return to near-normal at one year post-op
 - c) “triphasic response”: least common
 - phase 1: injury to pituitary reduces ADH levels for 4–5 days $\rightarrow$ DI (polyuria/polydipsia)
 - phase 2: cell death liberates ADH for the next 4–5 days $\rightarrow$ transient normalization or even SIADH-like water retention ($\square$ NB: there is a danger of inadvertently continuing vasopressin therapy beyond the initial DI phase into this phase causing significant hemodilution)
 - phase 3: reduced or absent ADH secretion $\rightarrow$ either transient DI (as in “A” above) or a “prolonged” DI (as in “B” above)
2. central herniation (p.303): shearing of pituitary stalk may occur
3. brain death: hypothalamic production of ADH ceases
4. with certain tumors:
 - a) pituitary adenomas: DI is rare even with very large macroadenomas. DI may occur with pituitary apoplexy (p.720)
 - b) craniopharyngioma: DI usually only occurs postoperatively since damage to pituitary or lower stalk does not prevent production and release of ADH by hypothalamic nuclei
 - c) suprasellar germ cell tumors
 - d) rarely with a colloid cyst
 - e) hypothalamic tumors: Langerhans cell histiocytosis
5. mass lesions pressing on hypothalamus: e.g. a-comm aneurysm
6. following head injury: primarily with basal (clival) skull fractures (p.884)
7. with encephalitis or meningitis
8. drug induced:
 - a) ethanol and phenytoin can inhibit ADH release
 - b) exogenous steroids may seem to “bring out” DI because they may correct adrenal insufficiency (below) and they inhibit ADH release
9. granulomatous diseases
 - a) Wegener’s granulomatosis (p.199): a vasculitis
 - b) neurosarcoidosis involving the hypothalamus (p.189)
10. inflammatory: autoimmune hypophysitis (p.1373)³³ or lymphocytic infundibuloneurohypophysitis³⁴ (distinct conditions)

Diagnosis

The following are usually adequate to make the diagnosis of DI, especially in the appropriate clinical setting:

1. dilute urine:
 - a) urine osmolality < 200 mOsm/L (usually 50–150) or specific gravity (SG) < 1.003 (may be 1.001 to 1.005). (Note that normally, urine osmolality averages between 500–800 mOsm/L; extreme range: 50–1400.)
 - b) or the inability to concentrate urine to > 300 mOsm/L in the presence of clinical dehydration
 - c) NB: large doses of mannitol as may be used in head trauma can mask this by producing a more concentrated urine
2. urine output (UO) > 250 cc/hr (peds: > 3 cc/kg/hr)
3. normal or above-normal serum sodium

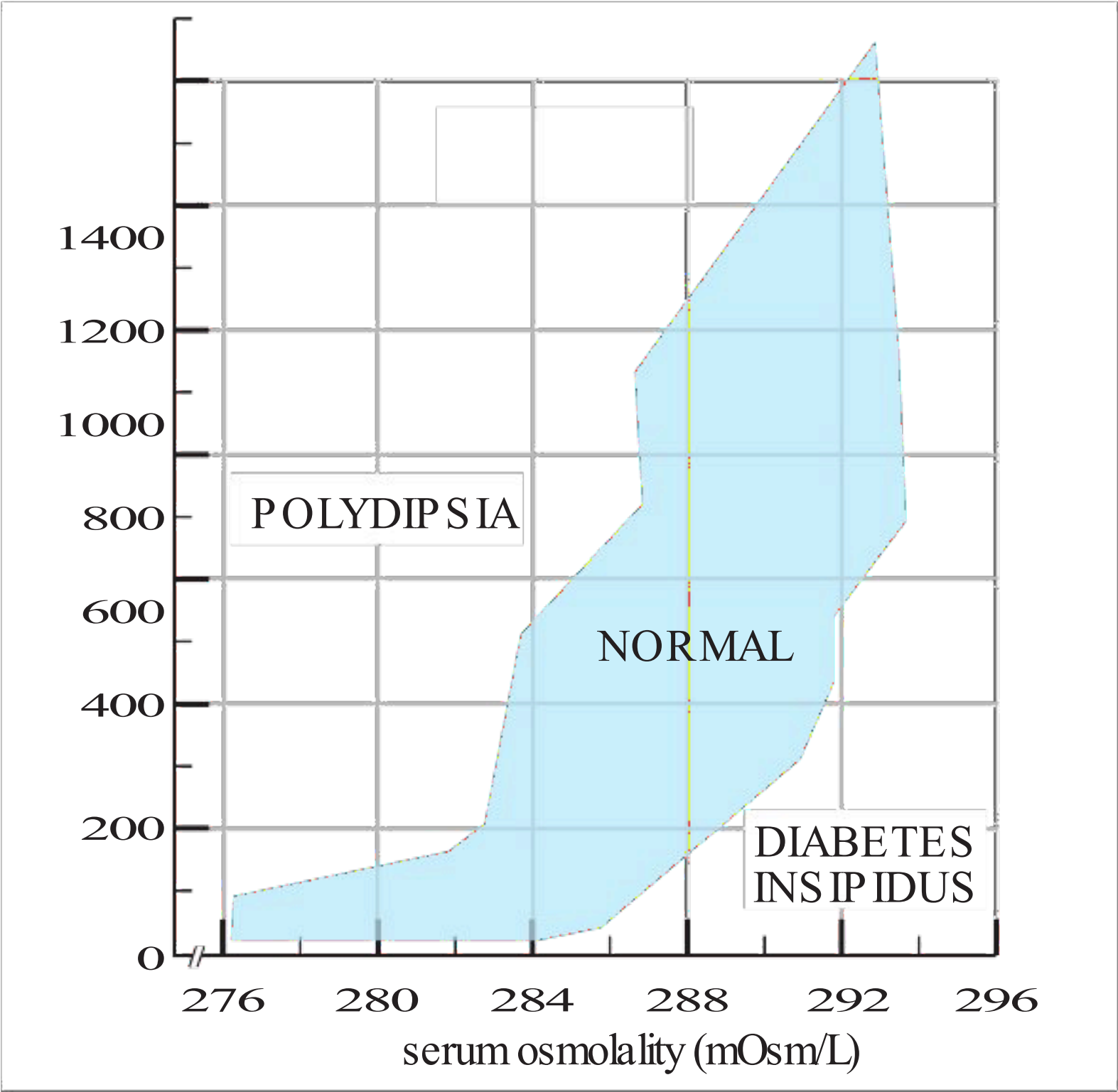


Fig. 5.5 Interpretation of simultaneous serum vs. urine osmolality (Provided by Arnold M. Moses, M.D., used with permission)

4. normal adrenal function: DI cannot occur in primary adrenal insufficiency because a minimum of mineralocorticoid activity is needed for the kidney to make free water, thus steroids may “bring out” underlying DI by correcting adrenal insufficiency

In uncertain cases, plot urine and simultaneous serum osmolality on the graph in ☐ Fig. 5.5.

1. low serum osmolality: usually indicates psychogenic polydipsia (pathological drinking of water)
2. if the point falls in the “normal” range, a supervised water deprivation test is needed to determine if the patient can concentrate their urine with dehydration (caution: see below)
3. high serum osmolality:
 - diagnosis of DI is established and no further testing to establish the diagnosis is required
 - further testing is only needed to differentiate central from nephrogenic DI, if desired to differentiate central from nephrogenic DI, give aqueous Pitressin® 5 U SQ
 - in central DI the urine osmolality should double within 1–2 hours
4. plotting more than one data point may help as some patients tend to “vacillate” around the border zones

Water deprivation test

If still unclear, the diagnosis of DI is confirmed by a water deprivation test (☐ CAUTION: perform only under close supervision as rapid and potentially fatal dehydration may ensue in DI). This test is rarely necessary if serum osmolality >298 mOsm/L. (Note that in compensated DI, serum osmolality is more likely to be lower and to overlap with normal.³⁵)

Table 5.6 Highest urinary osmolality after Pitressin in water deprivation test	
Δ in urinary Osm	Interpretation
<5% increase	normal
6–67% increase	partial ADH deficiency
>67% increase	severe ADH deficiency

- Stop IVs and make the patient NPO
- Monitoring:
 - check urine osmolality q hr
 - check patient weight q 1 hr
- continue the test until one of the following occurs:
 - normal response occurs: urine output decreases, and urine osmolality rises to 600–850 mOsm/L
 - 6–8 hours lapse
 - urine osmolality plateaus (i.e. changes <30 mOsm in 3 consecutive hours)
 - patient loses 3% of body weight
- if the patient fails to demonstrate the normal response, then:
 - give exogenous ADH (5 U aqueous Pitressin® SQ), which normally increases urine osmolality to >300 mOsm/L
 - check urine osmolality 30 and 60 minutes later
 - compare highest urine osmolality after Pitressin® to the osmolality just before Pitressin® according to □ Table 5.6

Treatment of DI

In conscious ambulatory patient

If DI is mild, and the patient’s natural thirst mechanism is intact, instruct patient to drink only when thirsty and they usually “keep up” with losses and will not become overhydrated.

If severe, the patient may not be able to maintain adequate intake of fluid or tolerate the frequent trips to bathroom. In these cases, treatment typically involves a vasopressin analogue. See below for a synopsis of vasopressin analogues. Typically start with either:

1. desmopressin (DDAVP®)
 - a) PO: 0.1 mg PO BID, adjust up or down PRN urine output (typical dosage range: 0.1–0.8 mg/d in divided doses)
 - b) nasal spray: 2.5 mcg (0.025 ml) by nasal insufflation BID, titrate up to 20 mcg BID as needed (the nasal spray may be used for doses that are multiples of 10 mcg)
- OR
2. ADH enhancing medications (works primarily in chronic partial ADH deficiency. Will not work in total absence of ADH)
 - clofibrate (Atromid S®) 500 mg PO QID
 - chlorpropamide: increases renal sensitivity to ADH
 - hydrochlorothiazide: thiazide diuretics may act by depleting Na⁺ which increases reabsorption in proximal tubules and shifting fluid away from distal tubules which is where ADH works. □: e.g. Dyazide® 1 PO q d (may increase up to 2 per day PRN)

In conscious ambulatory patient with impaired thirst mechanisms

If thirst mechanisms are not intact in conscious ambulatory patient, they run the risk of dehydration or fluid overload. For these patients:

1. have patient follow UO and daily weights, balance fluid intake and output using antidiuretic medication as needed to keep UO reasonable
2. check serial labs (approximately q weekly) including serum sodium, BUN

In non-ambulatory, comatose/stuporous, or brain-dead patient; see also Medical Management of the Potential Organ Donor (p.313)

1. follow I’s & O’s q 1 hr, with urine specific gravity (SG) q 4 hrs and whenever urine output (UO) > 250 ml/hr
2. labs: serum electrolytes with osmolality q 6 hrs

Table 5.7 Available preparations of vasopressin analogues					
Generic name	Trade name	Route	Concentration	Availability	Manufacturer
desmopressin	DDAVP®	SQ, IM, IV	4 mcg/ml	1 & 10 ml	Aventis
desmopressin	DDAVP® Nasal Spray	nasal spray	100 mcg/ml, each spray delivers 10 mcg	50 doses per bottle	Aventis
desmopressin	DDAVP® Tablets	PO		0.1 & 0.2 mg	Aventis
arginine vasopressin	aqueous Pitressin®	SQ, IM	20 U/ml (50 mcg/ml)	0.5 and 1 ml	Parke-Davis

Table 5.8 Mean time of hypertonic urine ^a (relative to plasma) ^b			
Generic name	Route	Dose	Mean duration of action ^c
desmopressin	SQ, IM, IV	0.5 mcg	8 hrs
desmopressin ^d	SQ, IM, IV	1.0 mcg	12 hrs
desmopressin	SQ, IM, IV	2.0 mcg	16 hrs
desmopressin	SQ, IM, IV	4.0 mcg	20 hrs
desmopressin	intranasal	10 mcg (0.1 ml)	12 hrs
desmopressin	intranasal	15 mcg (0.15 ml)	16 hrs
desmopressin	intranasal	20 mcg (0.2 ml)	20 hrs
arginine vasopressin	SQ, IM	5 U (12.5 mcg)	4 hrs (range: 4–8)

^aprovided by Arnold M. Moses, M.D., used with permission
^bonset of antidiuretic action of these preparations is 30–45 minutes following administration (except pituitary powder in oil which takes 2–4 hrs to start working)
^ctimes may vary from patient to patient, but are usually consistent in any individual
^dNote: 1 mcg BID of desmopressin is as effective as 4 mcg q d, but would obviously be less expensive

3. IV fluid management:
BASE IV: D5 1/2 NS+20 mEq KCl/L at appropriate rate (75–100 ml/hr)
PLUS: replace UO above base IV rate ml for ml with 1/2 NS
NB: for post-op patients, if the patient received significant intraoperative fluids, then they may have an appropriate post-op diuresis, in this case use 1/2 NS to replace only ≈ 2/3 of UO that exceeds the basal IV rate
4. if unable to keep up with fluid loss with IV (or NG) replacement (usually with UO>300 ml/hr), then EITHER
 - 5 U arginine vasopressin (aqueous Pitressin®) IVP/IM/SQ q 4–6 hrs (avoid tannate oil suspension due to erratic absorption and variable duration)
 - OR
 - vasopressin IV drip: start at 0.2 U/min & titrate (max: 0.9 U/min)
 - OR
 - desmopressin injection SQ/IV titrated to UO, usual adult dose: 0.5–1 ml (2–4 mcg) daily in 2 divided doses

Vasopressin analogues

□ Table 5.6, □ Table 5.7 and □ Table 5.8 shows dosing forms and duration of action of vasopressin analogues.

Pitressin® is aqueous solution of 8-arginine vasopressin and should be used with caution in patients with vascular disease (especially coronary arteries). □ Caution – soundalikes: sometimes pitocin is confused with pitressin because of similarities of the name.

DDAVP (1-deamino-8-D-arginine vasopressin) AKA desmopressin. More potent and longer acting than vasopressin.

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6 General Neurocritical Care

6.1 Parenteral agents for hypertension

Drug info: ☐ Nicardipine (Cardene®)

Calcium channel blocker (CCB) that may be given IV. Does not require arterial line, does not raise ICP. Does not reduce heart rate, but may be used in conjunction with e.g. labetalol or esmolol if that is desired. Side effects: H/A 15% nausea 5% hypotension 5% reflex tachycardia 3.5%
☐ start at 5 mg/hr IV (off label: 10 mg/hr may be used in situations where urgent reduction is needed). Increase by 2.5 mg/hr every 5–15 minutes up to a maximum of 15 mg/hr. Decrease to 3 mg/hr once control is achieved. ☐ Ampules contain 25 mg and must be diluted before administration.

Drug info: Nitroglycerin (NTG)

Raises ICP (less than with nitroprusside due to preferential venous action¹). Vasodilator, venous > arterial (large coronaries > small). Result: decreases LV filling pressure (pre-load). Does not cause “coronary steal” (cf nitroprusside).
☐ 10–20 mcg/min IV drip (increase by 5–10 mcg/min q 5–10 min). For angina pectoris: 0.4 mg SL q 5 min × 3 doses, check BP before each dose.

Drug info: Labetalol (Normodyne®, Trandate®)

Blocks α_1 selective, β non-selective (potency < propranolol). ICP reduces or no change.² Pulse rate: decreases or no change. Cardiac output does not change. Does not exacerbate coronary ischemia. May be used in controlled CHF, but not in overt CHF. Contraindicated in asthma. Renal failure: same dose. Side effects: fatigue, dizziness, orthostatic hypotension.

Intravenous (IV)

Onset 5 mins, peak 10 mins, duration 3–6 hrs.

☐ IV: patient supine; check BP q 5 min; give each dose slow IVP (over 2 min) q 10 minutes until desired BP achieved; dose sequence: 20, 40, 80, 80, then 80 mg (300 mg total). Once controlled, use $\approx$ same total dose IVP q 8 hrs.

☐ IV drip (alternative): add 40 ml (200 mg) to 160 ml of IVF (result: 1 mg/ml); run at 2 ml/min (2 mg/min) until desired BP (usual effective dose = 50–200 mg) or until 300 mg given; then titrate rate (bradycardia limits dose, increase slowly since effect takes 10–20 minutes).

Oral (PO)

Undergoes first pass liver degradation, therefore requires higher doses PO. PO onset: 2 hrs, peak: 4 hrs.

☐ PO: to convert IV $\rightarrow$ PO, start with 200 mg PO BID. To start with PO, give 100 mg BID, and increase 100 mg/dose q 2 day; max. = 2400 mg/day.

Drug info: Enalaprilat (Vasotec®)

An angiotensin-converting enzyme (ACE) inhibitor. The active metabolite of the orally administered drug enalapril (see below). Acts within $\approx$ 15 mins of administration.

Side effects: hyperkalemia occurs in $\approx$ 1% Do not use during pregnancy.

☐ IV: start with 1.25 mg slow IV over 5 mins, may increase up to 5 mg q 6 hrs PRN.

Drug info: Esmolol (Brevibloc®)

Cardioselective short-acting beta blocker.³ Being investigated for hypertensive emergencies. Metabolized by RBC esterase. Elimination half-life: 9 mins. Therapeutic response (>20% decrease in heart rate, HR<100, or conversion to sinus rhythm) in 72%. Side effects: dose related hypotension (in 20–50%), generally resolves within 30 mins of D/C. Bronchospasm less likely than other beta blockers. Avoid in CHF.

□ 500 mcg/kg loading dose over 1 min, follow with 4 min infusion starting with 50 mcg/kg/min. Repeat loading dose and increment infusion rate by 50 mcg/kg/min q 5 mins. Rarely >100 mcg/kg/min required. Doses >200 mcg/kg/min add little.

Drug info: Fenoldopam (Corlopam®)

Vasodilator. Onset of action <5 minutes, duration 30 mins.

□ IV infusion (no bolus doses): start with 0.1–0.3 mcg/kg/min, titrate by 0.1 mcg/kg/min q 15 min up to a maximum of 1.6 mcg/kg/min.

Drug info: Propranolol (Inderal®)

Main use IV is to counteract tachycardia with vasodilators (usually doesn't lower BP acutely when used alone), but esmolol and labetalol are more commonly used for this.

□ IV: load with 1–10 mg slow IVP, follow with 3 mg/hr. PO: 80–640 mg q d in divided doses.

6

6.2 Hypotension (shock)

6.2.1 Classification

- 1. hypovolemic: first sign usually tachycardia. >20–40% of blood volume loss must occur before perfusion of vital organs is impaired. Includes:
 - a) hemorrhage (external or internal)
 - b) bowel obstruction (with third spacing)
- 2. septic: most often due to gram negative sepsis
- 3. cardiogenic: includes MI, cardiomyopathy, dysrhythmias (including A-fib)
- 4. neurogenic: e.g. paralysis due to spinal cord injury. Blood pools in venous capacitance vessels
- 5. miscellaneous
 - a) anaphylaxis
 - b) insulin reaction

6.2.2 Cardiovascular agents for shock

Plasma expanders. Includes:

- 1. crystalloids: normal saline has less tendency to promote cerebral edema than others; see IV fluids (p.870), under control of elevated ICP
- 2. colloids: e.g. hetastarch (Hespan®). □ CAUTION: repeated administration over a period of days may prolong PT/PTT and clotting times and may increase the risk of rebleeding in aneurysmal SAH (p.1167).⁴
- 3. blood products: expensive. Risk of transmissible diseases or transfusion reaction

Drug info: Dopamine

See □ Table 6.1 for a summary of the effects of dopamine (DA) at various dosages. DA is primarily a vasoconstrictor (β_1 effects usually overridden by α -activity). 25% of dopamine given is rapidly converted to norepinephrine (NE). At doses > 10 mcg/kg/min one is essentially giving NE. May cause significant hyperglycemia at high doses.

□ Start with 2–5 mcg/kg/min and titrate.

Table 6.1 Dopamine dosage		
Dose (mcg/kg/min)	Effect	Result
0.5–2.0 (sometimes up to 5)	dopaminergic	renal, mesenteric, coronary, & cerebral vasodilatation, (+) inotrope
2–10	β_1	positive inotrope
> 10	α , β & dopaminergic	releases nor-epi (vasoconstrictor)

Drug info: Dobutamine (Dobutrex®)

Vasodilates by β_1 (primary) and by increased CO from (+) inotropy (β_2); result: little or no fall in BP, less tachycardia than DA. No alpha release nor vasoconstriction. May be used synergistically with nitroprusside. Tachyphylaxis after ≈ 72 hrs. Pulse increases $> 10\%$ may exacerbate myocardial ischemia, more common at doses > 20 mcg/kg/min. Optimal use requires hemodynamic monitoring. Possible platelet function inhibition.

□ usual range 2.5–10 mcg/kg/min; rarely doses up to 40 used (to prepare: put 50 mg in 250 ml D5 W to yield 200 mcg/ml).

Drug info: Amrinone (Inocor®)

Nonadrenergic cardiotonic. Phosphodiesterase inhibitor, effects similar to dobutamine (including exacerbation of myocardial ischemia). 2% incidence of thrombocytopenia.

□ 0.75 mg/kg initially over 2–3 min, then drip 5–10 mcg/kg/min.

Drug info: Phenylephrine (Neo-Synephrine®)

Pure alpha sympathomimetic. Useful in hypotension associated with tachycardia (atrial tachyarrhythmias). Elevates BP by increasing SVR via vasoconstriction, causes reflex increase in parasympathetic tone (with resultant slowing of pulse). Lack of β action means non-inotropic, no cardiac acceleration, and no relaxation of bronchial smooth muscle. Cardiac output and renal blood flow may decrease. Avoid in spinal cord injuries (p. 950).

□ pressor range: 100–180 mcg/min; maintenance: 40–60 mcg/min. To prepare: put 40 mg (4 amps) in 500 ml D5 W to yield 80 mcg/ml; a rate of 8 ml/hr = 10 mcg/min.

Drug info: Norepinephrine

Primarily vasoconstrictor (? counterproductive in cerebral vasospasm, ? decreases CBF). β -agonist at low doses. Increases pulmonary vascular resistance.

Drug info: Epinephrine (adrenalin globally)

□ 0.5–1.0 mg of 1:10,000 solution IVP; may repeat q 5 minutes (may bolus per ET tube). Drip: start at 1.0 mcg/min, titrate up to 8 mcg/min (to prepare: put 1 mg in 100 ml NS or D5W).

Drug info: Isoproterenol (Isuprel®)

Positive chronotropic and inotropic, → increased cardiac O₂ consumption, arrhythmias, vasodilatation (by β₁ action) skeletal muscle > cerebral vessels.

Drug info: Levophed

Direct β stimulation (positive inotropic and chronotropic).
□ start drip at 8–12 mcg/min; maintenance 2–4 mcg/min (0.5–1.0 ml/min) (to prepare: put 2 mg in 500 ml NS or D5 W to yield 4 mcg/cc).

6

6.3 Acid inhibitors

6.3.1 Stress ulcers in neurosurgery

See reference.⁵
The risk of developing stress ulcers (SU) AKA Cushing’s ulcers is high in critically ill patients with CNS pathology. These lesions are AKA Cushing’s ulcers due to Cushing’s classic treatise.⁶ 17% of SUs produce clinically significant hemorrhage. CNS risk factors include intracranial pathology: brain injury (especially Glasgow Coma scale score <9), brain tumors, intracerebral hemorrhage, SIADH, CNS infection, ischemic stroke, as well as spinal cord injury. The odds are increased with the coexistence of extra-CNS risk factors including: long-term use of steroids (usually >3 weeks), burns >25% of body surface area, hypotension, respiratory failure, coagulopathies, renal or hepatic failure and sepsis.
The pathogenesis of SUs is incompletely understood, but probably results from an imbalance of destructive factors (acid, pepsin & bile) relative to protective factors (mucosal blood flow, mucus-bicarbonate layer, endothelial cell replenishment & prostaglandins).⁵ CNS pathology, especially that involving the diencephalon or brain stem, can lead to reduction of vagal output which leads to hypersecretion of gastric acid and pepsin. There is a peak in acid and pepsin production 3–5 days after CNS injury.

6.3.2 Prophylaxis for stress ulcers

There is strong evidence that reduction of gastric acid (whether by antacids or agents that inhibit acid secretion) reduces the incidence of GI bleeding from stress ulcers in critically ill patients. Elevating gastric pH >4.5 also inactivates pepsin.
Other therapies that don’t involve alterations of pH that may be effective include sucralfate (see below) and enteral nutrition (controversial).⁵ Titrated antacids or sucralfate appear to be superior to H₂ antagonists in reducing the incidence of SUs.
Routine prophylaxis when steroids are used is not warranted unless one of the following risk factors are present: prior PUD, concurrent use of NSAIDs, hepatic or renal failure, malnourishment, or prolonged steroid therapy >3 weeks.

6.3.3 Possible increased pneumonia and mortality from altering gastric pH

Whereas bringing gastric pH to a more neutral level reduces the risk of SUs, pH >4 permits bacterial colonization of the normally sterile stomach. This may increase the risk of pneumonia from aspiration, and there is a suggestion that mortality may also be increased.⁷ Sucralfate may be as effective

in reducing bleeding, but may be associated with lower rates of pneumonia and mortality. There is insufficient data to determine the net result of sucralfate compared to no treatment.⁷

6.3.4 Histamine2 (H2) antagonists

Drug info: Ranitidine (Zantac®)

☐ Adult age ≤65 yrs: 150 mg PO BID, or 50 mg IVPB q 8 hrs. For age >65 with normal renal function: 50 mg IV q 12 hrs.

IV drip (provides a more consistently higher pH without peaks and troughs; some controversy that this may increase gastric bacterial concentration with increased risk of aspiration pneumonia has not been borne out): 6.25 mg/hr (e.g. inject 150 mg into 42 ml of IVF yielding 3.125 mg/ml, run at 2 ml/hr).

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Drug info: Famotidine (Pepcid®)

☐ Adult: 20 mg PO q hs for maintenance; 40 mg PO q hs for active ulcer therapy; IV: 20 mg q 12 hrs (for hypersecretory conditions, 20 mg IVPB q 6 hrs).⁸ Supplied: 20 & 40 mg tablets, 40 mg/5 ml suspension, and 20 & 40 mg orally disintegrating tablets as Pepcid RPD. Available OTC in 10 mg tablets as Pepcid AC. Available IV.

Drug info: Nizatidine (Axid®)

☐ 300 mg PO q d or 150 mg PO BID. Supplied: 150 & 300 mg pulvules. Available OTC in 75 mg tablets as Axid AR.

6.3.5 Gastric acid secretion inhibitors (proton pump inhibitors)

These agents reduce gastric acid by specific inhibition of the final step in acid secretion by gastric parietal cells (by inhibiting the (H⁺, K⁺)-ATPase enzyme system on the cell surface, the so-called “acid pump”). They block acid secretion regardless of the stimulus (Zollinger-Ellison syndrome, hypergastrinemia...). Full recovery of acid secretion upon discontinuation may not occur for weeks. ☐ Not indicated for long-term treatment as the trophic effects of the resultant elevated levels of gastrin may lead to gastric carcinoid tumors.

Drug info: Omeprazole (Prilosec®)

Inhibition of some hepatic P-450 enzymes results in reduced clearance of warfarin and phenytoin. Decreases the effectiveness of prednisone.

☐ Adult: for peptic ulcers and gastro-esophageal reflux disease (GERD) 20–40 mg PO daily. For Zollinger-Ellison syndrome: 20 mg PO q d to 120 mg PO TID (dose adjusted to keep basal acid output < 60 mEq/hr). Side effects: N/V, H/A, diarrhea, abdominal pain or rash in 1–5% of patients. Supplied: 10, 20 & 40 mg delayed-release capsules. Available OTC in 20.6 mg tablets as Prilosec OTC.

Drug info: □Lansoprazole (Prevacid®)

Found not to have an affect on a number of other drugs metabolized by cytochrome P-450 including: phenytoin, warfarin and prednisone.

□ Adult: 15 mg (for duodenal ulcer, GERD, or maintenance therapy) or 30 mg (for gastric ulcer or erosive esophagitis) PO q d, short-term treatment × 4 wks. Supplied: 15 & 30 mg delayed-release capsules.

Drug info: Pantoprazole (Protonix®)

□ PO: 40 mg PO q d for up to 8 wks. IV: 40 mg IV q d × 7–10 d. Supplied: PO: 40 mg delayed-release capsules.

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6.3.6 Miscellaneous

Drug info: Sucralfate (Carafate®)

Minimally absorbed from GI tract. Acts by coating ulcerated areas of mucosa, does not inhibit acid secretion. This may actually result in a lower incidence of pneumonia and mortality than agents that affect gastric pH (see above).

□ 1 gm PO QID on an empty stomach. Do not give antacids within one half-hour of sucralfate.

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7 Sedatives, Paralytics, Analgesics

7.1 Sedatives and paralytics

7.1.1 Richmond agitation-sedation scale (RASS)

A validated scale^{1,2} that uses positive numbers for agitation and negative numbers for sedation as shown in □ Table 7.1. Useful for quantitating the desired level of sedation when titrating sedatives for agitated patients.

Procedure for performing RASS assessment:

- 1. on observation, patient is alert, restless or agitated: score 0 to +4
- 2. if patient is not alert, state patient’s name and verbally instruct to open eyes and look at speaker: score –1 to –3
- 3. if no response to verbal stimulus, physically stimulate by shaking shoulder and/or sternal rub: score –4 or –5

7.1.2 Conscious sedation

Use of these agents requires ability to provide immediate emergency ventilatory support (including intubation). Agents include:

- 1. midazolam (Versed®) (p.870) with fentanyl
- 2. fentanyl
- 3. pentobarbital (Nembutal®): a barbiturate. □ for 70 kg adult: 100 mg slow IVP

Drug info: Methohexital (Brevital®)

More potent and shorter acting than thiopental (useful e.g. for percutaneous rhizotomy where patient needs to be sedated and awakened repeatedly). Lasts 5–7 min. Similar cautions with the added problem that methohexital may induce seizures. May no longer be available in the U.S.
□ Adult: 1 gm% solution (add 50 ml diluent to 500 mg to yield 10 mg/ml), 2 ml test dose, then 5–12 ml IVP at rate of 1 ml/5 secs, then 2 to 4 ml q 4–7 min PRN.

Table 7.1 Richmond agitation-sedation scale				
	Score	Term	Description	
Agitation	+4	combative	overly combative, violent, immediate danger to staff	
	+3	very agitated	pulls or removes tubes or catheters; aggressive	
	+2	agitated	frequent non-purposeful movements, fights ventilator	
	+1	restless	anxious, but movements not aggressive /vigorous	
	0	alert & calm		
Sedation	−1	drowsy	not fully alert, but has sustained awakening (eye-opening/contact) to voice (≥10 seconds)	verbal stimulation
	−2	light sedation	briefly awakens with eye contact to voice (<10 seconds)	
	−3	moderate sedation	movement or eye opening to voice (no eye contact)	
	−4	deep sedation	no response to voice, but movement or eye-opening to physical stimulation	physical stimulation
	−5	unarousable	no response to voice or physical stimulation	

7.1.3 Sedation

Generally requires intubation and mechanical ventilatory support in the ICU. Doses are generally lower than those used by anesthesiologists for general anesthesia.

Drug info: Thiopental (Pentothal®)

A short acting barbiturate. 1st dose causes unconsciousness in 20–30 secs (circulation time), depth increases up to 40 secs, duration = 5 mins (terminated by redistribution), consciousness returns over 20–30 mins.

Side effects: dose related respiratory depression, irritation if extravasated, intra-arterial injection → necrosis, agitation if injected slowly, an antianalgesic, myocardial depressant, hypotension in hypovolemic patients.

□ Adult: initial concentration should not exceed 2.5%
give 50 mg test dose moderately rapid IVP, then if tolerated give 100–200 mg IVP over 20–30 secs (500 mg may be required in large patient).

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Drug info: □ Remifentanyl (Ultiva®)

Ultrashort acting micro-opioid receptor agonist. Potency similar to fentanyl. Rapidly crosses BBB. Onset: < 1 min. Offset: 3–10 mins. Lowers ICP. Metabolism: non-hepatic hydrolysis by nonspecific blood and tissue esterases, □ no accumulation. Synergy with thiopental, propofol, isoflurane, midazolam requires reducing doses of these agents by up to 75% Side effects: bradycardia, hypotension (these side effects may be blunted by pretreatment with anticholinergics), N/V, muscle rigidity, pruritis (aspecially facial) dose dependent respiratory depression at doses > 0.05 mcg/kg/min.

□ Adult: avoid bolus doses. Start with drip of 0.05 mcg/kg/min. Titrate in 0.025 mcg/kg/min increments tp a maximum of 0.1–0.2 mcg/kg/min. Add a sedative if adequate sedation not achieved at maximum dose. Wean infusion in 25% decrements over 10 minues after extubation. Supplied: vials of 1, 2 or 5 mg powder to be reconstituted to 1 mg/ml solution.

Drug info: Fentanyl (Sublimaze®)

Narcotic, potency ≈ 100 × morphine. High lipid solubility → rapid onset. Offset (small doses): 20–30 mins. Unlike morphine and meperidine, does not cause histamine release. Lowers ICP. Side effects: dose dependent respiratory depression, large doses given rapidly may cause chest wall rigidity. Repeated dosing may cause accumulation. Diminished sensitivity to CO₂ stimulation, may persist longer than the depression of respiratory rate (up to 4 hours).

□ Adult: 25–100 mcg (0.5–2 ml) IVP, repeat PRN. Supplied: 50 mcg/ml; requires refrigeration.

Drug info: □ Propofol (Diprivan®)

A sedative hypnotic. Also useful in high doses during aneurysm surgery as a neuroprotectant (p. 1203). Protection seems to be less than with barbiturates. Offset time increases after ≈ 12 hours of use.

□ for sedation: start at 5–10 mcg/kg/min. Increase by 5–10 mcg/kg/min q 5–10 minutes PRN desired sedation (up to a max of 50 mcg/kg/min).

Side effects: include Propofol Infusion Syndrome: hyperkalemia, hepatomegaly, lipemia, metabolic acidosis, myocardial failure, rhabdomyolosis, renal failure and sometimes death.³ First identified in children, but may occur at any age. NB: metabolic acidosis of unknown etiology in a patient on propofol is propofol infusion syndrome until proven otherwise. Use with caution at doses > 50 mcg/kg/min or at any dose for > 48 hrs. Also note that the lipid carrier provides 1.1 kCal/ml and hypertriglyceridemia may occur.

Supplied: 500 mg suspended in a 50 ml bottle of fat emulsion. The bottle and tubing must be changed every 12 hours since it contains no bacteriostatic agent.

Drug info: □ Precedex® (Dexmedetomidine)

An alpha-2 adrenoceptor agonist. Acts in locus ceruleus and dorsal root ganglia. Has both sedative and analgesic properties and dramatically reduce the risk of respiratory depression and the amount of narcotic analgesics required. Reduces shivering.

□: usual loading dose is 1 mcg/kg IV over 10 minutes (loading dose not needed if patient already sedated with other agents), followed by continuous IV infusion of 0.2–1.0 mcg/kg/hr titrated to desired effect, not to exceed 24 hours (for short sedation or use as a “transition” drug). Side effects: clinically significant bradycardia and sinus arrest have occurred in young, healthy volunteers with increased vagal tone (anticholinergics such as atropine 0.2 mg IV or glycopyrrolate 0.2 mg IV may help). Use with caution in patients with advanced heart block, baseline bradycardia, using other drugs that lower heart rate, and hypovolemia. Supplied: 2 ml vials of 100 mcg/ml to be diluted in 48 ml NS for a final concentration of 4 mcg/ml for IV use.

7 7.2 Paralytics (neuromuscular blocking agents)

7.2.1 General information

CAUTION: requires ventilation (intubation or Ambu-bag/mask). Reminder: paralyzed patients may still be conscious and therefore able to feel pain, the simultaneous use of sedation is thus required for conscious patients.

Early routine use in head-injured patients lowers ICP (e.g. from suctioning⁴) and mortality, but does not improve overall outcome.⁵

Neuromuscular blocking agents (NMBAs) are classified clinically by time to onset and duration of paralysis as shown in □ Table 7.2. Additional information for some agents follows the table along with some considerations for neurosurgical patients.

Table 7.2 Onset and duration of muscle relaxants						
Clinical class	Agent	Trade name (®)	Onset (min)	Duration (min)	Spontaneous recovery (min)	Comment
Ultra-short	succinylcholine	Anectine	1	5–10	20	shortest onset and duration; plasma cholinesterase dependent; many side effects
Short	rocuronium	Zemuron	1–1.5	20–35	40–60	close to succinylcholine in onset in large doses; some vagolytic action in children
Intermediate	vecuronium	Norcuron	3–5	20–35	40–60	minimal cardiovascular side effects (bradycardia reported); no histamine release
	cisatracurium	Nimbex	1.5–2	40–60	60–80	no histamine release at recommended doses

7.2.2 Ultra-short acting paralytics

Drug info: Succinylcholine (Anectine®)

The only depolarizing ganglionic blocker (the rest are competitive blockers). Rapidly inactivated by plasma pseudocholinesterases. A single dose produces fasciculations then paralysis. Onset: 1 min. Duration of action: 5–10 min.

Indications

Due to significant side effects (see below), use is now limited primarily to the following indications. Adults: generally recommended only for emergency intubations where the airway is not controlled. In children: only when intubation is needed with a full stomach, or if laryngospasm occurs during attempted intubation using other agents.

Side effects

□ CAUTIONS: usually increases serum K⁺ by 0.5 mEq/L (on rare occasion causes severe hyperkalemia ([K⁺] up to 12 mEq/L) in patients with neuronal or muscular pathology, causing cardiac complications which cannot be blocked), therefore contraindicated in acute phase of injury following major burns, multiple trauma or extensive denervation of skeletal muscle or upper motor neuron injury. Do not use for routine intubations in adolescents and children (may cause cardiac arrest even in apparently healthy youngsters, many of whom have undiagnosed myopathies). Linked to malignant hyperthermia (p. 108).

May cause dysrhythmias, especially sinus bradycardia (treat with atropine). May get autonomic stimulation from ACh-like action → HTN, and brady- or tachycardia (especially in peds with repeated doses). The fasciculations may increase ICP, intragastric pressure, and intraocular pressure (contraindicated in penetrating eye injury, especially to anterior chamber; OK in glaucoma).

Precurarization with a “priming dose” of a nondepolarizing blocker (usually ≈ 10% of the intubating dose, e.g. pancuronium 0.5–1 mg IV 3–5 minutes prior to succinylcholine) in patients with elevated ICP or increased intraocular pressure (to ameliorate further pressure increases during fasciculation phase) and in patients who have eaten recently (controversial⁶). Phase II block (similar to nondepolarizing blocker) may develop with excessive doses or in patients with abnormal pseudocholinesterase.

Dosing

- Adult: 0.6–1.1 mg/kg (2–3 ml/70 kg) IVP (err on high side to allow time for procedure & to avoid multi-dosing complications), may repeat this dose × 1.
- Peds (CAUTION: Not recommended for routine use, see above) Children: 1.1 mg/kg. Infants (< 1 mos): 2 mg/kg.
- Supplied: 20 mg/ml concentration.

7.2.3 Short acting paralytics

Drug info: Rocuronium (Zemuron®)

In large doses, has speed of onset that approaches succinylcholine. However, in these doses, paralysis usually lasts ≈ 1–2 hrs. Expensive.

- Adult: initial dose 0.6–1 mg/kg. May be used as infusion of 10–12 mcg/kg/min.

7.2.4 Intermediate acting paralytics

Drug info: ☐ Vecuronium (Norcuron®)

Nondepolarizing (competitive) NMBA. Adequate paralysis for intubation within 2.5–3 minutes of administration. About one–third more potent than pancuronium, shorter duration of action (lasts ≈ 30 minutes after initial dose). Unlike pancuronium, very little vagal (i.e. cardiovascular) effects. No CNS active metabolites. Does not affect ICP or CPP. Hepatically metabolized. Due to active metabolites, paralysis has been reported to take 6 hrs to 7 days to recede following discontinuation of the drug after ≥ 2 days use in patients with renal failure.⁷ Must be mixed to use.

Dosing
Supplied: 10 mg freeze-dried cakes requiring reconstitution. Use within 24 hrs of mixing.

- ☐ Adult and children > 10 years age: 0.1 mg/kg (for most adults use 8–10 mg as initial dose). May repeat q 1 hr PRN. Infusion: 1–2 mcg/kg/min.
- ☐ Pediatric: children (1–10 yrs) require slightly higher dose and more frequent dosing than adult. Infants (7 weeks – 1 yr): slightly more sensitive on a mg/kg basis than adults, takes ≈ 1.5 × longer to recover. Use in neonates and continuous infusion in children is insufficiently studied.

Drug info: ☐ Cisatracurium (Nimbex®)

Nondepolarizing (competitive) blocker. This isomer of atracurium does not release histamine unlike its parent compound (see below). Provides about 1 hour of paralysis. Also undergoes Hofmann degradation, with laudanosine as one of its metabolites.

- ☐ Adult and children > 12 years age: 0.15 or 0.2 mg/kg as part of propofol/nitrous oxide/oxygen induction-intubation technique produces muscle paralysis adequate for intubation within 2 or 1.5 minutes, respectively. Infusion: 1–3 mcg/kg/min.
- ☐ Pediatric: children (2–12 yrs): 0.1 mg/kg given over 5–10 seconds during inhalational or opioid anesthesia.

7.2.5 Reversal of competitive muscle blockade

Reversal is usually not attempted until patient has at least 1 twitch to a train of 4 stimulus, otherwise reversal may be incomplete if patient is profoundly blocked and blockade may reoccur as the reversal wears off (a response of 1/4 indicates 90% muscle blockade)

- neostigmine (Prostigmin®): 2.5 mg (minimum) to 5 mg (maximum) IV
(start low, no efficacy from > 5 mg and can produce severe weakness especially if the maximum dose is exceeded in the absence of neuromuscular blockade)
- PLUS (to prevent bradycardia...),
 - EITHER
0.5 mg atropine for each mg of neostigmine
 - OR
0.2 mg glycopyrrolate (Robinul®) for each mg of neostigmine

7.3 Analgesics

7.3.1 General information

For a discussion of types of pain and pain procedures (p.476).

- Three types of pain medication
1. non-opioid pain medication (see below)
 - a) nonsteroidal anti-inflammatory drugs: aspirin, ibuprofen...
 - b) acetaminophen
 2. opioids (p.138)
 - a) agonists
 - b) partial agonists
 - c) mixed agonist/antagonists

3. drugs that are not strictly analgesics, but which act as adjuvants (p.143) when added to any of the above: tricyclic antidepressants, anticonvulsants, caffeine, hydroxyzine, corticosteroids (p.143)

7.3.2 Guiding principles

The key to good pain control is the early use of adequate levels of effective analgesics. For cancer pain, scheduled dosing is superior to PRN dosing, and “rescue” medication should be available.⁸ Non-opioid analgesics should be continued as more potent medications and invasive techniques are utilized.

7.3.3 Analgesics for some specific types of pain

Visceral or deafferentation pain

May sometimes be effectively treated with tricyclic antidepressants (p.477).
Tryptophan may be effective (p.143).
Carbamazepine (Tegretol®) may be useful for paroxysmal, lancinating pain.

Pain from metastatic bone disease

Steroids, aspirin, or NSAIDs are especially helpful, probably by reducing prostaglandin mediated sensitization of A-delta and C fibers, and therefore may be preferred to APAP.

7.3.4 Nonopioid analgesics

Acetaminophen

Table 7.3 Acetaminophen dosing	
Medication	Dosage
acetaminophen (APAP) (Tylenol®)	adult dose: 650 or 1000 mg PO/PR q 4–6 hrs, not to exceed 4000 mg/day ^a pediatric dose: infants: 10–15 mg/kg PO/PR q 4–6 hrs children: 1 grain/yr age (= 65 mg/yr up to 650 mg) PO/PR q 4–6 hrs not to exceed 15 mg/kg q 4 hrs
^a hepatic toxicity from APAP: usually with doses ≥ 10 gm/day, rare at doses <4000 mg. However, may occur at lower doses (even at high therapeutic doses) in alcoholics, fasting patients, and those taking cytochrome P-450 enzyme-inducing drugs	

Nonsteroidal anti-inflammatory drugs (NSAIDs)

The anti-inflammatory properties of NSAIDs is primarily due to inhibition of the enzyme cyclooxygenase (COX) which participates in the synthesis of prostaglandins and thromboxanes.⁹

- Characteristics of nonselective nonsteroidal anti-inflammatory drugs:
1. all are given orally except ketorolac tromethamine (Toradol®) (see below)
 2. no dependence develops
 3. additive effect improves the pain relief with opioid analgesics
 4. NSAIDs (and APAP) demonstrate a ceiling effect: a maximum dose above which no further analgesia is obtained. For aspirin and APAP, this is usually between 650–1300 mg, and is often higher for other NSAIDs which may also have a longer duration of action
 5. risk of GI upset is common, more serious risks of hepatotoxicity,¹⁰ or GI ulceration, hemorrhage, or perforation are less common
 6. taking medication with meals or antacids has not been proven effective in reducing GI side effects. Misoprostol (Cytotec®), a prostaglandin, may be effective in mitigating NSAID-induced gastric erosion or peptic ulcer. Contraindicated in pregnancy. □ 200 mcg PO QID with food as long as patient is on NSAIDs. If not tolerated, use 100 mcg. □ CAUTION: an abortifacient. Should not be given to pregnant women or women of childbearing potential
 7. most reversibly inhibit platelet function and prolong bleeding time (nonacetylated salicylates have less antiplatelet action, e.g. salsalate, trisalicylate, nabumetome). Aspirin, unlike all other

- NSAIDS, irreversibly binds to cyclooxygenase and thus inhibits platelet function for the 8–10 day life of the platelet
- all cause sodium and water retention and carry the risk of NSAID-induced nephrotoxicity¹¹ (by reducing synthesis of renal vasodilator prostaglandins →reduced renal blood flow which can → renal insufficiency, interstitial nephritis, nephrotic syndrome, hyperkalemia)
 - non-aspirin NSAIDs increase the risk of heart attack or stroke¹²

Drug info: Ketorolac tromethamine (Toradol®)

7

The only parenteral NSAID approved for use in pain control in the U.S. Analgesic effect is more potent than anti-inflammatory effect. Half-life ≈ 6 hrs. May be useful to control pain in the following situations:

- where the avoidance of sedation or respiratory depression is critical
- when constipation cannot be tolerated
- for patients who are nauseated by narcotics
- where narcotic dependency is a serious concern
- when epidural morphine has been used and further analgesia is needed without risk of respiratory depression (agonist type narcotics are contraindicated)
- cautions:
 - not indicated for use >72 hrs (complications have been reported primarily with prolonged use of the oral form)
 - use with caution in postoperative patients since (as with most NSAIDs) bleeding time is prolonged by platelet function inhibition (risk of GI or op-site hemorrhage is small, but is increased in patients >75 yrs old, when used >5 days, and when used in higher doses¹³)
 - even though IM dosing circumvents the GI system, gastric mucosal irritation and erosions may occur as with all NSAIDs (avoid use with PUD)
 - as with all NSAIDs, use with caution in patients at risk for renal side effects

□ Parenteral:For single dose administration: 30 mg IV or 60 mg IM in healthy adult. For multiple dosing: 30 mg IV or IM q 6 hrs PRN. Maximum dosage: 120 mg/day. Parenteral use should not exceed 5 days (3 days may be a better guideline).

For patient weight <50 kg, age >65 yrs, or reduced renal function (creatinine clearance <50 ml/min), all of the above dosages are halved (max daily dose: 60 mg). Creatinine clearance can be estimated using the Cockcroft-Gault equation¹⁴ shown in Eq (7.1), with normal values ≥60 ml/min.

$$\text{Creatinine clearance (ml/min)} = \frac{1}{140} \times \frac{(140 - \text{age in years}) \times \text{ideal wt (kg)}}{72 \times \text{serum creatinine (mg/dl)}} \times 0.85 \text{ for females}$$

□ PO: Indicated only as a continuation of IV or IM therapy, not for routine use as an NSAID. Switching from IM to PO: start with 10 mg PO q 4–6 hrs (combined PO and IM dose should be ≤120 mg on the day of transition). Supplied: 10 mg tablets.

7.3.5 Opioid analgesics

General information

Narcotics are most commonly used for moderate to severe acute pain or cancer pain (some experts characterize cancer pain as recurrent acute pain and not chronic pain).

- Characteristics of narcotics:
- no ceiling effect (p.137): i.e. increasing dosage increases the effectiveness (although with weak opioids for moderate pain, side effects may limit dosages to relatively low levels⁸)
 - with chronic use, tolerance develops (physical and psychological)
 - overdose possible (p.207), with the potential for respiratory depression with all, and seizures with some

Table 7.4 Nonsteroidal anti-inflammatory drugs (NSAIDs) ^a				
Generic name	Proprietary (®)	Typical adult oral dose ^b	Tabs/caps availability (mg) ^c	Daily maximum dose (mg)
aspirin ^d	(many)	500–1000 mg PO q 4–6 hrs (ceiling dose ≈ 1 gm)	325, 500	4000
diclofenac	Voltaren, Cataflam	start at 25 mg QID; additional dose q hs PRN; increase up to 50 mg TID or QID, or 75 mg BID	25, 50, 75	200
etodolac		for acute pain: 200–400 mg q 6–8 hrs	200, 300 caps, 400 tabs	1200
fenoprofen	Nalfon	200 mg q 4–6 hrs; for rheumatoid arthritis 300–600 mg TID-QID	200, 300, 600	3200
flurbiprofen		50 mg TID-QID or 100 mg TID	50, 100	300
ketoprofen	immediate release	start at 75 mg TID or 50 mg QID, ↑ to 150–300 mg daily DIV TID-QID	25, 50, 75	300
	extended release	150 mg q d	ER ^c 150	
ketorolac		see below	see below	
ibuprofen ^e	Motrin	400–800 mg QID (ceiling dose: 800 mg)	300, 400, 600, 800	3200
indomethacin		25 mg TID, ↑ by 25 mg total per day PRN	25, 50, SR 75	150–200
meclofenamate		50 mg q 4–6 hrs; ↑ to 100 mg QID if needed	50, 100	400
mefenamic	Ponstel	500 mg initial; then 250 mg q 6 hrs	250	
nabumetome ^f	Relafen	1000–2000 mg/d given in 1 or 2 doses	500, 750	2000
naproxen	Naprosyn	500 mg, then 250 mg q 6–8 hrs	250, 375, 500	<1250
naproxen sodium	Anaprox	550 mg, followed by 275 mg q 6–8 hrs	275, DS=550	1375
oxaprozin	Daypro	1200 mg q d (1st day may take 1800)	600	1800
piroxicam	Feldene	10–20 mg q d (steady state takes 7–12 d)	10, 20	
sulindac		200 mg BID; ↓ to 150 BID when pain controlled	150, 200	400
salsalate		3000 mg divided BID-TID (e.g. 500 mg 2-tabs TID)	500, 750	
tolmetin		400 mg TID (bioavailability is reduced by food)	200, DS=400, 600	1800
^a NSAIDs increase the risk of cardiovascular thrombotic events (heart attack or stroke) ¹² ^b when dosage ranges are given, use the smallest effective dose ^c abbreviations: DS=double strength; SR=slow release; ER=extended release; DOC=drug of choice ^d aspirin: has unique effectiveness in pain from bone metastases ^e ibuprofen: is available as a suspension (PediaProfen®) 100 mg/ml; dose for children 6 mos to 12 yrs age is 5–10 mg/kg with a maximum of 40 mg/kg/day (not FDA approved for children because of possible Reye’s syndrome) ^f unlike most NSAIDs, nabumetome does not interfere with platelet function				

Mild to moderate pain

Table 7.5 Weak opioids for mild to moderate pain		
Medication	Dosage	
codeine	usual adult dose: 30–60 mg IM/PO q 3 hrs PRN; use with caution in nursing mothers ^a and children (30 mg PO is equivalent to 300 mg aspirin) pediatric dose: 0.5–1 mg/kg/dose q 4–6 hrs PO or IV PRN	
pentazocine	pentazocine is a mixed agonist-antagonist	
	Talwin®	→ 12.5 mg pentazocine, 325 mg ASA. □: 2 PO TID-QID PRN
	with naloxone	→ 50 mg pentazocine, 0.5 mg naloxone. □: 1–2 PO q 3–4 hrs PRN up to 12 tabs/day
tramadol (Ultram®)	(see below)	

^a1–28% of women are ultrafast metabolizers of codeine and the resultant morphine may be passed on to the infant via the breast milk

Some useful medications are shown in □ Table 7.5.

Codeine and congener pentazocine, are usually no more effective that ASA or APAP and are usually combined with these drugs.

Drug info: Tramadol (Ultram®)

An oral opioid agonist that binds to μ-opioid receptors, and is also a centrally acting analgesic that inhibits reuptake of norepinephrine and serotonin. For acute pain, 100 mg is comparable to codeine 60 mg with ASA or APAP.^{15,16} There has been no report of respiratory depression when oral dosing recommendations are followed. Seizures and opioid-like dependence have been reported.¹⁶

□ 50 to 100 mg PO q 4–6 hrs PRN pain up to a maximum of 400 mg/day (or 300 mg/d for older patients). For moderately severe acute pain, an initial dose of 100 mg followed by 50 mg doses may suffice. Supplied: 50 mg tabs.

Moderate to severe pain

Table 7.6 Opioids for moderate to severe pain	
Medication	Dosage
hydrocodone	(Vicodin®, Lorcet®, Lortab®...): 5 mg hydrocodone + 500 mg acetaminophen; (Vicodin ES®, Lortab 7.5/500®): 7.5 mg hydrocodone + 500 mg APAP; □ 1 tab PO q 6 hrs PRN (may increase up to 2 tabs PO q 3–4 hrs not to exceed 8 pills/24 hrs).
	(Lorcet® Plus, Lorcet® 10/650): 7.5 or 10 mg hydrocodone (respectively)+ 650 mg APAP; □ 1 tab PO q 6 hrs PRN (not to exceed 6 tabs in 24 hrs).
	(Lortab® 10/500: 10 mg. hydrocodone + 500 mg APAP); □: 1–2 PO q 4 hrs PRN up to 6 tabs/day.
	(Norco®): 10 mg hydrocodone + 325 mg APAP scored tabs; □: 1 PO q 4 hrs PRN up to 6 tabs/day.
oxycodone	Supplied : usually available in combination as: aspirin 325 mg with oxycodone 5 mg (Percodan®) or acetaminophen (APAP) (Tylox® = APAP 500 mg + oxycodone 5 mg) (Percocet® = oxycodone/APAP in 2.5/325, 5/325, 7.5/500, 10/650) dose: 1 PO q 3–4 hrs PRN (may increase up to 2 PO q 3 hrs ^a)

Table 7.6 continued			
Medication	Dosage		
	Supplied :also available alone as OxyIR® 5 mg, OxyFast® oral solution of 20 mg/ ml, or in controlled-release tablets as OxyContin® 10, 20, 40, 80 ^b & 160 ^b mg (which last 12 hours, achieving steady state in 24–36 hours). □ Adult: OxyContin® tablets are taken whole and are not to be divided, chewed or crushed. It is intended for management of moderate to severe pain when continuous around-the-clock analgesic is needed for an extended period of time and is not intended for use as a PRN analgesic. For opiate naive patients, start with 10 mg PO q 12 hrs. For patients on narcotic medications, a conversion table is provided below for some medications. Titrate dose every 1–2 days, increasing dose by 25–50%q 12 hrs.		
	Conversion table for starting OxyContin®		
	Preparation currently being used	Dose	Suggested starting dose of OxyContin®
	oxycodone combina- tion pills (Tylox, Per- codan...) or Lortab, Vicodin or Tylenol #3	1–5 pills/day	10–20 mg PO q 12 hrs
		6–9 pills/day	20–30 mg PO q 12 hrs
		10–12 pills/day	30–40 mg PO q 12 hrs
	IV PCA morphine	determine total MSO4 dose used per 24 hrs	multiply total MSO4 dose in 24 hrs × 1.3 for total OxyContin dose in 24 hrs
hydromorphone	Dilaudid®: (see □ Table 7.7)		
morphine	used in low doses (see □ Table 7.7)		
a)not to exceed 4000 mg of acetaminophen/24 hrs (see footnote to □ Table 7.3)			
b)for use only in opioid-tolerant patients			

Severe pain

Table 7.7 Equianalgesic doses for SEVERE pain, AGONISTopioids (parenteral route is referenced to 10 mg IM morphine)					
Drug name: generic (proprietary®)	Route	Dose (mg)	Peak (hrs)	Duration (hrs)	Comments
morphine	IM	10	0.5–1	4–6	respiratory depression long acting PO forms: MS Contin®, Avinza® (see below)
	PO	20–60 ^a	1.5–2	4–7	
codeine (not rec- ommended at these doses)	IM	130		3–5	these high doses cause unacceptable side effects
	PO	200			
methadone ^b (Dolophine®)	IM	10	0.5–1	4–6	long half-life ^b
	PO	20	1.5–2	4–7	
oxycodone (e.g. Tylox® ^c) (OxyContin®)	IM	15			
	PO	30	1	3–4	combination (Tylox®) or liquid
	PO	30–40		12	OxyContin, see □ Table 7.6
oxymorphone	IM	1		3–5	available as suppository
	PR	10			
hydromorphone (Dilaudid®)	IM	1.5	0.5–1	3–4	

Table 7.7 continued

Drug name: generic (proprietary®)	Route	Dose (mg)	Peak (hrs)	Duration (hrs)	Comments
	PO	7.5	1.5–2	3–4	supplied: 1, 2, 3, & 4 mg tabs
fentanyl (Sublimaze®)	IV	0.1		1–2	not recommended for acute pain control, esp. in narcotic naive pts.
transdermal fentanyl patch (Duragesic®) ^d	transdermal	^e	12–24	72	patches of 25, 50, 75, 100 or 125 mcg/hr (use lowest effective)

^aIM:PO potency ratio for morphine is 1:6 for single doses, but changes to 1:2–3 with chronic dosing

^bdue to long half-life, repeated dosing can lead to accumulation and CNS depression (must reduce dose after ≈ 3 days, even though the analgesic half-life does not change), especially in the elderly or debilitated patient. Use should be limited to physicians with experience using these drugs

^cmay not be practical for use in severe pain since 1 Tylox® contains only 5 mg oxycodone (the acetaminophen limits the dosage), may use OxyContin® for higher doses of oxycodone

^d□ should not be used as routine post-op analgesic (risk of respiratory depression). Apply 1 patch to upper torso, replace q 72 hrs PRN.

^econversion from total daily parenteral morphine as follows:
8–27 mg MSO4/day → Duragesic 25 mcg/hr
28–37 mg MSO4/day → Duragesic 50 mcg/hr
38–52 mg MSO4/day → Duragesic 75 mcg/hr
53–67 mg MSO4/day → Duragesic 100 mcg/hr
68–82 mg MSO4/day → Duragesic 125 mcg/hr

Table 7.8 Equianalgesic doses for SEVERE pain, AGONIST/ANTAGONIST opioids (referenced to 10 mg IM morphine)

Drug name: generic (proprietary®)	Route	Dose (mg)	Peak (hrs)	Duration (hrs)	Comments
buprenorphine (Buprenex®)	IM	0.4			partial agonist
	SL	0.3			
Mixed agonist/antagonist ^a					
butorphanol	IM	2	0.5–1	4–6	
nalbuphine	IM	10	1	3–6	no sigma receptor occupation ^b
	IV	140 mcg/kg	0.5	2–5	
pentazocine (Talwin® ^c)	IM ^b	20–40	0.5–1	4–6	
	PO ^b	180 (start @ 50)	1.5–2	4–7	

^aall can precipitate withdrawal symptoms in patients physically dependent on agonists

^bmost agonist/antagonist drugs occupy sigma receptors (Stadol>Nubain), which may cause hallucinations

^cTalwin injectable (for IM use) contains only pentazocine. Talwin® Compound tablets contain ASA, therefore for high PO doses, use Talwin Nx which contains no ASA (□ Table 7.5)

Drug info: Avinza® (extended release morphine)

Once daily oral morphine formulation using a spherical oral drug absorption system (SODAS) (numerous ammonio-methacrylate copolymer beads, ≈ 1 mm dia.). Potential for overdose and/or abuse.

□: Dosage is titrated based on patient’s opioid tolerance and degree of pain. Taken as 1 capsule p. o. q d. Not to be taken “PRN.” Not for post-op pain. □ CAUTION: To prevent potentially fatal doses of morphine, capsules are to be swallowed whole, and are not to be chewed, crushed or dissolved. However, the contents of the capsule (the beads) may be sprinkled on apple-sauce for those unable to swallow the capsules, but the beads are not to be chewed or crushed. Side effects: Due to the potentially nephrotoxic effect of fumaric acid used in SODAS, the maximum dose of Avinza is 1600 mg/d. Doses ≥ 60 mg are for opioid tolerant patients only. Supplied: 30, 60, 90 & 120 mg capsules.

7.3.6 Adjuvant pain medications

The following may have efficacy in enhancing the effectiveness of opioid analgesics (and thereby may reduce the required dose).

Tricyclic antidepressants:

Tryptophan: an amino acid and a precursor of serotonin, may work by increasing serotonin levels. Requires high doses and has hypnotic effects, therefore 1.5–2 gm given usually q hs. Must give daily MVI as chronic tryptophan therapy depletes vitamin B6.

Antihistamines: histamines play a role in nociception. Antihistamines, which are also anxiolytic, antiemetic, and mildly hypnotic, are effective as analgesics or as adjuvants. Hydroxyzine (Atarax®, Vistaril®): □ start with 50 mg PO q AM and 100 mg PO q hs. May increase up to ≈ 200 mg daily.

Anticonvulsant-class drugs: carbamazepine, clonazepam, phenytoin, gabapentin or pregabalin tend to be more effective in neuropathic pain, e.g. from diabetic neuropathy, trigeminal neuralgia, post-herpetic neuralgia, glossopharyngeal neuralgia, and neuralgias due to nerve injury or infiltration with cancer.¹⁶ See index for entries.

Phenothiazines: some cause mild reduction in nociception. Most are tranquilizing and antiemetic. Best known for this use is fluphenazine (Prolixin®), usually given with a tricyclic antidepressant for neuropathic pain, Diabetic neuropathy, Treatment (p.545). Phenothiazines may reduce the seizure threshold.

Corticosteroids: in addition to the reduction of toxic effects of radiation or chemotherapy, they may potentiate narcotic analgesics. There are also a number of nonspecific beneficial effects: increased appetite, sense of well being, antiemetic. Side effects may limit usefulness (p.146).

Caffeine: although it possesses no intrinsic analgesic properties, doses of 65–200 mg enhance the analgesic effect of APAP, ASA or ibuprofen in for pain including: H/A, oral surgery pain and post-partum pain.

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8 Endocrinology

8.1 Corticosteroids

8.1.1 General information

Under normal, basal conditions, the zona fasciculata of the adrenal cortex secretes 15–25 mg/day of cortisol (hydrocortisone is the name for the identical pharmaceutical compound for administration), and 1.5–4 mg/day of corticosterone. Cortisol has a half-life of $\approx$ 90 minutes. The release of cortisol by the adrenal glands is stimulated by adrenocorticotrophic hormone (ACTH) from the pituitary which in turn is stimulated by corticotropin releasing hormone (CRH) from the hypothalamus.

8.1.2 Replacement therapy

In primary adrenocortical insufficiency (Addison’s disease), both glucocorticoids and mineralocorticoids must be replaced. In secondary adrenal insufficiency caused by deficient corticotropin (ACTH) release by the pituitary, mineralocorticoid secretion is usually normal and only glucocorticoids need to be replaced.

□ Table 8.1 shows equivalent daily corticosteroid doses for replacement therapy.

Physiologic replacement (in the absence of stress) can be accomplished with either:

1. hydrocortisone: 20 mg PO q AM and 10 mg PO q PM
2. or prednisone: 5 mg PO q AM and 2.5 mg PO q PM

Cortisol and cortisone are useful for chronic primary adrenocortical insufficiency or for Addisonian crisis. Because of mineralocorticoid activity, use for chronic therapy of other conditions (e.g. hypopituitarism) may result in salt and fluid retention, hypertension and hypokalemia.

8.1.3 Hypothalamic-pituitary-adrenal axis suppression

General information

Chronic steroid administration suppresses the hypothalamic-pituitary-adrenal (HPA) axis, and eventually causes adrenal atrophy. When the HPA is suppressed, if exogenous steroids are abruptly

Table 8.1 Equivalent corticosteroid doses ^a					
Steroid: generic (proprietary)	Equiv dose (mg)	Route	Dosing	Mineralocorticoid potency	Oral dosing forms
cortisone acetate	25	PO, IM	2/3 in AM, 1/3 in PM	2 +	tabs: 5, 10 & 25 mg
hydrocortisone AKA cortisol (Cortef®)	20	PO	2/3 in AM, 1/3 in PM	2 +	tabs: 5, 10 & 20 mg
(Solu-Cortef®)		IV, IM ^b			
prednisone	5	PO only	divided BID-TID	1 +	tabs: 1, 2.5, 5, 10, 20, 50 mg ^c
methylprednisolone	4	PO, IV, IM		0	Tabs ^d : 2, 4, 8, 16, 24, 32 mg
dexamethasone	0.75	PO, IV	divided BID-QID	0	scored tabs: 0.25, 0.5, 0.75, 1.5, 4, 6 mg
^a doses given are daily doses. Steroids listed are used primarily as glucocorticoids: equivalent glucocorticoid PO or IV dose is given; IM may differ					
^b IM route recommended only for emergencies where IV access cannot be rapidly obtained					
^c Sterapred Uni-Pak® contains 21 tabs of 5 mgs prednisone and tapers dosage from 30 to 5 mgs over 6 days; “DS” contains 10 mg tabs and tapers from 60 mg to 10 mg over 6 days; “DS 12-Day” contains 48 10 mg tabs and tapers from 60 mg to 20 mg over 12 days					
^d Medrol Dosepak® contains 21 tabs of 4 mgs methylprednisolone and tapers dosage from 24 mg/d to 4 mg/d over 6 days					

stopped or if acute illness develops (which increases the steroid requirements), symptoms of adrenocortical insufficiency (AI) may ensue □ Table 8.2. Severe cases of AI may progress to Addisonian crisis (p.147). Recovery of adrenal cortex lags behind the pituitary, so basal ACTH levels increase before cortisol levels do.

HPA suppression depends on the specific glucocorticoid used, the route, frequency, time, and duration of treatment. Suppression is unlikely with <40 mg prednisone (or equivalent) given in the morning for less than ≈ 7 days, or with every-other-day therapy of <40 mg for ≈ 5 weeks.¹ Some adrenal atrophy may occur after 3–4 days of high dose steroids, and some axis suppression will almost certainly occur after 2 weeks of 40–60 mg hydrocortisone (or equivalent) daily. After a month or more of steroids, the HPA axis may be depressed for as long as one year.

Measuring morning plasma hydrocortisone can evaluate the degree of recovery of basal adrenocortical function, but does not assess adequacy of stress response.

Steroid withdrawal

See reference.¹

In addition to the above dangers of hypocortisolism in the presence of HPA suppression, too rapid a taper may cause a flare-up of the underlying condition for which steroids were prescribed.

When the risk of HPA suppression is low (as is the case with short courses of steroids for less than ≈ 5–7 days² generally prescribed for most neurosurgical indications) abrupt discontinuation usually carries a low risk of AI. For up to ≈ 2 weeks of use, steroids are usually safely withdrawn by tapering over 1–2 weeks. For longer treatment, or when withdrawal problems develop, use the following conservative taper:

1. make small decrements (equivalent to 2.5–5 mg prednisone) every 3–7 d. Patient may experience mild withdrawal symptoms of³:
 - a) fatigue
 - b) anorexia
 - c) nausea
 - d) orthostatic dizziness
2. “backtrack” (i.e. increase the dose and resume a more gradual taper) if any of the following occur:
 - a) exacerbation of the underlying condition for which steroids were used
 - b) evidence of steroid withdrawal symptoms (□ Table 8.2)
 - c) intercurrent infection or need for surgery; see Stress doses (p. 146)
3. once “physiologic” doses of glucocorticoid have been reached (about 20 mg hydrocortisone/day or equivalent (□ Table 8.1):
 - a) the patient is switched to 20 mg hydrocortisone PO q AM (do not use long acting preparations)
 - b) after ≈ 2–4 weeks, a morning cortisol level is checked (prior to the AM hydrocortisone dose), and the hydrocortisone is tapered by 2.5 mg weekly until 10 mg/d is reached (lower limits of physiologic)
 - c) then, every 2–4 weeks, the AM cortisol level is drawn (prior to AM dose) until the 8 AM cortisol is > 10 mcg/100 ml, indicating return of baseline adrenal function
 - d) when this return of baseline adrenal function occurs:
 - daily steroids are stopped, but stress doses must still be given when needed (see below)
 - monthly cosyntropin stimulation (p.735) tests are performed until normal. The need for stress doses of steroids ceases when a positive test is obtained. The risk for adrenal insufficiency persist ≈ 2 years after cessation of chronic steroids (especially the first year)

Table 8.2 Symptoms of adrenal insufficiency (AI)
• fatigue • weakness • arthralgia • anorexia • nausea • hypotension • orthostatic dizziness • hypoglycemia • dyspnea • Addisonian crisis (p. 147) (if severe; with risk of death)

Steroid stress doses

During physiologic “stress” the normal adrenal gland produces ≈ 250–300 mg hydrocortisone/day. With chronic glucocorticoid therapy (either at present, or within last 1–2 yrs), suppression of the normal “stress-response” necessitates supplemental doses.

- In patients with a suppressed HPA axis:
- for mild illness (e.g. UTI, common cold), single dental extraction: double the daily dose (if off steroids, give 40 mg hydrocortisone BID)
 - for moderate stress (e.g. flu), minor surgery under local anesthesia (endoscopy, multiple dental extractions...): give 50 mg hydrocortisone BID
 - for major illness (pneumonia, systemic infections, high fever), severe trauma, or emergency surgery under general anesthesia: give 100 mg hydrocortisone IV q 6–8 hrs for 3–4 days until the stress is resolved
 - for elective surgery, see □ Table 8.3 for guidelines

8.1.4 Steroid side effects

Although deleterious side effects of steroids are more common with prolonged administration,⁴ some can occur even with short treatment courses. Some evidence suggests that low-dose glucocorticoids (≤10 mg/d of prednisolone or prednisone equivalent) for rheumatoid arthritis does not increase osteoporotic fractures, blood pressure, cardiovascular disease, or peptic ulcers,⁵ but weight gain and skin changes are common. Possible side effects include^{3,6}:

- cardiovascular and renal
 - hypertension
 - sodium and water retention
 - hypokalemic alkalosis
- CNS
 - progressive multifocal leukoencephalopathy (PML) (p.331)
 - mental agitation or “steroid psychosis”
 - spinal cord compression from spinal epidural lipomatosis (p.1408): rare
 - pseudotumor cerebri, Idiopathic intracranial hypertension (IIH) (p.766)
- endocrine
 - caution: because of growth suppressant effect in children, daily glucocorticoid dosing over prolonged periods should be reserved for the most urgent indications
 - secondary amenorrhea
 - suppression of hypothalamic-pituitary-adrenal axis: reduces endogenous steroid production → risk of adrenal insufficiency with steroid withdrawal (see above)
 - Cushingoid features with prolonged usage (iatrogenic Cushing’s syndrome): obesity, hypertension, hirsutism...
- GI: risk increased only with steroid therapy >3 weeks duration and regimens of prednisone >400–1000 mg/d or dexamethasone >40 mg/d⁷
 - gastritis and steroid ulcers: incidence lowered with the use of antacids and/or H2 antagonists (e.g. cimetidine, ranitidine...)
 - pancreatitis

Table 8.3 Steroid stress doses for elective surgery			
On day of surgery, 50 mg cortisone acetate IM, followed by 200 mg hydrocortisone IV infused over 24 hrs			
Post-op day	Hydrocortisone (mg)		
	8 AM	4 PM	10 PM
1	50	50	50
2	50	25	25
3	40	20	20
4	30	20	10
5	25	20	5
6	25	15	–
7	20	10	–

- intestinal or sigmoid diverticular perforation⁸: incidence $\approx 0.7\%$ Since steroids may mask signs of peritonitis, this should be considered in patients on steroids with abdominal discomfort, especially in the elderly and those with a history of diverticular disease. Abdominal x-ray usually shows free intraperitoneal air
- inhibition of fibroblasts
 - impaired wound healing or wound breakdown
 - subcutaneous tissue atrophy
- metabolic
 - glucose intolerance (diabetes) and disturbance of nitrogen metabolism
 - hyperosmolar nonketotic coma
 - hyperlipidemia
 - tend to increase BUN as a result of protein catabolism
- ophthalmologic
 - posterior subcapsular cataracts
 - glaucoma
- musculoskeletal
 - avascular necrosis (AVN) of the hip or other bones: usually with prolonged administration $\rightarrow$ cushingoid habitus and increased marrow fat within the bone⁹ (prednisone 60 mg/d for several months is probably the minimum necessary dose, whereas 20 mg/d for several months will probably not produce AVN¹⁰). Many cases blamed on steroids may instead be due to alcohol use, cigarette smoking,¹¹ liver disease, underlying vascular inflammation...
 - osteoporosis: may predispose to vertebral compression fractures which occur in 30–50% of patients on prolonged glucocorticoids. Steroid induced bone loss may be reversed with cyclical administration of etidronate¹² in 4 cycles of 400 mg/d $\square$ 14 days followed by 76 days of oral calcium supplements of 500 mg/d (not proven to reduced rate of VB fractures)
 - muscle weakness (steroid myopathy): often worse in proximal muscles
- infectious
 - immunosuppression: with possible superinfection, especially fungal, parasitic
 - possible reactivation of TB, chickenpox
- hematologic
 - hypercoagulopathy from inhibition of tissue plasminogen activator
 - steroids cause demargination of white blood cells, which may artifactually elevate the WBC count even in the absence of infection
- miscellaneous
 - hiccups: may respond to chlorpromazine (Thorazine®) 25–50 mg PO TID-QID $\square$ 2–3 days (if symptoms persist, give 25–50 mg IM)
 - steroids readily cross the placenta, and fetal adrenal hypoplasia may occur with the administration of large doses during pregnancy

8.1.5 Hypocortisolism

General information

AKA adrenal insufficiency.

Assessment: 8 A.M. serum cortisol level is the best way to test for hypocortisolism. Each lab should provide a lower limit of normal for their lab, which may be broken down further by age and gender.

Addisonian crisis

General information

AKA adrenal crisis. An adrenal insufficiency emergency.

Symptoms: mental status changes (confusion, lethargy, or agitation), muscle weakness.

Signs: postural hypotension or shock, hyperthermia (as high as 105° F, 45.6 C)

Labs

Hyponatremia, hyperkalemia, hypoglycemia.

Treatment of Addisonian crisis

If possible, draw serum for cortisol determination (do not wait for these results to institute therapy). Give fluids sufficient for dehydration and shock.

- For “glucocorticoid emergency”
- hydrocortisone sodium succinate (Solu-Cortef®): 100 mg IV STAT and then 50 mg IV q 6 hrs
- AND
- cortisone acetate 75–100 mg IM STAT, and then 50–75 mg IM q 6 hrs

- For “mineralocorticoid emergency”
- Usually not necessary in secondary adrenal insufficiency (e.g. panhypopituitarism)
- desoxycorticosterone acetate (Doca®): 5 mg IM BID
- OR
- fludrocortisone (Florinef®): 0.05- 0.2 mg PO q d

□ methylprednisolone is NOT recommended for emergency treatment.

8.2 Hypothyroidism

8.2.1 General information

Chronic primary hypothyroidism may result in (non-pathologic) enlargement of the pituitary gland. Plasma TSH determination will distinguish primary hypothyroidism (high TSH) from secondary hypothyroidism (low TSH). Wound healing and cardiac function may be compromised, and surgery under general anesthesia should be postponed if possible until thyroid levels are normalized. Effects of anesthesia may be markedly prolonged, and dosages should be adjusted accordingly.

8.2.2 Thyroid replacement

Caution in patients with adrenal insufficiency

Primary hypothyroidism may be associated with immunologic destruction of adrenal cortex (Schmidt syndrome). Secondary hypothyroidism may be associated with and may mask reduced adrenal function. □ Thyroid replacement without adrenal replacement in patients with adrenal insufficiency (as may occur in panhypopituitarism) may precipitate adrenal crisis (thus give ≈ 300–400 mg hydrocortisone IV over 24 hrs in addition to thyroid replacement).

8.2.3 Routine thyroid replacement dosing

Drug info: Levothyroxine (Synthroid®)

Almost pure T4 (contains no T3 as most T3 is produced peripherally from T4).

Dose required to prevent myxedema coma (not to achieve euthyroidism):

- Maintenance: □ 0.05 mg po q d
- when patient has been hypothyroid: □ start at 0.05 mg po q d and increase by 0.025 mg every 2–3 weeks

For euthyroidism (approximate dose, follow levels and clinical evaluation):

- for most adults <60 years age: □ 0.18 mg/day
- for elderly patients: □ 0.12 mg/day

Drug info: Desiccated thyroid (e.g. Armour thyroid®)

Typical dose: □ 60 mg (1 grain) to 300 mg daily.

Thyroid replacement in myxedema coma

Myxedema coma is an emergency of hypothyroidism and carries 50% mortality.
Symptoms: altered mental status or unresponsiveness.

Signs: hypotension, bradycardia, hyponatremia, hypoglycemia, hypothermia, hypoventilation, occasionally seizures.

Treatment

Drugs may need to be given IV due to reduced gastric motility.

1. general supportive care:
 - a) hypotension: treat with IV fluids (responds poorly to pressors until thyroid replacement accomplished)
 - b) hyponatremia: will correct with thyroid replacement; avoid hypertonic saline
 - c) hypoglycemia: IV glucose
 - d) symptoms of hypocortisolism: thyroid replacement may precipitate adrenal crisis (see caution above); give 300–400 mg hydrocortisone IV over 24 hrs
 - e) hypothermia: avoid active warming since this increases metabolic demand, use blankets to warm gradually
 - f) hypoventilation: check ABG, intubate if necessary
2. thyroid replacement (for average -sized adult):
 - a) IV replacement: □ 0.5 mg of levothyroxine IV, followed by 0.05–0.2 mg/d IV until patient able to tolerate PO or NG meds
 - b) nasogastric replacement: liothyronine (Cytomel®) is primarily T₃, has a rapid onset of action, much shorter half-life than T₄, and should be reserved for emergencies.
 - : liothyronine 0.05–0.1 mg per NG initially, followed by 0.025 mg BID per NG

8.3 Pituitary embryology and neuroendocrinology

8.3.1 Embryology and derivation of the pituitary gland

The posterior pituitary (neurohypophysis) derives from downward evagination of neural crest cells (brain neuroectoderm) from the floor of the third ventricle. The residual recess in the floor of the third ventricle is called the median eminence. The anterior pituitary gland (adenohypophysis) develops from an evagination of epithelial ectoderm of the oropharynx, the evagination is known as Rathke's pouch and is eventually separated from the oropharynx by the sphenoid bone. Cleft-like remnants of Rathke's pouch separates the adenohypophysis and neurohypophysis. The adenohypophysis is comprised of the pars distalis (anterior lobe), the pars intermedia (intermediate lobe) and the pars tuberalis (extension of adenohypophyseal cells on the anterior aspect of the pituitary stalk). The pituitary gland is functionally outside the blood-brain barrier.

8.3.2 Pituitary hormones, their targets and their controls

General information

The pituitary gland releases 8 hormones, 6 from the anterior pituitary, 2 from the posterior pituitary (□ Fig. 8.1). The anterior pituitary is one of only two sites in the body having a portal circulation (the other being the liver). 6 hypothalamic hormones released in a pulsatile fashion are conveyed in blood from hypothalamic capillaries through this portal circulation via the pituitary stalk to a second capillary bed in the anterior pituitary where they control release of hormones by adenohypophyseal gland cells.

Hormones released from the posterior pituitary (ADH & oxytocin) are synthesized in neurons in the hypothalamus (not gland cells) and are conveyed along their axons also in the pituitary stalk to the posterior pituitary gland where they are released.

The complete homeostatic loop (including negative feedback of the hypothalamic hormones) will not be covered here, and the reader is referred to physiology texts.

Proopiomelanocortin (POMC), AKA proopiometanocortin

241 amino acid polypeptide hormone precursor synthesized primarily in corticotroph cell of the anterior pituitary (but also found in the hypothalamus). Contains amino acid sequences for ACTH, alpha-melanocyte-stimulating hormone (α -MSH), β -lipotropin, γ -lipotropin, β -endorphin and met-enkephalin.

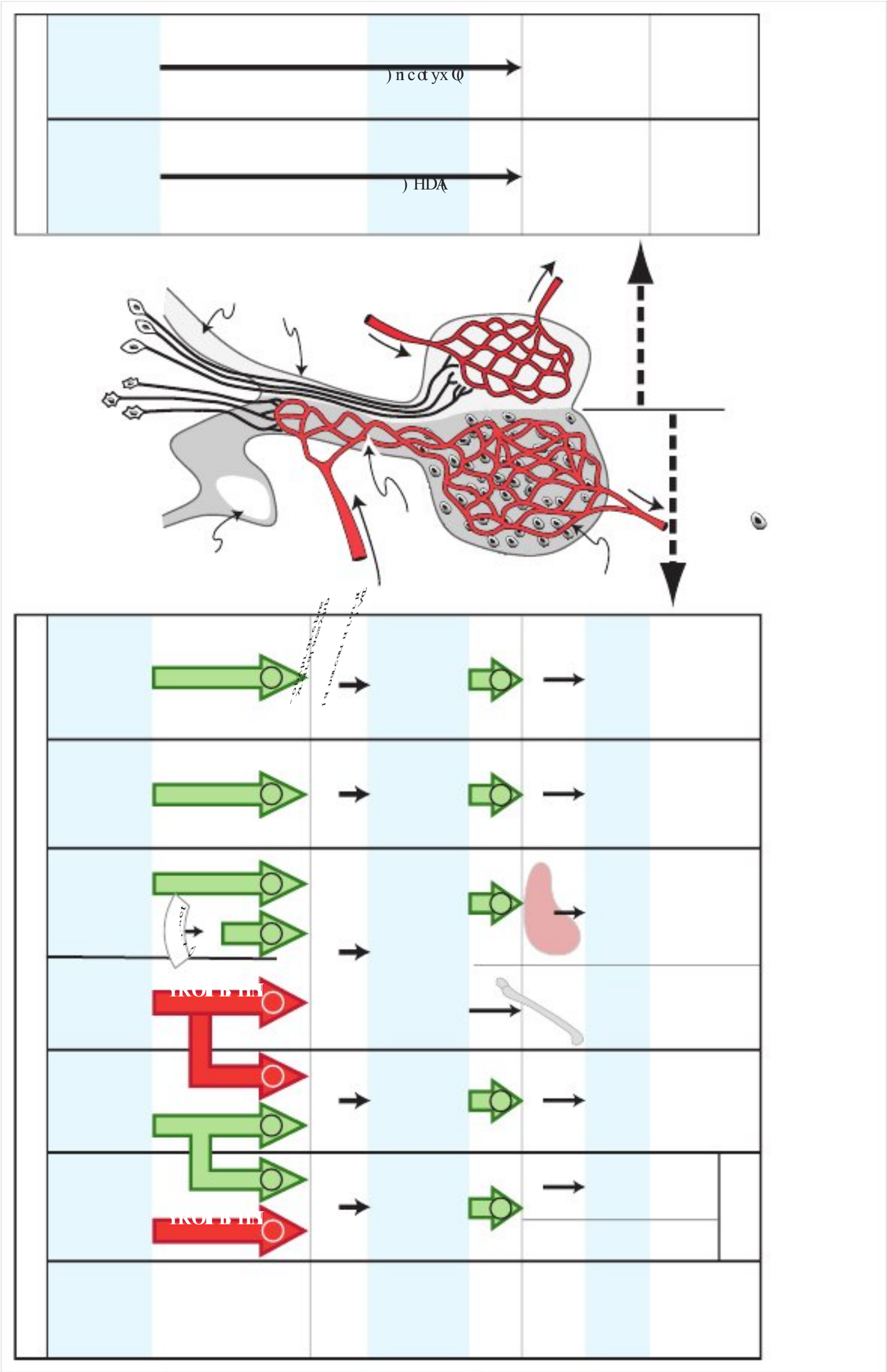


Fig. 8.1 Pituitary neuroendocrinology

Corticotropin AKA adrenocorticotrophic hormone (ACTH)

A 39 amino acid trophic hormone synthesized from POMC. The first 13 amino acids at the amino terminal of ACTH are identical to α -MSH. Active half-life is ≈ 10 minutes. Produces a diurnal peak in cortisol (the highest peak occurs in the early morning, with a second, lesser peak in the late afternoon) and also increases in response to stress.

Control: CRH from the hypothalamus stimulates the release of ACTH.

Prolactin (PRL)

AKA somatomammotropin. 199 amino-acid protein weighing 23,000 daltons. Levels are higher in females than males, and are higher still in pregnancy (see □ Table 46.3). Secreted in pulsatile fashion with a frequency and amplitude that varies during menstrual cycle (range: 5–27 ng/ml) (≈ 9 pulses/24 hours in the late luteal phase, ≈ 14 pulses/24 hours in the late follicular phase, the pulse amplitude increases from early to late follicular and luteal phases). There is also diurnal variation: levels begin to rise 1 hour after the onset of sleep, peak $\approx 5:00$ – $7:00$ AM, and nadir in midmorning after awakening. Heterogeneity of the molecule may produce different results between bioassays and immunoassays.

Control: PRL is the only pituitary hormone predominantly under inhibitory control from the hypothalamus by prolactin inhibitory factors (PRIFs), with dopamine being the primary PRIF. Prolactin releasing factors (PRFs) include: thyrotropin-releasing hormone (TRH) and vasoactive intestinal peptide (VIP). The physiologic role of PRFs is not established. For Ddx of hyperprolactinemia see □ Table 46.4.

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Growth hormone (GH)

A 191 amino-acid polypeptide trophic hormone. GH normally has pulsatile secretion (≈ 5 – 10 pulses/24 hours, primarily at night, up to 30 mcg/L), levels may be undetectable (<0.2 mcg/L) by standard assays between pulses.¹³ Insulin-like growth factor-1 (IGF-1) (formerly AKA somatomedin-C) is the protein secreted primarily by the liver in response to GH that is responsible for most of GH's systemic effects (see levels (p.736)). GH also acts directly on epiphyseal end-plates of long bone to stimulate chondrocyte proliferation.

Control: GH is under dual hypothalamic control via the hypophyseal portal system. GH-releasing hormone (GHRH) from the arcuate nucleus stimulates pituitary secretion and synthesis of GH and induces GH gene transcription. Somatostatin from the periventricular nucleus suppresses GH release only, and has no effect on synthesis. GH release is also stimulated by ghrelin,¹⁴ a peptide synthesized primarily in the GI tract in response to certain nutrients (may act partially or totally via hypothalamic GHRH).

Thyrotropin AKA thyroid stimulating hormone (TSH)

Glycoprotein trophic hormone secreted by thyrotroph cells of the anterior pituitary.

Control: TSH is also under dual hypothalamic control. TRH stimulates production and release of TSH. Somatostatin inhibits the release of TSH.

Gonadotropins

Follicle stimulating hormone (FSH) and luteinizing hormone (LH) (AKA lutropin) are released from the pituitary in response to gonadotropin releasing hormone 1 (GnRH, formerly luteinizing hormone releasing hormone LH-RH) synthesized primarily in the preoptic area of the hypothalamus.

Antidiuretic hormone (ADH)

AKA arginine vasopressin (AVP). The major source of this nanopeptide hormone is the magnocellular portion of the supraoptic nucleus of the hypothalamus. It is conveyed along axons in the supraoptic-hypophyseal tract to the posterior pituitary gland where it is released into the systemic circulation. All actions of ADH result from binding of the hormone to specific membrane bound receptors on the surface of target cells.¹⁵ One of the major effects of ADH is to increase the permeability of the distal renal tubules resulting in increased reabsorption of water, diluting the circulating blood and producing a concentrated urine. The most powerful physiologic stimulus for ADH release is an increase in serum osmolality, a less potent stimulus is a reduction of intravascular volume. ADH is also released in glucocorticoid deficiency, and is inhibited by exogenous glucocorticoids and adrenergic drugs. ADH is also a potent vasoconstrictor.

Oxytocin

A nonapeptide. Oxytocin is a neurotransmitter as well as a hormone. The hypothalamus is the main source of pituitary oxytocin which is stored in nerve endings in the neurohypophysis and is involved in the milk letdown reflex for breastfeeding as well as in uterine contraction during labor.

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9 Hematology

9.1 General information

Circulating blood volumes for adults and peds are shown in □ Table 9.1.

9.2 Blood component therapy

9.2.1 Massive transfusions

Definition: replacement of >1 blood volume (in average adult ≈ 20 U) in <24 hrs for adult, or >2 × circulating blood volume in peds, may cause dilution of effective platelets and coagulation factors. When operating on a pediatric patient, you can usually safely replace up to 1.5 × the circulating blood volume before problems with coagulopathy ensue.

Blood component therapy required for massive transfusions:

1. PRBCs
2. platelets (4 U in adult)
3. FFP

9.2.2 Cellular component

Red blood cell therapy

General information

Major histocompatibilities of blood are shown in □ Table 9.2.

Whole blood

1 U (≈ 510 cc)=450 cc blood +63 cc preservative.

Recommended transfusion criteria:

- exchange transfusions in neonates
- acute burn debridement and grafting in children

Packed red blood cells (PRBCs)

Recommended transfusion criteria:

1. acute blood loss ≥ 15% of patient’s blood volume
2. in asymptomatic patient: hemoglobin (Hb) ≤ 8 gm or Hct ≤ 24%

Table 9.1 Circulating blood volume	
Age	Vol (cc/kg ^a)
premature infant	85–100
term infant <1 month	85
infant >1 mos (& adult)	75
^a cc per kg of body weight	

Table 9.2 Blood compatibility (AB0)				
Blood type	Antibody present	Compatible blood (PRBC)	Compatible plasma	Compatible platelets or cryoprecipitate
A	B	A, 0	A, AB	0
B	A	B, 0	B, AB	
AB	none	AB, A, B, 0	AB	
0	A, B	0	AB, A, B, 0	

- 3. symptoms of anemia at rest
- 4. preoperative Hb ≤ 15 gm or Hct < 45% in the neonate

Amount to transfuse:
Adult: 1 U (250–300 cc) raises Hct by 3–4%
For peds, use Eq (9.1).

ml of PRBC to transfuse $\frac{1}{4} \frac{\text{estimated blood volume (ml)} \times \Delta \text{Hct increment desired (\%)}}{70\%}$ (where the Hct of PRBCs ranges 70–80%)

Give no faster than 2–3 cc/kg/hr.

Autologous blood transfusion

Predonated whole blood may be stored 35 days. PRBCs may be stored 42 days.
Patients may donate every 3 days to 1 week as long as they maintain Hct ≥ 34% (supplement with ferrous sulfate). The following patients require physician release before donating: patients with coronary artery disease, angina, cerebrovascular disease, seizure disorder, pregnancy (because of possible vasovagal episode) or patients with malignancy.
Try to time last donation > 72 hrs prior to surgery to allow patient to replenish some of the depleted RBCs before surgery.

9

9.2.3 Platelets

General information

Normal platelet count (PC) is 150K–400K (abbreviation used here: 150K = 150,000/mm³ = 150 × 10⁹/l). Thrombocytopenia is defined as PC < 150K. Bleeding (spontaneously or with invasive procedures) is rarely a problem with PC > 50K. Spontaneous hemorrhage is very likely with PC < 5K. Spontaneous intracranial hemorrhage is uncommon with PC > 30K, and is more common in adults than children. Based on patients with ITP, the risk of fatal hemorrhage in patients with PC < 30K is 0.0162–0.0389 cases per patient-year¹ (risk of death from infection is higher). Intracranial bleeding is usually subarachnoid or intraparenchymal, with petechial hemorrhages common.
1 unit of platelets contains 5.5 × 10¹⁰ (minimum) to 10 × 10¹⁰ platelets. The volume of 6 units is 250–300 ml. Platelets may be stored up to 5 days.

Recommended platelet transfusion criteria

- Indications for platelet transfusion²:
- 1. thrombocytopenia due to ↓ production (with or without increased destruction) (the most common causes are aplastic anemia and leukemia)
 - a) PC < 10K even if no bleeding (prophylactic transfusion to prevent bleeding)
 - b) PC < 20K and bleeding
 - c) PC < 30K and patient at risk for bleeding: complaints of H/A, presence of confluent (c.f. scattered) petechiae, continuous bleeding from a wound, increasing retinal hemorrhage
 - d) PC < 50K AND
 - major surgery planned within 12 hours
 - PC rapidly falling
 - patient < 48 hours post-op
 - patient requires lumbar puncture
 - acute blood loss of > 1 blood volume in < 24 hours
 - 2. platelet transfusions have limited usefulness when thrombocytopenia is due to platelet destruction (e.g. by antibodies as in ITTP) or consumption (if production is adequate or increased, platelet transfusion usually will not be useful)
 - 3. documented platelet dysfunction in a patient scheduled for surgery or in a patient with advanced hepatic and/or renal insufficiency (consider pharmacologic enhancement of platelet function, e.g. desmopressin³)

Other indications for platelet transfusion:

- 1. patients who have been on Plavix® or aspirin who need urgent surgery that cannot be postponed for ≈ 5 days to allow new platelets to be synthesized

Dosage

Approximately 25% of platelets are lost just with transfusion.

Peds: 1 U/m² raises PC by $\approx$ 10K, usually give 4 U/m².

Adult: 1 U raises platelet count by $\approx$ 5–10K. Typical dose for thrombocytopenic bleeding adult: 6–10 U (usual order: “8-pack”). Alternatively, 1 U of pheresed platelets may be given (obtained from a single donor by apheresis, equivalent to 8–10 U of pooled donor platelets).

Check PC 1–2 hrs after transfusion. The increase in PC will be less in DIC, sepsis, splenomegaly, with platelet antibodies, or if the patient is on chemotherapy. In the absence of increased consumption, platelets will be needed q 3–5 days.

9.2.4 Plasma proteins

FFP (fresh frozen plasma)

General information

1 bag = 200–250 ml (usually referred to as a “unit”, not to be confused with 1 unit of factor activity which is defined as 1 ml). FFP is plasma separated from RBCs and platelets, and contains all coagulation factors and natural inhibitors. FFP has an out-date period of 12 months. The risk of AIDS and hepatitis for each unit of FFP is equal to that of a whole unit of blood.

Recommended transfusion criteria

Recommendations (modified²):

1. history or clinical course suggestive of coagulopathy due to congenital or acquired coagulation factor deficiency with active bleeding or pre-op, with PT > 18 sec or APTT > 1.5 $\times$ upper limit of normal (usually > 55 sec), fibrinogen functioning normally and level > 1 g/l, and coagulation factor assay < 25% activity
2. proven coagulation factor deficiency with active bleeding or scheduled for surgery or other invasive procedure
 - a) congenital deficiency of factor II, V, VII, X, XI or XII
 - b) deficiency of factor VIII or IX if safe replacement factors unavailable
 - c) von Willebrand’s disease unresponsive to DDAVP
 - d) multiple coagulation factor deficiency as in hepatic dysfunction, vitamin K depletion or DIC
3. reversal of warfarin (Coumadin®) (p. 166) effect (PT > 18 sec, or INR > 1.6) in patient actively bleeding or requiring emergency surgery or procedure with insufficient time for vitamin K to correct (which usually requires > 6–12 hrs)
4. deficiency of antithrombin III, heparin cofactor II, or protein C or S
5. massive blood transfusion: replacement of > 1 blood volume ($\approx$ 5 L in 70 kg adult) within several hours with evidence of coagulation deficiency as in (1) and with continued bleeding
6. treatment of thrombotic thrombocytopenic purpura, hemolytic uremic syndrome
7. ☐ because of associated hazards and suitable alternatives, the use of FFP as a volume expander is relatively contraindicated

Dosage

Usual starting dose is 2 bags of FFP (400–600 ml). If PT is 18–22 secs or APTT is 55–70 secs, 1 bag may suffice. Doses as high as 10–15 ml/kg may be needed for some patients. Monitor PT/PTT (or specific factor assay) and clinical bleeding. Since factor VII has a shorter half-life ($\approx$ 6 hrs) than the other factors, PT may become prolonged before APTT.

Remember: if patient is also receiving platelets, that for every 5–6 units of platelets the patient is also receiving coagulation factors equivalent to $\approx$ 1 bag of FFP.

Albumin and plasma protein fraction (PPF, AKA Plasmanate®)

Usually from outdated blood, treated to inactivate hepatitis B virus. Ratio of albumin:globulin percentage in “albumin” is 96%:4% in PPF it is 83%:17% Available in 5% (oncotically and osmotically equivalent to plasma) and 25% (contraindicated in dehydrated patients). 25% albumin may be diluted to 5% by mixing 1 volume of 25% albumin to 4 volumes of D5 W or 0.9% NS (☐ caution: mixing with sterile water will result in a hypotonic solution that can cause hemolysis and possible renal failure).

Expensive for use simply as a volume expander ($\approx$ \$60–80 per unit). Indicated only when total protein < 5.2 gm% (otherwise, use crystalloid which is equally effective). Rapid infusion (> 10 cc/min) has been reported to cause hypotension (due to Na-acetate and Hageman factor fragments). Use in ARDS is controversial. In neurosurgical patients, may be considered as an adjunct for volume

expansion (along with crystalloids) for hyperdynamic therapy (p.1186) when the hematocrit is <40% following SAH where there is concern about increasing the risk of rebleeding e.g. with the use of hetastarch (p.1165).

Cryoprecipitate

Recommended transfusion criteria:

- 1. hemophilia A
- 2. von Willebrand disease
- 3. documented fibrinogen/factor VIII deficiency
- 4. documented disseminated intravascular coagulation (DIC): along with other modes of therapy

Prothrombin complex concentrate (PCC) (Kcentra® and others)

Derived from fresh-frozen human plasma, contains clotting factors II, VII, IX and X, with protein C & S to prevent thrombosis. Primary indication is to be given IV to reverse warfarin in emergency situations. However it is also used in other settings. Requires much lower volume than FFP to work. Also, when the INR gets down to about 1.4, PCC will continue to reduce the INR whereas FFP will have little or no benefit.

Optimal dosing is not known. Doses of 15-50 IU/kg have been given to hemophiliacs but the clotting deficit differs in vitamin-K depletion than in clotting factor absence. A reasonable dose that is often used is 25 IU/kg.

9

9.2.5 Anticoagulation considerations in neurosurgery

General information

Most of these issues have not been studied in a rigorous, prospective fashion. Yet, these questions frequently arise. The following is to be considered a framework of guidelines, and is not to be construed as a standard of care. □ Table 9.3 acts as an index to the topics discussed below.

Contraindications to heparin

Contraindications to heparin therapy are constantly being reevaluated. Massive PE producing hemodynamic compromise should be treated with anticoagulation in most cases despite intracranial risks. Contraindications to full anticoagulation with heparin include:

- recent severe head injury
- recent craniotomy: see below
- patients with coagulopathies
- hemorrhagic infarction
- bleeding ulcer or other inaccessible bleeding site

Table 9.3 Anticoagulation issues in neurosurgery
General neurosurgical contraindications to full anticoagulation with heparin (p.156)
Starting/continuing anticoagulation in the presence of the following neurosurgical conditions
• incidental aneurysm (p.157) • subarachnoid hemorrhage (p.157) • brain tumor (p.157) • following craniotomy (p.157) • acute epidural/subdural hematoma • chronic subdural hematoma • ischemic stroke ◦ after tPA (p.1287) ◦ for prevention of (p.1270) • intracerebral hemorrhage (p.1341)
Managing patients who are already anticoagulated who need a neurosurgical procedure
• warfarin (Coumadin®) (p.157) • heparin (p.160) • LMW-heparin (p.160) • antiplatelet drugs (aspirin, Plavix, NSAIDs) (p.160)
Recommendations for DVT prophylaxis in neurosurgical patients (p.168)

- uncontrollable hypertension
- severe hepatic or renal disease
- <4–6 hours before an invasive procedure (see below)
- brain tumor: see below

Patients with unruptured (incidental) cerebral aneurysms

Anticoagulation may not increase the risk of hemorrhage (i.e. rupture), however, should rupture occur, anticoagulation would most likely increase volume of hemorrhage and thus increase morbidity and mortality.

The decision to start/continue anticoagulant depends on the indication for the drugs, the size of the aneurysm (a small aneurysm <4 mm is not as worrisome). Patients needing Plavix® for drug-eluting cardiac stents should probably be left on their drugs.

Patients on anticoagulation/antiplatelet drugs who develop SAH

Coumadin and antiplatelet drugs are usually reversed.

In patients with brain tumor

Some authors are reluctant to administer full-dose heparin to a patient with a brain tumor,⁴ although a number of studies found no higher risk in these patients when treated with heparin or oral anticoagulation^{5,6,7} (PT should be followed very closely, one study recommended maintaining PT ≈ 1.25 × control⁷).

Post-operatively following craniotomy

Requires individualization based on the reason for the craniotomy. Surgery for parenchymal lesions where the surgery disrupts small vessels (e.g. brain tumor) is probably higher risk for hemorrhage than e.g. aneurysm surgery (expert opinion). Options:

Full anticoagulation: most neurosurgeons would probably not fully anticoagulate patients <3–5 days following craniotomy,⁸ and some recommend at least 2 weeks. However, one study found no increased incidence of bleeding when anticoagulation was resumed 3 days post craniotomy.⁹

Low-dose (prophylactic) anticoagulation: either with mini-dose heparin (5000 U SQ 2 hrs prior to craniotomy and continuing q 12 hrs post-op × 7 d) or enoxaparin (Lovenox) (30 mg SQ BID or as a single dose of 40 MG SQ q d). – RPDB study¹⁰: assessed safety (not efficacy), 55 patients undergoing craniotomy for tumor received mini-dose heparin as indicated had no increased bleeding tendency by any of the parameters measured. RPNB study¹¹: incidence of post-op hemorrhage increased to 11% with enoxaparin.

Management of anticoagulants prior to neurosurgical procedures

Pre-operative laboratory assessment of the coagulation pathway and platelet function is routinely used even though these studies rarely contribute critical information in the patient with a negative history for bleeding tendencies. There are no randomized studies to assess the value of coagulation laboratory measurements to patient care. This section encompasses the use of antiplatelet and anticoagulation medicines, their monitoring, and their reversal.

□ Table 9.4 summarizes this information.

Warfarin

Management guidelines

Patients on warfarin who must be anticoagulated as long as possible (e.g. mechanical heart valves) may be “bridged“ to LMW heparin injections, e.g. Lovenox (p.165), as follows: stop warfarin at least 3 days prior to the procedure, and begin self-administered LMW heparin injections which are discontinued as outlined in □ Table 9.4.

Patients with less critical anticoagulation needs (e.g. chronic a-fib) can usually stop the warfarin at least 4–5 days before the procedure, and a PT/INR is then checked on admission to the hospital. Patients must be advised that during the time that they are not anticoagulated, they are at risk of possible complications from the condition for which they are receiving the agents (annual risk for mechanical valve: ≈ 6% for a-fib: depends on several factors including age & history of prior stroke, an average for patients >65 years age is ≈ 5–6% see details (p.1304).

Table 9.5 Recommendations for holding new oral anticoagulants prior to invasive procedures related to renal function. ³¹			
	Dabigatran	Apixaban	Rivoraxaban
CrCl>80 ml/min	≥72 hr	≥48 hr	≥48 hr
CrCl 50–80 ml/min	≥72 hr	≥48 hr	≥48 hr
CrCl 30–49 ml/min	≥96 hr	≥72 hr	≥72 hr
CrCl<30 ml/min	≥120 hr	≥96 hr	≥96 hr
The recommended minimum interval between last dose and procedure is based on renal function and procedure risk. Generally, neurosurgical procedures including minor procedures such as IPs are considered interventions with a high bleeding risk			

For non-emergent neurosurgical procedures

For procedures where post-op mass effect from bleeding would pose serious risk (which includes most neurosurgical operations), it is recommended that the PT should be $\approx \leq 13.5$ sec (i.e. $\leq$ upper limits of normal) or the INR should be $\approx \leq 1.4$ (e.g. for reference, this INR is considered safe for performing a percutaneous needle liver biopsy). See also reversal of anticoagulation (p.166).

For emergent neurosurgical procedures

Give FFP (start with 2 units) and vitamin K (10–20 mg IV at ≤ 1 mg/min) as soon as possible; see also reversal of anticoagulation (p.166). The timing of surgery is then based on the urgency of the situation and the nature of the procedure (e.g. the decision might be to evacuate a spinal epidural hematoma in an acutely paralyzed patient before anticoagulation is fully reversed).

Heparin

For emergencies: if it would be deleterious to wait 4–6 hours after discontinuing heparin and then repeating the PTT to verify that anticoagulation has been corrected, then heparin can be reversed with protamine (p.166).

For non-emergencies

IV heparin: stop the drip ≈ 4 –6 hours prior to the planned procedure. Option: recheck PTT just prior to starting the procedure.

“Mini-dose” SQ heparin: not mandatory to stop for craniotomy, but if desired to discontinue, then give last dose ≥ 12 hours prior to surgery.

Low-molecular weight heparins (LMWH)

For emergencies: can be reversed with protamine (p.166).

Non-emergencies: See □ Table 9.4. Longer times are needed in renal failure. A factor Xa level can be used to check anticoagulation status, but this usually must be sent out, making it unsuitable for acute management.

Antiplatelet drugs and neurosurgical procedures

Platelet mechanistics and platelet function tests

Platelets are important for maintaining vascular endothelial integrity and are constantly involved with hemostasis in conjunction with coagulation factors. Severe thrombocytopenia can result in petechial hemorrhages or spontaneous intracerebral hemorrhage (ICH). Vascular wall disturbance is the initial stimulus for platelet deposition and activation. Platelets adhere to collagen via surface receptors GPIb-V-IX and von Willebrand factor. This adhesion sets off a cascade of reactions, which result in platelet aggregation forming a hemostatic plug. Historically, bleeding time (BT) was used as the screening test for abnormalities of platelet function. Due to unreliability, many institutions have replaced the BT with the platelet function assay (PFA) using the PFA-100 (platelet function analyzer). There are limited studies confirming its use according to the International Society of Thrombosis and Hemostasis.^{12,13}

In the PFA-100, primary hemostasis is simulated under “high-shear” flow by movement of citrated blood through a membrane-impregnated capillary in two collagen-coated cartridges; one stimulates platelets with adenosine diphosphate (ADP) and the other with epinephrine.¹⁴ This interaction with the collagen induces a platelet plug, which closes an aperture. Results are reported as

closure time in seconds. This method is eligible as a screening test for primary hemostatic disease such as von Willebrand disease as well as for monitoring the effect of antiplatelet therapy. The PFA-100 works for testing with aspirin but not with thienopyridine drug class (e.g. clopidogrel). Newly available PFA cartridges detect P2Y₁₂ receptor blockade in patients on theinopyridine drugs.¹⁵ VerifyNow® measures agonist-induced aggregation as an increase in light transmittance. The system contains a preparation of human fibrinogen-coated beads, which cause a change in light transmittance by ADP-induced platelet aggregation.¹⁵ There is little correlation between PFA-100 results and VerifyNow Assay.

Agents

□ Plavix® (clopidogrel) (p.1275) and aspirin. Cause permanent inhibition of platelet function that persists ≈ 5 days after discontinuation of the drug and can increase the risk of bleeding. For elective cases, 5–7 days off these drugs is recommended (surveys of German neurosurgeons^{16,17}: an average of 7 days was used for low-dose ASA, with a few who do spine surgery even while the patient is on ASA).

Cardiac stents: dual antiplatelet therapy (e.g. ASA+Plavix®) are mandatory for 4 weeks (90 days is preferable¹⁸) after placement of a bare metal cardiac stent, and for at least 1 year with drug-eluting stents (DES) (the risk declines from ≈ 6% to ≈ 3%).¹⁹ Even short gaps in drug therapy (e.g. to perform neurosurgical procedures) is associated with significant risk of acute stent occlusion (and therefore elective surgery during this time is discouraged²⁰). DES are so effective in suppressing endothelialization that lifetime dual antiplatelet therapy may be required. Bridging DES patients with antithrombin, anticoagulants, or glycoprotein IIb/IIIa agents has not been proven effective.²⁰

Reversal of antiplatelet drugs: While heparin and warfarin can be reliably and measurably reversed, the situation is less clear with antiplatelet agents.²¹ Agents used pre-op to reverse these drugs include: Desmopressin (p.166) (DDAVP®)^{16,17} and FFP.¹⁶

Reversal of Plavix for emergency surgery (p.154): platelets may be given, however, Plavix effects persists for up to a couple of days after the last dose, and can actually inhibit platelets given after the drug is discontinued (the half-life of aspirin is lower and should not be an issue after 1 day). In cases with continued oozing in the first day or so after discontinuing Plavix, the following regimen is an option:

1. recombinant activated coagulation factor VII (rFVIIa): even though the defect is in the platelets, rFVIIa works, via a mechanism not mediated by protein clotting factors. Very expensive (≈ \$10,000 per dose), but this must be balanced against the cost of repeat craniotomy, increased ICU stay and additional morbidity
 - a) initial dose²²: 90–120 mcg/kg
 - b) same dose 2 hrs later
 - c) 3rd dose 6 hrs after initial dose
2. platelets every 8 hours for 24 hours, either
 - a) 6 U of regular platelets
 - b) if patient is on fluid or volume restriction: 1 unit of pheresed platelets

□ Herbal products and supplements. Herbal products and supplements often affect platelet aggregation and the coagulation cascade by means that cannot be detected by laboratory tests. The increasing popularity of these unregulated products requires screening patients for their use. There are limited studies regarding the use of herbal supplements in neurosurgery and for an elective operation waiting 7–14 days after cessation of their use is warranted.

Fish Oil (Omega-3 Fatty Acids) is widely used among the cardiac and general population for treatment of dyslipidemia and hypertriglyceridemia. Fish oil may affect platelet aggregation by a reduction in arachnadic acid and thromboxane and adenosine diphosphate receptor blockade. Fish oil may also potentially lengthen bleeding times.^{23,24,25}

Garlic (*Allium sativum*) has increased in popularity as a supplement. Purported benefits include: lowering blood pressure, preventing infection and myocardial infarction, and treating hypercholesterolemia. Garlic has an antiplatelet effect through ADP receptor blockade, and reducing calcium and thromboxane.²⁶ There is particular concern with garlic as it may potentiate the antiplatelet or anticoagulant effect of aspirin or warfarin.²⁷

Ginkgo (*Ginkgo biloba*) has also become a popular supplement found in many formulations from capsules to energy drinks. Ginkgo has been used to treat a number of ailments including memory loss, depression, anxiety, dizziness, claudication, erectile dysfunction, tinnitus and headache. Ginkgo affects bleeding via an antiplatelet effect and antagonism of platelet-activating factor.^{28,29} See Ginkgo biloba under Spontaneous subdural hematoma (p.901).

Ginseng (*Panax ginseng*) has also been found to have antiplatelet activity through thromboxane inhibition and platelet-activating factor.³⁰

Some authors also advocate cautious use of ginger and vitamin E when planning surgery, but the exact antiplatelet mechanism is unclear.²⁵

Table 9.7 Recommended INRs ³⁹	
Indication	INR
- mechanical prosthetic heart valve - prevention of recurrent MI	2.5–3.5
antiphospholipid antibody syndrome (p. 1270) ⁴⁰	≥ 3
all other indications (DVT prophylaxis and treatment, PE, atrial fibrillation, recurrent systemic embolism, tissue heart valves)	2–3

Anticoagulants

See also platelet function inhibitors (p.164).

Warfarin

Drug info: Warfarin (Coumadin®)

An oral vitamin K antagonist. To anticoagulate average weight patient, give 10 mg PO q d × 2–4 days, then ≈ 5 mg q d. Follow coagulation studies, titrate to PT= 1.2–1.5 × control (or INR ≈ 2–3) for most conditions (e.g. DVT, single TIA). Higher PT ratios of 1.5–2 × control (INR ≈ 3–4) may be needed for recurrent systemic embolism, mechanical heart valves... (the recommended ranges for the International Normalized Ratio (INR) are shown in □ Table 9.7).

Starting warfarin: During the first ≈ 3 days of warfarin therapy, patients may actually be hypercoagulable (secondary to reduction of vitamin-K dependent anticoagulation factors protein C and protein S), putting them at risk of “Coumadin necrosis.” Therefore patients should be “bridged” by starting either Lovenox (p. 165) which can be self-administered as an outpatient, or heparin (with a therapeutic PTT).

Supplied: scored tabs of 1, 2, 2.5, 5, 7.5 and 10 mg. IV form: 5 mg/vial.

Heparin

Drug info: Heparin

□ Full anticoagulation in an average weight patient, give 5000 U bolus IV, follow with 1000 U/hr IV drip. Titrate to therapeutic anticoagulation of APTT= 2–2.5 × control (for DVT, some recommend 1.5–2 × control⁴¹).

□ prophylactic AKA low-dose (“mini-dose”) heparin: 5000 IU SQ q 8 or 12 hrs. Routine monitoring of APTT is usually not done, although occasionally patients may become fully anticoagulated on this regimen.

Side effects: (see Anticoagulant considerations in neurosurgery above): hemorrhage, thrombosis⁴² (heparin activates anti-thrombin III and can cause platelet aggregation) which can result in MIs, DVTs, PEs, strokes, etc. Heparin induced thrombocytopenia (HIT): transient mild thrombocytopenia is fairly common in the first few days after initiating heparin therapy, however severe thrombocytopenia occurs in 1–2% of patients receiving heparin > 4 days (usually has a delayed onset of 6–12 days, and is due to consumption in heparin-induced thrombosis or to antibodies formed against a heparin-platelet protein complex). The incidence of HIT in SAH is 5–6% and was similar with enoxaparin.⁴³ Consider use of fondaparinux in thrombocytopenic patients. Chronic therapy may cause osteoporosis.

Low molecular weight heparins

See references.^{44,45}

Low molecular weight heparins (LMWH) (average molecular weight = 3000–8000 daltons) are derived from unfractionated heparin (average MW = 12,000–15,000 daltons). LMWHs differ from unfractionated heparin because they have a higher ratio of anti-factor Xa to anti-factor IIa (antithrombin) activity which theoretically should produce antithrombotic effects with fewer hemorrhagic complications. Realization of this benefit has been very minor in clinical trials. LMWH have greater bioavailability after sub-Q injection leading to more predictable plasma levels which eliminates the need to monitor biologic activity (such as APTT). LMWH have a longer half-life and therefore require

fewer doses per day. LMWH have a lower incidence of thrombocytopenia. More effective in DVT prophylaxis than warfarin in orthopedic surgery.⁴⁶

Spinal epidural hematomas: There have been a number of case reports of spinal epidural hematomas occurring in patients on LMWH (primarily enoxaparin) who also underwent spinal/epidural anesthesia or lumbar puncture, primarily in elderly women undergoing orthopedic surgery. Some have had significant neurologic sequelae, including permanent paralysis.⁴⁷ The risk is further increased by the use of NSAIDs, platelet inhibitors, or other anticoagulants, and with traumatic or repeated epidural or spinal puncture.

- Available low molecular weight heparins. Drugs include:
- enoxaparin (Lovenox®): see below
- dalteparin (Fragmin®): □ 2500 anti-Xa U SQ q d
- ardeparin (Normiflo®): half-life = 3.3 hrs. □ 50 anti-Xa U/kg SQ q 12 hrs
- danaparoid (Orgaran®): a heparinoid. Even higher anti-Xa:anti-IIa ratio than LMWHs. Does not require laboratory monitoring. □ 750 anti-Xa U SQ BID
- tinzaparin (Logiparin®, Innohep®): not available in U.S. □ 175 anti-Xa U per kg SQ once daily

Drug info: Enoxaparin (Lovenox®)

□ dosage established following hip replacement is 30 mg SQ BID × 7–14 days (alternative: 40 mg SQ q d). Pharmacokinetics: After SQ injection, peak serum concentration occurs in 3–5 hrs. Half-life: 4.5 hrs.

Direct thrombin inhibitors

Drug info: Dabigatran (Pradaxa®, Rendix®)

An oral anticoagulant in the class of direct thrombin inhibitors. Administered as the prodrug dabigatran etexilate. Must be stopped 24 hrs prior to surgery.

Reversal of anticoagulation: Praxbind (idarucizumab) IV for emergencies. Reverses pradaxa within 4 hrs, lasts 24 hrs.⁴⁸

Drug info: Bivalirudin (Angiomax® or Angiox®)

A reversible direct thrombin inhibitor (DTI) that increases the rapidity of plasminogen activator-mediated recanalization. No effective reversal.
□: IV loading dose of 0.5 mg/kg IV, followed by continuous infusion of 1.75 mg/kg/hr. Intraarterial: inject 15 mg in 10 ml of heparinized saline via a microcatheter.

Factor Xa inhibitors

Drug info: Fondaparinux (Arixtra®)

A synthetic analog of the pentasaccharide binding sequence of heparin. Increases factor Xa inhibition without affecting factor IIa (thrombin).⁴⁹ Unlike heparin, fondaparinux does not bind to other plasma proteins or platelet factor-4 and does not cause heparin-induced thrombocytopenia (HIT) and can therefore be used in patients with HIT. May be more effective than enoxaparin (Lovenox®) for preventing post-op DVTs. Side effects: Bleeding is the most common side effect (may be increased by concurrent NSAID use). □ Contraindicated with severe renal impairment (CrCl < 30 ml/min).⁵⁰
□: 2.5 mg SQ injection q d. Supplied: 2.5 mg single-dose syringes. Pharmacokinetics: Peak activity occurs in 2–3 hrs. Half-life: 17–21 hrs. Anticoagulation effect lasts 3–5 half-lives. Elimination: in urine (in renal insufficiency reduce dose by 50% for CrCl 30–50 ml/min). STOP: 2–4 days pre-op (longer with kidney dysfunction)

Coagulopathies

Correction of coagulopathies or reversal of anticoagulants

Also refer to recommended normal values for coagulation studies in neurosurgery (p.157).

Platelets

See indications and administration guidelines (p.154).

Fresh frozen plasma

To reverse warfarin anticoagulation, use the following as a starting point and recheck PT/PTT afterwards:

- when patient is “therapeutically anticoagulated” start with 2–3 units FFP (approximately 15 ml/kg is usually needed)
- for severely prolonged PT/PTT, start with 6 units FFP

Prothrombin complex concentrate (PCC)

Warfarin induced anticoagulation may be reversed up to 4 or 5 times more quickly with PCC (Kcentra) (contains coag factors II, IX, and X) than with FFP.⁵¹ Patient may become hyperthrombotic with this.

Drug info: Vitamin K (Mephyton®)

To reverse elevated PT from warfarin , give aqueous colloidal solution of vitamin K1 (phytonadione, Mephyton®). Doses >10 mg may produce warfarin resistance for up to 1 week. FFP may be administered concurrently for more rapid correction (see above). See recommended levels of PT (p. 157).

□ adult: start with 10–15 mg IM; the effect takes 6–12 hrs (in absence of liver disease). Repeat dose if needed. The average total dose needed to reverse therapeutic anticoagulation is 25–35 mg.

IV administration has been associated with severe reactions (possibly anaphylactic), including hypotension and even fatalities (even with proper precautions to dilute and administer slowly), therefore IV route is reserved only for situations where other routes are not feasible and the serious risk is justified. □ IV (when IM route not feasible): 10–20 mg IV at a rate of injection not to exceed 1 mg/min (e.g. put 10 mg in 50 ml of D5 W and give over 30 minutes).

Drug info: Protamine sulfate

For heparin: 1 mg protamine reverses ≈ 100 U heparin (give slowly, not to exceed 50 mg in any 10 min period). Therapy should be guided by coagulation studies.

Reversal of low molecular weight heparins (LMWH): slow IV injection of a 1% solution of protamine can also be used to reverse LMWHs as follows:

Enoxaparin (Lovenox®): ≈ 60% of Lovenox can be reversed with 1 mg of protamine for every mg of Lovenox given (maximum dose = 50 mg) within the last 8 hrs, and 0.5 mg of protamine for every mg of Lovenox given from 8–12 hrs prior. Protamine is probably not needed for Lovenox given >12 hrs earlier.

Dalteparin (Fragmin®) or ardeparin (Normiflo®): 1 mg of protamine for every 100 anti-Xa IU of the LMWH (maximum dose = 50 mg) with a second infusion of 0.5 mg protamine for every 100 anti-Xa IU of LMWH if the APTT remains elevated 2–4 hours after the first dose is completed.

Danaparoid and Hirudin: no known reversing agent.

Drug info: Desmopressin (DDAVP®)

Causes an increase in factor III coagulant activity and von Willebrand factor which helps coagulation and platelet activity in hemophilia A and in von Willebrand’s disease Type I (where the factors are normal in makeup but low in concentration, □ but may cause thrombocytopenia in von Willebrand’s disease Type IIB where factors may be abnormal or missing).

□ 0.3 mcg/kg (use 50 ml of diluent for doses ≤ 3 mcg, use 10 ml for doses > 3 mcg) given over 15–30 minutes 30 minutes prior to a surgical procedure.

Elevated pre-op PTT

In a patient with no history of coagulopathy, a significantly elevated pre-op PTT is commonly due to either a factor deficiency or to lupus anticoagulant. Work-up:

1. mixing study
2. lupus coagulant

If the mixing study corrects the elevated PTT, then there is probably a factor deficiency. Consult a hematologist.

Lupus anticoagulant: If the test for lupus anticoagulant is positive, then the major risk to the patient with surgery is not bleeding, rather it is thromboembolism. Management recommendations:

1. as soon as feasible post-op, start patient on heparin (p.164) or LMW heparin (p.164), e.g. Lovenox
2. at the same time start warfarin, and maintain therapeutic anticoagulation for 3–4 weeks (the risk of DVT/PE is actually highest in the first few weeks post-op)
3. mobilize as soon as possible post-op
4. consider vena-cava interruption filter in patients for whom anticoagulation is contraindicated

Disseminated intravascular coagulation (DIC)

Abnormal intravascular coagulation which consumes clotting factors and platelets, coupled with abnormal activation of fibrinolytic system. Head trauma is an independent risk factor for DIC, possibly because the brain is rich in thromboplastin which may be released into systemic circulation with trauma.⁵² Other risk factors: shock, sepsis.

Presentation

Diffuse bleeding, cutaneous petechia, shock.

Labs

1. fibrinogen degradation products (FDP) > 16 mcg/ml (1–8 = normal; 8–16 = borderline; 32 = definitely abnormal; some labs require >40 for diagnosis of DIC) (the most common abnormality)
2. fibrinogen < 100 mcg/dl (some use 130)
3. PT > 16; PTT > 50
4. platelets < 50,000 (relatively uncommon)

Chronic DIC

PT & PTT may be normal; platelet & fibrinogen low, fibrin split products elevated.

Treatment

1. remove inciting stimulus if possible (treat infections, debride injured tissue, stop transfusions if suspected)
2. vigorous fluid resuscitation
3. anticoagulants, if not contraindicated (p. 156)
4. FFP if PT or PTT elevated, or fibrinogen < 130
5. platelet transfusion if platelet count < 100 K

Pseudo-DIC

Increased fibrin split products, normal fibrinogen.

Seen in conditions such as liver failure.

Thromboembolism in neurosurgery

Deep-vein thrombosis (DVT)

DVT is of concern primarily because of the potential for material (clot, platelet clumps...) to dislodge and form emboli (including pulmonary emboli, (PE)) which may cause pulmonary infarction, sudden death (from cardiac arrest), or cerebral infarction (from a paradoxical embolus, which may occur in the presence of a patent foramen ovale, see Cardiogenic brain embolism (p.1304)). The reported mortality from DVT in the LEs ranges from 9–50%⁵³ DVT limited to the calf has a low threat (<1%) of embolization, however, these clots later extend into the proximal deep veins in 30–50% of cases,⁵³ from where embolization may occur (in 40–50%), or they may produce postphlebotic syndrome.

Neurosurgical patients are particularly prone to developing DVTs (estimated risk: 19–50%) due at least in part to the relatively high frequency of the following:

1. long operating times of some procedures
2. prolonged bed rest pre- and/or post-op

3. alterations in coagulation status
 - a) in patients with brain tumors (see below) or head injury⁵⁴
 - related to the condition itself
 - due to release of brain thromboplastins during brain surgery
 - b) increased blood viscosity with concomitant “sludging”
 - from dehydration therapy sometimes used to reduce cerebral edema
 - from volume loss following SAH (cerebral salt wasting)
 - c) use of high-dose glucocorticoids

Specific “neurological” risk factors for DVT and PE include⁵³:

1. spinal cord injury (p.952)
2. brain tumor: autopsy prevalence of DVT=28% of PE=8.4% Incidence using 125I-fibrinogen⁵⁵: meningioma 72% malignant glioma 60% metastasis 20% Risk may be reduced by pre-op use of aspirin⁵⁶
3. subarachnoid hemorrhage
4. head trauma: especially severe TBI (p.918)
5. stroke: incidence of PE=1–19.8% with mortality of 25–100%
6. neurosurgical operation: risk is higher following craniotomy for supratentorial tumors (7% of 492 patients) than p-fossa tumors (0 out of 141)⁵⁷

Prophylaxis against DVT

Options include:

1. general measures
 - a) passive range of motion
 - b) ambulate appropriate patients as early as possible
2. mechanical techniques (minimal risk of complications):
 - a) pneumatic compression boots⁵⁸ (PCBs) or sequential compression devices (SCDs): reduces the incidence of DVTs and probably PEs. Do not use if DVTs already present. Continue use until patient able to walk 3–4 hrs per day
 - b) TED Stockings®: (TEDS) applies graduated pressure, higher distally. As effective as PCB. No evidence that the benefit is additive.⁵³ Care should be taken to avoid a tourniquet effect at the proximal end (note: TEDS® is a registered trademark. “TED” stands for thromboembolic disease)
 - c) electrical stimulation of calf muscles
 - d) rotating beds
3. anticoagulation; see also contraindications and considerations of anticoagulation in neurosurgery (p.156)
 - a) full anticoagulation is associated with perioperative complications⁵⁹
 - b) “low-dose” anticoagulation⁶⁰ (low-dose heparin): 5000 IU SQ q 8 or 12 hrs, starting 2 hrs pre-op or on admission to hospital. Potential for hazardous hemorrhage within brain or spinal canal has limited its use
 - c) low molecular weight heparins and heparinoids (p.164): not a homogeneous group. Efficacy in neurosurgical prophylaxis has not been determined
 - d) aspirin: role in DVT prophylaxis is limited because ASA inhibits platelet aggregation, and platelets play only a minor role in DVT
4. combination of PCBs and “mini-dose” heparin starting on the morning of post-op day 1 (with no evidence of significant complications)⁶¹

Recommendations

Recommended prophylaxis varies with the risk of developing DVT, as illustrated in □ Table 9.8.⁵³ See also details of prophylaxis in cervical spinal cord injuries (p.952).

Diagnosis of DVT

(For PE, see below). The clinical diagnosis of DVT is very unreliable. A patient with the “classic signs” of a hot, swollen, and tender calf, or a positive Homans’ sign (calf pain on dorsiflexion of the ankle) will have a DVT only 20–50% of the time.⁵³ 50–60% of patients with DVT will not have these findings.

Laboratory tests

- contrast venography: the “gold standard”, however it is invasive and carries risk of iodine reaction, occasionally produces phlebitis, not readily repeated

Table 9.8 Risk & prophylaxis of DVT in neurosurgical patients ^a			
Risk group	Estimated risk of calf DVT	Typical neurosurgical patients	Treatment recommendation
low risk	<10%	age <40 yrs, minimal general risk factors, surgery with <30 minutes general anesthesia	no prophylaxis, or PCB/TEDS
moderate risk	10–40%	age ≥40 yrs, malignancy, prolonged bed rest, extensive surgery, varicose veins, obesity, surgery >30 minutes duration (except simple lumbar discectomy), SAH, head injury	PCB/TEDs; or for patients without ICH or SAH, mini-dose heparin
high risk	40–80%	history of DVT or PE, paralysis ^b (para- or quadriplegia or hemiparesis), brain tumor (especially meningioma or malignant glioma)	PCB/TEDS+ (in patients without ICH or SAH) mini-dose heparin
^a abbreviations: DVT=deep venous thrombosis, PCB=pneumatic compression device, TEDS=TED (thromboembolic disease) Stockings®, ICH=intracerebral hemorrhage, SAH=subarachnoid hemorrhage			
^b see specifics regarding DVT prophylaxis in cervical SCI (p.952)			

- Doppler ultrasound with high-resolution real-time B-mode imaging: 95% sensitive and 99% specific for proximal DVT. Less effective for calf DVT.⁶² As a result, it is recommended that patients with initially negative studies undergo repeat studies over the next 7–10 days to R/O proximal extension. Requires more skill on the part of the tester than IPG. May be used in immobilized or casted LE (unlike IPG). Widely accepted as the non-invasive test of choice for DVT⁶³
- impedance plethysmography (IPG): looks for reduced electrical impedance produced by blood flow from the calf following relaxation of a pneumatic tourniquet. Good in detecting proximal DVT, not sensitive for calf DVT. A positive study indicates DVT that should be treated, a negative study can occur with non-occlusive DVT or with good collaterals, and should be repeated over a 2 week period
- 125I-fibrinogen: radiolabeled fibrinogen is incorporated into the developing thrombus. Better for calf DVT than proximal DVT. Expensive, and many false positives. Risk of HIV transmission has resulted in withdrawal of use
- D-dimer (a specific fibrin degradation product): high levels are associated with DVT and PE⁶⁴

Treatment of DVT

1. bed rest, with elevation of involved leg(s)
2. unless anticoagulation is contraindicated (p.156): start heparin as outlined in Anticoagulation (p.156), aim for APTT=1.5–2 × control; or fixed dose of LMW heparinoids, e.g. tinzaparin (Logiparin®,⁶⁵ or in the U.S. enoxaparin (Lovenox®) (p.165). Simultaneously initiate warfarin therapy. Heparin can be stopped after ≈ 5 days⁶⁶
3. in patients where anticoagulation is contraindicated, consider inferior vena cava interruption or placement of a filter (e.g. Greenfield filter)
4. in non-paralyzed patients, cautiously begin to ambulate after ≈ 7–10 days
5. wear anti-embolic stocking on affected LE indefinitely (limb is always at risk of recurrent DVT)

Pulmonary embolism (PE)

See reference.⁶⁷

Prevention of PE

Prevention of PE is best accomplished by prevention of DVT (p.168).⁶⁸

Presentation of PE

Post-op PE generally occurs 10–14 days following surgery.⁶⁸ The reported incidence⁶⁸ ranges from 0.4–5% A series (on a service with routine use of elastic stockings and, in high risk patients, “mini-dose” heparin) found a post-op incidence of ≈ 0.4% with a doubling of this number if only patients with major pathology (brain tumor, head trauma, or cerebrovascular or spinal pathology) were considered⁶⁸ (another series dealing only with brain tumors found a 4% incidence⁵⁷). Clinical diagnosis is nonspecific (differential diagnosis of symptoms is large, and ranges from atelectasis to MI or cardiac tamponade).

Common findings: sudden dyspnea (the most frequent finding), tachypnea, tachycardia, fever, hypotension, 3rd or 4th heart sound. Triad (rare): hemoptysis, pleuritic chest pain, dyspnea. Auscultation: pleuritic friction rub or rales (rare). Shock and CHF (mimics MI) indicates massive life-threatening PE. Mortality reported ranges from 9–60%⁶⁸ with a significant number of deaths within the first hour.

Diagnosis of PE

A negative D-dimer test (see above) reliably excludes PE in patients with a low clinical probability of PE⁶⁹ or in those with nondiagnostic VQ scan.⁶⁴

Alternatively, one can check for DVTutilizing IPG, Doppler, or venography (see above). If positive, this indicates a possible source of PE, and since the treatment is similar for both, no further search for PE need be made and treatment is started. If negative, further testing may be needed (e.g.VQ scan, see below).

Laboratory tests

D-dimer: see above.

General diagnostic tests

None are very sensitive or specific.

- EKG: “classic” S1Q3T3 is rare. Usually just nonspecific-ST & T changes occur. Tachycardia may be the only finding
- CXR: normal in 25–30% When abnormal, usually shows infiltrate and elevated hemidiaphragm
- ABG: not very sensitive. pO₂ >90 on room air virtually excludes massive PE

Specific radiographic evaluation

- Test of choice: contrast enhanced chest CT. Occasionally chest CTA may be employed. Can provide insight into alternate diagnoses
- pulmonary angiogram: historically, the “gold standard.” Invasive, expensive, and labor intensive. 3–4%risk of significant complications. Not indicated in most cases
- ventilation-perfusion scan (VQ scan): CXR is also needed. A perfusion defect with no ventilation defect in a patient with no previous history of PE strongly suggests acute PE. Equivocal studies occur when an area of malperfusion corresponds to an area of reduced ventilation (on ventilation scan) or infiltrate (on CXR). Probabilities of PE based on VQ scan are shown in □ Table 9.9.⁷⁰ A technically adequate normal VQ scan virtually rules out PE. Patients with low or intermediate probability scans should have a test for DVTor quantitative D-dimer (see above). If test for DVT is positive, treat; if it is negative, the choice is to follow serial IPG or Doppler studies for 2 weeks, or (rarely) to do a pulmonary angiogram
- thin-section contrast-enhanced chest CT: more accurate in patients with COPD who often have an indeterminate VQ scan

Treatment

If diagnosis is seriously entertained, start heparin – unless contraindicated (p.156) – without waiting for results of diagnostic studies. For an average 70 kg patient, begin with 5000–7500 unit IV bolus, followed by 1000 U/hr drip (less for smaller patient). Follow PTT and titrate drip rate for PTT 1.5 to 2 × control.

The use of heparin shortly after surgery and in patients with brain tumors is controversial, and vena caval interruption may be an alternate consideration (e.g. Greenfield filter).

Patients with massive PEs may be hemodynamically unstable. They usually require ICU care, often with PA catheter and pressors.

Table 9.9 Probability of PE based on VQ scan	
Scan results	Incidence of PE
high probability	90–95%
intermediate probability or indeterminate	30–40%
low probability	10–15%
normal	0–5%

9.3 Extramedullary hematopoiesis

9.3.1 General information

In chronic anemias (especially thalassemia major, AKA Cooley's anemia), low hematocrit results in chronic over-stimulation of bone marrow to produce RBCs. This results in systemic bony abnormalities, cardiomyopathy (due to hemochromatosis caused by increased breakdown of defective RBCs).

Pertinent to the CNS, there are three sites where extramedullary hematopoiesis (EMH) can cause findings:

- skull: produces “hair-on-end” appearance on skull x-ray
- vertebral bodies: may result in epidural cord compression⁷¹ (see below)
- choroid plexus

9.3.2 Epidural cord compression from EMH

The exuberant tissue is very radiosensitive, however, the patient may be somewhat dependent on the hematopoietic capacity of the tissue.

9.3.3 Treatment

Surgical excision followed by radiation therapy has been the recommended treatment. Repeated blood transfusions may help reduce EMH and may be useful post-op instead of RTX except for refractory cases.⁷¹

Surgery on these patients is difficult because of:

1. low platelet count
2. poor condition of bone
3. cardiomyopathy: increased anesthetic risk
4. anemia, coupled with the fact that most of these patients are “iron-toxic” from multiple previous transfusions
5. total removal of the mass is not always possible

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10 Neurology for Neurosurgeons

10.1 Dementia

□ Definition. Loss of intellectual abilities previously attained (memory, judgement, abstract thought, and other higher cortical functions) severe enough to interfere with social and/or occupational functioning.¹ Memory deficit is the cardinal feature, however, the DSM-IV definition requires impairment in at least one other domain (language, perception, visuospatial function, calculation, judgement, abstraction, problem-solving skills). Affects 3–11% of community-dwelling adults >65 yrs of age, with a greater presence among institutionalized residents.²

Risk factors: advanced age, family history of dementia, and apolipoprotein E-4 allele.

□ Delirium vs. dementia (critical distinction). Delirium AKA acute confusional state. Distinct from dementia, however, patients with dementia are at increased risk of developing delirium.^{3,4} A primary disorder of attention that subsequently affects all other aspects of cognition.⁵ Often represents life-threatening illness, e.g. hypoxia, sepsis, uremic encephalopathy, electrolyte abnormality, drug intoxication, MI. 50% of patients die within 2 yrs of this diagnosis.

Unlike dementia, delirium has acute onset, motor signs (tremor, myoclonus, asterixis), slurred speech, altered consciousness (hyperalert/agitated or lethargic, or fluctuations), hallucinations may be florid. EEG shows pronounced diffuse slowing.

□ Brain biopsy for dementia. Clinical criteria are usually sufficient for the diagnosis of most dementias. Biopsy should be reserved for cases of a chronic progressive cerebral disorder with an unusual clinical course where all other possible diagnostic methods have been exhausted and have failed to provide adequate diagnostic certainty.⁶ Biopsy may disclose CJD, low grade astrocytoma, and AD among others. The high incidence of CJD among patients selected for biopsy under these criteria necessitates appropriate precautions; see Creutzfeldt-Jakob disease (p.367). In a report of 50 brain biopsies performed to assess progressive neurodegenerative disease of unclear etiology,⁷ the diagnostic yield was only 20% (6% were only suggestive of a diagnosis, 66% were abnormal but nonspecific, 8% were normal). The yield was highest in those with focal MRI abnormalities. Among the 10 patients with diagnostic biopsies, the biopsy result led to a meaningful therapeutic intervention in only 4.

□ Recommendations. Based on the above, the following recommendations are made for patients with an otherwise unexplained neurodegenerative disease:

1. those with a focal abnormality on MRI: stereotactic biopsy
2. those without focal abnormality (possibly including SPECT or PET scan): brain biopsy should only be performed within an investigative protocol

□ Recommendations for specimen. Ideally the biopsy specimen should⁸:

1. be large enough (usually 1 cm³)
2. be taken from an affected area
3. include grey and white matter, pia and dura
4. be handled carefully to minimize artifact (electrocautery should not be used on the specimen side of the incision)

10.2 Headache

10.2.1 General information

Headache (H/A) may be broadly categorized as follows:

1. chronic recurring headaches
 - a) vascular type (migraine): see below
 - b) muscle contraction (tension) headaches
2. headache due to pathology
 - a) systemic pathology
 - b) intracranial pathology: a wide variety of etiologies including:
 - subarachnoid hemorrhage: sudden onset, severe, usually with vomiting, apoplexy, focal deficits possible; see differential diagnosis of paroxysmal H/A (p.1158)

- increased intracranial pressure from any cause (tumor, communicating hydrocephalus, inflammation, pseudotumor cerebri...)
 - irritation or inflammation of meninges: meningitis
 - tumor (p.590): with or without elevated ICP
- c) local pathology of the eye, nasopharynx, or extracranial tissues (including giant cell arteritis (p.195))
- d) following head trauma: postconcussive syndrome (p.923)
- e) following craniotomy: “syndrome of the trephined” (p.1431)

A severe new H/A, or a change in the pattern of a longstanding or recurrent H/A (including developing associated N/V, or an abnormal neurologic exam) warrants further investigation with CT or MRI.⁹

Unilateral H/A that never changes side over a period ≥ 1 year warrants an MRI; this would be atypical in migraine and may be a presentation of an occipital AVM (p.1239).

10.2.2 Migraine

General information

Migraine attacks usually occur in individuals predisposed to the condition, and may be activated by factors such as bright light, stress, diet changes, trauma, administration of radiologic contrast media (especially angiography) and vasodilators.

Classification

Based on the 1962 ad hoc committee on headache (H/A). See also index under Headache, e.g. for: crash migraine (thunderclap headache) (p.1158), post-myelogram headache (p.1508)...

Common migraine

Episodic H/A with N/V and photophobia, without aura or neurologic deficit.

Classic migraine

Common migraine + aura. May have H/A with occasional focal neurologic deficit(s) that resolve completely in ≤ 24 hrs.

Over half of the transient neurologic disturbances are visual, and usually consist of positive phenomena (spark photopsia, stars, complex geometric patterns, fortification spectra) which may leave negative phenomena (scotoma, hemianopia, monocular or binocular visual loss...) in their wake. The second most common symptoms are somatosensory involving the hand and lower face. Less frequently, deficits may consist of aphasia, hemiparesis, or unilateral clumsiness. A slow march-like progression of deficit is characteristic. The risk of stroke is probably increased in patients with migraine.¹⁰

Complicated migraine

Occasional attacks of classic migraine with minimal or no associated H/A, and complete resolution of neurologic deficit in ≤ 30 days.

Migraine equivalent

Neurologic symptoms (N/V, visual aura, etc.) without H/A (acephalgic migraine). Seen mostly in children. Usually develops into typical migraine with age. Aura may be shortened by opening and swallowing contents of a 10 mg nifedipine capsule.¹¹

Hemiplegic migraine

H/A typically precedes hemiplegia which may persist even after H/A resolves.

Cluster headache

AKA histaminic migraine. Actually a neurovascular event, distinct from true migraine. Recurrent unilateral attacks of severe pain. Usually oculofrontal or oculotemporal with occasional radiation into the jaw, usually recurring on the same side of the head. Ipsilateral autonomic symptoms (conjunctival injection, nasal congestion, rhinorrhea, lacrimation, facial flushing) are common. Partial Horner's

syndrome (ptosis and miosis) sometimes occurs. Male:female ratio is $\approx 5:1$. 25% of patients have a personal or family history of migraine.

Headaches characteristically have no prodrome, last 30–90 minutes, and recur one or more times daily usually for 4–12 weeks, often at a similar time of day, following which there is typically a remission for an average of 12 months.¹²

Prophylaxis for cluster H/A is only minimally effective:

1. β -adrenergic blockers are less effective
2. lithium: becoming drug of choice (response rate 60–80%). 300 mg PO TID and follow levels (desired: 0.7–1.2 mEq/L)
3. occasionally ergotamines are used
4. naproxen (Naprosyn®)
5. methysergide (Sansert®) 2–4 mg PO TID is effective in 20–40% of cases, must cycle patient off the drug to prevent retroperitoneal fibrosis, etc. (also see below)

Treatment for cluster H/A (prophylaxis is only minimally effective):

Treatment is difficult because there is no prodrome and the H/A often stop after 1–2 hrs. Treatment of acute attacks includes:

- 100%O₂ by face mask with patient sitting for ≤ 15 min or until attack aborted
- ergotamine: see below
- SQ sumatriptan: usually aborts attack within 15 minutes (see below)
- steroids: see below
- refractory cases may be considered for:
 - percutaneous radiofrequency sphenopalatine ganglion blockade¹³
 - occipital nerve stimulation¹⁴
 - hypothalamic deep brain stimulation

10

Basilar artery migraine

Essentially restricted to adolescence. Recurrent episodes lasting minutes to hours of transient neurologic deficits in distribution of vertebrobasilar system. Deficits include: vertigo (most common), gait ataxia, visual disturbance (scotomata, bilateral blindness), dysarthria, followed by severe H/A and occasionally nausea and vomiting.¹⁵ Family history of migraine is present in 86%

10.3 Parkinsonism

10.3.1 General information

Parkinsonism may be primary or secondary to other conditions. All result from a relative loss of dopamine mediated inhibition of the effects of acetylcholine in the basal ganglia.

10.3.2 Idiopathic paralysis agitans (IPA)

Clinical

Classical Parkinson’s disease AKA shaking palsy. Affects $\approx 1\%$ of Americans $>$ age 50 yrs,¹⁶ it is frequently underdiagnosed.¹⁷ Male:female ratio is 3:2. Not clearly environmentally or genetically induced, but may be influenced by these factors.

The classic triad is shown in □ Table 10.1. Other signs may include: postural instability, micrographia, mask-like facies. Gait consists of small, shuffling steps (marche á petits pas) or festinating gait.

Clinically distinguishing IPA from secondary parkinsonism (see below)

May be difficult early. IPA generally exhibits gradual onset of bradykinesia with tremor that is often asymmetrical, and initially responds well to levodopa. Other disorders are suggested with rapid progression of symptoms, when the initial response to levodopa is equivocal, or when there is early midline symptoms (ataxia or impairment of gait and balance, sphincter disturbance...) or the

Table 10.1 Classic triad of Parkinson’s disease
• tremor (resting, 4–7/second) • rigidity (cogwheel) • bradykinesia

presence of other features such as early dementia, sensory findings, profound orthostatic hypotension, or abnormalities of extraocular movements.^{18,19}

Pathophysiology

Degeneration primarily of pigmented (neuromelanin-laden) dopaminergic neurons of the pars compacta of the substantia nigra, resulting in reduced levels of dopamine in the neostriatum (caudate nucleus, putamen, globus pallidus). This decreases the activity of inhibitory neurons with predominantly D2 class of dopamine receptors which project directly to the internal segment of the globus pallidus (GPi), and also increases (by loss of inhibition) activity of neurons with predominantly D1 receptors which project indirectly to the globus pallidus externa (GPe) and subthalamic nucleus.²⁰ The net result is increased activity in GPi which has inhibitory projections to the thalamus which then suppresses activity in the supplemental motor cortex among other locations.

Histologically: Lewy bodies (eosinophilic intraneuronal hyaline inclusions) are the hallmark of IPA.

10.3.3 Secondary parkinsonism

General information

The differential diagnosis of Parkinson's disease includes the following etiologies of secondary parkinsonism or Parkinson-like conditions (sometimes of these are occasionally referred to as "Parkinson plus" syndromes or parkinsonian disorders) (see above for distinguishing features):

1. olivopontocerebellar degeneration (OPC)
2. striato-nigral degeneration (SND): more aggressive than parkinsonism
3. postencephalitic parkinsonism: followed an epidemic of encephalitis lethargica (von Economo disease) in the 1920s, victims are no longer living. Distinguishing features: oculogyric crisis, tremor involves not only extremities but also trunk and head, asymmetrical, no Lewy bodies
4. progressive supranuclear palsy (PSNP): impaired vertical gaze (see below)
5. multiple system atrophy (Shy-Drager syndrome): see below
6. drug induced: includes:
 - a) prescription drugs (elderly females seem more susceptible)
 - antipsychotics (AKA neuroleptics): haloperidol (Haldol®) which works by blocking post-synaptic dopamine receptors
 - phenothiazine antiemetics: prochlorperazine (Compazine®)
 - metoclopramide (Reglan®)
 - reserpine
 - b) MPTP (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine): a commercially available chemical intermediate which is also a by-product of the synthesis of MPPP (a meperidine analog) that was synthesized and self-injected by a graduate student,²¹ and later produced by illicit drug manufacturers to be sold as "synthetic heroin" and unwittingly injected by some IV drug abusers in northern California in 1983²² (there is also a case report of a chemist who worked with MPTP who developed parkinsonism).²³ MPTP was subsequently discovered to be a potent neurotoxin for dopaminergic neurons (with continued toxic effects that persisted for years²⁴). As a rule, the response to levodopa is dramatic, but short-lived with frequent side effects. In contrast to classic IPA, the locus coeruleus and dorsal motor vagus nucleus were essentially normal, and the symptoms differ slightly
 - c) there is an as yet unproven assertion that methylenedioxymethamphetamine (MDMA) AKA "ecstasy" (on the street), may hasten the onset of Parkinsonism (a study demonstrating a link had to be withdrawn because of a mislabeling of drugs)
7. toxic: poisoning with
 - a) carbon monoxide: symmetric low densities in the globus pallidus on CT
 - b) manganese: may be seen in miners, welders, and pyrotechnics workers. Manganese is excreted by the liver, □ people with hepatic insufficiency are more susceptible. Imaging: symmetrical high signal abnormalities on T1WI primarily in the globus pallidus with essentially no findings on T2WI or GRASS (almost pathognomonic)
8. ischemic (lacunes in basal ganglia): produces so-called arteriosclerotic parkinsonism AKA vascular parkinsonism: "lower-half" parkinsonism (gait disturbance predominates¹⁷). Also causes pseudobulbar deficits, emotional lability. Tremor is rare
9. posttraumatic: parkinsonian symptoms may occur in chronic traumatic encephalopathy, see dementia pugilistica (p.924). There are usually other features not normally present in IPA (e.g. cerebellar findings)

10. normal pressure hydrocephalus (NPH): urinary incontinence (p.404)...
11. neoplasm in the region of the substantia nigra
12. Riley-Day (familial dysautonomia)
13. parkinson-dementia complex of Guam: classic IPA+amyotrophic lateral sclerosis (ALS). Pathologically has features of parkinsonism and Alzheimer's disease but no Lewy bodies nor senile plaques
14. Huntington's disease (HD): whereas adults typically show chorea, when HD manifests in a young person it may resemble IPA
15. (spontaneous) intracranial hypotension (p.389) may present with findings mimicking IPA

Multiple system atrophy (MSA)

AKA Shy-Drager syndrome. Parkinsonism (indistinguishable from IPA), PLUS idiopathic orthostatic hypotension, PLUS other signs of autonomic nervous system (ANS) dysfunction (ANS findings may precede parkinsonism and may include urinary sphincter disturbance and hypersensitivity to norepinephrine or tyramine infusions). Degeneration of preganglionic lateral horn neurons of thoracic spinal cord. Unlike IPA, most do not respond to dopa therapy. NB: classic IPA may eventually produce orthostatic hypotension from inactivity or as a result of progressive autonomic failure.

Progressive supranuclear palsy (PSNP)

AKA Steele-Richardson-Olszewski syndrome.²⁵

Triad:

1. progressive supranuclear ophthalmoplegia (chiefly vertical gaze): paresis of voluntary vertical eye movement, but still moves to vertical doll's eyes maneuver
2. pseudobulbar palsy (mask-like facies with marked dysarthria and dysphagia, hyperactive jaw jerk, emotional incontinence usually mild)
3. axial dystonia (especially of neck and upper trunk)

Associated findings: subcortical dementia (inconstant), motor findings of pyramidal, extrapyramidal and cerebellar systems. Average age of onset: 60 yrs. Males comprise 60% Response to anti-parkinson drugs is usually very short lived. Average survival after diagnosis: 5.7 yrs.

Differentiating from Parkinson's disease (IPA):

Patients with PSNP have a pseudo-parkinsonism. They have mask facies, but do not walk bent forward (they walk erect), and they do not have a tremor. They tend to fall backwards.

Course

1. early:
 - a) many falls: due to dysequilibrium + downgaze palsy (can't see floor)
 - b) eye findings may be normal initially, subsequently may develop difficulty looking down (especially to command, less to following), caloric have normal tonic component but absent nystagmus (cortical component)
 - c) slurred speech
 - d) personality changes
 - e) difficulty eating: due to pseudobulbar palsy + inability to look down at food on plate
2. late:
 - a) eyes fixed centrally (no response to oculocephalics or oculovestibulars): ocular immotility is due to frontal lobe lesions
 - b) neck stiffens in extension (retrocollis)

Treatment for Parkinson's disease

Medical treatment for Parkinson's disease is beyond the scope of this book.

Surgical treatment

Before the introduction of L-dopa in the late 1960's, stereotactic thalamotomy was widely used for Parkinson's disease. The location ultimately targeted for lesioning was the ventrolateral nucleus. The procedure worked better for relieving the tremor than for the bradykinesia, however it was the latter symptom that was most disabling. This procedure cannot be done bilaterally without significant risk to speech function. The procedure fell out of favor when more effective drugs became available.²⁶

See Surgical treatment of Parkinson's disease (p.1524) for further information.

10.4 Multiple sclerosis

Key concepts

- an idiopathic demyelinating disease of the CNS producing exacerbating and remitting symptoms disseminated in space and time
- classic clinical findings: optic neuritis, paresthesias, INO and bladder symptoms
- diagnostic criteria (McDonald criteria) use clinical and/or lab results (MRI, CSF...) to stratify patients as: MS, probable MS, or not MS
- MRI: multiple usually enhancing lesions involving optic nerves & white matter of brain (especially periventricular white matter), cerebellum and spinal cord

10.4.1 General information

An idiopathic demyelinating disease (thus affecting only white matter) of the cerebrum, optic nerves, and spinal cord (especially the corticospinal tracts and the posterior columns). Does not affect peripheral myelin. Pathologically produces multiple plaques of various age in diffuse locations in the CNS, especially in the periventricular white matter. Lesions initially evoke an inflammatory response with monocytes and lymphocytic perivascular cuffing, but with age settle down to glial scars.

10.4.2 Epidemiology

Usual age of onset: 10–59 years, with the greatest peak between ages 20–40 years. The female:male incidence is approximately 2:1.²⁷
Prevalence varies with latitude, and is <1 per 100,000 near the equator, and is ≈ 30–80 per 100,000 in the northern U.S. and Canada.

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10.4.3 Classification

Typically causes exacerbations and remissions in various locations in the CNS (dissemination in space and time). Common symptoms: visual disturbances (diplopia, blurring, field cuts or scotoma), spastic paraparesis, and bladder disturbances. Nomenclature for the time course of MS is shown in □ Table 10.2.²⁸ Relapsing-remitting MS is the most common pattern (≥70%) at onset, and has the best response to therapy, but >50% of cases eventually become secondary progressive MS. Only 10% have primary progressive MS, and these patients tend to be older at onset (40–60 years) and frequently develop progressive myelopathy.²⁹ Progressive relapsing MS is very uncommon. Deficits present >6 months usually persist.

Table 10.2 Clinical categories of MS	
Category	Definition
relapsing-remitting	episodes of acute worsening with recovery and a stable course between relapses
secondary progressive	gradual neurologic deterioration ± superimposed acute relapses in a patient who previously had relapsing-remitting MS
primary progressive	gradual, nearly continuous neurologic deterioration from the onset of symptoms
progressive relapsing	gradual neurologic deterioration from the onset of symptoms, but with subsequent superimposed relapses

10.4.4 Clinical signs and symptoms

Visual disturbances

Disturbances of visual acuity may be caused by optic or retrobulbar neuritis which is the presenting symptom of MS in 15% of cases, and which occurs at some time in 50% of MS patients. The percentage of patients with an attack of optic neuritis and no prior attack that will go on to develop MS ranges from 17–87% depending on the series.³⁰ Symptoms: acute visual loss in one or both eyes with mild pain (often on eye movement).

Diplopia may be due to internuclear ophthalmoplegia (INO) (p.565) from a plaque in the MLF. INO is an important sign because it rarely occurs in other conditions besides MS or brainstem stroke.

Motor findings

Extremity weakness (mono, para, or quadriparesis) and gait ataxia are among the most common symptoms of MS. Spasticity of the LEs is often due to pyramidal tract involvement. Scanning speech results from cerebellar lesions.

Sensory findings

Posterior column involvement often causes loss of proprioception. Paresthesias of extremities, trunk, or face occur. Lhermitte's sign (electric shock-like pain radiating down the spine on neck flexion) is common, but is not pathognomonic. Trigeminal neuralgia occurs in $\approx 2\%$ and is more often bilateral and occurs at a younger age than the population in general.³¹

Mental disturbances

Euphoria (la belle indifference) and depression occur in $\approx 50\%$ of patients.

Reflex changes

Hyperreflexia and Babinski signs are common. Abdominal cutaneous reflexes disappear in 70–80%

GU symptoms

Urinary frequency, urgency, and incontinence are common. Impotence in males and reduced libido in either sex is often seen.

10.4.5 Differential diagnosis

The plethora of possible signs and symptoms in MS causes the differential diagnosis to extend to almost all conditions causing focal or diffuse dysfunction of the CNS. Conditions that may closely mimic MS clinically and on diagnostic testing include:

1. acute disseminated encephalomyelitis (ADEM) (p.182): generally monophasic. May also have CSF-OCB (p.182). Corpus callosum involvement is uncommon
2. CNS lymphoma (p.710)
3. other closely related demyelinating diseases: e.g. Devic syndrome (p.1409)
4. vasculitis
5. encephalitis: patients are usually very ill
6. chronic white matter changes: seen in older patients

10.4.6 Diagnostic criteria

No single clinical feature or diagnostic test is adequate for the accurate diagnosis of MS. Therefore, clinical information is integrated with paraclinical studies. Diagnosing MS after a single, acute relapsing clinically isolated syndrome (CIS) is very risky. 50–70% of patients with a CIS suggestive of MS will have multifocal MRI abnormalities characteristic of MS. The presence of these MRI abnormalities increases the risk of developing MS in 1–3 years (with greater prognostic significance than CSF-OCB). The more MRI lesions, the higher the risk.³² Criteria for the diagnosis of MS³³ follows.

Definitions

See references.^{33,34}

The following definitions are used in the classification system that follows:

1. attack (exacerbation, relapse): neurologic disturbance lasting >24 hrs³⁵ typical of MS when clinicopathological studies determine that the cause is demyelinating or inflammatory lesions
2. remission: ≥ 30 days should separate the onset of the first attack from the onset of a second
3. historical information: reporting of symptoms by the patient (confirmation by observer desirable), adequate to locate a lesion of MS, and has no other explanation (i.e. manifestations must not be attributable to another condition)
4. clinical evidence (signs): neuro dysfunction recorded by competent examiner
5. paraclinical evidence: tests or procedures demonstrating CNS lesion which has not produced signs; e.g. Uhthoff phenomenon or sign (worsening of symptoms with hot bath or shower), BAER, imaging procedures (CT, MRI), expert urological assessment
6. typical of MS: signs & symptoms (S/S) known to occur frequently in MS. Thus excludes gray matter lesions, peripheral nervous system lesions, and non-specific complaints such as H/A, depression, convulsive seizures, etc.
7. separate lesions: S/S cannot be explained on basis of single lesion (optic neuritis of both eyes simultaneously or within 15 days represents single lesion)
8. laboratory support: in this study, the only considerations were CSF oligoclonal bands (CSF-OCB) (see below) (OCB must not be present in serum) or increased CSF IgG production (CSF-IgG) (serum IgG must be normal). This assumes that syphilis, SSPE, sarcoidosis, etc. have been ruled out

Diagnosis of MS

The 2010 “McDonald MS Diagnostic criteria”³⁶ are shown in □ Table 10.3.

MRI

MRI is the preferred imaging study in evaluating MS³⁹ and can demonstrate dissemination of lesions in time and space. Recommended³³ brain MRI criteria for diagnosing MS are shown in □ Table 10.3.^{40,41} Lesions are normally >3 mm diameter.³³ MRI shows multiple white matter abnormalities in 80% of patients with MS (compared to 29% for CT).^{42,43} Lesions are high signal on T2, and acute lesions tend to enhance with gadolinium more than old lesions do. Periventricular lesions may blend in with the signal from CSF in the ventricles on T2, these lesions are shown to better advantage on FLAIR (fluid attenuation) MRI (p.229). These lesions are ovoid and are oriented perpendicular to the ependymal surface and are sometimes called Dawson’s fingers (after neuropathologist James Dawson).

Spinal cord lesions normally show little or no swelling, should be ≥ 3 mm but <2 vertebral segments, occupy only a portion of the cross-section of the cord, and must be hyperintense on T2.⁴⁴

Specificity of MRI is $\approx 94\%$ ⁴⁵ however, encephalitis as well as UBOs seen in aging may mimic MS lesions. DWI should be normal, however plaques can sometimes exhibit “shine through” (p.232) so the ADC map must be checked to rule-out infarct.

Focal tumefactive demyelinating lesions (TDL) may occur in isolation or, more commonly, in patients with established MS (concentric sclerosis of Baló). TDL may represent an intermediate position between MS and ADEM (p.182).⁴⁶ TDLs tend to be symmetric. TDLs may enhance, and show perilesional edema (but less than MS) and thus be mistaken for neoplasms. Biopsy results may be confusing. MRS may not be able to differentiate from neoplasm.⁴⁷

CSF

CSF analysis can support the diagnosis in some cases, but cannot document dissemination of lesions in time or space. The CSF in MS is clear and colorless. The OP is normal. Total CSF protein is <55 mg/dl in $\approx 75\%$ of patients, and <108 mg/dl in 99.7% (values near 100 should prompt a search for an alternative diagnosis). The WBC count is ≤ 5 cells/mcl in 70% of patients, and only 1% have a count >20 cells/mcl (high values may be seen in the acute myelitis).

In $\approx 90\%$ of patients with MS, CSF-IgG is increased relative to other CSF proteins, and a characteristic pattern occurs. Agarose gel electrophoresis shows a few IgG bands in the gamma region (oligoclonal bands) that are not present in the serum (higher resolution isoelectric focusing can demonstrate 10–15 bands). CSF-OCB are not specific for MS, and can occur in CNS infections and less commonly with strokes or tumors. The predictive value of the absence of IgG in a patient with suspected MS has not been satisfactorily elucidated.

Table 10.3 2010 McDonald MS Diagnostic Criteria ³⁶		
The diagnosis of MS requires elimination of more likely diagnoses and demonstration of lesions disseminated in space (DIS) and time (DIT)		
Clinical (attacks)	Lesions	Additional criteria to make the diagnosis
≥ 2	Objective clinical evidence of ≥ 2 lesions or objective clinical evidence of 1 lesion with reasonable historical evidence of a prior attack	None. If additional tests are done, results must still be consistent with MS
≥ 2	Objective clinical evidence of 1 lesion	DIS; or wait for further clinical attack implicating a different CNS location
1	Objective clinical evidence of ≥ 2 lesions	DIT; or wait for a second clinical attack
1	Objective clinical evidence of 1 lesion	DIS or wait for further clinical attack implicating a different CNS location and DIT; or wait for a second clinical attack
0 (progression from onset)		One year of disease progression (retrospective or prospective) and at least 2 of: • DIS in the brain based on ≥ 1 T2 MRI lesion in periventricular, juxtacortical or infratentorial regions • DIS in the spinal cord based on ≥ 2 T2 MRI lesions • or positive CSF
Paraclinical evidence in diagnosis of MS		
Evidence for DIS ³⁷	≥ 1 T2 MRI lesion in at least 2 out of 4 areas of the CNS: periventricular, juxtacortical, infratentorial or spinal cord • Gadolinium enhancement of lesions is not required • If the patient has a brainstem or spinal cord syndrome, the symptomatic lesions are excluded and do not contribute to lesion count	
Evidence for DIT ³⁸	• New T2 and/or gadolinium-enhancing MRI lesion(s) on follow-up MRI, with reference to a baseline study, irrespective of the timing of the baseline MRI or • Simultaneous presence of asymptomatic gadolinium-enhancing and non-enhancing lesions at any time	
Evidence for positive CSF	Oligoclonal bands in CSF (and not serum) or elevated IgG index	

Recommended criteria have been published,⁴⁸ most of which pertain to specifics of laboratory analysis, pertinent clinical excerpts are shown in □ Table 10.4.

10.5 Acute disseminated encephalomyelitis

AKA ADEM. Acute demyelinating condition, has been associated with relatively recent history of vaccination. Like MS, may also demonstrate oligoclonal bands in CSF. ADEM is generally monophasic, and lesions occur within a couple of weeks. There is usually a good response to high-dose IV corticosteroids.

10.6 Motor neuron diseases

10.6.1 General information

Degenerative diseases of motor neurons. See also comparison of upper motor neuron (UMN) with lower motor neuron (LMN) (p.504) and the paralysis they produce. Five subtypes of which ALS is the most common (see below).

Table 10.4 CSF criteria for MS	
1.	qualitative assessment of IgG is the most informative analysis and is best performed using IEF with some form of immunodetection (blotting or fixation)
2.	analysis should be performed on unconcentrated CSF and must be compared to simultaneously run serum sample in the same assay
3.	runs should use the same amount of IgG from serum and CSF
4.	each run should contain positive and negative controls
5.	quantitative analysis should be made in terms of one of the 5 recognized staining patterns for OCB
6.	an individual experienced in the techniques should report the results
7.	all other tests performed on the CSF (including WBC, protein & glucose, lactate) should be taken into consideration
8.	evaluation using light chains for immunodetection may be helpful in certain cases to resolve equivocal oligoclonal IgG patterns
9.	if clinical suspicion is high but CSF results are equivocal, negative or show only a single band, consider repeating the IP
10.	to measure IgG levels, nonlinear formulas that consider integrity of the blood-CSF barrier should be used (e. g. the ratio of CSF to serum albumin (AKA Qalb) is a measure of leakiness)
11.	labs analyzing CSF should have internal as well as external quality assessment controls
12.	quantitative IgG is a complementary test, but is not a substitute for qualitative IgG testing which has the highest sensitivity and specificity

Three patterns of involvement:

1. mixed UMN & LMN degeneration: amyotrophic lateral sclerosis (ALS) (see below). The most common of the motor neuron diseases
2. UMN degeneration: primary lateral sclerosis. Rare, onset after age 50. No LMN signs. Slower progression than ALS (yrs to decades). Pseudobulbar palsy is common.⁴⁹ Usually does not shorten longevity. May present with falling due to balance problems or low back and neck pain due to axial muscle weakness
3. LMN degeneration: progressive muscular atrophy (PMA) and spinal muscular atrophy (SMA)

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10.6.2 Amyotrophic lateral sclerosis

Key concepts

- degeneration of anterior horn cells and corticospinal tracts in the cervical spine and medulla (bulb) of unknown etiology
- a mixed upper and lower motor neuron disease (UMN → mild spasticity in IEs; LMN → atrophy and fasciculations in UEs)
- clinically: progressive muscle wasting, weakness, and fasciculations
- no cognitive, sensory, nor autonomic dysfunction

In the U.S. amyotrophic lateral sclerosis (ALS) is AKA Lou Gehrig’s disease named after the New York Yankee first baseman who announced that he had the disease in 1939. AKA motor neuron disease (singular).

Epidemiology

See reference.³⁰
Prevalence: 4–6/100,000. Incidence: 0.8–1.2/100,000.
Familial in 8–10% of cases. Familial cases usually follow autosomal dominant inheritance, but occasionally demonstrate a recessive pattern.
Onset usually after 40 years of age.

Pathology

Etiology is not known with certainty. Histology: degeneration of anterior horn alpha-motoneurons (in the spinal cord and in brain stem motor nuclei) (LMNs) and corticospinal tracts (UMNs). Produces mixed UMN & LMN findings, with a great deal of variability depending on which predominates at any given time.

Clinical

Characterized by progressive muscle wasting, weakness, and fasciculations. Involvement is of voluntary muscles, sparing the voluntary eye muscles and urinary sphincter. Classically, presents initially with weakness and atrophy of the hands (lower motor neuron) with spasticity and hyperreflexia of the lower extremities (upper motor neuron). However, LEs may be hyporeflexic if the lower motor neuron deficits predominate. Dysarthria and dysphagia are caused by a combination of upper and lower motor neuron pathology. Tongue atrophy and fasciculations may also occur. Although cognitive deficits are generally considered to be absent in ALS, in actuality 1–2% of cases are associated with dementia, and cognitive changes may occasionally predate the usual features of ALS.⁵⁰

Differential diagnosis

At times, it may be very difficult to distinguish ALS from cervical spondylotic myelopathy; see discussion of differentiating features (p.1086).

Diagnostic studies

EMG
Not absolutely necessary to make diagnosis in most cases. Fibrillations and positive sharp waves are found in advanced cases (may be absent early, especially if upper motor neuron pathology predominates). LMN findings in the LE in the absence of lumbar spine disease, or fibrillation potentials in the tongue are suggestive of ALS.

LP (CSF)
May have slightly elevated protein.

10 Treatment

Much of care is directed towards minimizing disability:

1. risk of aspiration may be reduced with
 - a) tracheostomy
 - b) gastrostomy tube to allow continued feeding
 - c) vocal cord injection with Teflon
2. noninvasive ventilation: e.g. BiPAP spasticity that occurs when upper motor neuron deficits predominate may be treated (usually with short-lived response) with:
 - a) baclofen (p.1530): also may relieve the commonly occurring cramps
 - b) diazepam
3. riluzole (Rilutek®): inhibits presynaptic release of glutamate. Doses of 50-200 mg/d increases tracheostomy-free survival at 9 & 12 months, but the improvement is more modest or may be non-existent by ≈ 18 months ^{51,52}

Prognosis

Most patients die within 5 years of onset (median survival: 3–4 yrs). Those with prominent oropharyngeal symptoms may have a shorter life-span usually due to complications of aspiration.

10.7 Guillain-Barré syndrome

10.7.1 General

Key concepts

- acute onset of peripheral neuropathy with progressive muscle weakness (more severe proximally) with areflexia, reaches maximum over 3 days to 3 weeks
- cranial neuropathy: also common, may include facial diplegia, ophthalmoplegia
- little or no sensory involvement (paresthesias are not uncommon)
- onset often 3 days-5 weeks following viral URI, immunization, Campylobacter jejuni enteritis, or surgery
- pathology: focal segmental demyelination with endoneurial monocytic infiltrate
- elevated CSF protein without pleocytosis (albuminocytologic dissociation)

Guillain-Barré syndrome (GBS) AKA acute polyradiculoneuritis, among others, is actually a collection of syndromes having inflammatory polyradiculoneuropathy in common. Its most frequent form is acute inflammatory demyelinating polyradiculoneuropathy (AIDP). First described as an ascending paralysis, most forms are characterized by symmetric weakness and areflexia. Mild cases may present only with ataxia, whereas fulminant cases may ascend to complete tetraplegia with paralysis of respiratory muscles and cranial nerves. There are also a number of variants (p.186).

GBS is the most common acquired demyelinating neuropathy. Incidence is $\approx 1\text{--}3/100,000$. The lifetime risk for any one individual getting GBS is $\approx 1/1,000$.

GBS is triggered by both humoral and cell mediated autoimmune response to an immune sensitizing event. Frequent (but not essential) antecedents: viral infection, surgery, immunization, mycoplasma infection, enteric infection with *Campylobacter jejuni* (≈ 4 days of intense diarrhea). Higher frequency in the following conditions than in general population: Hodgkin's disease, lymphoma, lupus.

Most cases involve antibodies to gangliosides and glycolipids in peripheral myelin (axon antibodies occur in some forms). For unknown reasons serum creatine kinase can be mildly elevated, and may correlate with muscle type pain.⁵³

10.7.2 Diagnostic criteria

See reference.⁵⁴

features required for diagnosis:

- progressive motor weakness of more than 1 limb (from minimal weakness $\pm$ ataxia to paralysis, may include bulbar or facial or EOM palsy). Unlike most neuropathies, proximal muscles are affected more than distal
- areflexia (usually universal, but distal areflexia with definite hyporeflexia of biceps and knee jerks suffices if other features consistent)

features strongly supportive of diagnosis:

- clinical features (in order of importance)
 - progression: motor weakness peaks at 2 wks in 50% by 3 wks in 80% and by 4 wks in $>90\%$
 - relative symmetry
 - mild sensory symptoms/signs (e.g. mild paresthesias in hands or feet)
 - cranial nerve involvement: facial weakness in 50% usually bilateral. GBS presents initially in EOMs or other Cr. N. in $<5\%$ of cases. Oropharyngeal muscles may be affected
 - recovery usually by 2–4 wks after progression stops, may be delayed by months (most patients recover functionally)
 - autonomic dysfunction (may fluctuate): tachycardia and other arrhythmias, postural hypotension, HTN, vasomotor symptoms
 - afebrile at onset of neuritic symptoms
 - variants (not ranked):
 - fever at onset of neuritic symptoms
 - severe sensory loss with pain
 - progression >4 wks
 - cessation of progression without recovery
 - sphincter dysfunction (usually spared): e.g. bladder paralysis
 - CNS involvement (controversial): e.g. ataxia, dysarthria, Babinski signs
- CSF: albuminocytologic dissociation ($\uparrow$ protein without pleocytosis)
 - protein: elevated after 1 wk of symptoms, >55 mg/dl
 - cells: 10 or fewer mononuclear leukocytes/ml
 - variants
 - no CSF protein rise 1–10 wks after onset (rare)
 - 11–50 monocytes/ml
 - electrodiagnostics: 80% have NCV slowing or block at some time (may take several weeks in some). NCV usually $<60\%$ of normal, but not in all nerves

features casting doubt on diagnosis:

- marked, persistent, asymmetry of weakness
- persistent bowel or bladder dysfunction
- >50 monocytes/ml CSF
- PMNs in CSF
- sharp sensory level

features of conditions in the differential diagnosis (see below)

10.7.3 Guillain-Barré variants

General information

A number of variants have been described (some may simply be incomplete forms of typical Guillain-Barré). Autonomic dysfunction may occur in some.

Miller-Fisher variant of GBS

Ataxia, areflexia and ophthalmoplegia. May also have ptosis. 5% of cases of GBS. Serum marker: anti-GQ1b antibodies.

Acute motor axonal neuropathy (AMAN)

This variant and AIDP are the most common to follow *Campylobacter jejuni* enteritis.

Pharyngeal-cervical-brachial variant

Facial, oropharyngeal, cervical, and UE weakness, sparing the LEs.

Pure sensory variant

Sensory loss accompanied by areflexia.

Atypical GBS

May be accompanied by rhabdomyolysis.⁵⁵

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10.7.4 Differential diagnosis

Also see conditions in the differential diagnosis under Myelopathy (p. 1407)

1. Guillain-Barré syndrome (including one of its variants)
2. critical illness polyneuropathy (p. 542): EMG: ↓ CMAP & SNAP
3. current hexacarbon abuse: volatile solvents (n-hexane, methyl n-butyl ketone), glue sniffing
4. acute intermittent porphyria (AIP): a disorder of porphyrin metabolism. CSF protein is not elevated in AIP. Recurrent painful abdominal crises are common. Check urine delta-aminolevulinic acid or porphobilinogen
5. recent diphtheritic infection: diphtheritic polyneuropathy has a longer latency and a slower crescendo of symptoms
6. lead neuropathy: UE weakness with wrist drop. May be asymmetrical
7. poliomyelitis: usually asymmetric, has meningeal irritation
8. hypophosphatemia (may occur in chronic IV hyperalimentation)
9. botulism: difficult to distinguish clinically from GBS. Normal NCV and a facilitating response to repetitive nerve stimulation on electrodiagnostics
10. toxic neuropathy (e.g. from nitrofurantoin, dapsone, thallium or arsenic)
11. tick paralysis: may cause an ascending motor neuropathy without sensory impairment. Careful examination of the scalp for tick(s)
12. chronic immune demyelinating polyradiculoneuropathy (CIDP) AKA chronic relapsing GBS, chronic relapsing polyneuritis.⁵⁶ ICD9: 357.81 chronic inflammatory demyelinating polyneuritis. Similar to GBS, but long time course (symptoms must be present >2 mos). CIDP produces progressive, symmetrical, proximal & distal weakness, depression of muscle stretch reflexes, and variable sensory loss. Cranial nerves are usually spared (facial muscles may be involved). Balance difficulties are common. Need for respiratory support is rare. Peak incidence: age 40–60 yrs. Electrodiagnostics and nerve biopsy findings are indicative of demyelination. CSF findings are similar to GBS (see above). Most respond to immunosuppressive therapy (especially prednisolone & plasmapheresis) but relapses are common. Refractory cases may be treated with IV gamma-globulin, cyclosporin-A,⁵⁷ total body lymphoid irradiation or interferon- α ⁵⁸
13. critical illness myopathy: ICD9: 359.81 chronic illness myopathy. Muscles not excitable with direct stimulation. EMG: low or normal CMAP with normal SNAP. Muscle biopsy: abnormalities may range from Type II fiber atrophy to necrosis (severe necrosis may not recover)
14. motor neuron disease (p. 182): AKA ALS. Hyperreflexia in LEs
15. myasthenia gravis: weakness worsens towards the end of the day and with repeat efforts. Positive assay for circulating anti-acetylcholine receptor antibodies
16. spinal cord injury

10.7.5 Imaging

No characteristic finding, however, diffuse enhancement of cauda equina and nerve roots occurs in up to 95% of cases.⁵⁹ Thought to be due to disruption of the blood-nerve barrier from inflammation. Conspicuous nerve root enhancement correlates with pain, GBS disability grade, and duration of recovery.⁵⁹

10.7.6 Treatment

Immunoglobulins may be helpful. In severe cases, early plasmapheresis hastens the recovery and reduces the residual deficit. Its role in mild cases is uncertain. Steroids are not helpful.⁶⁰ Mechanical ventilation and measures to prevent aspiration are used as appropriate. In cases of facial diplegia, the eyes must be protected from exposure (neuromuscular) keratitis.

10.7.7 Outcome

Recovery may not be complete for several months. 35% of untreated patients have residual weakness and atrophy. Recurrence of GBS after achieving maximal recovery occurs in $\approx 2\%$

10.8 Myelitis

10.8.1 General information

AKA acute transverse myelitis (ATM). The terminology is confusing: myelitis overlaps with “myelopathy.” Both are pathologic conditions of the spinal cord. Myelitis indicates inflammation, and etiologies include: infectious/post-infectious, autoimmune, and idiopathic. Myelopathy is generally reserved for compressive, toxic, or metabolic etiologies⁶¹; see also differential diagnosis (p. 1407).

10.8.2 Etiology

Many so-called “causes” remain unproven. Immunologic response against the CNS (most likely via cell mediated component) is the probable common mechanism. Animal model: experimental allergic encephalomyelitis (requires myelin basic protein of CNS, not peripheral).

Generally accepted etiologies include (items with an asterisk * may be more properly associated with myelopathy rather than myelitis):

1. infectious and post-infectious
 - a) primary infectious myelitis
 - viral: poliomyelitis, myelitis with viral encephalomyelitis, herpes zoster, rabies
 - bacterial: including tuberculoma of spinal cord
 - spirochetal: AKA syphilitic myelitis. Causes syphilitic endarteritis
 - fungal (aspergillosis, blastomycosis, cryptococcosis)
 - parasitic (Echinococcus, cysticercosis, paragonimiasis, schistosomiasis)
 - b) post-infectious: including post-exanthematous, influenza
2. post-traumatic
3. physical agents
 - a) decompression sickness (dysbarism)
 - b) electrical injury*
 - c) post-irradiation
4. paraneoplastic syndrome (remote effect of cancer): most common primary is lung, but prostate, ovary and rectum have also been described⁶²
5. metabolic
 - a) diabetes mellitus*
 - b) pernicious anemia*
 - c) chronic liver disease*
6. toxins
 - a) cresyl phosphates*
 - b) intra-arterial contrast agents*
 - c) spinal anesthetics
 - d) myelographic contrast agents
 - e) following chemonucleolysis⁶³
7. arachnoiditis
8. autoimmune
 - a) multiple sclerosis (MS), especially Devic syndrome (p. 1409)
 - b) following vaccination (smallpox, rabies)

9. collagen vascular disease
- a) systemic lupus erythematosus
- b) mixed connective tissue disease

10.8.3 Clinical

Presentation

34 patients with ATM⁶⁴: age of onset ranged 15–55 yrs, with 66% occurring in 3rd and 4th decade. 12 patients (35%) had a viral-like prodrome. Presenting symptoms are shown in □ Table 10.5, with other presenting symptoms of unspecified frequency including⁶⁵: fever and rash.

Presenting level

The levels at presentation in 62 patients with ATM are shown in □ Table 10.6.⁶⁵ The thoracic level is the most common sensory level. Infrequently, ATM is the presenting symptom of MS (≈ 3–6% of patients with ATM develop MS).

Progression

Progression is usually rapid, with 66% reaching maximal deficit by 24 hrs, however the interval between first symptom and maximal deficit varies from 2 hrs-14 days.⁶⁵ Findings at the time of maximal deficit are shown in □ Table 10.7.

10.8.4 Evaluation

Imaging should be done to rule-out a compressive lesion. Myelogram, CT & MRI: no characteristic finding. One paper reports 2 patients with fusiform cord enlargement.⁶⁷ MRI may be able to demonstrate the area of involvement within the cord. MRI may show the “central dot sign”,⁶⁸ an area of high signal on axial T2WI usually centrally located with a small dot of isointense signal in the core of the hyperintensity.

CSF: normal during acute phase in 38% of LPs. Remainder (62%) had elevated protein (usually >40 mg%) or pleocytosis (lymphocytes, PMNs, or both) or both.

Table 10.5 Presenting symptoms in myelitis		
Symptom	Series A ^a	Series B ^b
pain (back or radicular)	35%	35%
muscle weakness	32%	13%
sensory deficit or paresthesias	26%	46%
sphincter disturbance	12%	6%
^a series A: 34 patients with ATM ⁶⁴		
^b series B: 52 patients with acute or subacute transverse myelitis ⁶⁶		

Table 10.6 Level of sensory deficit	
Level	%
cervical	8%
high thoracic	36%
low thoracic	32%
lumbar	8%
unknown	16%

Table 10.7 Symptoms at time of maximal deficit (62 patients with ATM ⁶⁵)	
Symptom	%
sensory deficit or paresthesias	100%
muscle weakness	97%
sphincter disturbance (hesitancy, retention, overflow incontinence)	94%
pain in back, abdomen, or limbs	34%
fever	27%
nuchal rigidity	13%

Evaluation scheme

In a patient developing acute myelopathy/paraplegia, especially when ATM is considered likely, the first test of choice is an emergency MRI. If not readily available, a myelogram (with CT to follow) directed at the region of the sensory level is performed (CSF may be sent in this circumstance once block is ruled out).

10.8.5 Treatment

- No treatment has been studied in a randomized controlled trial.
1. steroids: not beneficial for all causes of myelitis,⁶⁹ especially with ASIA A (complete loss of spinal cord function).⁷⁰ Rx: high-dose IV methylprednisolone 3-5 days (doses quoted include 500 mg/d, and 1000 mg/d⁷¹). The decision to introduce additional treatment measures is based on the response to steroids and the MRI appearance after ≈ 5 days of steroids
 2. plasma exchange (PLEX) for patients that do not respond to steroids within 3-5 days
 3. other forms of immune suppression may be attempted for failure of above therapies, including: cyclophosphamide (usually under the direction of an oncologist)
 4. in cases of focal spinal cord enlargement, surgical decompression may be considered in cases that fail to respond to the above

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10.8.6 Prognosis

In a series of 34 ATM patients with ≥5 yrs follow-up (F/U)⁶⁴: 9 patients (26%) had good recovery (ambulate well, mild urinary symptoms, minimal sensory and UMN signs); 9 (26%) had fair recovery (functional gait with some degree of spasticity, urinary urgency, obvious sensory signs, paraparesis); 11 (32%) poor (paraplegic, absent sphincter control); 5 (15%) died within 4 mos of illness. 18 patients (62%of survivors) became ambulatory (in these cases, all could walk with support by 3–6 mos).
In a series of 59 patients⁶⁵ (F/U period unspecified): 22 (37%) had good recovery; 14 (24%) poor; 3 died in acute stage (respiratory insufficiency in 2, sepsis in 1). Recovery occurred between 4 weeks and 3 mos after onset (no improvement occurred after 3 mos).

10.9 Neurosarcoidosis

10.9.1 General

Key concepts

- neurologic involvement of sarcoidosis (a systemic granulomatous disease)
- may produce multiple cranial nerve palsies
- the most common neurologic manifestation is diabetes insipidus
- corticosteroids are beneficial for systemic as well as neurologic involvement

Sarcoidosis is a granulomatous disease that is usually systemic, and may include the CNS (so-called neurosarcoidosis AKA neurosarcoid). Only 1–3%of cases have CNS findings without systemic manifestations.⁷² The cause of the disease is unknown. An exaggerated cellular immune response for

unknown reasons is the currently favored hypothesis. Organs commonly involved include lungs, skin, lymph nodes, bones, eyes, muscles, and parotid glands.³⁰

10.9.2 Pathology

CNS sarcoidosis primarily involves the leptomeninges, however parenchymal invasion often occurs. Adhesive arachnoiditis with nodule formation may also occur (nodules have a predilection for the posterior fossa). Diffuse meningitis or meningoencephalitis may occur, and may be most pronounced at the base of the brain (basal meningitis) and in the subependymal region of the third ventricle (including the hypothalamus).

Constant microscopic features of neurosarcoidosis include noncaseating granulomas with lymphocytic infiltrates. Langhans giant cells may or may not be present.

10.9.3 Epidemiology

Incidence of sarcoidosis is $\approx 3\text{--}50$ cases/100,000 population; neurosarcoidosis occurs in $\approx 5\%$ of cases (reported range: 1–27%). In one series, the median age of onset of neurologic symptoms was 44 years.

10.9.4 Clinical findings

Clinical findings include multiple cranial nerve palsies in 50–70% (particularly facial n., including diplegia), peripheral neuropathy, and myopathy.⁷³ Occasionally the lesions may produce mass effect,⁷⁴ and hydrocephalus may result from adhesive basal arachnoiditis. Patients may have low grade fever. Intracranial hypertension is common and may be dangerous. Hypothalamic involvement may produce disorders of ADH (diabetes insipidus, disordered thirst). Rare involvement of the pituitary may produce pituitary insufficiency. Seizures occur in 15%

0.4% of patients with sarcoidosis develop spinal cord involvement,⁷⁵ and in 16% of these, the spinal cord was the only identifiable site of involvement.

10.9.5 Laboratory

CBC: mild leukocytosis and eosinophilia may occur.

Serum angiotensin-converting enzyme (ACE): abnormally elevated in 83% of patients with active pulmonary sarcoidosis, but in only 11% with inactive disease.⁷⁶ False positive rate: 2–3% may also be elevated in primary biliary cirrhosis.

CSF: similar to any subacute meningitis: elevated pressure, mild pleocytosis (10–200 cells/mm³) mostly lymphocytes, elevated protein (up to 2,000 mg/dl), mild hypoglycorrhachia (15–40 mg/dl), CSF ACE is elevated in $\approx 55\%$ of cases with neurosarcoidosis (normal in patients with sarcoidosis not involving the CNS).⁷⁷ No organisms are recovered on culture or gram stain.

10.9.6 Imaging

CXR

Usually demonstrates characteristic findings of sarcoidosis (hilar adenopathy, mediastinal lymph nodes...).

MRI

Gadolinium enhancement of the leptomeninges and/or optic nerve may be the only abnormal finding(s). Meningeal enhancement was seen in 38% of neurosarcoidosis patients.⁷⁸ Lesions may be solitary or multiple, and may be located intra- or extraparenchymal, periventricular, and/or in basal cisterns. Lesions may be seen on FLAIR that would otherwise have been missed. Hydrocephalus may occur.

Gallium scan

Nuclear medicine scan with ⁶⁷Ga citrate (p.236). Described findings include:

1. Panda sign⁷⁹: uptake in lacrimal glands, parotid glands & nasopharynx (normal). Not specific for sarcoidosis
2. lambda distribution⁸⁰: uptake in hilar lymph nodes
3. leopard man sign⁸¹: diffuse dappled pattern due to uptake in soft tissues, skin, muscles, mediastinum, and lacrimal glands

Table 10.8 Differential diagnosis of neurosarcoidosis

- 1. Hodgkin’s disease
- 2. chronic granulomatous meningitis:
 - a) Hansen’s disease (leprosy)
 - b) syphilis
 - c) cryptococcosis
 - d) tuberculosis
- 3. multiple sclerosis
- 4. CNS lymphoma
- 5. pseudotumor cerebri
- 6. granulomatous angiitis

10.9.7 Differential diagnosis

Differentiating granulomatous angiitis (GA) from neurosarcoidosis that involves only the CNS can be done on histologic criteria: the inflammatory reaction in sarcoidosis is not limited to the region immediately surrounding blood vessels as it is in GA, where extensive disruption of the vessel wall may occur.

10.9.8 Diagnosis

Making the diagnosis is relatively easy when systemic involvement occurs: characteristic findings on CXR, biopsy of skin or liver nodules, muscle biopsy, serum ACE assay.
Isolated neurosarcoidosis may be more difficult to diagnose, and may require biopsy (see below).

10.9.9 Biopsy

In uncertain cases, biopsy may be indicated. Whenever possible, MRI should be used to localize a supratentorial region of involvement. If a mass lesion cannot be biopsied, a meningeal biopsy may be done and should include all layers of meninges and cerebral cortex. Cultures and stains for fungus and acid-fast bacteria (TB) should be performed in addition to microscopic examination.

10.9.10 Treatment

Antibiotics have not been proven to be of benefit. Immunosuppression primarily with corticosteroids are beneficial for systemic as well as neurologic involvement. Therapy may be initiated with prednisone 60 mg PO qd in adults, and tapered based on response. Therapy with cyclosporine may allow a reduction in steroid dosage in refractory cases.⁸² Treatment for unresponsive cases include: methotrexate, cytoxan, cyclophosphamide, azathioprine, low dose XRT. CSF shunting is indicated if hydrocephalus develops.

10.9.11 Prognosis

Usually a benign disease. Peripheral and cranial nerve palsies recover slowly.

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11 Neurovascular Disorders and Neurotoxicology

11.1 Posterior reversible encephalopathy syndrome (PRES)

11.1.1 General information

AKA reversible posterior leukoencephalopathy syndrome (RPLS). A group of encephalopathies with characteristic pattern of widespread vasogenic brain edema seen on CT or MRI with some predominance in the parietal and occipital regions.¹ The most common PRES pattern involves watershed zones with involvement of the cortex, subcortical and deep white matter to a variable extent.¹ A small number of patients with PRES will go on to infarction.

Patients may present with headache, seizures, mental status changes and focal neurologic deficit. Intracerebral hemorrhage (ICH) and SAH may occur in up to 15%¹

11.1.2 Associated findings and conditions

Includes:

1. hypertensive encephalopathy: commonly seen in the setting of subacute blood pressure elevations (as may occur with malignant hypertension). Imaging studies show symmetric confluent lesions with mild mass effect and patchy enhancement primarily in the subcortical white matter of the occipital lobes² which may produce cortical blindness
 - a) moderate to severe hypertension is seen in $\approx 75\%$ of patients with PRES¹ although the upper limits of autoregulation are often not reached
 - b) in addition to hemispheric patterns of edema isolated brain stem and cerebellar edema have been described. Posterior fossa edema has been reported to cause obstructive hydrocephalus in a severe case³
2. preeclampsia/eclampsia associated with cerebral edema.⁴ The condition is often temporary, but (permanent) infarctions also occur. Restricted diffusion on MR imaging is seen in 11–26% of cases. Abnormal DWI areas on MRI may be associated with a worse prognosis⁵
 - a) may present (e.g. with blindness) during pregnancy complicated by preeclampsia or eclampsia⁶
 - b) may develop 4–9 days post-partum and may be associated with vasospasm⁷ even in patients not meeting clinical criteria for the diagnosis of eclampsia
 - c) toxemia is attributed to the placenta. Delivery and removal of the placenta is felt to be curative⁸
3. infection, sepsis and shock: blood pressure was normal in 40% (edema was greater in the normotensive patients). Gram positive organisms predominate⁹
4. autoimmune disease: PRES has been described in patients with lupus, scleroderma, Wegener's granulomatosis and polyarteritis nodosa.¹ These patients often receive regimens of immunosuppressive medications (tacrolimus, cyclosporine), which have also been linked to cases of PRES
5. cancer chemotherapy: PRES occurs in patients receiving multi-drug high dose chemotherapy most commonly for hematopoietic malignancies
6. transplantation: PRES has been reported both with bone marrow and solid organ transplantation
 - a) incidence: 3–16% with bone marrow transplantation depending on the preconditioning regimen and whether or not it is myeloablative¹
 - b) highest incidence in the first month following allogeneic bone marrow transplant¹
 - c) lower incidence following solid organ transplants. Occurs earlier following liver transplantation, usually within 2 months. Occurs later in renal transplants¹
7. cyclosporine post-transplant neurotoxicity⁹

11.1.3 Treatment

Disordered autoregulation mandates tight control of blood pressure to reduce the risk of ICH. The underlying cause needs to be addressed (i.e. control HTN, hold immunosuppressives or chemotherapeutics, delivery of the placenta, etc.).

11.2 Crossed cerebellar diaschisis

Hypometabolism of cerebellar cortex contralateral to a cerebral hemispheric lesion (lesions include: stroke, brain tumor...). Lesions in the motor cortex, anterior corona radiata, and thalamus produce the most marked suppression of metabolism. Theory: hypometabolism is due to disconnection of

cerebro-ponto-cerebellar pathways → decreased oxygen and glucose consumption → decreased CO₂ production → local arterial constriction (down-regulation of cerebellar blood flow).

11.3 Vasculitis and vasculopathy

11.3.1 General information

The vasculitides are a group of disorders characterized by inflammation and necrosis of blood vessels. Vasculitis may be primary or secondary. Those that may affect the CNS are listed in □ Table 11.1, all of these cause tissue ischemia (even after the inflammation is quiescent) that may range in effect from neuropraxia to infarction.

11.3.2 Giant cell arteritis (GCA)

Key concepts

- formerly often referred to as temporal arteritis
- a chronic vasculitis of large and medium caliber vessels, primarily involving cranial branches of the arteries arising from the aortic arch
- age >50 years; affects women twice as often as men
- important possible late complications: blindness, stroke, thoracic aortic aneurysms and aortic dissections
- temporal artery biopsy is recommended for all patients suspected of GCA
- corticosteroids are the drug of choice for treatment

AKA temporal arteritis (TA), AKA cranial arteritis. A chronic granulomatous arteritis of unknown etiology involving primarily the cranial branches of the aortic arch (especially the external carotid artery (ECA)),¹¹ which if untreated, may lead to blindness. Takayasu’s arteritis is similar to GCA, but tends to affect large arteries in young women; it has 2 phases: inflammatory (treated with corticosteroids) and stenotic (treated with arterial bypasses).

11

Epidemiology

Seen almost exclusively in Caucasians >50 yrs age (mean age of onset is 70). Incidence: 17.8 per 100,000 people ≥50 years old¹² (range: 0.49–23). Prevalence: ≈ 223 (autopsy incidence may be much

Table 11.1 Vasculitides that may affect the CNS ¹⁰						
Vasculitis	Frequency of neuro involvement	Type of CNS involvement ^a				
		Acute encephalopathy	Seizure	Cranial nerve	Spinal cord	ICH or SAH
periarteritis nodosa ^b (PAN) ^c	20–40%	++	++	+	+	+
hypersensitivity vasculitis ^b	10%	+	+	0	0	+
giant cell (temporal) arteritis ^b	10%	+	0	++	0	0
Takayasu’s arteritis	10–36%	+	++	++	+	+
Wegener’s granulomatosis ^b	23–50%	+	++	++	+	+
lymphomatoid granulomatosis ^b	20–30%	++	+	++	+	0
isolated angiitis of the CNS ^b	100%	++	+	++	++	+
Behçet’s disease ^b	10–29%	++	+	++	+	+
^a KEY: 0 = uncommon or unreported; + = not uncommon; ++ = common; ICH = intracerebral hemorrhage; SAH = subarachnoid hemorrhage						
^b see section that follows for these topics						
^c PAN: a group of disorders, frequencies may vary by subgroup						

higher).¹³ More common in northern latitudes and among individuals of Scandinavian descent suggesting a genetic and environmental causes.¹¹ Female:male ratio is $\approx$ 2:1 (reported range: 1.05–7.4:1). 50%of GCA patients also have polymyalgia rheumatica (PMR) (p.198).

Pathology

Discontinuous (so-called “skip lesions”) inflammatory reaction of lymphocytes, plasma cells, macrophages, $\pm$ giant cells (if absent, intimal proliferation may be prominent); predominantly in media of involved arteries. Arteries preferentially involved include the ophthalmic and posterior ciliary branches and the entire distribution of the external carotid system (of which the STA is a terminal branch). Other arteries in the body may be involved (reported involvement of abdominal aorta, femoral, brachial and mesenteric arteries are rarely symptomatic). Unlike PAN, GCA generally spares the renal arteries.

Clinical

Various combinations of symptoms of giant cell arteritis are listed in □ Table 11.2. Onset is usually insidious, although occasionally it may be abrupt.¹⁴

- Details of some findings:
1. H/A: the most common presenting symptom. May be nonspecific or located in one or both temporal areas, forehead, or occiput. May be superficial or burning with paroxysmal lancinating pain
 2. symptoms relating to ECA blood supply (strongly suggestive of GCA, but not pathognomonic¹⁶): jaw claudication, tongue, or pharyngeal muscles
 3. ophthalmologic symptoms: due to arteritis and occlusion of branches of ophthalmic artery or posterior ciliary arteries
 - a) symptoms include: amaurosis fugax (precedes permanent visual loss in 44%), blindness, visual field cuts, diplopia, ptosis, ocular pain, corneal edema, chemosis
 - b) blindness: incidence is $\approx$ 7% and once it occurs, recovery of sight is unlikely
 4. systemic symptoms
 - a) nonspecific constitutional symptoms: fever (may present as FUO in 15%of cases), anorexia, weight loss, fatigue, malaise
 - b) 30%have neurologic manifestations. 14%are neuropathies including mononeuropathies and peripheral polyneuropathies of the arms or legs¹⁷
 - c) musculoskeletal symptoms
 - PMR (p.198) is the most common (occurs in 40%of patients):
 - peripheral arthritis, swelling & pitting edema of hands & feet in 25%
 - arm claudication from stenosis of subclavian and axillary arteries
 - d) thoracic aortic aneurysms: 17 times as likely in GCA. Annual CXRs are adequate for screening
 5. temporal arteries on physical examination may exhibit tenderness, swelling, erythema, reduced pulsations, or nodularity. Normal in 33%
 6. the presence of systemic symptoms correlates with a lower incidence of blindness or stroke

Differential diagnosis

1. periarteritis nodosa (PAN) (p.199)
2. hypersensitivity vasculitis
3. atherosclerotic occlusive disease
4. malignancy: shares systemic also has symptoms of low grade fever, malaise and weight loss
5. infection
6. trigeminal neuralgia (p.479)
7. ophthalmoplegic migraine
8. dental problems

Table 11.2 Signs and symptoms of GCA ^{15,11}		
Frequent (>50% of cases)	Occasional (10–50% of cases)	Rare (<10% of cases)
H/A: 66% temporal artery tenderness	visual symptoms weight loss fever (low grade) proximal myalgias jaw claudication facial pain scalp tenderness	blindness extremity claudication tongue claudication ear pain synovitis stroke angina

Evaluation

Laboratory studies

- 1. ESR>40 mm/hr (usually >50) by Westergren method (if>80 mm/hr with above clinical syndromes, highly suggestive of GCA). ESR is normal in up to 22.5%⁸
- 2. C-reactive protein: another acute phase reactant that is more sensitive than ESR. Has the advantage that it can be performed on frozen sera
- 3. CBC: may show mild normochromic anemia¹⁹
- 4. rheumatoid factor, ANA, and serum complement usually normal
- 5. LFTs abnormal in 30%(usually elevated alkaline phosphatase)
- 6. tests for rheumatoid factor and ANA are usually negative
- 7. temporal artery angiography not helpful (angiography elsewhere indicated if suspicion of large artery involvement exists)
- 8. CT: usually not helpful, one report described calcified areas corresponding to the temporal arteries²⁰
- 9. temporal artery biopsy: see below

Temporal artery biopsy

Sensitivity and specificity are shown in □ Table 11.3.

Indications and timing

Current recommendations: temporal artery biopsy in all patients suspected of having GCA.¹¹ May be controversial. Arguments for: toxicity of a long course of steroids in an elderly patient, and a high rate of false initial responses of other illnesses to steroids. Arguments against: since a negative biopsy cannot exclude the diagnosis, cases with a negative biopsy but a strong clinical suspicion are often treated as though they have GCA.²² In general, however, biopsy is considered prudent before embarking on a long course of high-dose steroid therapy.¹⁶ Complications of biopsy are rare and include bleeding, infection, and only in the setting of active vasculitis has scalp necrosis been reported (not linked to biopsy).

In general, perform biopsy before starting steroids if biopsy can be done immediately.¹¹ Otherwise, start steroids to preserve vision and perform biopsy usually within 1 week (pathologic changes can be seen after more than 2 weeks of therapy,²³ therefore do not withhold steroids to await biopsy).

Technique of temporal artery biopsy

Biopsy side of involvement if laterality exists. The yield is increased by removing a portion of artery that is involved clinically (a tender or inflamed segment).²⁴ Mark the frontal branch of the STA with a skin marker (spare the main trunk and parietal branch if possible). Infiltrate local anesthetic. The incision is made parallel to the artery and if possible behind the hairline. The incision is taken down to the fascia of the temporalis muscle, to which the STA is superficial.²⁵ Optimal length of STA biopsy: 4–6 cm (if an abnormal segment of STA can be palpated, some say that a smaller biopsy to include this area may be sufficient, but this is probably unreliable as the muscle may be tender, etc.). Step-sectioning by pathologist through the entire length of the biopsy specimen also increases the yield.

Frozen sections can be performed. Biopsy of the contralateral side if the first side is negative in cases where clinical suspicion is high increases the yield only by 5–10%

Treatment

No known cure. Steroids can produce symptomatic relief and usually prevent blindness (progression of ocular problems 24–48 hrs after institution of adequate steroids is rare). Totally blind patients or those with longstanding partial visual loss are unlikely to respond to any treatment.

- 1. for most cases:
 - a) start with prednisone , 40–60 mg/d PO divided BID-QID (qod dosing is usually not effective in initial management)
 - b) if no response after 72 hrs, and diagnosis certain, ↑ to 10–25 mg QID

Table 11.3 Temporal artery biopsy	
sensitivity	≈ 90%(reported range ^{15,21} is 9–97%)
specificity	near 100%
predictive value	≈ 94%

- c) once response occurs (usually within 3–7 days), give entire dose as q AM dose for 3–6 weeks until symptoms resolved and ESR normalizes (occurs in 87% of patients within ≈ 4 weeks) or stabilizes at <40 – 50 mm/hr
- d) once quiescent, a gradual taper is performed to prevent exacerbations: reduce by 10 mg/d q 2–4 weeks to 40 mg/d, then by 5 mg/d q 2–4 wks to 20 mg/d, then by 2.5 mg/d q 2–4 wks to 5–7.5 mg/d which is maintained for several months, followed by 1 mg/d decrements q 1–3 mos (usual length of treatment is 6–24 mos; do not D/C steroids when ESR normalizes)
- e) if symptoms recur during treatment, prednisone dose is temporarily increased until symptoms resolve (isolated rise in ESR is not sufficient reason to increase steroids¹¹)
- f) patients should be followed closely for ≈ 2 years
2. in severely ill patients: methylprednisolone, 15–20 mg IV QID
3. anticoagulant therapy: controversial
4. acute blindness (onset within 24–36 hrs) in a patient with giant cell arteritis:
 - a) consider up to 500 mg methylprednisolone IV over 30–60 mins (no controlled studies show reversal of blindness)
 - b) some have used intermittent inhalation of 5% carbon dioxide and oxygen

Outcome

Complications of steroid therapy occur in $\approx 50\%$ of patients. Most are not life threatening, and include vertebral compression fractures in $\approx 36\%$, peptic ulcer disease in $\approx 12\%$, proximal myopathy, cataracts, exacerbation of diabetes; also see Possible deleterious side effects of steroids (p. 146).

30–50% of patients will have spontaneous exacerbations of GCA (especially during the first 2 years) regardless of the corticosteroid regimen.¹¹

Survival parallels that of the general population. Onset of blindness after initiation of steroid therapy is rare.

11.3.3 Polymyalgia rheumatica (PMR)

General information

PMR and giant cell arteritis (GCA) (p. 195) may be different points on a continuum of the same disease. Both have an increased frequency of HLA-DR4 and systemic monocyte activation. 15% of patients with PMR eventually develop GCA.

Epidemiology

See reference.¹¹

Both GCA & PMR occur in people ≥ 50 years old. The incidence increases with age and peaks between 70–80 years and is higher at higher latitudes.¹¹

PMR is more common than GCA. Prevalence: 500/100,000.²⁶ Incidence: 52.5 per 100,000 people $\geq$ age 50, higher in females (61.7) than males (39.9).²⁷

Features

See reference.¹¹

- an inflammatory condition of unknown etiology
- clinical characteristics
 - a) aching and morning stiffness in the cervical region and shoulder & pelvic girdles lasting > 1 month. The pain usually increases with movement
 - shoulder pain: present in 70–95% of patients. Radiates toward elbow
 - hip & neck pain: 50–70% Hip pain radiates towards knees
 - b) age ≥ 50 years
 - c) ESR ≥ 40 mm/hr (7–20% have normal ESR²⁸)
 - d) usually responds rapidly to low dose corticosteroids (≤ 20 mg prednisone/day) see below
 - e) systemic symptoms (present in $\approx 33\%$): fever, malaise or fatigue, anorexia and weight loss
- favorable prognosis: usually remits in 1–3 years

Treatment

PMR responds to either low doses of steroids²⁶ (10–20 mg prednisone/day) or sometimes to NSAIDs (response to steroids is much more rapid). The initial dose of steroids is maintained for 2–4 weeks, and then by $\leq 10\%$ of the daily dose every 1–2 weeks¹¹ while observing for signs of GCA.

11.3.4 Other vasculitides

Periarteritis nodosa

AKA polyarteritis nodosa. Actually a group of necrotizing vasculitides, including:

- classic periarteritis nodosa (PAN): a multisystem disease with inflammatory necrosis, thrombosis (occlusion), and hemorrhage of arteries and arterioles in every organ except lung & spleen. Nodules may be palpated along medium sized muscular arteries. Commonly produces mononeuritis multiplex, weight loss, fever, and tachycardia. Peripheral nerve manifestations are attributed to arteritic occlusion of vasa nervorum. CNS manifestations are uncommon and include H/A, seizures, SAH, retinal hemorrhages, and stroke in $\approx 13\%$
- allergic angiitis and granulomatosis (Churg-Strauss syndrome)
- systemic necrotizing vasculitis

These patients do better when treated with cyclophosphamide rather than steroids.

Wegener's granulomatosis

General information

A systemic necrotizing granulomatous vasculitis involving the respiratory tract (lung $\rightarrow$ cough/hemoptysis, and/or nasal airways $\rightarrow$ serosanguinous nasal drainage $\pm$ septal perforation $\rightarrow$ characteristic "saddle nose deformity") and frequently the kidneys (no reported cases of kidney involvement without respiratory).²⁹

Nasal obstruction and crusting are the usual initial findings. Arthralgia (not true arthritis) is present in $>50\%$

Neurologic involvement usually consists of cranial nerve dysfunction (usually II, III, IV, & VI; less often V, VII, & VIII; and least commonly IX, X, XI, & XII) and peripheral neuropathies, with diabetes insipidus (occasionally preceding other symptoms by up to 9 months). Focal lesions of the brain and spinal cord occur less frequently.

Differential diagnosis

Differential diagnosis includes:

- "lethal midline granuloma" (may be similar or identical to polymorphic reticulosis) may evolve into lymphoma. May cause fulminant local destruction of the nasal tissue. Differentiation is crucial as this condition is treated by radiation; one should avoid immune suppression (e.g. cyclophosphamide). Probably does not involve true granulomas. Renal and tracheal involvement do not occur
- fungal disease: *Sporothrix schenckii* & *Coccidioides* may cause identical syndrome
- other vasculitides: especially Churg-Strauss syndrome (asthma and peripheral eosinophilia usually seen), and PAN (granulomas usually lacking)

Evaluation

Biopsy of upper airways consists of removing all crusts, and obtaining as much friable mucosa as possible. This tissue should be fixed in formaldehyde and examined pathologically within 24 hrs (do not freeze). A sample should also be cultured (including fungal and acid-fast cultures). Renal biopsy should not be done when more specific tissue is available from the upper airway.

Treatment

Untreated, Wegener's granulomatosis is rapidly fatal, with a median survival of 5 months, and more than 90% of patients are dead within 2 years of diagnosis.³⁰ For fulminating disease: prednisone 60–80 mg/d until disease is controlled (documented by decreasing ESR and improvement of serum creatinine).

When disease is stable: cyclophosphamide (Cytoxan®) ≈ 2 mg/kg daily (takes 2–3 weeks to have an effect). Continue 1 year beyond last evidence of active disease. Low-dose weekly methotrexate may be an acceptable alternative to cyclophosphamide in selected patients.³⁰

11.3.5 Lymphomatoid granulomatosis

Rare; affects mainly the lungs, skin (erythematous macules or indurated plaques in 40%) and nervous system (CNS in 20% peripheral neuropathies in 15%). Sinuses, lymph nodes, and spleen are usually spared.

11.3.6 Behçet's syndrome

Relapsing ocular lesions and recurrent oral and genital ulcers, with occasional skin lesions, thrombophlebitis, and arthritis.¹⁰ H/A occur in >50% Neurologic involvement includes pseudotumor, cerebellar ataxia, paraplegia, seizures, and dural sinus thrombosis. Only 5% have neurologic symptoms as the presenting complaint.

86% have CSF pleocytosis and protein elevation. Cerebral angiography is usually normal. CT may show focal areas of enhancing low density.

Steroids usually ameliorate ocular and cerebral symptoms, but usually have no effect on skin and genital lesions. Uncontrolled trials of cytotoxic agents → some benefit. Thalidomide may be effective (uncontrolled studies), but carries risk of serious adverse effects (teratogenicity, peripheral neuropathy...).³¹

Although painful, the disease is usually benign. Neurologic involvement portends a worse prognosis.

11.3.7 Isolated CNS vasculitis

General information

AKA isolated angiitis of the CNS. Rare ($\approx$ 20 cases reported³² as of 1983); limited to vessels of CNS. Small vessel vasculitis is $\approx$ always present → segmental inflammation and necrosis of small leptomeningeal and parenchymal blood vessels with surrounding tissue ischemia or hemorrhage.¹⁰

Presentation

Combinations of H/A, confusion, dementia, and lethargy. Occasionally seizures. Focal and multifocal brain disturbance occurs in >80% Visual symptoms are frequent (secondary either to involvement of choroidal and retinal arteries, or to involvement of visual cortex → visual hallucinations).

Evaluation

ESR & WBC count are usually normal. CSF may be normal or have pleocytosis and/or elevated protein. CT may show enhancing areas of low density.

Angiography (required for diagnosis): characteristically shows multiple areas of symmetrical narrowing ("string of pearls" configuration). If normal, it does not exclude diagnosis.

Histological diagnosis (recommended): all biopsy material should be cultured. Brain parenchyma biopsy infrequently shows vasculitis. Leptomeningeal biopsy invariably shows involvement.

Treatment and outcome

Reportedly fatal if untreated, but may smolder for years.

Rarity of this condition makes treatment uncertain. Recommended: cyclophosphamide (Cytoxan®) 2 mg/kg/d and prednisone 1 mg/kg/d qod therapy.

NB: this condition is thought to be T-cell mediated, but prednisone causes more B-cell suppression, therefore breakthrough during prednisone therapy is not uncommon.

11.3.8 Hypersensitivity vasculitis

Neurologic involvement is not a prominent feature of this group of vasculitides, which include:

- drug induced allergic vasculitis: A number of drugs are associated with the development of cerebral vasculitis. These include methamphetamines ("speed"), cocaine (frank vasculitis occurs³³ but is rare), heroin and ephedrine
- cutaneous vasculitis
- serum sickness: may → encephalopathy, seizures, coma, peripheral neuropathy and brachial plexopathy
- Henoch-Schönlein purpura

11.3.9 Fibromuscular dysplasia

General information

A vasculopathy (angiopathy) affecting primarily branches of the aorta, with renal artery involvement in 85% of cases (the most common site) and commonly associated with hypertension. The disease has an incidence of $\approx$ 1% and results in multifocal arterial constrictions and intervening regions of aneurysmal dilatation.

The second most commonly involved site is the cervical internal carotid (primarily near C1–2), with fibromuscular dysplasia (FMD) appearing on 1% of carotid angiograms, making FMD the second

most common cause of extracranial carotid stenosis.³⁴ Bilateral cervical ICA involvement occurs in $\approx$ 80%of cases. 50%of patients with carotid FMD have renal FMD. Patients with FMD have an increased risk of intracranial aneurysms and neoplasms, and are probably at higher risk of carotid dissection.

Aneurysms and fibromuscular dysplasia: The reported incidence of aneurysms with FMD³⁵ ranges from 20–50%

Etiology

The actual etiology remains unknown, although congenital defects of the media (muscular layer) and internal elastic layer of the arteries has been identified which may predispose the arteries to injury from otherwise well-tolerated trauma. A high familial rate of strokes, HTN, and migraine have supported the suggestion that FMD is an autosomal dominant trait with reduced penetrance in males.³⁶

Presentation

Most patients have recurrent, multiple symptoms shown in □ Table 11.4.

Up to 50%of patients present with episodes of transient cerebral ischemia or infarction. However, FMD may also be an incidental finding and some cases have been followed for 5 years without recurrence of ischemic symptoms suggesting that FMD may be a relatively benign condition.

Headaches are commonly unilateral and may be mistaken for typical migraine. Syncope may be caused by involvement of the carotid sinus.

Horner’s syndrome occurs in $\approx$ 8%of cases. T-wave changes on EKG may be seen in up to one–third of cases, and may be due to involvement of the coronary arteries.

Diagnosis

The “gold-standard” for the diagnosis of FMD is the angiogram. The three angiographic types of FMD³⁷ are shown in □ Table 11.5.

Treatment

Medical therapy including antiplatelet medication (e.g. aspirin) has been recommended.

Direct surgical treatment is problem ridden due to the difficult location (high carotid artery, near the base of the skull), and the friable nature of the vessels making anastamosis or arteriotomy closure difficult.

Transluminal angioplasty has achieved some degree of success. Carotid cavernous fistulas and arterial rupture have been reported as complications.

Table 11.4 Previous symptoms in 37 cases of aortocranial FMD ³⁶	
Symptom	%
H/A	78%
mental distress	48%
tinnitus	38%
vertigo	34%
cardiac arrhythmia	31%
TIA	31%
syncope	31%
carotidynia	21%
epilepsy	15%
hearing impairment	12%
abdominal angina	8%
angina/MI	8%

Table 11.5 Angiographic classification of FMD	
Type	Findings
1	most common (80–100% of reported cases). Multiple, irregularly spaced, concentric narrowings with normal or dilated intervening segments giving rise to the so-called “string of pearls” appearance. Corresponds with arterial medial fibroplasia
2	focal tubular stenosis, seen in $\approx 7\%$ of cases. Less characteristic for FMD than Type 1, and may also be seen in Takayasu’s arteritis and other conditions
3	“atypical FMD.” Rare. May take on various appearances, most commonly consisting of diverticular outpouchings of one wall of the artery

11.3.10 Miscellaneous vasculopathies

CADASIL

Key concepts

- clinical: migraines, dementia, TIAs, psychiatric disturbances
- MRI: white matter abnormalities
- autosomal dominant inheritance
- anticoagulants controversial, generally discouraged

11

An acronym for Cerebral Autosomal Dominant Arteriopathy with Subcortical Infarcts and Leukoencephalopathy.³⁸ A familial disease with onset in early adulthood (mean age at onset: 45 ± 11 yrs), mapped to chromosome 19. Clinical and neuroradiologic features are similar to those seen with multiple subcortical infarcts from HTN, except there is no evidence of HTN. The vasculopathy is distinct from that seen in lipohyalinosis, arteriosclerosis and amyloid angiopathy, and causes thickening of the media (by eosinophilic, granular material) of leptomeningeal and perforating arteries measuring 100–400 μ m in diameter.

Clinical involvement

Recurrent subcortical infarcts (84%), progressive or stepwise dementia (31%), migraine with aura (22%), and depression (20%). All symptomatic and 18% of asymptomatic patients had prominent subcortical white-matter and basal ganglia hyperintensities on T2WI MRI.

Treatment

Warfarin (Coumadin®) is used by some.

11.3.11 Paraneoplastic syndromes affecting the nervous system

General information

Paraneoplastic syndromes (PNS), AKA “remote effects of cancer.” Develop acutely or subacutely. May mimic or be mimicked by metastatic disease. The neurologic disability is usually severe, and may precede other manifestations of the cancer by 6–12 mos. Often one particular neural cell type is predominantly affected. The presence of a PNS may portend a more benign course of the cancer.

16% of patients with lung Ca, and 4% with breast Ca will develop a PNS.

Pathogenesis unknown. Theories: ? toxin; ? competition for essential substrate; ? opportunistic infection; ? auto-immune process.

Types of syndromes

1. affecting cerebrum or cerebellum
- a) encephalitis
 - diffuse
 - limbic and brainstem: usually due to small-cell lung Ca or testicular Ca³⁹ as a result of serum antineuronal antibodies

- b) “limbic encephalitis” (mesial): dementia (decreased memory, psychiatric symptoms, hallucinations)
- c) pan-cerebellar degeneration (PCD) AKA subacute cerebellar degeneration*: (see below)
- d) opsoclonus-myoclonus syndrome*: in peds, usually indicates neuroblastoma
- 2. affecting spinal cord
 - a) poliomyelitis (anterior horn syndrome): mimics ALS (weakness, hyporeflexia, fasciculations)
 - b) subacute necrotizing (transverse) myelitis: rapid necrosis of spinal cord
 - c) ganglionitis* (dorsal root ganglion): chronic or subacute. Pure sensory neuronopathy (not neuropathy)
- 3. affecting peripheral nervous system
 - a) chronic sensory-motor: typical neuropathy (as in DM or EtOH abuse)
 - b) pure sensory (p.1268) ⁴⁰
 - c) pure motor: rare. Almost always due to lymphoma (mostly Hodgkin’s)
 - d) acute inflammatory demyelinating polyradiculopathy, AKA Guillain-Barré (p. 184)
 - e) Eaton-Lambert (EL) myasthenic syndrome*: rare. 66%of patients with this syndrome will have cancer, most common primary is oat cell Ca of lung. Pre-synaptic neuromuscular junction (PSNMJ) blockade due to antibodies against the PSNMJ; NB: true myasthenia gravis (MG) is a post-synaptic block. Worse in AM, improves during day with recruitment (opposite of MG, which is worse at night or with exercise due to depletion). Mostly motor, but often accompanied by paresthesias. MG is affects mostly nicotinic receptors, but EL also affects muscarinic receptors, and therefore autonomic symptoms may occur: dry mouth, males may have impotence. Repetitive nerve stimulation on EMG: for MG use 2–5 Hz stimulation, for EL use > 10 Hz, MG: decremental response with low frequency, with EL there is incremental response (more response with repeat stimulation).
 - f) myasthenia gravis
 - g) polymyositis: in age > 60 yrs, 25%of patients with this have a malignancy*, most often linked to bronchogenic Ca
 - h) type IIb muscle fiber atrophy: the most common paraneoplastic syndrome; mainly proximal muscle weakness (same as in other endocrine myopathies, e.g. hypothyroid, steroid)

* “classic” neurologic PNS. In a patient without previous cancer history presenting with one of these syndromes with an asterisk, work-up for occult malignancy has high yield.

Pan-cerebellar degeneration

Severe Purkinje cell loss (due to anti-Purkinje cell antibody) → severe pan-cerebellar dysfunction. Presents with vertigo, gait and upper and lower extremity ataxia, dysarthria, N/V, diplopia, oscillopsia, nystagmus, oculomotor dysmetria. Usually not treatable nor remitting even with immune suppression. 20%of patients improve with treatment of the primary cancer. CT is WNL early, late → cerebellar atrophy. In 70%of cases, cerebellar findings precede diagnosis of cancer.

The most common primary malignancies in pan-cerebellar degeneration are shown in □ Table 11.6.

Evaluation

- LP: CSF for cell count, cytology and IgG. Typically WBCs and IgG are elevated
- Evaluation for primary
 - CTof chest/abdomen/pelvis
 - lymph node exam
 - pelvic exam and mammogram in women

Table 11.6 Common primaries with pan-cerebellar degeneration	
Women	Men
ovarian Ca breast uterus Hodgkin’s lymphoma	lung Ca Hodgkin’s lymphoma

11.4 Neurotoxicology

11.4.1 Ethanol

General information

The acute and chronic effects of ethyl alcohol (ethanol, EtOH) abuse on the nervous system (not to mention the effects of EtOH on other organ systems) are protean,⁴¹ and are beyond the scope of this text). Neuromuscular effects include:

- 1. acute intoxication: see below
- 2. effects of chronic alcohol abuse
 - a) Wernicke’s encephalopathy (p.206)
 - b) cerebellar degeneration: due to degeneration of Purkinje cells in the cerebellar cortex, pre-dominantly in the anterior superior vermis
 - c) central pontine myelinolysis (p.115)
 - d) stroke: increased risk of
 - intracerebral hemorrhage (p.1330)
 - ischemic stroke⁴²
 - possibly aneurysmal SAH
 - e) peripheral neuropathy (p.541)
 - f) skeletal myopathy
- 3. effects of alcohol withdrawal: usually seen in habituated drinkers with cessation or reduction of ethanol intake
 - a) alcohol withdrawal syndromes: see below
 - b) seizures: up to 33% of patients have a generalized tonic-clonic seizure 7–30 hrs after cessation of drinking – Alcohol withdrawal seizures (p.464)
 - c) delirium tremens (DTs): see below

Acute intoxication

The primary effect of EtOH on the CNS is depression of neuronal excitability, impulse conduction, and neurotransmitter release due to direct effects on the cell membranes. □ Table 11.7 shows the clinical effects associated with specific EtOH concentrations. Mellanby effect: the severity of intoxication is greater at any given level when blood alcohol levels are rising than when falling.

In most jurisdictions, individuals with blood ethanol levels ≥21.7 mmol/l (100 mg/dl) are defined as legally intoxicated, and a number of states have changed this to 80 mg/dl. However, even levels of 10.2 mmol/l (47 mg/dl) are associated with increased risk of involvement in motor vehicle accidents. Chronic alcoholism leads to increased tolerance; in habituated individuals survival with levels exceeding 1000 mg/dl has been reported.

Alcohol withdrawal syndrome

General information

Compensation for the CNS depressant effects of EtOH occurs in chronic alcoholism. Consequently, rebound CNS hyperactivity may result from falling EtOH levels. Clinical signs of EtOH withdrawal are classified as major or minor (the degree of autonomic hyperactivity and the presence/absence of DTs differentiates these), as well as early (24–48 hrs) or late (>48 hrs).

Signs/symptoms include: tremulousness, hyperreflexia, insomnia, N/V, autonomic hyperactivity (tachycardia, systolic HTN), agitation, myalgias, mild confusion. If EtOH withdrawal seizures (p.464) occur, they tend to be early. Perceptual disturbances or frank hallucinosis may also occur early.

Table 11.7 Blood ethanol concentrations		
[blood EtOH]		Clinical effect
mmol/liter	mg/dl	
5.4	25	mild intoxication: altered mood, impaired cognition, incoordination
>21.7	100	vestibular and cerebellar dysfunction: increased nystagmus, diplopia, dysarthria, ataxia
>108.5	500	usually fatal from respiratory depression

Hallucinosi s consists of visual and/or auditory hallucinations with an otherwise clear sensorium (which distinguishes this from the hallucinations of DTs). DTs can occur 3–4 days after cessation of drinking (see below).
Suppressed by benzodiazepines, resumption of drinking, β -adrenergic antagonists, or α 2-agonists.

Prevention of and treatment for alcohol withdrawal syndrome

See reference.⁴³

Mild EtOH withdrawal is managed with a quiet, supportive environment, reorientation and one-to-one contact. If symptoms progress, institute pharmacologic treatment.

Benzodiazepines

Benzodiazepines (BDZs) are the mainstay of treatment. They reduce autonomic hyperactivity, and may prevent seizures and/or DTs. All BDZs are effective. Initial doses are shown in □ Table 11.8 and are higher than those used for treating anxiety. Symptom triggered dosing with repeated evaluation utilizing a standardized protocol (e.g. CIWA-Ar⁴⁴) may be more efficacious than fixed-dose schedules.⁴⁵ Avoid IM administration (erratic absorption).

Adjunctive medications

Associated conditions commonly seen in patients experiencing alcohol withdrawal syndrome include dehydration, fluid and electrolyte disturbances, infection, pancreatitis, and alcoholic ketoacidosis, and should be treated accordingly.

Other medications used for EtOH withdrawal itself include:

1. drugs useful for controlling HTN (caution: these agents should not be used alone because they do not prevent progression to more severe levels of withdrawal, and they may mask symptoms of withdrawal)
 - a) β -blockers: also treat most associated tachyarrhythmias
 - atenolol (Tenormin®): reduces length of withdrawal and BDZ requirement
 - □ avoid propranolol (psychotoxic reactions)
 - b) α -agonists: do not use together with β -blockers
2. phenobarbital: an alternative to BDZs. Long acting, and helps prophylax against seizures
3. baclofen: a small study⁴⁷ found 10 mg PO q d X 30 days resulted in rapid reduction of symptoms after the initial dose and continued abstinence
4. “supportive” medications
 - a) thiamine : 100 mg IM q d $\times$ 3 d (can be given IV if needed, but there is risk of adverse reaction).
Rationale: high-concentration glucose may precipitate acute Wernicke’s encephalopathy in patients with thiamine deficiency
 - b) folate 1 mg IM, IV or PO q d $\times$ 3 d
 - c) MgSO4 1 gm $\times$ 1 on admission: helpful only if magnesium levels are low, reduces seizure risk.
Be sure renal function is normal before administering
 - d) vitamin B12 for macrocytic anemia: 100 mcg IM (do not give before folate)
 - e) multivitamins: of benefit only if patient is malnourished

Table 11.8 Guidelines for BDZ doses for EtOH withdrawal ^a		
Drug	Dose	
	Oral	IV
chlordiazepoxide (Librium®)	100 mg initially, then 25–50 mg PO TID-QID, gradually taper over $\approx$ 4 days). Additional doses may be needed for continuing agitation, up to 50 mg PO hourly ⁴⁶	–
lorazepam (Ativan®)	4 mg initially, then 1–2 mg PO q 4 hrs	1–2 mg q 1–2 hrs
diazepam (Valium®)	20 mg PO initially, then 10 mg PO BID-QID	5–10 mg initially
midazolam (Versed®)		titrate drip to desired effect
^a modify as appropriate based on patient response		

5. seizures: see indications for treatment (p.464)
 - a) phenytoin (Dilantin®) (p.446): load with 18 mg/kg = 1200 mg/70 kg
 - b) continued seizures may sometimes be effectively treated with paraldehyde if available
6. ethanol drip: not widely used. 5%EtOH in D5 W, start at 20 cc/hr, and titrate to a blood level of 100–150 mg/dl

Delirium tremens (DTs)

When DTs occur, they usually begin within 4 days of the onset of EtOH withdrawal, and typically persist for 1–3 days.

Signs and symptoms include: profound disorientation, agitation, tremor, insomnia, hallucinations, severe autonomic instability (tachycardia, HTN, diaphoresis, hyperthermia).⁴⁸ Mortality is 5–10% (higher in elderly), but can be reduced with treatment (including treating associated medical problems and treatment for seizures).

Haloperidol and phenothiazines may control hallucinations, but can lower the seizure threshold. HTN and tachyarrhythmias should be treated as outline above under alcohol withdrawal syndrome.

Wernicke's encephalopathy (WE)

General information

AKA Wernicke-Korsakoff encephalopathy (not to be confused with Korsakoff's syndrome or Korsakoff's psychosis). Classic triad: encephalopathy (consisting of global confusion), ophthalmoplegia, and ataxia (NB: all 3 are present in only 10–33% of cases).

Due to thiamine deficiency. Body stores of thiamine are adequate only for up to $\approx$ 18 days. May be seen in:

1. a certain susceptible subset of thiamine deficient alcoholics. Thiamine deficiency here is due to a combination of inadequate intake, reduced absorption, decreased hepatic storage, and impaired utilization
2. hyperemesis (as in some pregnancies)
3. starvation: including anorexia nervosa, rapid weight loss
4. gastroplication (bariatric surgery)
5. hemodialysis
6. cancers
7. AIDS
8. prolonged IV hyperalimentation

Clinical

Oculomotor abnormalities occur in 96% and include: nystagmus (horizontal > vertical), lateral rectus palsy, conjugate-gaze palsies.

Gait ataxia is seen in 87% and results from a combination of polyneuropathy, cerebellar dysfunction, and vestibular impairment.

Systemic symptoms may include: vomiting, fever.

Diagnostic testing

MRI: May show high signal in T2WI and FLAIR images in the paraventricular (medial) thalamus, the floor of the 4th ventricle, and periaqueductal gray of the midbrain. These changes may resolve with treatment.⁴⁹ Atrophy of the mammillary bodies may also be seen. Normal MRI does not R/O the diagnosis.

Treatment

Wernicke's encephalopathy (WE) is a medical emergency. When WE is suspected, 100 mg thiamine should be given IM or IV (oral route is unreliable, see above) daily for 5 days. □ IV glucose can precipitate acute WE in thiamine deficient patients, □ give thiamine before glucose.

Thiamine administration improves eye findings within hours to days; ataxia and confusion improve in days to weeks. Many patients that survive are left with horizontal nystagmus, ataxia, and 80% have Korsakoff's syndrome (AKA Korsakoff's psychosis), a disabling memory disturbance involving retrograde and anterograde amnesia.

11.4.2 Opioids

Includes heroin (which is usually injected IV, but the powder can be snorted or smoked) as well as prescription drugs. Opioids produce small pupils (miosis).

Overdose may produce:

1. respiratory depression
2. pulmonary edema
3. coma
4. hypotension and bradycardia
5. seizures
6. fatal overdose may occur with any agent, but is more likely with synthetic opioids such as fentanyl (Sublimaze®) among users unfamiliar with their high potency

Reversal of intoxication⁵⁰

A test dose of naloxone (Narcan®) 0.2 mg IV avoids sudden complete reversal of all opioid effects. If no significant reaction occurs, an additional 1.8 mg (for a total dose of 2 mg) will reverse the toxicity of most opioids. If needed, the dose may be repeated q 2–3 minutes up to a total of 10 mg, although even larger doses may be needed with pentazocine or buprenorphine (Buprenex®). Naloxone may precipitate narcotic withdrawal symptoms in opioid dependent patients, with anxiety or agitation, piloerection, yawning, sneezing, rhinorrhea, nausea, vomiting, diarrhea, abdominal cramps, muscle spasms... which are uncomfortable but not life threatening. Clonidine (Catapres®) may be helpful for some narcotic withdrawal symptoms.

With longer acting opioids, especially methadone (Dolophine®), repeat doses of naloxone may be obviated by the use of nalmefene (Revex®), a long-acting narcotic antagonist which is not appropriate for the initial treatment of opioid overdosage.

11.4.3 Cocaine

Cocaine is extracted from *Erythroxylon coca* leaves (and other *Erythroxylon* species) and is thus unrelated to opioids. It blocks the re-uptake of nor-epinephrine by presynaptic adrenergic nerve terminals. It is available in 2 forms: cocaine hydrochloride (heat labile and water soluble, it is usually taken PO, IV or by nasal insufflation) and as the highly purified cocaine alkaloid (free base or crack cocaine, which is heat stable but insoluble in water and is usually smoked).

Peak toxicity occurs 60–90 minutes after ingestion (except for “body packers”), 30–60 minutes after snorting, and minutes after IV injection or smoking (freebase or crack).⁵⁰

11

Acute pharmacologic effects of cocaine

Effects on body systems outside the nervous system include: tachycardia, acute myocardial infarction, arrhythmias, rupture of ascending aorta (aortic dissection), abruptio placenta, hyperthermia, intestinal ischemia, and sudden death.

Acute pharmacologic effects pertinent to the nervous system include:

1. mental status: initial CNS stimulation that first manifests as a sense of well-being and euphoria. Sometimes dysphoric agitation results, occasionally with delirium. Stimulation is followed by depression. Paranoia and toxic psychosis may occur with overdosage or chronic use. Addiction may occur
2. pupillary dilatation (mydriasis)
3. hypertension: from adrenergic stimulation

Non-pharmacologic effects related to the nervous system

1. pituitary degeneration: from chronic intranasal use
2. cerebral vasculitis: less common than with amphetamines
3. seizures: possibly related to the local anesthetic properties of cocaine
4. stroke⁵¹
 - a) intracerebral hemorrhage: see Intracerebral hemorrhage, Etiologies (p.1332)
 - b) subarachnoid hemorrhage^{52,53}: possibly as a result of HTN in the presence of aneurysms or AVMs, however, sometimes no lesion is demonstrated on angiography.⁵⁴ May possibly be due to cerebral vasculitis
 - c) ischemic stroke⁵⁵: may result from vasoconstriction
 - d) thrombotic stroke⁵⁰
 - e) TIA⁵⁶
5. anterior spinal artery syndrome⁵⁶
6. effects of maternal cocaine use on the fetal nervous system include⁵⁷: microcephaly, disorders of neuronal migration, neuronal differentiation and myelination, cerebral infarction, subarachnoid and intracerebral hemorrhage, and sudden infant death syndrome (SIDS) in the postnatal period

Treatment of toxicity

Most cocaine toxicity is too short-lived to be treated. Anxiety, agitation or seizures may be treated with IV benzodiazepines, e.g. lorazepam (p.471). Refractory HTN may be treated with nicardipine (p.126) or phentolamine (Regitine®) (p.655). IV lidocaine used to treat cardiac arrhythmias may cause seizures.⁵⁰

11.4.4 Amphetamines

Toxicity is similar to that of cocaine (see above), but longer in duration (may last up to several hours). Cerebral vasculitis may occur with prolonged abuse which may lead to cerebral infarction.

Elimination of amphetamines requires adequate urine output. Antipsychotic drugs such as haloperidol (Haldol®) should not be used because of risk of seizures.

11.4.5 Carbon monoxide

General information

Carbon monoxide (CO) is the largest source of death from poisoning in the U.S.A.

Normal cellular function requires ≈ 5 ml O₂/100 ml blood. Blood normally contains ≈ 20 ml O₂/100 ml.

CO binds to hemoglobin (Hb) with an affinity ≈ 250 times that of O₂, and it causes a left shift of the Hb/O₂ dissociation curve. It also binds to intracellular myoglobin.

Only $\approx 6\%$ of patients show the classic “cherry-red” color of blood.

Clinical findings

Clinical findings related to CO-Hb levels are shown in □ Table 11.9.

Diagnostic studies

EKG changes are common, usually non-specific ST-T wave changes.

In cases of severe intoxication, CT may show symmetrical low attenuation in the globus pallidus; see differential diagnosis (p.1386).

Outcome

Prognosticators

1. outcome is more closely correlated with hypotension than with actual CO-Hb level
2. coma
3. metabolic acidosis
4. EEG
5. CT/MRI changes: in one study, the presence of MRI lesions after 1 month did not accurately predict subsequent outcome
6. CO-Hb level
7. other factors probably have an effect, including: age, severity of exposure

Approximately 40% of patients exposed to significant levels of CO die. 30–40% have transient symptoms but make a full recovery. 10–30% have persistent neurological sequelae including CO-encephalopathy (may be delayed in onset) – impaired memory, irritability, parietal lobe symptoms including various agnosias.

Brain lesions:

1. white matter lesions:
 - a) multifocal small necrotic lesions in deep hemispheres
 - b) extensive necrotic zones along lateral ventricles
 - c) Grinker’s myelinopathy (not necrosis)
2. grey matter lesions:
 - a) bilateral necrosis of globus pallidus
 - b) lesions of hippocampal formation and focal cortical necrosis

Table 11.9 Levels of CO-Hb	
CO-Hb level (%)	Signs/symptoms ^a
0–10	none
10–20	mild H/A, mild DOE
20–30	throbbing H/A
30–40	severe H/A, dizziness, dimming of vision, impaired judgement
40–50	confusion, tachypnea, tachycardia, possible syncope
50–60	syncope, seizures, coma
60–70	coma, hypotension, respiratory failure, death
>70	rapidly fatal
^a NB: smokers may have CO-Hb levels of 15%without signs or symptoms	

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Part III

Imaging and Diagnostics

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12 Plain Radiology and Contrast Agents

12.1 C-Spine x-rays

12.1.1 Normal findings

For radiographic signs of cervical spine trauma, see □ Table 63.2, and for guidelines for diagnosing clinical instability, see □ Table 65.4.

Contour lines

On a lateral C-spine x-ray, there are 4 contour lines (AKA arcuate lines). Normally each should form a smooth, gentle curve (□ Fig. 12.1):

1. posterior marginal line (PML): along posterior cortical surfaces of vertebral bodies (VB). Marks the anterior margin of spinal canal
2. anterior marginal line (AML): along anterior cortical surfaces of VBs

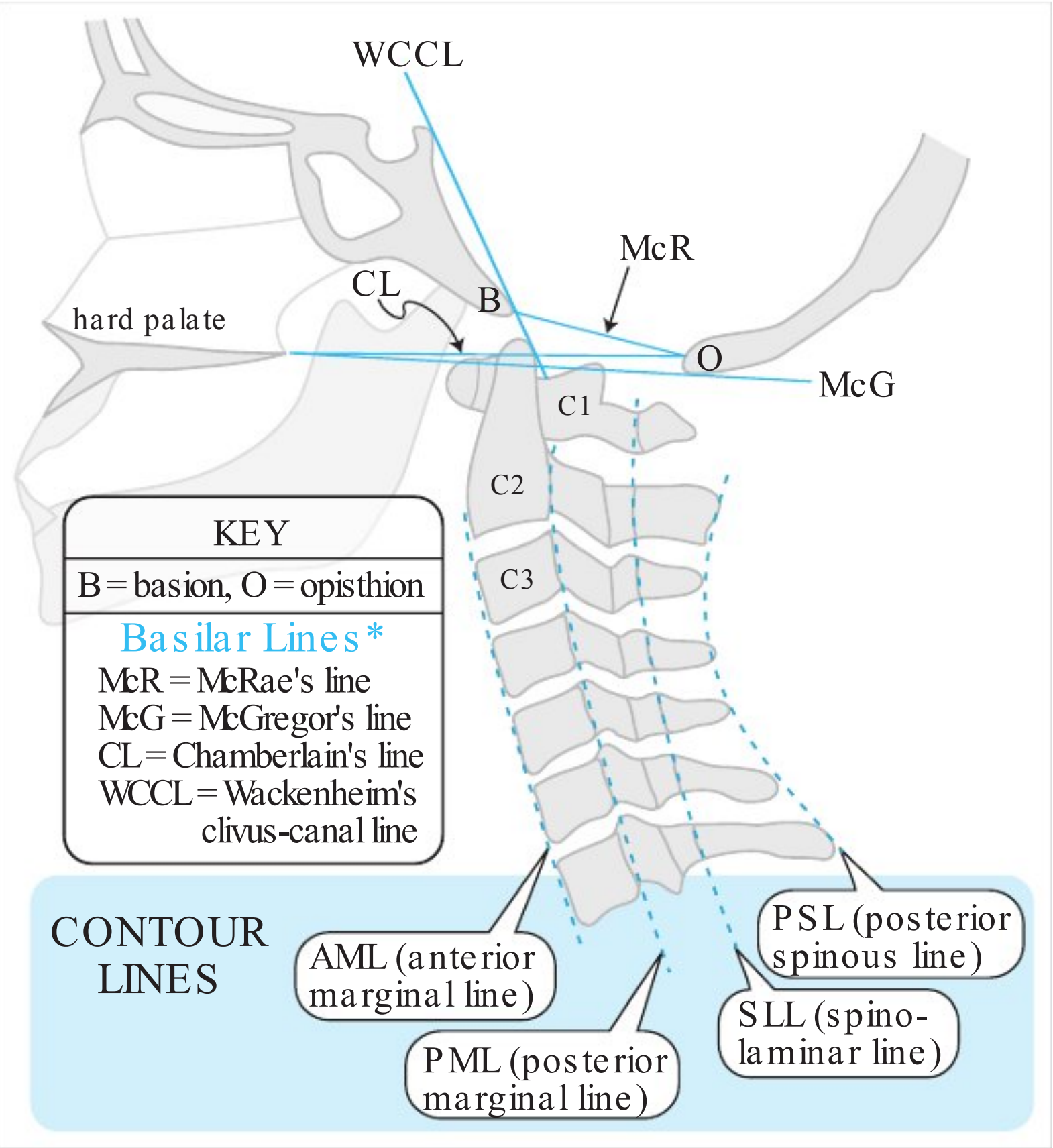


Fig. 12.1 Spinal contour lines and lines used to diagnose basilar invagination
Lateral view through craniocervical junction.
* See discussion of the basilar lines (p. 218)

Table 12.1 Normal ADI		
Patient		ADI
adults	males	≤ 3 mm
	females	≤ 2.5 mm
pediatrics ⁷ (≤ 15 yrs)		≤ 4 mm

3. spinolaminar line (SL): along base of spinous processes. The posterior margin of the spinal canal
4. posterior spinous line (PSL): along tips of spinous processes

Relation of atlas to occiput

See also criteria for atlantooccipital dislocation (AOD) (p.965).

Relation of atlas to axis

These measurements are useful for atlantoaxial subluxation/dislocation (p.968) e.g. in trauma, rheumatoid arthritis (p.1134) or Down syndrome (p.1138).

12.1.2 Rule of Spence

On AP or open-mouth odontoid x-ray, if the sum total overhang of both C1 lateral masses on C2 is ≥ 7 mm (x + y in □ Fig. 12.7), the transverse atlantal ligament (TAL) is probably disrupted^{1,2} (when corrected for an 18% magnification factor, it has been suggested that the criteria be increased to ≥ 8.2 mm³)

12.1.3 (Anterior) atlantodental interval (ADI)

Note: the term ADI usually refers to the anterior atlantodental interval (there is also a posterior ADI (p.213) and a lateral ADI which can be seen on AP radiographs).

AKA predental space. The distance between the anterior margin of the dens and the closest point of the anterior arch of C1 (“C1 button”) on a lateral C-spine x-ray (□ Fig. 12.2). The normal maximal ADI is variously given in the range of 2 to 4 mm.^{4,5} Commonly accepted upper limits are shown in □ Table 12.1. An abnormally increased ADI is a surrogate marker for TAL disruption⁶

12.1.4 Posterior atlantodental interval (PADI)

AKA the neural canal width (NCW).⁸ The PADI is the AP diameter of the bony canal at C1 and is measured from the back of the odontoid to the anterior aspect of the posterior C1 ring (□ Fig. 12.2). It is more useful than the ADI for some conditions, e.g. AAS in rheumatoid arthritis (p.1134) or Down syndrome (p.1138).

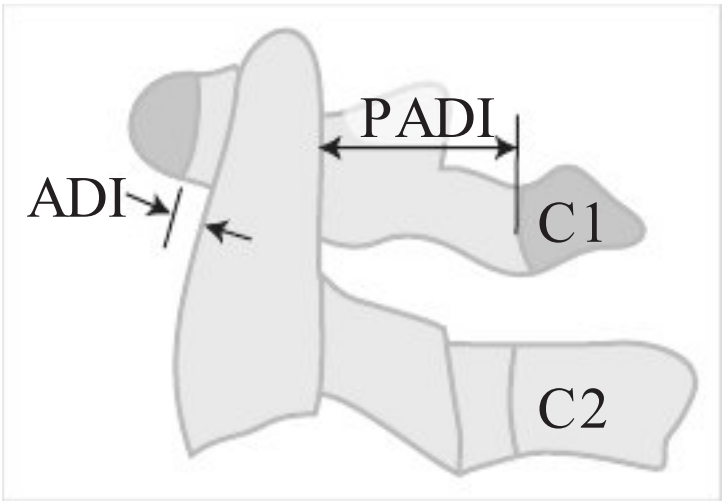


Fig. 12.2 The atlantodental interval (ADI) (p.213) and posterior atlantodental interval (PADI) on a lateral C-spine x-ray

Table 12.2 Normal prevertebral soft tissue				
Space	Level	Maximum normal width (mm)		
		Adults		Peds
		MDCT	Lateral X-Ray	
retropharyngeal	C1	8.5	10	unreliable
	C2–4	6–7 ^a	5–7	
retrotracheal	C5–7	18	22	14

^aCT data was deemed unreliable at C4¹¹

Canal diameter

Normal canal diameter on lateral C-spine x-ray (from spinolaminar line (SLL) to posterior vertebral body with 6 foot tube to film distance)⁹: 17 ± 5 mm. In the presence of osteophytic spurs, measure from the back of the spur to the SLL.

Cervical spinal stenosis: various cutoffs for the normal minimum AP diameter have been suggested.¹⁰ On a plain lateral C-spine x-ray this is usually measured from the posterior vertebral body (or the posterior aspect of an osteophyte) to the spinolaminar line. Some use 15 mm. Most agree that stenosis is present when the AP diameter is <12 mm in an adult. This measurement is less critical than it once was, it is a surrogate marker for stenosis severe enough to compress the spinal cord, which now may be demonstrated directly with MRI (or myelography).

Prevertebral soft tissue

Abnormally increased prevertebral soft tissue (PVST) may indicate the presence of a vertebral fracture, dislocation, or ligamentous disruption.¹² Normal values for lateral C-spine x-ray and CT scan are shown in □ Table 12.2. Plain films are subject to errors due to magnification and rotation. Multi-detector CT (MDCT) eliminates these shortcomings.¹¹

Increased PVST is more likely with anterior than posterior injuries.¹³ NB: the sensitivity of these measurements is only ≈ 60%at C3 and 5%at C6.¹² False positives may occur with basal skull/facial fractures, especially with fracture of the pterygoid plates. An ET-tube may allow fluid to accumulate in the posterior oropharynx which can obscure this measurement. In this setting, one can look for a thin fat layer between the prevertebral muscles and the posterior pharynx on cervical CT; the pre-vertebral tissue (posterior to this line) will be thickened (no measurements available at this time). MRI can also demonstrate abnormal signal within the prevertebral tissue.

Interspinous distances

C-spine AP: a fracture/dislocation or ligament disruption may be diagnosed if the interspinous distance is 1.5 times that at both adjacent levels (measured from center of spinous processes).¹⁴ Also look for a malalignment of spinous processes below a certain level which may be evidence of rotation due to a unilaterally locked facet.

C-spine lateral: look for “fanning” or “flaring” which is an abnormal spread of one pair of spinous processes that may also indicate ligament disruption.

12.1.5 Pediatric C-spine

C1 (atlas)

- Ossification centers¹⁵: usually 3 (□ Fig. 12.3)
- 1 (sometimes 2) for body (not ossified at birth; appears on x-ray during 1st yr)
 - 1 for each neural arch (appear bilaterally ≈ 7th fetal week)
- Synchondroses¹⁵:
- synchondrosis of the spinous process: fuses by ≈ 3 yrs age
 - 2 neurocentral synchondroses: fuse by ≈ age 7 yrs

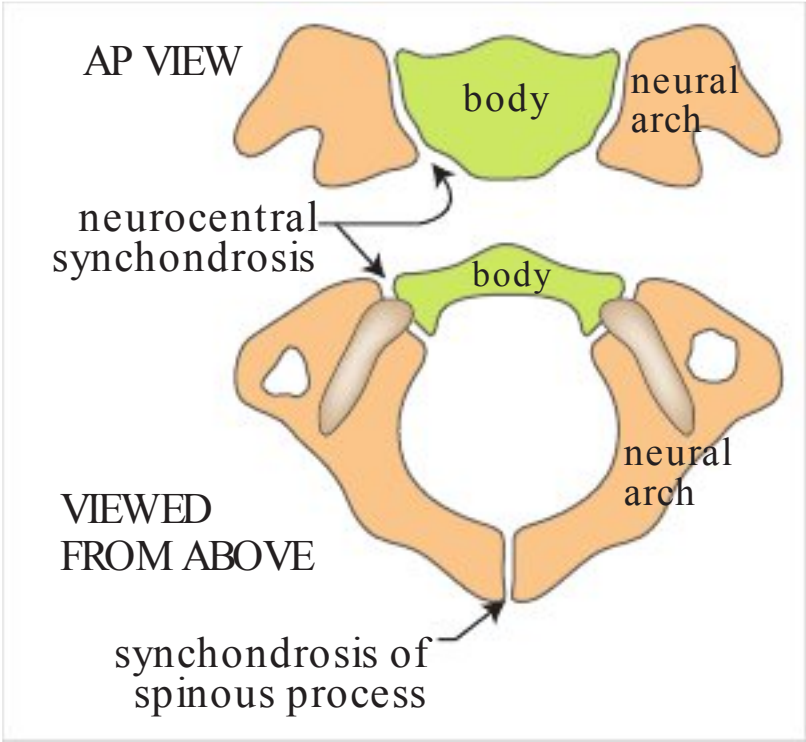


Fig. 12.3 Pediatric C1 (atlas)

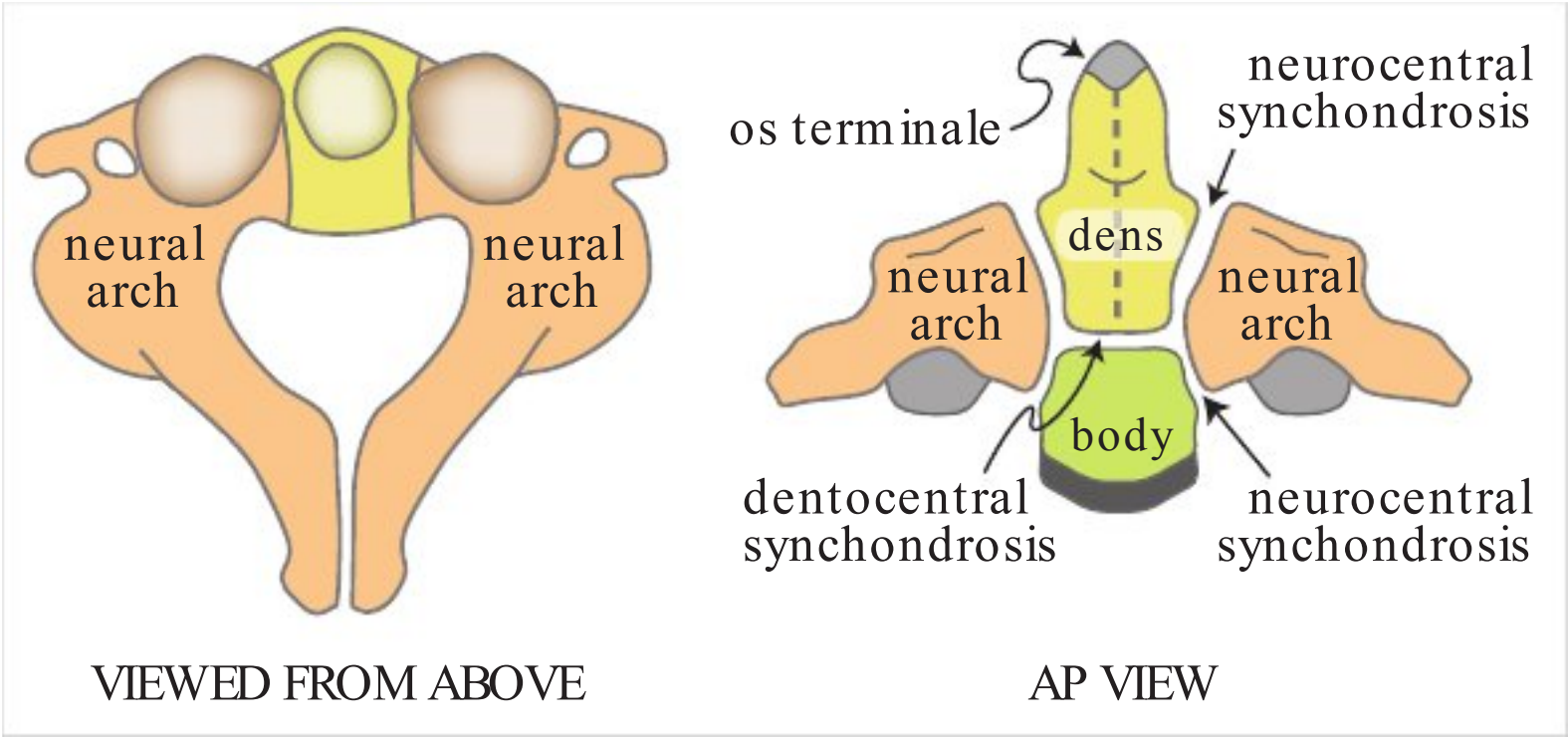


Fig. 12.4 Pediatric C2 (axis)

The ossification centers of C1 fail to completely close in 5% of adults (usually posteriorly). When present, the rare anterior defect is usually associated with a posterior defect.

C2 (axis)

Developmentally there are 5 ossification centers. The two halves of the odontoid fuse together in the midline (dashed line in Fig. 12.4) at 7 months development, so that at birth there are 4 primary ossification centers (Fig. 12.4):

- odontoid process
- vertebral body
- 2 neural arches

The posterior arches fuse together by 2-3 years of age. The anterior synchondroses normally fuse between 3-6 years of age. However, the dentocentral synchondrosis (AKA subdental synchondrosis) may be visible on x-ray until ≈ 11 years of age. A secondary ossification center (os terminale) appears at the summit of the dens between 3-6 years of age, and fuses with the dens by age 12 years.¹⁵

C3-7

3 ossification centers at birth¹⁶ (see Fig. 12.5).

- vertebral body
- 2 neural arches

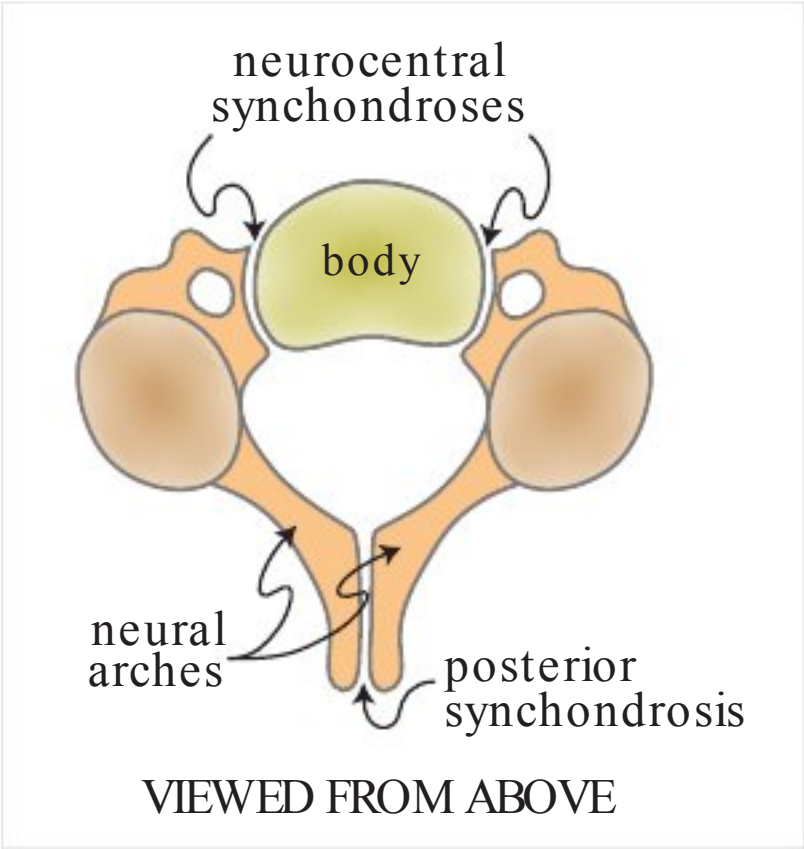


Fig. 12.5 Pediatric C3-7

The 2 neural arches fuse together posteriorly by 2-3 years age.
The neural arches each fuse to the body by 3-6 years age.
Cervical bodies are normally slightly wedge shaped in pediatric population (narrower anteriorly).
Wedging decreases with age.

12.2 Lumbosacral (LS) spine x-rays

L4–5 is normally the lumbar disc space with the greatest vertical height. Also see Normal LS spine measurements (p.1102).

AP view: look for defect or non visualization of the “owl’s eyes” which is due to pedicle erosion which may occur with lytic tumors (common with metastatic disease).

Oblique views: look for discontinuity in neck of “Scotty dog” for defect in pars interarticularis.

Butterfly vertebra: An uncommon congenital anomaly thought to arise from failure of fusion of the lateral halves of the VB due to persistent notochord tissue, producing a “butterfly” appearance on AP x-rays or coronal CT scan reconstructions. The involved VB is widened, and adjacent vertebrae may show a compensatory deformity as if to fill in some of the gap. May be associated with other spinal and rib malformations.¹⁷ On lateral views may simulate compression fracture. In severe cases, there may be significant kyphosis and/or scoliosis. Often asymptomatic, requiring no treatment. May be associated with lipomyelomeningocele (p.269).

12.3 Skull x-rays

Water’s view: AKA submental vertex view. X-ray tube angled up 45° (perpendicular to clivus).
Towne’s view: x-ray tube angled down 45°, to view occiput.

12.3.1 Sella turcica

Normal adult dimensions on skull x-ray

Technique: true lateral, 91 cm target to film distance, central ray 2.5 cm anterior and 1.9 cm superior to EAM. □ Table 12.3 shows normal values (□ Fig. 12.6 shows how measurements are made).

Depth (D): defined as the greatest measurement from floor to diaphragma sellae.

Length (L): defined as the greatest AP diameter.

Abnormal findings

Pituitary adenomas tend to enlarge the sella, in contrast to craniopharyngiomas which erode the posterior clinoids. Empty sella syndrome tends to balloon the sella symmetrically, and also does not erode the clinoids. Tuberculum meningiomas usually do not enlarge the sella, and may be associated with enlargement of the sphenoid sinus; see sphenoid pneumosinus dilatans (p.1372).

Table 12.3 Normal sella turcica dimensions (□ Fig. 12.6)			
Dimension	Max	Min	Avg
D (depth) (mm)	12	4	8.1
L (length) (mm)	16	5	10.6

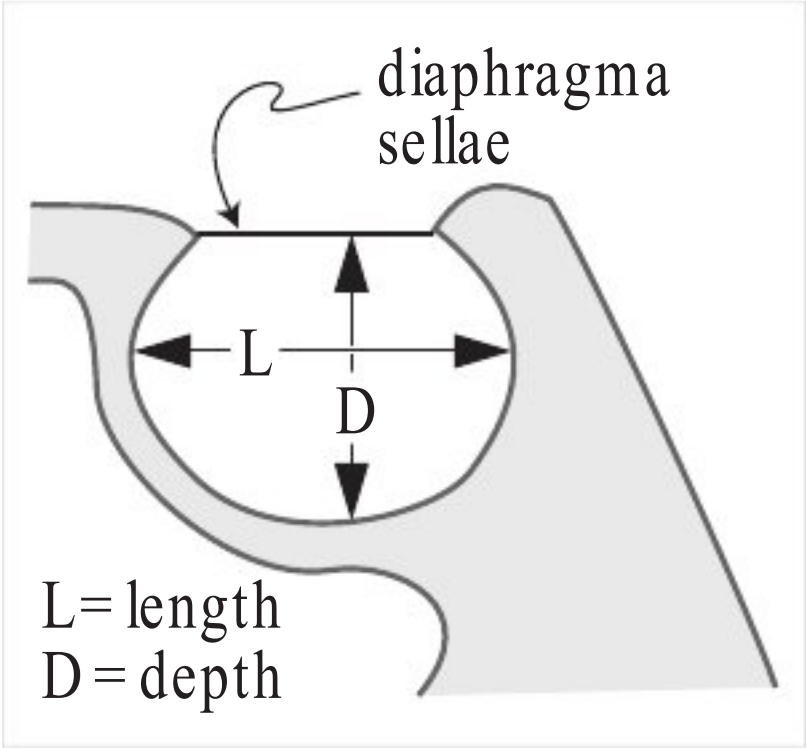


Fig. 12.6 Measurements of the sella turcica (lateral view)

“J” shaped sella suggests optic nerve glioma. It can also occur congenitally in Hurler syndrome (a mucopolysaccharidosis).

12.3.2 Basilar invagination and basilar impression (BI)

Terminology

The terms basilar impression and basilar invagination are often used interchangeably in the literature: historically, basilar invagination (AKA cranial settling) denoted upward indentation of skull base usually due to acquired softening of bone (see below), often associated with atlanto-occipital fusion, while basilar impression implied normal bone. Making a distinction seems pointless (the abbreviation (BI) will be used for either). Common feature: upward displacement of the upper cervical spine (including odontoid process, AKA cranial migration of the odontoid) through the foramen magnum into the p-fossa.

Platybasia: flattening of the skull base. Originally assessed on plain x-rays (which are subject to error due to skull rotation or difficulty identifying landmarks), now more commonly evaluated on CT or MRI. May or may not be associated with BI, and may occur in association with craniofacial abnormalities, Chiari malformation, Paget’s disease...

Quantitated by measuring the basal angle, which on plain x-rays, measured the angle between lines drawn from the nasion to center of sella and then to the anterior foramen magnum,¹⁸ but on MRI was felt to be better represented by the angle between a line drawn along the floor of the anterior fossa to the dorsum sellae and a second line drawn along the posterior clivus.¹⁹ Normal mean basal angle: 130°. Platybasia: >145° (abnormally obtuse basal angle).

Two subtypes of BI

See reference.²⁰

Type I: BI without Chiari malformation. Tip of odontoid tends to be above CL, McR, and WCCL in □ Fig. 12.7. Brainstem compression is due to odontoid process invagination. 85% can be reduced with traction. Treatment: transoral surgery is recommended, usually accompanied by posterior fusion

Type II: BI+Chiari malformation. Odontoid tip tends to be above CL, but not McR or WCCL. Brainstem compression is due to reduced p-fossa volume. Only 15% can be reduced with traction. Foramen magnum decompression is appropriate

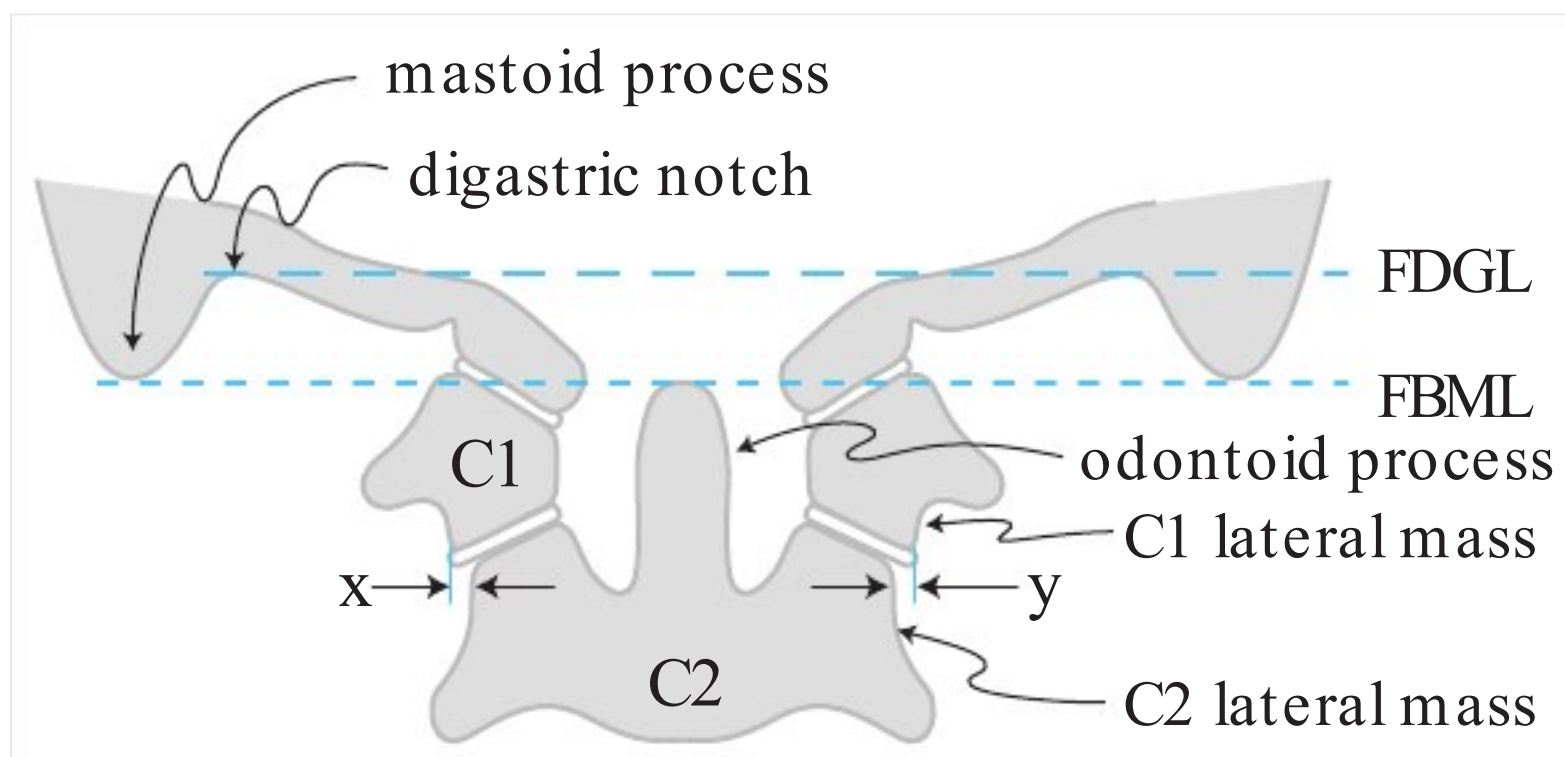


Fig. 12.7 AP view through craniocervical junction
FDGL=Fischgold's digastric line, FBML=Fischgold's bimastoid line, $x+y$ =total overhang of C1 on C2; see Rule of Spence (p.970)

Measurements used in BI

(□ Fig. 12.1 and □ Fig. 12.7):

1. McRae's line ("McR" in □ Fig. 12.1): drawn across foramen magnum (tip of clivus (basion) to opisthion).²¹ The mean position of the odontoid tip below the line is 5 mm (± 1.8 mm SD) on CT and 4.6 mm (± 2.6 mm SD) on MRI.²² No part of odontoid should be above this line (the most accurate measure for BI).
2. Chamberlain's line ("CL" in □ Fig. 12.1)²³: posterior hard palate to posterior margin of foramen magnum (opisthion). Less than 3 mm or half of dens should be above this line, with 6 mm being definitely pathologic. Seldom used because the opisthion is often hard to see on plain film and may also be invaginated. On CT²⁴ and MRI²² the normal odontoid tip is 1.4 mm (± 2.4) below the line
3. McGregor's baseline ("McG" in □ Fig. 12.1)²⁵: posterior margin of hard palate to most caudal point of occiput. No more than 4.5 mm of dens should be above this. On CT²⁴ and MRI²² the normal odontoid tip is 0.8 mm (± 2.4) above the line
4. Wackenheim's clivus-canal line ("WCCL" in □ Fig. 12.1): the odontoid should be tangential to or below the line that extends the course of the clivus (the clivus baseline). If the clivus is concave or convex, this baseline is drawn to connect the basion to the base of the posterior clinoids on the clivus²⁶
5. (Fischgold's) digastric line ("FDGL" in □ Fig. 12.7): joins the digastric notches. The normal distance from this line to the middle of the atlanto-occipital joint is 10 mm (decreased in BI).²⁷ No part of odontoid should be above this line. More accurate than the bimaxillary line (FBML)
6. Fischgold's bimaxillary line ("FBML" in □ Fig. 12.7): joins tips of mastoid processes. The odontoid tip averages 2 mm above this line (range: 3 mm below to 10 mm above) and this line should cross the atlanto-occipital joint

Conditions associated with BI

1. congenital conditions (BI is the most common congenital anomaly of the craniocervical junction, it is often accompanied by other anomalies^{28 (p 148–9)})
 - a) Down syndrome
 - b) Klippel-Feil syndrome (p.271)
 - c) Chiari malformation (p.277): in a series of 100 patients, 92 had BI²⁹
 - d) syringomyelia
2. acquired conditions
 - a) rheumatoid arthritis (in part due to incompetence of transverse ligament, see Basilar impression in rheumatoid arthritis (p.1137))
 - b) post-traumatic
3. conditions with BI associated with softening of bone include³⁰:
 - a) Paget's disease

- b) osteogenesis imperfecta: patients have blue discolored sclera and early hearing loss and due to a genetic defect that causes defective Type 1 collagen. Bones are weak ("brittle-bone disease"). Autosomal dominant inheritance. There are 4 common types of OI and some uncommon ones
- c) osteomalacia
- d) rickets
- e) hyperparathyroidism

12.4 Contrast agents in neuroradiology

Also see Intraoperative dyes (p.1426) for visible dyes useful in the operating room.

12.4.1 Iodinated contrast agents

General precautions

Water-soluble contrast agents have superseded non-water-soluble ones such as Pantopaque® (ethyl iodophenylundecylate or iophendylate meglumine).

□ Caution: iodinated contrast (IV or intra-arterial) may delay excretion of metformin (Glucophage®, Avandamet®), an oral hypoglycemic agent used in diabetes type II, and can be associated with lactic acidosis and renal failure (particularly in patients with CHF or those consuming alcohol). The manufacturer recommends withholding metformin 48 hrs prior to and following contrast administration (or longer if there is evidence of declining renal function following use of contrast). Metformin should also be held $\approx$ 48 hours before any surgery, and should not be restarted post-op until the patient has fully recovered and is eating and drinking normally.

Maximum dose of iodine with normal renal function is $\approx$ 86 gm in a 24 hour period.

Intrathecal contrast agents

Inadvertent intrathecal injection of unapproved contrast agents

□ Caution: serious reactions can occur with inadvertent intrathecal injection (e.g. for myelography, cisternography, ventriculography...) of iodinated contrast media that are not specifically indicated for intrathecal use (including ionic contrast agents as well as some non-ionic agents (e.g. Optiray®, Reno-60...)). This can cause uncontrollable seizures, intracerebral hemorrhage, cerebral edema, coma, paralysis, arachnoiditis, myoclonus (tonic-clonic muscle spasms), rhabdomyolysis with subsequent renal failure, hyperthermia, and respiratory compromise, with a significant fatality rate.³¹

Management suggestions for inadvertent intrathecal injection include:

1. immediately remove CSF+contrast if the error is recognized when the opportunity is available (e.g. withdraw fluid through myelography needle)
2. elevate head of bed $\approx$ 45° (to keep contrast out of head)
3. if there is a question about what may have occurred (i.e. it is not certain if an inappropriate contrast agent was used) send blood and CSF with contrast for high-performance liquid chromatography for identification of agent³²
4. antihistamines: e.g. diphenhydramine (Benadryl®) 50 mg deep IM
5. respiration: supplemental oxygen, and if needed, intubation
6. control HTN
7. IV hydration
8. IV steroids
9. sedation if patient is agitated
10. treat fever with acetaminophen and if needed with a cooling blanket
11. pharmacologic paralysis if necessary to manage muscle activity
12. anticonvulsant medication: more than one agent may be required (e.g. phenytoin + phenobarbital + a benzodiazepine)
13. consider unenhanced brain CT scan: may help assess if contrast has diffused intracranially, but this requires placing patient flat and may not be advisable
14. insert lumbar subarachnoid drain with CSF drainage (e.g. 10 cc q hr)
15. monitor: electrolytes, anticonvulsant levels, creatine kinase (CK)
16. repeat EEGs to assess seizure activity while sedated/paralyzed

Iohexol (Omnipaque®)

The primary approved agent employed for intrathecal use today is iohexol (Omnipaque®).

A non-ionic triiodinated compound. Concentration is expressed as follows: e.g. Omnipaque 300 contains the equivalent of 300 mg of organic iodine per ml of media (300 mgI/ml).

Used for myelography, cisternography as well as IV contrasted CT. Uses and concentrations are shown in □ Table 12.4.

Intrathecal use

NB: only Omnipaque 180, 210, 240 and 300 are labeled for intrathecal use. 140 and 350 are not FDA approved for intrathecal use, however, some neuroradiologists will use Omnipaque 140 or diluted 180 e.g. for CT ventriculography (off-label usage).

Consider discontinuing neuroleptic drugs (including: phenothiazines, e.g. chlorpromazine, prochlorperazine, and promethazine) at least 48 hours prior to procedure. Elevate HOB ≥ 30° for the first few hours after the procedure. Hydrate orally or IV.

Use with caution in patients with seizure history, severe cardiovascular disease, chronic alcoholism or multiple sclerosis.

Iohexol undergoes slow diffusion from the intrathecal space to the systemic circulation and is eliminated by renal excretion with no significant metabolism or deiodination.

Maximum dosage: a total dose of 3060 mg iodine should not be exceeded in an adult during a single myelogram (some say up to 4500 mg is OK) (e.g. 15 cc of Omnipaque 300 = 15 ml × 300 mgI/ml = 4500 mg of iodine).

Iopamidol (e.g. Isovue 300, Isovue 370®)

Triiodinated, non-ionic, water-soluble. Used for intravascular and intrathecal radiographic contrast. Isovue 300 and 370 contains 300 and 270 mg iodine/ml, respectively.

Table 12.4 Iohexol concentrations for adults		
Procedure	Concentration (mgI/ml)	Volume (ml)
lumbar myelography via IP	180	10–17
	240	7–12.5
thoracic myelography via IP or cervical injection	240	6–12.5
	300	6–10
cervical myelography via IP	240	6–12.5
	300	6–10
cervical myelography via C1–2 puncture	180	7–10
	240	6–12.5
	300	4–10
complete myelography via IP	240	6–12.5
	300	6–10
cerebral arteriography ^a	300	≈ 6–12 ml/vessel
IV contrast enhanced CT scan of the brain	240	120–250 ml IV drip
	350	70–150 ml bolus ^b
CT cisternography via IP or C1–2 puncture	300	12
	350	12
CT ventriculography via ventricular catheter	180 ^c	2–3
plain film ventriculography via ventricular catheter	180	2–3
plain film “shunt-o-gram” injected via shunt into ventricles	180	2–3
plain film “shunt-o-gram” injected via shunt distal to valve so as not to enter into ventricles (to check distal shunt function)	300	10–12
	350	10–12
^a most centers use Optiray®, see text		
^b follow with 250 ml bolus of 0.45% NS to rehydrate patient		
^c 180 will be very dense on CT, and some use 1–3 ml of 140 or diluted 180% (dilute approximately 2 parts contrast to 1 part preservative-free normal saline)		

Non-intrathecal contrast agents

For inadvertent intrathecal injection of contrast agents not intended for intrathecal use, see above.

Ioversol (Optiray®)

□ Not for intrathecal use (see above).

Uses and concentrations include:

- arteriography: Optiray 300 (ioversol 64%) or Optiray 320 (ioversol 68%). Total procedural dose should not usually exceed 200 ml
- IV contrast enhanced CT scan of brain:
 - a) adult: 50–150 ml of Optiray 300, 320, or 100–250 ml of Optiray 240.
Typically: 100 ml of Optiray 320
 - b) pediatrics: 1–3 ml/kg of Optiray 320

Iopromide (Ultravist®)

□ Not for intrathecal use (see above). Available in 150, 240, 300 & 370 mg iodine/ml. Osmolality of Ultravist 300 is 607.

Cerebral angiography (300 mg/ml): maximum dose is 150 ml per procedure.

Contrast enhanced CT (CECT) (300 mg/ml). □ Pediatrics (>2 years age): typical dose is 1–2 ml/kg IV, maximum dose is 3 ml/kg per procedure. Adult: typical dose is 50–200 ml, maximum dose is 200 ml.

Iodixanol (Visipaque®)

□ Not for intrathecal use (see above). Triiodinated, non-ionic, isosmolar to blood. For intravascular use. FDA approved for CECT, some angiographers use Visipaque 270 for cerebral angiography (slightly lower opacification, but also slightly lower iodine dose). Available in 270 and 320 mg iodine/ml.

Iodinated contrast with allergies or renally insufficiency

Allergy prep

Indicated for patients with previous history of reaction to IV iodinated contrast material. Minor previous reactions such as hives and itching merit preparation with this regimen whenever possible. Patients with anaphylactic shock or severe edema causing compromise of the airway should probably not receive IV iodine even with this prep, unless absolutely necessary. □ Caution: in spite of this regimen, the patient may still have serious reaction (modified³³). This prep has also been used for the rare gadolinium allergy.

1. utilize non-ionic contrast medium (e.g. iohexol) whenever possible
2. have emergency equipment available during study
3. medications:
 - a) steroid (□ Table 8.1 for further details of steroid dosing)
 - prednisone 50 mg PO: 20–24 hrs, 8–12 hrs & 2 hrs before study
 - equivalent dose of IV Solumedrol® (methylprednisolone): ≈ 25 mg
 - b) diphenhydramine (Benadryl®) 50 mg, EITHER IM 1 hr before, OR IV 5 min before study
 - c) optional: H2 antagonist, e.g. cimetidine 300 mg PO or IV 1 hr before study

Medications for an emergency scan when 24 hour prep is not possible:

- hydrocortisone 100 mg IV then scan within 2 hours

Prep for renal insufficiency or patients with DM

For patients with DM or mild renal insufficiency (e.g. slight serum creatinine elevation, >1.2 mg/dl (U.S.) which is >100 μmol/L) 1 mg/dl of creatinine (used in the U.S.)=88.4 μmol/L, to mitigate against iodine contrast-induced nephropathy:

- N-acetyl cysteine (Mucomyst; the actual efficacy of NAC has not been proven, and may be no better than hydration alone): regimens all accompany hydration and include:
 - a) 800 mg PO q 8 hrs for 24 hours before the study,³⁴ followed by 600 mg PO BID for 24 hours after the study
 - b) 600 mg PO BID X 2 days before the study, 600 mg PO BID for 24 hours after
 - c) 600-mg IV bolus before the study, and 600 mg PO BID for 48 hours after³⁵

- hydration: 1 L of sterile water with 3 amps of sodium bicarbonate IV at 100 ml/hr, start 1 hour prior to the study, and continue until entire L given

12.4.2 Reactions to intravascular contrast media

General information

See also treatment of inadvertent intrathecal injection of ionic contrast agents (p.219)

□ Beta blockers

Beta blockers can increase the risk of contrast media reactions, and may mask some manifestations of an anaphylactoid reaction. They also make use of epinephrine inadvisable since the alpha effects of epinephrine will predominate (bronchospasm, vasoconstriction, increased vagal tone). If treatment is required for hypotension after beta-blocker administration, may try glucagon 2–3 mg IV bolus, followed by 5 mg IV drip over 1 hour (glucagon has positive inotropic and chronotropic effect that is not mediated through adrenergic pathways).

Idiosyncratic reactions and treatment

Hypotension with tachycardia (anaphylactoid reaction)

1. mild: Trendelenburg position. IV fluids
2. if no response but remains mild:
 - epinephrine (use with caution in patients with coronary artery disease, limited cardiac reserve, hypertension, or unclipped cerebral aneurysm)
 - a) 0.3–0.5 ml of 1:1000 SQ (0.3–0.5 mg) q 15–20 mins (peds: 0.01 mg/kg)
 - b) or, ASEP recommendations (especially for elderly or patients in shock): 10 ml of 1:100,000 IV over 5 to 10 min (put 0.1 ml of 1:1000 in 10 ml of NS, or dilute 1 amp of 1:10,000 to 10 ml with NS)
3. moderate to severe or worsening (anaphylaxis): add:
 - a) IV colloidal fluids, e.g. hetastarch (Hespan®) 6% (colloids are required since there is extravascular shift of fluids due to see page, these agents also carry a small risk of allergic reaction)
 - b) epinephrine (see above). May repeat $\times 1$
 - c) O₂ 2–6 L/min per NC. Intubate if necessary
 - d) EKG to R/O ischemic changes
4. if shock develops: add dopamine (p.128), start at 5 mcg/kg/min

Hypotension with bradycardia (vasovagal reaction)

1. mild:
 - a) Trendelenburg position
 - b) IV fluids
2. if no response, add:
 - a) atropine 0.75 mg IV, may repeat up to 2–3 mg over 15 mins PRN. Use with caution in patients with underlying heart disease
 - b) EKG and/or cardiac monitor: especially if atropine or dopamine are used
3. if no response: add dopamine (p.128), start at 5 mcg/kg/min

Urticaria

1. mild: self limited. No treatment necessary
2. moderate:
 - a) diphenhydramine (Benadryl®) 50 mg PO or deep IM (avoid IV, can cause anaphylaxis itself)
 - b) cimetidine (Tagamet®) 300 mg PO or IV diluted to 20 ml and given over 20 mins. H₂ receptors contribute to wheal and flare of reaction
3. severe: treat as above for moderate reaction, and add:
 - a) epinephrine (see above)
 - b) maintain IV line

Facial or laryngeal angioedema

1. epinephrine: see above. May repeat up to 1 mg
2. if respiratory distress: O₂ 2–6 L/min. Intubate if necessary (orotracheal may be very difficult due to swelling of tongue, nasotracheal intubation or emergency cricothyrotomy may be required)

3. diphenhydramine: see above
4. cimetidine: see above
5. if angioedema is accessible, add ice pack
6. maintain IV line
7. steroids are usually effective only for chronic angioedema

Bronchospasm

1. mild to moderate:
 - a) epinephrine: see above. May repeat up to 1 ml
 - b) if respiratory distress: O₂ 2–6 L/min. Intubate if necessary
 - c) maintain IV line
 - d) inhalational therapy with a β -adrenergic agonist, e.g. albuterol (Proventil®) if respiratory therapy is available, otherwise, metered dose inhaler e.g. pirbuterol (Maxair®) or metaproterenol (Metaprel®), 2 puffs
2. severe: treat as above for moderate reaction, and add:
 - a) aminophylline 250–500 mg in 10–20 cc NS slow IV over 15–30 mins. Monitor for hypotension and arrhythmias
 - b) intubate
3. prolonged: add the following (will not have immediate effect):
 - a) hydrocortisone 250 mg IV
 - b) diphenhydramine: see above
 - c) cimetidine: see above

Pulmonary edema

1. O₂ 2–6 L/min per NC. Intubate if necessary
2. raise head and body
3. furosemide (Lasix®) 40 mg IV
4. EKG
5. if hypoxia develops (may manifest as agitation or combativeness), add:
 - a) morphine 8–15 mg IV. May cause respiratory depression, be prepared to intubate
 - b) epinephrine: see above. ☐ CAUTION: use only if MI can be R/O as cause of the pulmonary edema. Patients with acute intracranial pathology may be at risk of neurogenic pulmonary edema (p.1178)

Seizures

If seizure is not self limited, start with lorazepam (Ativan®) 2–4 mg IV for an adult. Take precautions for status epilepticus (p.470) and proceed to other drugs as indicated (p.471).

12.5 Radiation safety for neurosurgeons

12.5.1 General information

Radiation exposure has both a deterministic component (exposure over a certain threshold will cause a specific injury) as well as a stochastic component (any dose increases the chances of an adverse event, and the higher the cumulative dose, the higher the chances).

12.5.2 Units

See reference.³⁶

Absorbed dose: the amount of energy absorbed per unit mass. Expressed in Gray or rads.

Gray (Gy): the SI unit. 1 Gy = 100 cGy = 100 rads = an absorbed dose of 1 Joule/kg.

Rad: 1 rad = an absorbed dose of 100 ergs/gram = 0.01 joule/kg = 0.01 Gy = 1 cGy.

The biological effect (dose equivalent) of radiation: can be expressed in rem or Sieverts.

Sievert (Sv): the SI unit. The dose equivalent in sieverts is equal to the absorbed dose in grays multiplied by a “quality factor” (Q) which differs for different sources of radiation, e.g. high-energy protons have a Q of 10, x-rays have a Q of 1. 1 Sv = 100 rems.

Roentgen-equivalent man (rem): the absorbed dose in rads multiplied by Q. 1 rem is estimated to cause $\approx$ 300 additional cases of cancer per million persons (one third of which are fatal). 1 rem = 0.01 sievert.

12.5.3 Typical radiation exposure

The average annual exposure to radiation is 360 mrem (about 30 mrem are due to background cosmic radiation, $\approx$ 20% of the total dose is due to radioactive potassium-40 which is in every cell). Exposure from a transcontinental airline flight is $\approx$ 5 mrem.

CXR: causes about 0.01–0.04 rem of exposure to the chest.

Spine x-ray with obliques: 5 rem.

CAT scan (brain, noncontrast): median effective dose to the head = 0.2 rem, but the range varied 13 fold within and across institutions.³⁷

Spine CT: 5 rem.

Cerebral arteriogram: $\approx$ 10–20 rem (including fluoroscopy).³⁸

Cerebral embolization: 34 rem.

Bone scan: 4 rem.

C-arm fluoroscopy³⁹: exposure is shown in □ Table 12.5.

Doses during a minimally invasive TLIF⁴⁰:

Patient exposure: mean 60 mGy to the skin in the AP plane (range: 8–250 mGy), 79 mGy in the lateral plane.

Surgeon exposure: 76 mrem to dominant hand, 27 mrem at the waist under a lead apron, and 32 mrem to an unprotected thyroid level detector.

12.5.4 Occupational exposure

The U.S. Nuclear Regulatory Commission (NRC) maximal recommended annual occupational dose limits for radiation are shown in □ Table 12.6.⁴¹ The 1990 recommendations of the International Commission on Radiological Protection (ICRP) was to keep exposure $\leq$ 2 rem/year averaged over 5 years.⁴²

ALARA: an acronym for “As Low As Reasonably Achievable” by which the NRC means making every reasonable effort to keep radiation dose as far below the limits as possible consistent with the purpose for which the licensed activity is undertaken.⁴³

Steps to reduce occupational radiation dose (to staff) during surgery:

1. increase the distance from the radiation source: radiation exposure is proportional to the inverse square of the distance. Conventional wisdom is to try to keep 6 feet away. In a AANS publication, 3 m (10 ft) was recommended⁴⁴
Lead aprons/shields may or may not work. Distance ALWAYS works⁴⁵ (inverse square law – double the distance and get 1/4 the radiation).
2. shielding: shielding is less effective at higher kV (used with larger patients). Portable lead “doors” are more effective than aprons. Wrap-around 2-piece aprons are better than front side aprons. With front aprons the wearer must always face the x-ray source, otherwise the apron can actually reflect some radiation back onto the wearer. Non-lead aprons may not provide the rated protection at levels > 100 keV⁴⁶
3. don’t overuse magnification: most fluoro systems increase the radiation emitted $\times \approx 4$ to compensate for the associated reduction in image brightness
4. “boost” mode can double the radiation output. Use should be kept to a minimum
5. use live fluoro only when absolutely necessary
6. for lateral imaging, when possible stand on the “downstream” (image intensifier (ImI)) side of the C-arm: scatter is the most significant cause of exposure here and is higher on the source side⁴⁷ (this asymmetry is not as significant for C-spine⁴⁸)
7. keep the ImI as close to the patient as possible (reduces patient & staff exposure and improves image quality)
8. on AP images (with the patient prone or supine): position the x-ray tube under the table with the ImI over the patient (lowers scatter exposure to staff)⁴⁹
9. collimate the beam as much as possible: reduces radiation to patient and to staff, and results in less image degradation
10. keep hands, arms, etc. out of the primary beam at all times (consider using leaded gloves if hands need to be within the beam or nearby for an extended time)
11. minimize number of images: plan your shot, avoid frequent “checks” or peeks
12. use image guided navigation when possible and practical
13. leaded glasses are recommended only for personnel with very high fluoro times: cataracts can be induced by single doses of 200 rads (very high), cumulative doses of 750 rads have not been associated with cataracts

Table 12.5 Radiation exposure with fluoroscopy ^{39a}				
Distance from beam		Typical team member	Deep exposure	Superficial exposure
feet	meters		(mrem/min)	
Direct beam		patient	4000	
1	0.3	surgeon	20	29
2	0.6	assistant	6	10
3	0.9	scrub tech	0	≤2
5	1.5	anesthesiologist	0 ^b	0 ^b
^a in a mock OR set up for maximal scatter				
^b after 10 minutes of exposure				

Table 12.6 Annual occupational radiation dose limits	
Target organ	Recommended MAXIMAL dose (rem/yr)
whole body	5
lens of eye	15
skin, hands, feet	50
other organs (including thyroid)	15

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13 Imaging and Angiography

13.1 CATscan (AKA CTscan)

13.1.1 General information

CATscans employ ionizing radiation (x-rays) with the attendant risks; see Radiation safety for neurosurgeons (p.223).

Attenuation of the x-ray beam on a CT scan is defined in Hounsfield units. These units are not absolute, and vary between CTscanner models. Some sample values are shown in □ Table 13.1.

13.1.2 Noncontrast vs. IVcontrast enhanced CTscan (CECT)

Noncontrast CT scans are often employed in emergency situations (to quickly rule-out most acute abnormalities), to evaluate bone in great detail, or as a screening test. It excels in demonstrating acute blood (EDH, SDH, IPH, SAH), fractures, foreign bodies, pneumocephalus and hydrocephalus. It is weak in demonstrating acute stroke (DWI MRI is preferred), and often has poor signal quality in the posterior fossa (due to bone artifact).

IVenhanced CTscans are used primarily for imaging neoplasms or vascular malformations, especially in patients with contraindications to MRI. All CT contrast agents contain iodine.

Typical IV dose of contrast: 60–65 ml of e.g. Isovue 300® (p.220) which delivers 18–19.5 grams of iodine.

13.1.3 CT angiography (CTA)

Employs rapid injection of iodinated contrast at 3–4 cc/sec, typically 65–75 ml of e.g. Isovue 300®. Optimal results in patients who can hold their breath for 30–40 seconds (for spiral CT).

Table 13.1 Hounsfield units for a sample CT scanner		
Definitions	Hounsfield units	Comment
no attenuation (air)	−1000	definition
water	0	definition
dense bone	+1000	definition
Cranial CT		
brain (grey matter)	30 to 40	
brain (white matter)	20 to 35	
cerebral edema	10 to 14	
CSF	+5	
bone	+600	
blood clot ^a	75 to 80	acute SDH or EDH, fresh SAH
fat	−35 to −40	
calcium	100 to 300	
enhanced vessels	90–100	
Spine CT		
disc material	55–70	disc density is $\approx 2 \times$ thecal sac
thecal sac	20–30	
^a Hct <23% will cause an acute SDH to be isodense with brain		

Various methods may be used to determine timing of CT after injection: may be based on time to peak in aorta after a small test injection, or can be based empirically on time, or give injection and look for peak in the region of interest.

Accuracy is diminished for vessels that are perpendicular to the axial CT plane. Also in the vicinity of dense clot, CTA has trouble resolving the adjacent vessels.

13.1.4 CT perfusion (CTP)

Requires use of iodinated contrast. Areas of interest are selected from an unenhanced CT scan in the 3 supratentorial vascular territories. Contrast is given at a standard rate (e.g. 40 ml IV at 5 ml/sec). Scans through the regions of interest are repeated at intervals, e.g. every 2 seconds for 1 minute.

Acetazolamide (ACZ) (Diamox®) challenge: after the above, a bolus of 1000 mg of IV ACZ is given, and scans are repeated at intervals for approximately 10 minutes, with a final scan usually at 15 minutes.

Parameters then calculated from the images: cerebral blood volume (CBV), CBF, meant transit times (MTT), and time to peak (TTP). In ischemic stroke, MTT is almost always increased and CBF is decreased.

Abnormalities that can be demonstrated:

1. flow significant stenosis: decreased CBV & CBF, increased MTT and TTP
2. steal: after ACZ challenge (see above), CBV & CBF decrease, often with increases in the corresponding contralateral territory; MTT increases

In comparison to perfusion weighted MRI (PWI) (p.232):

1. PWI acquires multiple slices of the whole brain over and over. CTP is limited to a given slice or several slices (usually 10–20 mm thick), and one has to choose where to place that slice
2. PWI has more artifact than CTP

13.2 Magnetic resonance imaging (MRI)

13.2.1 General information

Definitions¹

Abbreviations:

- TR: time to repetition
- TE: time to echo
- T_i: time to inversion
- T₁: spin-lattice relaxation time (“time to magnetize”) (regrowth)
- T₂: spin-spin relaxation time (“time to demagnetize”) (decay)

13.2.2 T₁ weighted image (T1WI)

Short T₁ → high signal (bright). “Anatomic image”, somewhat resembles CT. Shorter acquisition time than T2WI. Proton rich tissue (e.g. H₂O) has long T₁.

Table 13.2 Range of acquisition data		
	short TE (te <50)	long TE (te >80)
short TR (TR<1000)	T1WI	
long TR (TR>2000)	proton density or spin density	T2WI

Table 13.3 T1WI, MRI intensity change				
fat (including bone marrow), blood >48 hrs old, melanin	white matter	grey matter	calcium	CSF, bone
(note: grey-bar illustrates direction of intensity change and does not show actual grey on MRI)				

Clues to recognizing T1WI: CSF is black, subcutaneous fat is white, TR and TE are short (hundreds and double digits, respectively).

□ The only objects that appear white on T1WI are: fat, melanin, Onyx® (p.1589), and subacute blood (>48 hrs old). White matter is higher signal than grey matter (myelin has a high fat content). Most pathology is low signal on T1WI.

13.2.3 T₂ weighted image (T2WI)

Long T₂ → high signal (bright). “Pathological image.” Most pathology shows up as high signal, including surrounding edema.

Clues to recognizing T2WI: CSF is white, TR & TE are long (thousands and hundreds, respectively).

13.2.4 Spin density image

AKA balanced image, AKA proton density image. Partway between T1WI and T2WI. CSF=grey, approximately isodense with brain. Becoming less commonly used.

13.2.5 FLAIR

Acronym: FLuid-Attenuated Inversion Recovery. Long TR and TE. Resembles a T2WI except the CSF is nulled out (appears dark). The grey/white intensity pattern is reversed from T1WI and is more prominent. Most abnormalities including MS plaques, other white matter lesions, tumors, edema, encephalomalacia, gliosis and acute infarcts appear bright. Periventricular lesions such as MS plaques become more conspicuous. Also good for demonstrating abnormalities in CSF.

Differential diagnosis of increased signal in subarachnoid spaces on FLAIR:

1. subarachnoid hemorrhage (SAH): □ the best sequence for detecting acute SAH on MRI
2. meningitis: occurs in some cases
3. meningeal carcinomatosis
4. superior sagittal sinus thrombosis
5. stroke
6. adjacent tumor: ? if related to higher protein
7. previous administration of gadolinium
8. high levels of FIO₂ especially at levels nearing 100%as may be used in patients getting MRI under general anesthesia.² Shows up in basal cisterns and in sulci over the convexity, but not in ventricles

13.2.6 Echo train (AKA fast spin echo (FSE))

tr is held constant, te is progressively increased utilizing multiple echoes (8–16) rather than one. Image approaches T2WI but with substantially reduced acquisition time (fat is brighter on FSE, which may be rectified by fat suppression techniques).

13.2.7 Gradient echo

AKA T2* (called T2-star), and some manufacturers have trademarked names for this, e.g. “GRASS” (a GE trademarked acronym for Gradient Recalled Acquisition in a Steady State) or FISP. A “fast” T2WI utilizing a partial flip angle. CSF and flowing vessels appear white. Bone, calcium and heavy metals are dark. Typical acquisition data: TR=22, TE=11, angle 8°. Used e.g. in cervical spine to produce a “myelographic” image, improves MRI’s ability to delineate bony spurs. Also shows small old cerebral hemorrhages (seen in 60% of patients presenting with hemorrhagic infarction, and in 18% with ischemic infarcts³); these patients may be at increased risk of hemorrhage from anticoagulation.

□ Gradient-echo T2WI MRI is the 3–4 × more sensitive test than FLAIR for demonstrating intraparenchymal blood (which appears dark) due to high sensitivity to paramagnetic artifact. It is not as sensitive as the more recently introduced SWI.

Table 13.4 T1WI, MRI intensity change				
brain edema/water	CSF	grey matter	white matter	bone, fat
(note: grey-bar illustrates direction of intensity change and does not show actual grey level on MRI)				

13.2.8 “STIR” image

Acronym for “Short Tau Inversion Recovery.” Summates T1 & T2 signals. Causes fat to drop out – sometimes also called fat suppression or “fat sat” (for fat saturation), allows gadolinium enhancement to show up better in areas of fat. Useful primarily in spine and orbit. Very good for showing bone edema (can help in dating spine fractures). The dorsal root ganglion may enhance on fat suppression images.

13.2.9 Contraindications to MRI

General information

An extensive reference⁴ details safety issues. Web sites for MRI safety include: www.MRIsafety.com and www.IMRSE.org. Some issues that come up frequently in neurosurgical patients follows.

Pregnancy and MRI

During the first trimester, MRI can cause reabsorption of products of conception (miscarriage). There are no studies to determine the long term effects of MRI on a fetus after the first trimester (the low risk of MRI in this situation is probably preferable to the known dangers of ionizing radiation of x-rays (including CT)⁵). Gadolinium contrast is contraindicated during all of pregnancy, and is not approved for use in age <2 years. Breast-feeding must be interrupted for 2 days after administration of gadolinium to the mother.

Common contraindications to MRI

1. cardiac pacemakers/defibrillator, implanted neurostimulators, cochlear implants, infusion pumps: may cause temporary or permanent malfunction
2. ferromagnetic aneurysm clips (see below): some centers exclude all patients with any type of aneurysm clip
3. metallic implants or foreign bodies with large component of iron or cobalt (may move in field, or may heat up)
4. Swann-Ganz catheter (pulmonary artery catheter)
5. metallic fragments within the eye
6. placement of a vascular stent, coil or filter within the past 6 weeks
7. shrapnel: BB's (some bullets are OK)
8. relative contraindications:
 - a) claustrophobic patients: may be able to sedate adequately to perform study
 - b) critically ill patients: ability to monitor and access to patient are impaired. Specially designed non-magnetic ventilator may be required. Cannot use most brands of electronic IV pumps/regulators
 - c) obese patients: may not physically fit into many closed bore MRI scanners. Open bore scanners may circumvent this but many utilize lower field strength magnets and generally produce inferior quality images in large patients
 - d) non-MRI compatible metal implants in the region of interest (or previous surgery with high speed drills which may leave metal filings): may produce susceptibility artifact which can distort the image in that area
 - e) programmable shunt valve (p.418): most will tolerate up to a 3 T MRI without permanent damage, however, the pressure setting may be altered and therefore should be rechecked after having an MRI for any reason

Aneurysm clips and MRI

MRI concerns in patients with a cerebral aneurysm clip:

1. the danger of the MRI magnetic field causing the aneurysm clip to be pulled or torqued off of the aneurysm or to tear the neck
2. the artifact produced by the metal of the clip in the magnetic field
3. heat generated in the region of the clip: not clinically significant

The more ferromagnetic the clip, the larger the force exerted on it by the magnetic field and the greater the image distortion near the clip.

Stainless steel (SS) is classified as martensitic (ferromagnetic) or austenitic (non-ferromagnetic). Cobalt-based superalloys are non-ferromagnetic and include Elgiloy (Sugita clips), Phynox (Yasargil),

Table 13.5 Magnetic remanence of aneurysm clips ⁶		
Clip	Type of steel	Magnetic remanence (no units)
MRI compatible: no		
Drake DR 12	martensitic SS	100
Heifetz	17–7PH	44
Mayfield	martensitic SS	74
Scoville	EN-58J	64
MRI compatible: yes		
Olivecrona		0
Sugita	Elgiloy	0
Sugita with loop	gold plated	1
McFadden	Vari-Angle	0
Yasargil	316	0
Yasargil	Phynox	0
Yasargil (old)		1
silver clip		0

and Vari-Angle (McFadden). Essentially all modern aneurysm clips are MRI compatible, but patients clipped before the 1990s may have ferromagnetic clips.

□ Table 13.5 shows the magnetic remanence of various clips which is related to their ferromagnetic properties. If in doubt at the time of aneurysm surgery, apply the following simple test: non-ferromagnetic clips cannot be lifted or dragged with a small magnet.

13.2.10 Hemorrhage on MRI

Because its signal characteristics change with time (and location), blood is one of the most complex entities to interpret on MRI. A mnemonic for the changes in appearance of blood on MRI with time is shown in □ Table 13.6. See also intracerebral hemorrhage (p.1330). Blood, hemosiderin and calcium are dark on GRASS images. FLAIR (p.229) is the best sequence for detecting SAH on MRI (p.229).

13.2.11 MRI contrast

Current agents are mostly based on gadolinium (a rare earth metal which is paramagnetic in solutions) include: gadopentatate dimeglumine (Magnevist®), gadodiamide (Omniscan®), gadoversetamide (OptiMARK®), gadobenate dimeglumine (MultiHance®) and gadoteridol (ProHance®).

Adverse reactions:

1. anaphylactic reactions: rare (prevalence: 0.03–0.1%)
2. nephrotoxicity: incidence is lower than with iodinated agents used with angiograms or CT and x-ray contrast
3. nephrogenic systemic fibrosis (NSF): a rare, but serious illness characterized by fibrosis of skin, joints and other organs, which is associated with certain gadolinium containing agents given to patients with severe renal failure (most were on dialysis). □ Gadolinium is now relatively contraindicated with a GFR of 30–60 ml/min, and is contraindicated with GFR<30.⁷ Safest agents: Dotarem, Gadovist and ProHance.⁸ Contrast agents with a linear structure appear to be associated with a higher risk of NSF and include: Omniscan, Multihance, Magnevist and OptiMARK. In patients with end-stage renal disease, the risk is ≈ 2.4%per gadolinium study⁹
4. gadolinium allergy: use the same allergy prep as for iodine allergy (p.221)
5. as of the time of publication of this book, the FDA is investigating the risk of accumulation of gadolinium in brain tissue after repeated MRI scans using gadolinium based contrast agents (GBCAs).¹⁰ In the meantime, the FDA is recommending that health-care professionals consider

Table 13.6 Variation of brain MRI signal characteristics of intraparenchymal blood over time				
Phase	Approximate time after onset	T1 MRI	T2 MRI	Mnemonic ^a
Hyperacute	0-6 hrs ^b	I	B	I be
Acute	6-72 hrs	I	D	iddy
Early subacute	3-7 d	B	D	biddy
Late subacute	7-14 d	B	B	baby
Chronic	>2 weeks	D	D	doodoo
^a Mnemonic: B (bright or hyperintense compared to brain), D (dark or hypointense), I (isointense)				
^b Some authors consider up to about 24 hrs as hyperacute				

limiting the use of GBCAs to situations where the additional information provided by the contrast is absolutely necessary

6. see also issues related to pregnancy (p.230)

13.2.12 Magnetic resonance angiography (MRA)

There are 2 ways to obtain an MRA

- Gadolinium enhanced: usually for extracranial vessels (e.g. carotids)
- Noncontrast images using flow related enhancement techniques (most common: 2D time of flight (2D TOF)). Usually for intracranial vessels. Anything that appears bright on T1 WI will also show up on MRA, but doesn't necessarily represent blood flow. This includes fat and fat-laden macrophages in an area of old stroke. Using fat-sat T1 WI can mitigate this. Has some utility in screening for aneurysms (p.1161), and for angiographically occult vascular malformations (p.1246). High-flow AVMs are hard to resolve because arterialized veins can appear similar to arteries.

13.2.13 Diffusion-weighted imaging (DWI) and perfusion-imaging (PWI)

13

Diffusion-weighted imaging

Primary uses: early detection of ischemia (stroke) and differentiating active MS plaques from old ones. DWI is sensitive to random Brownian motion of water molecules.

Two images are generated, an apparent diffusion coefficient (ADC) map (based on a number of variables (time, slice orientation...)), and a trace image (the actual DWI).¹¹ Freely diffusing water (e.g. in CSF) appears dark on DWI.

The DWI is based on a T2WI, and anything that is bright on T2WI can also be bright on DWI (so-called "shine-through"). Since bright areas on DWI can represent either restricted diffusion or T2 "shine-through," □ check the ADC map: if the area is black on the ADC map, then this likely represents true restricted diffusion (recent infarct is the most common etiology).

□ Intraparenchymal areas of bright signal on DWI that are not bright on the ADC map are abnormal and represent regions of restricted diffusion such as acute stroke.

Differential diagnosis of areas of increased signal (bright) on DWI:

1. ischemic brain: acute stroke and areas with hypoperfusion (penumbra). While restricted diffusion usually indicates irreversible cell injury (death), it can sometimes indicate tissue that is just near cell death (penumbra). Acute brain ischemia can light up within minutes.^{12,11} The DWI abnormality will persist for ≈ 1 month. The ADC map usually normalizes after ≈ 1 week
2. cerebral abscess (p.323): DWI=bright, ADC=dark
3. active MS plaque (old plaques will not be bright)
4. some tumors: most tumors are dark on DWI, but highly cellular tumors may have decreased diffusion (bright on DWI) (e.g. epidermoids, some meningiomas...)

Other possible uses of DWI:

TIAs: some, but not all,¹³ are associated with DWI abnormalities. However, factors other than focal ischemia (e.g. global ischemia, hypoglycemia, status epilepticus...) can produce ADC decline and the DWI images must therefore be interpreted in relation to the clinical setting.¹¹

DWI may also be able to distinguish cytotoxic from vasogenic edema (p.90).^{14,15}

Perfusion-weighted MRI

Provides information related to the perfusion status of the microcirculation. PWI is the most sensitive study for ischemia of the brain (more sensitive than DWI and FLAIR which primarily show infarcted tissue). There are several methods currently in use; the bolus-contrast approach is the most widely employed.¹¹ Ultrafast gradient imaging is used to follow the gradual reduction to normal following administration of contrast (usually gadolinium). A signal wash-out curve is derived and is compared to contrast in an artery. In practical terms, PWI is not widely used because of technical challenges. Time-to-peak and mean-transit-time are 2 common parameters that are displayed (higher signal=longer times beyond normal).

DWI and PWI mismatch

DWI and PWI may be combined to locate areas of diffusion-perfusion mismatch (deficit on PWI that exceeds the zone of diffusion deficit on DWI), thus identifying salvageable brain tissue at risk of infarction – “penumbra” (p.1202) – e.g. to screen for potential candidates for thrombolytic therapy.¹⁶

13.2.14 Magnetic resonance spectroscopy (MRS)

General information

This section specifically covers proton (H+) MRS which can be performed on almost any MRI scanner (especially units ≥1.5 T) with the appropriate software. Spectroscopy of other nuclei (e.g. phosphorous) can be evaluated only with specialized equipment.

Single voxel MRS

General information

A small area is selected on the “scout” MRI and the spectroscopic peaks for that region are displayed in resonance as a function of parts-per-million (ppm). Since only small regions are selected, may be subject to “sampling” error.

Clinically important characteristic peaks are delineated in □ Table 13.7.

Table 13.7 Important peaks on proton MRS		
Moiety	Resonance (ppm)	Description
lipid	0.5–1.5	slightly overlaps lactate peak at TE ≈ 35
lactate	1.3	a couplet peak. Not present in normal brain. End product of anaerobic glycolysis, □ a marker of hypoxia. Present in: ischemia, infection, demyelinating disease, inborn errors of metabolism... At higher TE (e.g. TE= 144), the peak inverts which can help distinguish it from the lipid peak
N-acetyl aspartate (NAA)	2	a neuronal marker. Normally the tallest peak (higher than Cr or Cho). ↓ in ≈ all focal and regional brain abnormalities (tumor, MS, epilepsy, Alzheimer’s disease, abscess, brain injury...)
creatine (Cr)	3 ^a	useful primarily as a reference for choline. Higher in grey matter than white matter
choline (Cho)	3.2	marker of membrane synthesis. ↑ in neoplasms and some rare conditions of increased cell growth & in the developing brain. □ Stroke is low in choline
^a Cr has another less important peak		

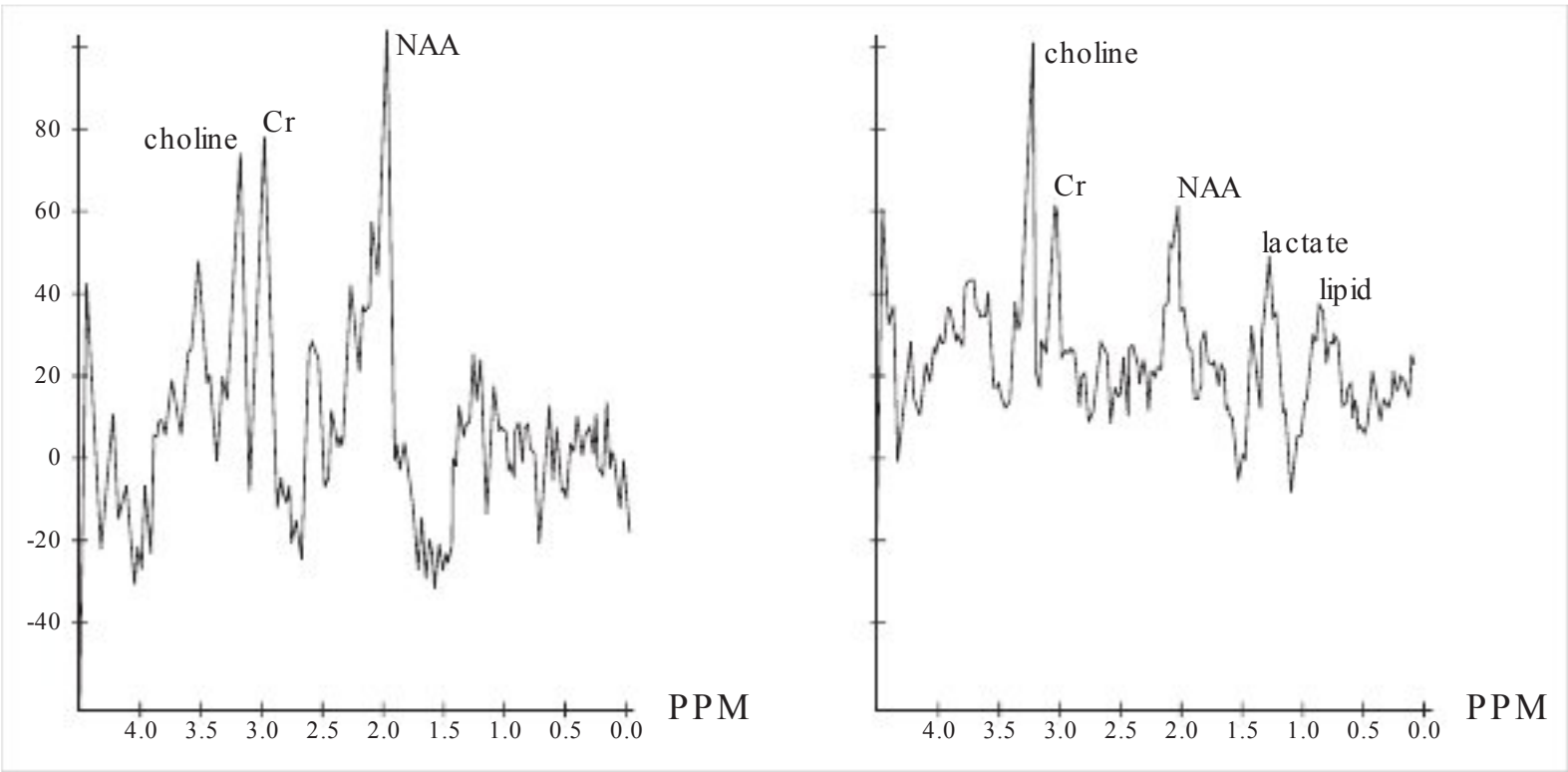


Fig. 13.1 Proton MRS of (A) normal brain, and (B) high grade glioma

Illustrative patterns

Normal brain: See □ Fig. 13.1.

Tumor: See □ Fig. 13.1. ↓ NAA, ↑ lactate, ↑ lipid, ↑ choline (rule of thumb: with gliomas, the higher the choline, the higher the grade up to grade 3, thereafter necrosis reduces relative choline levels and the lipid peak may be utilized).

Stroke: ↑ lactate peak predominates. Choline is characteristically low.

Abscess¹⁷: Reduced NAA, Cr & choline peaks, and “atypical peaks” (succinate, acetate...) from bacterial synthesis is pathognomonic for abscess (not always present). Lactate may be elevated.

Multiple sclerosis: Bland pattern. NAA slightly reduced. Lactate and lipid slightly elevated. Choline not elevated.

Possible uses of MRS

1. differentiating abscess from neoplasm
2. post-op enhancement vs. recurrence of tumor
3. distinguishing tumor from MS plaques: occasionally cannot be differentiated
4. in AIDS: may be able to help differentiate toxo from lymphoma from PML(PML: ↓ NAA, no significant increase in choline, lactate or lipid)
5. the promise of differentiating tumor infiltration from edema has not materialized
6. some utility in distinguishing tumor from radiation necrosis (p.1560)
7. large inositol peak may distinguish hemangiopericytoma from meningioma¹⁸)

Multi-voxel MRS

Color coded scan with selected overlay for NAA, choline... one at a time. May reduce risk of sampling error.

13.2.15 Diffusion tensor imaging (DTI) MRI and white matter tracts

AKA diffusion tensor tractography (DTT) MRI. An MRI technique that demonstrates white matter tracts by exploiting the difference in diffusion parallel to the nerve axons that comprise white matter tracts from diffusion perpendicular to their course.

Available only with specialized software for specific MRI scanners.

Contraindications are same as for MRI in general (p.230).

Probably most useful to permit planning surgical approaches that minimize disruption of critical white matter tracts during intraparenchymal brain surgery for deep lesions, especially when a lesion (e.g. tumor, AVM, cerebral hemorrhage...) may displace these tracts from their expected position.

The major divisions of white matter tracts demonstrable by DTT MRI are (□ Fig. 13.2):

- Projection fibers: tend to be oriented rostro-caudally
 - Corticospinal tract coalesces as corona radiata funnels into internal capsule and forms pyramidal tract

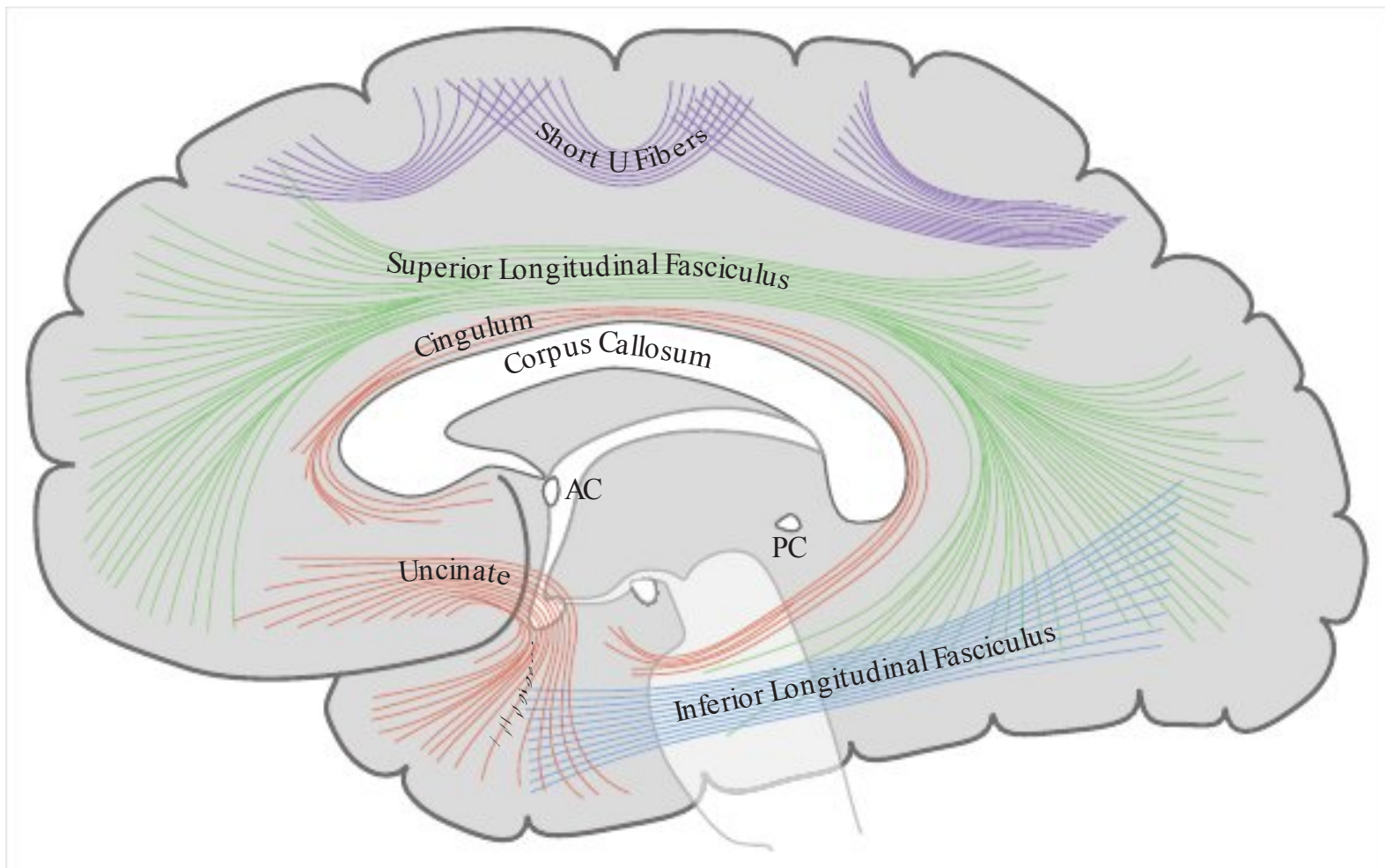


Fig. 13.2 White matter tracts (color conventions for DTI are not used in this anatomic diagram)

- Commissural fibers: medio-laterally oriented, connecting the cerebral hemispheres
 - Corpus callosum
 - Anterior commissure
 - Posterior commissure
- Association fibers: connect regions within the same hemisphere
 - U-fibers: connect adjacent gyri
 - Long association fibers: connect more distant areas
 - Optic radiations: connect lateral geniculate bodies to visual cortex. Pass lateral to the body of the lateral ventricles.
 - Uncinate fasciculus: connects the anterior temporal lobe to the inferior frontal gyrus. Damage can cause language deficits.
 - Superior longitudinal fasciculus (SLF): connects regions of frontal lobe to temporal and occipital lobes. Injury can cause language deficits.
 - Arcuate fasciculus: part of SLF. Classic neuroanatomy teaching: connects the superior and middle frontal gyri (Broca's area (motor speech)) to the superior temporal gyrus (Wernicke's area (language comprehension)). DTI has suggested broader connections, including premotor cortex. Injury causes conduction aphasia
 - Inferior longitudinal fasciculus (ILF): connects temporal and occipital lobes at the level of the optic radiation. Injury can cause deficits in object recognition, visual agnosias, prosopagnosia (face blindness)
 - Cingulum: project from cingulate gyrus to the entorhinal cortex as part of the limbic system

The convention for color coding tracts on DTI images¹⁹:

- Blue: superior-inferior tracts
- Red: mediolateral (horizontal) tracts
- Green: anterior-posterior tracts

Owing to a number of technical considerations, DTI is somewhat more operator- dependent than conventional MRI.

For surgical planning, the goal is to keep the surgical trajectory roughly parallel (at $<30^\circ$ angle) to the long axis of the white matter tract that one is trying to preserve (unproven hypothesis²⁰).

Surgical “corridors” have been described taking into consideration preservation of white matter tracts:

- Anterior corridor: parallel to association fibers, between the SLF and the cingulum
- Posterior corridor: enters at the parieto-occipital sulcus, passes adjacent to the optic radiations
- Lateral corridor

13.3 Angiography

See Endovascular Neurosurgery section (p.1575).

13.4 Myelography

Contraindications:

1. anticoagulation
2. allergy to iodinated contrast: requires iodine allergy prep (p.221). NB: risk of adverse reaction still persists
3. infection at the desired puncture site
4. extensive midline lumbar spinal fusion may preclude needle access to the subarachnoid space

Lumbar myelogram

Using iohexol (Omnipaque® 140 or 180) as shown in □ Table 12.4.

Cervical myelogram with water soluble contrast via LP

Use iohexol (Omnipaque® 300 or 240) as shown in □ Table 12.4. Insert spinal needle into lumbar subarachnoid space, tilt the head of the myelogram table down with the patient's neck extended and then inject dye. If a complete cervical block is seen, have patient flex neck. If the block cannot be traversed, patient may need C1–2 puncture or MRI (first obtain a CT which may show dye above the block that cannot be appreciated on myelography alone).

Post myelographic CT

Increases sensitivity and specificity of myelography (p.1031). In cases of complete block on myelogram, CT will often show dye distal to the apparent site of the block.

13.5 Radionuclide scanning

13.5.1 Three phase bone scan

Technetium-99 (99mTc) pertechnetate is a radioisotope that may be attached to various substrates for use in bone scanning. It may be used to label polyphosphate (rarely used today), diphosphonate²¹ (MDP), or phosphorous (HDP) (the most widely agent used currently). Accumulates in areas of osteoblastic activity.

With technetium 99m-HDP, images are obtained immediately after injection (flow phase), at 15 min (blood pooling) and in 4 hours (bone imaging). Cellulitis shows up as increased activity in the first 2 phases, and there is little or diffuse increased activity in the 3rd. Osteomyelitis causes increased uptake in all 3 phases.

Used in evaluation of acute osteomyelitis with sensitivity and specificity of $\approx 95\%$ each, and is usually positive within 2–3 days. False positives can occur in conditions involving increased bone turnover, e.g. fracture, septic arthritis, tumors. False negative can occur in cases with associated bone infarction.

Applications for bone scans include:

1. infection
 - a) osteomyelitis of the spine – vertebral osteomyelitis (p.355) – or skull
 - b) discitis (p.356)
2. tumor
 - a) spine metastases (p.818)
 - b) primary bone tumors of the spine (p.792)
 - c) skull tumors (p.775)
3. diseases involving abnormal bone metabolism
 - a) Paget's disease: of the skull or spine (p.1120)
 - b) hyperostosis frontalis interna (p.780)
4. craniosynostosis (p.252)
5. fractures: spine or skull
6. "low back problems" (p.1032): to help identify some of the above conditions

13.5.2 Gallium scan

Nuclear medicine scan with ⁶⁷Ga citrate which accumulates in areas of inflammation and some malignancies. Utility in neurosurgery for: sarcoidosis (p.189), chronic vertebral osteomyelitis; see also comparison to bone scan (p.355).

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14 Electrodiagnostics

14.1 Electroencephalogram (EEG)

14.1.1 General information

The primary use of EEG is in the diagnosis and management of seizure disorders. Non-convulsive use of EEG is essentially limited to monitoring for burst suppression (see below) (e.g. induced barbiturate coma) or for differential diagnosis of diffuse encephalopathy, including:

1. differentiating psychogenic unresponsiveness from organic: a normal EEG indicates either psychiatric unresponsiveness or locked-in syndrome
2. non-convulsive status epilepticus (seizures): absence or complex partial status
3. subclinical focal abnormalities: e.g. PLEDs (see below), focal slowing...
4. specific patterns diagnostic for certain pathologies: e.g.:
 - a) periodic lateralizing epileptiform discharges (PLEDs): may occur with any acute focal cerebral insult (e.g. herpes simplex encephalitis (HSE), abscess, tumor, embolic infarct): seen in 85% of cases of HSE (onset 2–5 d after presentation), if bilateral is $\approx$ diagnostic of HSE
 - b) subacute sclerosing panencephalitis (SSPE) (pathognomonic pattern): periodic high voltage with 4–15 secs separation with accompanying body jerks, no change with painful stimulation (differential diagnosis includes PCP overdose)
 - c) Creutzfeldt-Jakob disease (p.367): myoclonic jerks. EEG $\rightarrow$ bilateral sharp wave 1.5–2 per second (early $\rightarrow$ slowing; later $\rightarrow$ triphasic). May resemble PLEDs, but are reactive to painful stimulation (most PLEDs are not)
 - d) triphasic waves: not really specific. May be seen in hepatic encephalopathy, post-anoxia, and hyponatremia
5. objective measure of severity of encephalopathy: usually used for anoxic encephalopathy (e.g. periodic spikes with seizures indicates $<5\%$ chance of normal neurologic outcome, with high mortality). Alpha coma, burst suppression, and electrocerebral silence are all poor prognosticators
6. differentiating hydranencephaly (p.288) from severe hydrocephalus
7. as a clinical confirmatory test in the determination of brain death (p.310)

14.1.2 Common EEG rhythms.

Common EEG rhythms are shown in [Table 14.1](#).

14.1.3 Burst suppression

Isoelectric intervals interrupted by bursts of 8–12 Hz electrical activity that diminish to 1–4 Hz prior to electrical silence.¹ Often used as an endpoint for titrating neuroprotective drugs such as barbiturates, propofol...

14.2 Evoked potentials

14.2.1 General information

Evoked potentials are averaged EEG waveforms recorded following repetitive stimulation. The process of averaging nulls-out EEG activity that is not time -locked to the stimulus. Resultant waveforms contain peaks that are named N (negative – upward deflection) or P (positive – downward deflection) followed by the latency in milliseconds to the onset of the peak.

Table 14.1 Common EEG rhythms		
Rhythm	Symbol	Frequency
delta	Δ	0–3 Hz
theta	θ	4–7 Hz
alpha	α	8–13 Hz
beta	β	>13 Hz

14.2.2 Sensory evoked potentials (SEP)

General information

May use electrical stimulation of peripheral nerves (somatosensory or (SSEP)), auditory clicks through earphones (auditory or AEP, AKA BAER (brainstem auditory evoked response)) or flashing lights through goggles (visual EP or VEP).
Evoked potentials are most commonly used by neurosurgeons for intraoperative monitoring purposes. SSEP (especially from median nerve stimulation) also has prognostic significance in cervical spondylotic myelopathy² although use for this purpose is limited.

Typical waveforms

Abbreviations

Abbreviations used below: BAER=brain stem auditory evoked response; UE/LE SSEP=upper/lower extremity somatosensory evoked potential; PR VER=pattern reversal visual evoked response which requires patient cooperation and visual attention as opposed to flash VER which may even be done through closed eyelids. See also references.^{3,4}

14.2.3 Intra-operative evoked potentials

General information

EPs may be used for intra-operative monitoring (e.g. monitoring hearing during resection of vestibular schwannomas, or monitoring SSEPs during some spine surgery), however, their delayed nature often makes them of limited usefulness in avoiding acute intra-operative injury. A 10%increased latency of

Table 14.2 Typical stimulus values for intra-op evoked potentials				
Test	Stimulus			Comment
	Freq (Hz)	Duration (mcS)	Magnitude	
BAER	23.5	150	85–100 dB	rarefaction usually better than compression
UE SSEP (median nerve at wrist)	4.7	300–700	up to 50 mA	supramaximal stimulus (sensory threshold + motor threshold)
LE SSEP (posterior tibial at ankle)	4.7	300–700	up to 100 mA	supramaximal stimulus
PR VER	1.97			16 × 16 checks, 1.6 cm each, at 1 meter (subtends 55" arc visual angle)

Table 14.3 Input characteristics for acquiring evoked potentials					
Test	Analysis				Electrode derivations
	Input filter (Hz)	Sensitive (mcV)	Duration (mS)	Reps	
BAER	150–3000	25	15	1500	M ₁ ^a -C _Z Z, M ₂ [*] -C _Z , ground = FZ
UE SSEP	30–3000	50	55–60	600	F _Z -Erb’s point, C _{V7} -F _{PZ} , C ₃ -F _{PZ} , C ₃ ’-NC (non-cephalic, e.g. shoulder)
LE SSEP	30–3000	50	60	600	popliteal fossa (front to back), C _Z -F _{PZ} , back (L ₅ -T ₁₂) (difficult in obese or elderly), C _r -C _c (optional: somatosensory ipsilateral to contralateral)
PR VER	5–100	50	500	100	O ₁ -A ₁ , O _Z -A ₁ , O ₂ -A ₁ , O _Z -C _Z
^a M= mastoid (“i” is ipsilateral to stimulus, and “c” is contralateral)					

a major EP peak, or a drop in amplitude of $\geq 50\%$ is significant and should cause the surgeon to assess all variables (retractors, instruments, blood pressure...). Intra-operative SSEPs may also be used to localize primary sensory cortex in anesthetized patients (as opposed to using brain mapping techniques in awake patients) by looking for phase reversal potentials across the central sulcus.^{5,6}

Brainstem auditory evoked responses (BAER)

AKA auditory brainstem response (ABR), AKA auditory evoked potential (AEP). Auditory clicks are delivered to the patient by earphones. Peaks (Table 14.4). Once used for assistance in diagnosing vestibular schwannomas, their use for intraoperative monitoring is limited and has been largely replaced by direct eighth cranial nerve monitoring which provides more rapid information for the surgeon.

SSEP monitoring during spine surgery

Paralytics actually improve SSEP recordings by reducing muscle artifact, but will abolish the visible twitch that confirms that the stimulus is being received.

Typical stimulus sites: median, ulnar, and tibial nerves. Impulses ascend in the ipsilateral posterior column. UE SSEPs travel primarily in the dorsal columns, but LE SSEPs are carried mostly in the dorsolateral fasciculus (p.72) which is supplied by the anterior spinal artery. Thus, UE and LE SSEPs are more sensitive to direct mechanical effects primarily on the posterior spinal cord (sensory) and may remain unchanged with some injuries to the anterior cord (motor), however, LE SSEPs can detect ischemic effects on the anterior cord by virtue of involvement of the anterior spinal artery.

In a personal series of 809 patients,⁷ 17 had SSEP degradation, 14 of these (82%) responded to intra-op interventions (see below), and in 13 of these 14 (93%) there were no new deficits. In the 3 that did not respond, 2 had significant new neurologic deficits.

Table 14.4 Evoked potential waveforms (note: values may differ from lab to lab)		
Test	Figure	Possible generators
BAER		CM cochlear microphonic P ₁ 1 distal VIII nerve, P ₂ 2 proximal VIII or cochlear nucleus, P ₃ 3 lower pons (? superior olivary complex), P ₄ 4 mid-upper pons, P ₅ 5 upper pons or inferior colliculus
UE SSEP		N ₉ 9 (on F _Z -E _p P where E _p is Erb's point) AKA EP: entry of volley into distal brachial plexus, N ₁₁ 11 (on C _{V7} -F _{PZ}): root entry zone (cervical region), N ₁₃ 13 cervicomedullary junction (recorded from C2), N ₁₉ 19 primary sensory cortex, P ₂₂ 22 (early) motor cortex, P ₂₂ 22 (late) IPSP "reaction" to N ₁₈ 18
LE SSEP		P ₂₂ 22 (on L ₅ -T ₁₂): lumbo-sacral plexus, P ₄₀ 40 (on C _Z -F _{PZ}): sensory cortex (analogous to N18 in UE SSEP, reversed in polarity for ? reason), N ₂₇ 27 (on C _{V7} -F _{PZ}): ? dorsal column nucleus N ₃₀ 30
PR VER		P ₁₀₀ striate & pre-striate occipital cortex, with contributions from thalamocortical volleys

Table 14.5 Normal values for evoked potentials ^a (note: values may differ from lab to lab)				
Test	Parameters measured	— — Normal values — —		Comment
		Mean	+2.5 std dev	
BAER	I-V peak latency	4.01 mS	4.63 mS	
	I-III peak latency	2.15 mS	2.66 mS	prolongation suggests lesion between pons & colliculus, often vestibular schwannoma
	V absolute latency	5.7 mS	6.27 mS	
	III-V peak latency			prolongation suggests lesion between lower pons & midbrain, may be seen in M.S.
UE SSEP	N ₉ 9-N ₁₈ 18 peak latency	9.38 mS	11.35 mS	
IE SSEP	P ₂₂ 22-P ₄₀ 40 peak latency	15.62 mS	20.82 mS	
	P ₄₀ 40 absolute latency	37.20 mS	44.16 mS	
PR VER	P ₁₀₀ 10 absolute latency		+3 S.D.	
	P ₁₀₀ inter-eye difference	8–10 mS		Inter-eye difference is more sensitive with full field stimulation. Monocular defect suggests conduction defect in that optic nerve anterior to chiasm (e.g. M.S., glaucoma, compression retinal degeneration). Bilateral defect does not localize.
^a normal values in boldface are critical values used as cutoff for abnormal results				

Transcranial motor evoked potentials (TCMEPs)

Anesthetic requirements: In addition to EP anesthetic requirements, neuromuscular blockade must be minimized to permit ≥2 out of 4 twitches.

AKA motor evoked potentials (MEP): transcranial electrical or magnetic stimulation of motor cortex and descending motor axons with recording of motor potentials from distal spinal cord or muscle groups. Using direct electrical stimulation is limited in awake patients by local pain. Due to the large potentials, the acquisition time is shorter and feedback to the surgeon is almost immediate. However, due to patient movement from the muscle contractions, continuous recording is usually not possible (except with monitoring the response over the spinal cord). Useful for surgery involving the spinal cord (cervical or thoracic), no utility for lumbar spine surgery. Seizures occur rarely, usually in patients with increased seizure risk and with high-rate stimulation frequency.

- Contraindications to MEP:
1. history of epilepsy/seizures
 2. past surgical skull defects
 3. metal in head or neck
 4. use special care with implanted electronic devices

Descending evoked potentials (DEP)

(Formerly referred to by the misleading term “neurogenic motor evoked potentials”). Rostral stimulation of the spinal cord with recording of a caudal neurogenic response from the spinal cord or peripheral nerve, or a myogenic response from a distal muscle. DEPs can be mediated primarily by sensory nerves and therefore do not represent true motor potentials. However, shown to be sensitive to spinal cord changes and may be useful when TCMEPs cannot be obtained.

14.2.4 Electrophysiologic monitoring criteria to trigger notification of surgeon

Any of the following:

1. SSEP:
 - a) 50% decrease in peak signal amplitude from baseline
 - b) increase in peak latency > 10%
 - c) complete loss of a waveform
2. TCMEP: sustained 50% decrease in signal amplitude
3. DEP: decrease in signal of > 60%

Interventions for loss or degradation of monitoring signal during spine surgery

When compression is the culprit, the prognosis may be good. Vascular injuries generally do not fare as well.

Options/suggestions include (adapted/excerpted from the “Vitale checklist”⁸):

1. verify that the change is real (check connections, equipment...)
2. place OR on alert status
 - a) announce intraoperative pause and stop the case
 - b) eliminate possible distractions (music, unnecessary conversations...)
 - c) “muster the troops”: the attending anesthesiologist, senior neurologist or neurophysiologist and experienced nurse are called to the room. Consult a surgical colleague if necessary
3. anesthetic/metabolic considerations
 - a) optimize mean arterial pressure (MAP usually > 85 mm Hg preferred)
 - b) check hematocrit for anemia (could contribute to cord ischemia)
 - c) optimize blood pH (rule out acidemia) and pCO₂
 - d) normalize patient body temperature
 - e) check anesthetic technical factors: assess extent of paralytics
 - f) discuss possibility of “Stagnara wake up test” (see below) with attending anesthesiologist and scrub nurse
4. technical/neurophysiologic considerations
 - a) rule-out 60 Hz. interference from other equipment (OR table, C-arm, microscope... anything with a plug)
 - b) verify that stimulating electrodes recording leads are making good contact

14.3 NCS/EMG

14

14.3.1 General information

Electrodiagnostic studies of peripheral nerves consist of two parts:

1. conduction measurements: typically referred to as “NCV” (nerve conduction velocity) but technically should be called NCS (nerve conduction studies) since amplitude, latency and duration of motor & sensory nerves are also evaluated
2. electromyogram (EMG) AKA “needle exam” (see below)

14.3.2 Electromyography

General information

There are 3 phases of an EMG exam:

Phase 1 – insertional activity: the electric response of the muscle to mechanical irritation caused by small movements of the needle

Phase 2 – spontaneous activity: in muscle at rest

1. normal: silent with stationary needle once insertional activity has subsided
2. spontaneous activity: independently produced electrical activity. Usually abnormal (although sometimes seen in normal volunteers).
 - a) after denervation (secondary to a nerve injury) or muscle injury:
 - positive sharp waves (PSW)
 - fibrillation potentials (AKA fibrillations or fibs): action potentials arising from single muscle fibers. Detectable on EMG; not visible to the naked eye, c.f. fasciculations (p.505). Earliest onset 7–10 days after denervation, sometimes not for 3–4 weeks. If the nerve recovers, it

may reinnervate the muscle, but with larger motor units resulting in longer duration and decreased numbers

- b) myotonic discharges (“dive bomber” sound on speaker monitor)
- c) complex repetitive discharge (CRD): ephaptic conduction of groups of adjacent muscle fibers. Occurs in neuropathic or myopathic disorders
- d) fasciculation potentials: nonspecific, but typically associated with motor neuron disease (ALS) (p.183)
- e) other less common spontaneous activity includes: myokymic, neuromyotonic and cramp discharges

Phase 3 – volitional activity: evaluated with minimal volitional effort and maximal effort

1. motor unit action potential (MUAP) analysis: includes evaluation of motor unit amplitude, duration, polyphasia and stability. Generally increase amplitude and duration suggest a disorder of the LMN, and reduction of amplitude and duration suggests a primary myopathic disorder
2. with minimal volitional effort. Two possible abnormal findings
 - a) reduced recruitment (or fast firing) is always indicative of a neuropathic process
 - b) early or increased recruitment: indicative of a myopathic process
3. with maximal effort

Definitions

SNAP: sensory nerve action potential. Key concept: since the ganglion of the sensory nerves lies within the neural foramen, preganglionic lesions (injury to the nerve root proximal to the neural foramen, e.g. root compression by herniated disc or root avulsion) does not affect the cell body, and therefore the distal SNAP is unaffected.⁹ Postganglionic lesions (distal to the neural foramen, e.g. peripheral nerve injury) reduces SNAP amplitudes and/or slows the sensory conduction velocity.

F-wave: stimulation of a nerve causes orthodromic and antidromic conduction. Some anterior horn cells that are stimulated antidromically will fire orthodromically producing the F-wave. F-wave latency may be prolonged in multilevel radiculopathy (not sensitive). Most helpful in evaluating proximal root slowing, e.g. GBS (p.184).

H-reflex: practical $\approx$ only in S1 nerve root, similar information to the ankle jerk. Stimulation of Ia afferent fibers passes through a monosynaptic connection causing an orthodromic alpha-motor action potential that can be measured in the triceps surae.

Volitional activity: the motor unit action potential (MUAP) can be assessed only with voluntary muscle contraction by the patient. Components of the MUAP measured include: amplitude, rise time, duration, and number of phases (crossings of the baseline).

Polyphasic potentials: MUAPs with >4 phases. Normally comprise $<15\%$ of MUAPs. Following a nerve injury, abnormally increased polyphasic potentials can be seen 6–8 weeks after reinnervation begins, gradually increase over several months, and then begin to wane (as firing becomes more synchronous).

Myotonia: there are a number of myotonic conditions, including myotonic dystrophy in which there is sustained contraction of the muscle. Classic EMG finding: “dive bomber” sound due to myotonic discharges.

EMG for radiculopathy

EMG pearls for the neurosurgeon

General principles:

- EMG – if a reliable motor exam can be done, the EMG will not likely add any information. A normal motor exam will usually be associated with a normal EMG
- EMG is not extremely sensitive for radiculopathy (e.g. irritative radiculopathy might not be picked up), this is more likely in the cervical region than lumbar. However, when abnormal, EMG is very specific
- EMG is best reserved for cases with documented weakness where additional localizing/prognostic information is needed, or when the patient’s strength cannot be reliably assessed (inability to cooperate, functional overlay...)
- Timing
 - It takes about 3 weeks after onset of radiculopathy for the EMG to reliably show any findings
 - “Acute changes” begin at about 3 weeks and can last up to about 6 months
 - Chronic changes can be seen starting at about 6 months, and may persist indefinitely

Cervical EMG:

- EMG is most helpful for nerve roots C5-T1. There are no good muscles to reliably test C3-4, and compression here may cause findings in lower nerve roots

Lumbar EMG:

- If lumbar MRI is normal in a patient with evidence of motor weakness (e.g. foot drop), do an EMG to look for peripheral neuropathy (again, a good motor exam can give the same info). If the EMG is negative for peripheral neuropathy (e.g. peroneal nerve palsy) then do an MRI (or CT) of abdomen and pelvis to look for pelvic floor tumor

Findings

Include spontaneous activity (fibs & PSWs, see above).

The earliest possible finding (within 2–3 d) is reduced recruitment with volitional activity, but this occurs only with significant compression of motor fibers.

EMG is useful if there is a concern about possible overlapping peripheral neuropathy (e.g. carpal tunnel syndrome vs. C6 radiculopathy).

EMG criteria for radiculopathy

1. fibrillations and/or positive sharp waves in at least 2 muscles innervated by a single nerve root in question, but by 2 different peripheral nerves
2. abnormal paraspinals: this supports the diagnosis, but is not required since paraspinals will be normal in $\approx 50\%$

Lumbar radiculopathy from herniated disc

With radiculopathy, SNAP is usually normal (see above). Paraspinal muscle fibrillations may occur. Accuracy in predicting level of involvement¹⁰ is $\approx 84\%$

Foot drop: the short head of the biceps femoris in the LE is the first muscle innervated by the peroneal division of the sciatic nerve at or just above the popliteal fossa just after the nerve splits off from the sciatic nerve. In cases e.g. of foot drop it is a good muscle to test to determine if there is a peroneal neuropathy vs. a more proximal lesion (i.e. above the popliteal fossa).

Findings with healing radiculopathy (e.g. following discectomy or spontaneous healing):

- motor potentials return first (if nerve injury were “complete”, it would take a month to return)
- if lost, sensory potentials return last or may not return
- following laminectomy, paraspinal potentials may no longer be useful for EMG because cutting the muscles during surgery alters their electrical signals resulting in effective denervation due to muscle injury. Fibs and PSWs decrease in amplitude over time but may remain present indefinitely

EMG in plexopathy

Reduction of SNAP with no paraspinal muscle fibrillations (the dorsal rami exit proximally to innervate the paraspinals, and are involved $\approx$ only with root lesions).

EMG in nerve root avulsion

Produces muscles weakness and sensory loss with normal SNAP since the lesion is proximal to the dorsal root ganglion (where the cell bodies for the sensory nerves are located).

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Part IV

Developmental Anomalies

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IV

15 Primary Intracranial Anomalies

15.1 Arachnoid cysts, intracranial

15.1.1 General information

Key concepts

- a congenital abnormality, most common in middle fossa, cerebellopontine angle (CPA), suprasellar region, and posterior fossa
- most are asymptomatic (i.e. an incidental finding) except in the suprasellar region
- imaging often shows remodeling of bone; imaging characteristics exactly mimic CSF on CT or MRI in most cases
- recommendation for incidentally discovered arachnoid cyst in adults: a single follow-up imaging study in 6–8 months is usually adequate to rule-out any increase in size. Subsequent studies only if concerning symptoms develop

AKA leptomeningeal cysts, distinct from posttraumatic leptomeningeal cysts (AKA growing skull fractures) (p.915), and unrelated to infection. Arachnoid cysts (AC) are congenital lesions that arise during development from splitting of arachnoid membrane (thus they are technically intra-arachnoid cysts) and contain fluid that is usually identical to CSF. They do not communicate with the ventricles or subarachnoid space. May be uniloculated or may have septations. Typically lined with meningotheial cells positive for epithelial membrane antigen (EMA) and negative for carcinoembryonic antigen (CEA). AC may also occur in the spinal canal.

“Temporal lobe agenesis syndrome” is a label that had been used to describe the findings with middle cranial fossa ACs. This term is now obsolete since brain volumes on each side are actually the same,¹ bone expansion and shift of brain matter account for the parenchyma that appears to be replaced by the AC.

- Two types of histological findings²:
1. “simple arachnoid cysts”: arachnoid lining with cells that appear to be capable of active CSF secretion. Middle fossa cysts seem to be exclusively of this type
 2. cysts with more complex lining which may also contain neuroglia, ependyma, and other tissue types

15.1.2 Epidemiology of intracranial arachnoid cysts

Incidence: 5 per 1000 in autopsy series. Comprise ≈ 1% of intracranial masses.
Male:female ratio is 4:1. More common on the left side.
Bilateral arachnoid cysts may occur in Hurler syndrome (a mucopolysaccharidosis).

15.1.3 Distribution

Almost all occur in relation to an arachnoid cistern (exception: intrasellar, the only one that is extradural, □ Table 15.1).

Epidermoid cysts in the cerebellopontine angle (CPA) may mimic an arachnoid cyst, but are high signal on DWI MRI.

See also differential diagnosis of midline posterior fossa arachnoid cysts (p.256).

15.1.4 Presentation

Most ACs are asymptomatic. Those that become symptomatic usually do so in early childhood.⁴ The presentation varies with location of the cyst, and oftentimes appear mild considering the large size of some.

- Typical presentations are shown in □ Table 15.2 ⁴ and include:
1. symptoms of intracranial hypertension (elevated ICP): H/A, N/V, lethargy
 2. seizures
 3. sudden deterioration:

Table 15.1 Distribution of arachnoid cysts ³	
Location	%
sylvian fissure	49%
CPA	11%
supracollicular	10%
vermian	9%
sellar & suprasellar	9%
interhemispheric	5%
cerebral convexity	4%
clival	3%

Table 15.2 Typical presentations of arachnoid cysts		
Middle fossa cysts	Suprasellar cysts with hydrocephalus	Diffuse supra- or infratentorial cysts with hydrocephalus
seizures headache hemiparesis	intracranial hypertension craniomegaly developmental delay visual loss precocious puberty bobble-head doll syndrome	intracranial hypertension craniomegaly developmental delay

- a) due to hemorrhage (into cyst or subdural compartment): middle fossa cysts are notorious for hemorrhage due to tearing of bridging veins. Some sports organizations do not allow participation in contact sports for these patients
- b) due to rupture of the cyst
- 4. as a focal protrusion of the skull
- 5. with focal signs/symptoms of a space occupying lesion
- 6. incidental finding discovered during evaluation for unrelated condition
- 7. suprasellar cysts may additionally present with⁵:
 - a) hydrocephalus (probably due to compression of the third ventricle)
 - b) endocrine symptoms: occurs in up to 60% Includes precocious puberty
 - c) head bobbing (the so-called “bobble-head doll syndrome”⁶): considered suggestive of suprasellar cysts, but occurs in as few as 10%
 - d) visual impairment

15.1.5 Evaluation

General information

Routine evaluation with CT or MRI is usually satisfactory. Further evaluation with CSF contrast or flow studies (cisternograms, ventriculograms...) are only occasionally necessary for the diagnosis of midline suprasellar and posterior fossa lesions⁴; for Differential diagnosis, see Intracranial cysts (p.1374); see □ Fig. 15.1 for the classification scheme of Galassi et al for middle fossa cysts.

CTscan

Smooth bordered non-calcified extraparenchymal cystic mass with density similar to CSF and no enhancement with IV contrast. Expansion of nearby bone by remodelling is usually seen, confirming their chronic nature. Often associated with ventriculomegaly (in 64% of supratentorial and 80% of infratentorial cysts).

Convexity or middle fossa cysts exert mass effect on adjacent brain and may compress ipsilateral lateral ventricle and cause midline shift. Suprasellar, quadrigeminal plate, and midline posterior-fossa cysts may compress the third and fourth ventricle and cause hydrocephalus by obstructing the foramina of Monro or the Sylvian aqueduct.

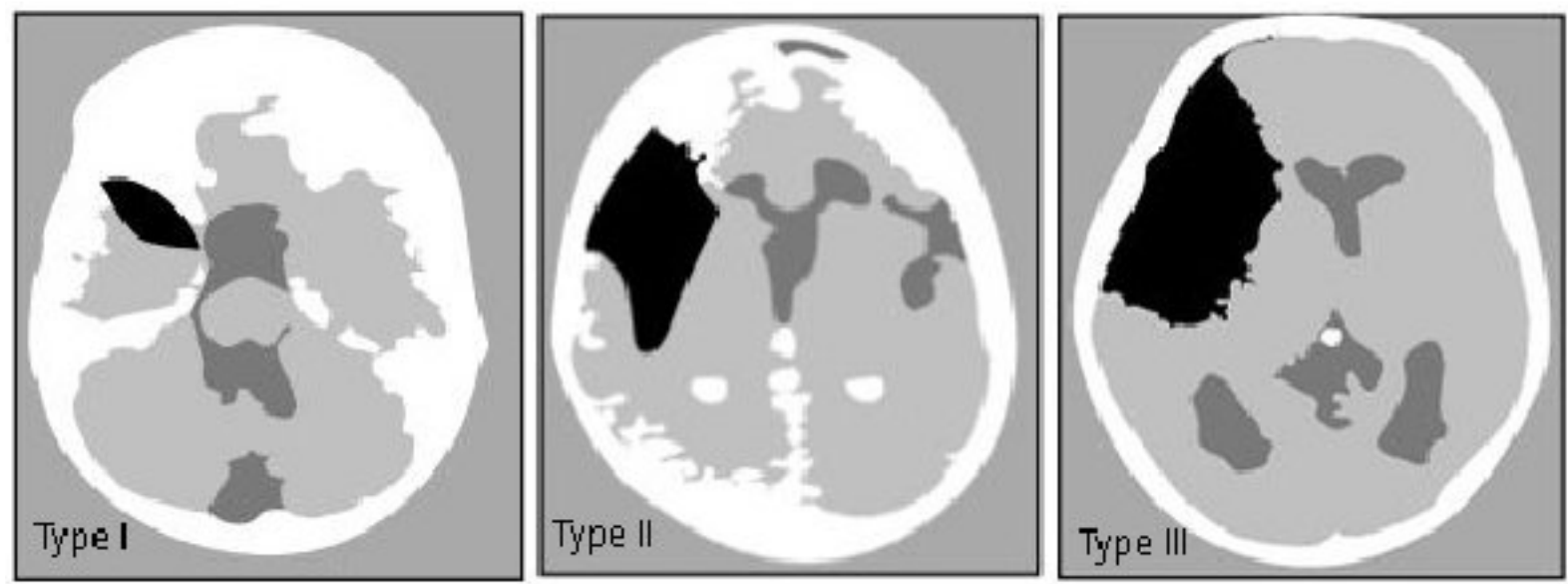


Fig. 15.1 CT Classification of Sylvian fissure arachnoid cysts.⁷
Type I: small, biconvex, located in anterior temporal tip. No mass effect. Communicates with subarachnoid space on water-soluble contrast CT cisternogram (WS-CTC).
Type II: involves proximal and intermediate segments of Sylvian fissure. Completely open insula gives rectangular shape. Partial communication on WS-CTC.
Type III: involves entire Sylvian fissure. Marked midline shift. Bony expansion of middle fossa (elevation of lesser wing of sphenoid, outward expansion of squamous temporal bone). Minimal communication on WS-CTC. Surgical treatment usually does not result in total reexpansion of brain (may approach type II lesion).

MRI

Better than CT in differentiating the CSF contained in arachnoid cysts from the fluid of neoplastic cysts. May also show cyst walls.

Cisternograms and/or ventriculograms

Using either iodinated contrast or radionuclide tracers. Variable rate of opacification has resulted in difficulty correlating results with operative findings. Some cysts are actually diverticula, and may fill with radiotracer or contrast.

15.1.6 Treatment

General information

Many (but not all) authors recommend not treating arachnoid cysts that do not cause mass effect or symptoms, regardless of their size and location. For incidentally discovered arachnoid cyst in an adult not considered for surgery: a single follow-up imaging study in 6–8 months is usually adequate to rule-out any changes (since they may grow in size). Subsequent studies may be done if concerning symptoms develop. Pediatric patients may need to be followed until adulthood.

Treatment considerations for cysts (excluding suprasellar cysts)

Surgical treatment options are summarized in □ Table 15.3.

Cyst shunting

Probably the best overall treatment. For shunting into peritoneum, use a low pressure valve. If there is concurrent ventriculomegaly, one may simultaneously place a ventricular shunt (e.g. through a “Y” connector). Ultrasound, ventriculoscope, or image guidance may assist in locating suprasellar cysts. Shunting of middle fossa ACs may also be accomplished through the lateral ventricle, thus shunting both compartments.⁹

□ NB: in running the distal shunt tubing from the middle fossa, it should be routed behind the ear (do not tunnel in front of ear to avoid injury to facial nerve – if this anterior route is unavoidable, it may help to solicit the services of a plastic surgeon to help avoid the facial nerve).

Treatment of Suprasellar cysts

These cysts present with some unique treatments options which include:

- transcallosal cystectomy¹⁰

Table 15.3 Surgical treatment options for arachnoid cysts		
Procedure	Advantages	Disadvantages
drainage by needle aspiration or burr hole evacuation	• simple • quick	• high rate of recurrence of cyst and neurologic deficit
craniotomy, excising cyst wall and fenestrating it into basal cisterns	• permits direct inspection of cyst (may help with diagnosis) • loculated cysts (rare) treated more effectively • avoids permanent shunt (in some cases) • allows visualization of bridging vessels (small advantage)	• subsequent scarring may block fenestration allowing reaccumulation of cyst • flow through subarachnoid space may be deficient; many patients develop shunt dependency post-op • significant morbidity and mortality (may be due to abrupt decompression)
endoscopic cyst fenestration through a burr hole ⁸	• as above	• as above
shunting of cyst into peritoneum or into vascular system	• definitive treatment • low morbidity/mortality • low rate of recurrence	• patient becomes “shunt dependent” • risk of infection of foreign body (shunt)

- percutaneous ventriculo-cystostomy: procedure of choice of Pierre-Kahn et al.⁵ Performed via a paramedian coronal burr hole through the lateral ventricle and foramen of Monroe (may be facilitated by using a ventriculoscope⁸)
- subfrontal approach (for fenestration or removal): dangerous and ineffective⁵

□ ventricular drainage is ineffective (actually promotes cyst enlargement) and should not be routinely considered

15.1.7 Outcome

Even following successful treatment a portion of the cyst may remain due to the remodeling of the bone and chronic shift of brain contents. Hydrocephalus may develop following treatment. Endocrinopathies tend to persist even after successful treatment of suprasellar cysts.

15.2 Craniofacial development

15.2.1 Normal development

Fontanelles

Anterior fontanelle: the largest fontanelle. Diamond shaped, 4 cm (AP) × 2.5 cm (transverse) at birth. Normally closes by age 2.5 yrs.

Posterior fontanelle :triangular. Normally closes by age 2–3 mos.

Sphenoid and mastoid fontanelles: small, irregular. Normally, former closes by age 2–3 mos, latter by age 1 yr.

Cranial vault

Growth: largely determined by growth of brain. 90%of adult head size is achieved by age 1 yr; 95% by age 6 yrs. Growth essentially ceases at age 7 yrs. By end of 2nd yr, bones have interlocked at sutures and further growth occurs by accretion and absorption.

Skull is unilaminar at birth. Diplöe appear by 4th yr and reach a maximum by age 35 yrs (when diploic veins form).

Mastoid process: formation commences by age 2 yrs, air cell formation occurs during 6th yr.

15.2.2 Craniosynostosis

General

Originally called craniostenosis. Incidence: $\approx 0.6/1000$ live births.

Primarily a prenatal deformity, postnatal craniosynostosis (CSO) occurs uncommonly (postnatal causes consist primarily of positional alterations which may not represent true synostosis). CSO is rarely associated with hydrocephalus (HCP).¹¹ The assertion that CSO may follow CSF shunting for HCP is unproven. Other causes for failure of normal skull growth include lack of brain growth due to any of the causes of arrested development of the cerebral hemispheres (lissencephaly, micropolygyria, some cases of hydranencephaly...).

Treatment is usually surgical. In most instances, the indication for surgery is for cosmesis and to prevent the severe psychological effects of having a disfiguring deformity. However, with multiple CSO, brain growth may be impeded by the unyielding skull. Also, ICP may be pathologically elevated, and although this is more common in multiple CSO,¹² elevated ICP occurs in $\approx 11\%$ of cases with a single stenotic suture. Coronal synostosis can cause amblyopia. Most cases of single suture involvement can be treated with linear excision of the suture. Involvement of multiple sutures or the skull base usually requires the combined efforts of a neurosurgeon and craniofacial surgeon, and may need to be staged in some cases. Risks of surgery include: blood loss, seizures, stroke.

Diagnosis

Many cases of “synostosis” are really due to positional flattening (e.g. “lazy lambdoid”, see below). If this is suspected, instruct parents to keep head off of flattened area and recheck patient in 6–8 weeks: if it was positional, it should be improved, if it was CSO then it usually declares itself. The diagnosis of CSO may be aided by:

1. palpation of a bony prominence over the suspected synostotic suture (exception: lambdoidal synostosis, see below)
2. gentle firm pressure with the thumbs fails to cause relative movement of the bones on either side of the suture
3. plain skull x-rays:
 - a) lack of normal lucency in center of suture. Some cases with normal x-ray appearance of the suture (even on CT) may be due to focal bony spicule formation¹³
 - b) beaten copper calvaria (p.254), sutural diastasis and erosion of the sella may be seen in cases of increased ICP¹⁴
4. CTscan:
 - a) helps demonstrate cranial contour
 - b) may show thickening and/or ridging at the site of synostosis
 - c) will demonstrate hydrocephalus if present
 - d) may show expansion of the frontal subarachnoid space¹⁵
 - e) three-dimensional CT may help better visualize abnormalities
5. in questionable cases, a technetium bone scan can be performed¹⁶:
 - a) there is little isotope uptake by any of the cranial sutures in the first weeks of life
 - b) in prematurely closing sutures, increased activity compared to the other (normal) sutures will be demonstrated
 - c) in completely closed sutures, no uptake will be demonstrated
6. MRI: usually reserved for cases with associated intracranial abnormalities. Often not as helpful as CT
7. measurements, such as occipito-frontal-circumference may not be abnormal even in the face of a deformed skull shape

Increased ICP

Evidence of increased ICP in the newborn with craniosynostosis include:

1. radiographic signs (on plain skull x-ray or CT, see above)
2. failure of calvarial growth (unlike the non-synostotic skull where increased ICP causes macrocrania in the newborn, here it is the synostosis that causes the increased ICP and lack of skull growth)
3. papilledema
4. developmental delay

Types of craniosynostosis

Sagittal synostosis

General information

The most common CSO affecting a single suture; 80% male. Results in dolichocephaly or scaphocephaly (boat shaped skull) with frontal bossing, prominent occiput, palpable keel-like sagittal ridge. OFC remains close to normal, but the biparietal diameter is markedly reduced. As many as 44% of patients with nonsyndromic sagittal synostosis have elevated ICP.¹⁷

Surgical treatment

Skin incision may be longitudinal or transverse. A linear “strip” craniectomy is performed, excising the sagittal suture from the coronal to the lambdoid suture, preferably within the first 3–6 months of life. The width of the strip should be at least 3 cm, no proof exists that interposing artificial substances (e.g. silastic sheeting over the exposed edges of the parietal bone) retards the recurrence of synostosis. Great care is taken to avoid dural laceration with potential injury to the underlying superior sagittal sinus. The child is followed and reoperated if fusion recurs before 6 months age. After $\approx$ 1 yr age, more extensive cranial remodelling is usually required.

Coronal synostosis

General information

Accounts for 18% of CSO, more common in females. In Crouzon’s syndrome this is accompanied by abnormalities of sphenoid, orbital and facial bones (hypoplasia of midface), and in Apert’s syndrome is accompanied by syndactyly.¹⁸ Unilateral coronal CSO $\rightarrow$ plagiocephaly with forehead on affected side flattened or concave above eye (normal side falsely appears to bulge abnormally), supra-orbital margin higher than normal side (on skull x-ray $\rightarrow$ harlequin eye sign). The orbit rotates out on the abnormal side, and can produce amblyopia. Without treatment, flattened cheeks develop and the nose deviates to the normal side (root of nose tends to rotate towards deformity).

Bilateral coronal CSO (usually in craniofacial dysmorphism with multiple suture CSO, e.g. Apert’s) $\rightarrow$ brachycephaly with broad, flattened forehead (acrocephaly). When combined with premature closure of frontosphenoidal and frontoethmoidal sutures, results in foreshortened anterior fossa with maxillary hypoplasia, shallow orbits, progressive proptosis.

Surgical treatment

Simple strip craniectomy of the involved suture has been used, often with excellent cosmetic result. However, some argument that this may not be adequate has been presented. Therefore, a more current recommendation is to do frontal craniotomy (uni- or bi-lateral) with lateral canthal advancement by taking off orbital bar.

Metopic synostosis

At birth, the frontal bone consists of two halves separated by the frontal or metopic suture. Abnormal closure results in a pointed forehead with a midline ridge (trigonocephaly). Many of these have a 19p chromosome abnormality and are retarded.

Lambdoid synostosis

Epidemiology

Long considered a clinical rarity with a reported incidence range 1–9% of CSO,¹⁹ recent reports suggest a higher incidence of 10–20%²⁰ which may be due to an actual increased incidence, or simply to increased awareness or changing diagnostic criteria. More common in males (male:female = 4:1), and the right side is involved in 70% of cases. Usually presents between 3–18 months of age, but may be seen as early as 1–2 months of age.

Controversy exists regarding the actual criteria for this condition, and some authors differentiate between those cases which appear to have a primary abnormality of the lambdoid suture from those which may be due to positional flattening, the so-called “lazy lambdoid.” Others do not make this distinction, and sometimes refer to the condition as occipital plagiocephaly to avoid the need to implicate abnormalities of the lambdoid suture.

Positional flattening (or molding) may be produced by:

1. decreased mobility: patients who constantly lie supine with the head to the same side, e.g. cerebral palsy, mental retardation, prematurity, chronic illness
2. abnormal postures: congenital torticollis,²¹ congenital disorders of the cervical spine

3. intentional positioning: trend since 1992 to place newborns in a supine sleeping position to reduce the risk of sudden infant death syndrome (SIDS),²² sometimes with a foam wedge to tilt the child to one side to reduce the risk of aspiration
4. intrauterine etiologies²³: intrauterine crowding (e.g. from multiparous births or large fetal size), uterine anomalies

Clinical findings

Flattening of the occiput. May be unilateral or bilateral. If unilateral, it is sometimes termed lambdoid plagiocephaly which when severe also produces bulging of the ipsilateral forehead resulting in a “rhomboid” skull with the ipsilateral ear located anterior and inferior to the contralateral ear. The contralateral orbit and forehead may also be flattened. This may be confused with hemifacial microsomia or with plagiocephaly seen in unilateral coronal craniosynostosis. Bilateral lambdoid synostosis produces brachycephaly with both ears displaced anteriorly and inferiorly.¹⁹ Unlike the palpable ridge of sagittal or coronal synostosis, an indentation may be palpated along the synostotic lambdoid suture (although a perisutural ridge may be found in some).

Diagnostic evaluation

The physical exam is the most important aspect of diagnosis. Skull x-ray may help differentiate (see below). If the skull x-ray is equivocal, prevent the infant from laying on the affected side for several weeks. A bone scan should be obtained if no improvement occurs (see below). In definite cases of synostosis, and for some cases of refractory positional flattening (which usually corrects with time, but may take up to 2 years) surgical treatment may be indicated.

Skull x-ray: Shows a sclerotic margin along one edge of the lambdoid suture in 70% of cases. Local “beaten copper cranium” (BCC) occasionally may be seen due to indentations in the bone from underlying gyri which may be due to locally increased ICP. BCC produces a characteristic mottled appearance of the bone with lucencies of varying depth having round and poorly margined edges. BCC correlates with generalized ↑ ICP only when it is seen with sellar erosion and sutural diastasis.¹⁴

CT scan: Bone windows may show eroded or thinned inner table in the occipital region in 15–20% of cases,²⁰ >95% are on the side of the involvement. The suture may appear closed. Brain windows show parenchymal brain abnormalities in <2% heterotopias, hydrocephalus, agenesis of the corpus callosum; but ≈ 70% will have significant expansion of the frontal subarachnoid space (may be seen in synostosis of other sutures, see above).

Bone scan: Isotope uptake in the lambdoid suture increases during the first year, with a peak at 3 months of age²⁴ (following the usual inactivity of the first weeks of life). The findings with synostosis are those typical for CSO (p.252).

Treatment

Early surgical treatment is indicated in cases with severe craniofacial disfigurement or those with evidence of increased ICP. Otherwise, children may be managed nonsurgically for 3–6 months. The majority of cases will remain static or will improve with time and simple nonsurgical intervention. Approximately 15% will continue to develop a significant cosmetic deformity.

Nonsurgical management²⁵:

Although improvement can usually be attained, some degree of permanent disfigurement is frequent.

Repositioning will be effective in ≈ 85% of cases. Patients are placed on the unaffected side or on the abdomen. Infants with occipital flattening from torticollis should have aggressive physical therapy and resolution should be observed within 3–6 months.

More severe involvement may be treated with a trial of molding helmets²⁶ (however, no controlled study has proven the efficacy).

Surgical treatment:

Required in only ≈ 20% of cases. The ideal age for surgery is between 6 and 18 months. The patient is positioned prone on a well-padded cerebellar headrest (the face should be lifted and gently massaged every ≈ 30 minutes by the anesthesiologist to prevent pressure injuries).

Surgical options range from simple unilateral craniectomy of the suture to elaborate reconstruction by a craniofacial team.

Linear craniectomy extends from the sagittal suture to the asterion is often adequate for patients ≤ 12 weeks of age without severe disfigurement. Great care is taken to avoid dural laceration near the asterion which is in the region of the transverse sinus. The excised suture demonstrates an internal ridge. Better results are obtained with earlier surgery, more radical surgery may be necessary after the age of 6 months.

Average blood loss for uncomplicated cases is 100–200 ml and therefore transfusion is often required.

Multiple synostoses

Fusion of many or all cranial sutures → oxycephaly (tower skull with undeveloped sinuses and shallow orbits). These patients have elevated ICP.

Craniofacial dysmorphic syndromes

Over 50 syndromes have been described, □ Table 15.4, shows a few selected ones.

A number of craniosynostosis syndromes are due to mutations in the FGFR (fibroblast growth factor receptor) genes. FGFR gene-related craniosynostosis syndromes include some classic syndromes (Apert, Crouzon, Pfeiffer...) as well as several newer entities (Beare-Stevenson, Muenke, Jackson-Weiss syndromes). All exhibit autosomal dominant inheritance.

15.2.3 Encephalocele

General information

Cranium bifidum is a defect in the fusion of the cranial bone, it occurs in the midline, and is most common in the occipital region. If meninges and CSF herniate through the defect, it is called a meningocele. If meninges and cerebral tissue protrude, it is called an encephalocele.

Encephalocele AKA cephalocele is an extension of intracranial structures outside of the normal confines of the skull. One case was seen for every five cases of spinal myelomeningoceles.²⁸ A nasal polypoid mass in a newborn should be considered an encephalocele until proven otherwise. See also Differential diagnosis (p.1388).

Classification

System based on Suwanwela and Suwanwela²⁹:

- 1. occipital: often involves vascular structures
- 2. cranial vault: comprises ≈ 80% of encephaloceles in Western hemisphere
 - a) interfrontal
 - b) anterior fontanelle
 - c) interparietal: often involves vascular structures
 - d) temporal
 - e) posterior fontanelle
- 3. fronto-ethmoidal: AKA sincipital; 15% of encephaloceles; external opening into face in one of the following 3 regions:
 - a) nasofrontal: external defect in the nasion
 - b) naso-ethmoidal: defect between nasal bone and nasal cartilage
 - c) naso-orbital: defect in the antero-inferior portion of medial orbital wall
- 4. basal: 1.5% of encephaloceles; (see below)
 - a) transethmoidal: protrudes into nasal cavity through defect in cribriform plate
 - b) spheno-ethmoidal: protrudes into posterior nasal cavity

Table 15.4 Selected craniofacial dysmorphic syndromes (modified ²⁷ (p 123–4))				
Syndrome	Genetics		Craniofacial findings	Associated findings
	Sporadic	Inherited		
Crouzon (craniofacial dysostosis)	yes (25%)	FGFR AD	CSO of coronal & basal skull sutures, maxillary hypoplasia, shallow orbits, proptosis	HCP rare
Apert (acrocephalosyndactyly)	yes (95%)	FGFR AD	same as Crouzon	syndactyly of digits 2,3,4; shortened UE, HCP common
Kleeblattschadel	yes	AD	CSO with trilobular skull	isolated, or with Apert's or thanatophoric dwarfism
^a Abbreviations: AD=autosomal dominant; FGFR=fibroblast growth factor receptor gene-related; CSO=craniosynostosis; HCP=hydrocephalus; UE=upper extremities				

- c) transsphenoidal: protrudes into sphenoid sinus or nasopharynx through patent craniopharyngeal canal (foramen cecum)
- d) fronto-sphenoidal or sphenoid-orbital: protrudes into orbit through superior orbital fissure
- 5. posterior fossa: usually contains cerebellar tissue and ventricular component

Basal encephalocele

The only group that does not produce a visible soft tissue mass. May present as CSF leak or recurrent meningitis. May be associated with other craniofacial deformities, including: cleft lip, bifid nose, optic-nerve dysplasia, coloboma and microphthalmia, hypothalamic-pituitary dysfunction.

Encephalocele is characterized by defects around the foramen magnum, rachischisis and retrocollis. Most are stillborn, some survive up to age 17.

Etiology

Two main theories:

1. arrested closure of normal confining tissue allows herniation through persistent defect
2. early outgrowth of neural tissue prevents normal closure of cranial coverings

Treatment

Occipital encephalocele

Surgical excision of the sac and its contents with water-tight dural closure. It must be kept in mind that vascular structures are often included in the sac. Hydrocephalus is often present and may need to be treated separately.

Basal encephalocele

Caution: a transnasal approach to a basal encephalocele (even for biopsy alone) may be fraught with intracranial hemorrhage, meningitis, or persistent CSF leak. Usually a combined intracranial approach (with amputation of the extracranial mass) and transnasal approach is used.

Outcome

Occipital encephalocele

The prognosis is better in occipital meningocele than in encephalocele. The prognosis is worse if a significant amount of cerebral tissue is present in the sac, if the ventricles extend into the sac, or if there is hydrocephalus. Less than $\approx 5\%$ of infants with encephalocele develop normally.

15.3 Dandy Walker malformation

15

15.3.1 General information

Definition: an enlarged posterior fossa with complete or partial agenesis of the cerebellar vermis and cystic dilatation of the fourth ventricle which is distorted and encased in a membrane. The anomaly was first described by Dandy & Blackfan in 1914, and was named Dandy Walker malformation forty years later by Benda to acknowledge Taggart and Walker's contributions in 1942.³⁰

15.3.2 Differential diagnosis

Disorders with posterior fossa CSF collections include³¹:

1. Dandy Walker malformation (DWM)
2. Dandy Walker variant (DWV): vermian hypoplasia and cystic dilatation of the fourth ventricle, without enlargement of the posterior fossa
3. persistent Blake's pouch cyst (BPC): tetraventricular hydrocephalus, communicating 4th ventricle and posterior fossa cyst, with or without hypoplasia of both the cerebellar vermis and the medial aspects of the cerebellar hemispheres
4. retrocerebellar arachnoid cyst: anteriorly displaces the 4th ventricle and cerebellum, which can produce significant mass effect
5. Joubert's syndrome: absence or underdevelopment of the cerebellar vermis
6. mega cisterna magna: enlarged posterior fossa secondary to an enlarged cisterna magna with a normal vermis and fourth ventricle)

Differentiating features: DWM and DWV are difficult to distinguish, and may represent a continuum of developmental anomalies that are grouped together as Dandy Walker complex.³²

Retrocerebellar arachnoid cysts and BPCs may mimic DWM, but these do not have vermian agenesis and the cyst does not open into the 4th ventricle. The position of the choroid plexus of the fourth ventricle is normal in arachnoid cysts, absent in Dandy Walker malformations, and displaced into the superior cyst wall in BPC. An intrathecal enhanced CT scan (performed after instilling iodinated contrast into a ventricular catheter) would identify a mega cisterna magna which communicates with the ventricles, while DWM and most but not all arachnoid cysts do not.

15.3.3 Pathophysiology

The etiology of DWM is unknown. Multiple unsatisfactory theories have been abandoned. DWM is likely due to dysembryogenesis, secondary to insults of varying severity to the cerebellum and 4th ventricle. This results in agenesis of the cerebellar vermis with a large posterior fossa cyst communicating with an enlarged 4th ventricle.^{32,30}

Hydrocephalus occurs in 70–90% of cases, and Dandy Walker malformation is present in 2–4% of all cases of hydrocephalus.

15.3.4 Risk factors and epidemiology

Gestational exposure to rubella, CMV, toxoplasmosis, warfarin, alcohol, and isotretinoin are thought to be predisposing factors. Autosomal recessive inheritance has been identified in a few cases, but a genetic basis is lacking in most. Incidence: 1 per 25,000–35,000 live births.³⁰ Male:female=1:3.

15.3.5 Associated abnormalities

CNS abnormalities include agenesis of the corpus callosum in 17%³³ and occipital encephalocele in 7%. Other findings include heterotopias, spina bifida, syringomyelia, microcephaly, dermoid cysts, porencephaly, and Klippel-Feil deformity. Most have an enlarged posterior fossa with elevation of the torcular herophili. Atresia of the foramina of Magendie and Luschka may occur.³⁴

Systemic abnormalities include³³: facial abnormalities (e.g. angiomas, cleft palates, macroglossia, facial dysmorphism), ocular abnormalities (e.g. coloboma, retinal dysgenesis, microphthalmia), and cardiovascular anomalies (e.g. septal defects, patent ductus arteriosus, aortic coarctation, dextrocardia). Note: be aware of the possibility of a cardiac abnormality when considering surgery on these patients.

15.3.6 Treatment

Early decompression of ventriculomegaly is recommended to achieve maximum cognitive development. In the absence of hydrocephalus, DWM may be followed. When treatment is necessary, the posterior fossa cyst must be shunted. Shunting the lateral ventricles alone is contraindicated because of the risk of upward herniation.³⁵ However, it is important to confirm patency of the cerebral aqueduct, otherwise the supratentorial ventricles need to be shunted concomitantly. Varying reports exist regarding rates of associated aqueductal stenosis, although it is widely believed to be rare.

Another option once used commonly is excision of the obstructing membrane. This has fallen out of favor due to its associated risks of morbidity and mortality. However, it remains an option for patients with frequent shunt malfunctions.

Newer treatments include endoscopic third ventriculostomy in cases where the aqueduct is patent, however further study is necessary.^{36,37}

15.3.7 Prognosis

Prognosis ranges widely as there are various levels of severity of the malformation. Some pediatric neurosurgical literature quotes 12–50% mortality rates, although this is improving with modern shunting techniques. Only 50% have normal IQ. Ataxia, spasticity, and poor fine motor control are common. Seizures occur in 15%

15.4 Aqueductal stenosis

15.4.1 General information

Aqueductal stenosis (AqS) produces what is sometimes called triventricular hydrocephalus, characterized by a normal sized 4th ventricle and enlarged third and lateral ventricles on MRI or CT. Most cases occur in children, however some present for the first time in adulthood.

15.4.2 Etiologies

1. a congenital malformation: may be associated with Chiari malformation or neurofibromatosis
2. acquired
 - a) due to inflammation (following hemorrhage or infection, e.g. syphilis, T.B.)
 - b) neoplasm: especially brainstem astrocytomas – including tectal gliomas (p.634), lipomas
 - c) quadrigeminal plate arachnoid cysts

15.4.3 Aqueductal stenosis in infancy

AqS is a frequent cause of congenital hydrocephalus (HCP) (up to 70% of cases²⁷), but occasionally may be the result of HCP. Patients with congenital AqS usually have HCP at birth or develop it within $\approx$ 2–3 mos. Congenital AqS may be due to an X-linked recessive gene.²⁸ Four types of congenital AqS described by Russell (summarized³⁸):

1. forking: multiple channels (often narrowed) with normal epithelial lining that do not meet, separated by normal nervous tissue. Usually associated with other congenital abnormalities (spina bifida, myelomeningocele)
2. periaqueductal gliosis: luminal narrowing due to subependymal astrocytic proliferation
3. true stenosis: aqueduct histologically normal
4. septum

15.4.4 Aqueductal stenosis in adulthood

General information

AqS may be an overlooked cause of “normal pressure hydrocephalus” in the adult.³⁹ It is unknown why some cases of AqS would remain occult, and manifest only in adulthood. In one series of 55 cases,⁴⁰ 35% had duration of symptoms <1 year, 47% for 1–5 years; the longest was 40 yrs. Although most follow this longstanding benign course, there are reports of elevated ICP and sudden death.

Symptoms

See □ Table 15.5. Headache was the most common symptom, and had characteristics of H/A associated with elevated ICP. Visual changes were next, and usually consisted of blurring or loss of acuity. Endocrine changes included menstrual irregularities, hypothyroidism, and hirsutism.

Signs

Papilledema was the most common finding (53%). Visual fields were normal in 78% the remainder having reduced peripheral vision, increased blind spots, quadrant or hemianopic field cuts, or scotomata. Intellectual impairment was present in at least 36%. Other signs included: ataxia (29%), “pyramidal tract signs” in 44% (mild hemi- or para-paresis (22%), spasticity (22%), or Babinski’s (20%)), anosmia (9%).

Evaluation

MRI is the test of choice. MRI will show the absence of the normal flow void in the Sylvian aqueduct. Contrast should be given to rule-out tumor.

Treatment (of non-tumoral AqS)

Although treatments of the primary lesion have been attempted (e.g. lysis of aqueductal septum), this has fallen into disfavor with the improved efficacy of CSF shunting and endoscopic third ventriculostomy (ETV).

Table 15.5 Symptoms of aqueductal stenosis presenting in adulthood (55 patients > 16 years age ⁴⁰)		
Symptom	No.	%
H/A	32	58%
visual disturbances	22	40%
mental deterioration	17	31%
gait disturbance	16	29%
frequent falling	13	24%
endocrine disturbance	10	18%
nausea/vomiting	9	16%
seizures	8	15%
incontinence	7	13%
vertigo	6	11%
IE weakness	4	7%
hemiparesis or hemianesthesia	4	7%
diplopia	3	5%
dysarthria	1	
deafness	1	

1. shunting: CSF is usually shunted to the peritoneum or the vascular system, however shunting to subarachnoid space is also feasible (once obstruction at the level of the arachnoid granulations has been ruled out)
2. a Torkildsen shunt (shunting a lateral ventricle to the cisterna magna⁴¹) may work in adult cases,³⁸ however pediatric patients with obstructive hydrocephalus may not have an adequately developed subarachnoid space for this to function properly
3. endoscopic third ventriculostomy (p.415)

Follow-up of at least two years to rule out tumor is recommended.

15.5 Agensis of the corpus callosum

15.5.1 General information

A failure of commissuration occurring $\approx$ 2 weeks after conception. Results in expansion of the third ventricle and separation of the lateral ventricles (which develop dilated occipital horns and atria, and concave medial borders).

The corpus callosum (CC) forms from rostrum (genu) to splenium,⁴² □in agenesis there may be an anterior portion with absence of the posterior segment (the converse occurs infrequently). Absence of the anterior CC with presence of some posterior CC is indicative of some form of holoprosencephaly.

15.5.2 Incidence

1 in 2,000–3,000 neuroradiological examinations.

15.5.3 Associated neuropathologic findings

See reference.⁴³

- porencephaly

- microgyria
- interhemispheric lipomas and lipomas of the corpus callosum (p.260)
- arhinencephaly
- optic atrophy
- colobomas
- hypoplasia of the limbic system
- bundles of Probst: aborted beginnings of corpus callosum, bulge into lateral ventricles
- loss of horizontal orientation of cingulate gyrus
- schizencephaly (p.288)
- anterior and hippocampal commissures may be totally or partially absent⁴⁴
- hydrocephalus
- cysts in the region of the corpus callosum
- spina bifida with or without myelomeningocele
- absence of the septum pellucidum (p.260)

15.5.4 Possible presentation

- hydrocephalus
- microcephaly
- seizures (rare)
- precocious puberty
- disconnection syndrome: more likely with acquired CC defect than in congenital

May be an incidental finding, and by itself may have no clinical significance. However, may be occur as part of a more complex clinical syndrome or chromosomal abnormality (e.g. Aicardi syndrome: agenesis of CC, seizures, retardation, patches of retinal pigmentation).

15.6 Absence of the septum pellucidum

Etiologies⁴⁵ (p 178)

1. holoprosencephaly (p.289)
2. schizencephaly (p.288)
3. agenesis of the corpus callosum (p.259)
4. Chiari type 2 malformation (p.284)
5. basal encephalocele
6. porencephaly/hydranencephaly
7. may occur in severe hydrocephalus: thought to be due to necrosis with resorption of the septum
8. septo-optic dysplasia: see below

15

Septo-optic dysplasia^{46,45} (p 175–8)

AKA de Morsier syndrome. Incomplete early morphogenesis of anterior midline structures produces hypoplasia of the optic nerves and possibly optic chiasm (affected patients are blind) and pituitary infundibulum. The septum pellucidum is absent in about half the cases. About half the cases also have schizencephaly (p.288).

Presentation may be due to secondary hypopituitarism manifesting as dwarfism, isolated growth hormone deficiency, or panhypopituitarism. Occasionally hypersecretion of growth hormone, corticotropin or prolactin may occur, and precocious puberty may occur. Most patients are of normal intelligence although retardation may occur. Septo-optic dysplasia may be a less severe form of holoprosencephaly (p.289), and occasionally may occur as part of this anomaly (with its attendant poorer prognosis for function or survival). The ventricles may be normal or dilated. May be seen by the neurosurgeon because of concerns of possible hydrocephalus.

15.7 Intracranial lipomas

15.7.1 General information

Intracranial and intraspinal lipomas are felt to be of maldevelopmental origin⁴⁷ (p 706) and may arise from failure of involution of the primitive meninges.⁴⁸

15.7.2 Epidemiology of intracranial lipomas

Incidence: 8 in 10,000 autopsies. Usually found in or near the midsagittal plane, particularly over the corpus callosum; lipomas in this region are frequently associated with agenesis of the corpus callosum (p.259). The tuber cinereum and quadrigeminal plate are less frequently affected.⁴³ Rarely, the CP angle or cerebellar vermis may be involved. May occur in isolation, but also has been described in association with a number of congenital anomalies, including: trisomy 21, Pai’s syndrome, frontal encephalocele, facial anomalies.... Other midline abnormalities may also be found: agenesis of the corpus callosum, myelomeningocele, and spina bifida.⁴⁸

15.7.3 Evaluation

May be diagnosed by CT, MRI (study of choice), and by ultrasound in infants.
CT: Low density, may have peripheral calcification (difficult to appreciate on MRI).⁴⁸ Differential diagnosis on CT: primarily between dermoid cyst, teratoma⁴⁹ and germinoma.⁴⁸
MRI: characteristic finding is a midline lesion with signal characteristics of fat (high intensity on T1WI, low intensity on T2WI).

15.7.4 Presentation

Often discovered incidentally. Large lipomas may be associated with seizures, hypothalamic dysfunction, or hydrocephalus (possibly from compression of the aqueduct). Associated findings that may or may not be directly related: mental retardation, behavioral disorders and headache.

15.7.5 Treatment

Direct surgical approach is seldom necessary for intracranial lipomas.⁴⁹ Shunting may be required for cases where hydrocephalus results from obstruction of CSF circulation.⁴⁹

15.8 Hypothalamic hamartomas

15.8.1 General information

Key concepts

- rare, non-neoplastic congenital malformation, usually occurs in tuber cinereum
- may be parahypothalamic (pedunculated) or intrahypothalamic (sessile)
- presentation: precocious puberty, seizures (usually starting with gelastic seizures (brief unprovoked laughter)), developmental delay
- treatment: GnRH analogs for precocious puberty. Latero-basal craniotomy for pedunculated lesions, transcallosal interforniceal approach for intrahypothalamic lesions, option of endoscopic approach for lesions ≤ 1.5 cm dia, stereotactic radiosurgery may be an alternative

Hypothalamic hamartomas (HH; hamartoma: an abnormal conglomeration of cells normally found in the same area) AKA diencephalic hamartomas or hamartomas of the tuber cinereum. Rare, non-neoplastic congenital malformations arising from inferior hypothalamus or tuber cinereum (floor of the third ventricle between the infundibular stalk and the mammillary bodies). May occur as part of Pallister-Hall syndrome (genetics: AD inherited defect in GLI3 gene resulting in abnormally short GLI3 protein which participates in normal shaping of many organs).

15.8.2 Clinical findings

1. specific types of seizures:
 - a) gelastic seizures (brief episodes of unprovoked laughter⁵⁰) are the most characteristic type and are the earliest seizure manifestation. Present in up to 92% of patients.⁵¹ They are resistant to medical management and can lead to cognitive and behavioral deficits.⁵² Not pathognomonic. A neocortical origin has been described⁵³

- b) epileptic encephalopathy: gelastic fits gradually increase in frequency and other seizure types accrue: complex partial seizures, drop attacks, tonic seizures, tonic-clonic seizures, and secondarily generalized seizures. This phase is associated with marked deterioration of cognitive and behavioral abilities. Develops in 52% by a mean age of 7 years⁵¹
- 2. precocious puberty: believed to be due to release of gonadotropin-releasing hormone (GnRH) found within hamartoma cells.⁵⁴ HH are the most common CNS tumor to cause precocious puberty, other causes include: other CNS tumors – astrocytoma, ependymoma, pineal tumors (p. 658), optic/hypothalamic gliomas (especially in NFT patients) -, CNS XRT, hydrocephalus, CNS inflammation, septo-optic dysplasia (p. 260), and chronic hypothyroidism
- 3. developmental delay: primarily in patients with seizure disorder (severity correlates with duration of seizures). 46% of patients have borderline intellectual function (mental retardation)
- 4. behavioral disturbances⁵⁵: aggressive behavior, rage attacks...

15.8.3 Imaging

MRI: nonenhancing, isointense on T1WI, slightly hyperintense or isointense on T2WI.⁵⁶

15.8.4 Pathology

Two subtypes of hypothalamic hamartomas^{56,51}:

- 1. pedunculated or parahypothalamic: narrower base attached to the floor of the hypothalamus (not arising within hypothalamus). No distortion of 3rd ventricle. Generally associated with precocious puberty more than seizures
- 2. intrathalamic or sessile: within hypothalamus (distorting the 3rd ventricle) or broad attachment to hypothalamus. More often associated with seizures. 66% have developmental delay, 50% have precocious puberty

Microscopic pathology: Clusters of disorganized small neurons surrounded by large pyramidal like neurons in an astrocyte-rich neuropil⁵⁷ (in contrast to the usual ganglion cells surrounded by oligodendrocytes found in the hypothalamus).

15.8.5 Treatment

Precocious puberty usually responds well to GnRH analogs.⁵⁸

□ Indications for surgery:

- 1. precocious puberty that fails to respond to medical therapy (GnRH analogs)
- 2. seizures that cannot be adequately controlled medically. Post-op seizure control is related to completeness of resection
- 3. neurologic deficit from mass effect of the tumor

□ Options:

- 1. surgical resection
 - a) pedunculated lesions: approaches include⁵⁹ subtemporal, subfrontal, pterional, orbitozygomatic (most commonly recommended). Risks: cranial neuropathy, stroke⁵⁹
 - b) sessile lesions with intraventricular component: transcallosal anterior interforaminal approach.^{60,61,62} Risks: memory impairment (forniceal injury), endocrine disturbances, weight gain^{60,62}
 - c) neuroendoscopic approach: considered for HH ≤ 1.5 cm diameter.⁶³ Risks: 25% incidence of thalamic cerebrovascular injury
- 2. stereotactic radiosurgery: especially for small sessile lesions, subtotal resection, or patients refusing or not candidates for surgery. In small series, 3-year outcome showed improvement similar to surgical resection with less neurologic and endocrinologic morbidity^{64,65}

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16 Primary Spinal Anomalies

16.1 Spinal arachnoid cysts

16.1.1 General information

Almost always dorsal, most common in thoracic spine. With a ventral cyst, consider a neurenteric cyst (see below). Most are actually extradural and these are sometimes referred to as arachnoid diverticula – these may be associated with kyphoscoliosis in juveniles or with spinal dysraphism. Intradural arachnoid cysts may be congenital or may follow infection or trauma.

Usually asymptomatic, even if large.

16.1.2 Treatment

When indicated, treatment options include:

1. percutaneous procedures: may be done under MRI¹ or CT guidance. CT guidance usually requires use of intrathecal contrast to delineate the cyst
 - a) needle aspiration
 - b) needle fenestration¹
2. open surgical resection or fenestration

16.2 Spinal dysraphism (spina bifida)

16.2.1 Definitions

See reference.²

□ Spina bifida occulta. Congenital absence of a spinous process and variable amounts of lamina. No visible exposure of meninges or neural tissue.

The following two entities are grouped together under the term spina bifida aperta (aperta from the Latin for “open”) or spina bifida cystica.

□ Meningocele. Congenital defect in vertebral arches with cystic distension of meninges, but no abnormality of neural tissue. One third have some neurologic deficit.

□ Myelomeningocele. Congenital defect in vertebral arches with cystic dilatation of meninges and structural or functional abnormality of spinal cord or cauda equina.

16.2.2 Spina bifida occulta (SBO)

Reported prevalence range of SBO: 5–30% of North Americans (5–10% is probably more realistic). The defect may be palpable, and there may be overlying cutaneous manifestations (in □ Table 16.3).

Often an incidental finding, usually of no clinical importance when it occurs alone. Numerous reviews have shown no statistical association of SBO with nonspecific LBP.^{3,4} An increased incidence of disc herniation was shown in one study.⁵

SBO may occasionally be associated with diastematomyelia, tethered cord, lipoma, or dermoid tumor. When symptomatic from one of these associated conditions, the presentation is usually that of tethered cord; gait disturbance, leg weakness and atrophy, urinary disturbance, foot deformities..., see Tethered cord syndrome (p.272).

16.2.3 Myelomeningocele

Embryology

The anterior neuropore closes at gestation day 25. The caudal neuropore closes at day 28.

Epidemiology/genetics

Incidence of spina bifida with meningocele or myelomeningocele (MM) is 1–2/1000 live births (0.1–0.2%). Risk increases to 2–3% if there is one previous birth with MM, and 6–8% after two affected children. The risk is also increased in families where close relatives (e.g. siblings) have given birth to

MM children, especially when on the mother's side of the family. Incidence may increase in times of war, famine or economic disasters, but it may be gradually declining overall.⁶ Transmission follows non-Mendelian genetics, and is probably multifactorial. Prenatal folate (in the form of folic acid) lowers the incidence of MM (p.290).

Hydrocephalus in myelomeningocele

Hydrocephalus (HCP) develops in 65–85% of patients with MM, and 5–10% of MM patients have clinically overt HCP at birth.⁷ Over 80% of MM patients who will develop HCP do so before age 6 mos. Most MM patients will have an associated Chiari type 2 malformation (p.284). Closure of the MM defect may convert a latent HCP to active HCP by eliminating a route of egress of CSF.

Latex allergy in myelomeningocele

Up to 73% of MM patients are allergic to proteins present in latex (the milky sap from the rubber tree *Hevea brasiliensis*), found only in naturally occurring rubber products (and which are not present in synthetics such as: silicone, vinyl, plastic, neoprene, nitrile...). The allergy is thought to arise from early and frequent exposure to latex products during medical care for these patients, and there is a suggestion that latex-free surgery on these infants may reduce the risk of the development of latex allergy.⁸

Prenatal diagnosis

See Prenatal detection of neural tube defects (p.290).

Intrauterine closure of MM defect

Controversial. Does reduce incidence of Chiari II defect, but it has not been determined if this is clinically significant. Argued whether this reduces incidence of hydrocephalus. Does not improve distal neurologic function.

General management

Assessment and management of lesion

- measure size of defect
- assess whether lesion is ruptured or unruptured
 - ruptured: start antibiotics (e.g. nafcillin and gentamicin; D/C 6 hrs after MM closure, or continue if shunt anticipated in next 5 or 6 days)
 - unruptured: no antibiotics necessary
- cover lesion with telfa, then sponges soaked in lactated ringers or normal saline (form a sterile gauze ring around the lesion if it is cystic and protruding) to prevent desiccation
- Trendelenburg position, patient on stomach (keeps pressure off lesion)
- perform surgical closure within 36 hrs unless there is a contraindication to surgery (simultaneous shunt is not usually done except if overt hydrocephalus (HCP) at birth): see below

16

Neurological assessment and management

- items related to spinal lesion
 - watch for spontaneous movement of the LEs (good spontaneous movement correlates with better later functional outcome⁹)
 - assess lowest level of neurologic function □ Table 16.1) by checking response of LEs to painful stimulus: although some infants will have a clear demarcation between normal and abnormal levels, at least 50% show some mixture of normal, reflex, and autonomous activity (arising from uninhibited anterior horn motor neurons)⁹
 - differentiating reflex movement from voluntary may be difficult. In general, voluntary movement is not stereotyped with repetitive stimulus and reflex movement usually only persists as long as the noxious stimulus is applied
- items related to the commonly associated Chiari type 2 malformation:
 - measure OFC: risk of developing hydrocephalus (see above). Use OFC graphs (p.395), and also look for abnormal rate of growth (e.g. > 1 cm/day)
 - head U/S within ≈ 24 hrs
 - check for inspiratory stridor, apneic episodes

Table 16.1 Findings in various levels of MM lesion ¹⁰	
Paralysis below	Findings
T12	complete paralysis of all muscles in IEs
L1	weak to moderate hip flexion, palpable contraction in sartorius
L2	strong hip flexion and moderate hip adduction
L3	normal hip adduction & almost normal knee extension
L4	normal hip adduction, knee extension & dorsiflexion/inversion of foot; some hip abduction in flexion
L5	normal adduction, flexion & lateral rotation of hip; moderate abduction; normal knee extension, moderate flexion; normal foot dorsiflexion; hip extension absent; • produces dorsiflexed foot and flexed thigh
S1	normal hip flexion & abduction/adduction, moderate extension and lateral rotation; strong knee flexion & inversion/eversion of foot; moderate plantarflexion of foot; extension of all toes, but flexion only of terminal phalanx of great toe; normal medial & lateral hip rotation; complete paralysis of foot intrinsic (except abductor and flexor hallucis brevis); • produces clawing of toes and flattening of sole of foot
S2	difficult to detect abnormality clinically; • with growth this produces clawing of the toes due to weakness of intrinsic muscles of sole of foot (innervated by S3)

Ancillary assessment and management

- evaluation by neonatologist to assess for other abnormalities, especially those that may preclude surgery (e.g. pulmonary immaturity). There is an average incidence of 2–2.5 additional anomalies in MM patients
- bladder: start patient on regular urinary catheterizations, obtain urological consultation (non-emergent)
- AP & lat spine films: assess scoliosis (baseline)
- orthopedic consultation for severe kyphotic or scoliotic spine deformities and for hip or knee deformities

Surgical management

Timing of MM closure

Early closure of MM defect is not associated with improvement of neurologic function, but evidence supports lower infection rate with early closure. MM should be closed within 24 hrs whether or not membrane is intact (after $\approx$ 36 hrs the back lesion is colonized and there is increased risk of postoperative infection).

Simultaneous MM defect closure and VP shunting

In patients without hydrocephalus, most surgeons wait at least $\approx$ 3 days after MM repair before shunting. In MM patients with clinically overt HCP at birth (ventriculomegaly with enlarged OFC and/or symptoms), MM repair and shunting may be performed in the same sitting without increased incidence of infection, and with shorter hospitalization.^{11,12} It may also reduce the risk of MM repair breakdown previously seen during the interval before shunting. Patient is positioned prone, head turned to right (to expose the right occiput), right knee and thigh flexed to expose right flank (consider using left flank to prevent confusion with appendectomy scar later in life).

Surgical technique of myelomeningocele repair

Key concepts

- critical goals: 1) free placode from dura (to avoid tethering), 2) water-tight dural closure, 3) skin closure (can be accomplished in essentially all cases). Closure does not restore any neurologic function
- timing goal: surgical closure with latex-free setup ideally $\leq$ 36 hours after birth
- helpful tips: start at normal dura, open as wide as the defect, trim placode if necessary to close dura, undermine skin to achieve closure (avoid trapping skin $\rightarrow$ dermoid tumor)
- post-op CSF leak usually means a shunt is required

General principles¹³: prevent desiccation – keep the exposed neural tissue moist. Use latex-free environment (reduces development of latex allergy, as well as attack by maternal antibodies that may have crossed through the placenta). Do not allow scrub solutions or chemical antimicrobials to contact neural placode. Do not use monopolar cautery. At every point during the closure, avoid placing tension on the neural placode.

Multiple layer closure is advocated, 5 layers should be attempted, although occasionally only 2 or so layers may be closed. There is no evidence that multiple layer closure either improves neurologic function or prevents later tethering, but there is a suggestion that when tethering does occur, it may be easier to release when a previous multilayered closure was performed. Silastic does not prevent adherence in series with long follow-up (>6 yrs), and may even render untethering procedures more difficult.

Begin by dividing the abnormal epithelial covering from the normal skin. The pia-arachnoid may be separated from the neural tissue. The placode is folded into a tube and the pia-arachnoid is then approximated around it with 7–0 suture (absorbable suture, e.g. PDS, may make future re-operation easier). It often helps to start with normal dura above, and then work down. The dura can then be isolated around the periphery and followed deep to the spinal canal superiorly. The dura is then also formed into a tube and approximated in a water-tight closure. If the dura cannot be closed, the placode may be judiciously trimmed. The filum terminale should be divided if it can be located. The skin is then mobilized and closed. Dermoid tumors may result from retained skin during the closure, but alternatively dermoids may also be present congenitally.¹⁴

If there is a kyphotic deformity, it is repaired at the same sitting as the MM defect closure. The kyphotic bone is rongeured, and 2–0 Vicryl is used to suture the adjacent bones. Some surgeons use a brace post-op, some do not.

Post-op management of MM repair

1. keep patient off all incisions
2. bladder catheterization regimen
3. daily OFC measurements
4. avoid narcotics (midbrain malformation renders these patient more sensitive to respiratory depression from narcotics)
5. if not shunted
 - a) regular head U/S (twice weekly to weekly)
 - b) keep patient flat to ↓ CSF pressure on incision
6. if a kyphectomy was done, use of a brace is optional (surgeon preference)

Late problems/issues

Include:

1. hydrocephalus: may mimic ≈ anything listed below. ALWAYS RULE OUT SHUNT MALFUNCTION when a MM patient deteriorates
2. syringomyelia (and/or syringobulbia) (p.1144):
3. Tethered cord syndrome (p.272) as many as 70% of MM patients have a tethered cord radiographically (some quote 10–20%, but only a minority are symptomatic. Unfortunately there is no good test to check for symptomatic retethering (SSEPs may deteriorate,¹⁵ myelography may help)
 - a) scoliosis: early untethering of cord may improve scoliosis; see Scoliosis in tethered cord (p.272)
 - b) symptomatic tethering may manifest as delayed neurological deterioration¹⁶
4. dermoid tumor at the MM site (p.784)¹⁷: incidence ≈ 16%
5. medullary compression at foramen magnum, see symptomatic Chiari II malformation (p.284)
6. use of growth hormone to increase stature is controversial

Outcome

Without any treatment, only 14–30% of MM infants survive infancy; these usually represent the least severely involved; 70% will have normal IQ's. 50% are ambulatory.

With modern treatment, ≈ 85% of MM infants survive. The most common cause of early mortality are complications from the Chiari malformation (respiratory arrest, aspiration...), where late mortality is usually due to shunt malfunction. 80% will have normal IQ. Mental retardation is most closely linked to shunt infection. 40–85% are ambulatory with bracing, however, most choose to use wheelchairs for ease. 3–10% have normal urinary continence, but most may be able to remain dry with intermittent catheterization.

16.2.4 Lipomyeloschisis

General information

Dorsal spinal dysraphism with lipoma. Six forms are described,¹⁸ the following 3 are clinically important as possible causes of progressive neurologic dysfunction via tethering (p.272) and/or compression:

1. (intra)dural lipoma
2. lipomyelomeningocele (see below)
3. fibrolipoma of the filum terminale

Lipomyelomeningocele

General information

A subcutaneous lipoma that passes through a midline defect in the lumbodorsal fascia, vertebral neural arch, and dura, and merges with an abnormally low tethered cord.¹⁸ These may be terminal, dorsal, or transitional (between the two).

The intradural fatty tumor may also be known as lipoma of the cauda equina. In addition to being abnormally low, the conus medullaris is split in the midline dorsally usually at the same level as the bifid spine, and this dorsal myeloschisis may extend superiorly under intact spinal arches.¹⁹ There is a thick fibrovascular band that joins the lamina of the most cephalic vertebrae with the bifid lamina. This band constricts the meningocele sac and neural tissue, causing a kink in the superior surface of the meningocele. Asymptomatic lipomas of the filum terminale occur in 0.2–4%^{20,21} of MRIs.

The dura is dehiscent at the level of the dorsal myeloschisis, and reflects onto the placode. The lipoma passes through this dehiscence to become attached to the dorsal surface of the placode, and may continue cephalad under intact arches with the possibility of extension into the central canal superiorly to levels without dorsal myeloschisis. The lipoma is distinct from the normal epidural fat which is looser and more areolar. The subarachnoid space typically bulges to the side contralateral to the lipoma. These lipomas account for 20% of covered lumbosacral masses.

Presentation

In a pediatric series, 56% presented with a back mass, 32% with bladder problems, and 10% because of foot deformities, paralysis or leg pain.²²

Physical examination

Almost all patients have cutaneous stigmata of the associated spina bifida: fatty subcutaneous pads (located over the midline and usually extends asymmetrically to one side) with or without dimples, port-wine stains, abnormal hair, dermal sinus opening, or skin appendages.²³ Clubbing of the feet (talipes equinovarus) may occur.

The neurologic exam may be normal in up to 50% of patients (most presenting with skin lesion only). The most common neurologic abnormality was sensory loss in the sacral dermatomes.

Evaluation

Plain LS spine x-rays will show spina bifida in most cases. Present in almost all by definition, but some may have segmentation anomalies instead such as butterfly vertebra (p.216). Abnormalities of fusion and sacral defects may also be seen.

The abnormally low conus can be demonstrated on myelogram/CT or on MRI. MRI also demonstrates the lipomatous mass (high signal on T1WI, low signal on T2WI).

All patients should have pre-op urological evaluation to document any deficit.

Treatment

Since symptoms are due to (1) tethering of the spinal cord, especially during growth spurts, and (2) compression due to progressive deposition of fat, especially during periods of rapid weight gain; the goals of surgery are to release the tethering and reduce the bulk of fatty tumor. Simple cosmetic treatment of the subcutaneous fat pad does not prevent neurologic deficit, and may make later definitive repair more difficult or impossible.

Surgical treatment is indicated when the patient reaches 2 months of age, or at the time of diagnosis if the patient presents later in life. Adjuncts to surgical treatment include evoked potential monitoring and laser. Overall, with surgery, 19% will improve, 75% will be unchanged, and 6% will worsen. Foot deformities often progress regardless.

Surgical technique (modified)

See reference.¹⁹

1. mobilize the subcutaneous mass, it funnels down through the deep fascia
2. open last intact vertebral arch (work from normal dura)
3. identify the fibrovascular band that crosses the most cephalic widely bifid lamina
4. sectioning the fibrovascular band frees the dural tube and releases the sharp kink in the superior surface of the meningocele
5. taking care to preserve dorsal nerve roots, the dura is incised anterior to the dura-lipoma junction
6. similar procedure is carried out with arachnoid membrane
7. dural/arachnoid incisions are continued around entire extent of tethered conus
8. cord and placode are untethered; monitoring techniques described in Tethered cord syndrome (p.272) are an option
9. □ subtotal removal of lipoma: lipoma is then trimmed as completely as possible, intentionally leaving some fat behind to avoid injury to dorsal surface of placode. Superior extension along dorsal surface of cord or into central canal is debulked as much as is safely possible
10. the placode is reformed into a closed neural tube
11. close the pial margins
12. the dura is closed (primarily if possible, or using fascia lata graft if too much tension is placed on folded placode)

16.2.5 Dermal sinus

General information

A tract beginning at the skin surface, lined with epithelium. Usually located at either end of neural tube: cephalic or caudal. Most common location is lumbosacral. Probably results from failure of the cutaneous ectoderm to separate from the neuro-ectoderm at the time of closure of the neural groove.²

Spinal dermal sinus

General information

May appear as a dimple or as a sinus, with or without hairs, usually very close to midline, with an opening of only 1–2 mm. Surrounding skin may be normal, pigmented (“port wine” discoloration), or distorted by an underlying mass.

The sinus may terminate superficially, may connect with the coccyx, or may traverse between normal vertebrae or through bifid spines to the dural tube. It may widen at any point along its path to form a cyst; called an epidermoid cyst if lined with stratified squamous epithelium and containing only keratin from desquamated epithelium, or called a dermoid cyst if also lined with dermis (containing skin appendages, such as hair follicles and sebaceous glands) and also containing sebum and hair.

Although innocuous in appearance, they are a potential pathway for intradural infection which may result in meningitis (sometimes recurrent) and/or intrathecal abscess. Less serious, a local infection may occur. The lining dermis contains normal skin appendages which may result in hair, sebum, desquamated epithelium and cholesterol, within the tract. As a result, the contents of the sinus tract are irritating and can cause a sterile (chemical) meningitis with possible delayed arachnoiditis if it enters the dural space.

Incidence of a presumed sacral sinus (a dimple whose bottom could not be seen on skin retraction): 1.2% of neonates.²⁴

Dermal sinuses are similar but distinct from pilonidal cysts which may also be congenital (although some authors say they are acquired), contain hair, are located superficial to the postsacral fascia, and may become infected.

If the tract expands intrathecally to form a cyst, the mass may present as a tethered cord or as an intradural tumor. Bladder dysfunction is usually the first manifestation.

The tract from a spinal dermal sinus always courses cephalad as it dives inward from the surface. An occipital sinus may penetrate the skull and can communicate with dermoid cysts as deep as the cerebellum or fourth ventricle.

Evaluation

These tracts are NOT to be probed or injected with contrast as this can precipitate infection or sterile meningitis.

Exam is directed towards detecting abnormalities in sphincter function (anal and urinary), lumbosacral reflexes, and lower extremity sensation and function.

Radiologic evaluation

When seen at birth, ultrasound is the best means to evaluate for spina bifida and a possible mass inside the canal.

If seen initially following birth, an MRI should be obtained. Sagittal images may demonstrate the tract and its point of attachment. MRI also optimally demonstrates masses (lipomas, epidermoids...) within the canal.

Plain x-rays and CT are unable to demonstrate the fine tract which may exist between the skin and the dura.

Plain x-rays must be done when embarking on surgery as part of operative planning, as preparation for the possibility of a complete laminectomy.

Treatment

Sinuses above the lumbosacral region should be surgically removed. More caudally located sinuses are slightly controversial. Although $\approx 25\%$ of presumed sacral sinuses seen at birth will regress to a deep dimple on follow-up (time not specified), it is recommended that all dermal sinuses should be surgically explored and fully excised prior to the development of neurologic deficit or signs of infection. The results following intradural infection are never as good as when undertaken prior to infection. Surgery within the week of diagnosis is appropriate. Sinuses that terminate on the tip of the coccyx rarely penetrate the dura, and may not need to be treated unless local infection occurs.

Surgical technique

An ellipse is cut around the opening, and the sinus is followed deep until the termination of the tract is encountered. Careful insertion of a lacrimal duct probe under direct vision may facilitate excision without violating the tract. If the tracts penetrates the spine, laminectomy must be performed and the tract followed to its full extent (even if necessary to extend the laminectomy to T12). An extradural cyst may be present. If the tract enters the dura, it usually does so in the midline, and in these cases the dura should be opened and inspected. Extreme care is taken to prevent spilling the contents into the subdural space.

Cranial dermal sinus

General information

Stalk begins with a dimple in the occipital or nasal region. Cutaneous stigmata of hemangioma, subcutaneous dermoid cyst, or abnormal hair formation may occur. Occipital sinuses extend caudally, and if they enter the skull, they do so caudal to the torcular herophili. Presentation may include recurrent bacterial (usually *S. aureus*) or aseptic meningitis. Evaluation should include MRI to look for intracranial extension and associated anomalies, including an intracranial dermoid cyst.

Treatment

When operating on a cranial dermal sinus, use a sagittally based incision to permit deep exploration. The tract must be followed completely. Be prepared to enter the posterior fossa.

16.3 Klippel-Feil syndrome

16.3.1 General information

Congenital fusion of two or more cervical vertebrae. Ranges from fusion of only the bodies (congenital block vertebrae) to fusion of the entire vertebrae (including posterior elements). Results from failure of normal segmentation of cervical somites between 3–8 weeks gestation. Involved vertebral bodies are often flattened and associated disc spaces are absent or hypoplastic. Hemivertebrae may also occur. Neural foramina are smaller than normal and oval. Cervical stenosis is rare. Complete absence of the posterior elements with an enlarged foramen magnum and fixed hyperextension posture is called iniencephaly and is rare. Incidence of Klippel-Feil is unknown due to its rarity and the fact that it is frequently asymptomatic.

May occur in conjunction with other congenital cervical spine anomalies such as basilar impression and atlanto-occipital fusion.

16.3.2 Presentation

Classic clinical triad (all 3 are present in <50%):

1. low posterior hairline
2. shortened neck (brevicollis)
3. limitation of neck motion (may not be evident if <3 vertebrae are fused, if fusion is limited only to the lower cervical levels,²⁵ or if hypermobility of nonfused segments compensates). Limitation of movement is more common in rotation than flexion-extension or lateral bending

Other clinical associations include scoliosis in 60% facial asymmetry, torticollis, webbing of the neck (called pterygium colli when severe), Sprengel's deformity in 25–35% (raised scapula due to failure of the scapula to properly descend from its region of formation high in the neck to its normal position about the same time as the Klippel-Feil lesion occurs), synkinesis (mirror motions, primarily of hands but occasionally arms also) and less commonly facial nerve palsy, ptosis, cleft or high arched palate. Systemic congenital abnormalities may also occur including: genitourinary (the most frequent being unilateral absence of a kidney), cardiopulmonary, CNS, and in $\approx$ 30% deafness²⁶ (due to defective development of the osseous inner ear).

No neurologic symptoms have ever been directly attributed to the fused vertebrae, however symptoms may occur from nonfused segments (less common in short-segment fusions) which may be hypermobile possibly leading to instability or degenerative arthritic changes.

16.3.3 Treatment

Usually directed at detecting and managing the associated systemic anomalies. Patients should have cardiac evaluation (EKG), CXR, and a renal ultrasound. Serial examinations with lateral flexion-extension lateral C-spine x-rays to monitor for instability. Occasionally, judicious fusion of an unstable nonfused segment may be needed at the risk of further loss of mobility. See also recommendations regarding athletic competition (p.937).

16.4 Tethered cord syndrome

16.4.1 General information

Abnormally low conus medullaris. Usually associated with a short, thickened filum terminale, or with an intradural lipoma (other lesions, e.g. lipoma extending through dura, or diastematomyelia are considered as separate entities). Most common in myelomeningocele (MM). Diagnosis must be made clinically in MM, as almost all of these patients will have tethering radiographically.

16.4.2 Presentation

Presenting signs and symptoms in patients with tethered cord are shown in □ Table 16.2.

16.4.3 Myelomeningocele patients

If a MM patient has increasing scoliosis, increasing spasticity, worsening gait (in those previously ambulatory), or deteriorating urodynamics²⁸:

- always make sure that there is a working shunt with normal ICP
- if painful, should be considered tethered cord until proven otherwise
- if painless, should be considered syringomyelia until proven otherwise
- may be due to brainstem compression – see symptomatic Chiari II malformation (p.284) – requiring posterior fossa decompression

16.4.4 Scoliosis in tethered cord

Progressive scoliosis may be seen in conjunction with tethered cord. Early untethering of the cord may result in improvement of scoliosis, however, untethering must be done when the scoliosis is mild. When cases of $\leq 10^\circ$ scoliosis were untethered, 68% had neurologic improvement and the remaining 32% were stabilized, whereas when scoliosis is severe ($\geq 50^\circ$) $\approx$ 16% deteriorated.

Table 16.2 Presenting signs and symptoms ²⁷ (p 1331–2)	
Finding	%
cutaneous findings • hypertrichosis • sub-Q lipoma (no intraspinal extension) • miscellaneous (hemangiomatous discoloration, dermal sinus, multiple manifestations)	54% • 22% • 15% • 17%
gait difficulty with LE weakness	93%
visible muscle atrophy, short limb, or ankle deformity	63%
sensory deficit	70%
bladder dysfunction	40%
bladder dysfunction as only deficit	4%
pain in back, leg, or foot arches	37%
scoliosis or kyphosis ^a	29%
posterior spina bifida (lumbar or sacral)	98%
^a high incidence of scoliosis and kyphosis due to inclusion of series by Hoffman	

16.4.5 Tethered cord in adults

General information

Although most cases of tethered cord present in childhood, cases of adult tethered cord also occur (≈ 50 published cases as of 1982). For comparison of adult and childhood forms, □ Table 16.3

Evaluation

Radiographically: low conus medullaris (below L2) and thickened filum terminale (definition of thickened filum: normal diameter < 1 mm; diameters > 2 mm are pathological). NB: apparent filum diameter on CT-myelogram may vary with concentration of contrast material.
It is difficult to differentiate a tethered cord from a congenitally low lying conus (filum diameter is generally normal in the latter).

16.4.6 Pre-op evaluation

Pre-operative cystometrogram is strongly recommended, especially if the patient seems continent (postoperative changes in bladder function are not uncommon, possibly due to stretching of the lower fibers of the cauda equina).

Surgical treatment

If the only abnormality is a thickened, shortened filum, then a limited lumbosacral laminectomy may suffice, with division of the filum once identified.
If a lipoma is found, it may be removed with the filum if it separates easily from neural tissues.

Distinguishing features of the filum terminale intraoperatively

The filum is differentiated from nerve roots by presence of characteristic squiggly vessel on surface of filum. Also, under the microscope, the filum has a distinctively whiter appearance than the nerve roots, and ligamentous-like strands can be seen running through it. NB: intra-op electrical stimulation and recording of anal sphincter EMG are more definitive.

Table 16.3 Comparison of childhood and adult tethered cord syndrome ²⁹		
Finding	Childhood tethered cord	Adult tethered cord
pain	uncommon; usually in back and legs, not peri-anal nor perineal	present in 86% often peri-anal & perineal; diffuse & bilateral; occasionally shock-like
foot deformities	common early; usually progressive cavovarus deformity (club foot)	not seen
progressive spinal deformity	common; usually progressive scoliosis	uncommon (<5%)
motor deficits	common; usually gait abnormalities & regression of gait training	usually presents as leg weakness
urological symptoms	common; usually continuous urinary dribbling, delayed toilet training, recurrent UTIs, enuresis	common; usually urinary frequency, urgency, sensation of incomplete emptying, stress incontinence, overflow incontinence
trophic ulcerations	relatively common in IEs	rare
cutaneous stigmata of dysraphism	present in 80–100%(tuft of hair, dimple, capillary angioma (naevus flammeus)	present in <50%
aggravating factors	growth spurts	trauma, maneuvers associated with stretching conus, lumbar spondylosis, disc herniation, spinal stenosis
From J Neurosurg, D. Pang and J.E. Wilberger, Vol. 57, pp. 40, 1982, with permission.		

Outcome

In MM, it is usually impossible to permanently untether a cord, however, in a growing MM child, it may be that after 2–4 untetherings that the child will be finished growing and tethering may cease to be a problem. Cases that are untethered early in childhood may recur later, especially during the adolescent growth-spurt. Incidence of post-op CSF leak: 15%

Adult form: surgical release is usually good for pain relief. However, it is poor for return of bladder function.

16.5 Split cord malformation

16.5.1 General information

There is no uniformly accepted nomenclature for malformations characterized by duplicate or split spinal cords. Pang et al.³⁰ have proposed the following.

The term split cord malformation (SCM) should be used for all double spinal cords, all of which appear to have a common embryologic etiology.

16.5.2 Type I SCM

Defined as two hemicords, each with its own central canal and surrounding pia, each within a separate dural tube separated by a dural-sheathed rigid osseocartilaginous (bony) median septum. This has often (but not consistently) been referred to as diastematomyelia. There are abnormalities of the spine at the level of the split (absent disc, dorsal hypertrophic bone where the median “spike” attaches).³¹ Two-thirds have overlying skin abnormalities including: nevi, hypertrichosis (tuft of hair), lipomas, dimples or hemangiomas. These patients often have and an orthopedic foot deformity (neurogenic high arches).

Treatment: symptoms are most commonly due to tethering of the cord; and are usually improved by untethering. In addition to untethering, the bony septum must be removed and the dura reconstituted as a single tube (these spines are often very distorted and rotated, therefore start at normal anatomy and work towards defect). □ DO NOT cut the tethered filum until after the median septum is removed to avoid having the cord retract up against septum.

16.5.3 Type II SCM

Consists of two hemicords within a single dural tube, separated by a nonrigid fibrous median septum. This has sometimes been referred to as diplomyelia. Each hemicord has nerve roots arising from it. There is usually no spine abnormality at the level of the split, but there is usually spina bifida occulta in the lumbosacral region.

Treatment: consists of untethering the cord at the level of the spina bifida occulta, and occasionally at the level of the split.³¹

16.6 Lumbosacral nerve root anomalies

Congenital anomalies of nerve roots are rare. This possibility should be considered in cases of failed back surgery for herniated disc.

Classification system of Cannon et al.³²

1. Type 1 anomalies: include conjoined nerve root (2 nerve roots arise from a common dural sheath). They separate at various distances from the thecal sac, and exit through the same or separate neural foramina. Neurosurgeons need to be aware of this anomaly to avoid inadvertent injury e.g. during surgery for herniated disc
2. Type 2 anomalies: 2 nerve roots exit through one foramen. Variants³³:
 - a) leaves an unoccupied neural foramen
 - b) all foramina occupied, but one foramen has 2 nerve roots
3. Type 3 anomalies: adjacent nerve roots are connected by an anastomosis

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17 Primary Craniospinal Anomalies

17.1 Chiari malformations

17.1.1 General information

The term “Chiari malformation” (after pathologist, Hans Chiari) is preferred for type 1 malformations, with the commonly used term “Arnold-Chiari malformation” reserved for type 2 malformation.

The Chiari malformations consists of four types of hindbrain abnormalities, probably unrelated to each other. The majority of Chiari malformations are types 1 or 2 (Table 17.1), a very limited number of cases comprise the remaining types. Chiari zero is a novel condition (p.286).

17.1.2 Type 1 Chiari malformation

General information

Key concepts

- a heterogeneous entity with the common feature of impaired CSF circulation through the foramen magnum
- may be congenital or acquired
- evaluation: MRI of brain and cervical spine (to R/O syringomyelia). Cine MRI to evaluate CSF flow through foramen magnum in uncertain cases
- cerebellar tonsillar herniation on MRI: criteria vary, >5 mm below the foramen magnum is often cited, but is neither essential nor diagnostic of the condition
- treatment, when indicated, is surgical, but aspects of what that surgery should entail are controversial (enlargement of foramen magnum is usually involved)
- associated with syringomyelia in 30–70% which almost always improves with treatment of the Chiari malformation

AKA primary cerebellar ectopia,² AKA adult Chiari malformation (since it tends to be diagnosed in the 2nd or 3rd decade of life). A heterogeneous group of conditions, with the underlying commonality of disruption of normal CSF flow through the foramen magnum (FM). Some cases are congenital, but others are acquired (this section is kept here under developmental for historical and organizational reasons).

Table 17.1 Comparisons of Chiari type 1 and 2 anomalies (adapted ¹)		
Finding	Chiari type 1 (see below)	Chiari type 2 (p.284)
caudal dislocation of medulla	unusual	yes
caudal dislocation into cervical canal	tonsils	inferior vermis, medulla, 4th ventricle
spina bifida (myelomeningocele)	may be present	rarely absent
hydrocephalus	may be absent	rarely absent
medullary “kink”	absent	present in 55%
course of upper cervical nerves	usually normal	usually cephalad
usual age of presentation	young adult	infancy
usual presentation	cervical pain, suboccipital H/A	progressive hydrocephalus, respiratory distress

Classically described as a rare abnormality restricted to caudal displacement of cerebellum with tonsillar herniation below the foramen magnum (below for criteria) and “peg-like elongation of tonsils.” Unlike Chiari type 2, the medulla is not caudally displaced (some authors disagree on this point³), the brainstem is not involved, lower cranial nerves are not elongated, and upper cervical nerves do not course cephalad. Syringomyelia of the spinal cord is present in 30–70%⁴ True hydromyelia probably doesn’t occur; CSF flow has not been documented in man, and it is generally not possible to find communication between the syrinx and the central canal in Chiari 1 patients. Hydrocephalus occurs in 7–9% of patients with Chiari type 1 malformation and syringomyelia.⁴

Cerebellar tonsil descent below FM with impaction, while common, is no longer a sine qua non of diagnosis.

Associations

May be associated with

1. a small posterior fossa
 - a) underdevelopment of the occipital bone due to a defect in the occipital somites originating from the para-axial mesoderm
 - b) low lying tentorium (the roof of the p-fossa)
 - c) thickened or elevated occipital bone (the floor of the p-fossa)
 - d) space occupying lesion in p-fossa: arachnoid cyst (retrocerebellar or supracerebellar⁵), tumor (e.g. FM meningioma or cerebellar astrocytoma), hypervascular dura
2. has been described with just about anything that takes up intracranial space
 - a) chronic subdural hematomas
 - b) hydrocephalus
3. following lumboperitoneal shunt (p.418) or multiple (traumatic) LPs⁶: acquired Chiari 1 malformation (may be asymptomatic)
4. arachnoid web or scar or fibrosis around brainstem and tonsils near FM
5. abnormalities of the upper cervical spine
 - a) hypermobility of the craniovertebral junction
 - b) Klippel-Feil syndrome
 - c) occipitalization of the atlas
 - d) anterior indentation at foramen magnum: e.g. basilar invagination or retroversion of the odontoid process
6. Ehlers-Danlos syndrome
7. craniosynostosis: especially cases involving all sutures
8. retained rhomboid roof: rare

Epidemiology

Average age at presentation is 41 years (range: 12–73 yrs). Slight female preponderance (female: male=1.3:1). Average duration of symptoms clearly related to Chiari malformation is 3.1 yrs (range: 1 month-20 yrs); if nonspecific complaints, e.g. H/A, are included, this becomes 7.3 years.⁷ This latency is probably lower in the MRI era.

Clinical

17

Clinical correlates

Patients with Chiari type 1 malformation may present due to any or all of the following:

1. compression of brain stem at the level of the foramen magnum
2. hydrocephalus
3. syringomyelia
4. isolation of the intracranial pressure compartment from the spinal compartment causing transient elevations of ICP intracranial pressure
5. 15–30% of patients with adult Chiari malformation are asymptomatic⁸

Symptoms

The most common symptom is pain (69%), especially headache which is usually felt in the suboccipital region □ Table 17.2). H/A are often brought on by neck extension or valsalva maneuver. Weakness is also prominent, especially unilateral grasp. Lhermitte’s sign may also occur. Lower extremity involvement usually consists of bilateral spasticity.

Table 17.2 Presenting symptoms in Chiari 1 malformation (71 cases ³)	
Symptom	%
pain	69%
H/A	34%
neck (suboccipital, cervical)	13%
girdle	11%
arm	8%
leg	3%
weakness (1 or more limbs)	56%
numbness (1 or more limbs)	52%
loss of temperature sensation	40%
painless burns	15%
unsteadiness	40%
diplopia	13%
dysphasia	8%
tinnitus	7%
vomiting	5%
dysarthria	4%
miscellaneous	
dizziness	3%
deafness	3%
fainting	3%
facial numbness	3%
hiccough	1%
facial hyperhidrosis	1%

Signs

Downbeat nystagmus is considered a characteristic of this condition. 10%will have a normal neurologic exam with occipital H/A as their only complaint. Some patients may present primarily with spasticity.

- See □ Table 17.3. Three main patterns of clustering of signs³:
1. foramen magnum compression syndrome (22%): ataxia, corticospinal and sensory deficits, cerebellar signs, lower cranial nerve palsies. 37%have severe H/A
 2. central cord syndrome (65%): dissociated sensory loss (loss of pain & temperature sensation with preserved touch & JPS), occasional segmental weakness, and long tract signs (syringomyelic syndrome⁹). 11%have lower cranial nerve palsies
 3. cerebellar syndrome (11%): truncal and limb ataxia, nystagmus, dysarthria

Natural history

The natural history is not known with certainty (only 2 reports on “natural history”). A patient may remain stable for years, with intermittent periods of deterioration. Rarely, spontaneous improvement may occur (debated).

Table 17.3 Presenting signs in Chiari I malformation (127 patients ⁷)	
Sign	%
hyperactive lower extremity reflexes	52%
nystagmus ^a	47%
gait disturbance	43%
hand atrophy	35%
upper extremity weakness	33%
“cape” sensory loss	31%
cerebellar signs	27%
hyperactive upper extremity reflexes	26%
lower cranial nerve dysfunction	26%
Babinski sign	24%
lower extremity weakness	17%
dysesthesia	17%
fasciculation	11%
Horner’s sign	6%
^a classically: downbeat nystagmus on vertical movement, and rotatory nystagmus on horizontal movement; also includes oscillopsia ¹⁰	

Evaluation

Plain x-rays

Of 70 skull x-rays, only 36%were abnormal (26%showed basilar impression, 7%platybasia, and 1 patient each with Paget’s and concave clivus); in 60 C-spine x-rays, 35%were abnormal (including assimilation of atlas, widened canal, cervical fusions, agenesis of posterior arch of atlas).

MRI

MRI of brain and C-spine is the diagnostic test of choice. Easily shows many of the classic abnormalities described earlier, including tonsillar herniation, as well as hydrosyringomyelia which occurs in 20–30%of cases. Also demonstrates ventral brain stem compression when present. Other findings include: hydrocephalus, empty sella.

Tonsillar herniation: Criteria for the descent of the tonsillar tips below the foramen magnum (FM) to diagnose Chiari type 1 malformation have gone through a number of reconsiderations.

Σ

Tonsillar herniation identified radiographically is of limited prognostic value in diagnosing Chiari I malformation, and requires clinical correlation.

Initially, >5 mm was defined as clearly pathologic¹¹ (with 3–5 mm being borderline). Barkovich¹² found tonsillar positions as shown in □ Table 17.4, and □ Table 17.5 shows the effect of utilizing 2 vs. 3 mm as the lowest normal position.

The tonsils normally ascend with age¹³ as shown in □ Table 17.6

Table 17.4 Location of cerebellar tonsils below foramen magnum ¹²		
Group	Mean ^a	Range
normal	1 mm above	8 mm above to 5 mm below
Chiari I	13 mm below	3–29 mm below
^a based on measurements in 200 normals and 25 Chiari I patients taken in relation to the lower part of the foramen magnum		

Table 17.5 Criteria for Chiari I ¹²		
Criteria for lowest extent of tonsils accepted as normal	Sensitivity for Chiari I	Specificity for Chiari I
2 mm below FM	100%	98.5%
3 mm below FM	96%	99.5%

Table 17.6 Tonsillar position relative to FM at various ages ¹³		
Age (years)	Normal (mm) ^a	2 S.D. ^b (mm)
0–9	–1.5	–6
10–19	–0.4	–5
20–29	–1.1	
30–39	0.0	–4
40–49	0.1	
50–59	0.2	
60–69	0.2	
70–79	0.6	
80–89	1.3	–3
^a negative number indicates distance below FM		
^b S.D. = standard deviation. Descent >2 S.D. beyond normal is suggested as a criteria for tonsillar ectopia		

Patients with syringohydromyelia without hindbrain herniation that responded to p-fossa decompression have been described¹⁴ (so-called “Chiari zero malformation”). Conversely, 14% of patients with tonsillar herniation >5 mm are asymptomatic¹⁵ (average extent of ectopia in this group was 11.4 ± 4.86 mm).

Potentially more significant than the absolute tonsillar descent is the amount of compression of the brainstem at the FM, best appreciated on axial T2WI MRI through the FM. Complete obliteration of CSF signal and compression of the brainstem at the FM by impacted tonsils is a common significant finding.

Cine MRI

AKA CSF flow study. May demonstrate blockage of CSF flow at FM. Not widely available. Accuracy is not high, therefore usually does not alter management.

Myelography

Generally used only when MRI cannot be obtained. Only 6% false negative. It is critical to run the intrathecal contrast (dye) all the way up to the foramen magnum. Usually combined with CT scan.

CT

Unenhanced CT is poor for evaluating the foramen magnum region due to bony artifact. It is very good at demonstrating hydrocephalus (as is MRI). When combined with intrathecal iodinated contrast (myelogram), reliability improves. Findings: tonsillar descent with possible complete blockage of dye at foramen magnum.

Treatment

Indications for surgery

Since patients respond best when operated on within 2 years of the onset of symptoms (below), early surgery is recommended for symptomatic patients. Asymptomatic patients may be followed and operated upon if and when they become symptomatic. Patients who have been symptomatic and stable for years may be considered for observation, with surgery indicated for signs of deterioration.

Surgical techniques

The most frequently performed operation is posterior fossa decompression (suboccipital craniectomy), with or without other procedures (usually combined with dural patch grafting and cervical laminectomy of C1, sometimes to C2 or C3). Options for grafts: same incision (pericranium), separate incision (e.g. or fascia lata), and allograft (avoided by many authors because of dissatisfaction with ability to provide water-tight closure and because of infectious risks).

Goals of surgery: decompress the brain stem and reestablish normal flow of CSF at the craniocervical junction.

The patient is positioned prone on chest rolls with the head in a Mayfield head-holder or in a horseshoe headrest. Flex the neck to open the interspace between the occiput and posterior arch of C1. The shoulders are retracted inferiorly with adhesive tape. If a fascia lata graft is to be taken, elevate one thigh on a sandbag. A midline incision from inion to $\approx$ C2 spinous process is made. The removal of bone above the foramen magnum should be $\approx$ 3 cm high by $\approx$ 3 cm wide (keep the posterior-fossa part of these operations small, the main thrust is to open the foramen magnum to decompress the tonsils and an upper cervical laminectomy; the compression is not in the p-fossa). Excessive removal of occipital bone may allow the cerebellar hemispheres to herniate through the opening ("cerebellar ptosis"), and create additional problems. If a pericranial graft is to be taken, it should be harvested at this time to reduce the amount of blood entering the subsequent dural opening.¹⁶ Pericranial graft can be procured without extending the incision about the inion using the technique of Dr. Robert Ojemann¹⁶ with subgaleal dissection and using a monopolar cautery with a bent tip to incise the periosteum and then a Penfield #1 dissector to free it from the bone surface.

Open the dura in a "Y" shaped incision, and excise the triangular top flap. CAUTION: the transverse sinuses are usually abnormally low in Chiari malformations. Suture the patch graft to provide more room for the contents (tonsils+medulla).

An option that is sometimes used in pediatrics is to not initially open the dura but to lyse constricting bands over the dura at the foramen magnum and then and use intraoperative ultrasound to determine if there is adequate room for CSF flow, the dura is then opened only if there is not.

Historical procedures that have been appended to the above: plugging the obex (with muscle or teflon), drainage of syrinx if present (fenestration, usually through dorsal root entry zone, with or without stent or shunt), 4th ventricular shunting, terminal ventriculostomy, and opening foramen of Magendie if obstructed (see reference for illustrations⁹). Current recommendations are that these or other additional procedures beyond dural patch grafting are usually not warranted.

Some authors repeatedly admonish not to attempt to remove adhesions binding the tonsils together (to avoid injuring vital structures, including PICAs). Others recommend cautiously separating the tonsils and even shrinking them down with bipolar cautery.

In cases with ventral brain-stem compression, some authors advocate performing a transoral clivus-odontoid resection as they feel these patients may potentially deteriorate with posterior fossa decompression alone.¹⁷ Since this deterioration was reversible with odontoidectomy, it may be reasonable to perform this procedure on patients who show signs of deterioration or progression of basilar impression on serial MRIs after posterior fossa decompression.

Booking the case: Chiari malformation

Also see defaults & disclaimers (p. 27).

1. position: prone
2. equipment:
 - a) optional microscope
 - b) intra-op Doppler, if used
3. consent:
 - a) procedure: surgery through the back of the neck to open the bone at the base of the skull and to insert a “patch” to make more room for the brainstem
 - b) alternatives: non-surgical management is usually not effective
 - c) complications: CSF leak, brainstem injury/stroke, apnea, failure to improve syrinx (if present)

Operative findings

See ☐ Table 17.7.

Tonsillar herniation is present in all cases (by definition); the most common position being at C1 (62%). Fibrous adhesions between dura, arachnoid and tonsils with occlusion of foramina of Luschka and Magendie in 41% The tonsils separated easily in 40%

Surgical complications

After suboccipital craniectomy plus C1–3 laminectomy in 71 patients, with dural patch grafting in 69, one death due to sleep apnea occurred 36 hrs post-op. Respiratory depression was the most common post-op complication (in 10 patients), usually within 5 days, mostly at night. Close respiratory

Table 17.7 Operative findings in Chiari I (71 patients ³)	
Finding	%
tonsillar descent	100%
below foramen magnum	4%
C1	62%
C2	25%
C3	3%
unspecified level	6%
adhesions	41%
syringomyelia	32%
dural band (at foramen magnum or C1 arch)	30%
vascular abnormalities ^a	20%
skeletal abnormalities	
inverted foramen magnum	10%
keel of bone	3%
C1 arch atresia	3%
occipitalization of C1 arch	1%
cervicomedullary “hump”	12%
^a vascular abnormalities: PICA dilated or abnormal course in 8 patients (PICA often descends to lower margin of tonsils ⁹); large dural venous lakes in 3	

Table 17.8 Long-term follow-up after surgery for Chiari I malformation (69 patients, 4 years mean F/U ³)	
early improvement of pre-op symptoms	82%
percent of above that relapsed ^a	21%
early improvement of pre-op signs	70%
no change from pre-op status	16%
worse than pre-op	0
^a these patients deteriorated to pre-op status (none deteriorated further) within 2–3 years of surgery; relapse occurred in 30% with foramen magnum compression syndrome, and in 21% with central cord syndrome	

monitoring is therefore recommended.³ Other risks of the procedure include: CSF leak, herniation of cerebellar hemispheres, vascular injuries (to PICA...).

Operative results

See □ Table 17.8.

Patients with pre-op complaints of pain generally respond well to surgery. Weakness is less responsive to surgery, especially when muscle atrophy is present.¹⁷ Sensation may improve when the posterior columns are unaffected and the deficit is due to spinothalamic involvement alone.

Rhoton feels that the main benefit of operation is to arrest progression.

The most favorable results occurred in patients with cerebellar syndrome (87%showing improvement, no late deterioration). Factors that correlate with a worse outcome are the presence of atrophy, ataxia, scoliosis, and symptoms lasting longer than 2 years.¹⁷

17.1.3 Type 2 (Arnold)-Chiari malformation

General information

Key concepts

- usually associated with myelomeningocele, often accompanied by hydrocephalus pathology includes: caudally displaced cervicomedullary junction, small posterior fossa, tectal beaking. Is probably not due to tethering
- major clinical findings: swallowing difficulties, apnea, stridor, opisthotonos, downbeat nystagmus
- when symptomatic: always check the shunt first! Then, consider surgical decompression (which cannot correct intrinsic brainstem abnormalities)
- cranial and cervical MRI is the diagnostic test of choice.

17

Usually associated with myelomeningocele (MM), or rarely spina bifida occulta.

Pathophysiology

Probably does not result from tethering of the cord by the associated MM. More likely due to primary dysgenesis of the brainstem with multiple other developmental anomalies.¹⁸

Major findings

Caudally dislocated cervicomedullary junction, pons, 4th ventricle and medulla. Cerebellar tonsils located at or below the foramen magnum. Replacement of normal cervicomedullary junction flexure with a “kink-like deformity.”

Other possible associated findings:

1. beaking of tectum
2. absence of the septum pellucidum with enlarged interthalamic adhesion: absence of the septum pellucidum is thought to be due to necrosis with resorption secondary to hydrocephalus, and not a congenital absence¹⁹ (p 178)

3. poorly myelinated cerebellar folia
4. hydrocephalus: present in most
5. heterotopias
6. hypoplasia of falx
7. microgyria
8. degeneration of lower cranial nerve nuclei
9. bony abnormalities:
 - a) of cervicomedullary junction
 - b) assimilation of atlas
 - c) platybasia
 - d) basilar impression
 - e) Klippel-Feil deformity (p.271)
10. hydromyelia
11. craniolacunia of the skull (see below)

Presentation

Findings are due to brain stem and lower cranial nerve dysfunction. Onset is rare in adulthood. The presentation of neonates differs substantially from older children, and neonates were more likely to develop rapid neurological deterioration with profound brain stem dysfunction over a period of several days than were older children in whom symptoms were more insidious and rarely as severe.²⁰

Findings include^{21,20}:

1. swallowing difficulties (neurogenic dysphagia) (69%).²² Manifests as poor feeding, cyanosis during feeding, nasal regurgitation, prolonged feeding time, or pooling of oral secretions. Gag reflex often decreased. More severe in neonates
2. apneic spells (58%): due to impaired ventilatory drive. More common in neonates
3. stridor (56%): more common in neonates, usually worse on inspiration (abductor and occasionally adductor vocal cord paralysis seen on laryngoscopy) due to 10th nerve paresis; usually transient, but may progress to respiratory arrest
4. aspiration (40%)
5. arm weakness (27%) that may progress to quadriparesis²³
6. opisthotonos (18%)
7. nystagmus: especially downbeat nystagmus
8. weak or absent cry
9. facial weakness

Diagnostic evaluation

Skull films

May demonstrate cephalofacial disproportion from congenital HCP. Craniolacunia (AKA lückenschädel) in 85%(round defects in the skull with sharp borders, separated by irregularly branching bands of bone; not due to increased ICP). Low lying internal occipital protuberance (foreshortened posterior fossa). Enlarged foramen magnum in 70% elongation of upper cervical lamina.¹

CT and/or MRI findings

Cranial and cervical MRI is the diagnostic test of choice.

- primary findings
 - a) “Z” bend deformity of medulla*
 - b) cerebellar peg
 - c) tectal fusion (“tectal beaking”)
 - d) enlarged massa intermedia (interthalamic adhesion)*
 - e) elongation/cervicallization of medulla
 - f) low attachment of tentorium
- associated findings
 - a) hydrocephalus
 - b) syringomyelia in the area of the cervicomedullary junction (reported incidence in pre MRI era¹⁷ ranges from 48–88%)
 - c) trapped fourth ventricle
 - d) cerebellomedullary compression
 - e) agenesis/dysgenesis of corpus callosum*

* items with an asterisk are best appreciated on MRI

Laryngoscopy

Performed in patients with stridor to rule out croup or other upper respiratory tract infection.

Treatment

General information

- insert CSF shunt for hydrocephalus (or check function of existing shunt)
- if neurogenic dysphagia, stridor, or apneic spells occur, expeditious posterior fossa decompression is recommended (see below) (required in 18.7% of MM patients²¹); before recommending decompression, always make sure the patient has a functioning shunt!

Surgical decompression

NB: it has been argued that part of the explanation for the poor operative results in infants is that many of the neurological findings may be due in part to intrinsic (uncorrectable) abnormalities which surgical decompression cannot improve.^{24,25} A dissenting view is that the histologic lesions are due to chronic brain stem compression and concomitant ischemia, and that expeditious brain stem decompression should be carried out when any of the following critical warning signs develop: neurogenic dysphagia, stridor, apneic spells.²⁰

Surgical technique

Decompression of cerebellar tonsils, usually with dural graft to decompress dura. Patients is placed prone, with the neck flexed. A suboccipital craniectomy is combined with a cervical laminectomy which must be carried down to the bottom of the tonsillar tip.²³ A thick constricting dural band is usually found between the C1 arch and foramen magnum. The dura is opened in a “Y” shaped incision. Caution when opening the dura above the level of the foramen magnum in infants as they have a well-developed occipital sinus and may have large dural lakes.²¹ DO NOT attempt to dissect tonsils from underlying medulla. In cases with a significant syringomyelic cavity, a syringo-subarachnoid shunt is placed.²⁰

Tracheostomy (usually temporary) is recommended if stridor and abductor laryngeal palsy were present pre-op. Close post-op respiratory monitoring is needed for obstruction and reduced ventilatory drive (mechanical ventilation is indicated for hypoxia or hypercarbia).

Outcome

68% had complete or near complete resolution of symptoms, 12% had mild to moderate residual deficits, and 20% had no improvement (in general, neonates fared worse than older children).²⁰

Respiratory arrest is the most common cause of mortality (8 of 17 patients who died), with the rest due to meningitis/ventriculitis (6 patients), aspiration (2 patients), and biliary atresia (1 patient).²¹

In follow-up ranging 7 mos-6 yrs, 37.8% mortality in operated patients.

Pre-op status and the rapidity of neurologic deterioration were the most important prognosticators. Mortality rate is 71% in infants having cardiopulmonary arrest, vocal cord paralysis or arm weakness within 2 weeks of presentation; compared to 23% mortality in patients with a more gradual deterioration. Bilateral vocal cord paralysis was a particularly poor prognosticator for response to surgery.²⁰

17.1.4 Other Chiari malformations

Chiari type 0

Patients with syringohydromyelia without hindbrain herniation that respond to p-fossa decompression have been described¹⁴ (so-called “Chiari zero malformation”).

Chiari type 1.5

Obex situated below foramen magnum, does not respond to suboccipital decompression with or without duroplasty.

Chiari type 3

Rare. Both the definition and even the existence are controversial. Most descriptions are based on 1 or 2 cases. Original description cited dislocation of the cerebellum below the foramen magnum into

an occipital encephalocele.²⁶ Some have added herniation of the medulla, fourth ventricle and all of the cerebellum into an occipital and high cervical encephalocele. Some have sided with Raimondi who included occipital encephaloceles associated with caudal displacement of the cerebellum and medulla.^{27,28}

Prognosis is poor for most, as it is usually incompatible with life.

Chiari type 4

Originally described as cerebellar hypoplasia without cerebellar herniation.²⁹ Existence as a distinct clinical entity is debated.²⁶

17.2 Neural tube defects

17.2.1 Classification

General information

There is no universally accepted classification system. Two are presented below.

Lemire classification

A system adapted from Lemire.³⁰

1. neurulation defects: non-closure of the neural tube results in open lesions
 - a) craniorachischisis: total dysraphism. Many die as spontaneous abortion
 - b) anencephaly: AKA exencephaly. Due to failure of fusion of the anterior neuropore. Neither cranial vault nor scalp covers the partially destroyed brain. Uniformly fatal. Risk of recurrence in future pregnancies: 3%
 - c) meningocele: most common in lumbar region
 - myelomeningocele (MM) (p.265)
 - myelocele
2. postneurulation defects: produces skin-covered (AKA closed) lesions (some may also be considered “migration abnormalities”, see below)
 - a) cranial
 - microcephaly: see below
 - hydranencephaly: loss of most of cerebral hemispheres, replaced by CSF (see below). Must R/O maximal hydrocephalus (see below)
 - holoprosencephaly: see below
 - lissencephaly: see below
 - porencephaly: see below to distinguish from schizencephaly
 - agenesis of corpus callosum: see below
 - cerebellar hypoplasia/Dandy Walker syndrome (p.256)
 - macrocephaly AKA megalencephaly: see below
 - b) spinal
 - diastatomyelia, diplomyelia: see Split cord malformation (p.274)
 - hydromyelia/syringomyelia (p.1144)

Migration abnormalities

A slightly different classification scheme defines the following as abnormalities of neuronal migration (some are considered postneurulation defects, see above):

1. lissencephaly: The most severe neuronal migration abnormality. Maldevelopment of cerebral convolutions (probably an arrest of cortical development at an early fetal age). Infants are severely retarded and usually don’t survive >2 yrs
 - a) agyria: completely smooth surface
 - b) pachygyria: few broad & flat gyri with shallow sulci
 - c) polymicrogyria: small gyri with shallow sulci. May be difficult to diagnose by CT/MRI, and may be confused with pachygyria
2. heterotopia: abnormal foci of (nonenhancing) gray matter which may be located anywhere from the subcortical white matter to (most commonly) the subependymal lining of the ventricles. May manifest as nodules or as a band of cortex. An early migration defect that results from arrest of radial migration. Almost always presents with seizures
3. cortical dysplasia: a cleft that does not communicate with the ventricle. Heterotopias are common. A migration abnormality not quite as severe as schizencephaly

4. schizencephaly:

- a) sine qua non: cleft that communicates with the ventricle (communication may be confirmed with CT cisternogram if necessary)
- b) cleft lined with cortical grey matter (often abnormal, may have polymicrogyria). This distinguishes it from porencephaly, a cystic lesion lined with connective or glial tissue that may communicate with the ventricular system, often caused by vascular infarcts or following intracerebral hemorrhage or penetrating trauma (including repeated ventricular punctures)
- c) two forms:
 - open lipped: large cleft to ventricle. Very severe forms may mimic hydranencephaly (see below)
 - close lipped (walls fused): □ look for a dimple in the lateral wall of the lateral ventricle immediately under the cortical cleft (the appearance of which may mimic an enlarged sulcus)
- d) may be unilateral or bilateral
- e) pia and arachnoid fuse
- f) there may be an “abnormal” vein that represents a cortical vein that now looks medullary because it follows the cortex into the cleft)
- g) absence of septum pellucidum in 80–90%
- h) presentation may range from seizures to hemiparesis depending on size and location

17.2.2 Examples of neural tube defects

Hydranencephaly

General information

A post-neurulation defect. Total or near-total absence of the cerebrum (small bands of cerebrum may be consistent with the diagnosis³¹), with intact cranial vault and meninges, the intracranial cavity being filled with CSF. There is usually progressive macrocrania, but head size may be normal (especially at birth), and, occasionally, microcephaly may occur. Facial dysmorphism is rare.

May be due to a variety of causes, the most commonly cited is bilateral ICA infarcts (which results in absence of brain tissue supplied by the anterior and middle cerebral arteries with preservation in the distribution of the PCA). May also be due to infection (congenital or neonatal herpes, toxoplasmosis, equine virus).

Less affected infants may appear normal at birth, but are often hyperirritable and retain primitive reflexes (Moro, grasp, and stepping reflex) beyond 6 mo. They rarely progress beyond spontaneous vowel production and social smiling. Seizures are common.

Differentiation from hydrocephalus

Progressive enlargement of CSF spaces may occur which can mimic severe (“maximal”) hydrocephalus (HCP). It is critical to differentiate the two since true HCP may be treated by shunting which may produce some re-expansion of the cortical mantle. Many means to distinguish hydranencephaly and HCP have been described, including:

1. EEG: shows no cortical activity in hydranencephaly (maximal HCP typically produces an abnormal EEG, but background activity will be present throughout the brain³¹) and is one of the best ways to differentiate the two
2. CT,^{32,31} MRI or ultrasound: majority of intracranial space is occupied by CSF. Usually do not see frontal lobes or frontal horns of lateral ventricles (there may be remnants of temporal, occipital or subfrontal cortex). A structure consisting of brainstem nodule (rounded thalamic masses, hypothalamus) and medial occipital lobes sitting on the tentorium occupies a midline position surrounded by CSF. Posterior fossa structures are grossly intact. The falx is usually intact (unlike alobar holoprosencephaly), and is not thickened, but may be displaced laterally. In HCP, some cortical mantle is usually identifiable
3. transillumination of the skull: in a darkened room, a bright light is placed against the surface of the skull. To transilluminate, the patient must be <9 mos old and the cortical mantle under the light source must be <1 cm thick,^{33 (p 215)} can also occur if fluid displaces the cortex inward (e.g. subdural effusions). Too insensitive to be very helpful
4. angiography: in “classic” cases resulting from bilateral ICA occlusion, no flow through supraclinoid carotids and a normal posterior circulation is expected

Treatment

Shunting may be performed to control head size, but unlike the case with maximal hydrocephalus, there is no restitution of the cerebral mantle.

Holoprosencephaly

AKA arhinencephaly. Failure of the telencephalic vesicle to cleave into two cerebral hemispheres. The degree of cleavage failure ranges from the severe alobar (single ventricle, no interhemispheric fissure) to semilobar and lobar (less severe malformations). The olfactory bulbs are usually small and the cingulate gyrus remains fused. Median faciocerebral dysplasia is common, and the degree of severity parallels the extent of the cleavage failure □ Table 17.9). 80%are associated with trisomy (primarily trisomy 13, and to a lesser extent trisomy 18). Survival beyond infancy is uncommon, most survivors are severely retarded, a minority are able to function in society. Some develop shunt dependent hydrocephalus. The risk of holoprosencephaly is increased in subsequent pregnancies of the same couple.

Microcephaly

Definition: head circumference more than 2 standard deviations below the mean for sex and gestational age. Terms that are sometimes used synonymously: microcrania, microcephalus. Not a single entity, many of the conditions in □ Table 17.9 may be associated with microcephaly. It may also result from maternal cocaine abuse.³⁵ It is important to differentiate microcephaly from a small skull resulting from craniosynostosis in which surgical treatment may provide opportunity for improved cerebral development.

Macroencephaly

Adapted.³⁶ (p 109) AKA macrencephaly, AKA megalencephaly. Not to be confused with macrocephaly (p.1403), which is enlargement of the skull. Not a single pathologic entity. An enlarged brain which may be due to: hypertrophy of gray matter alone, gray and white matter, presence of additional structures (glial overgrowth, diffuse gliomas, heterotopias, metabolic storage diseases...).

Conditions in which macrocephaly may be seen include:

- neurocutaneous syndromes (especially neurofibromatosis)
- megalencephaly-capillary malformation syndrome (MCAP): an overgrowth syndrome with megalencephaly (often with hydrocephalus, Chiari malformation, polymicrogyria and seizures), and capillary malformations in the skin (usually on the face)

Brains may weigh up to 1600–2850 grams. IQ may be normal, but developmental delay, retardation, spasticity and hypotonia may occur. Head circumference is 4–7 cm above mean. The usual signs of hydrocephalus (frontal bossing, bulging fontanelle, “setting sun” sign, scalp vein engorgement) are absent. Imaging studies (CT or MRI) show normal sized ventricles and can be used to rule out extra-axial fluid collections.

Table 17.9 The five facies of severe holoprosencephaly ³⁴		
Type of face	Facial features	Cranium and brain findings
cyclopia	single eye or partially divided eye in single orbit; arhinia with proboscis	microcephaly; alobar holoprosencephaly
ethmocephaly	extreme orbital hypotelorism; separate orbits; arhinia with proboscis	microcephaly; alobar holoprosencephaly
cebocephaly	orbital hypotelorism; proboscis-like nose; no median cleft lip	microcephaly; usually has alobar holoprosencephaly
with median cleft lip	orbital hypotelorism; flat nose	microcephaly; sometimes has trigonocephaly; usually has alobar holoprosencephaly
with median philtrum-premaxilla anlage	orbital hypotelorism; bilateral lateral cleft lip with median process representing philtrum-premaxillary anlage; flat nose	microcephaly; sometimes has trigonocephaly; semilobar or lobar holoprosencephaly

17.2.3 Risk factors

1. lack of prenatal folic acid: early administration of folic acid^{37,38,39} (0.4 mg/d if no history of neural tube defects; 4 mg/d in a carrier or with previous child with NTD was associated with a 71% reduction in recurrence of NTD⁴⁰) (confirm that vitamin B12 levels are normal)
2. folate antagonists (e.g. carbamazepine) doubles the incidence of MM
3. mothers with 5, 10-methylenetetrahydrofolate reductase (MTHFR) gene polymorphism. The common variant, C677 T, substitutes an alanine residue for valine at position 222 in the folate dependent MTHFR enzyme → decreased enzyme activity → reduced levels of tissue folate, and increased levels of homocysteine in the plasma. This polymorphism may be homozygous (TT genotype) or heterozygous (CT genotype); present in ≈10% and 38% of the population, respectively. The effects with the TT genotype are more pronounced than with the heterozygous CT form, and there is an increased risk of neural tube defects, as well as a lesser increased risk of cardiovascular disease⁴¹
4. use of valproic acid (Depakene®) during pregnancy is associated with a 1–2% risk of NTD⁴²
5. maternal heat exposure in the form of hot-tubs, saunas or fever (but not electric blankets) in the first trimester was associated with an increased risk of NTDs⁴³
6. obesity (before and during pregnancy) increases the risk of NTD^{44,45}
7. maternal cocaine abuse may increase the risk of microcephaly, disorders of neuronal migration, neuronal differentiation and myelination³⁵

17.2.4 Prenatal detection of neural tube defects

Serum alpha-fetoprotein (AFP)

See Alpha-fetoprotein (p.600) for background. A high maternal serum AFP (≥ 2 multiples of the median for the appropriate week of gestation) between 15–20 weeks gestation carries a relative risk of 224 for neural tube defects, and an abnormal value (high or low) was associated with 34% of all major congenital defects.⁴⁶ The sensitivity of maternal serum AFP for spina bifida was 91% (10 of 11 cases), it was 100% for 9 cases of anencephaly. However, other series show a lower sensitivity. Closed lumbosacral spine defects, accounting for ≈ 20% of spina bifida patients,⁴⁷ will probably be missed by serum AFP screening, and may also be missed on ultrasound. Since maternal serum AFP rises during normal pregnancy, an overestimate of gestational age may cause an elevated AFP to be interpreted as normal, and an underestimate may cause a normal level to be interpreted as elevated.⁴⁸

Ultrasound

Prenatal ultrasound will detect 90–95% of cases of spina bifida, and thus in cases of elevated AFP, it can help differentiate NTDs from non-neurologic causes of elevated AFP (e.g. omphalocele), and can help to more accurately estimate gestational age.

Amniocentesis

For pregnancies subsequent to a MM, if prenatal ultrasound does not show spinal dysraphism, then amniocentesis is recommended (even if abortion is not considered, it may allow for optimal post-partum care if MM is diagnosed). Amniotic fluid AFP levels are elevated with open neural tube defects, with a peak between weeks 13–15 of pregnancy. Amniocentesis also carries a ≈ 6% risk of fetal loss in this population.

17.3 Neurenteric cysts

17.3.1 General information

No uniformly accepted nomenclature. Working definition: CNS cyst lined by endothelium primarily resembling that of the GI tract, or less often, respiratory tract. Congenital. Not true neoplasms. Most common alternate term: enterogenous cyst. Less common terms include: teratomatous cyst, intestinoma, archenteric cyst,⁴⁹ enterogene cyst, and endodermal cyst. Usually affect the upper thoracic and lower cervical spine.⁵⁰ Associated developmental vertebral anomalies (e.g. diastematomyelia) are common.⁵¹ Rarely intracranial (see below). Spinal neurenteric cysts (NEC) may have a fistulous or fibrous connection to the GI tract (through a spinal dysraphism) and some call these endodermal

sinus cysts. Occurs as a result of persistence of the neurenteric canal (temporary duct between the notochord and the primitive gut (amniotic and yolk sacs) formed during week 3 of embryogenesis by breakdown of the floor of the notochordal canal).

17.3.2 Intracranial neurenteric cysts

General information

Rare, most common in p-fossa. Initially, may be difficult to rule-out metastasis from an extremely well-differentiated primary adenocarcinoma of unknown origin (absence of progressive disease suggests NEC). Locations:

1. posterior fossa
 - a) cerebellopontine angle (CPA)⁴⁹: usually intradural, extraaxial (case report of extradural lesion with bone destruction⁵²)
 - b) in midline anterior to brainstem⁵⁰
 - c) cisterna magna⁵³
2. supratentorial: only 15 case reports as of 2004.⁵⁴ Locations: suprasellar⁵⁵ (possible confusion with Rathke's cleft cyst), frontal lobe intraparenchymal,⁵⁴ quadrigeminal plate region, dural-based extra-axial. Source of endoderm is controversial since the primitive foregut extends cranially only to the midbrain.⁵⁶ Theory: colloid cysts, Rathke cleft cysts, and supratentorial NECs may all arise from remnants of Seessel's pouch, a transient endodermally derived diverticulum of the cranial end of the embryonic foregut⁵⁷

Clinical

Most commonly present during the first decade of life.⁵¹ Pain or myelopathy from the intraspinal mass are the most common presentations in older children and adults. Neonates and young children may present with cardiorespiratory compromise from an intrathoracic mass, or with cervical spinal cord compression.⁵¹ Meningitis may occur from the fistulous tract, especially in newborns and infants.

Imaging

Intracranial NEC:

- CT: usually low density, nonenhancing⁵⁸
- T1 WI MRI: isointense or sl hyperintense to CSF (may be hyperintense if there are blood products). T2 WI isointense to CSF.⁵⁸ Nonenhancing

Histology

Most are simple cysts lined by cuboidal-columnar epithelium and mucin secreting goblet cells. Less common types of epithelium described include: stratified squamous and pseudostratified columnar, and ciliated epithelial cells. Mesodermal components may be present, including smooth muscle and adipose tissue, and some have called these teratomatous cysts^{59,60} which is not to be confused with teratomas which are true germinal cell neoplasms. May be histologically identical to colloid cysts.

Treatment

Spinal NEC

Surgical removal usually reverses the symptoms. Recurrence is uncommon with complete removal of cyst wall.

Intracranial NEC

Capsule adherent to brainstem may prevent complete resection, which predisposes to delayed recurrence. Apparently successful treatment by evacuation of contents and marsupialization has been reported (5 cases, mean follow-up: 5 yrs⁶¹). Incomplete removal requires long-term follow-up. Hydrocephalus is shunted if indicated.

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Part V

Coma and Brain Death

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18 Coma

18.1 General information

Consciousness has two components: arousal and content. Impairment of arousal can vary from mild (drowsiness or somnolence), to obtundation, to stupor to coma. Coma is the severest impairment of arousal, and is defined as the inability to obey commands, speak, or open the eyes to pain.

The Glasgow Coma Scale (GCS) is a widely used scoring system with good repeatability and is shown in □ Table 18.1 (note: the scale is used to assess level of consciousness and is not designed for following neurologic deficits). General practice is to record a “T” (for “intubated”) next to the verbal score and the total score for patients whose verbal axis cannot be assessed because of intubation.² No single GCS score defines a cutoff for coma, however, 90% of patients with $GCS \leq 8$ and none with $GCS \geq 9$ meet the above definition of coma. Thus, $GCS \leq 8$ is a generally accepted operational definition of coma.

- A number of scales for use in children have been proposed. One is shown in □ Table 18.2.³
- Coma results from one or more of the following:
- dysfunction of high brainstem (central upper pons) or midbrain
 - bilateral diencephalic dysfunction
 - diffuse lesions in both cerebral hemispheres (cortical or subcortical white matter)

Table 18.1 Glasgow coma ^a scale ¹ (recommended for age ≥ 4 yrs)			
Points ^b	Best eye opening	Best verbal	Best motor
6	–	–	obeys
5	–	oriented	localizes pain
4	spontaneous	confused	withdraws to pain
3	to speech	inappropriate	flexion (decorticate)
2	to pain ^c	incomprehensible	extensor (decerebrate)
1	none	none	none ^d
^a technically, this is a scale of impaired consciousness, whereas “coma” implies unresponsiveness			
^b range of total points: 3 (worst) to 15 (normal)			
^c when testing eye opening to pain, use peripheral stimulus (the grimace associated with central pain may cause eye closure)			
^d if no motor response, important to exclude spinal cord transection			

Table 18.2 Children’s coma scale ^a (for age <4 yrs)					
Points ^b	Best eye		Best verbal		Best motor
6	–		–		obeys
5	–		smiles, oriented to sound, follows objects, interacts		localizes pain
Crying					
Interaction					
4	spontaneous		consolable	inappropriate	withdraws to pain
3	to speech		inconsistently consol- able	moaning	flexion (decor- ticate)
2	to pain		inconsolable	restless	extensor (de- cerebrate)
1	none		none	none	none
^a same as adult Glasgow coma scale except for verbal response ³					
^b range of total points: 3 (worst) to 15 (normal)					

18.2 Posturing

18.2.1 General information

The following terms do not accurately localize the site of the lesion. Decorticate posturing implies a more rostral lesion than extensor posturing, and prognosis may be slightly better.

18.2.2 Decorticate posturing

Classically attributed to disinhibition by removal of corticospinal pathways above the midbrain.

Overview: abnormal flexion in UE and extension in LE.

Detail:

- UE: slow flexion of arm, wrist and fingers with adduction
- LE: extension, internal rotation, plantarflexion

18.2.3 Decerebrate posturing

Classically attributed to disinhibition of vestibulospinal tract (more caudal) and pontine reticular formation (RF) by removing inhibition of medullary RF (transection at intercollicular level, between vestibular and red nuclei).

Overview: abnormal extension in UE and LE.

Detail:

- Head & trunk: opisthotonos (head and trunk extended), teeth clenched
- UE: arms extended, adducted and hyperpronated (internally rotated), wrists flexed, fingers flexed
- LE: extended and internally rotated, feet plantarflexed and inverted, toes plantarflexed.

18.3 Etiologies of coma

18.3.1 Toxic/metabolic causes of coma

1. electrolyte imbalance: especially hypo- or hypernatremia, hypercalcemia, renal failure with elevated BUN & creatinine, liver failure with elevated ammonia
2. endocrine: hypoglycemia, nonketotic hyperosmolar state, DKA (diabetic ketoacidosis, AKA diabetic coma), myxedema coma, Addisonian crisis (hypoadrenalism)
3. vascular: vasculitis, DIC, hypertensive encephalopathy (p. 194)
4. toxic: EtOH, drug overdose (including narcotics, iatrogenic polypharmacy, barbiturates), lead intoxication, carbon monoxide (CO) poisoning, cyclosporine (causes an encephalopathy that shows white-matter changes on MRI that is often reversible with discontinuation of the drug)
5. infectious/inflammatory: meningitis, encephalitis, sepsis, lupus cerebritis, neurosarcoidosis (p. 189), toxic-shock syndrome
6. neoplastic: leptomeningeal carcinomatosis, rupture of neoplastic cyst
7. nutritional: Wernicke's encephalopathy, vitamin B₁₂ deficiency
8. inherited metabolic disorders: porphyria, lactic acidosis
9. organ failure: uremia, hypoxemia, hepatic encephalopathy, Reye's syndrome, anoxic encephalopathy (e.g. post-resuscitation from cardiac arrest), CO₂ narcosis
10. epileptic: status epilepticus (including non-convulsive status), post-ictal state (especially with unobserved seizure)

18.3.2 Structural causes of coma

1. vascular:
 - a) bilateral cortical or subcortical infarcts (e.g. with cardioembolism due to SBE, mitral stenosis, A-fib, mural thrombus...)
 - b) occlusion of vessel supplying both cerebral hemispheres (e.g. severe bilateral carotid stenosis)
 - c) bilateral diencephalic infarcts: well described syndrome. May be due to occlusion of a thalamo-perforator supplying both medial thalamic areas or with "top-of-the-basilar" occlusion. Initially resembles metabolic coma (including diffuse slowing on EEG), patient eventually arouses with apathy, memory loss, vertical gaze paresis
2. infectious: abscess with significant mass effect, subdural empyema, herpes simplex encephalitis
3. trauma: hemorrhagic contusions, edema, hematoma (see below)
4. neoplastic: primary or metastatic

- 5. herniation from mass effect: presumably brainstem compression causes dysfunction of reticular activating system or mass in one hemisphere causing compression of the other results in bilateral hemisphere dysfunction
- 6. increased intracranial pressure: reduces CBF
- 7. acute lateral shift (midline shift) of the brain: e.g. due hematoma (subdural or epidural) □ Table 18.3)

18.3.3 Pseudocoma

Differential diagnosis:

- 1. locked-in syndrome: ventral pontine infarction
- 2. psychiatric: catatonia, conversion reaction
- 3. neuromuscular weakness: myasthenia gravis, Guillain-Barré

18.3.4 Approach to the comatose patient

General information

This section addresses nontraumatic coma. See Head trauma (p.824) for that topic.
Initial evaluation: includes measures to protect brain (by providing CBF, O₂, and glucose), assesses upper brainstem (Cr. N. VIII), and rapidly identifies surgical emergencies. Keep “pseudocoma” as a possible etiology in back of mind.

Outline of approach to comatose patient

- 1. cardiovascular stabilization: establish airway, check circulation (heartbeat, BP, carotid pulse), CPR if necessary
- 2. obtain blood for tests
 - a) STAT: electrolytes (especially Na, glucose, BUN), CBC+diff, ABG
 - b) others as appropriate: toxicology screen (serum & urine), calcium, ammonia, antiepileptic drug (AED) levels (if patient is taking AEDs)
- 3. administer emergency supportive medications
 - a) glucose: at least 25 ml of D50 IVP. Due to potentially harmful effect of glucose in global ischemia, if possible check fingerstick glucose first, otherwise glucose is given without exception, unless it is known with certainty that serum glucose is normal
 - b) naloxone (Narcan®): in case of narcotic overdose. 1 amp (0.4 mg) IVP
 - c) flumazenil (Romazicon®): in case of benzodiazepine overdose. Start with 0.2 mg IV over 30 seconds, wait 30 secs, then give 0.3 mg over 30 secs at 1 minute intervals up to 3 mg or until patient arouses
 - d) thiamine: 50–100 mg IVP (3% of Wernicke’s present with coma)
- 4. core neuro exam (assesses midbrain/upper pons, allows emergency measures to be instituted rapidly, more thorough evaluation possible once stabilized): see Core neuro exam below
- 5. if herniation syndrome or signs of expanding p-fossa lesion with brainstem compression □ Table 18.4): initiate measures to lower ICP – see Treatment measures for elevated ICP (p.866) -, then get a CT scan if patient begins improving, otherwise emergency surgery. □ Do NOT do LP
- 6. if meningitis suspected (altered mental status + fever, meningeal signs...)
 - a) if no indication of herniation, p-fossa mass (□ Table 18.4), focal deficit indicating mass effect or papilledema: perform LP, start antibiotics immediately (do not wait for CSF results); see Meningitis (p.318)
 - b) if evidence of possible mass effect, coagulopathy or herniation, CT to R/O mass. If significant delay anticipated, consider empiric antibiotics or careful LP with small gauge needle (≤22

Table 18.3 Effect of lateral shift on level of consciousness ⁴	
Amount of midline shift	Level of consciousness
0–3 mm	alert
3–4 mm	drowsy
6–8.5 mm	stuporous
8–13 mm	comatose

Table 18.4 Signs of herniation syndrome or posterior fossa lesion	
Herniation syndromes	Signs of P-fossa lesion
also see Herniation syndromes (p.302)	also see Posterior fossa (infratentorial) tumors (p.592)
• unilateral sensory or motor deficit • progressive obtundation → coma • unilateral 3rd nerve palsy • decorticate or decerebrate posturing (especially if unilateral)	• initial symptoms of diplopia, vertigo, bilateral limb weakness, ataxia, occipital H/A • rapid onset of deterioration/coma • bilateral motor signs at onset • miosis • absent caloric to horizontal movement, possibly with preserved vertical movements • ocular bobbing • ophthalmoplegia • multiple cranial nerve abnormalities with long tract signs • apneustic, cluster or ataxic respirations

- Ga.), measure opening pressure (OP), remove only a small amount of CSF if OP high, replace CSF if patient deteriorates; LP in this setting may be risky, see Lumbar puncture (p.1504).
7. treat generalized seizures if present. If status epilepticus is suspected, treat as indicated (p.470); obtain emergency EEG if available
 8. treat metabolic abnormalities
 - a) restore acid-base balance
 - b) restore electrolyte imbalance
 - c) maintain body temperature
 9. obtain as complete history as possible once stabilized
 10. administer specific therapies

Core neuro exam for coma

Respiratory rate and pattern

The most common disorder in impaired consciousness (this information is often lacking in patients that are intubated early in their course):

- Cheyne-Stokes (□ Fig. 18.1 a): breathing gradually crescendos in amplitude and then trails off, followed by an expiratory pause, and then the pattern repeats. Hyperpneic phase is usually longer than apneic. Usually seen with diencephalic lesions or bilateral cerebral hemisphere dysfunction (non-specific), e.g. early increased ICP or metabolic abnormality. Results from an increased ventilatory response to CO₂
- hyperventilation: usually in response to hypoxemia, metabolic acidosis, aspiration, or pulmonary edema. True central neurogenic hyperventilation is rare, and usually results from dysfunction within the pons. If no other brainstem signs are present, may suggest psychiatric disorder
- cluster breathing (□ Fig. 18.1 b): periods of rapid irregular breathing separated by apneic spells, may appear similar to Cheyne-Stokes, may merge with various patterns of gasping respirations. High medulla or lower pons lesion. Often an ominous sign
- apneustic (rare; □ Fig. 18.1 c): a pause at full inspiration. Indicates pontine lesion, e.g. with basilar artery occlusion
- ataxic (Biot’s breathing; □ Fig. 18.1 d): no pattern in rate or depth of respirations. Seen with medullary lesion. Usually preterminal

Pupil

Record size (in mm) in ambient light, and in reaction to direct/consensual light

1. □ equal and reactive pupils indicates toxic/metabolic cause with few exceptions (see below) (may have hippus). The light reflex is the most useful sign in distinguishing metabolic from structural coma
 - a) the only metabolic causes of fixed/dilated pupil: glutethimide toxicity, anoxic encephalopathy, anticholinergics (including topically applied atropine), occasionally with botulism toxin poisoning
 - b) narcotics cause small pupils (miosis) with a small range of constriction and sluggish reaction to light (in severe overdose, the pupils may be so small that a magnifying glass may be needed to see reaction)

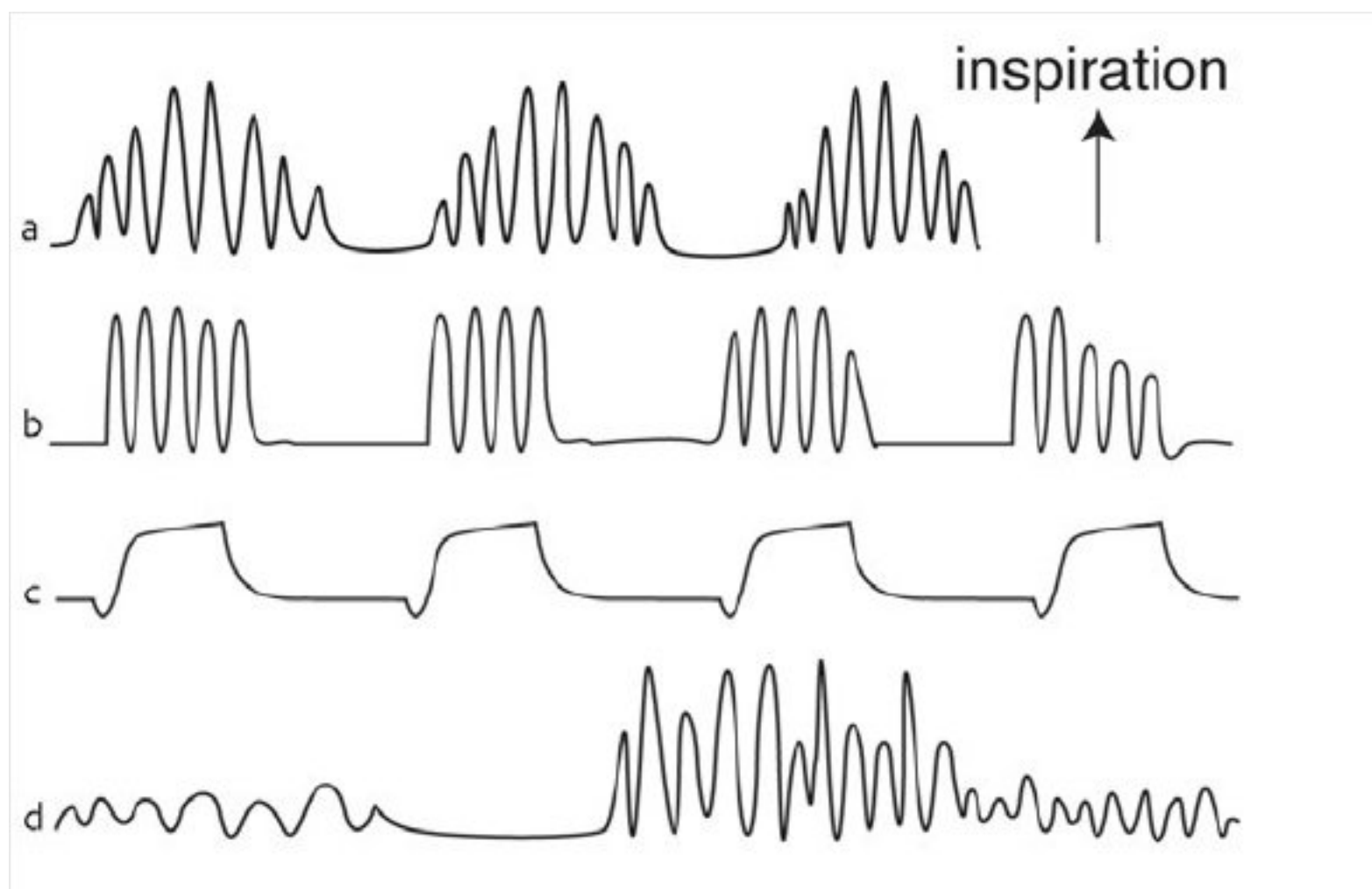


Fig. 18.1 Respiratory rate and pattern.

- a) Cheyne-Stokes respiratory pattern
- b) Cluster breathing
- c) Apneustic respiratory pattern
- d) Ataxic respirations

2. unequal; note: an afferent pupillary defect does not produce anisocoria, see Alterations in pupillary diameter (p.561)
 - a) fixed and dilated pupil: usually due to oculomotor palsy. Possible herniation, especially if larger pupil associated with ipsilateral 3rd nerve EOM palsy (eye deviated “down and out”)
 - b) possible Horner’s syndrome: consider carotid occlusion/dissection (NB: in Horner’s syndrome, the miotic (smaller) pupil is the abnormal one)
3. bilateral pupil abnormalities
 - a) pinpoint with minute reaction that can be detected with magnifying glass⁵: pontine lesion (sympathetic input is lost; parasympathetics emerge at Edinger-Westphal nucleus and are unopposed)
 - b) bilateral fixed and dilated (7–10 mm): subtotal damage to medulla or immediate post-anoxia or hypothermia (core temperature <90° F (32.2° C))
 - c) midposition (4–6 mm) and fixed: more extensive midbrain lesion, presumably due to interruption of sympathetics and parasympathetics

Extraocular muscle function

1. deviations of ocular axes at rest
 - a) bilateral conjugate deviation:
 - frontal lobe lesion (frontal center for contralateral gaze): looks toward side of destructive lesion (away from hemiparesis). Looks away from side of seizure focus (looks at jerking side), may be status epilepticus. Reflex eye movements (see below) are normal
 - pontine lesion: eyes look away from lesion and towards hemiparesis; caloric impaired on side of lesion
 - “wrong way gaze”: medial thalamic hemorrhage. Eyes look away from lesion and towards hemiparesis (an exception to the axiom that the eyes look towards a destructive supratentorial lesion)⁵
 - downward deviation: may be associated with unreactive pupils, Parinaud’s syndrome (p.99). Etiologies: thalamic or midbrain pretectal lesions, metabolic coma (especially barbiturates), may follow a seizure
 - b) unilateral outward deviation on side of larger pupil (III palsy): uncal herniation
 - c) unilateral inward deviation: VI (abducens) nerve

- d) skew deviation
 - III or IV nerve/nucleus lesion
 - infratentorial lesion (frequently dorsal midbrain)
2. spontaneous eye movements
 - a) “windshield wiper eyes”: random roving conjugate eye movements. Non-localizing. Indicates an intact III nucleus and medial longitudinal fasciculus
 - b) periodic alternating gaze, AKA “ping-pong gaze”: eyes deviate side to side with frequency of $\approx 3-5$ per second (pausing 2–3 secs in each direction). Usually indicates bilateral cerebral dysfunction
 - c) ocular bobbing (p.570): repetitive rapid vertical deviation downward with slow return to neutral position.
3. internuclear ophthalmoplegia (INO) (p.565): due to lesion in medial longitudinal fasciculus (MLF) (fibers crossing to contralateral III nucleus are interrupted). Eye ipsilateral to MLF lesion does not adduct on spontaneous eye movement or in response to reflex maneuvers (e.g. calorics)
4. reflex eye movements (maneuvers to test brainstem)
 - a) oculovestibular reflex^a, AKA ice water calorics: first rule-out TM perforation and occlusion of the EAC by cerumen. Elevate the HOB 30°, irrigate one ear with 60–100 ml of ice water^b. NB: response is inhibited by neuromuscular blocking agents (NMBA)
 - a comatose patient with an intact brainstem will have tonic conjugate eye deviation to side of cold stimulus which may be delayed up to one minute or more. There will be no fast component (nystagmus) (the cortical component) even if the brainstem is intact. (NB: oculocephalic reflex^c (doll’s eyes) provides similar information as oculovestibular reflex^d, but poses a greater risk to the spinal cord if C-spine not cleared)
 - no response: symmetrical, could be specific toxin (e.g. neuromuscular block or barbiturates), metabolic cause, brain death or possibly massive infratentorial lesion
 - asymmetric: infratentorial lesion, especially if response inconsistent with 3rd nerve palsy (herniation). Usually maintained in toxic/metabolic coma
 - nystagmus without tonic deviation (i.e. eyes remain in primary position) virtually diagnostic of psychogenic coma
 - contralateral eye fails to adduct: INO (MLF lesion)
 - b) optokinetic nystagmus presence strongly suggests psychogenic coma

Notes:

^a Oculovestibular reflexes (calorics): the anticipated response is commonly misunderstood. In a normal awake patient there is slow deviation towards the side of the cold stimulus with nystagmus (which is named for the rapid, cortical phase) in the opposite direction (hence the mnemonic “COWS” (cold-opposite, warm-same)). Nystagmus will be absent in the comatose patient.

^b HOB at 30° places the horizontal semicircular canal (SCC) vertically for maximal response.⁶ (p 56) – Cold water → downward endolymphatic currents, away from the ampulla of the horizontal SCC.⁶ (p 57)

^c Oculocephalic reflex (“doll’s eyes” or “doll’s head”): do not perform if there is any uncertainty about cervical-spine stability. In an awake patient, the eyes will either move with the head, or, if the movement is slow enough and the patient is fixating on an object, there will be contraversive conjugate eye movement⁷ (c.f. oculovestibular reflex which does not depend on patient’s level of cooperation). In a comatose patient with an intact brainstem & cranial nerves, there will also be contraversive conjugate eye movement (a positive doll’s eyes response).

^d Oculovestibular reflexes are absent but oculocephalic are maintained only when vestibular inputs are interrupted, e.g. streptomycin toxicity of labyrinths or bilateral vestibular schwannomas.

Motor

Record muscle tone and reflexes, response to pain, plantar reflex (Babinski). Note asymmetries

1. appropriate: implies corticospinal tracts and cortex intact
2. asymmetric: supratentorial lesion (tone usually increased), unlikely in metabolic
3. inconsistent/variable: seizures, psychiatric
4. symmetric: metabolic (usually decreased). Asterixis, tremor, myoclonus may be present in metabolic coma
5. hyporeflexia: consider myxedema coma, especially in patient presenting weeks after transsphenoidal surgery
6. patterns
 - a) decorticate posturing: arms flex, legs extend: large cortical or subcortical lesion
 - b) decerebrate posturing: arms and legs extend: brainstem injury at or below lower midbrain

- c) arms flexed, legs flaccid: pontine tegmentum
- d) arms flaccid, legs appropriate (“man-in-the-barrel syndrome”): anoxic injury (poor prognosis)

Cilio-spinal reflex

Pupillary dilatation to noxious cutaneous stimuli: tests integrity of sympathetic pathways

1. bilaterally present: metabolic
2. unilaterally present: possible 3rd nerve lesion (herniation) if on side of larger pupil. Possible pre-existing Horner’s syndrome if on side of smaller pupil
3. bilaterally absent: usually not helpful

18.4 Herniation syndromes

18.4.1 General information

Classic teaching has been that shifts in brain tissue (e.g. caused by masses or increased intracranial pressure) through rigid openings in the skull (herniation) compress other structures of the CNS producing the observed symptoms. In actuality it may be that herniation is an epiphenomenon that occurs late in the process and is not actually the cause of the observations.⁸ However, herniation models still serve as useful models.

The five most common herniation syndromes are:

- Supratentorial herniation
 - central (transtentorial) herniation (p.303)
 - uncal herniation (p.304)
- cingulate herniation: cingulate gyrus herniates under falx (AKA subfalcine herniation). Usually asymptomatic unless ACA kinks and occludes causing bifrontal infarction. Usually warns of impending transtentorial herniation
- Infratentorial herniation
 - upward cerebellar (see below)
 - tonsillar herniation (see below)

18.4.2 Coma from supratentorial mass

See reference.⁶

General information

Central and uncal herniation each causes a different form of rostral-caudal deterioration. Central herniation results in sequential failure of: diencephalon, midbrain, pons, medulla (p.303). See also uncal herniation (p.304). “Classic” signs of increased ICP (HTN, bradycardia, altered respiratory pattern) usually seen with p-fossa lesions may be absent in slowly developing supratentorial masses.

Distinction between central and uncal herniation is difficult when dysfunction reaches the midbrain level or below. Predicting the location of the lesion based on the herniation syndrome is unreliable.

Clinical characteristics differentiating uncal from central herniation

- decreased consciousness occurs early in central herniation, late in uncal
- uncal herniation syndrome rarely gives rise to decorticate posturing

Differential diagnosis of supratentorial etiologies

1. vascular: stroke, intracerebral hemorrhage, SAH
2. inflammatory: cerebral abscess, subdural empyema, herpes simplex encephalitis
3. neoplastic: primary or metastatic
4. traumatic: epidural or subdural hematoma, depressed skull fracture

18.4.3 Coma from infratentorial mass

General information

NB: it is essential to identify patients with primary posterior fossa lesions (□ Table 18.4) as they may require emergent surgical intervention.

Etiologies of infratentorial masses:

1. vascular: brainstem infarction (including basilar artery occlusion), cerebellar infarction or hematoma
2. inflammatory: cerebellar abscess, central pontine myelinolysis, brainstem encephalitis
3. neoplasms: primary or metastatic
4. traumatic: epidural or subdural hematoma

Hydrocephalus

Infratentorial masses can produce obstructive hydrocephalus by compressing the Sylvian aqueduct and/or 4th ventricle.

Upward cerebellar herniation

Occasionally seen with p-fossa masses, may be exacerbated by ventriculostomy. Cerebellar vermis ascends above tentorium, compressing the midbrain, and possibly occluding SCAs → cerebellar infarction. May compress sylvian aqueduct → hydrocephalus.

Tonsillar herniation

Cerebellar tonsils “cone” through foramen magnum, compressing medulla → respiratory arrest. Usually rapidly fatal.

Occurs with either supra- or infra-tentorial masses or with elevated ICP. May be precipitated by LP. In many cases, there may simply be pressure on the brainstem without actual herniation.⁹ There are also cases with significant cerebellar herniation through the foramen magnum with the patient remaining alert.⁸

18.4.4 Central herniation

General information

AKA transtentorial herniation AKA tentorial herniation. Usually more chronic than uncal herniation, e.g. due to tumor, especially of frontal, parietal or occipital lobes.

The diencephalon is gradually forced through the tentorial incisura. The pituitary stalk may be sheared, resulting in diabetes insipidus. PCAs may be trapped along the open edge of the incisura, and may occlude producing cortical blindness; see Blindness from hydrocephalus (p.396). The brainstem suffers ischemia from compression and shearing of perforating arteries from basilar artery → hemorrhages within the brainstem (Duret hemorrhages).

Imaging

MRI or CT: the perimesencephalic cisterns may be compressed.

Skull x-rays: downward displacement of the pineal gland may be identified.¹⁰

Stages of central herniation

Diencephalic stage

Early. May be due to diffuse bilateral hemisphere dysfunction (e.g. from decreased blood flow from increased ICP) or (more likely) from bilateral diencephalic dysfunction due to downward displacement. This stage warns of impending (irreversible) midbrain damage but is frequently reversible if the cause is treated.

Consciousness: Altered alertness is first sign; usually lethargy, agitation in some. Later: stupor → coma.

Respiration: Sighs, yawns, occasional pauses. Later: Cheyne-Stokes.

Pupils: Small (1 – 3 mm), small range of contraction.

Oculomotor: Conjugate or slightly divergent roving eyes; if conjugate then brainstem intact. Usually positive DOLL'S EYES and conjugate ipsilateral response to cold water caloric (CWC). Impaired upgaze due to compression of superior colliculi and diencephalic pretectum: Parinaud's syndrome (p.99)

Motor: Early: appropriate response to noxious stimuli, bilateral Babinski, gegenhalten (paratonic resistance). If previously hemiparetic contralateral to lesion: may worsen. Later: motionlessness & grasp reflexes, then DECORTICATE (initially contralateral to lesion in most cases).

Midbrain – upper pons stage

When midbrain signs fully developed (in adults), prognosis is very poor (extreme ischemia of midbrain). Fewer than 5% of cases will have a good recovery if treatment is successfully undertaken at this stage.

Respiration: Cheyne-Stokes → sustained tachypnea.

Pupils: Moderately dilated midposition (3–5 mm), fixed. Note: in pontine hemorrhage pinpoint pupils appear because the loss of sympathetics leaves the parasympathetics unopposed, whereas in herniation, the parasympathetics are usually lost, too (3rd nerve injury).

Oculomotor: Doll's eyes & CWC impaired, may be dysconjugate. MLF lesion → internuclear ophthalmoplegia (when doll's or CWC elicited and dysconjugate, medially moving eye moves less than laterally moving eye).

Motor: Decorticate → bilaterally DECEREBRATE (occasionally spontaneously).

Lower pons – upper medullary stage

Respiration: Regular, shallow and rapid (20–40/min).

Pupils: Midposition (3–5 mm), fixed.

Oculomotor: Doll's eyes and CWC unelicitable.

Motor: Flaccid. Bilateral Babinski. Occasionally LE flexion to pain.

Medullary stage (terminal stage)

Respiration: Slow, irregular rate and depth, sighs/gasps. Occasionally hyperpnea alternating with apnea

Pupils: Dilate widely with hypoxia.

Outcome after central herniation

In a series of 153 patients with signs of central herniation (altered level of consciousness, anisocoria or fixed pupils, abnormal motor findings) 9% had good recovery, 18% had functional outcome, 10% were severely disabled, and 60% died.¹¹

Factors associated with a better result were young age (especially age ≤ 17 yrs), anisocoria with deteriorating Glasgow Coma Score and nonflaccid motor function. Factors associated with poor outcome were bilaterally fixed pupils, with only 3.5% of these patients having a functional recovery.

18.4.5 Uncal herniation

General information

Usually occurs in rapidly expanding traumatic hematomas, frequently in the lateral middle-fossa or temporal lobe pushing medial uncus and hippocampal gyrus over edge of tentorium, entrapping third nerve and directly compressing midbrain. PCA may be occluded (as with central herniation). For CT criteria see below.

Impaired consciousness is NOT a reliable early sign. Earliest consistent sign: unilaterally dilating pupil. However, it is unlikely that a patient undergoing early uncal herniation would be completely neurologically intact except for anisocoria (do not dismiss confusion, agitation, etc.). Once brainstem findings appear, deterioration may be rapid (deep coma may occur within hours).

CT and/or MRI criteria

See reference.¹²

The tentorial incisura surrounds the interpeduncular and pre-pontine cisterns and brainstem. There is great interpersonal variability in the amount of space in the incisura.

Impending uncal or hippocampal herniation may be indicated by encroachment on lateral aspect of suprasellar cistern → flattening of normal pentagonal shape. Once herniation occurs CT may show: brainstem displacement and flattening, compression of contralateral cerebral peduncle, midbrain rotation with slight increase of ipsilateral subarachnoid space. Also, contralateral hydrocephalus may occur.¹³

Obliteration of parasellar and interpeduncular cisterns occurs as uncus and/or hippocampus are forced through hiatus. Brainstem is elongated in AP direction due to lateral compression. Since dural

structures enhance with IV contrast, this may be used to help delineate tentorial margins when necessary.

Stages of uncal herniation

Early third nerve stage

This is not a brainstem finding, it is due to 3rd nerve compression.

Pupils: Approach to the comatose patient

Oculomotor: Doll's eyes (oculocephalic reflex)=normal or dysconjugate. CWC (oculovestibular reflex)=slow ipsilateral deviation, impaired nystagmus, may be dysconjugate if external oculomotor ophthalmoplegia (EOO).

Respirations: Normal.

Motor: Appropriate response to nociceptive stimulus. Contralateral Babinski.

Late third nerve stage

Midbrain dysfunction occurs almost immediately after symptoms extend beyond those due to focal cerebral lesion (i.e. may skip diencephalic stage, due to lateral pressure on midbrain). Treatment delays may result in irreversible damage.

Pupils: Pupil fully dilates.

Oculomotor: Once pupil blown, then external oculomotor ophthalmoplegia (EOO).

Consciousness: Once EOO occurs: stupor→coma

Respirations: Sustained hyperventilation, rarely Cheyne-Stokes.

Motor: Usually produces contralateral weakness. However, the contralateral cerebral peduncle may be compressed against the tentorial edge causing ipsilateral hemiplegia (Kernohan's phenomenon, a false localizing sign). Then bilateral decerebration (decortication unusual).

Midbrain – upper pons stage

Contralateral pupil fixes in midposition or full dilation. Eventually, both midposition (5–6 mm) and fixed.

Oculomotor: Impaired or absent.

Respirations: Sustained hyperpnea.

Motor: Bilateral decerebrate rigidity.

Following the midbrain – upper pons stage

From this point onward, the uncal syndrome is indistinguishable from central herniation (see above).

18.5 Hypoxic coma

Anoxic encephalopathy may be due to anoxic anoxia (drop in pO_2) or anemic anoxia (following exsanguination or cardiac arrest). Myoclonus is common.

Vulnerable cells:

1. cerebral grey matter: lesions predominate in 3rd cortical layer (white matter is usually better preserved due to lower O_2 requirements)
2. Ammon's horn is also vulnerable, especially the Sommer section
3. in the basal ganglia (BG):
 - a) anoxic anoxia severely affects globus pallidus
 - b) anemic anoxia affects the caudate nucleus and putamen
4. in the cerebellum: Purkinje cells, dentate nuclei, and inferior olives are affected

Multivariate analysis yields outcome prognosticators shown in □ Table 18.5 and □ Table 18.6. NB: this analysis applies only to hypoxic-ischemic coma; and is based retrospectively on 210 patients, most S/P cardiac arrest with many medical complications.¹⁴ More recent studies confirm the poor prognosis of unreactive pupils and lack of motor response to pain¹⁵; if either of these findings are seen within a few hours after cardiac arrest there is an 80% risk of death or permanent vegetative state, and if present at 3 days these this rate rose to 100%

Glucocorticoids (steroids) have been shown to have no beneficial effect on survival rate or neurological recovery rate after cardiac arrest.¹⁶

Table 18.5 Patients with BEST chance of regaining independence ^a	
Time of exam	Finding
<6 hrs from onset	(pupillary light reflex present) AND (GCS-motor>1) AND (spontaneous EOM WNL, i.e. orienting or conjugate roving)
1 day	(GCS-motor>3) AND (GCS-eye improved ≥2 from initial)
3 days	(GCS-motor>3) AND (spontaneous EOM WNL)
1 week	GCS-motor=6
2 weeks	oculocephalic WNL
^a abbreviations: WNL=within normal limits, GCS=Glasgow Coma Scale (“GCS-motor” refers to the motor score...); EOM=extraocular muscle;	

Table 18.6 Patients with virtually NO chance of regaining independence ^a	
Time of exam	Finding
<6 hrs	no pupillary light reflex
1 day	(GCS-motor<4) AND (spontaneous eye movements not orienting nor conjugate roving)
3 days	GCS-motor<4
1 week	(GCS-motor<6) AND (at <6 hrs spontaneous EOM not orienting nor conjugate roving) AND (at 3 d GCS-eye<4)
2 week	(oculocephalic not WNL) AND (at 3 d GCS-motor<6) AND (at 3 d GCS-eye<4) AND (at 2 wk GCS-eye not improved at least 2 points from initial)
^a abbreviations: WNL=within normal limits, GCS=Glasgow Coma Scale (“GCS-motor” refers to the motor score...); EOM=extraocular muscle;	

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19 Brain Death and Organ Donation

19.1 Brain death in adults

The President’s Commission for the Study of Ethical Problems in Medicine first published guidelines for the determination of death in 1981¹ which contributed to the approval of the Uniform Determination of Death Act (UDDA; policy statement, see box).²

Uniform determination of death act, 1980 (verbatim quote)

“An individual who has sustained either

1. irreversible cessation of circulatory and respiratory functions, or

2. irreversible cessation of all functions of the entire brain, including the brain stem,

is dead.

A determination of death must be made with accepted medical standards.”

Most states have adopted the UDDA, although some have enacted amendments stipulating qualifications of the determining clinician(s). Individual hospitals may also mandate that certain protocols be followed.

As reaffirmed in 2010,³ when the clinical determination of brain death is made in accordance with the original published guidelines,⁴ there has been no report of recovery of neurologic function in adults.

19.2 Brain death criteria

19.2.1 General information

This section deals with brain death in adults. For individuals <5 years age, see Brain death in children (p.312).

When the cause of death is other than natural causes the Medical Examiner or Coroner (depending on the authority in your jurisdiction) will be contacted per hospital policy.

Key point: Criteria shown below may be used to determine the clinical absence of brain and brainstem function. Then to ensure that the total cessation of brain function is irreversible, the clinician must take into consideration the cause of the absence, and exclude conditions that can mimic the clinical appearance of brain death. This may require ancillary confirmatory tests and observation for a period of time.

Waiting periods: There is insufficient evidence to support a specific observation period to ensure that the cessation of neurologic function is irreversible.³ This requires that the determination of brain death take into consideration all of the available information and circumstances.

19.2.2 Establishing the cause of cessation of brain activity

The cause of the cessation of brain activity (CBA) can usually be determined by a combination of history, physical examination, laboratory tests and imaging studies.

19.2.3 Clinical criteria

See □ Table 19.1 for a summary of basic requirements and clinical findings that may be used in determining brain death. Details follow below.

- Recommendations^{1,5,3}:
1. Absence of brainstem reflexes:
 - a) ocular examination:
 - fixed pupils: no response to bright light (caution after resuscitation: see below). The size of the pupils is unimportant (they are usually midposition (4–6 mm) but may vary to dilated range (9 mm): dilated pupils can be compatible with brain death because cervical sympathetic pathways may remain intact)
 - absent corneal reflexes (corneal reflex: eye closing to corneal, not scleral, stimulation)

Table 19.1 Summary of findings in brain death (see text for details) ³	
Vital signs & general criteria	
• Core temp >36° C (96.8° F)	
• SBP≥ 100 mm Hg	
• No drugs that could simulate brain death. Blood alcohol content (BAC) should be <0.08%	
Absence of brainstem reflexes	
• Fixed pupils	No pupillary reaction to light
• Absent corneal reflexes	Touching cornea with a gauze does not cause eye closure
• Absent oculovestibular reflex (calorics)	No eye movement of any sort to ice water in ear with HOB elevated to 30°
• Absent oculoccephalic reflex: “Doll’s eyes” (p.301)	Turning the head does not cause contralateral eye deviation (clear C-spine first)
• Absent gag reflex	No gagging reaction to movement of ET tube
• Absent cough reflex	No coughing in response to bronchial suctioning
No response to deep central pain	Stimulate areas like supraorbital ridge. No limb movement, no eye movement, no facial movement
Failed apnea challenge	No respirations with pCO ₂ >60 mm Hg

- absent oculoccephalic “doll’s eyes” reflex (p.301), contraindicated if C-spine not cleared
 - absent oculovestibular reflex (cold water calorics): instill 60–100 ml ice water into one ear (□ do not do if TM perforated) with HOB at 30°. Brain death is excluded if any eye movement. Wait at least 1 minute for response, and ≥ 5 min before testing the opposite side (to prevent cancellation of opposing response)
- b) absent oropharyngeal reflex (gag) to stimulation of posterior pharynx
- c) no cough response to bronchial suctioning
2. apnea test AKA apnea challenge: no spontaneous respirations after disconnection from ventilator (assesses function of medulla). Respirations are defined as abdominal or chest excursions that produce adequate tidal volumes; if there is any question, a spirometer may be connected to the patient.⁴ Since elevating PaCO₂ increases ICP which could precipitate herniation and vasomotor instability, this test should be reserved for last and only used when the diagnosis of brain death is reasonably certain. Guidelines^{6,7}:
- a) Apnea for >2 minutes with PaCO₂ >60 mm Hg or PaCO₂ >20 mm Hg over baseline or pH <7.3 (CO₂ is the most potent stimulus for respirations). If patient does not breathe by this point, they won’t breathe at a higher PaCO₂. Not as valid with severe COPD
- b) to prevent hypoxemia during the test (with the danger of cardiac arrhythmia or myocardial infarction):
- preoxygenate for ≥ 10 minutes before the test with 100%FIO₂ to PaO₂ >200 mm Hg
 - prior to the test, reduce the ventilator rate to bring the PaCO₂ to 35-40 mm Hg (to shorten the test time and thus reduce the risk of hypoxemia)
 - during the test, have passive O₂ flow administered at 6 L/min through either a pediatric oxygen cannula or a No. 14 French tracheal suction catheter (with the side port covered with adhesive tape) passed to the estimated level of the carina
- c) starting from normocapnea, the average time to reach PaCO₂ =60 mm Hg is 6 minutes (classic teaching is that PaCO₂ rises 3 mm Hg/min, but in actuality this the rate at which PaCO₂ rises varies widely, with an average 3.7 ± 2.3⁶; or 5.1 mm Hg/min if starting at normocarb⁷). Sometimes as long as 12 minutes may be necessary
- d) the test is aborted if:
- the patient breathes (chest or abdominal movement, gasps): incompatible with brain death
 - SBP <90 mm Hg (hypotension)
 - if O₂ saturation drops <80%for >30 seconds (on pulse oximeter)
 - significant cardiac arrhythmias occur

- e) if patient does not breathe, send ABG at regular intervals and at the completion of test regardless of reason for termination. If the patient does not breathe for at least 2 minutes after a $\text{PaCO}_2 > 60$ mm Hg is documented, then the test is valid and is compatible with brain death (if the patient is stable and ABGs results are available within a few minutes, the apnea challenge may be continued while waiting for results in case the PaCO_2 is < 60)
 - f) if PaCO_2 stabilizes below 60 mm Hg and the pO_2 remains adequate, try reducing the passive O_2 flow rate slightly (O_2 flow may be washing out CO_2 from lungs)
 - g) the test is positive (i.e. compatible with brain death) if there are no respirations and PaCO_2 is ≥ 60 mm Hg (or there is a 20 mm Hg rise in PaCO_2 above baseline)
3. no motor function
 - a) no response to deep central pain: there should be no movement of limbs, no eye opening or eye movement, no facial movement
 - b) true decerebrate or decorticate posturing or seizures are incompatible with the diagnosis of brain death
 - c) spinal cord mediated reflex movements (including flexor plantar reflexes, flexor withdrawal, muscle stretch reflexes,⁸ and even abdominal and cremasteric reflexes) can be compatible with brain death, and may occasionally consist of complex movements,⁹ including bringing one or both arms up to the face,¹⁰ or sitting up (the “Lazarus” sign¹¹) especially with hypoxemia (thought to be due to spinal cord ischemia stimulating surviving motor neurons in the upper cervical cord). If complex integrated motor movements occur, it is recommended that confirmatory testing be performed prior to pronouncement of brain death¹²
 4. absence of complicating conditions (that could simulate brain death on exam):
 - a) hypothermia: core temp should be $> 36^\circ \text{C}$ (96.8°F). Below this temp, pupils may be fixed and dilated,¹³ respirations may be difficult to detect, and recovery is possible¹⁴
 - b) no evidence of remediable exogenous or endogenous intoxication, including drug or metabolic (blood alcohol level should be $< 0.08\%$ barbiturates, benzodiazepines, meprobamate, methaqualone, trichloroethylene, paralytics, hepatic encephalopathy, hyperosmolar coma...). If there is doubt, depending on circumstances, lab tests including drug levels (serum and urine) may be sent. Pseudocholinesterase deficiency is present in 1/3000 patients which can cause succinylcholine to last up to 8 hours (instead of 5 mins). A twitch monitor can rule-out NMB (place the electrodes immediately behind the eye or across the zygomatic arch)
 - c) shock (neuro exam should be reliable if $\text{SBP} \geq 100$ mm Hg) and anoxia. Loss of $> 45\%$ of circulating blood volume can produce lethargy
 - d) immediately post-resuscitation: shock or anoxia may cause fixed and dilated pupils. Atropine (p. 311) may cause slight dilatation but not unreactivity. Neuromuscular blockage (e.g. for intubation) does not affect pupils because the iris lacks nicotinic receptors
 - e) patients coming out of pentobarbital coma (wait until level $\approx \leq 10$ mcg/ml)
 5. confirmation of brain death by use of Ancillary confirmatory tests (preferred tests³: EEG, CRAG or angiography, see below) is not required. May be used at the discretion of the physician, generally if there is uncertainty about the reliability of other parts of the exam
 6. recommended observation periods: there is insufficient evidence to determine a minimal observation period to ensure that neurologic function has irreversibly ceased³:
 - a) in situation where overwhelming brain damage from an irreversible condition is well established (e.g. massive intracerebral hemorrhage, gunshot wound traversing the brain...), and no uncertainty in the clinical exam, an ancillary confirmatory tests would usually not be necessary
 - b) in clear-cut situations as outlined above, if several hours have passed since the onset of the brain insult, a single neurologic examination consistent with brain death should suffice although many states require two examinations by statute³ (see state and local laws below)
 - c) in less clear-cut situations (e.g. anoxic brain injury, hypothermia...) longer observation periods are appropriate and ancillary confirmatory tests may be considered (see below)

19.2.4 State and local laws

Most states have adopted the Uniform Determination of Death Act (UDDA) regarding brain death. State amendments and local regulations or hospital policies may dictate that more than 1 practitioner must concur on the diagnosis. It is incumbent that the practitioner know the applicable regulations before making the diagnosis.

19.2.5 Ancillary confirmatory tests

General information

There is insufficient evidence that any ancillary test can accurately determine brain death.³ Preferred tests³: EEG, CRAG or angiography.

Cerebral angiography

Requires absence of cerebral blood flow, which is incompatible with brain survival. Pros: highly sensitive for determining death of cerebral hemispheres. Cons: costly, time-consuming, requires transport of the patient to x-ray department, invasive, potentially damaging to organs that may be used for donation, and is not optimal for detecting small amount of blood flow to brainstem. Requires a radiologist and technician. Criteria: absence of intracranial flow at the level of the carotid bifurcation or circle of Willis⁵). Filling of the superior sagittal sinus may occur in a delayed fashion and is not incompatible with brain death. Interobserver validity has not been studied. Not routinely used in the diagnosis of brain death, but may be employed in difficult situations.

EEG

Can be done at bedside. Requires experienced interpreter. Does not detect brainstem activity, and electrocerebral silence (ECS) (i.e. isoelectric EEG) does not exclude the possibility of reversible coma. □ Use ECS as a clinical confirmatory test only in patients without drug intoxication, hypothermia, or shock, and not in patients where brainstem activity might be preserved (i.e. situations where the clinical brainstem exam cannot be performed). Note: a practical problem with EEG for brain-death determination is that it is often difficult to get a tracing that is totally free of electrical signal even in patients who are brain dead by other criteria.

Definition of electrocerebral silence on EEG: no electrical activity > 2 mV with the following requirements:

1. recording from scalp or referential electrode pairs ≥ 10 cm apart
2. 8 scalp electrodes and ear lobe reference electrodes
3. inter-electrode resistance < 10,000 Ω (or impedance < 6,000 Ω) but over 100 Ω
4. sensitivity of 2 mV/mm
5. time constants 0.3–0.4 sec for part of recording
6. no response to stimuli (pain, noise, light)
7. record > 30 mins
8. repeat EEG in doubtful cases
9. qualified technologist and electroencephalographer with ICU EEG experience
10. telephone transmission not permissible

Cerebral radionuclide angiogram (CRAG)

General information

Can be done using a gamma camera, or more contemporary HMPAO SPECT (for 99mTechnetium hexamethylpropyleneamine oxime single-photon emission CT). May not detect minimal blood flow to the brain, especially brainstem. Necessitates transport to the radiology/nuclear medicine department and requires an experienced interpreter.

May be useful to confirm clinical brain death in the following settings:

1. where complicating conditions are present, e.g. hypothermia, hypotension (shock), drug intoxication... (e.g. patients emerging from barbiturate coma), metabolic abnormalities
2. in patients with severe facial trauma where evaluation of ocular findings may be difficult or confusing
3. in patients with severe COPD or CHF where apnea testing may not be valid
4. to shorten the observation period

Technique

Using gallium camera

1. scintillation camera is positioned for an AP head and neck view
2. inject 20–30 mCi of 99mTc-labeled serum albumin or pertechnetate in a volume of 0.5–1.5 ml into a proximal IV port, or a central line, followed by a 30 ml NS flush
3. perform serial dynamic images at 2 second intervals for ≈ 60 seconds
4. then, obtain static images with 400,000 counts in AP and then lateral views at 5, 15 & 30 minutes after injection

5. if a study needs to be repeated because of a previous non-diagnostic study or a previous exam incompatible with brain death, a period of 12 hours should lapse to allow the isotope to clear from the circulation

Findings

No uptake in brain parenchyma = “hollow skull phenomenon” (□ Fig. 19.1). Termination of carotid circulation at the skull base, and lack of uptake in the ACA and MCA distributions (absent “candelabra effect”). There may be delayed or faint visualization of dural venous sinuses even with brain death¹⁵ due to connections between the extracranial circulation and the venous system.

MRI and MR angiography (MRA)

MRA is very sensitive for detecting loss of blood flow in cavernous ICA, however the specificity has not been accurately evaluated³ (i.e. might give false positives for brain death in comatose patients) and is not considered a valid confirmatory test.

CT angiography (CTA)

Blood flow on CTA (i.e. not consistent with brain death) was seen in patients with isoelectric EEG. False positive rate had not been determined in comatose non-brain dead patients. CTA is not considered to be a valid confirmatory test for brain death.³

Transcranial doppler

See reference.⁴

Not widely used.

1. small peaks in early systole without diastolic flow or reverberating flow (indicative of significantly increased ICP)
2. initial absence of doppler signals cannot be used as criteria for brain death since 10% of patients do not have temporal insonation windows

SSEPs

One protocol requires bilateral absence of N20-P22 response with median nerve stimulation. An alternative criteria is disappearance of the P14 peak¹⁶ (substrate: medial lemniscus and nucleus cuneatus) on nasopharyngeal electrode recordings. Studies were judged as Class III data and that P14 recordings could be a valuable confirmatory test but that this test has not been used routinely and that interobserver variability needed to be studied.³

Atropine

In brain death, 1 amp of atropine (1 mg) IV should not affect the heart rate due to the absence of vagal tone (the normal response to atropine of increased heart rate rules out brain death, but

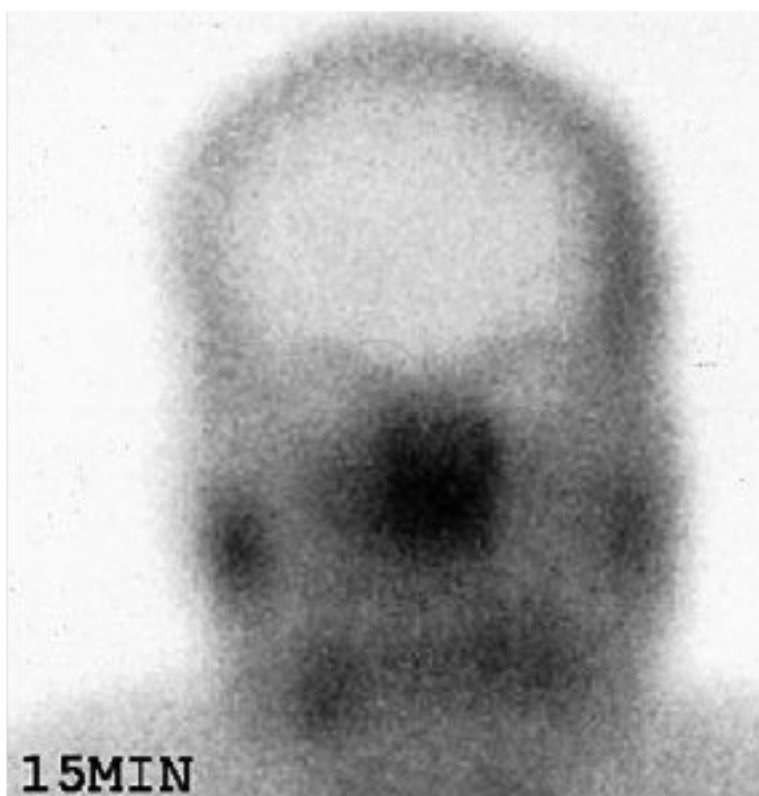


Fig. 19.1 “Hollow-skull” sign on CRAG radionuclide CBF study (static AP view taken 15 minutes after injection)

absence of the response is not helpful since some conditions such as Guillain-Barre may blunt the response).

Systemic atropine in usual doses causes slight pupillary dilatation,^{17,18} but does not eliminate reaction to light (therefore, to eliminate uncertainty, it is prudent to examine the pupils before giving the atropine).

19.2.6 Pitfalls in brain death determination

The following pitfalls may complicate the determination of brain death:

- Movement of body parts after brain death. Movements are sometimes complex in nature, and may occur as long as 32 hours after brain death. Many are mediated by spinal cord discharges as it undergoes cell death. Documented observations include: facial movements, finger tremor, repetitive leg movements, and even sitting up. These movements are often repetitive, usually stereotypical, and do not change with changing stimuli.
- The appearance of breathing. This typically occurs with a ventilator that is set to trigger on detecting respiratory effort. Ventilators may be sensing air movement created by transmission of arterial pulses of the great vessels to the lung or actions of a chest tube.

19.3 Brain death in children

19.3.1 General information

The following is based on 2011 guidelines¹⁹ that are endorsed by the Society of Critical Care Medicine, The Section for Critical Care and Section of Neurology of the American Academy of Pediatrics and the American College of Critical Care Medicine.

Key points¹⁹:

- The diagnosis of brain death in term newborns, infants and children is a clinical diagnosis requiring absence of neurologic function and a known irreversible cause of loss of function.
- These guidelines are not supported for infants <37 weeks gestational age because of insufficient data
- Ancillary tests are not required and are not a substitute for a correctly performed neurologic exam
- Two examinations that include apnea testing separated by an observation period is recommended
- Treat and correct conditions that can interfere with the neurologic exam including: hypothermia, hypotension, interfering drugs (high levels of sedatives, analgesics, paralytics, high doses of anti-convulsant drugs) and metabolic disturbances

19.3.2 Clinical examination

Two examinations each including apnea testing, each consistent with brain death, performed by different attending physicians separated by an observation period are required. Apnea testing may be performed by the same physician.

Apnea testing requires documentation of arterial PaCO₂ that is 20 mm Hg above the baseline and ≥ 60 mm Hg with no respiratory effort. If apnea testing cannot be safely completed, an ancillary study should be done.

Recommended observation periods between exams:

- For term newborns (37 weeks gestational age) through 30 days of age: 24 hours
- For infants and children (>30 days to 18 years): 12 hours
- Following cardiopulmonary resuscitation, the diagnosis of brain death should be deferred ≥24 hrs if there are concerns or inconsistencies in the examination

19.3.3 Ancillary studies

These tests are not required to make the determination of brain death. Use may be considered:

- When apnea testing cannot be safely completed e.g. due to underlying medical conditions or desaturation to <85% or inability to achieve paCO₂ ≥ 60 mm Hg
- If there is uncertainty about results of the neurologic examination
- If drugs that interfere with the neurologic exam may be present
- To reduce the inter-examination observation period

When ancillary tests are employed, a second neurologic examination and apnea test should be performed to the extent possible, and there should not be any finding inconsistent with brain death.

19.4 Organ and tissue donation

19.4.1 General considerations

The Center for Medicare Services (CMS) conditions of participation require all hospitals that receive Medicare funds to refer all imminent deaths to the local Organ Procurement Organization (OPO).²⁰ The OPO is responsible for the determination of suitability and for discussion of donation with the legal next of kin. The discussion must be by trained personnel. The OPO is also responsible for organ donor management, allocation and facilitating recovery of organs in the OR.²⁰

19.4.2 Referral of the potential organ donor

Most OPO's have developed a process for referral of the potential organ donor by educating the critical care nurses to referral by a set of "triggers." The triggers usually include patients with a neurologic injury (anoxia, hemorrhage, trauma etc.), on a ventilator and either losing brain stem reflexes, GCS < 5 or for discussion of withdrawal of support. This set of triggers results in the referral of many patients not suitable for donation but allows for the early notification of the OPO and reduces the risk of missed referrals.

19.4.3 Medical management of the potential organ donor

Brain death results in several predictable physiologic aberrations. Many hospitals have developed "Catastrophic Brain Injury" order sets to address these predictable consequences.

Hypotension

With hypovolemia due to diabetes insipidus and destruction of the pontine and medullary vasomotor centers most brain dead patients are hypotensive. Treatment requires restoration of a euvolemic state and support with vasopressors. Usually norepinephrine to supply inotropic support and neosynephrine to increase peripheral vascular resistance is sufficient to support the blood pressure.

Diabetes insipidus

With loss of hypothalamic function brain dead individuals frequently have posterior pituitary dysfunction and diabetes insipidus. This is manifest by large volume dilute urinary output, hypernatremia and hyperosmolar serum. Management options include DDAVP injection (1-2 mcg SC/IV q 12 hours) or a vasopressin drip (0.01 – 0.04 units/min IV). The vasopressin drip may be preferable because the shorter duration of action can help avoid oliguria due to overdosing.

Hypothermia

Loss of temperature regulation frequently causes hypothermia which can worsen coagulopathy and invalidate brain death testing. Application of a warming blanket to support temperature will help restore normal physiology.

19.4.4 Organ Procurement Organization (OPO) process

Authorization

OPO staff will respond to referrals and after discussion with medical staff and nursing staff engage the family in a discussion involving authorization for organ donation. United Network for Organ Sharing (UNOS) data has demonstrated that OPO trained staff have higher authorization rates than medical staff. This is thought to be due to the position of the OPO staff only advocating for donation and if the treating staff advocates for donation there may be a sense of abandonment.

Donor evaluation

The OPO staff will evaluate donor suitability. Donors will be ruled out if there is thought to be a high potential for transmission of malignancy. The OPO will screen for blood-borne pathogens (HIV, HCV, HBV). Each organ will be evaluated for suitability.

- Heart: EF > 50% no LVH, no CAD
- Lungs: P/F Ratio > 300, normal bronchoscopy
- Liver: ALT, AST, GGTP and bilirubin WNL or returning to normal and no known liver disease

- Kidneys: BUN and Creatinine WNL
- Pancreas: normal lipase, amylase and HgbA1c

Allocation and recovery

Once brain death occurs in a patient authorized for organ donation, the OPO will allocate organs according to UNOS allocation policy and UNOS generated allocation lists. When the transplant centers have accepted organs an OR time will be set and the teams will come to the donor hospital for organ recovery. The time frame from authorization until organ recovery frequently takes 24 to 36 hours or longer.

19.4.5 Organ donation after cardiac death

General information

Key concepts

- candidates: ventilator dependent patients (typically with brain or spinal cord injury) where the family has decided to withdraw support and the medical team expects the patient would progress to asystole less than 60 minutes after withdrawal
- consent from legal next of kin for: organ donation, heparin, and femoral lines
- clearance from medical examiner when applicable (usually, cases of unnatural death)
- counsel the family that the procedure cannot be done in $\approx 20\%$ of cases. They are to be notified immediately if this happens and end-of-life care continues
- the transplant team cannot participate in end-of-life care or declaration of death, and should not be in O.R. until after cardiac death is declared

Candidates for organ donation after cardiac death are typically ventilator dependent patients with brain or spinal cord injuries who are so near death that further treatment is futile, but who do not meet brain death criteria. Organs typically recovered in this manner: kidneys, liver, pancreas, lungs, and rarely the heart.²¹

Ethical concerns related to DCD organ recovery have been raised.²² The Institute of Medicine has reviewed DCD twice (1997 and 2000) and determined DCD to be ethically sound and OPO’s have been encouraged to pursue DCD donation.²³

Consent

Prior to any discussion of donation, the family should have made their decision to withdraw support and allow the patient to progress to death. After the family has had this discussion with the treating physician, the OPO can discuss DCD with the legal next of kin. Consent must also be obtained for any donation-related procedures prior to death (which typically includes heparin infusion to prolong organ viability²⁴ and the possibility of femoral catheters). The discussion should also include process to return to ICU if the patient does not progress to asystole.

Clearance from the medical examiner must be obtained in applicable cases (including deaths due to accident, homicide, suicide...).

Procedure

Life sustaining measures are discontinued (typically consisting of extubation) usually in the operating room. Death is pronounced typically ≈ 2 to 5 minutes after cardiac activity becomes insufficient to generate a pulse, because limited data indicates that circulation will not spontaneously return²⁵ (NB: EKG activity does not need to cease). After declaration of death, cold perfusion of organs is performed and they are procured.

To avoid potential conflicts of interest, no member of the transplant team can participate in end-of-life care nor the declaration of death.²¹ About 20% of the time, the progression to cardiac death does not occur in a timeframe that permits organ retrieval. In these cases, organ donation is cancelled, the family must be immediately notified, and end-of-life care continues.

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Part VI
Infection

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VI

20 Bacterial Infections of the Parenchyma and Meninges and Complex Infections

20.1 Meningitis

20.1.1 General information

Community acquired meningitis (CAM) is generally more fulminant than meningitis following neurosurgical procedures or trauma (CAM tends to occur with more virulent organisms or in individuals with impaired host defenses). Waterhouse-Friderichspageen syndrome: occurs in 10–20% of children with meningococcal infection (usually disseminated infection in age <10 yrs), produces large petechial hemorrhages in the skin and mucous membranes, fever, septic shock, adrenal failure (due to hemorrhage into adrenal glands) and DIC. Focal neurologic signs are rare in acute purulent meningitis. Community acquired meningitis is a medical emergency, and should be treated immediately. See Lumbar puncture (p.323) for a discussion about when to perform an LP.

The remainder of this chapter discusses meningitis that is not community acquired.

20.1.2 Post-neurosurgical procedure meningitis

1. usual organisms: coagulase-negative staphylococci, *S. aureus*, Enterobacteriaceae, *Pseudomonas* sp., pneumococci usually with basilar skull fractures and otorhinologic surgery
2. empiric antibiotics: vancomycin (to cover MRSA), adult □ 15 mg/kg q 8-12 hours to achieve trough 15-20 mg/dl + cefipime 2 gm IV q 8 hrs
3. if severe PCN allergy, use aztreonam 2gm IV q6-8 H or ciprofloxacin 400 mg IV q8h
4. if severe infection, consider intrathecal therapy delivered daily (use only preservative free drug)
 - vancomycin
 - tobramycin/gentamicin
 - amikacin
 - colistin
5. streamline ABX based on sensitivities, e.g. if organism turns out to be MSSA, change vancomycin oxacillin or nafcillin

For suspected CSF fistula

1. usual organisms: streptococci; see CSF fistula (cranial) (p.384)
2. treatment/work-up; see CSF fistula (cranial) (p.384)
3. immunocompromised host (e.g. AIDS)
 - a) usual organisms: as above PLUS *Cryptococcus neoformans*, *M. tuberculosis*, HIV aseptic meningitis, *L. monocytogenes*
 - b) empiric antifungal agents for cryptococcal meningitis: Induction therapy: Liposomal amphotericin B 3-4 mg/kg IV daily + flucytosine 25 mg/kg PO QID for at least 2 weeks followed by
 - c) Consolidation therapy: fluconazole 400 mg PO daily for at least 8 weeks followed by
 - d) Chronic maintenance therapy: fluconazole 200 mg PO daily

20.1.3 Post craniospinal trauma meningitis (post-traumatic meningitis)

Epidemiology

Occurs in 1–20% of patients with moderate to severe head injuries.¹ Most cases occur within 2 weeks of trauma, although delayed cases have been described.² 75% of cases have demonstrable basal skull fracture (p.884), and 58% had obvious CSF rhinorrhea.

20

Pathogens

As expected from above, there is a high rate of infection with organisms indigenous to the nasal cavity. The most common organisms in a series from Greece were Gram-positive cocci (*Staph. hemolyticus*, *S. warneri*, *S. cohnii*, *S. epidermidis*, and *Strep. pneumonia*) and Gram-negative bacilli (*E. coli*, *Klebsiella pneumonia*, *Acinetobacter anitratus*).¹

Treatment

1. See also CSF Fistula, Treatment (p.388)
2. antibiotics: appropriate antibiotics are selected based on CSF penetration and organism sensitivities (adapted to the pathogens common in the patient's locale; in the above series, all Gram-negative strains appeared resistant to ampicillin and third-generation cephalosporins, but were sensitive to imipenem and ciprofloxacin; Gram-positive strains were all sensitive to vancomycin). For empiric antibiotics: Vancomycin 15 mg/kg IV q8-12 hours to achieve trough 15-20 mg/dl + meropenem 2gm IV q8h
3. surgical treatment vs. "conservative treatment": controversial. Some feel that any case of post-traumatic CSF rhinorrhea should be explored,^{3,4} and that cases of spontaneous cessation often represent obscuration by incarcerated brain, so-called "sham healing" with the potential for later CSF leak and/or meningitis.² Others support the notion that cessation (possibly with the assistance of lumbar spinal drainage) is acceptable
4. continue antibiotics for 1 week after CSF is sterilized. If rhinorrhea persists at this time, surgical repair is recommended

20.1.4 Recurrent meningitis

Patients with recurrent meningitis must be evaluated for the presence of abnormal communication with the intraspinal/intracranial compartment. Etiologies include dermal sinus (p.270) (either spinal or cranial), CSF fistula (p.384), or neurenteric cyst (p.290).

20.1.5 Chronic meningitis

Usually due to one of the following etiologies:

1. tuberculosis
2. fungal infections
3. cysticercosis, neurocysticercosis (p.371)

Differential diagnosis includes:

1. sarcoidosis
2. meningeal carcinomatosis

20.1.6 Antibiotics for specific organisms in meningitis

Specific antibiotics

See reference.⁵

Route is IV unless specified otherwise.

1. *S. pneumoniae*: PCN G (2nd choice: chloramphenicol)
 - a) if $MIC \leq 0.06$: PCN G or ampicillin, alternative: third generation cephalosporin (ceftriaxone)
 - b) if $MIC \geq 0.12$: third generation cephalosporin (ceftriaxone)
 - c) if cephalosporin resistance: vancomycin
 - d) alternative: moxifloxacin
2. *N. meningitidis*: PCN G (2nd choice: chloramphenicol)
 - a) if $MIC \leq 0.1$ PCN G or ampicillin
 - b) if $MIC \geq 0.1$: third generation cephalosporin (ceftriaxone)
 - c) alternative: moxifloxacin, meropenem
3. *H. influenza*:
 - a) beta lactamase negative: ampicillin
 - b) beta lactamase positive
 - third generation cephalosporin (ceftriaxone)
 - alternative: aztreonam, ciprofloxacin
4. Group B strep
 - a) ampicillin
 - b) alternative: vancomycin
5. *L. monocytogenes*
 - a) ampicillin ± IV gentamicin
 - b) alternative: IV sulfamethoxazole/trimethoprim
6. *S. aureus*
 - a) if methicillin susceptible
 - oxacillin or nafcillin
 - PCN allergy: vancomycin

- b) if methicillin resistant
 - vancomycin ± rifampin
 - alternative: linezolid ± rifampin
- 7. aerobic Gram negative bacilli (GNB)
 - a) ceftriaxone, or cefotaxime, or moxifloxacin (in order of preference, make alterations based on sensitivities)
 - b) if aminoglycoside required, intraventricular therapy may be indicated after the newborn period
- 8. *P. aeruginosa*
 - a) ceftazidime or cefepime
 - b) alternative meropenem or aztreonam
 - c) if ventriculitis: consider IT gentamicin or tobramycin
- 9. *Candida* spp: Liposomal amphotericin B 3-4 mg/kg IV daily + flucytosine 25 mg/kg PO QID

Length of treatment for meningitis

Generally continue antibiotics for 10 – 14 days total. Duration is dependent on organism and clinical response. Treatment should be 21 days for listeria, group B strep and some GN bacilli.

20.2 Cerebral abscess

20.2.1 General information

Key concepts

- may arise from hematogenous spread, contiguous spread, or direct trauma
- risk factors: pulmonary abscess or AV fistulas, congenital cyanotic heart disease, immune compromise, chronic sinusitis/otitis, dental procedures
- symptoms are similar to any other mass lesions but tend to progress rapidly
- peripheral WBC may be normal or slightly ↑ , CRP usually ↑
- organisms: Streptococcus is most common, up to 60% are polymicrobial
- imaging: usually round with thin enhancing ring on CT or MRI. T2WI → high signal lesion with thin rim of low intensity surrounded by high signal (edema). Unlike with tumor, DWI often shows core of restricted diffusion (not reliable)
- treatment: IV antibiotics, needle drainage for some, excision infrequently (for fungal or resistant abscess)

20.2.2 Epidemiology

Approximately 1500–2500 cases per year in the U.S. Incidence is higher in developing countries. Male:female ratio is 1.5–3:1.

20.2.3 Risk factors

Risk factors include: pulmonary abnormalities (infection, AV-fistulas..., see below), congenital cyanotic heart disease (see below), bacterial endocarditis, penetrating head trauma (see below), chronic sinusitis or otitis media, and immunocompromised host (transplant recipients on immunosuppressants, HIV/AIDS).

20.2.4 Vectors

General information

Prior to 1980, the most common source of cerebral abscess was from contiguous spread. Now, hematogenous dissemination is the most common vector. In 10–60% no source can be identified.⁶

Hematogenous spread

Abscesses arising by this means are multiple in 10–50% of cases.⁷ No source can be found in up to 25% of cases. The chest is the most common origin:

1. in adults: lung abscess (the most common), bronchiectasis and empyema
2. in children: congenital cyanotic heart disease (CCHD) (estimated risk of abscess is 4–7% which is $\approx$ 10-fold increase over general population), especially tetralogy of Fallot (which accounts for $\approx$ 50% of cases). The increased Hct and low PO₂ in these patients provides an hypoxic environment suitable for abscess proliferation. Those with right-to-left (veno-atrial) shunts additionally lose the filtering effects of the lungs (the brain seems to be a preferential target for these infections over other organs). Streptococcal oral flora is frequent, and may follow dental procedures. Coexisting coagulation defects often further complicate management⁸
3. pulmonary arteriovenous fistulas: $\approx$ 50% of these patients have Osler-Weber-Rendu syndrome (AKA hereditary hemorrhagic telangiectasia), and in up to 5% of these patients a cerebral abscess will eventually develop
4. bacterial endocarditis: only rarely gives rise to brain abscess.⁹ More likely to be associated with acute endocarditis than with subacute form
5. dental abscess
6. GI infections: pelvic infections may gain access to the brain via Batson's plexus

In patients with septic embolization, the risk of cerebral abscess formation is elevated in areas of previous infarction or ischemia.¹⁰

Contiguous spread

1. from purulent sinusitis: spreads by local osteomyelitis or by phlebitis of emissary veins. Virtually always singular. Rare in infants because they lack aerated paranasal and mastoid air cells. This route has become a less common source of cerebral abscess due to improved treatment of sinus disease (with antibiotics and, especially with surgery for chronic otitis media and mastoiditis)
 - a) middle-ear and mastoid air sinus infections $\rightarrow$ temporal lobe and cerebellar abscess. The risk of developing a cerebral abscess in an adult with active chronic otitis media is $\approx$ 1/10,000 per year¹¹ (this risk appears low, but in a 30 year-old with active chronic otitis media the lifetime risk becomes $\approx$ 1 in 200)
 - b) ethmoidal and frontal sinusitis $\rightarrow$ frontal lobe abscess
 - c) sphenoid sinusitis: the least common location for sinusitis, but with a high incidence of intracranial complications due to venous extension to the adjacent cavernous sinus $\rightarrow$ temporal lobe
2. odontogenic $\rightarrow$ frontal lobe. Rare. Associated with a dental procedure in the past 4 weeks in most cases.¹² May also spread hematogenously

Following penetrating cranial trauma or neurosurgical procedure

Following penetrating trauma: The risk of abscess formation following civilian gunshot wounds to the brain is probably very low with the use of prophylactic antibiotics, except in cases with CSF leak not repaired surgically following traversal of an air sinus. An abscess following penetrating trauma cannot be treated by simple aspiration as with other abscesses, open surgical debridement to remove foreign matter and devitalized tissue is required.

Post-neurosurgical: especially with traversal of an air sinus. Abscess has been reported following use of intracranial pressure monitors and halo traction.¹³

20.2.5 Pathogens

1. cultures from cerebral abscesses are sterile in up to 25% of cases
2. organisms recovered varies with the primary source of infection
3. in general: Streptococcus is the most frequent organism, 33–50% are anaerobic or microaerophilic. Multiple organisms may be cultured to varying degrees (depends on care of technique), usually in only 10–30% of cases, but can approach 60%⁶ and usually includes anaerobes (Bacteroides sp. common)
4. when secondary to fronto-ethmoidal sinusitis: Strep. milleri and Strep. anginosus may be seen
5. from otitis media, mastoiditis, or lung abscess: usually multiple organisms, including anaerobic strep., Bacteroides, Enterobacteriaceae (Proteus)
6. post traumatic: usually due to S. aureus or Enterobacteriaceae
7. odontogenic (dental) source: may be associated with Actinomyces

- 8. following neurosurgical procedures: Staph. epidermidis and aureus may be seen
- 9. immunocompromised hosts including transplant patients (both bone marrow and solid organ) and AIDS: fungal infections are more common than otherwise would be seen. Organisms include:
 - a) Toxoplasma gondii (p.334); see also treatment (p.333)
 - b) Nocardia asteroides (p.336)
 - c) Candida albicans
 - d) Listeria monocytogenes
 - e) mycobacterium
 - f) Aspergillus fumigatus often from a primary pulmonary infection
- 10. infants: Gram negatives are common because IgM fraction of antibodies don't cross the placenta

20.2.6 Presentation

Adults: no findings are specific for abscess, and many are due to edema surrounding the lesion. Most symptoms are due to increased ICP (H/A, N/V, lethargy). Hemiparesis and seizures develop in 30–50%of cases. Symptoms tend to progress more rapidly than with neoplasms.

Newborns: patent sutures and poor ability of infant brain to ward off infection → cranial enlargement. Papilledema is rare before 2 yrs of age. Common findings: seizures, meningitis, irritability, increasing OFC, and failure to thrive. Most newborns with abscess are afebrile. Prognosis is poor.

20.2.7 Stages of cerebral abscess

□ Table 20.1 shows the four well recognized histologic stages of cerebral abscess, and correlates this with the resistance to insertion of an aspirating needle at the time of surgery. It takes at least 2 weeks to progress through this maturation process, and steroids tend to prolong it.

20.2.8 Evaluation

Bloodwork

Peripheral WBC: may be normal or only mildly elevated in 60–70%of cases (usually >10,000).

Blood cultures: should be obtained when abscess is suspected, usually negative.

ESR: may be normal (especially in congenital cyanotic heart disease CCHD where polycythemia lowers the ESR).

C-reactive protein (CRP): hepatic synthesis increases with inflammatory conditions, however, infection anywhere in body (including brain abscess and dental abscess) can raise the CRP level. May also be elevated in noninfectious inflammatory conditions and brain tumor. Sensitivity for abscess is ≈ 90% specificity is ≈ 77%¹⁴ See also normal values (p.347).

Table 20.1 Histologic staging of cerebral abscess		
Stage	Histologic characteristics (days shown are general estimates)	Resistance to aspirating needle
1	early cerebritis: (days 1–3) early infection & inflammation, poorly demarcated from surrounding brain, toxic changes in neurons, perivascular infiltrates	intermediate resistance
2	late cerebritis: (days 4–9) reticular matrix (collagen precursor) & developing necrotic center	no resistance
3	early capsule: (days 10–13) neovascularity, necrotic center, reticular network surrounds (less well developed along side facing ventricles)	no resistance
4	late capsule: (>day 14) collagen capsule ^a , necrotic center, gliosis around capsule	firm resistance, “pop” on entering
^a abscess is ≈ the only process in the brain that leaves a collagen scar, all other scars are glial scars		

Lumbar puncture (LP)

The role of LP is very dubious in abscess. Although LP is abnormal in >90% there is no characteristic finding diagnostic of abscess. The OP is usually increased, and the WBC count and protein may be elevated. The offending organism can rarely be identified from CSF obtained by LP (unless abscess ruptures into ventricles) with positive cultures in $\approx 6\text{--}22\%$ ¹⁵ There is a risk of transtentorial herniation, especially with large lesions.

Σ

□ Due to the risk involved and the low yield of useful information, avoid the use of LP in evaluating patients with suspected cerebral abscess if not already done.

Brain imaging

CT

Ring enhancing. Sensitivity $\approx 100\%$ For CT staging of abscess see below.

MRI

See □ Table 20.2 for findings. Enhanced T1WI → thin-walled ring enhancement surrounding low intensity central region (□ Fig. 89.1). Fluid-fluid levels may be seen. Occasionally gas producing organisms may cause pneumocephalus.

Diffusion MRI: DWI → bright, ADC → dark (restricted diffusion suggesting viscous fluid)¹⁶ (□ Fig. 89.1). Unlike most tumors which are dark on DWI (□ Fig. 89.2). More reliable with pyogenic abscess, less reliable e.g. with fungal¹⁷ or TB abscess).

MR-spectroscopy: presence of amino acids and either acetate or lactate are diagnostic for abscess.

Infrequently used imaging

Leukocyte scan with 99mTc-HMPAO: patient’s own WBCs are tagged and reinjected. Close to 100% sensitivity and specificity (sensitivity will be reduced if patient is treated with steroids within 48 hrs prior to the scan).¹⁴

Staging cerebral abscess on imaging

CT staging

Late cerebritis (stage 2) has similar features to early capsule (stage 3) on routine contrast and non-contrast CT. There is some therapeutic importance in differentiating these two stages; the following aids in distinguishing¹⁸:

- 1. cerebritis: tends to be more ill-defined
 - a) ring-enhancement: usually appears by late cerebritis stage, usually thick
 - b) further diffusion of contrast into central lumen, and/or lack of decay of enhancement on delayed scan 30–60 min after contrast infusion
- 2. capsule:
 - a) faint rim present on pre-contrast CT (necrotic center with edematous surrounding brain cause collagen capsule to be seen)
 - b) thin ring enhancement AND (more importantly) delayed scans → decay of enhancement

NB: Thin ring enhancement but lack of delayed decay correlates better with cerebritis

NB: Steroids reduce degree of contrast enhancement (especially in cerebritis)

Table 20.2 MRI findings with cerebral abscess		
Stage	T ₁ WI	T ₂ WI
cerebritis	hypointense	hi signal
capsular	lesion center → low signal, capsule → mildly hyperintense, perilesional edema → low signal	center → iso- or hyperintense, capsule → dark (collagen), perilesional edema → hi signal

MRI staging

□ Table 20.2 shows MRI findings in cerebral abscess. In the cerebritis stage, the margins are ill defined.

Additional evaluation

CXR and chest CT (if indicated) to look for pulmonary source.

Cardiac echo (including TEE, Doppler and/or echo with agitated saline injection (bubble study)): for suspected hematogenous spread, to look for patent foramen ovale or cardiac vegetations.

20.2.9 Treatment

General information

There is no single best method for treating a brain abscess. Treatment usually involves:

- Surgical treatment: needle drainage or excision
- correction of the primary source
- long-term use of antibiotics: often IV x 6-8 weeks and possibly followed by oral route x 4-8 weeks. Duration should be guided by clinical and radiographic response

Surgical vs. pure medical management

General information

In a patient with suspected cerebral abscess, tissue should be obtained in almost every case to confirm diagnosis and to isolate pathogens (preferably before initiation of antibiotics).

Medical treatment

In general, surgical drainage or excision is employed in the treatment. Purely medical treatment of early abscess (cerebritis stage)¹⁹ is controversial. NB: pathogens were cultured from well encapsulated abscesses despite adequate levels of appropriate antibiotics in 6 patients who failed medical therapy.²⁰ Failure may be due to poor blood supply and acidic conditions within the abscess (which may inactivate antibiotics in spite of concentrations exceeding the MIC).

Medical therapy alone is more successful if:

1. treatment is begun in cerebritis stage (before complete encapsulation), even though many of these lesions subsequently go on to become encapsulated
2. small lesions: diameter of abscesses successfully treated with antibiotics alone were 0.8–2.5 cm (1.7 mean). Those that failed were 2–6 cm (4.2 mean).
□ 3 cm is suggested as a cutoff,²¹ above this diameter surgery should be included
3. duration of symptoms ≤ 2 wks (correlates with higher incidence of cerebritis stage)
4. patients show definite clinical improvement within the first week

Medical management alone considered if:

1. poor surgical candidate (NB: with local anesthesia, stereotactic biopsy can be done in almost any patient with normal blood clotting)
2. multiple abscesses, especially if small
3. abscess in poorly accessible location: e.g. brain stem²²
4. concomitant meningitis/ependymitis

Indications for surgical treatment

Indications for initial surgical treatment include:

1. significant mass effect exerted by lesion (on CT or MRI)
2. difficulty in diagnosis (especially in adults)
3. proximity to ventricle: indicates likelihood of intraventricular rupture which is associated with poor outcome^{23,8}
4. evidence of significantly increased intracranial pressure
5. poor neurologic condition (patients responds only to pain, or does not even response to pain)
6. traumatic abscess associated with foreign material
7. fungal abscess
8. multiloculated abscess
9. follow-up CT/MRI scans cannot be obtained every 1–2 weeks
10. failure of medical management: neurological deterioration, progression of abscess towards ventricles, or after 2 wks if the abscess is enlarged. Also considered if no decrease in size by 4 wks.

Management

General outline

- obtain blood cultures
- initiate antibiotic therapy (preferably after biopsy specimen obtained), regardless of which mode of treatment (medical vs. surgical) is chosen (see below)
- LP (p.323): avoid in most cases of cerebral abscess
- anticonvulsants: indicated for seizures, prophylactic use is optional
- steroids: controversial. Reduces edema, but may impede therapy (see below)

Antibiotic selection

1. initial antibiotics of choice when pathogen unknown, and especially if *S. aureus* suspected (if there is no history of trauma or neurosurgical procedure, then the risk of MRSA is low):
 - vancomycin: covers MRSA. 15 mg/kg IV q8-12 hours to achieve trough 15-20 mg/dl PLUS
 - a 3rd generation cephalosporin (ceftriaxone); utilize cefepime if post surgical PLUS
 - metronidazole (Flagyl®). Adult: 500 mg q6-8 hours
 - alternative to cefepime + metronidazole: meropenem 2gm IV q8h
 - make appropriate changes as sensitivities become available
2. if culture shows only strep, may use PCN G (high dose) alone or with ceftriaxone
3. if cultures show methicillin sensitive staph aureus and the patient does not have a beta lactam allergy, can change vancomycin to nafcillin (adult: 2 gm IV q 4 hrs. peds: 25 mg/kg IV q 6 hrs)
4. *Cryptococcus neoformans*, *Aspergillus* sp., *Candida* sp.: Liposomal amphotericin B 3-4 mg/kg IV daily + flucytosine 25 mg/kg PO QID.
5. in AIDS patients: *Toxoplasma gondii* is a common pathogen, and initial empiric treatment with sulfadiazine + pyrimethamine + leucovorin is often used (p.332)
6. for suspected or confirmed *nocardia asteroides*, see details (p.336)

Antibiotic duration

IV antibiotics for 6–8 wks (most commonly 6), may then D/C even if the CT abnormalities persist (neovascularity remains). NB: CT improvement may lag behind clinical improvement. Duration of treatment may be reduced if abscess and capsule entirely excised surgically. Oral antibiotics may be used following IV course.

Glucocorticoids (steroids)

Reduces edema and decreases likelihood of fibrous encapsulation of abscess. May reduce penetration of antibiotics into abscess.²¹ Immune suppression may also be deleterious.

□ Reserved for patients with clinical and imaging evidence of deterioration from marked mass effect, and duration of therapy should be minimized.

Follow-up imaging

If therapy is successful, imaging should show decrease in:

1. degree of ring enhancement
2. edema
3. mass effect
4. size of lesion: takes 1 to 4 wks (2.5 mean). 95% of lesions that will resolve with antibiotics alone decrease in size by 1 month

Surgical treatment

Options

See reference.²⁴

1. needle aspiration: the mainstay of surgical treatment. Especially well-suited for multiple or deep lesions (see below) may also be used with thin walled or immature lesions
2. surgical excision: Shortens length of time on antibiotics and reduces risk of recrudescence. Recommended in traumatic abscess to debride foreign material (especially bone), and in fungal abscess because of relative antibiotic resistance (see below)
3. external drainage: controversial. Not frequently used
4. instillation of antibiotics directly into the abscess: has not been extremely efficacious, although it may be used as for refractory *Aspergillus* abscesses

Needle aspiration

Most often implemented with stereotactic localization especially for deep lesions.²⁵ May be performed under local anesthesia if necessary (e.g. in patients who are poor surgical candidates for general anesthesia). May be combined with irrigation with antibiotics or normal saline. Repeated aspirations are required in up to 70% of cases. May be the only surgical intervention required (in addition to antibiotics), but sometimes must be followed with excision (especially with multiloculated abscess).

Performed through a trajectory chosen to:

- 1. minimize the path length through the brain
- 2. avoid traversing the ventricles or vital neural or vascular structures
- 3. avoid traversing infected structures outside the intracranial compartment (infected bone, paranasal sinuses, and scalp wounds)
- 4. in cases of multiples abscesses, target⁷:
 - a) when the diagnosis is unknown: the largest lesion or the one causing the most symptoms
 - b) once the diagnosis of abscess is confirmed
 - any lesion ≥ 2.5 cm diameter
 - lesions causing significant mass effect
 - enlarging lesions

Cultures

Send aspirated material for the following:

- 1. stains
 - a) Gram stain
 - b) acid-fast stain for Mycobacterium (AFB stain, acid-fast resist decolorization with an acid-alcohol mixture and retain the initial dye carbolfuchsin and appear red. The genus Mycobacterium and the genus Nocardia are acid-fast, all other bacteria will be decolorized and stain blue, the color of the counterstain methylene blue)
 - c) modified acid-fast stain (for Nocardia, see below) looking for branching acid fast bacillus
 - d) special fungal stains (e.g., methenamine silver, mucicarmine)
- 2. culture
 - a) routine cultures: aerobic and anaerobic
 - b) fungal culture: this is not only helpful for identifying fungal infections, but since these cultures are kept for longer period and any growth that occurs will be further characterized, fastidious or indolent bacterial organisms may sometimes be identified
 - c) TB culture
 - d) molecular testing: PCR (mycobacteria, EBV, JC virus)

Excision

Can only be performed during the “chronic” phase (late capsule stage). Abscess is removed as any well encapsulated tumor. The length of time on antibiotics can be shortened to ≈ 3 days in some cases following total excision of an accessible, mature abscess (e.g. located in pole of brain). Recommended for abscesses associated with foreign body and most Nocardia abscesses (see below). May also be needed necessary for: fungal abscess, multiloculated or resistant lesions.

20.2.10 Outcome

In the pre-CT era, mortality ranged from 40–60% With advances in antibiotics, surgery, and the improved ability to diagnose and follow response with CT and/or MRI, mortality rate has been reduced to ≈ 10% but morbidity remains high with permanent neurologic deficit or seizures in up to 50% of cases. Current outcomes are shown in □ Table 20.3. A worse prognosis is associated with poor neurologic function, intraventricular rupture of abscess, and almost 100% mortality with fungal abscesses in transplant recipients.

Table 20.3 Outcomes with cerebral abscess	
mortality (CT era data) ^{26,7}	0–10%
neurologic disability	45%
late focal or generalized seizures	27%
hemiparesis	29%

20.3 Subdural empyema

20.3.1 General information

Referred to as subdural abscess prior to 1943.²⁷ Subdural empyema (SDE) is a suppurative infection that forms in the subdural space, which has no anatomic barrier to spread over the convexity and into the interhemispheric fissure²⁸ (and occasionally to the opposite hemisphere and posterior fossa). Antibiotic penetration into this space is poor. Distinguished from abscess which forms within brain substance, surrounded by tissue reaction with fibrin and collagen capsule formation. Hence, SDE tends to be more emergent.

SDE may be complicated by cerebral abscess (seen in 20–25% of imaging studies in patients with SDE), cortical venous thrombosis with risk of venous infarction, or localized cerebritis.

20.3.2 Epidemiology

Less common than cerebral abscess (ratio of abscess:empyema is $\approx$ 5:1). Found in 32 cases in 10,000 autopsies. Male:female ratio is 3:1.

Location: 70–80% are over the convexity, 10–20% are parafalcine.

20.3.3 Etiologies

See □ Table 20.4 for etiologies. Most often occurs as a result of direct extension of local infection (rarely following septicemia). Spread of the infection to the intracranial compartment may occur through the valveless diploic veins, often with associated thrombophlebitis.²⁹

Chronic otitis media was the leading cause of SDE in the preantibiotic era, but in the U.S. this has now been surpassed by paranasal sinus disease especially with frontal sinus involvement³⁰ (may also follow mastoid sinusitis). SDE is a rare but sometimes fatal complication of cranial traction devices.^{31,30} Infection of preexisting subdural hematomas (both treated and untreated, in infants and adults) have been reported³⁰ (bacteremic seeding of an unoperated SDH is very rare).

Trauma includes compound skull fractures and penetrating injuries. Other etiologies include: osteomyelitis, pneumonia, unrelated infection (e.g. foot cellulitis) in diabetics.

20.3.4 Organisms

The causative organism varies with the specific source of the infection. SDE associated with sinusitis is often caused by aerobic and anaerobic streptococci □ Table 20.5). Following trauma or neurosurgical procedures, staphylococci and Gram-negative species predominate (whereas *S. aureus* was not a common pathogen in sinusitis related SDE). Sterile cultures occur in up to 40% (some of which may be due to fastidious anaerobes and/or previous exposure to antibiotics).

Table 20.4 Etiologies of SDE	
Location	%
paranasal sinusitis (especially frontal) ^a	67–75
otitis (usually chronic otitis media) ^b	14
post surgical (neuro or ENT)	4
trauma	3
meningitis (more common in peds ³²)	2
congenital heart disease	2
misc. (including pulmonary suppuration)	4
undetermined	3
^a more common in adults	
^b no cases from otitis in a recent series ³⁰	

Table 20.5 Organisms in SDE associated with sinusitis	
Organisms	%
Adult cases	
aerobic streptococcus	30–50%
staphylococci	15–20%
microaerophilic and anaerobic strep	15–25%
aerobic Gram-negative rods	5–10%
other anaerobes	5–10%
Childhood	
Organisms are similar to meningitis for the same age group. Antibiotics choice is the same as for meningitis	

Table 20.6 Findings on presentation with SDE ^a	
Finding	%
fever	95
H/A	86
meningismus (nuchal rigidity...)	83
hemiparesis	80
altered mental status	76
seizures	44
sinus tenderness, swelling or inflammation	42
nausea and/or vomiting	27
homonymous hemianopsia	18
speech difficulty	17
papilledema	9
^a from a review of multiple articles ³⁰	

20.3.5 Presentation

Neurologic findings are shown in □ Table 20.6. Symptoms are due to mass effect, inflammatory involvement of the brain and meninges, and thrombophlebitis of cerebral veins and/or venous sinuses. SDE should be suspected in the presence of meningismus+unilateral hemisphere dysfunction. Marked tenderness to percussion or pressure over affected air sinuses is common.²⁸ Forehead or eye swelling (from emissary vein thrombosis) may occur.

Focal neurologic deficit and/or seizures usually occur late.

20.3.6 Evaluation

- CT: IV contrast is usually helpful. CT may miss some cases (related to early generation scanners, failure to give IV contrast, poor scan quality...). If normal, repeat the CT at a later time or do an MRI if clinical suspicion persists. Findings: hypodense (but denser than CSF) crescentic or lenticular extracerebral lesion with dense enhancement of medial membrane; inward displacement of gray-white interface; ventricular distortion and effacement of basal cisterns are common findings³³
- MRI: low signal on T1 WI, high signal on T2 WI. Pial ependymal line: a non-specific MRI finding in CNS infection

- LP: □ potentially hazardous (risk of herniation). Organisms are usually present only in cases originating from meningitis. If no meningitis, usually see findings consistent with a parameningeal inflammatory process: moderate sterile pleocytosis (150–600 WBC/mm³) with PMNs predominating; glucose normal; opening pressure is usually high²⁸; protein is usually elevated (range: 75–150 mg/dl)

20.3.7 Treatment

1. surgical drainage: indicated in most cases (nonsurgical management has been reported,³⁴ but should only be considered with minimal neurologic involvement, limited extension and mass effect of SDE, and early favorable response to antibiotics) usually done relatively emergently
2. early in the course, the pus tends to be more fluid and may be more amenable to burr hole drainage; later, loculations develop which may necessitate craniotomy
3. there has been controversy over the optimal surgical treatment. Early studies indicated a better outcome with craniotomy than with burr holes. Recent studies show less difference
 - a) critically ill patients with localized SDE may be candidates for burr-hole drainage (usually inadequate if loculations are present). Repeat procedures may be needed, and up to 20% will later require a craniotomy
 - b) craniotomy: to debride and, if possible, drain. A wide craniotomy is often required because of septations. The dura appears white because of pus underneath. Open and wash out subdural space. Do not try to remove material adherent to cortex (may cause infarction)
4. antibiotics: similar to treatment for cerebral abscess
5. anticonvulsants: usually used prophylactically, mandatory if seizures occur

20.3.8 Outcome

See □ Table 20.7. Mortality has dropped from near 100% in the pre-antibiotic era to ≈ 10% Neurologic deficits tend to improve following treatment, but were present in 55% of patients at the time of discharge from the hospital.³⁰ Age ≥ 60 years, obtundation or coma at presentation, and SDE related to surgery or trauma (rather than sinusitis) carry a worse prognosis.³⁰ Burr-hole drainage may be associated with a worse outcome than with craniotomy, but this may have been influenced by the poorer condition of these patients. Fatal cases may have associated venous infarction of the brain.

20.4 Neurologic involvement in HIV/AIDS

20.4.1 Types of neurologic involvement

General information

40–60% of all patients with acquired immunodeficiency syndrome (AIDS) will develop neurologic symptoms, with one-third of these presenting initially with their neurologic complaint.^{35,36} Only ≈ 5% of patients that die with AIDS have a normal brain on autopsy. One study found the CNS complications of AIDS shown in □ Table 20.8.

The most common conditions producing focal CNS lesions in AIDS³⁸:

1. toxoplasmosis
2. primary CNS lymphoma
3. progressive multifocal leukoencephalopathy (PML)
4. cryptococcal abscess
5. TB (tuberculoma)

Table 20.7 Outcome with SDE	
Outcome	%
persistent seizures	34%
residual hemiparesis	17%
mortality	10–20%

Table 20.8 CNS complications of AIDS (320 patients ³⁵)	
Complication	%
viral syndromes	
subacute encephalitis ^a	17
atypical aseptic meningitis	6.5
herpes simplex encephalitis	2.8
□ progressive multifocal leukoencephalopathy (PML)	1.9 ^b
viral myelitis	0.93
varicella zoster encephalitis	0.31
non-viral infections	
□ Toxoplasma gondii	>32
Cryptococcus neoformans	13
Candida albicans	1.9
coccidiomycosis	0.31
Treponema pallidum (neurosyphilis)	0.62
atypical Mycobacteria	1.9
Mycobacterium tuberculosis	0.31
Aspergillus fumigatus	0.31
bacteria (E. coli)	0.31
neoplasms	
□ primary CNS lymphoma	4.7
systemic lymphoma with CNS involvement	3.8
Kaposi's sarcoma (including brain mets)	0.93
stroke (stroke)	
infarct	1.6
intracerebral hemorrhage	1.2
miscellaneous/unknown	7.8
^a CMV encephalitis occasionally occurs	
^b more recent estimate ³⁷ of the incidence of PML in AIDS: 4%	

Primary effects of HIV infection

Aside from opportunistic infection and tumors caused by the immunodeficient state, infection with the Human Immunodeficiency Virus (HIV) itself can cause direct neurologic involvement including:

1. AIDS encephalopathy: the most common neurologic involvement, occurs in ≈ 66% of patients with AIDS involving the CNS
2. AIDS dementia AKA HIV dementia complex
3. aseptic meningitis
4. cranial neuropathies: including “Bell’s palsy” (occasionally bilateral)
5. AIDS related myelopathy: vacuolization of spinal cord; see Myelopathy (p.1407)
6. peripheral neuropathies

CNS toxoplasmosis in AIDS

May present as:

1. mass lesion (toxoplasmosis abscess): the most common lesion causing mass effect in AIDS patients (70–80% of cerebral mass lesions in AIDS³⁹) (see below for CT/MRI findings)
2. meningoencephalitis
3. encephalopathy

CNS toxoplasmosis occurs late in the course of HIV infection, usually when CD4 counts are <200 cells/mm³.

PML in HIV/AIDS

Progressive multifocal leukoencephalopathy (PML):

1. Is caused by a ubiquitous polyomavirus (a subgroup of papova virus, small nonenveloped viruses with a closed circular double DNA-stranded genome) called “JC virus” (JCV, named after the initials of the patient in whom it was first discovered, not to be confused with Jakob-Creutzfeldt – a prion disease – nor with Jamestown Canyon virus, also confusingly called JC virus, a single-stranded RNA virus that occasionally causes encephalitis in humans). 60–80% of adults have antibodies to JCV⁴⁰
2. frequently manifests in patients with suppressed immune systems, including
 - a) AIDS: currently the most common underlying disease associated with PML
 - b) prior to AIDS, the most common associated diseases were chronic lymphocytic leukemia & lymphoma
 - c) allograft recipients: due to immunosuppression⁴¹
 - d) chronic steroid therapy
 - e) PML also occurs with other malignancies, and with autoimmune disorders (e.g. SLE)
3. pathologic findings: focal myelin loss (demyelination, □ effects white matter) with sparing of axon cylinders, surrounded by enlarged astrocytes and bizarre oligodendroglial cells with eosinophilic intranuclear inclusion bodies. EM can detect the virus. Sometimes occurs in brainstem and cerebellum
4. clinical findings: mental status changes, blindness, aphasia, progressive cranial nerve, motor, or sensory deficits and ultimately coma. Seizures are rare
5. imaging findings: see below
6. clinical course: usually rapidly progressive to death within a few months, occasionally longer survival occurs inexplicably.⁴² There is no effective treatment. Some promise initially with anti-retroviral therapy⁴³
7. definitive diagnosis requires brain biopsy (sensitivity: 40–96%) although it is infrequently employed. JCV has been isolated from brain and urine. Polymerase chain reaction (PCR) of JCV DNA from CSF has been reported, and is specific but not sensitive for PML

Primary CNS lymphoma (PCNSL) in AIDS

Occurs in $\approx 10\%$ of patients with AIDS.⁴⁴ PCNSL is associated with the Epstein-Barr virus (p. 711).

Neurosyphilis

1. AIDS patients can develop neurosyphilis in as little as 4 mos from infection⁴⁵ (unlike the 15–20 yrs usually required in non-immunocompromised patients)
2. neurosyphilis can develop in spite of what would otherwise be adequate treatment for early syphilis with benzathine PCN^{45,46}
3. CDC recommendations⁴⁷: treat patients having symptomatic or asymptomatic neurosyphilis with:
 - penicillin G 3–4-million units IV q 4 hrs (total of 24-million units/d) for 10–14 days or
 - penicillin G procaine 2.4 million units IM daily + probenecid 500 mg QID orally, both for 10–14 days
 - alternative: Rocephin 2gm IV once daily for 10–14 days for patients with a mild beta-lactam allergy
 - for severe beta-lactam allergy: PCN desensitization

20.4.2 Neuroradiologic findings in AIDS

Overview

MR with gadolinium is recommended as the initial screening procedure of choice for AIDS patients with CNS symptoms (lower false negative rate than CT³⁸).

See □ Table 20.9 for a comparison of neuroradiologic findings in toxoplasmosis, PCNSL and PML.

Table 20.9 Comparison of neuroradiologic lesions in AIDS ^a			
Feature	Toxo	PCNSL	PML
Multiplicity	usually > 5 lesions	multiple but < 5 lesions	may be multiple
Enhancement	ring	homogeneous	none
Location	basal ganglia and grey-white junction	subependymal	usually limited to white matter
Mass effect	mild-moderate	mild	none-minimal
Miscellaneous	lesions surrounded by edema	may extend across corpus callosum	high signal on T2 WI, low on T1 WI
^a abbreviations: Toxo = toxoplasmosis, PCNSL=primary CNS lymphoma, PML=progressive multifocal leukoencephalopathy			

CT/MRI findings in toxoplasma abscess

- See □ Table 20.9.
1. most common findings: large area (low density on CT) with mild to moderate edema, ring enhancement with IV contrast in 68% compatible with abscess (of those that did not ring enhance, many showed hypodense areas with less mass effect with slight enhancement adjacent to lesion), well circumscribed margins⁴⁸
 2. most commonly located in basal ganglia , are also often subcortical
 3. often multiple (typically > 5 lesions⁴⁹) and bilateral
 4. usually with little to moderate mass effect³⁸ (in BG, may compress third ventricle and sylvian aqueduct causing obstructive hydrocephalus)
 5. most patients with toxoplasmosis had evidence of cerebral atrophy

CT/MRI findings in PML

- See □ Table 20.9. Note: the appearance of PML may differ in AIDS patients from its appearance in non-AIDS patients.
1. CT: diffuse areas of low density. MRI: high intensity on T2 WI
 2. normally involves only white matter (spares cortex), however in AIDS patients gray matter involvement has been reported
 3. no enhancement (on either CT or MRI), unlike most toxoplasmosis lesions
 4. no mass effect
 5. no edema
 6. lesions may be solitary on 36% of CTs and on 13% of MRIs
 7. borders are usually more ill-defined than in toxoplasmosis⁴⁸

CT/MRI findings in primary CNS lymphoma (PCNSL)

- See □ Table 20.9. NB: the appearance of PCNSL may differ in AIDS patients from non-AIDS patients.
1. multiple lesions with mild mass effect and edema that tend to ring-enhance on CT, or appear as areas of hypointensity surrounding central area of high intensity (target lesions) on T2 WI MRI (unlike non-AIDS cases which tend to enhance homogeneously⁵⁰)
 2. there is a greater tendency to multicentricity in AIDS patients than in the nonimmunosuppressed population⁵¹

20.4.3 Management of intracerebral lesions

Neurosurgical consultation is often requested for biopsy in an AIDS patient with questionable lesion (s). The diagnostic dilemma is usually for low density lesions on CT, and in the United States is primarily between the following:

- toxoplasmosis: treated with pyrimethamine and sulfadiazine + leucovorin (see below)
- PML: no proven effective treatment (initiate or optimize antiretroviral therapy may help⁴³)
- CNS lymphoma: usually treated with RTX; see CNS lymphoma (p. 710)

- note: cryptococcus is more common than PML or lymphoma, but usually manifests as cryptococcal meningitis (p.376), and not as a ring enhancing lesion

Recommendations

PML can usually be identified radiographically. However, radiographic imaging alone cannot reliably differentiate toxoplasmosis from lymphoma or from some other concurrent conditions (patients with toxoplasmosis may have other simultaneous diseases). Therefore, the following recommendations are made:

1. obtain baseline toxoplasmosis serology (IgG) on all known AIDS patients (NB: 50% of the general population have been infected by toxo and have positive titer by age 6 years, 80–90% will be positive by middle adulthood).
2. multiple enhancing lesions with basal ganglion involvement in a patient whose toxo titer is positive have a high probability of being toxo
3. primary CNS lymphoma (PCNSL): with a single lesions, lymphoma is more likely than toxo. If the possibility of PCNSL is strong
 - a) consider LP (contraindicated in presence of mass effect)
 - high volume LP for cytology: PCNSL can be diagnosed in ≈ 10 –25% of cases using ≈ 10 ml of CSF
 - or send CSF for polymerase chain reaction (PCR) amplification of viral DNA of Epstein-Barr virus or JC-virus⁵² (the agents responsible for AIDS-related PCNSL and PML, respectively)
 - b) some centers recommend early biopsy to identify PCNSL cases to avoid delaying RTX for 3 weeks while assessing response to antibiotics³⁸; instead of biopsy, a few centers advocate empiric radiation treatment (for possible lymphoma)
4. in patient with possible toxoplasmosis (i.e. positive toxo serology and imaging findings typical for toxo) even if other conditions have not been excluded:
 - a) initial therapy: sulfadiazine 1000 mg four times daily for patients < 60 kg or 1500 mg four times daily for patients ≥ 60 kg + pyrimethamine 200 mg loading dose, then 50 mg daily for patients < 60 kg or 75 mg daily for patients ≥ 60 kg + folinic acid 10–25 mg daily to prevent pyrimethamine induced hematologic toxicity
 - b) for patients who cannot take sulfadiazine (including those who develop sulfa allergy), change sulfadiazine to clindamycin 600 mg IV or PO q 6 hrs
 - c) alternative regimens:
 - atovaquone 1500 mg PO BID + pyrimethamine 200 mg loading dose, then 50 mg daily for patients < 60 kg or 75 mg daily for patients ≥ 60 kg + folinic acid 10–25 mg daily
 - atovaquone 1500 mg PO BID + sulfadiazine 1000 mg four times daily for patients < 60 kg or 1500 mg four times daily for patients ≥ 60 kg
 - d) there should be a clinical and radiographic response within 2–3 weeks⁵³
 - e) if no response to therapy after 3 weeks (some recommend 7–10 days⁵⁴), then consider alternative diagnosis (brain biopsy should be considered)
 - f) if response is good, reduce dosage of sulfadiazine after 6 to 50% of the above dose for chronic maintenance therapy: sulfadiazine 1000 mg twice daily for patients < 60 kg or 1500 mg four times daily for patients ≥ 60 kg + pyrimethamine 25–50 mg daily + folinic acid 10–25 mg daily
 - g) chronic maintenance therapy can be discontinued in asymptomatic patients who have completed initial therapy if they are receiving antiretroviral therapy (ART), have a suppressed HIV viral load, and have maintained a CD4 count > 200 cells/mcl for at least six months
5. perform biopsy in the following settings:
 - a) in patient with a negative toxo titer (note: patients occasionally have negative titer because of anergy)
 - b) accessible lesion(s) atypical for toxo (i.e. non-enhancing, sparing basal ganglia, periventricular location)
 - c) in the presence of extraneural infections or malignancies that may involve the CNS
 - d) lesion that could be either lymphoma or toxo (e.g. single lesion, see 3. A.)
 - e) in patients who have lesions not inconsistent with toxo but fail to respond to appropriate anti-toxo medications in the recommended time (see above)
 - f) the role of biopsy for non-enhancing lesions is less well defined as the diagnosis does not influence therapy (most are PML or biopsies are non-diagnostic), it may be useful only for prognostic purposes⁵⁴
 - g) note: the risk of open biopsy in AIDS patients may be higher than nonimmunocompromised patients. Stereotactic biopsy may be especially well suited, with up to 96% efficacy, fairly low morbidity (major risk: significant hemorrhage, $\approx 8\%$ incidence) and low mortality^{55,56}
6. stereotactic biopsy guidelines:
 - a) if multiple lesions are present, choose the most accessible lesion in the least eloquent brain area, or the lesion not responding to treatment

- b) biopsy the center of non-enhancing lesions, or the enhancing portion of ring-enhancing lesions
- c) recommended studies on biopsy: histology; immunoperoxidase stain for *Toxoplasma gondii*; stains for TB and fungus; culture for TB, fungi, pyogens

20.4.4 Prognosis

Patients with CNS toxo have a median survival of 446 days, which is similar to that with PML but longer than AIDS-related PCNSL.⁴⁹

Patients with CNS lymphoma in AIDS survive on average a shorter time than similarly treated CNS lymphoma in nonimmunosuppressed patients (3 months vs. 13.5 mos). Median survival is <1 month with no treatment. CNS lymphoma in AIDS tends to occur late in the disease, and patients often die of unrelated causes (e.g. *Pneumocystis carinii* pneumonia).⁵⁴

20.5 Lyme disease – neurologic manifestations

20.5.1 General information

Lyme disease (LD) is a complex multisystem disease caused by various species of *Borrelia* spirochetes (in North America: *Borrelia burgdorferi*) transmitted to humans by the *Ixodes scapularis* or *pacificus* ticks (the American dog tick is not involved). It was first recognized in Lyme, Connecticut in 1975, and is now the most common arthropod-borne infection in the U.S.⁵⁷

20.5.2 Clinical findings

There are 3 clinical stages which can overlap or occur separately.

□ Stage 1 (early localized disease, erythema migrans and flu-like illness). Systemic signs of infection usually begin with a flu-like illness within days to weeks of infection, symptoms include: fever, chills, malaise, fatigue or lethargy, backache, headache, arthralgia, and myalgia. Regional or generalized lymphadenopathy may occur.

The hallmark of LD is erythema chronicum migrans (ECM) (classically a “bulls-eye rash”) which begins 3–30 days after the tick bite, and occurs in 60–75% of patients. ECM usually begins in the thigh, inguinal region, or axilla, and consists of an expanding macular rash with bright red borders and central clearing and induration that usually fades without scarring in 3–4 weeks. In addition to ECM, other dermatologic findings include: malar rash (13%), diffuse erythema, and urticaria. Within 30 days of the tick bite, spirochetes may be demonstrated in acellular spinal fluid.

□ Stage 2 (early disseminated disease). Several weeks to months after infection, untreated patients develop signs of more serious organ involvement. Cardiac and neurologic involvement may occur. Manifestations include:

1. cardiac: occurs in 8% Conduction defects (usually A-V block, generally brief and mild) and myopericarditis
2. ocular: panophthalmitis, ischemic optic atrophy, and interstitial keratitis occur rarely
3. neurologic: occurs in 10–15% of patients with stage 2 disease
 - a) the clinical triad of neurologic manifestations of Lyme disease is⁵⁸:
 - cranial neuritis (especially that mimicking Bell’s palsy: Lyme disease is the most common cause of bilateral “Bell’s palsy” (facial diplegia) in endemic areas)
 - meningitis
 - radiculopathy
 - b) other possible neurologic involvement includes: encephalitis, myelitis, peripheral neuritis

Neurologic findings are frequently migratory, and ≈ 60% of patients have multiple neurologic findings simultaneously. In Europe, Bannwarth’s syndrome (chronic lymphocytic meningitis, peripheral neuropathy, and radiculopathy) is the most common manifestation, and primarily affects the peripheral nervous system.⁵⁹ Neurologic symptoms usually resolve gradually.

□ Stage 3 (late disease). Arthritis and chronic neurologic syndromes may occur in this stage. Arthralgias are common in stage 1, but true arthritis usually does not begin for months to years after infection, and is seen in ≈ 60% of cases.⁶⁰ When arthritis occurs, it may affect the knee (89%), hip (9%), shoulder (9%), ankle (7%) and/or elbow (2%).⁶¹ Neurologic involvement includes⁶²:

1. encephalopathy (chronic, manifestation may be subtle)
2. encephalomyelitis (chronic, manifestation may be subtle)
3. peripheral neuropathy (chronic, manifestation may be subtle)
4. ataxia
5. dementia
6. sleep disorder
7. neuropsychiatric disease and fatigue syndromes

20.5.3 Diagnosis

Diagnostic criteria

There is no test indicative of active infection. The spirochete is difficult to culture from infected humans. Diagnosis is easy if a history of travel to endemic areas, tick bite, and ECM are identified. □ Table 20.10 shows the CDC criteria for diagnosis.

Serology

It takes 7–10 days from initial infection to develop antibodies to *B. burgdorferi*, but it takes $\approx$ 2–3 wks before antibodies can reliably be detected in untreated patients (antibiotics can reduce the immune response).⁶⁴ If the first serum test is negative, it should be repeated in 4–6 weeks if the clinical suspicion of LD is strong (seroconversion from negative to positive is supportive of *B. burgdorferi* infection). False positives can occur with other borrelial and treponemal infections (e.g. syphilis, however, VDRL test will differentiate the two).

Enzyme-linked immunosorbent assay (ELISA) detects IgM or IgG. Antibodies to *B. burgdorferi* is the usual test method. IgM is elevated acutely, and IgG gradually rises and is elevated in almost all patients at 4–6 weeks and is usually highest in patients with arthritis.⁵⁷ Western blot may help identify false-positive ELISA results (more sensitive and specific than ELISA, however, results may vary between labs). Amplification of *B. burgdorferi* DNA by polymerase chain reaction (PCR) yields a more very sensitive test that may have significant false positives, and can be positive even if the DNA is from dead organisms.

CSF

Elevated CSF IgG antibody titer to *B. burgdorferi* may occur with neurologic involvement.⁶⁵ CSF findings in late disease are usually compatible with aseptic meningitis. Oligoclonal bands and increased ratio of IgG to albumin may occur.⁶⁶

20.5.4 Treatment

See references.^{67,68,62}
Antibiotic therapy is more effective early in the illness.

20.6 Nocardia brain abscess

20.6.1 General information

Nocardia infections may involve the CNS in multiple ways.

Table 20.10 CDC criteria for diagnosis of Lyme disease ⁶³	
Area	Criteria
In endemic area	• erythema chronicum migrans (ECM) • antibody titer $\geq$ 1:256 by IFA^a and involvement of $\geq$ 1 organ system^b
In non-endemic area	• ECM with antibody titer $\geq$ 1:256 • ECM with involvement of $\geq$ 2 organ systems^b • antibody titer $\geq$ 1:256 by IFA^a and involvement of $\geq$ 1 organ system^b
^a IFA=immunofluorescence antibody	
^b either musculoskeletal, neurologic or cardiac	

Nocardiosis is caused primarily by *Nocardia asteroides* (other *Nocardia* species such as *N. brasiliensis* are less common), a soil-born aerobic actinomycete (a bacteria, not a fungus) that is usually inoculated through the respiratory tract and produces a localized or disseminated infection. Hematogenous spread frequently results in cutaneous lesions and CNS involvement. *Nocardia* is responsible for 2% of all brain abscesses, the majority of these are *N. asteroides*.

Nocardiosis occurs primarily in patients with chronic debilitating illnesses including:

1. neoplasms: leukemia, lymphoma...
2. conditions requiring long-term corticosteroid treatment
3. Cushing's disease
4. Paget's disease of bone
5. AIDS
6. renal or cardiac organ transplant recipients

The diagnosis is suspected in high-risk patients presenting with soft-tissue abscesses and CNS lesions. CNS involvement occurs in about one-third and includes:

1. cerebral abscess: often multiloculated
2. meningitis
3. ventriculitis in patients with CSF shunt⁶⁹
4. epidural spinal cord compression from vertebral osteomyelitis⁷⁰

20.6.2 Diagnosis

Brain biopsy may not be needed in high-risk patients with confirmed nocardia infection in other sites,⁶⁹ except possibly in AIDS patients where the risk of multiple organism infections or infection plus tumor (particularly lymphoma) is considerable.

20.6.3 Treatment

General information

Surgical indications (p.324) are the same as for other abscesses.

Antibiotics

See references.^{71,72}

- Primary choice: trimethoprim-sulfamethoxazole (TMP-SMZ 15 mg/kg IV of trimethoprim component per day in two to four divided doses PLUS imipenem 500 mg IV q6h ± amikacin 7.5 mg/kg IV q12h (if CNS disease with multiorgan involvement)
- Alternative if sulfa allergy: imipenem 500 mg IV q6h PLUS amikacin 7.5 mg/kg IV q12h

Antimicrobial susceptibility testing should be conducted on all isolates.

Duration: because of risks of relapse or hematogenous spread, treatment is recommended for at least one year with CNS involvement, possibly indefinitely in for immunocompromised hosts.

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21 Skull, Spine, and Post-Surgical Infections

21.1 Shunt infection

21.1.1 Epidemiology

Acceptable shunt infection rate¹: <5–7%(although many published series have a rate near 20%² possibly due to different patient population).

Risk of early infection after shunt surgery: reported range is 3–20%per procedure (typically $\approx$ 7%).

Over 50%of staph infections occur within 2 weeks post-shunt, 70%within 2 mos. Source is often the patient's own skin.¹ It is estimated that in $\approx$ 3%of operations for shunt insertion the CSF is already infected (therefore CSF during shunt insertion is recommended).

21.1.2 Morbidity of shunt infections in children

Children with shunt infections have an increased mortality rate and risk of seizure than those without shunt infection. Those with myelomeningocele who develop ventriculitis after shunting have a lower IQ compared to those without infection.³ Mortality ranges from 10–15%

21.1.3 Risk factors for shunt infection

Many factors have been blamed. Some that seem to be better documented include:

- young age of patient²: in myelomeningocele (MM) patients, waiting until the child is 2 weeks old may significantly lower the infection rate
- length of procedure
- open neural tube defect

21.1.4 Pathogens

Early infection

Most commonly:

- Staph. epidermidis (coagulase-negative staph): 60–75%of infections (most common)
- S. aureus
- Gram-negative bacilli (GNB): 6–20%(may come from intestinal perforation)

In neonates E. coli and Strep. hemolyticus dominate.

Late infection (>6 months after procedure)

Risk: 2.7–31%per patient (typically 6%). Almost all are S. epidermidis. 3.5%of patients account for 27%of infections.⁴

“Late” shunt infections may be due to:

- an indolent infection due to Staph. epidermidis
- seeding of a vascular shunt during episode of septicemia (probably very rare)
- colonization from an episode of meningitis

Fungal infections

Candida spp. infections

Candida spp. are responsible for the majority of fungal ventricular shunt infections. Usually occurs in children <1 year age. Incidence: 1–7%⁵ The 4th leading pathogen causing meningitis in neurosurgical patients in 1 study,⁶ possibly related to the use of prophylactic antibiotics used for ICP monitoring and CSF drainage. Higher incidence in VP shunt patients with abdominal infections and shunts placed in patients with previous bacterial meningitis.⁷ CSF typically shows: elevated WBCs and protein, normal glucose. Management recommendations:

1. completely remove the contaminated shunt (may be more important than with bacterial infections)
2. place a fresh external ventricular drain (if patient is shunt dependent)
3. treat with antifungal therapy

4. place a fresh shunt after $\geq 5-7$ days of therapy and clinical response is apparent
5. continue antifungal agents for 6–8 weeks

21.1.5 Presentation

Signs and symptoms

Non-specific syndrome: fever, N/V, H/A, lethargy, anorexia, irritability. May also present as shunt malfunction; 29% of patients presenting with shunt malfunction had positive cultures.

Erythema and tenderness along shunt tubing may occur.

Distal infection of ventriculoperitoneal shunts may mimic acute abdomen.

In neonates may manifest as apneic episodes, anemia, hepatosplenomegaly, and stiff neck.⁸ *S. epidermidis* infections tend to be indolent (smoldering). GNB infections usually cause more severe illness; abdominal findings are more common; main clinical manifestation is fever, usually intermittent and low grade.

Shunt nephritis⁹: may occur with chronic low level infection of a ventriculovascular shunt causing immune complex deposition in renal glomeruli, characterized by proteinuria and hematuria.

Blood tests

WBC: $<10K$ in one fourth of shunt infections. It is $>20K$ in one-third.

ESR: rarely normal in shunt infections.

Blood cultures: positive in less than one-third of cases.

CSF: WBC is usually not >100 cells/mm³. Gram stains may be positive $\approx 50\%$ (yield with *S. epidermidis* is much lower). Protein is often elevated, glucose may be low or normal. Rapid antigen tests used for community acquired meningitis are usually not helpful for the organisms that tend to cause shunt infections. CSF cultures are negative in 40% of cases (higher culture yield if CSF WBC count is $>20K$).

Evaluation of shunt for infection

1. history and physical directed at determining presence of above signs and symptoms with emphasis on
 - a) history suggestive of infection at another site
 - exposure to others with viral syndromes, including sick siblings
 - GI source (e.g. acute gastroenteritis). Often associated with diarrhea.
 - otitis media (check tympanic membranes)
 - tonsillitis/pharyngitis
 - appendicitis (peritoneal inflammation may impede VP shunt outflow)
 - URI
 - UTI
 - pneumonia
 - b) physical exam to R/O meningismus (stiff neck, photophobia...)
2. bloodwork
 - a) serum WBC count with differential
 - b) acute phase reactants: ESR & CRP
 - c) blood cultures
3. shunt tap: should be done in cases of suspected shunt infection. Clip hair (do not shave) and prep carefully to avoid introducing infection. GNB requires different therapy and has higher morbidity than staph, thus it is desirable to identify these rare patients: $>90\%$ of these had positive Gram-stained CSF smear (only a few Gram-positive infections have positive results). GNB have higher protein and lower glucose, and neutrophils predominate in differential (unpublished data¹)
4. imaging
 - a) CT: usually not helpful for diagnosing infection. Ependymal enhancement when it occurs is diagnostic of ventriculitis. CT may demonstrate shunt malfunction
 - b) abdominal U/S or CT: the presence of an abdominal pseudocyst is suggestive of infection
5. LP: usually NOT recommended. ☐ May be hazardous in obstructive hydrocephalus (HCP) with a nonfunctioning shunt. Often does not yield the pathogen even in communicating HCP, especially if the infection is limited to a ventriculitis. If positive, may obviate a shunt tap

21.1.6 Treatment

Antibiotics alone (without removal of shunt hardware)

Although eradication of shunt infections without removal of hardware has been reported,^{10 (p 595–7),11} this has a lower success rate than with shunt removal,¹² may require protracted treatment (up to 45 days in some), risks problems associated with draining infected CSF into the peritoneum (reduced CSF absorption, abdominal signs/symptoms including tenderness to full-blown peritonitis^{10 (p 235)}) or vascular system – shunt nephritis (p.340), sepsis... – and often requires at least partial shunt revision at some point in most cases. Treatment with antibiotics without shunt removal is therefore recommended only in cases where the patient: is terminally ill, is a poor anesthetic risk, or has slit ventricles that might be difficult to catheterize.

Removal of shunt hardware

In most instances, during the initial treatment with antibiotics the shunt is either externalized (i.e. tubing is diverted at some point distal to the ventricular catheter and connected to a closed drainage system), or sometimes the entire shunt may be removed. In the latter case, some means of CSF drainage must be provided in shunt dependent cases; either by insertion of an external ventricular drain (EVD), or by intermittent ventricular taps (rarely employed) or LPs (with communicating HCP). EVD allows easy monitoring of CSF flow, control of ICP, and repeated sampling for signs of resolution of infection (normalization of WBC count and surveillance cultures). In addition, EVD allows for possible administration of intrathecal antibiotics. In symptomatic patients or those with a positive CSF culture,¹³ any hardware removed should be cultured as only $\approx 8\%$ are sterile in shunt infections. Skin organisms are fastidious and may take several days to grow.

If there is an abdominal pseudocyst, the fluid should be drained through the peritoneal catheter before removing it.

Antibiotics

Empiric antibiotics

See reference.¹⁴

1. Vancomycin (adult) 15 mg/kg IV q8-12 hours to achieve trough 15-20 mg/dl for MRSA coverage + cefepime 2gm IV q8h or meropenem 2gm q8h to cover gram negative pathogens. Streamline therapy based on culture and sensitivity results
2. intraventricular injection of preservative-free antibiotics may be used in addition to IV therapy. Clamp EVD for one hour after injection

Treatment for specific organisms

Positive cultures from shunt hardware removed at the time of shunt revision in the absence of clinical symptoms or a positive CSF culture may be due to contamination and do not always require treatment.¹³

1. *S. aureus* and *S. epidermidis*
 - a) if methicillin sensitive: nafcillin or oxacillin $\pm$ IT vancomycin
 - b) if methicillin resistant: continue IV vancomycin + PO rifampin $\pm$ IT vancomycin
2. Enterococcus: IV ampicillin $\pm$ IT gent
3. other streptococci: either antistreptococcal or above enterococcal regimen
4. aerobic GNR: base on susceptibilities. Both IV beta-lactam & IT aminoglycoside indicated
5. *Serratia marcescens*: a rare cause of VP shunt infection¹⁵ but the high morbidity may warrant aggressive antibiotic therapy (IV ceftriaxone + IT aminoglycoside) and surgical therapy
6. *Corynebacterium* spp. & *Propionibacterium* spp. (diphtheroids)
 - a) if PCN sensitive: use enterococcal regimen above
 - b) if PCN resistant: IV + IT vancomycin
7. *Candida* spp.: see protocol and drugs (p.320). Systemic antifungal therapy and removal of shunt is warranted. Avoid echinocandins (antifungal drugs that inhibit synthesis of glucan in the fungal cell wall) as they have poor CNS penetration

Subsequent management

Once the CSF is sterile $\times 3$ days, convert the EVD to a shunt (if an EVD was not used, it is still recommended that the shunt be replaced with new hardware). Continue antibiotics an additional 10–14 days.

Managing ventriculoperitoneal shunts in patients with peritonitis

Peritonitis may occur as a result of:

1. perforation of a viscus (sometimes as a result of penetration by the peritoneal catheter tip,¹⁶ more common with obsolete Raimondi wire-reinforced tubing)
2. spontaneous bacterial peritonitis (SBP): absence of an identifiable intra-abdominal source. Most commonly diagnosed in patients with cirrhotic ascites¹⁷
3. or as a result of seeding through a VP shunt in a patient with a shunt infection: predominantly gram-positive, cutaneous organisms¹⁸

Concerns following an episode of peritonitis in a patient with VP shunt:

1. ascending infection into the CNS: uncommon, especially in the acute setting while on appropriate antibiotics with shunts containing a 1-way valve (as most do). CSF grows predominantly mixed, gram-negative intestinal flora¹⁸
2. contamination of the distal shunt: prevents permanent eradication of infection (appendicitis in the absence of peritonitis does not produce shunt infection¹⁸)
3. shunt malfunction due to distal shunt obstruction: often as a result of walling off of the catheter tip, usually by omentum in reaction to the infection

Management recommendations following an episode of peritonitis (many viable options):

1. immediate appropriate treatment of peritonitis, usually managed by general surgeon (e.g. for ruptured appendix: appendectomy and appropriate antibiotics), with initial attempt to address shunt not being mandatory
2. anecdotally, cases have been managed successfully by cleaning off the peritoneal catheter with bacitracin solution, and then wrapping the catheter in a bacitracin soaked lap sponge until the time to close the abdomen
3. if the peritonitis was diffuse or if the shunt catheter is believed to have been contaminated, an option is to externalize the distal catheter, preferably once the patient has stabilized from the peritonitis (afebrile, stable vital signs, normal WBC)
 - a) externalization is done in a manner to avoid pulling contaminated catheter up towards hopefully sterile portions of the shunt. This can be accomplished by reopening the skin incision used for inserting the peritoneal catheter, and making a second incision over the shunt tubing, well above this entry point. The catheter is then divided at the upper incision. The catheter is grasped at the lower incision and is pulled, extracting both ends (the peritoneal end and the end just cut). The remaining catheter coming from above is connected to an external drainage system
 - b) CSF cultures are monitored daily
 - c) if 3 consecutive cultures are negative, a new distal catheter may be implanted
 - d) if cultures continuously grow organisms, then the shunt may be contaminated and should then be replaced with an entirely new shunt system
 - e) when it is time to replace the shunt, some authors^{19,20} recommend using an alternative site other than the peritoneum, but this is not mandatory¹⁸

21.2 External ventricular drain (EVD)-related infection

21.2.1 General information

Key concepts

- Common organisms: *S. epidermidis* and *S. aureus* followed by gram negative bacilli and propionibacterium acne.
- Diagnosis: Hypoglycorrhea (CSF glucose/Blood glucose < 0.2), rising cell index, and CSF pleocytosis > 1000 in the presence of positive CSF culture suggests EVD-related infection

21.2.5 Clinical presentation

Clinical signs and symptoms may include the following; however, these symptoms are nonspecific as they are commonplace in the neuro-ICU as a result of the underlying pathology (e.g. intracranial hemorrhage or hydrocephalus)²⁸

- change in mental status
- fever: alternative sources of fever may include intracranial hemorrhage, central fever, thrombotic episodes, drug fevers, in addition to non-CNS infection like blood-borne infections, hospital-acquired pneumonias and urinary tract infections
- meningismus: stiff neck, Brudzinski or Kernig's sign.

21.2.6 Diagnosis

□ Blood parameters. These parameters can suggest the diagnosis but should not be relied upon exclusively.

- A prospective study showed likely EVD infections in the presence of these parameters²⁹: peripheral WBC count > 15 in EVD infections (vs. < 11 in non-infected)
- Serum inflammatory markers: there is very limited literature on the diagnostic utility of ESR and CRP. Procalcitonin alone was found not to be helpful.³⁰

□ CSF parameters. There are limited studies on the diagnostic accuracy of CSF parameters in post craniotomy meningitis and ventriculitis. Often the surgery itself can cause “chemical meningitis” or postoperative meningitis, particularly posterior fossa surgeries or in the presence of intraventricular hemorrhage. The CSF leucocyte and CSF glucose values can look very similar to infectious meningitis making it hard to distinguish these entities based on these parameters. The following could help in confirming underlying EVD infection.

- Hypoglycorrhachia (low CSF glucose): ratio of [CSF glucose]/[blood glucose] < 0.2
- CSF pleocytosis > 1000 or rising Cell index (p. 343)
- CSF protein was not a reliable predictor for incipient ventricular catheter infection³¹

Routine CSF sampling: CSF sampling should be performed only when symptoms appear. There is no evidence of benefit to obtaining CSF cultures or cell count at the time of EVD insertion (false positive cultures may occur from contamination).³²

21.2.7 Principles of management

- It is difficult to achieve high CSF antimicrobial levels due to blood-CSF barrier.
- Some hospital-acquired organisms have higher MICs (minimal inhibitory concentration) for antimicrobials than community-acquired organisms.
- Organisms often form biofilms on the catheters which resist antimicrobial penetration. For this reason, catheter should be removed if it is safe to do so.
- Empiric antibiotics: initiate if ventriculitis is suspected once the appropriate sampling has been obtained
 - If no penicillin allergy:
 - vancomycin as a continuous infusion or divided doses (2–3) of 60 mg per kg of body weight per day after a loading dose of 15 mg per kg of body weight, aiming for trough (15 – 25 mcg/ml) PLUS
 - ceftazidime 2 g IV Q 8 hrs or cefepime 2 g IV Q 8hrs
 - For penicillin allergy:
 - vancomycin as a continuous infusion or divided doses (2–3) of 60 mg per kg of body weight per day after a loading dose of 15 mg per kg of body weight PLUS
 - meropenem 2 g IV q8 or aztreonam 2 g IV Q 6 hrs
- Switch to more selective agents as appropriate based on culture and susceptibility when they become available (see □ Table 21.1)
- Duration of treatment should be individualized to the patient but as a rule of thumb: treat for 2 weeks if the infection was with *S. aureus* and *S. epidermidis*, and 3 weeks if it was gram negatives.³³
- Failing to respond to systemic treatment or infection with a resistant organism might require Intrathecal/Intraventricular antibiotic administration. Choose the antimicrobial based on susceptibility. Dosages for intraventricular antibiotics:
 - Vancomycin: for slit ventricles 5 mg, 10 mg with normal-sized ventricles, 15–20 mg for patients with enlarged ventricles.

Table 21.1 Selective antibacterial agents (based on culture and susceptibility)

Bacteria	Specific antimicrobial regimen
MRSA and MRSE (with an MIC≤1 mcg/ml)	Vancomycin continuous infusion or divided doses (2–3/d) of 60 mg per kg of body weight per day after a loading dose of 15 mg per kg of body weight. If the catheter is retained can add rifampin 300 mg IV q 12 hrs
MRSA and MRSE (with MIC>1 mcg/ml) or patient with vancomycin allergy	Linezolid 600 mg IV or PO q 12 hrs
MSSA and MSSE	Nafcillin 2 g IV q4
Propionobacter acne	Penicillin G 2 MU IV q4
Pseudomonas	Ceftazidime 2 g IV Q8, Cefepime 2 g IV Q8, or Meropenem 2 g q 8 hrs
E. coli or other enterobacteriaceae	Ceftriaxone 2 g IV Q 12 hrs or meropenem 2 g IV q 8 hrs
Enterobacter or Citrobacter	Cefepime 2 g IV Q 8 hrs or meropenem 2 g IV q 8 hrs

- Aminoglycoside: Dosing can also be tailored to ventricular size. Frequency can be adjusted based on drain output as well: once daily for drain output >100 ml/day, every other day if drain output =50–100 ml/day, every third day if drainage <50 ml/day
 - Gentamicin: 4–8 mg
 - Tobramycin: 5–20 mg
 - Amikacin: 5–30 mg
- Colistimethate sodium: 10 mg CMS, which is 125,000 IU or 3.75 mg CBA (Colistin Base units)
- Daptomycin: 2–5 mg
- After IT administration of an antimicrobial, clamp the drain for 15–60 minutes to allow the antimicrobial concentration to equilibrate in the CSF before opening the drain³⁴
- Expert opinion: wait at least 7–10 days after the CSF cultures become sterile to implant a shunt if needed.

21.2.8 Prevention

- Tunneling >5 cm away from the burr hole.³⁵
- Antibiotic coated catheters (e.g., Rifampin + minocycline) significantly reduce the risk of EVD infection.^{36,37,38,39,40}
- Routine catheter exchange at day 5 did not show reduction in the rate of infection.^{41,42,43} Therefore a single catheter may be employed as long as clinically required.⁴⁴
- Prolonged antibiotic prophylaxis while the EVD is in place does not decrease the risk of infection and may select for resistant organisms. However one dose pre-procedure antimicrobial may be administered.

21.3 Wound infections

21.3.1 Laminectomy wound infection

General information

Occurs in 0.9–5%of cases.⁴⁵ May range from superficial to severe dehiscent wound infection to deeper infection (discitis/osteomyelitis ± epidural abscess). The risk is increased with age, long term steroid use, obesity, and possibly DM. Intraoperative mild hypothermia (as commonly occurs in the operating room) may also increase the risk of wound infection (as demonstrated with colorectal resection⁴⁶). Most are caused by S. aureus.

Superficial wound infection

Management

1. culture the wound and/or any purulent drainage

- 2. start the patient empirically on vancomycin plus cefepime or meropenem
- 3. modify antibiotics appropriately when culture and sensitivity results available
- 4. debride wound of all necrotic and devascularized tissue and any visible suture material (foreign bodies). Superficial wound infections may be debrided in the office or treatment room, deep infections must be done in OR
- 5. shallow defects may be allowed to heal by secondary intention, and the following is one possible regimen
 - a) pack the wound defect with 1/4" Iodophor® gauze
 - b) dressing changes at least BID (for hospitalized patients, change q 8 hrs), remove and trim ≈ 0.5–1" of packing with each dressing change
 - while wound is purulent, utilize 1/2 strength Betadine® wet to dry dressings
 - when purulence subsides, switch to normal saline wet to dry
 - c) antibiotics, may be useful as an adjunct to wound treatment initially, switch to oral antibiotics as early as possible, a duration of 10–14 days total is probably adequate if local wound care is being done
- 6. some prefer to close wound by primary intention,⁴⁷ it is critical that there be no tension on the wound for healing to occur. Some close over an irrigation system or antibiotic beads. Retention sutures may be helpful⁴⁸
- 7. with large defects or when bone and/or dura becomes exposed, the use of a muscle flap (often performed by a plastic surgeon) is probably required⁴⁵
- 8. CSF leakage requires exploration in the OR with watertight dural closure to prevent meningitis

Postoperative discitis

Epidemiology

Incidence after lumbar discectomy⁴⁹: 0.2–4%(realistic estimate is probably at the lower end of this range). May also occur after LP, myelogram, cervical laminectomy, lumbar sympathectomy, discography, fusions (with or without instrumentation) and other procedures. Very rare after ACDF. Risk factors include: advanced age, obesity, immunosuppression, systemic infection at the time of surgery.

Pathophysiology

There is some controversy as to whether some cases of post-op discitis are not infectious,⁵⁰ an auto-immune process has been implicated in some of these so-called “avascular” or “chemical” or “aseptic” discitis cases. These cases are less common than infectious ones. ESR and CRP abnormalities may be less pronounced in these patients, and biopsy of the disc space fails to grow organisms or show signs of infection (infiltrates of lymphocytes or PMNS) on microscopy.⁵⁰

In septic cases, various mechanisms for infection have been proposed: direct inoculation at the time of surgery, infection following aseptic necrosis of disc material...

Pathogens

See □ Table 21.2. Most studies report *S. aureus* as the most commonly identified organism, accounting for ≈ 60%of positive cultures,⁴⁹ followed by other staph species. Also reported: Gram-negative organisms (including *E. coli*), *Strep viridans*, *Streptococcus* species anaerobes, TB and fungi. Enteric flora in post-op discitis may due to unrecognized breach of the anterior longitudinal ligament with bowel perforation.

Blood cultures were positive in 2 of 6 patients (both *S. aureus*) in one series.⁵¹

For culture techniques, see surgical management section below.

Clinical

- 1. interval from operation to onset of symptoms: 3 days to 8 mos (most commonly 1–4 wks post-op, usually after an initial period of pain relief and recovery from surgery). 80%present by 3 wks

Table 21.2 Culture results (14 patients, Craig needle biopsy)	
Organism	No. of patients
Staphylococcus epidermidis	4
S. aureus	3
No growth	7

- 2. symptoms:
 - a) moderate to (usually) severe back pain at the site of operation was the most common symptom, exacerbated by virtually any motion of the spine, often accompanied by paraspinal muscle spasms. Back pain is usually out of proportion to the findings
 - b) fever ($>38^{\circ}\text{C}$ in 9 patients; literature reports only 30–50%are febrile) and chills
 - c) pain radiating to hip, leg, scrotum, groin, abdomen or perineum (true sciatica is uncommon)
- 3. signs: in 27 patients,⁵¹ all had paravertebral muscle spasm and limited range of motion of the spine. 13 (48%) were virtually immobilized by pain. Point tenderness over the infected level occurred in 9, expressible pus in 2 (literature reports 0–8%). No new neurologic deficits were noted. Only 10–12%have associated wound infection⁵²
- 4. lab findings:
 - a) ESR: in a series of 27 patients,⁵¹ 96%had $\text{ESR}>20\text{ mm/hr}$ (60 =ave.; >40 in 17 patients; >100 in 5 patients; the single patient <20 was on steroids). ESR increases after uncomplicated discectomy, peaking at 2–5 days, and can fluctuate for 3–6 weeks before normalizing.⁵³ An elevated ESR that never decreases after surgery is a strong indicator of discitis. NB: ESR in anemic patients is unreliable and no reference range can be established (use CRP in these cases)
 - b) C-reactive protein (CRP)⁵³: an acute phase protein synthesized by hepatocytes that because of rapid decomposition may be a more specific indicator of post-op infection than ESR. Values vary from lab to lab, but CRP is normally not detectable in the blood (i.e. $<0.6\text{ mg/dl}=6\text{ mg/L}$). After uncomplicated discectomy (i.e. in the absence of discitis), CRP peaks $\approx 2\text{--}3$ days post-op (to $4.6 \pm 2.1\text{ mg/dl}$ after lumbar microdiscectomy, 9.2 ± 4.7 after conventional lumbar discectomy, 7.0 ± 2.3 after anterior lumbar fusion, and 17.3 ± 3.9 after PLIF), and returns to normal between 5–14 days post-op
 - c) WBC: $>10,000$ in only 8/27 patients⁵¹ (prevalence in literature: 18–30%)

Radiographic evaluation

In postoperative discitis (POD), the average time from surgery to changes on plain x-ray is 3 mos (range: 1–8 mos). Average time from first change to spontaneous spinal fusion: 2 yrs.

MRI: The triad of gadolinium enhancement shown in □ Table 21.3 is strongly suggestive of discitis (some asymptomatic patients may have some of these findings, but they rarely have all).⁵⁴

MRI also rules-out other causes of post-op pain (epidural abscess, recurrent/residual disc herniation...).

Management

- 1. initial labs (in addition to routine): ESR, C-reactive protein, CBC, blood cultures
- 2. analgesics +muscle relaxants (e.g. diazepam (Valium®) 10 mg PO TID)
- 3. antibiotics:
 - a) IV antibiotics for 1-6 wks then PO for 1-6 mos
 - b) most start with anti-staphylococcal antibiotics (initial empiric therapy: vancomycin ± PO rifampin) and cefepime or meropenem. Modify based on sensitivities if positive cultures are obtained
 - c) duration of therapy depends on depth of infection and presence of hardware
 - superficial infection: 1-2 weeks
 - deep infection 4-8 weeks, possibly up to 12 weeks in complex cases
 - consider chronic PO therapy if hardware not removed
- 4. activity restriction (one of the following used, usually until significant pain relief):
 - a) spinal immobilization with brace
 - b) strict bed rest
 - c) advance activity with brace as tolerated
- 5. some authors recommend steroid therapy initially to assist pain relief

Table 21.3 Gadolinium enhancement in discitis		
Location of gadolinium enhancement	Number (out of 15 post-op patients without discitis)	Number (out of 7 patients with discitis)
1. vertebral bone marrow	1	7
2. disc space	3	5
3. posterior annulus fibrosus	13	7

6. cultures: performed if radiographs suspicious, usually performed utilizing percutaneous CT-guided technique
 - a) sites
 - disc aspiration if evidence of disc space involvement
 - needling of paraspinal mass if present
 - b) send culture for the following
 - Stains: (a) Gram stain (b) fungal stain (c) AFB stain
 - Cultures: (a) routine cultures: aerobic and anaerobic (b) fungal culture: this is not only helpful for fungus, but since these cultures are kept for longer period and any growth that occurs will be further characterized, fastidious or indolent bacterial organisms may sometimes be identified (c) AFB (TB) culture
7. 3 out of 27 patients underwent anterior discectomy and fusion after unsuccessful medical therapy⁵¹

Outcome

9 patients developed bony bridging in 12–18 mos; 10 developed bony fusion in 18–24 mos.⁵¹

All patients eventually become pain free (or significantly improve). This is not the case in all series, where some report 60% were pain free at F/U, others found slight back pain in most patients, and yet others report severe chronic LBP in 75%⁴⁹ 67–88% return to their previous work, and 12–25% received disability pension; these numbers are similar to the outcome from disc surgery in general.

No difference in outcome was found for the various activity restrictions specified, except for earlier pain relief with first two types listed above.

21.3.2 Craniotomy wound infection

Also; see under meningitis, post-neurosurgical procedure (p.318).

C-reactive protein

Following uncomplicated craniotomy for microsurgery for brain tumors, C-reactive protein (CRP) peaked on post-op day (POD) 2 with a mean value of 32 ± 38 mg/l.⁵⁵ Values declined from POD 3 through 5, reaching a mean of 6.7 ± 11 on POD 5. These values may be lower than with most post-op infections.

21.4 Osteomyelitis of the skull

21.4.1 General information

The skull is normally very resistant to osteomyelitis, and hematogenous infection is rare. Most infections are due to contiguous spread (usually from an infected air sinus, occasionally from scalp abscess) or to penetrating trauma (including surgery and fetal scalp monitors⁵⁶). With longstanding infection, edema and swelling in the area may become visible (usually over the forehead, but also may occur over the mastoids), and is called “Pott puffy tumor” (after Percival Pott).

21.4.2 Pathogens

Staphylococcus is the most common organism, with *S. aureus* predominating, followed by *S. epidermidis*. In neonates, *E. coli* may be the infecting organism.

21.4.3 Imaging

Imaging findings may include: bony resorption, periosteal reaction, contrast enhancement.

21.4.4 Treatment

Antibiotics alone are rarely curative. Treatment usually involves surgical debridement of infected skull, biting off infected bone with rongeurs until a normal snapping sound replaces the more muted sound made by rongeur-ing infected bone. In the case of an infected craniotomy bone flap, the flap usually must be removed and discarded, and the edges of the skull rongeur-ed back to healthy bone. Bone suspected of infection should be sent for cultures.

Closure of the scalp is then performed either leaving an bone defect (for later cranioplasty) or cranioplasty can be performed using titanium mesh.

Debridement surgery is followed by at least 6–12 weeks of antibiotics.⁵⁷ Until MRSA is ruled out: vancomycin +cefepime or meropenem. Culture results guide choice of antibiotic. Once MRSA is ruled out, vancomycin may be changed to a penicillinase resistant synthetic penicillin (e.g. nafcillin). Most treatment failures occurred in patients treated with <4 weeks of antibiotics following surgery.

Cranioplasty may be performed ≈ 6 months post-op if there are no signs of residual infection.

21.5 Spine infections

Spine infections may be divided into the following major categories:

1. vertebral osteomyelitis (p.353) (spondylitis):
 - a) pyogenic
 - b) nonpyogenic, granulomatous
 - tuberculous spondylitis
 - brucellosis
 - aspergillosis
 - blastomycosis
 - coccidiomycosis
 - infection with Candida tropicalis
2. discitis (p.356): usually associated with vertebral osteomyelitis (spondylodiscitis) (p.353)
 - a) spontaneous
 - b) post-operative/post-procedure
3. spinal epidural abscess (see below)
4. spinal subdural empyema
5. meningitis
6. spinal cord abscess

MRI experience suggests that patients with infectious spondylitis will develop an associated epidural abscess if untreated, and that epidural empyema is unusual in the absence of vertebral osteomyelitis.⁵⁸ Thus, the discovery of one of these conditions should prompt a search for the other.

21.5.1 Spinal epidural abscess

General information

Key concepts

- should be considered in a patient with back pain, fever, and spine tenderness
- major risk factors: diabetes, IV drug abuse, chronic renal failure, alcoholism
- may produce progressive myelopathy, sometimes with precipitous deterioration, therefore early surgery has been advocated by some even if no neuro deficit
- fever, sweats or rigors are common, but normal WBC and temperature can occur
- classical presentation of a skin boil (furuncle) occurs in only ≈ 15% somewhere on the body
- treatment: controversial. Many patients improve with antibiotics alone, but some may deteriorate precipitously

Epidemiology

Incidence: 0.2–1.2 per 10,000 hospital admissions annually,⁵⁹ possibly on the rise.⁶⁰ Average age: 57.5 ± 16.6 years.⁶¹

Thoracic level is the most common site (≈ 50%), followed by lumbar (35%) then cervical (15%).⁶¹ 82%were posterior to the cord, and 18%anterior in one series.⁵⁹ SEA may span from 1 to 13 levels.⁶²

Spinal epidural abscess (SEA) is often associated with vertebral osteomyelitis (in one series of 40 cases, osteomyelitis occurred in all cases of anterior SEA, in 85%of circumferential SEA, and no cases of posterior SEA) and intervertebral discitis.

Co-morbid conditions

Chronic diseases associated with compromised immunity were identified in 65% of 40 cases.⁶³ Associated conditions included diabetes mellitus (32%), IV drug abuse (18%), chronic renal failure (12%), alcoholism (10%), and the following in only 1 or 2 patients: cancer, recurrent UTI, Pott's disease, and positivity for HIV. Chronic steroid use and recent spinal procedure or trauma (e.g. GSW) are also risk factors.⁶² Skin infection (e.g. furuncle).

Clinical features

Usually presents with excruciating pain localized over spine with tenderness to percussion. Radicular symptoms follow with subsequent distal cord findings, often beginning with bowel/bladder disturbance, abdominal distension, weakness progressing to para- and quadriplegia. Average time is 3 days from back pain to root symptoms; 4.5 days from root pain to weakness; 24 hrs from weakness to paraplegia.

Fever, sweats or rigors are common, but are not always present.⁶²

A furuncle (skin boil) somewhere on the body may be identified in 15%

Patients may be encephalopathic. This may range from mild to severe and may further delay diagnosis. Meningismus with a positive Kernig's sign may occur.

Patients with post-operative SEA may demonstrate surprisingly few signs or symptoms (including lack of leukocytosis, lack of fever) aside from local pain.⁶⁴

Pathophysiology of spinal cord dysfunction

Although some cord symptoms may be due to mechanical compression (including that due to vertebral body collapse), this is not always found.⁶⁵ A vascular mechanism has also been postulated, and various combinations of arterial and venous pathology have been described⁵⁹ (one autopsy series showed little arterial compromise, but did show venous compression and thrombosis, thrombophlebitis of epidural veins, and venous infarction and edema of the spinal cord⁶⁶). Occasionally, there may be infection of the spinal cord itself, possibly by extension through the meninges.

Differential diagnosis

SEA should be considered in any patient with backache, fever, and spine tenderness,⁶⁷ especially diabetics, IV drug abusers or immunocompromised patients. Also see Differential diagnosis, Myelopathy (p.1407).

Differential diagnosis:

1. meningitis
2. acute transverse myelitis (paralysis is usually more rapid, radiographic studies are normal)
3. intervertebral disc herniation
4. spinal cord tumors
5. post-op SEA may appear similar to pseudomeningocele⁶⁴

Source site of infection

1. hematogenous spread is the most common source (26–50% of cases) either to the epidural space or to the vertebra with extension to epidural space. Reported foci include:
 - a) skin infections (most common): furuncle may be found in 15% of cases
 - b) parenteral injections, especially with IV drug abuse⁶⁸
 - c) bacterial endocarditis
 - d) UTI
 - e) respiratory infection (including otitis media, sinusitis, or pneumonia)
 - f) pharyngeal or dental abscess
2. direct extension from:
 - a) decubitus ulcer
 - b) psoas abscess: see below
 - c) penetrating trauma, including: abdominal wounds, neck wounds, GSW
 - d) pharyngeal infections
 - e) mediastinitis
 - f) pyelonephritis with perinephric abscess
 - g) dermal sinus

3. following spinal procedures (3 of 8 of these patients had readily identified perioperative infections of periodontia, UTI, or AV-fistula⁶³)
 - a) open procedures: especially lumbar discectomy (incidence⁶⁴ $\approx 0.67\%$)
 - b) closed procedures: e.g. epidural catheter insertion for spinal epidural anesthesia,^{69,70,71} lumbar puncture⁷²...
4. a history of recent back trauma is common (in up to 30%)
5. no source can be identified in up to 50% of patients in some series⁷³

Organisms

Operative cultures are most useful in identifying the responsible organism, these cultures may be negative (possibly more common in patients previously on antibiotics) and in these cases blood cultures may be positive. No organism may be identified in 29–50% of cases.

1. Staph. aureus: the most common organism (cultured in $>50\%$) possibly due to its propensity to form abscesses, its ubiquity, and its ability to infect normal and immunocompromised hosts (these facts help explain why many SEA arise from skin foci)
2. aerobic & anaerobic streptococcus: second most common
3. E.coli
4. Pseudomonas aeruginosa
5. Diplococcus pneumoniae
6. Serratia marcescens
7. Enterobacter
8. chronic infections:
 - a) TB is the most common of these, and although it has become less widespread in the U.S. it is still responsible for 25% of cases of SEA,⁷⁴ it is usually associated with vertebral osteomyelitis, see Pott's disease (p.354)
 - b) fungal: cryptococcosis, aspergillosis, brucellosis
 - c) parasitic: Echinococcus
9. multiple organisms in $\approx 10\%$
10. anaerobes cultured in $\approx 8\%$

Laboratory tests

CBC: leukocytosis common in acute group (average WBC = $16,700/\text{mm}^3$), but usually normal in chronic (ave. WBC = $9,800/\text{mm}^3$).⁵⁹

ESR elevated in most,⁷⁵ usually >30 ,⁶³ CRP.

LP: performed cautiously in suspected cases at a level distant to the clinically suspected site (C1–2 puncture may be needed to do myelogram) with constant aspiration while approaching thecal sac to detect pus (danger of transmitting infection to subarachnoid space); if pus is encountered, stop advancing, send the fluid for culture, and abort the procedure. CSF protein & WBC usually elevated; glucose normal (indicative of parameningeal infection). 5 of 19 cases grew organisms identical to abscess.

Blood cultures: may be helpful in identifying organism in some cases.

Anergy battery: (e.g. mumps and Candida) to assess immune system.

Radiographic studies

Plain films

Usually normal unless there is osteomyelitis of adjacent vertebral bodies (more common in infections anterior to dura). Look for lytic lesions, demineralization, and scalloping of endplates (may take 4–6 weeks after onset of infection).

MRI

Imaging study of choice. Differentiates other conditions (especially transverse myelitis or spinal cord infarction) better than myelo/CT, and doesn't require LP.

Typical findings: T1 WI $\rightarrow$ hypo- or iso-intense epidural mass, vertebral osteomyelitis shows up as reduced signal in bone. T2 WI $\rightarrow$ high intensity epidural mass that often enhances with gadolinium (3 patterns of enhancement: 1) dense homogeneous, 2) inhomogeneous with scattered areas of sparse or no uptake, and 3) thin peripheral enhancement⁷⁶) but may show minimal enhancement in the acute stage when comprised primarily of pus with little granulation tissue. Vertebral osteomyelitis shows up as increased signal in bone, associated discitis produces increased signal in disc and loss of intranuclear cleft. Unenhanced MRI may miss some SEA,⁷⁷ gadopentetate dimeglumine enhancement may slightly increase sensitivity.⁷⁸

Myelogram-CT

Usually shows findings of extradural compression (e.g. “paintbrush appearance” when complete block is present). In the event of complete block, C1–2 puncture may be needed to delineate upper extent (unless post-myelographic CT shows dye above the lesion). See cautions above regarding LP.

CTscan

Intraspinal gas has been described on plain CT.⁷⁹ Post-myelographic CT is more sensitive.

Treatment

General information

Controversial. In most cases, treatment consists of early surgical evacuation combined with antibiotics as the treatment of choice. Argument: although there are reports of management with antibiotics alone^{80,81,82} ± immobilization,⁵⁸ rapid and irreversible deterioration has occurred even in patients treated with appropriate antibiotics who were initially neurologically intact.^{63,61} 86% of those who deteriorated were initially treated with antibiotics alone.⁶² Therefore it has been recommended that nonsurgical management be reserved for the following patients (reference⁸⁰ modified⁶²):

1. those with prohibitive operative risk factors
2. involvement of an extensive length of the spinal canal
3. complete paralysis for >3 days

To add fuel to the argument, in many cases, at the time of surgery, instead of a true abscess, inflammatory tissue that is not easily or effectively debrided is encountered.

Surgery

Goals are establishing diagnosis and causative organism, drainage of pus and debridement of granulation tissue, and bony stabilization if necessary. Most SEA are posterior to the dura and are approached with extensive laminectomy. For posteriorly located SEA and no evidence of vertebral osteomyelitis, instability will usually not follow simple laminectomy and appropriate postoperative antibiotics.⁷³ Thorough antibiotic irrigation is employed intra-operatively. Primary closure is often employed. Post-op drainage is not necessary in cases with only granulation tissue and no pus. For recurrent infections, reoperation and post-op suction-irrigation may be needed.⁸³

Patients with associated osteomyelitis of the vertebral body may develop instability after laminectomy alone⁸⁴ especially if significant bony destruction is present. Thus for anterior SEA, usually with osteomyelitis (especially Pott’s disease), a posterolateral extracavitary approach is utilized whenever possible (to avoid transabdominal or transthoracic approach in these debilitated patients) with removal of devitalized bone usually followed by posterior instrumentation and fusion. Strut grafting with autologous bone (rib or fibula) can be done acutely in Pott’s disease with little risk of graft infection. With purulent osteomyelitis, metal hardware is not contraindicated (titanium is more resistant to harboring bacteria than stainless steel for several reasons, including the fact that titanium does not permit bacteria to form a glycocalyx on its surface), but bone grafting may run the risk of perpetuating the infection. In this situation, some surgeons use beads of calcium sulfate bone void filler impregnated with antibiotic (e.g. Stimulan® Rapid Cure™ antibiotic beads).

Specific antibiotics

If organism and source unknown, *S. aureus* most likely. Empiric antibiotics:

- Ceftriaxone or cefepime (use when pseudomonas is a concern)
PLUS
- metronidazole
PLUS
- vancomycin:
 - until methicillin resistant *S. aureus* (MRSA) can be ruled out
 - once MRSA is ruled out switch to synthetic penicillin (e.g. nafcillin or oxacillin)
- ± rifampin PO

Modify antibiotics based on culture results or knowledge of source (e.g. IV drug abusers have a higher incidence of Gram-negative organisms).

Modify antibiotics based on culture results or knowledge of source (e.g. IV drug abusers have a higher incidence of Gram-negative organisms).

Duration of treatment

For spinal epidural abscess (SEA), treatment should continue for a minimum of 6 weeks. Longer therapy may be warranted in complicated infections and for patients who have spinal implants or hardware. Immobilization for at least 6 weeks during antibiotic therapy is recommended.

Outcome

Fatal in 4–31%⁸⁵ (the higher end of the range tends to be in older patients and in those paralyzed before surgery⁶³). Patients with severe neurologic deficit rarely improve, even with surgical intervention within 6–12 hrs of onset of paralysis, although a few series have shown a chance for some recovery with treatment within 36 hrs of paralysis.^{67,86} Reversal of paralysis of caudal spinal cord segments if present for more than a few hours is rare (exception: Pott's disease has 50% return). Mortality is usually due to original focus of infection or as a complication of residual paraplegia (e.g. pulmonary embolism).

21.5.2 Vertebral osteomyelitis

General information

Key concepts

- presentation and risk factors similar to spinal epidural abscess (p. 349)
- percutaneous needle biopsy for C&S and to rule-out tumor can usually be done by neurosurgeon or interventional radiologist
- treatment: most cases can be managed nonsurgically with long-term antibiotics
- surgery is considered for instability, and infrequently for severe resistance to Abx

For differential diagnosis, see Destructive lesions of the spine (p. 1392). Often associated with discitis, which may be grouped together under the term spondylodiscitis. VO has features similar to spinal epidural abscess (SEA) (p. 349).

Vertebral body collapse and kyphotic deformity may occur with possible retropulsion of necrotic bone and disc fragments, compressing the spinal cord or cauda equina.

Complications that may accrue

1. spinal epidural abscess
2. subdural abscess
3. meningitis
4. bony instability
5. progressive neurologic impairment
6. unique to cervical spine involvement: pharyngeal abscess
7. unique to thoracic spine involvement: mediastinitis

Epidemiology

Vertebral osteomyelitis (VO) comprises 2–4% of all cases of osteomyelitis.⁸⁷ Incidence is 1:250,000 in general population. Incidence appears to be rising. Male:female ratio is 2:1. Incidence increases with age, most patients are >50 years old. The lumbar spine is the most common site, followed by thoracic, cervical and sacrum.⁸⁸ Thoracic VO may →empyema.

Risk factors

1. IV drug abuse⁸⁹
2. diabetes mellitus: susceptible to unusual bacterial infections and even fungal osteomyelitis
3. hemodialysis: a diagnostic challenge since radiographic changes of osteomyelitis can occur even in the absence of infection, see Destructive lesions of the spine (p. 1392)
4. immunosuppression
 - a) AIDS
 - b) chronic corticosteroid use
 - c) ethanol abuse
5. infectious endocarditis
6. following spinal surgery or invasive diagnostic or therapeutic procedures
7. may occur in elderly patients with no other identifiable risk factors⁹⁰

Clinical

Signs/symptoms: localized pain (90%), fever (52% with fever spikes and chills being rare), weight loss, paraspinal muscle spasm, radicular symptoms (50–93%) or myelopathy. VO sometimes produces few systemic effects (e.g. WBC and/or ESR may be normal). $\approx 17\%$ of patients with VO have neurologic symptoms. The risk of paralysis may be higher in the older patient, in cervical VO (vs. thoracic or lumbar), in those with DM or rheumatoid arthritis, and in those with VO due to *S. aureus*.⁸⁴ Neurologic findings are uncommon initially, which may delay the diagnosis.⁹¹ Sensory involvement is less common than motor and long-tract signs because compression is primarily anterior.

Pathogenesis

Source of infection

Sources of spontaneous VO: UTI (the most common), respiratory tract, soft-tissues (e.g. skin boils, IV drug abuse...), dental flora. In 37% of cases a source is never identified.⁹²

Potential routes of spread

Three main routes: arterial, venous, and direct extension:

1. hematogenous: hematogenously disseminated spondylodiscitis in adults usually involves bone initially, and once infection is established in the subchondral space, spread is to the adjacent disc and thence to the next VB⁹³
 - a) arterial
 - b) via spinal epidural venous plexus (Batson's plexus⁹⁴)
2. direct extension (e.g. following surgery/LP, trauma, or local infection)

Organisms

1. *Staphylococcus aureus* is the most common pathogen (>50%) as in SEA
2. *E. coli* is a distant second
3. organisms associated with some primary infection sites⁹⁵:
 - a) IV drug abusers: *Pseudomonas aeruginosa* and *S. aureus* are common
 - b) urinary tract infections: *E. coli* & *Proteus* spp. are common
 - c) respiratory tract infections: *Streptococcus pneumoniae*
 - d) alcohol abuse: *Klebsiella pneumoniae*
 - e) endocarditis:
 - acute endocarditis: *Staph. aureus*
 - subacute endocarditis: *Streptococcus* spp.
4. tuberculous VO: *Mycobacterium tuberculosis* (see below)
5. unusual organisms include: *nocardia* (p.335)
6. *Mycobacterium avium* complex (*M. Avium* and *M. intracellulare*) (MAC) can cause pulmonary disease in nonimmunocompromised patients (usually elderly or on chronic steroids), but can also cause VO similar to TB⁹⁶ as part of disseminated disease which usually occurs in HIV patients
7. polymicrobial infections: rare (<2.5% of pyogenic VO infections)

Tuberculous vertebral osteomyelitis: AKA tuberculous spondylitis, AKA Pott's disease. More common in third world countries. Typically symptomatic for many months. Usually affects more than one level. The most common levels involved are the lower thoracic and upper lumbar levels. Has a predilection for the vertebral body, sparing the posterior elements. Psoas abscess is common (the psoas major muscle attaches to the bodies and intervertebral discs from T12-L5). Sclerosis of the involved vertebral body may occur. Definitive diagnosis requires the identification of acid fast bacilli on culture or Gram stain of biopsy material (may be done percutaneously).

Neurologic deficit develops in 10–47% of patients,⁹⁷ and may be due to medullary and radicular artery inflammation in most cases. The infection itself rarely extends into the spinal canal,⁹⁸ however, epidural granulation tissue or fibrosis or a kyphotic bony deformity may cause cord compression.⁹⁷

The role of surgical debridement and fusion with TB is controversial, and good results may be obtained with either medical treatment or surgery. Surgery may be more appropriate when definite cord compression is documented or for complications such as abscess or sinus formation⁹⁹ or spinal instability.

Diagnostic tests

Laboratories

WBC: elevated in only ≈ 35%(rarely > 12,000), associated with poor prognosis.
ESR: elevated in almost all. Usually >40 mm/hr. Mean: 85.
CRP: may be more sensitive than ESR, and may tend to normalize more quickly with appropriate treatment.¹⁰⁰ See also normal values (p.347).

Cultures/biopsy

Culture: blood (positive in ≈ 50%), urine and any focal suppurative process.
Needle biopsy with cultures: can usually be done percutaneously via transpedicular approach with CT or fluoroscopic guidance. May be helpful even if blood cultures are positive (different organisms retrieved in 15%⁰¹) □ an attempt at direct culture from the involved site should be made. Ideally, cultures should be done before antibiotics are started. The yield of needle biopsy cultures ranges from 60–90% Open biopsy is more sensitive, but morbidity is higher.

Imaging

A comparison of sensitivities and specificities of various imaging modalities is shown in □ Table 21.4. NB: CT may be negative if done too early in the course.
MRI: T1WI → confluent low signal in vertebral bodies and intervertebral disc space. T2WI → increased intensity of involved VBs and disc space.¹⁰² Contrast: enhancement of VB and disc, also look for paraspinal and epidural mass.
CT scan: helpful for demonstrating bony involvement as well as detailed bony anatomy in case instrumentation is required during treatment.
Plain x-ray: changes take from 2–8 weeks from the onset of infection to develop. Earliest changes are loss of cortical endplate margins and loss of disc space height.
Bone scan: three phase bone scan (p.236) has reasonably good sensitivity and specificity. Gallium scan (p.236) has better accuracy, findings include increased uptake in the 2 adjacent VBs with loss of intervening disc.¹⁰³ Indium-111 labeled WBC scan: low sensitivity for vertebral osteomyelitis.

Work-up

- In patients with suspected vertebral osteomyelitis (VO) (see text above for details):
1. clinical: history of IV drug abuse, DM, immunocompromise, skin boil
 2. physical exam: R/O radiculopathy & myelopathy, point tenderness over spine
 3. diagnostic tests:
 - a) bloodwork: WBC, ESR & CRP (a normal ESR is almost incompatible with VO), blood cultures
 - b) imaging:
 - MRI without and with contrast
 - If MRI is contraindicated: CT-myelogram assesses bony anatomy and can demonstrate spinal canal compromise. Bone scan may occasionally be helpful if the diagnosis is still uncertain when suspicion is high
 - c) percutaneous needle biopsy with cultures: usually by radiologist. Cultures should include: fungal, aerobic and anaerobic bacterial, and TB

Treatment

Also see more details (p.352). 90% of cases can be managed non-surgically with antibiotics and immobilization. Characteristics of potential candidates for non-surgical treatment are listed in □ Table 21.5. Must also take into account level(s) involved and patient’s condition.

Table 21.4 Accuracy of various imaging modalities for vertebral osteomyelitis ¹⁰²			
Modality	Sensitivity	Specificity	Accuracy
plain x-rays	82%	57%	73%
bone scan	90%	78%	86%
gallium scan	92%	100%	93%
bone scan + gallium scan	90%	100%	94%
MRI	96%	92%	94%

Table 21.5 Candidates for non-surgical treatment in pyogenic spontaneous spondylodiscitis⁹⁵

- organism identified
- antibiotic sensitivity
- single disc space involvement with little VB involvement
- minimal or no neurologic deficit
- minimal or no spinal instability

In cases with high suspicion for VO, antibiotics may be started as soon as biopsy has been performed (some treat even earlier). For details of antimicrobials, see Treatment (p.352) under spinal epidural abscess.

Improvement on imaging can lag behind clinical response and ESR/CRP.

Indications for neurosurgical intervention (note: intervention by a general surgeon may be indicated for empyema, psoas abscess...):

1. progression of disease despite adequate best-case antibiotic therapy
2. spinal instability
3. spinal epidural abscess (p.349)
4. chronic infection refractory to medical management

For patients not being treated surgically:

1. percutaneous biopsy to obtain ID & sensitivity of organism
2. antibiotics:
 - a) IV antibiotics for at least 6 weeks (the rate of treatment failure is increased when IV antibiotics are given for <4 weeks⁹⁵; longer, e.g. 12 weeks, if ESR not normalizing or if extensive bone involvement and paravertebral infection)
 - b) followed by 6–8 weeks of oral agents⁹⁵
3. pain medication as appropriate for pain
4. TLSO brace: to reduce pain (due to movement at involved site) and to reduce stress on weakened bone until healing
5. check upright films in the TLSO to verify stability in the brace
6. follow-up at approximately 8 and 12 weeks with x-rays in brace, then consider discontinuing brace if infection and pain are under control

Surgical treatment

Decompression of neural elements, removal of inflammatory tissue and infected bone to decrease bioburden. Use of instrumented fusion is not contraindicated even for pyogenic infections. Although not routinely used, bone morphogenic protein (rhBMP-2) in 14 patients undergoing circumferential fusion for refractory infections did not produce complications.¹⁰⁴

21.5.3 Discitis

General information

Infection of the nucleus pulposus. May start in the cartilaginous endplate and spread to the disc and vertebral body (VB). Similar to vertebral osteomyelitis, except osteomyelitis primarily involves the VB and spreads secondarily to the disc space.

Setting: may be “post-op“ or “spontaneous“

- Spontaneous discitis: Occurs in the absence of any procedure. Discussed below
- Post-op discitis: Can occur following a number of procedures; see Postoperative discitis (p.346).

This is covered under post-op infections

Many radiographic features of spondylodiscitis and tumor (metastatic and primary) are similar, but tumors rarely involve the disc space, whereas most infections begin in, or before too long, involve the disc space; for more details, see Differentiating factors (p.1392).

Two distinct types:

1. juvenile: age usually <20 yrs (see below)
2. adult: usually occurs in susceptible patients (diabetics, IV drug abusers)

Juvenile discitis

Age usually <20 yrs, with a peak between 2–3 years. Probably due to the presence of primordial feeding arteries that nourish the nucleus pulposus and which involute at $\approx$ 20–30 yrs age. Lumbar spine is more commonly involved than thoracic or cervical.

Common presentation in young children: refusal to walk or stand progressing to refusal to sit. Back pain is most common in children >9 yrs age. Low grade fever may be present. ESR is usually 2–3 $\times$ normal. WBC is sometimes elevated. H. flu is a more commonly seen pathogen in this group.

In most cases, there is complete resolution in 9–22 weeks without recurrence in long-term follow-up studies.^{97 (p 365–71)} Surgery is reserved for the rare case that progresses in spite of antibiotics, for spinal instability, or for recurrent cases.

Most authors reserve antibiotics for patients with^{97 (p 365–71)}:

1. positive cultures (blood cultures or biopsy cultures)
2. elevated WBC count, constitutional symptoms, or high fever
3. poor response to rest or immobilization
4. neurologic sequelae (very rare)

Antibiotics should be given for a total of 4–6 weeks. Start with IV antibiotics, and when clinical symptoms improve convert to PO for the remainder of therapy.

Clinical

1. symptoms:
 - a) pain (the primary symptom)
 - local pain, moderate to severe, exacerbated by virtually any motion of the spine, usually well localized to the level of involvement
 - radiating to abdomen,¹⁰⁵ hip, leg, scrotum, groin, or perineum
 - radicular symptoms: occurs in 50%² to 93%¹⁰⁶ depending on the series
 - b) fever and chills: up to 70% are afebrile
2. signs:
 - a) localized tenderness
 - b) paravertebral muscle spasm
 - c) limitation of movement

Work-up

Overview

See following sections for details.

- blood tests
 - WBC
 - ESR & CRP
 - blood cultures
- imaging
 - MRI of the region of concern without and with contrast: the diagnostic test of choice.
 - If MRI is contraindicated: CT-myelogram, bone scan
- percutaneous needle biopsy: usually performed by interventional radiologist
- a source of the infection should be sought
 - thorough history for possible risk factors: skin lesions, IV drug abuse, immunocompromise
 - echocardiogram TEE: rule-out endocarditis or valvular vegetations

Radiographic evaluation

General information

A characteristic radiographic finding that helps distinguish infection from metastatic disease is that destruction of the disc space is highly suggestive of infection, whereas in general, tumor does not cross the disc space; see Differentiating factors (p. 1392).

Plain x-rays

Usually not helpful for early diagnosis. Sequence of changes on plain films:

- earliest changes: interspace narrowing with some demineralization of the VB. Not seen <2–4 wks following onset of clinical symptoms, nor later than 8 wks

- sclerosis (eburnation) of adjacent cortical margins with increased density of adjacent areas of VB representing new bone formation, starting 4–12 weeks following onset of clinical symptoms
- irregularity of the adjacent vertebral endplates, with sparing of the pedicles (except for tuberculosis, which may involve the pedicles)
- in 50% of cases, the infection remains confined to the disc space, in the other 50% it spreads to adjacent VB
- a late finding is widening (ballooning) of the disc space with erosion of the VB
- circumferential bone formation may lead to exuberant spur formation between VBs 6–8 months into course of illness
- spontaneous fusion of the VB may occur

MRI

Without and with gadolinium contrast. Demonstrates involvement of disc space and of VBs. MRI can R/O paravertebral or epidural spinal abscess but is poor in assessing bony fusion and bone integrity. As sensitive as radionuclide bone scan. Characteristic finding: decreased signal from the disc and adjacent portion of VBs on T1WI, and increased signal from these structures on T2WI. Enhancement is common. Characteristic findings may occur 3–5 days after onset of symptoms.

CT and myelo-CT

Like MRI, may also R/O paravertebral or epidural spinal abscess, and while it is better for assessing bony fusion and integrity, by itself it is poor for demonstrating canal compromise. With the addition of water soluble intrathecal contrast (myelo-CT), also assesses the spinal canal for compromise.

Diagnostic criteria

Three basic changes on CT¹⁰⁷ (if all 3 are present, pathognomonic for discitis; if only the 1st 2 are present, then only 87% specific for discitis):

1. endplate fragmentation
2. paravertebral soft-tissue swelling with obliteration of fat planes
3. paravertebral abscess

Nuclear medicine

Very sensitive for discitis and vertebral osteomyelitis (85% sensitivity), but may be negative in up to 85% of patients with Pott's disease. Uses either technetium-99 (abnormal as early as 7 days following onset of clinical symptoms) or gallium-67 (abnormal within 14 days). A positive scan shows focal increased uptake in adjacent endplates, and may be differentiated from osteomyelitis which will involve only one endplate. A positive scan is not specific for infection, and may also occur with neoplasms, fractures, and degenerative changes.

Laboratory studies

ESR: In non-immunocompromised patients, ESR will be elevated in almost all cases with an average value of 60 mm/hr (although discitis with a normal ESR occurs rarely, it should call the diagnosis into question. ESR may be useful to follow as an indicator of response to treatment.

C-reactive protein (CRP): Often used in conjunction with ESR

WBC: Peripheral WBC is often normal, and rarely is elevated above 12,000.

PPD (Purified Protein Derivative, AKA Mantoux screening test): May be helpful to R/O Pott's disease in cases of spontaneous discitis (p. 354), may be negative in 14% of cases.¹⁰⁸

Cultures: An attempt should be made to obtain direct cultures from the involved disc space. These may be obtained percutaneously (e.g. Craig needle biopsy) with CT or other radiographic guidance (reported up to 60% positive culture rate), or from intra-operative specimen (NB: surgery for open biopsy alone is usually not indicated). Staining for acid fast bacilli (AFB) to identify *Mycobacterium tuberculosis* (TB) should be done in all cases.

Blood cultures may be positive in $\approx$ 50% of cases, and can be helpful in guiding choice of antimicrobial agent when positive.

Pathogens

Staphylococcus aureus is the most common organism when direct cultures are obtained, followed by *S. albus* and *S. epidermidis*. Gram negative organisms may also be found, including *E. coli* and *Proteus* species.

Pseudomonas aeruginosa may be more common in IV drug abusers.

H. flu is common in juvenile discitis (see below).

Mycobacterium tuberculosis: Tuberculous spondylitis (Pott's disease) may also occur.

Treatment

General information

Outcome is generally good, and antibiotics together with spinal bracing (immobilization) are adequate treatment in $\approx 75\%$ of cases. Occasionally surgery is required. See also under postoperative discitis for other aspects of management (p.347).

Most patients are started on strict bed rest, and are then mobilized with or without a brace as tolerated

Spinal bracing

Probably does not affect final outcome. Affords earlier pain relief for some, and may allow return to activity at an earlier time. For thoracic or upper lumbar discitis, the patient is fitted with a clam-shell-type body jacket which is used for 6–8 weeks on the average. As a practical matter, most patients find that the discomfort from the brace is worse than that without the brace. Alternative forms of immobilization include spica cast (provides better immobilization for lower lumbar discitis) or a corset-type brace (less immobilization but better tolerated).

Antibiotics

Choice of antibiotics is guided by the results of direct cultures when positive. In the 40–50% of cases where no organism is isolated, broad spectrum antibiotics should be used. Positive blood culture results may also help guide choice of antibiotics.

Two alternative treatment plans suggested:

1. treat with IV antibiotics for an arbitrary period of time, usually ≈ 4 –6 weeks, followed with oral antibiotics for an additional 4–6 weeks
2. treat with IV antibiotics until the ESR normalizes, then change to PO

Surgery

Required in only $\approx 25\%$ of cases.

Indications for surgery:

1. situations where the diagnosis is uncertain, especially when neoplasm is a strong consideration (CT guided needle biopsy usually helps here)
2. decompression of neural structures, especially with associated spinal epidural abscess or compression by reactive granulation tissue. Ascending numbness, weakness, or onset of neurogenic bladder herald cauda equina syndrome
3. drainage of associated abscess, especially septated abscesses that might be recalcitrant to CT guided percutaneous needling
4. rarely, to fuse an unstable spine. Most cases go on to spontaneous fusion

Approaches

- anterior approaches: generally used in the cervical or thoracic regions. Removes some or most of the offending infected tissue
 - cervical spine: anterior discectomy and fusion for limited involvement; corpectomy with strut graft and plating with posterior instrumentation (360° fusion) for more extensive involvement
 - thoracic spine: a posterolateral approach (e.g. transpedicular or costotransversectomy approach) or lateral approach (e.g. trans-thoracic or retro-coelomic) may be used
- posterior laminectomy
 - may be used in the lumbar region (below the conus medullaris)
 - ☐ laminectomy alone is not appropriate in the thoracic or cervical spine when there is anterior compression of the spinal cord.

21.5.4 Psoas abscess

General information

1. Applied anatomy of psoas muscle:
 - a) one of 2 heads of the iliopsoas muscle (the other head is iliacus)
 - b) origin: inner surface of ilium, base of sacrum, and transverse processes, vertebral bodies (VB) and intervertebral discs of spinal column starting from the inferior margin of T12 VB, extending to the upper part of L5 VB. Insertion: lesser trochanter of the femur. Psoas is the primary hip flexor
 - c) 30% of people also have a psoas minor which lies anterior to the psoas major

Table 21.6 Conditions associated with secondary psoas abscess ¹¹¹	
Organ system	Condition
gastrointestinal	diverticulitis, appendicitis, Crohn’s disease, colorectal cancer
genitourinary	UTI, cancer
musculoskeletal infections	vertebral osteomyelitis, infectious sacroiliitis, septic arthritis
other	endocarditis, femoral artery catheterization, infected abdominal aortic aneurysm graft, hepatocellular Ca, intrauterine contraceptive device, trauma, sepsis, dialysis (peritoneal or long-term hemodialysis)

- d) innervation: branches of L2–4 nerve roots proximal to the formation of the femoral nerve
- e) susceptibility to infection
 - rich vascular supply makes it vulnerable to hematogenous spread
 - proximity to structures that may be a source of infection: sigmoid colon, jejunum, vermiform appendix, ureters, aorta, renal pelvis, pancreas, iliac lymph nodes and spine
- 2. may be primary (no identifiable underlying disease) or secondary in which it may be associated with one of the conditions shown in □ Table 21.6
- 3. risk factors: IV drug abuse, HIV/AIDS, age > 65 years, DM, immunosuppression, renal failure

Clinical findings

Physical findings: signs of iliopsoas inflammation include:

- 1. active: pain on flexing the hip against resistance
- 2. passive: with the patient lying on the unaffected side, hyperextension of the affected hip stretches the psoas muscle and produces pain.

Diagnostic tests

- 1. routine infection work-up: WBC (often elevated), blood cultures, U/A + C&S (pyuria may be seen)
- 2. AP abdominal x-ray: psoas shadow may be obliterated
- 3. CT: sensitivity is 80–100% (MRI is not better).¹⁰⁹ Enlargement of psoas muscle on affected side best seen inside iliac wing.

Treatment often includes drainage of the psoas abscess either surgically or percutaneously with CT-guidance.

Mortality rates associated with psoas abscess: 2.4% with primary, 19% with secondary, with sepsis being the usual cause of death.¹¹⁰

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22 Other Nonbacterial Infections

22

22.1 Viral encephalitis

Encephalitides that come to the attention of the neurosurgeon usually cause imaging findings that may mimic mass lesions, for cases where biopsy may be helpful, and shunting for hydrocephalus is needed. Those covered in this book:

1. herpes simplex encephalitis: see below
2. multifocal herpes varicella-zoster virus leukoencephalitis (p.366)
3. progressive multifocal leukoencephalopathy (PML) (p.331)

22.1.1 Herpes simplex encephalitis

General information

Key concepts

- a hemorrhagic viral encephalitis with a predilection for temporal lobes
- definitive diagnosis requires brain biopsy
- optimal treatment: early administration of IV acyclovir

Herpes simplex encephalitis (HSE) AKA multifocal necrotizing encephalomyelitis, is caused by the herpes simplex virus (HSV) type I. It produces an acute, often (but not always) hemorrhagic, necrotizing encephalitis with edema. There is a predilection for the temporal and orbitofrontal lobes and limbic system.

Epidemiology

Estimated incidence of HSE: 1 in 750,000 to 1 million persons/yr. Equally distributed between male and females, in all races, in all ages (over 33%of cases occur in children 6 mos to 18 yrs), throughout the year.¹

Presentation

Patients are often confused and disoriented at onset, and progress to coma within days. Adult presentations are shown in □ Table 22.1, and for pediatrics in □ Table 22.2. Other symptoms include headache.

Diagnostic studies

Diagnosis can often be made on the basis of history, CSF, and MRI. Treatment should be instituted rapidly without waiting for biopsy, before the onset of coma.

1. CSF: leukocytosis (mostly monos), RBCs 500–1000/mm³, (NB: 3%have no pleocytosis), protein rises markedly as disease progresses. HSV antibodies may appear in the CSF but takes at least ≈ 14 days and is thus not useful for early diagnosis

Table 22.1 Herpes simplex encephalitis – adult presentation	
Symptom	%
altered consciousness	97%
fever	90%
seizures (usually focal onset)	67%
personality changes	71%
hemiparesis	33%

Table 22.2 Herpes simplex encephalitis – presentation in age <10 yrs
irritability
altered mentation
malaise
seizure
disorientation
dysphasia
hemiparesis
fever
papilledema (except in age ≤2 yrs)

- 2. EEG: periodic lateralizing epileptiform discharges (PLEDs) (triphasic high-voltage discharges every few seconds) usually from the temporal lobe. EEG may vary rapidly over few days (unusual in conditions mimicking HSE)
- 3. CT: edema predominantly localized in temporal lobes (poorer prognosis once hemorrhagic lesions visible). In one review, 38%of initial CTs were normal² (many were on early generation CTscanners or were done within 3 days of onset). Hemorrhages were apparent in only 12%of the initially abnormal CTs
- 4. MRI: more sensitive than CT,³ demonstrates edema as high signal on T2WI, primarily within the temporal lobe, with some extension across sylvian fissure (“transsylvian sign”),² especially suggestive of HSE if bilateral. Differentiate from MCA infarct (which may also span sylvian fissure) by typical arterial distribution of the latter. Enhancement doesn’t occur until the 2nd week
- 5. technetium brain scan: process localized to temporal lobes
- 6. brain biopsy: false negatives may occur⁴; see details below

Brain biopsy

Indications: reserved for questionable cases. May not be necessary in patients with fever, encephalopathy, compatible CSF findings, focal neuro findings (focal seizure, hemiparesis, or cranial nerve palsy), and supporting evidence of at least one of the following: focal EEG, CT, MRI or technetium brain scan abnormality.

Should be performed within ≤48 hrs of starting acyclovir (otherwise false negatives may occur).

Biopsy results: of 432 brain biopsies performed using the technique below, 45%had HSE, 22%had identifiable but non HSE pathology (e.g. vascular disease, other viral infection, adrenal leukodystrophy, bacterial infection...), and 33%remained without a diagnosis.⁵

Technique

- 1. anterior inferior temporal lobe is preferred site
 - a) the side chosen for biopsy is the one showing maximal involvement based on clinical information (e.g. localizing seizures), EEG and/or imaging studies⁶
 - b) 10 × 10 × 5 mm deep specimen obtained from anterior portion of the inferior temporal gyrus with NO COAGULATION on specimen side (cut surface with # 11 blade, then cauterize pial surface on non-specimen side)
 - c) 2nd specimen obtained from beneath surface specimen with fenestrated pituitary biopsy forceps
- 2. virus isolation is the most specific (100%) and sensitive (96–97%) test for HSE. Other findings (less accurate): perivascular cuffing, lymphocytic infiltration, hemorrhagic necrosis, neuronophagia, intranuclear inclusions (present in 50%)
- 3. if electron microscopy (EM) or immunohistofluorescence is available, 70%may be diagnosed within ≈ 3 hrs of biopsy
- 4. biopsy tissue handling
 - a) avoid macerating specimens for histology
 - b) tissue for EM: placed in glutaraldehyde
 - c) tissue for permanent histology: placed in formalin
 - d) tissue for culture:

- handling: specimen is placed in sterile specimen container and sent directly to virology lab. If lab is closed, tissue may be placed in regular refrigerator for up to 24 hrs or placed in –70° C freezer for indefinite time (virus remains viable for up to 5 yrs). ☐ DO NOT place specimen in regular freezer (destroys virus)
- cultures generally take at least 1 week to become positive
- cultures checked for 3 weeks before being declared negative

Treatment

General treatment measures

General supportive measures: to control elevated ICP from edema, includes: elevate HOB, mannitol, hyperventilation (dexamethasone unproven efficacy); also see Treatment measures for elevated ICP (p.866). Anticonvulsants are used for seizure prophylaxis.

Antiviral medications

Acyclovir is the drug of choice for HSE.

Drug info: Acyclovir (Zovirax®)

- ☐ Adult: 30 mg/kg/day, in divided q 8 hr doses in minimum volume of 100 ml IV fluid over 1 hr (caution: this fluid load may be hazardous, especially since cerebral edema is already usually problematic) for 14–21 days (some relapses have been reported after only 10 days of treatment).
 - ☐ Children >6 mos age: 500 mg/m² IV q 8 hrs × 10 days.
 - ☐ Neonatal: 10 mg/kg IV q 8 hrs for 10 days.

Drug info: Vidarabine (Vira-A®)

Six month mortality following treatment with acyclovir was influenced by:

- age (6%under age 30, 36%over age 30)
- Glasgow coma score (GCS) at time of treatment initiation (25%for GCS ≤ 10, 0%for GCS > 10)
- duration of disease before therapy (0%for initiating therapy within 4 days of onset of symptoms, 35%if after 4 days)

22.1.2 Multifocal varicella-zoster leukoencephalitis

Caused by the herpes varicella-zoster virus (VZV) which is responsible for varicella (chickenpox), herpes zoster (HZ) (shingles), and post-herpetic neuralgia (p.493). VZV is a herpesvirus that is distinct from the herpes simplex virus.

Symptomatic zoster-related encephalitis occurs in <5% of immunocompromised patients (including AIDS patients) with cutaneous zoster.⁷ It typically follows cutaneous HZ by a short time (average time: 9 days) although cases have been reported where many months have lapsed.⁸

Manifestations include: altered level of consciousness, headache, photophobia, meningismus. Although focal neurologic deficits may occur, these are uncommon.

Recently, vasculopathy following VZV reactivation has been increasingly recognized.⁹

MRI may show multiple, discrete, round and oval lesions with minimal edema (best seen on T2WI) and minimal enhancement.

Unlike herpes simplex virus, VZV is difficult to isolate in culture. On brain biopsy, look for multiple discrete lesions within grey and white matter, with Cowdry type A intranuclear inclusion bodies in oligodendrocytes, astrocytes, and neurons, and a positive direct fluorescent antibody test directed against VZV.

There is a case report of VZV encephalitis treated with IV acyclovir.⁷

22.2 Creutzfeldt-Jakob disease

22.2.1 General information

22

Key concepts

- an invariably fatal encephalopathy characterized by rapidly progressive dementia, ataxia and myoclonus
- death usually occurs within 1 yr of onset of symptoms
- 3 forms: 1) transmissible (possibly via prions), 2) autosomal dominant inherited, 3) sporadic
- characteristic EEG finding: bilateral sharp wave (0.5–2 per second)
- pathology: status spongiosus without inflammatory response

Creutzfeldt-Jakob disease (CJD) is one of 4 known rare human diseases associated with transmissible spongiform encephalopathy agents, also called prions (proteinaceous infectious particles). Although sometimes also referred to as a “slow virus”, these agents contain no nucleic acids and are also resistant to processes that inactivate conventional viruses (□ Table 22.4). Prions do not provoke an immune response. The other human prion diseases are kuru, Gerstmann-Sträussler-Scheinker disease, and fatal familial insomnia (described in 2 families^{10,11,12}). The protease-resistant protein associated with disease is designated PrPres or PrPSc, and is an isoform of a naturally occurring protease-sensitive protein designated PrPsen or PrPC. In the abnormal state, PrPsen which is a predominantly alpha-helical structure, undergoes a post-translational conformation change to PrPres which has large beta-sheets, and which accumulates in neural cells, disrupting function and leading to cell death and vacuolization.¹³ The famous choreographer George Balanchine died of CJD in 1983.

CJD occurs in 3 forms: transmissible, inherited and sporadic.

22.2.2 Epidemiology

Annual incidence of CJD: 0.5–1.5 per million population¹³ with little change over time and no geographic clustering (except in locations with large numbers of familial cases). Over 200 people die of CJD in the U.S. each year.

22.2.3 Acquired prion diseases

Natural route of infection is unknown and virulence appears low, with lack of significant dissemination by respiratory, enteric, or sexual contact. There is no increased incidence in spouses (only a single conjugal pair of cases has been verified), physicians or laboratory workers. There is no evidence of transplacental transmission. The only known cases of horizontal transmission of CJD have occurred iatrogenically (see below). Kuru has been transmitted via handling and ingestion of infected brains in ritualistic funereal cannibalism practiced among the Fore (pronounced: “fore-ay”) linguistic group in the eastern highlands of Papua, New Guinea,¹⁴ a practice which was generally abandoned in the 1950s. Kuru is a subacute, uniformly fatal disease involving cerebellar degeneration (the word “kuru” means “to tremble” in the local language^{15 (p 6)}).

Most noniatrogenically transmitted cases of CJD occur in patients >50 yrs old, and is rare in age < 30. The incubation period can range from months to decades. The onset of symptoms following direct inoculation is usually faster (common range: 16–28 mos), but still may be much longer (up to 30 years with corneal transplant,¹⁶ and 4–21 yrs with hGH transmission). In experimental models of CJD, higher inoculation doses produce shorter incubation periods.¹⁷

22.2.4 Inherited CJD

5–15% of cases of CJD occur in an autosomal dominant inheritance pattern with abnormalities in the amyloid gene¹⁸ on chromosome 20 with a penetrance of 0.56.¹⁹ Since familial CJD is dominantly inherited, analysis for the PrP gene is not indicated unless there is a history of dementia in a first degree relative.

22.2.5 Sporadic CJD

In $\approx 90\%$ of cases of CJD, no infectious or familial source can be identified,¹⁸ and these cases are considered sporadic. 80% occur in persons 50–70 yrs old.¹³ Sporadic cases show no abnormality in the PrP gene.

There appears to be a genetic susceptibility in the sporadic and iatrogenically transmitted CJD cases, with the majority of these showing specific changes in the human prion protein.

22.2.6 New variant CJD

Cases of atypical CJD are well-recognized. A new variant of CJD (vCJD) was identified in 10 cases of unusually young individuals (median age at death: 29 yrs) during 1994–95 in the United Kingdom,²⁰ and has been strongly linked to the 1980s epidemic of bovine spongiform encephalopathy (BSE), dubbed (“mad cow” disease) by the lay press. The BSE epidemic may have been exacerbated by the practice of feeding discarded sheep organs (called offals) to the cows (a practice banned since 1989). This raises the question of possible transmission and mutation of the sheep slow-virus disease, scrapie (which resembles kuru in man) to cows. None of the vCJD patients had periodic spikes on EEG characteristic of classic CJD, the clinical course was atypical (having prominent psychiatric symptoms and early cerebellar ataxia, somewhat similar to kuru), and brain plaques showed unusual features also reminiscent of amyloid plaques seen in kuru. A comparison of vCJD to sporadic CJD is shown in □ Table 22.3.

22.2.7 Iatrogenic transmission of CJD

Described only in cases of direct contact with infected organs, tissues or surgical instruments. Has been reported with: corneal transplants,^{21,16} intracerebral EEG electrodes sterilized with 70% alcohol and formaldehyde vapor after use on a CJD patient,²² operations in neurosurgical O.R.s after procedures on CJD patients, in recipients of pituitary-derived human growth hormone (hGH)²³ (most cases have occurred in France¹⁷; there is no longer a risk of CJD with growth hormone in the U.S. since distribution of pituitary derived hGH was halted in 1985 and current hGH is obtained from recombinant DNA technology), and dural graft with cadaveric dura mater (Lyodura®) (most cases have occurred in Japan¹⁷). Ethylene oxide, autoclaving, formalin and ionizing radiation do not inactivate the CJD agent^{24,25,26} (see □ Table 22.4 for other ineffective procedures). Recommended sterilization procedures for suspected CJD tissues and contaminated materials also appear in □ Table 22.4.

22.2.8 Pathology

The typical form of CJD produces the classic histologic triad of neuronal loss, astrocytic proliferation, and cytoplasmic vacuoles in neurons and astrocytes (status spongiosus), all in the absence of an inflammatory response. There is a predilection for cerebral cortex and basal ganglia, but all parts of the CNS may be involved. In 5–10% of cases, these changes are accompanied by the deposition of amyloid plaques (plaques are common in kuru, vCJD and some familial spongiform encephalopathies). Immunostaining for PrPres is definitive.

Table 22.3 Comparison of vCJD to sporadic CJD ¹³		
Characteristic	vCJD	sporadic
mean age at onset (yr)	29	60
mean duration of disease (mo)	14	5
most consistent and prominent early signs	psychiatric abnormalities, sensory symptoms	dementia, myoclonus
cerebellar signs (%)	100	40
periodic complexes on EEG (%)	0	94
pathological changes	diffuse amyloid plaques	sparse plaques in 5–10%

Table 22.4 Operating room sterilization procedures for CJD ²⁷
Fully effective (recommended) procedures • steam autoclaving for 1 hr at 132°C, or • immersion in 1N sodium hydroxide (NaOH) for 1 hr at room temperature
Partially effective procedures • steam autoclaving at either 121° C or 132° C for 15–30 mins, or • immersion in 1N NaOH for 15 mins, or lower concentrations (<0.5N) for 1 hr at room temp, or • immersion in sodium hypochlorite (household bleach) undiluted or up to 1:10 dilution (0.5%) for 1 hr²⁸
□ Ineffective procedures: • boiling, UV or ionizing radiation, ethylene oxide, ethanol, formalin, beta-propiolactone, detergents, quaternary ammonium compounds, Lysol®, alcoholic iodine, acetone, potassium permanganate, routine autoclaving

Table 22.5 Major clinical signs in sporadic CJD ¹³	
Sign	Freq (%)
cognitive deficits ^a	100
myoclonus	>80
pyramidal tract signs	>50
cerebellar signs	>50
extrapyramidal signs	>50
cortical visual deficits	>20
abnormal extraocular movements	>20
lower motor-neuron signs	<20
vestibular dysfunction	<20
seizures	<20
sensory deficits	<20
autonomic abnormalities	<20
^a dementia, psychiatric and behavioral abnormalities	

22.2.9 Presentation

One-third initially express vague feelings of fatigue, sleep disorders, or reduced appetite. Another third have neurologic symptoms including memory loss, confusion, or uncharacteristic behavior. The last third have focal signs including cerebellar ataxia, aphasia, visual deficits (including cortical blindness), or hemiparesis.

The typical course is inexorable, progression of dementia, often noticeably worse week by week, with subsequent rapid development of pyramidal tract findings (limb weakness and stiffness, pathologic reflexes), and late extrapyramidal findings (tremor, rigidity, dysarthria, bradykinesia) and myoclonus (often stimulus triggered). Clinical signs of sporadic CJD are shown in □ Table 22.5.

Supranuclear gaze palsy is an occasional finding, also usually late.¹⁹ In early stages, CJD may resemble Alzheimer’s disease (SDAT). 10% of cases present as ataxia without dementia or myoclonus. Cases with predominant spinal cord findings may be initially mistaken for ALS.

Myoclonus subsides in the terminal phases, and akinetic mutism ensues.

22.2.10 Diagnosis

Diagnostic criteria

The complete “diagnostic triad” (dementia, myoclonus and periodic EEG activity) may be absent in up to 25% of cases. Diagnostic criteria have been published²⁹ as shown in □ Table 22.6. No patients in

Table 22.6 Diagnostic criteria ^a of CJD ²⁹					
Pathologically confirmed (with unequivocal spongiform changes)					
• clinically: requires brain biopsy (see text)					
• found at autopsy					
Clinical criteria	Mental deterior- ation	Myoclonus	1–2 Hz peri- odic EEG complexes	Any movement disorder or pe- riodic EEG ac- tivity	Duration of illness (months)
clinically definite	+	+	+		<12
clinically probable	+	+OR+			<18
clinically possible	+			+	<24
^a in patients with normal metabolic status and spinal fluid. If there are early cerebellar or visual symptoms and then muscular rigidity, or if another family member has died of pathologically verified CJD, then upgrade the degree of certainty to the next higher category					

their series with a diagnosis other than CJD fulfilled the criteria for clinically definite CJD. The most common condition other than CJD fulfilling the criteria for clinically probable CJD was SDAT (especially difficult to distinguish in the early stages). There is a CSF immunoassay for the 14–3-3 brain protein (see below).

Differential diagnosis

CSF examination to exclude infections such as tertiary syphilis or SSPE is recommended. Toxicity from bismuth, bromides and lithium must be ruled-out. Myoclonus is usually more prominent early in toxic/metabolic disorders than in CJD, and seizures in CJD are usually late.¹³

Diagnostic tests

1. imaging: no characteristic CT or MR finding. These studies are frequently normal, but are essential to rule-out other conditions (e.g. herpes-simplex encephalitis, recent stroke...). Diffuse atrophy may be present, especially late. MRI may show increased intensity on T2WI in areas typically involved (basal ganglion, striatum) in up to 79% of cases (retrospectively).³⁰ This is nonspecific but may help differentiate CJD from SDAT³¹
2. blood tests: serum assays for S-100 protein are so insensitive and nonspecific³² that it can only be used as an diagnostic adjunct
3. CSF:
 - a) routine labs: usually normal, although protein may occasionally be elevated
 - b) abnormal proteins:
 - abnormal proteins (designated 130 & 131) have been identified in the CSF of patients with CJD,³³ but the assay is technically difficult and is therefore not practical for routine clinical use
 - proteins 130/131 were identified as the normal neuronal protein 14–3-3, and a relatively simple immunoassay for this was developed for use on as little as 50 mcl of CSF.³⁴ Detection of the 14–3-3 protein in the CSF has 96% sensitivity and specificity for CJD among patients with dementia. False positives may occur in other conditions involving extensive neuronal destruction including: acute stroke, herpes encephalitis, multi-infarct dementia, primary CNS lymphoma and rarely SDAT (most cases of SDAT test negative). Requires CSF (cannot be done on blood)
4. EEG: characteristic finding of bilateral, symmetrical, periodic bi- or triphasic synchronous sharp-wave complexes, AKA periodic spikes, AKA pseudoperiodic sharp-wave complexes (0.5–2 per second) have ≈ 70% sensitivity and 86% specificity.³⁵ They resemble PLEDs (p.238), but are responsive to noxious stimulus (may be absent in familial CJD¹⁹ and in the recent UK variant (see above))
5. SPECT scan: may be abnormal in vCJD even when EEG is normal,³⁶ however the findings are not specific for vCJD
6. brain biopsy: see below
7. tonsillar biopsy: patients with variant CJD (vCJD) may have detectable levels of variant type 4 of the abnormal prion protein (PrPSc) in their lymphoreticular system, which may be accessed by a 1 cm wedge-biopsy of one palatine tonsil (using careful aseptic precautions)³⁷

Brain biopsy

Due to lack of an effective treatment and the potential for iatrogenic infection in surgery, biopsy is reserved for cases where establishing the diagnosis is deemed important, or as part of a research study,⁶ or when diagnostic tests are equivocal and other potentially treatable etiologies are suspected.

Technique: to prevent aerosolization of the infectious agent, a manual saw is recommended over a power craniotome, and every effort should be made to avoid cutting the dura with the saw. Recommended decontamination procedures should be followed (□ Table 22.4 and references). Specimens should be clearly labeled as being from suspected CJD patients to alert laboratory personnel to the hazard. Tissue should be fixed in a saturated 15%phenolized formalin (15 g of phenol per dl of 10% neutral buffered formalin with the undissolved phenol layering at the bottom of the solution).³⁸

Analysis for classic histologic findings (see above) and/or immunostaining for PrPres are the gold standards of diagnosis.

22.2.11 Treatment and prognosis

Given the lack of demonstrated infectivity (with tissues other than brain or CSF), isolation precautions such as gowns or masks are felt to be unnecessary.¹³

There is no known treatment. The disease is rapidly progressive. Median survival is 5 months, and 80%of patients with sporadic CJD die within 1 year of diagnosis.¹³

22.3 Parasitic infections of the CNS

22.3.1 General information

A number of parasitic infections may involve the central nervous system. Immunosuppression (including HIV) increases the susceptibility.³⁹ CNS parasitic infections include (those that potentially involve neurosurgical intervention have a dagger (†):

1. cysticercosis†: see Neurocysticercosis below
2. toxoplasmosis†: may occur as a congenital TORCH infection, or in the adult usually with AIDS; see Neurologic manifestations of AIDS (p.329). *Toxoplasma gondii* is an obligate intracellular protozoan that is ubiquitous but does not cause clinical infection except in immunocompromised hosts. Histologic features: necrosis containing 2–3 nm tachyzoites (cysts)
3. echinococcus† (p.375)
4. amebiasis†: ≈ exclusively *Naegleria fowleri* (p.377)
5. schistosomiasis
6. malaria
7. African trypanosomiasis.

† parasitic infections with a dagger are those that are more likely to come to neurosurgical attention.

22.3.2 Neurocysticercosis

General information

Key concepts

- intracranial encystment of larva of *Taenia solium* (pork tapeworm)
- the most common parasitic infection of the CNS
- neurological symptoms: seizures or progressive intracranial hypertension
- occurs from ingesting the parasite's eggs, not from eating infested meat
- characteristic imaging finding: low density cysts with eccentric punctate high density (the scolex= tapeworm head). Hydrocephalus is common
- medical treatment: all patients get steroids. Start antihelminthic drugs (praziquantel or albendazole) when no signs of intracranial hypertension
- biopsy sometimes needed for diagnosis. Surgery: may be required for spinal, intraventricular or subarachnoid cysts (more refractory to medical therapy) or for giant cysts (>50 mm) when intracranial hypertension persists despite steroids

Cysticercosis is the most common parasitic infection involving the CNS⁴⁰ and in some low-income countries it is the most common cause of acquired epilepsy.⁴¹ It is caused by *Cysticercus cellulosae*, the larval stage of the pork tapeworm *Taenia solium*, which has a marked predilection for neural tissue. Cysticercosis is endemic in areas of Mexico, Eastern Europe, Asia, Central and South America, and Africa. The incidence of neurocysticercosis (encystment of larva in the brain) may reach 4% in some areas.⁴² The incubation period varies from months to decades, but 83% of cases show symptoms within 7 years of exposure.

Life cycle of *T. Solium*

Stages

There are 3 stages to the life cycle: larva, embryo (or oncosphere) and adult. *T. solium* can infect man in two different ways: as the adult worm or as the larva.

Infection with the adult worm (taeniasis – a parasitic infection)

Human intestinal tapeworm infection (taeniasis) results from eating undercooked infested (measly) pork. The encysted larvae are released in the small bowel and can then mature within the intestine into an adult over about 2 months. The scolex (head) of the segmented adult worm attaches by means of four suckers and two rows of hooklets to the wall of the small intestine where the worm absorbs food directly through its cuticle. Man is the only known definitive host for the adult tapeworm, for which the GI tract is the sole habitat. Proglottids (mature segments, each containing reproductive organs) produce eggs which are liberally excreted along with gravid proglottid segments in the feces.

Infection with the larva

The disease cysticercosis occurs when animals or humans become an intermediate host for the larval stage by ingesting viable eggs produced by the proglottid. The most common routes of ingestion of viable eggs are:

1. food (usually vegetables) or water contaminated with human feces containing eggs or gravid proglottids (this is the means whereby pigs acquire the disease)
2. fecal-oral autoinoculation in an individual harboring the adult form of the tapeworm due to lack of good sanitary habits or facilities
3. autoinfection by reverse peristalsis of gravid proglottids from the intestine into the stomach (unproven theoretical possibility)

In the duodenum of man and pig, the shell of the ova dissolves and the thusly hatched embryos (oncospheres) burrow through the small bowel wall to enter the lymphatics or systemic circulation and gain access to the following commonly involved sites:

- brain: involved in 60–92% of cases of cysticercosis. Latency from ingestion of eggs to symptomatic neurocysticercosis: 2–5 years⁴³
- skeletal muscle
- eye: immunologically privileged, like brain
- subcutaneous tissue
- heart

Once in the tissue of the intermediary host, embryos develop a cyst wall in ≈ 2 months (immature cyst) which matures in ≈ 4 months to a larva. Larval cysts are usually rapidly eliminated by the immune system. Many larvae die naturally within 5–7 yrs or with cysticidal therapy producing an inflammatory reaction with collapse of the cyst (granular nodular stage), these sometimes calcify (nodular calcified stage). In pigs, the larva lie dormant in the muscle, “waiting” to be eaten after which the cycle repeats.

Types of neurologic involvement

Spinal cord and peripheral nerves involvement is rare.

Giant cysts: definition: cyst with diameter > 50 mm.⁴⁴

Two types of cysts tend to develop in the brain⁴⁵:

1. *cysticercus cellulosae*: regular, round or oval thin-walled cyst, ranging in size from ≈ 3 to 20 mm tending to form in the parenchyma or narrow subarachnoid spaces. This cyst contains a scolex (head), is usually static, and produces only mild inflammation during the active phase

2. *cysticercus racemosus*: larger (4–12 cm), grows actively producing grape-like clusters in the basal subarachnoid spaces and produces intense inflammation. There are no larvae in these cysts. These cysts usually degenerate in 2–5 years, in which the capsule thickens and the clear cyst contents are replaced by a whitish gel which undergoes calcium deposition with concomitant shrinkage of the cyst

Location of the cysts tends to fall into 1 of 4 groups:

1. meningeal: found in 27–56% of cases with neural involvement. Cysts are adherent or free-floating and are located either in:
 - a) dorsolateral subarachnoid space: usually *C. cellulosae* type, causing minimal symptoms
 - b) basal subarachnoid space: usually the expanding *C. racemosus* form producing arachnoiditis and fibrosis comprising a chronic meningitis with hypoglycorrhachia. Can obstruct foramina of Luschka and Magendie producing hydrocephalus, or can cause entrapment of basal cisterns → cranial neuropathies (including visual disturbance). Extremely high mortality with this form
2. parenchymal: found in 30–63% focal or generalized seizures occurs in ≈ 50% of cases (up to 92% in some series)
3. ventricular: found in 12–18% possibly gaining access via the choroid plexus. Pedunculated or free floating cysts occur, can block CSF flow and cause hydrocephalus with intermittent intracranial hypertension (Brun syndrome). There may be adjacent ependymal enhancement (ependymitis)
4. mixed lesions: found in ≈ 23%

Clinical

Presentation: seizures, signs of elevated ICP (36% of patients with CNS cysticercosis present with increased ICP⁴⁶), focal deficits related to the location of the cyst, and altered mental status are the most common findings. Increased ICP may be due to hydrocephalus or to giant cysts. Symptoms may also be produced by the immunologic reaction to the infestation (cysticercotic encephalitis). Cranial nerve palsies can occur with basal arachnoiditis. Subcutaneous nodules may sometimes be felt.

Diagnosis

General information

Diagnosis is usually made by imaging studies and confirmatory serologic tests.

Laboratory evaluation

Mild peripheral eosinophilia can occur, but is inconsistent and thus unreliable.

CSF may be normal. Eosinophils are seen in 12–60% of cases and suggests parasitic infection. Protein may be elevated.

Stool: less than 33% of cases have *T. solium* ova in the stool.

Serology

Most centers use enzyme-linked immunoelectrotransfer blot (EIBT) against glycoprotein antigens (western blot) which is ≈ 100% specific and 98% sensitive,⁴⁷ although sensitivity is less (70%) in cases with a solitary cyst.⁴⁸ May be used on serum or CSF. EIBT has effectively superseded ELISA where titer is considered significant at 1:64 in serum, and 1:8 in the CSF; checking for titer exceeding these thresholds in the serum produces a test that is more sensitive and in the CSF is more specific for cysticercosis. False negative rates are higher in cases without meningitis.

Radiographic evaluation

Soft-tissue x-rays may show calcifications in subcutaneous nodules, and in thigh and shoulder muscles.

Skull x-rays show calcifications in 13–15% of cases with neurocysticercosis. May be single or multiple. Usually circular or oval in shape.

CT

The following findings on brain CT have been described (modified^{45,49}):

1. ring enhancing cysts of various sizes representing living cysticerci. Little inflammatory response (edema) occurs as long as larva is alive. Characteristic finding: small (<2.5 cm) low density cysts with eccentric punctate high density that may represent the scolex

2. low density with ring enhancement seen as an intermediate stage between living cyst and calcified remnant representing intermediate stage in granuloma formation. Resultant inflammatory reaction can cause edema, and basal arachnoiditis in cysts located in basal subarachnoid space. Often ring enhancing
3. intraparenchymal punctate calcifications (granuloma) sometimes with, but usually without surrounding enhancement, seen with dead parasites
4. hydrocephalus. Sometimes with intraventricular cysts which may be isointense with CSF on plain CT⁵⁰ and may require contrast CT ventriculography⁵¹ or MRI to be demonstrated

MRI

Early findings: nonenhancing cystic structure(s) with eccentric T1WI hyperintensity (scolex) with no inflammatory response. Lesions may be seen in parenchyma, ventricle, and subarachnoid space. The cyst collapses in later stages of parasitic evolution, with initial edema that gradually resolves with time.

Treatment

Overview

Combination of:

1. antihelmintic medication: antiparasitic and/or cysticidal regimens
2. antiepileptics: to treat seizures, which may sometimes be medically refractory
3. steroids (see below)
4. surgery:
 - a) surgical resection of lesions when appropriate
 - b) ventricular CSF diversionary procedures

Steroids

Corticosteroids should be used in all patients. May temporarily relieve symptoms, and may help decrease edema that tends to occur initially during treatment with antihelmintic drugs. If possible, start 2–3 d before antihelmintics (e.g. dexamethasone 8 mg q 8 hours⁴⁴), on day 3 decrease to 4 mg q 8 hours, on day 6 change to prednisone 0.4 mg/kg per day divided TID. Taper steroids after antihelmintics are discontinued. In patients with symptoms of intracranial hypertension: antihelmintic treatment is started after symptoms subside (usually after 3 doses). □ Any cysticercocidal drug may cause irreversible damage when used to treat ocular or spinal cysts, even with corticosteroid use.

Antiepileptics

Seizures usually respond to a single AED. However, the risk of seizures may be lifelong. Risk factors for recurrent seizures: calcified brain lesions, multiple seizures, multiple brain cysts.⁵²

Antihelmintic drugs

Since many lesions resolve on their own, and there are significant side effects to these drugs, their use is controversial.⁵³

Praziquantel (Biltricide®) is an antihelmintic with activity against all known species of schistosomes. Several regimens have been published:

- 50 mg/kg/d divided in 3 doses (same dose for pediatrics) for 15 days (doses of 100 mg/kg/d have been recommended⁴⁴ because steroids reduce serum concentration by 50%⁵⁴). Produces a significant reduction in symptoms and in number of cysts seen on CT⁴⁰
- 10–100 mg/kg/d × 3–21 days
- high dose single day regimen: 25–30 mg/kg q 2 hrs × 3 doses
- for intestinal infestation: single oral dose of 5–10 mg/kg

Albendazole (Zentel®) 15 mg/kg per day divided in 2–3 doses, taken with a fatty meal to enhance absorption (same dose for pediatrics), may be given for 3 months,^{55,56} can be stopped sooner if imaging shows resolution.⁴⁴ More parasiticidal than praziquantel and may have fewer side effects.

Niclosamide (Niclocide® and others) may be given orally to treat adult tapeworms in the GI tract. □ 1 gm (2 tablets) chewed PO, repeated in 1 hour (total=2 gm).

Intraventricular disease: There is no consensus on the efficacy of medical treatment for intraventricular cysts.^{46,44,57}

Surgery

Surgery may sometimes be necessary to establish the diagnosis. Stereotactic biopsy may be well suited for some cases, especially with deep lesions.

CSF diversion is necessary for patients with symptomatic hydrocephalus, although tubing may become obstructed by granulomatous inflammatory debris.⁵⁸

Surgery may be indicated for spinal cysts⁴² and for intraventricular cysts which may be less responsive to medical therapy. The latter may sometimes be dealt with using stereotactic techniques and/or endoscopic instrumentation,⁵¹ however, shunting and antihelmintics may suffice.⁵⁷ Surgery may also be needed for giant cysts when intracranial hypertension does not respond to steroids.⁴⁴ Antihelmintics may be required even after complete surgical removal because of possibility of relapse.⁴⁴

Follow-up

CT or MRI scan every 6 months until lesions disappear or calcify.⁴⁴

Contacts

Both patients with cysticercosis and their personal contacts should be screened for tapeworm infection since a single dose of niclosamide or praziquantel will eliminate the tapeworm.⁵⁹ Close contacts of persons with tapeworms should have screening by medical history and serologic testing for cysticercosis; if suggestive of cysticercosis a neurologic exam and CT or MRI should be done.

22.3.3 Echinococcosis

General information

AKA hydatid (cyst) disease. Caused by encysted larvae of the dog tapeworm *Echinococcus granulosus* in endemic areas (Uruguay, Australia, New Zealand...). The dog is the primary definitive host of the adult worm. Intermediate hosts for the larval stage include sheep and man. Ova are excreted in dog feces and contaminate herbage eaten by sheep. After ingestion, the embryos hatch and the parasite burrows through the duodenal wall to gain hematogenous access to multiple organs (liver, lungs, heart, bone, brain). Dogs eat these infested organs and the parasite enters the intestine where it remains.

Man is infected either by eating food contaminated with ova, or by direct contact with infected dogs. CNS involvement occurs in only $\approx 3\%$. Produces cerebral cysts that are confined to the white matter. Primary cysts are usually solitary, secondary cysts (e.g. from embolization from cardiac cysts that rupture or from iatrogenic rupture of cerebral cysts) are usually multiple. The CT density of the cyst is similar to CSF, it does not enhance (although rim enhancement may occur if there is an inflammatory reaction), and there is little surrounding edema. It contains germinating parasitic particles called “hydatid sand” containing $\approx 400,000$ scoleces/ml. The cyst enlarges slowly (rates of ≈ 1 cm per year are quoted, but this is variable and may be higher in children), and usually does not present until quite large with findings of increased ICP, seizures, or focal deficit. Patients often have eosinophilia and may have positive serologic tests for hydatid disease.

Treatment

Treatment is surgical removal of the intact cyst. Every effort must be made to avoid rupturing these cysts during removal, or else the scoleces may contaminate the adjacent tissues with possible recurrence of multiple cysts or allergic reaction. May use adjunctive medical treatment with albendazole (Zentel®) 400 mg PO BID (pediatric dose: 15 mg/kg/d) $\times 28$ days, taken with a fatty meal, repeated as necessary.⁵⁶

The Dowling technique is recommended⁶⁰:

1. the head is positioned so that the cyst points straight up towards the ceiling when the OR table is 30° head up
2. drilling burr holes and performing craniotomy must be done very carefully to avoid rupturing the cyst or tearing the dura which is thin and under tension
3. do not coagulate with anything but low-power bipolar (to avoid cyst rupture)
4. open the dura circumferentially away from the dome of the cyst as it may be adherent to the dura
5. keep the surface of the cyst moist to prevent desiccation and rupture
6. open the thinned overlying cortex gently, separating it from the cyst with irrigation and cottonoids. The cortical opening need only be $\approx 3/4$ the cyst diameter but no less

7. insert a soft rubber catheter between the cyst and the brain, and gently irrigate with saline as the head of the OR table is slowly lowered 45° while the surgeon supports the adjacent cortex with his/her fingers
8. continue irrigating more saline and float the cyst out and into a saline filled receptacle
9. if the cyst is ruptured during the procedure, immediately place a sucker in the cyst to aspirate the contents, remove the capsule, and wash the cavity with saline for 5 minutes. Change instruments and gloves. Placing 10% formalin soaked cottonoids on the cavity for a few minutes is controversial⁶¹ (p.3750)

22.4 Fungal infections of the CNS

22.4.1 General information

Most are medically treated conditions that do not require neurosurgical intervention. They tend to present either with chronic meningitis or brain abscess. Some of the more common ones or those of particular relevance to neurosurgery include:

1. cryptococcosis: see below
 - a) cryptococcal meningitis
 - b) cryptococcoma (mucinous pseudocyst): rare
2. candidiasis: the most common fungal infection of the CNS, but rarely diagnosed before autopsy. Very rare in healthy individuals. Most are *C. albicans*
 - a) candidal meningitis (p.320): the most common CNS infection; see □ (p.320)
 - b) parenchymal infection: candida brain abscesses are rare
 - c) following ventricular shunt placement: almost all fungal VP shunt infections are due to *Candida* spp. (p.339)⁶²
3. aspergillosis (p.322): may be associated with cerebral abscess in organ transplant patients
4. coccidiomycosis: caused by the dimorphic fungus *Coccidioides immitis*. Endemic in southwestern U.S. (including San Joaquin Valley in southern California), Mexico, and Central America. Usually presents as meningitis, with rare reports of parenchymal lesions⁶³
5. mucormycosis (phycomycosis) (p.568): usually occurs in diabetics

22.4.2 Cryptococcal involvement of the CNS

General information

Cryptococcal CNS involvement with *Cryptococcus* is diagnosed more frequently in living patients than any other fungal disease. Occurs in healthy or immunocompromised patients. In HIV, *Cryptococcus neoformans* is the typical agent.

1. cryptococcoma (mucinous pseudocyst): a parenchymal collection which occurs almost exclusively in AIDS patients. Much less common than cryptococcal meningitis. No enhancement of the lesion or the meninges. Usually 3–10 mm in diameter and are frequently located in the basal ganglia (due to spread by small perforating vessels)
2. cryptococcal meningitis (p.376)
 - a) occurs in 4–6% of patients with AIDS.⁶⁴ Typical symptoms: fever, malaise and H/A.⁶⁵ Meningeal signs (nuchal rigidity, photophobia...) occur in only ≈ 25% Encephalopathic symptoms (lethargy, altered mentation...) usually from increased ICP occur in a minority
 - b) can also occur without AIDS: gatti variety can infect the brain of immunocompetent hosts⁶⁶
 - c) may be associated with increased ICP (with or without hydrocephalus on CT/MRI), decreased visual acuity, and/or cranial nerve deficits. Dilation of Virchow-Robbins spaces may be seen on imaging; on MRI the signal is similar to CSF on T1WI & T2WI but will be higher signal on FLAIR
 - d) late deterioration in the absence of documented infection may respond to decadron 4 mg q 6 hrs transitioned to prednisone 25 mg p.o. q d⁶⁷

Diagnosis

LP

Opening pressure (OP) should be measured in the lateral decubitus position.⁶⁸ OP is usually elevated, and is >20 cm H₂O in up to 75%

CSF: cryptococcal antigen titer is invariably high with cryptococcal meningitis or meningoencephalitis.

Serum cryptococcal antigen: almost always elevated with CNS involvement.⁶⁸

Management

2009 CDC guidelines for CNS cryptococcal infection in HIV-infected adolescents/adults⁶⁸:

1. antifungal agents: the recommended initial standard treatment⁶⁸ is amphotericin B deoxycholate (Amphocin®) 0.7 mg/kg IV q d, plus fluconazole (an oral triazole) 100 mg/kg po q d in 4 divided doses
2. patients with clinical signs of increased ICP (confusion, blurred vision, papilledema, LE clonus...) should have LP to measure ICP
3. management of intracranial hypertension (ICHT) ($OP \geq 25$ cm H₂O) with or without hydrocephalus; corticosteroids, acetazolamide and mannitol have not been shown to be effective⁶⁹:
 - a) daily LPs: drain enough CSF to reduce ICP by 50% (typically 20–30 ml)⁷⁰
 - b) daily LPs may be suspended when pressures are normal for several consecutive days
 - c) lumbar drain: occasionally needed for extremely high OPs (>40 cm H₂O) when frequent LPs are required to or fail to control symptoms⁶⁹
 - d) CSF shunt: considered when daily LPs are no longer tolerated or when signs and symptoms of ICHT are not being relieved (neither dissemination of infection through the distal shunt nor creation of a nidus of infection refractory to medical therapy has been described⁷¹). Options:
 - lumboperitoneal shunt
 - VP or VA shunt^{72,73}
4. antifungal treatment is continued for ≥ 2 weeks if renal function is normal (most immunocompetent patients will be successfully treated with 6 weeks of therapy⁶⁹)
5. after 2 weeks of treatment, repeat the LP to look for clearance of the organism from the CSF. Positive CSF cultures after 2 weeks of treatment are predictive of future relapse and are associated with worse outcome
6. treatment failures: defined as lack of clinical improvement after 2 weeks of appropriate therapy including management of ICHT, or relapse after an initial response, defined as either a positive CSF culture and/or rising CSF cryptococcal Ag titer with a compatible clinical picture.
Management:
 - a) optimal management has not been defined
 - b) trials with alternative antifungals (e.g. flucytosine) or higher doses of fluconazole
7. maintenance therapy (secondary prophylaxis): HIV patients who have completed 10 weeks of treatment should be maintained on fluconazole 200 mg q d until immune reconstitution occurs, otherwise lifetime treatment is indicated⁶⁸
8. the risk of recurrence is low for patient who remain asymptomatic after a complete course of therapy and have sustained increase (>6 months) of CD4 + counts to ≥ 200 cells/mcl. Some experts perform an LP to document negative CSF culture and antigen before stopping maintenance therapy

22.5 Amebic infections of the CNS

22.5.1 General information

Naegleria fowleri: the only ameba (alternative spelling: amoeba) known to cause CNS infection in humans → primary amebic meningoencephalitis (PAM): diffuse encephalitis with hemorrhagic necrosis and purulent meningitis involving brain and spinal cord. Rare (only 95 cases in the U.S. as of 2002, and ≈ 200 cases worldwide as of 2004). The ameba lives in fresh water and soil and typically gains entry to the CNS by invading nasal olfactory mucosa. PAM usually occurs within 5 days of exposure, usually from diving in warm freshwater.

Associated cerebral edema may cause increased ICP and, ultimately, herniation. Fatal in $\approx 95\%$ of cases, usually within 1 week.

CSF: cloudy and often hemorrhagic, $\uparrow$ leukocytes, $\uparrow$ protein, normal or $\downarrow$ glucose, Gram stain negative (no bacteria or fungi), wet prep → motile trophozoites (may be confused with WBCs).

22.5.2 Treatment

Drug of choice: amphotericin B (lipid preparations (Abelcet®) achieve higher MICs (minimal inhibitory concentrations) than other amphotericin preparations). Miconazole may be synergistic with amphotericin B.

Surgical intervention: ventriculostomy with CSF drainage may be indicated when findings are suggestive of increased ICP. In one survivor, surgical drainage of a brain abscess was performed in addition to treatment with a 6-week course of amphotericin B, rifampicin, and chloramphenicol.

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Part VII

Hydrocephalus and
Cerebrospinal Fluid (CSF)

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VII

23 Cerebrospinal Fluid

23.1 General information

23

Cerebrospinal fluid (CSF) surrounds the brain and spinal cord, and may function as a shock absorber for the CNS. It may also serve an immunological function analogous to the lymphatic system.¹ It circulates within the subarachnoid space, between the arachnoid and the pial membranes.

CSF is normally a clear colorless fluid with a specific gravity of 1.007 and a pH of $\approx 7.33\text{--}7.35$.

23.2 Production

23.2.1 Location

80% of CSF is produced by the choroid plexuses, located in both lateral ventricles (accounts for $\approx 95\%$ of CSF produced in the choroid plexuses) and in the 4th ventricle. Most of the rest of intracranial production occurs in the interstitial space.² A small amount may also be produced by the ependymal lining of the ventricles. In the spine, it is produced primarily in the dura of the nerve root sleeves.

□ Table 23.1 shows properties of CSF production, volumes and pressures.

23.2.2 Production rate

In the adult, CSF is produced at a rate of about 0.3 ml/min (□ Table 23.1). In terms that are clinically relevant, this approximates 450 ml/24hrs, which means that in an adult (where the average total CSF volume in the body is 150 ml), the CSF is “turned over” ≈ 3 times every day. The rate of formation is independent of the intracranial pressure³ (except in the limiting case when ICP becomes so high that cerebral blood flow is reduced⁴).

23.3 Absorption

CSF is absorbed primarily by arachnoid villi (granulations) that extend into the dural venous sinuses. Other sites of absorption include the choroid plexuses and lymphatics. The rate of absorption is pressure dependent.⁵

23.4 CSF constituents

23.4.1 Cellular components of CSF

In normal adult CSF, there are 0–5 lymphocytes or mononuclear cells per mm^3 , and no polys (PMNs) or RBCs. In the absence of RBCs, 5–10 WBCs per mm^3 is suspicious, and >10 WBCs per mm^3 is definitely abnormal.

Table 23.1 Normal CSF production, volumes, and pressure			
Property	Peds		Adult
	Newborn	1–10 yrs	
total volume (ml)	5		150 (50% intracranial, 50% spinal)
formation rate	25 ml/d		$\approx 0.3\text{--}0.35$ ml/min ($\approx 450\text{--}750$ ml/d)
pressure ^a (cm of fluid)	9–12	mean: 10 normal: <15	adult: 7–15 (>18 usually abnormal) young adult: $<18\text{--}20$
^a as measured in the lumbar subarachnoid space, with the patient relaxed in the lateral decubitus position			

23.4.2 Noncellular components of CSF

Table 23.2 CSF solutes. ^{6 (p 169),7} For CEA, AFP, & hCG, see Tumor markers (p. 598)				
Constituent	Units	CSF	Plasma	CSF:plasma ratio
osmolarity	mOsm/L	295	295	1.0
H ₂ O content		99%	93%	
sodium	mEq/L	138	138	1.0
potassium	mEq/L	2.8	4.5	0.6
chloride	mEq/L	119	102	1.2
calcium	mEq/L	2.1	4.8	0.4
pCO ₂	mm Hg	47	41 ^a	1.1
pH		7.33	7.41	
pO ₂	mm Hg	43	104 ^a	0.4
glucose	mg/dl	60	90	0.67
lactate	mEq/L	1.6	1.0 ^a	1.6
pyruvate	mEq/L	0.08	0.11 ^a	0.73
lactate:pyruvate		26	17.6 ^a	
total protein ^b	mg/dl	35	7000	0.005
albumin	mg/L	155	36600	0.004
IgG	mg/L	12.3	9870	0.001
^a arterial plasma				
^b Note: CSF protein is lower in ventricular fluid than in lumbar subarachnoid space				
Data from Table 6–1 of “Cerebrospinal Fluid in Diseases of the Nervous System” by Robert A. Fishman, M.D., © 1980, W. B. Saunders Co., Philadelphia, PA, used with permission				

23.4.3 Variation with site

The composition of CSF differs slightly in the ventricles where the majority of it is produced compared to the lumbar subarachnoid space.

23.4.4 CSF variations with age

Table 23.3 Variations with age					
Age group	WBC /mm ³	RBC /mm ³	Protein (mg/dl)	Glucose (mg/dl)	Glucose ratio (CSF:plasma)
newborn					
preemie	10	many	150	20–65	0.5–1.6
term	7–8	mod	80	30–120	0.4–2.5
infants					
1–12 mos	5–6	0	15–80		
1–2 yrs	2–3	0	15		

Table 23.3 continued

Age group	WBC /mm ³	RBC /mm ³	Protein (mg/dl)	Glucose (mg/dl)	Glucose ratio (CSF:plasma)
young child	2–3	0	20		
child 5–15 yrs	2–3	0	25		
adolescent & adult	3	0	30	40–80	0.5
senile	5	0	40 ^a		

^anormal CSF protein rises ≈ 1 mg/dl per year of age in the adult

23.5 Cranial CSF fistula

Key concepts

- suspect in posttraumatic otorrhea/rhinorrhea or recurrent meningitis
- management strategy: 1) confirm the fluid is CSF, 2) identify the site of origin of the leak, 3) determine etiology/mechanism
- most bedside tests are unreliable and include: “reservoir sign”, target/halo sign, qualitative glucose
- most accurate confirmatory test is β2-transferrin
- CT cisternography is the test of choice for localizing site of fistula

23.5.1 General information

AKA CSF leak. CSF fistula should be suspected in patients with otorrhea or rhinorrhea after head trauma, or in patients with recurrent meningitis.

23.5.2 Possible routes of egress of CSF

1.

mastoid air cells (especially after p-fossa surgery, e.g. for vestibular schwannoma (VS))
2.

sphenoid air cells (especially post-transsphenoidal surgery)
3.

cribriform plate/ethmoidal roof (floor of frontal fossa)
4.

frontal sinus air cells
5.

herniation into empty sella and then into sphenoid air sinus
6.

along path of internal carotid artery
7.

Rosenmüller’s fossa: located just inferior to cavernous sinus, may be exposed by drilling off anterior clinoids to allow access to ophthalmic artery aneurysms
8.

site of the opening of the transient lateral craniopharyngeal canal
9.

percutaneously through a surgical or traumatic wound
10.

petrous ridge or internal auditory canal: following temporal bone fracture or vestibular schwannoma surgery. Then either:

a)

rhinorrhea: through middle ear →eustachian tube →nasopharynx

b)

otorrhea: via perforated tympanic membrane →external auditory canal

23.5.3 Traumatic vs. nontraumatic etiology

Description

Two major subgroups of CSF fistula (omitting the ambiguous category of “spontaneous”)⁸:

1.

traumatic (or posttraumatic): may occur acutely or may be delayed

a)

post-procedure (iatrogenic). Including: post-transsphenoidal surgery and post skull base surgery

b)

posttraumatic (more common): 67–77%of cases
2.

nontraumatic

a)

high pressure
 - hydrocephalus
 - tumor

Table 23.4 CSF findings in various pathologic conditions (adult values) ^a						
Condition	OP (cm H ₂ O)	Appearance	Cells (per mm ³)	Protein (mg%)	Glucose (% se- rum)	Miscellaneous
normal	7–18	clear colorless	0 PMN, 0 RBC 0–5 monos	15–45	50	
acute purulent meningitis	Freq ↑	turbid	few-20K (WBCs mostly PMNs)	100–1000	<20	few cells early or if treated
viral meningitis & encephalitis	nl	nl	few-350 WBCs (mostly monos)	40–100	nl	PMNs early
Guillain-Barré	nl	nl	nl	50–1000	nl	protein↑ freq. IgG
polio	nl	nl	50–250 (monos)	40–100	nl	
TB meningitis ^b	freq ↑	opalescent, yellow, fibrin clot on standing	50–500 (lympho-cytes and monocytes)	60–700	20–40	PMN early, (+) AFB culture, (+) Ziel-Neelson stain
fungal meningitis	Freq ↑	opalescent	30–300 (monos)	100–700	<30	(+) India ink prep with cryptococcus
amebic meningoencephalitis (Naegleria)	freq ↑	cloudy, may be hemorrhagic	↑ WBCs (400–26K), ↑ RBCs	↑	nl or ↓	negative Gram stain; wet mount → motile trophozoites (p.377)
parameningeal infection	↑ if block	nl	WBCs nl or ↑ (0–800)	↑	nl	e.g. spinal epidural abscess
traumatic ^c (bloody) tap	nl	bloody; supernatant colorless	RBC:WBC ratio ≈ as in peripheral	slight ↑	nl	RBCs ↓ in successive tubes; no xanthochromia
SAH ^c	↑	bloody; supernatant xanthochromic	early: ↑ RBCs	50–400	nl or ↓	RBCs disappear in 2 wks, xanthochromia may persist for weeks
			late: ↑ WBCs	100–800		
multiple sclerosis (MS) ^d	nl	nl	5–50 monos	nl-800	nl	usually↑ gamma globulins (oligoclonal)
^a abbreviations: OP=opening pressure; nl=normal; ↑ =increased; ↓ =decreased; freq =frequently						
^b the CSF findings in TB meningitis are almost pathognomonic when they occur in combination; 20–30% have acid-fast bacilli in their CSF sediment smears						
^c to differentiate traumatic tap from SAH, see Differentiating SAH from traumatic tap (p.1506)						
^d see more information on the CSF in MS (p.181)						

- b) normal pressure
- congenital defects
 - bony erosion from infection or necrosis
 - focal atrophy (olfactory or sellar)

Traumatic fistula

Occur in 2–3% of all patients with head injury, 60% occur within days of trauma, 95% within 3 months.⁹ 70% of cases of CSF rhinorrhea stop within 1 wk, and usually within 6 mos in the rest. Non-traumatic cases cease spontaneously in only 33%. Adult:child ratio is 10:1, rare before age 2 yrs. In children the incidence of CSF leaks is less than 1% of closed head injuries.¹⁰ Anosmia is common in traumatic leaks (78%), rare in spontaneous.¹¹ Most (80–85%) CSF otorrhea ceases in 5–10 days.

CSF fistula occurred in 8.9% of 101 cases of penetrating trauma, and increases the infection rate over those penetrating injuries without fistula (50% vs. 4.6%).¹² It is reported to occur post-op in up to 30% of cases of skull-base surgery.¹³

Nontraumatic CSF fistula

General information

Nontraumatic leaks primarily occur in adults >30 yrs. Often insidious. May be mistaken for allergic rhinitis. Unlike traumatic leaks, these tend to be intermittent, the sense of smell is usually preserved, and pneumocephalus is uncommon.¹⁴

Sometimes associated with the following¹⁵

1. agenesis of the floor of the anterior fossa (cribriform plate) or middle fossa
2. empty sella syndrome (p. 773): primary or post transsphenoidal surgery
3. increased ICP and/or hydrocephalus
4. infection of the paranasal sinuses
5. tumor: including pituitary adenomas (p. 718), meningiomas
6. a persistent remnant of the craniopharyngeal canal¹⁶
7. AVM¹⁴
8. congenital anomalies: most involve dehiscence of bone
 - a) dehiscence of the footplate of the stapes (a congenital abnormality) which can produce CSF rhinorrhea via the eustachian tube¹⁴
 - b) dehiscence below foramen rotundum

Spontaneous posterior fossa CSF fistula

1. pediatric: usually presents with either meningitis or hearing loss
 - a) preserved labyrinthine function (hearing and balance): these usually present with meningitis. 3 usual routes of fistula:
 - facial canal: can fistulize into middle ear
 - petromastoid canal: along path of arterial supply to mucosa of mastoid air sinuses
 - Hyrtl's fissure (AKA tympanomeningeal fissure): links p-fossa to hypotympanum
 - b) anomalies of labyrinth (hearing lost): one of several types of Mondini dysplasias, usually presenting with rounded labyrinth/cochlea that permits CSF to erode through oval or round window into auditory canal
2. adult: usually presents with conductive hearing loss with serous effusion, meningitis (often following an episode of otitis media), or cerebral abscess. Occurs most commonly through middle fossa. May be due to arachnoid granulations eroding into air sinus compartment

23.6 Spinal CSF fistula

Often presents with postural headache associated with neck stiffness and tenderness.¹⁷

23.7 Meningitis in CSF fistula

Incidence with posttraumatic CSF leak: 5–10% increases as leak persists >7 days. Meningitis is more common with spontaneous fistula. Risk may be higher in fistula following a neurosurgical procedure than in post-traumatic fistula possibly due to elevated ICP common in latter (forces CSF outward).

Meningitis may promote inflammatory changes at the site of the leak, with a resultant cessation of the leak. However, this often proves to be a temporary resolution, providing a false sense of security.

Pneumococcal meningitis is the most common pathogen (83% of cases¹⁸), mortality is lower than in pneumococcal meningitis without underlying fistula (<10% vs. 50%), possibly because the latter is frequently seen in elderly debilitated patients. Prognosis in children is worse.⁹

23.8 Evaluation of the patient with CSF fistula

23.8.1 Determining if rhinorrhea or otorrhea is due to a CSF fistula

1. characteristics of the fluid suggesting the presence of CSF
 - a) fluid is as clear as water (unless infected or admixed with blood)
 - b) fluid does not cause excoriation within or outside the nose
 - c) patients with CSF rhinorrhea describe the taste as salty
2. confirmatory tests
 - a) β_2 -transferrin: present in CSF, but absent in tears, saliva, nasal exudates and serum (except for newborns and patients with liver disease).^{19,20} The only other source is the vitreous fluid of the eye. It may be detected by protein electrophoresis. A minimum of ≈ 0.5 ml needs to be collected, placed in a sterile urine container, packed in dry ice, and shipped to a lab that can perform this study. Very sensitive & specific
 - b) collect fluid and obtain quantitative glucose (urine glucose detection strips are too sensitive, and may be positive even with excess mucus). Test the fluid shortly after collection to minimize fermentation. Normal CSF glucose is > 30 mg% (usually lower with meningitis) whereas lacrimal secretions and mucus are usually < 5 mg%. A negative test is more helpful since it rules out CSF (except in hypoglycorrhachia (low glucose in the CSF)), but there is a 45–75% chance of false positive²¹ (p 1638)
 - c) “ring sign”: when a CSF leak is suspected but the fluid is blood tinged, allow the fluid to drip onto linen (sheet or pillowcase). A ring of blood with a larger concentric ring of clear fluid (so called “double ring” or halo sign) suggests the presence of CSF. An old, but unreliable, sign
 - d) reservoir sign: a gush of fluid that occurs with a certain head position. Most commonly when first sitting up after a period of recumbency. Thought to indicate drainage of CSF pooled in the sphenoid sinus. Not reliable²²
3. radiographic signs: pneumocephalus on CT or skull x-ray. Pneumocephalus occurs in $\approx 20\%$ of patients with CSF leaks²³ (p 280)
4. cisternogram: intrathecal injection of radionuclide tracer followed by scintigram or injection of radiopaque contrast followed by CT scan (see below)
5. anosmia is present in $\approx 5\%$ of CSF leaks
6. following skull-base surgery (especially involving greater superficial petrosal nerve) there may be a pseudo-CSF rhinorrhea possibly due to nasal hypersecretion from imbalanced autonomic regulation of the nasal mucosa¹³ ipsilateral to the surgery. Often accompanied by nasal stuffiness and absent ipsilateral lacrimation, and occasionally by facial flushing

Localizing the site of CSF fistula

90% of the time, localization does not require water-soluble contrast CT cisternography (WS-CTC) (see below).

1. CT: to detect pneumocephalus, fractures, skull base defects, hydrocephalus and obstructive neoplasms. Include thin coronal cuts or reconstructions through anterior fossa all the way back to the sella turcica
 - a) non-contrast (optional): to demonstrate bony anatomy
 - b) with IV contrast: leak site is usually associated with abnormal enhancement of adjacent brain parenchyma (possibly from inflammation)
2. water-soluble contrast CT cisternography (procedure of choice): see below
3. plain skull x-ray (helpful in only 21%)
4. MRI: may provide additional information for localization and can R/O p-fossa mass, tumor, and empty sella better than CT. Both CT and MRI can R/O hydrocephalus. T2WI fast spin-echo sequences with fat suppression and video image reversal have been used to visualize CSF flow (sensitivity and specificity are 0.87 and 0.57 respectively)²⁴
5. older tests (abandoned in favor of above):
 - a) radionuclide cisternography (RNC): not a contemporary test. Poor localization. Some of the studied radiopharmaceuticals are no longer available
 - b) intrathecal (visible) dye studies: some success with indigo carmine or fluorescein (p.1426) with little or no complications.
 - methylene blue (p.1426) is neurotoxic and should not be used

Water-soluble contrast CT cisternography

Procedure of choice. This test is performed if:

1. no site identified on plain CT (with coronals)

2. when patient is leaking clinically (the site is only sometimes identified in the absence of an active leak)
3. when multiple bony defects are identified, and it is essential to determine which site is actively leaking
4. if a bony defect seen on plain CT does not have associated changes of abnormal enhancement of adjacent brain parenchyma

Technique²⁵: Use iohexol (p.219) 6–7 ml of 190–220 mg/ml injected into lumbar subarachnoid space via 22 gauge spinal needle (or 5 ml via C1–2 puncture). Patient positioned in -70° Trendelenburg × 3 min prone with neck gently flexed, in CT they are kept prone with head hyperextended with 5 mm coronal cuts with 3 mm overlap (use 1.5 mm cuts if necessary). May need provocative maneuvers (coronal scans prone (brow up) or in position of leak, intrathecal saline infusion (requires Harvard pump)²⁶...).

Look for accumulation of contrast in air sinuses. Apparent discontinuity of bone on CT without extravasation of contrast is probably not the site of leakage (bone discontinuities may be mimicked by partial volume averaging on CT).

23.9 Treatment for CSF fistula

23.9.1 Initial treatment

Acutely after trauma, observation is justified as most cases cease spontaneously.

Prophylactic antibiotics: Controversial. There was no difference in the incidence or morbidity of meningitis between treated and untreated patients.²⁷ Furthermore, the risk of selecting resistant strains appears real⁹ and is therefore usually avoided.

23.9.2 For persistent posttraumatic or post-op leaks

Non-surgical treatment

1. measures to lower ICP:
 - a) bed rest: although recumbency may ameliorate symptoms, there is no other benefit from bed rest²⁸
 - b) avoid straining (stool softeners) and avoid blowing nose
 - c) acetazolamide (250 mg PO QID) to reduce CSF production
 - d) modest fluid restriction; caution post-transsphenoidal because of possible DI (p.120): 1500 ml/day in adults, 75% of maintenance/day in peds
2. if leak persists (caution: first R/O obstructive hydrocephalus with CT or MRI)
 - a) LP: q d to BID (lower pressure to near atmospheric or until H/A)
OR
 - b) continuous lumbar drainage (CLD): via percutaneous catheter. Two (of many) management options:
 - keep HOB elevated 10–15° and place drip chamber at shoulder level (lower the chamber if leak persists) and leave open to drain (uses pressure to regulate drainage – may be dangerous e.g. if drainage bag falls to floor)
 - allow 15–20 cc to drain, then clamp tubing. Repeat q 1 hour
 - c) CLD may require ICU monitoring. If patient deteriorates with drain in place: immediately stop drainage, place patient flat in bed (or slight Trendelenburg), start 100%O₂, get CT or bedside cross-table skull x-ray (to R/O tension pneumocephalus due to drawing in of air)
3. surgical treatment in persistent cases (see below)

Surgical treatment

General information

If the site of leakage is not identified prior to attempted surgical treatment, 30% develop a recurrent leak post-op, with 5–15% of these developing meningitis before leak is stopped.²⁶

Indications for surgical intervention

1. traumatic CSF leak that persists >2 weeks in spite of non-surgical measures
2. spontaneous leaks and those of delayed onset following trauma or surgery: usually require surgery because of a high incidence of recurrence
3. leaks complicated by recurrent meningitis

Leaks through cribriform plate/ethmoidal roof

Extradural approach: Generally preferred by ENT surgeons.²⁹ If a frontal craniotomy is being performed, an intradural approach should be used since problems may arise in dissecting the dura off of the floor of the frontal fossa, wherein the dura almost always tears and then it is difficult to know if an identified tear is the cause of the leak or if it is iatrogenic. Fluorescein dye mixed with CSF injected intrathecally may help demonstrate the leak intraoperatively. CAUTION: must be diluted with CSF to reduce risk of seizures (p.1426).

Intradural approach

Generally the procedure of choice.³⁰ If the fistula site is unidentified preoperatively, use a bifrontal bone flap.

General techniques of intradural approach:

Close bone defects with fat, muscle, cartilage, or bone.

Close dural defect with fascia lata, temporalis muscle fascia, or pericranium. Fibrin glue may be used to help hold tissue in place.

If the leak is unidentified pre-op and intra-op, then pack both cribriform plates and sphenoid sinus (incise dura over tuberculum sellae, drill through bone to reach sphenoid sinus, remove mucosa or pack it inferiorly, pack with fat).

Post op: lumbar drain after craniotomy is controversial. Some feel CSF pressure may help enhance the seal.³¹ If used, place the drip chamber at the level of shoulder for 3–5 days (for precautions, see above).

Consider shunt (LP or VP) if elevated ICP or hydrocephalus is demonstrated.

Leaks into sphenoid sinus (including post-transsphenoidal surgery leak)

1. LP BID or CLD: as long as pressure >150 mm H₂O or CSF xanthochromic
 - a) if leak persists >3 days: repack sphenoid sinus and pterygoid recesses with fat, muscle, cartilage and/or fascia lata (must reconstruct floor of sella, packing alone is inadequate). Some recommend against muscle since it putrefies and shrinks. Continue LP or CLD as above for 3–5 days post-op
 - b) if leak persists >5 days: lumboperitoneal shunt (first R/O obstructive hydrocephalus)
2. more difficult surgical approach: intracranial (intradural) approach to medial aspect of middle cranial fossa
3. consider transnasal sellar injection of fibrin glue under local anesthesia³²

Petrous bone

May present as otorrhea or as rhinorrhea (via the eustachian tube).

1. following posterior fossa surgery: see also treatment following vestibular schwannoma surgery (p.685)
2. following mastoid bone fractures: may be approached via extensive mastoidectomy¹⁴
3. due to dehiscence of the footplate of the stapes: may require obliteration of the middle ear and eustachian tube through a tympanomeatal flap¹⁴

23.10 Intracranial hypotension (spontaneous)

23.10.1 General information

Intracranial hypotension may be spontaneous or post-traumatic (including iatrogenic, e.g. post-LP). This chapter addresses spontaneous intracranial hypotension (SIH).

Key concepts

- orthostatic headache (headache that is improved in recumbency)
- diagnosed clinically with assistance of LP
- characteristic findings on imaging not required for dx : “SEEPS” (sagging brain, pachymeningeal enhancement, engorged veins, pituitary hyperemia, subdural fluid)
- excludes patients with history of dural puncture, penetrating spinal trauma, spinal surgery or procedures
- epidural blood patch provides relief in the majority of patients

Epidemiology

Incidence 5:100,000, prevalence 1:50,000.^{33,34} More common in females^{33,34,35,36} with mean age of presentation in the 40's.^{34,35}

Clinical

The syndrome of spontaneous intracranial hypotension is characterized by the following in the absence of antecedent trauma or dural puncture:

1. orthostatic headache: dramatically worse when upright, improved in recumbency
2. low CSF pressure
3. diffuse pachymeningeal enhancement (cerebral and/or spinal) on MRI

Most patients have orthostatic headache with sudden onset, but other headaches have been described such as thunderclap, non-positional, exertional headaches, headaches at the end of the day, and even paradoxical headaches with worsening upon lying.^{36,37} Atypical patients have been described without headache, without pachymeningeal enhancement on MRI,³⁸ with clinical signs of encephalopathy, cervical myelopathy or parkinsonism.³⁹ Since some patients may have normal intracranial pressure, the term “CSF hypovolemia” has been suggested.⁴⁰

Diagnosis

Diagnostic criteria per IHS Classification (ICHD-III)⁴¹:

1. any headache fulfilling criterion C (see below)
2. low CSF pressure (<6 cm of water) and/or evidence of CSF leakage on imaging
3. headache has developed in temporal relation to the low CSF pressure or CSF leakage, or has led to its discovery
4. not better accounted for by another ICHD-III

Compared to older criteria,^{36,42} current diagnostic criteria exclude the need for the headache to worsen within 15 minutes from sitting or standing, eliminate the need for specific associated symptoms used previously to aid in diagnosis although some of these symptoms are often seen (neck stiffness, tinnitus, hypacusia, photophobia, nausea), exclude the need for MRI as there is no change seen in 20-25% of patients,^{33,36,37,43} and exclude the effectiveness of epidural blood patch within 72hrs, as it has no effect in approximately 25% of patients.^{33,36,44}

Median delay from presentation to diagnosis of SIH is 4 months.^{34,35} This delay may be detrimental to patient outcomes. Therefore, brain MRI without and with contrast is recommended in patients with new onset orthostatic headaches.³⁵

Pathophysiology

The underlying cause of SIH is a spontaneous CSF leak.³⁴ Evidence supports an underlying weakness of the meninges as a contributing factor, for instance connective tissue disorders like Marfan syndrome, and Ehlers-Danlos syndrome.^{17,33,34,45} Spinal diverticula, at the cervicothoracic junction or thoracic spine, thoracic being more common,^{34,36,37} and excluding lumbosacral perineural cysts, are thought to be the source of CSF leak in most patients. No relationship has been found between cranial leaks and SIH.^{33,46} Other causes of dural injury are degenerative disc disease, osteophytes and bony spurs.³⁴ The orthostatic headache is believed to be caused by the descent of the brain, causing strain on intracranial structures sensitive to pain.^{33,37,47}

Evaluation

1. radiographic studies
 - a) MRI Brain: findings (mnemonic SEEPS)
 - Sagging of the brain caused by the loss of buoyancy from low CSF volume.^{33,37} Associated with low lying cerebellar tonsils occurred, seen in 36% of patients,³⁹ effacement of perichiasmatic and prepontine cisterns, bowing of the optic chiasm, flattening of pons, and ventricular collapse^{33,36,37,48}
 - Enhancement of the pachymeninges, sparing the leptomeninges, is common from dilation of subdural blood vessels^{33,46,48}
 - Engorgement of veins. Can also see venous distension sign as transverse sinus becomes dilated and convex⁴⁹
 - Pituitary hyperemia
 - Subdural fluid collections seen in 50% of patients.^{34,50} Can be hygromas versus hematomas, with hygromas being twice as frequent as hematomas. Occasionally may require intervention
 - b) CT Brain is not as conclusive but can help identify these changes. 11% of SIH patients also have pseudo-SAH finding on CT caused by effacement of basal cisterns due to sagging of the brain^{51,52}
 - c) CT Myelogram with iodinated contrast: Study of choice for diagnosis and localization of CSF leak. Timed images immediately after contrast injection or at delayed intervals after injections can help localize intermittent leaks^{34,37}
 - d) MRI with intrathecal gadolinium: Alternative to CT myelogram. Injection of 0.5 ml of gadolinium followed by full spine T1 with fat suppression imaging an hour after injection. Contrast remains for 24 hours, hence it can aid in detection of intermittent leaks. Prospective cohort study localized leak in 67% of patients with SIH. In another study, MRI 15 minutes after gadolinium injection identified CSF leak in 21% of SIH patients with a negative CT myelogram. No side effects were reported, but intrathecal gadolinium is not FDA approved (off-label use)^{44,53,54}
 - e) spinal MRI: May show evidence of CSF leak, but is more likely to help localize extrathecal fluid collections for patients with local symptoms.³³ If there is focal spine pain, the leak will often be near this location. Other findings include: dural enhancement, dilated veins, deformed dural sac, meningeal diverticula, syringomyelia, and retrospinal fluid collections at C1-C2.^{34,55,56,57,58,59,60,61,62}
 - f) radioisotope cisternography: Poor resolution, may leave as many as one third of leaks unidentified.^{34,37} Can be used especially if CT myelogram fails
2. lumbar puncture: CSF pressure < 6 cm of water is part of the diagnostic criteria. Patients have been identified with normal CSF pressure.^{34,43,63} Associated CSF findings that have been identified include: lymphocytic pleocytosis, high protein level and xanthochromia^{37,34,51}
3. positive response to EBP can also be used to support the diagnosis.

Treatment

None of these treatments have been evaluated by randomized clinical trials.

- conservative medical management: bed rest, hydration, analgesics, caffeine, and abdominal binder. Limited effect has been reported with intravenous caffeine, steroids and theophylline^{33,34,36,37}
- epidural blood patch (EBP): see technique (p. 1509). Injection of autologous blood (10-20 ml) into epidural space. Evidence shows patients respond well and usually immediately.^{34,52} However, some patients require more than one EBP and relief in headache may not be permanent.⁴¹ If unsuccessful, can repeat blood patch with same or larger amount of blood. Positioning the patient in Trendelenburg position after injection aids in movement of blood to cover more segments for increased effectiveness.^{37,34} May not be effective in up to 25-33%^{33,34,36,44,64}
- directed epidural blood patch to site of leak if the above fails
- percutaneous placement of fibrin sealant at site of leak: can provide relief in patients that fail to improve with conservative measures and epidural blood patch^{37,34,64,65}
- surgical intervention: last resource for patients without relief with conservative measures, EBP or fibrin sealant in which the exact site of leak has been identified. Meningeal diverticula can be ligated with suture, aneurysm clips or muscle pledget with gel foam and fibrin sealant, technique that may also be effective if a dural defect is identified^{37,34}

Outcome

After appropriate treatment, clinical improvement is seen and precedes radiographic improvement. MRI usually takes days to weeks to normalize. Complete resolution of HA was achieved in 70% of

patients (usually in days to weeks). Resolution was more likely if receiving EBP and less likely if they had multiple sites of CSF leak.³⁹ Patients with MRI changes characteristic of SIH and an identifiable focal CSF leak have better outcomes when compared to patients with multilevel CSF leaks.^{37,34,66} CSF leak can recur in approximately 10% of patients. An association between longer interval from symptom onset to diagnosis and poorer outcome has been reported.³⁵

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24 Hydrocephalus – General Aspects

24.1 Basic definition

An abnormal accumulation of cerebrospinal fluid within the ventricles of the brain.

24

24.2 Epidemiology

Estimated prevalence: 1–1.5%
Incidence of congenital hydrocephalus is $\approx$ 0.9–1.8/1000 births (reported range from 0.2 to 3.5/1000 births¹).

24.3 Etiologies of hydrocephalus

24.3.1 General information

- Hydrocephalus (HCP) is either due to subnormal CSF reabsorption or, rarely, CSF overproduction.
- subnormal CSF reabsorption. Two main functional subdivisions:
 1. obstructive hydrocephalus (AKA non-communicating): block proximal to the arachnoid granulations (AG) On CT or MRI: enlargement of ventricles proximal to block (e.g. obstruction of aqueduct of Sylvius → lateral and 3rd ventricular enlargement out of proportion to the 4th ventricle, sometimes referred to as triventricular hydrocephalus)
 2. communicating hydrocephalus (AKA non-obstructive): defect in CSF reabsorption by the AG
 - CSF overproduction: rare. As with some choroid plexus papillomas; even here, reabsorption is probably defective in some as normal individuals could probably tolerate the slightly elevated CSF production rate of these tumors.

24.3.2 Specific etiologies of hydrocephalus

- The etiologies in one series of pediatric patients is shown in □ Table 24.1.
1. congenital
 - a) Chiari Type 2 malformation and/or myelomeningocele (MM) (usually occur together)
 - b) Chiari Type 1 malformation: HCP may occur with 4th ventricle outlet obstruction
 - c) primary aqueductal stenosis (usually presents in infancy, rarely in adulthood)
 - d) secondary aqueductal gliosis: due to intrauterine infection or germinal matrix hemorrhage³
 - e) Dandy Walker malformation (p.256): atresia of foramina of Luschka & Magendie. The incidence of this in patients with HCP is 2.4%
 - f) X-linked inherited disorder (p.401): rare
 2. acquired
 - a) infectious (the most common cause of communicating HCP)
 - post-meningitis; especially purulent and basal, including TB, cryptococcus (p.376)
 - cysticercosis
 - b) post-hemorrhagic (2nd most common cause of communicating HCP)
 - post-SAH
 - post-intraventricular hemorrhage (IVH): many will develop transient HCP. 20–50% of patients with large IVH develop permanent HCP requiring a shunt

Table 24.1 Etiologies of HCP in 170 pediatric patients with HCP ²	
congenital (without myelomeningocele)	38%
congenital (with MM)	29%
perinatal hemorrhage	11%
trauma/subarachnoid hemorrhage	4.7%
Tumor	11%
previous infection	7.6%

- c) secondary to masses
 - non neoplastic: e.g. vascular malformation
 - neoplastic: most produce obstructive HCP by blocking CSF pathways, especially tumors around aqueduct, e.g. medulloblastoma. A colloid cyst can block CSF flow at the foramen of Monro. Pituitary tumor: suprasellar extension of tumor or expansion from pituitary apoplexy
- d) post-op: 20% of pediatric patients develop permanent hydrocephalus (requiring shunt) following p-fossa tumor removal. May be delayed up to 1 yr
- e) neurosarcoidosis (p.189)
- f) “constitutional ventriculomegaly”: asymptomatic. Needs no treatment
- g) associated with spinal tumors⁴: ? due to ↑ protein?, ↑ venous pressure?, previous hemorrhage in some?

24.3.3 Special forms of hydrocephalus

These are covered in subsequent sections

- normal pressure hydrocephalus (NPH) (p.403)
- entrapped fourth ventricle (p.402)
- arrested hydrocephalus (p.402)

24.4 Signs and symptoms of HCP

24.4.1 In older children (with rigid cranial vault) and adults

Symptoms are those of increased ICP, including: papilledema, H/A, N/V, gait changes, upgaze and/or abducens palsy. Slowly enlarging ventricles may initially be asymptomatic.

24.4.2 In young children

Overview

- abnormalities in head circumference (OFC) (see below)
- cranium enlarges at a rate > facial growth
- irritability, poor head control, N/V
- fontanelle full and bulging
- enlargement and engorgement of scalp veins: due to reversal of flow from intracerebral sinuses due to increased intracranial pressure⁵
- Macewen’s sign: cracked pot sound on percussing over dilated ventricles
- 6th nerve (abducens) palsy: the long intracranial course is postulated to render this nerve very sensitive to pressure
- “setting sun sign” (upward gaze palsy) (p.99): Parinaud’s syndrome (p.99) from pressure on region of suprapineal recess
- hyperactive reflexes
- irregular respirations with apneic spells
- splaying of cranial sutures (may be seen on plain skull x-ray)

Occipital-frontal circumference

The occipital-frontal circumference (OFC) should be followed in every growing child (as part of a “well-baby” check-up, and especially in infants with documented or suspected hydrocephalus (HCP)). As a rule of thumb, the OFC of a normal infant should equal the distance from crown to rump.⁶ (rule No. 335) for the differential diagnosis of macrocephaly (p.1403).

Measurement technique⁷: use a non-stretchable tape, measure the circumference of the head just above the supraorbital ridge anteriorly and around the most prominent part of the occiput posteriorly (keeping above the ears). Pull the tape snug to compress hair (exclude any braids or hairclips). Take 2 separate measurements (reposition the tape each time), if the measurements are within 2 mm, record the largest value. If the measurements disagree by >2 mm, take a third measurement and record the average of the 2 closest measures.

Normal head growth: parallels normal curves as seen on the graphs on the inside front cover of this book, or in □ Fig. 24.1 and □ Fig. 24.2 for preemies. Any of the following may signify treatable conditions such as active HCP, subdural hematoma, or subdural effusions, and should prompt an evaluation of the intracranial contents (e.g. CT, head U/S ...):

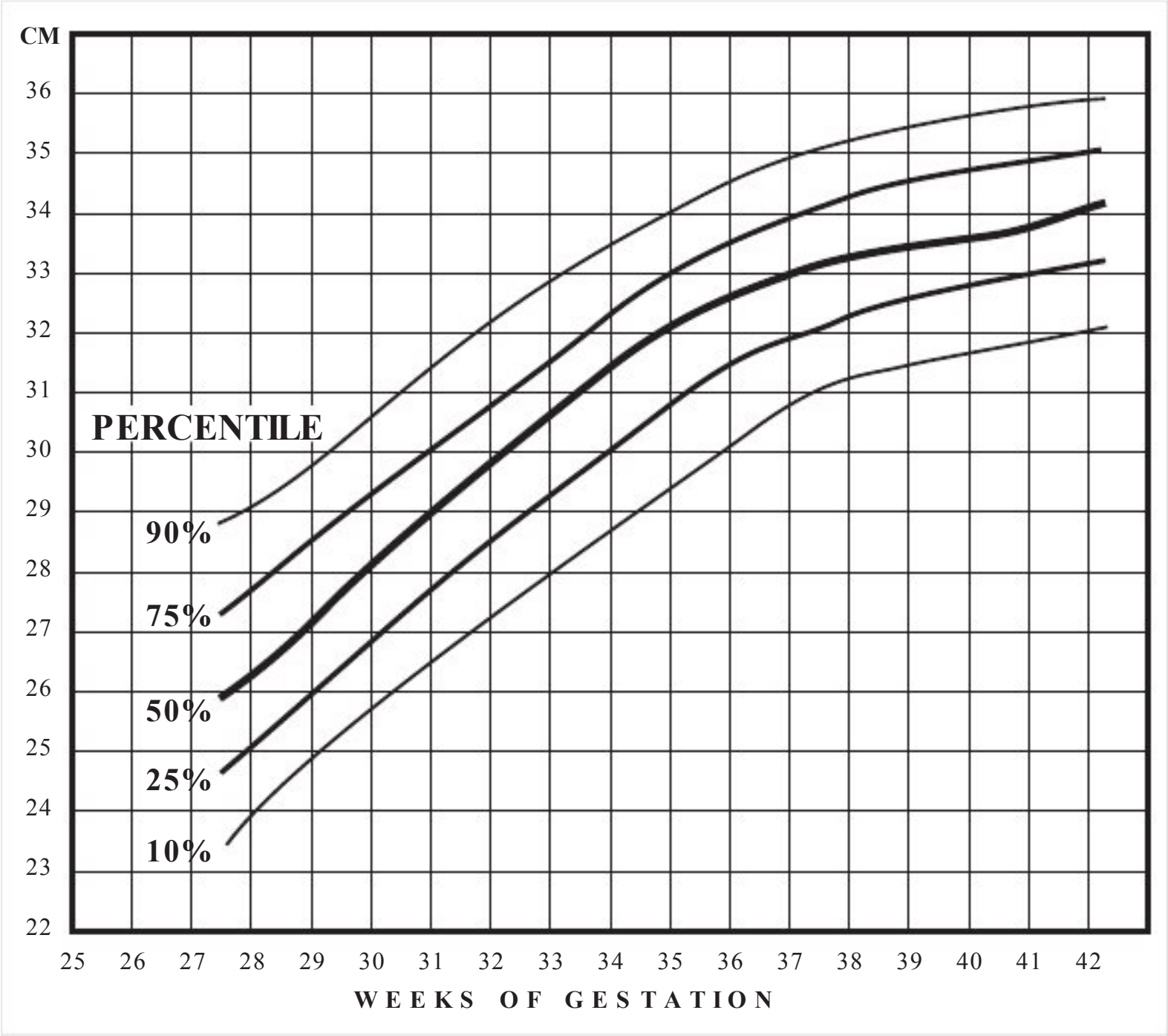


Fig. 24.1 OFC for premature infants as a function of gestational age

1. Progressive upward deviations from the normal curve (crossing curves)
2. continued head growth of more than 1.25 cm/wk
3. OFC approaching 2 standard deviations (SD) above normal
4. head circumference out of proportion to body length or weight, even if within normal limits for age (□ Fig. 24.2)⁸

These conditions may also be seen in the “catch-up phase” of brain growth in premature infants after they recover from their acute medical illnesses, see Catch-up phase of brain growth (p.1351). Deviations below the curves or head growth in the premature infant in the neonatal period of less than 0.5 cm/wk (excluding the first few weeks of life) may indicate microcephaly (p.289).

Technique: measure circumference around forehead and occiput (excluding ears) three consecutive times, and use the largest value. OFC is then plotted on a graph of average values as a function of age⁹ and followed for each individual patient. Use the graphs on the inside front cover for most children and adolescents. The graph in □ Fig. 24.1 shows the OFC for premature infants as a function of gestational age up to term.

The graph in □ Fig. 24.2 shows the relationship of head circumference, weight and length for various gestational ages.

24.4.3 Blindness from hydrocephalus

General information

Blindness is a rare complication of hydrocephalus and/or shunt malfunction. Possible causes include:

1. occlusion of posterior cerebral arteries (PCA caused by downward transtentorial herniation
2. chronic papilledema causing injury to optic nerve at the optic disc
3. dilatation of the 3rd ventricle with compression of optic chiasm

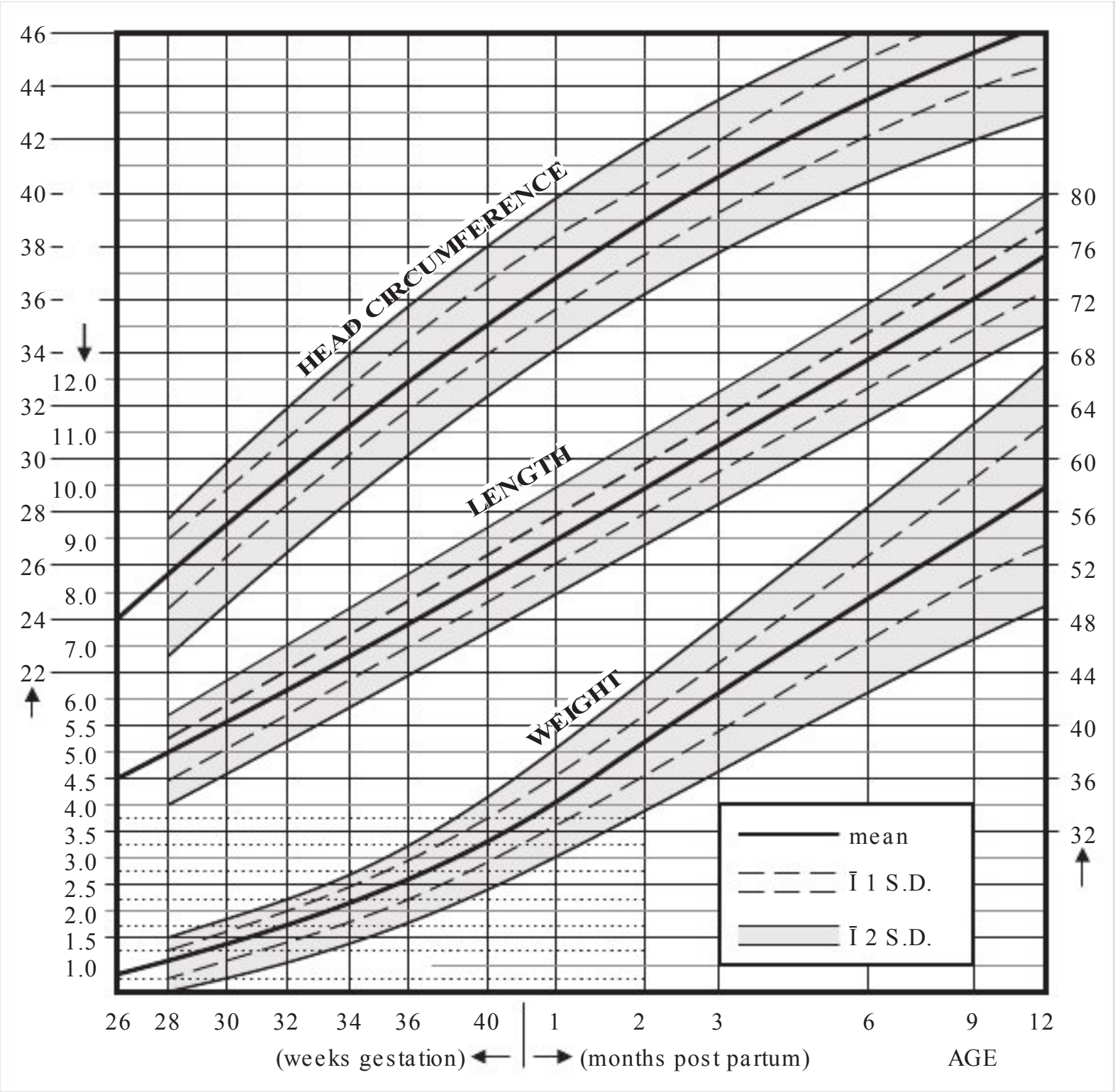


Fig. 24.2 Head circumference, weight and length
(Redrawn from Journal of Pediatrics, “Growth Graphs for the Clinical Assessment of Infants of Varying Gestational Age”, Babson S G, Benda G I, vol 89, pp 815, with permission)

Ocular motility or visual field defects are more common with shunt malfunction than is blindness.^{10,11,12,13} One series found 34 reported cases of permanent blindness in children attributed to shunt malfunction with concomitant increased ICP¹⁴ (these authors were based in a referral center for visually impaired children, thus incidence not estimated). Another series of 100 patients with tentorial herniation (most from acute EDH and/or SDH) proven by CT; 48 patients operated; only 19 of 100 survived >1 month (all were in operated group); 9 of 100 developed occipital lobe infarct (2 died, 3 vegetative state, remaining 4 moderate to severe disability).¹⁵

Types of visual disturbance

9 of 14 had pregeniculate (anterior visual pathway) blindness with marked optic nerve atrophy (early), and reduced pupillary light reflexes. 5 of 14 had postgeniculate (cortical) blindness with normal light responses and minimal or no optic nerve atrophy (or atrophy late). A few patients had evidence of damage in both sites.

Cortical blindness: due to lesions posterior to lateral geniculate bodies (LGB), may also be seen with hypoxic injuries or trauma.¹⁶ Occasionally associated with Anton’s syndrome (denial of visual deficit) and with Ridoch’s phenomenon (appreciation of moving objects without perception of stationary stimuli).

Pathophysiology

In patients with occipital lobe infarction

Occipital lobe infarctions (OLI) in PCA distribution are seen either bilaterally, or if unilateral are associated with other injuries to optic pathways posterior to LGB. The most often cited mechanism is compression of PCA resulting from brain herniating downward through tentorial notch where the PCA or its branches lie on the surface of the hippocampal gyrus and tend to cross the free edge of the tentorium¹⁷ (some authors implicate parahippocampal gyral compression in tentorial notch directly injuring LGBs; this may never produce permanent blindness). Alternatively, upward cerebellar herniation (e.g. from ventricular puncture in face of a p-fossa mass) may impinge on PCA or branches with the same results.¹⁸

OLIs are more likely with a rapid rise in ICP (doesn't allow compensatory shifts and collateral circulation to develop).¹⁹ Macular sparing is common, possibly due to potential dual blood supply of occipital poles (sometimes filled both by PCA and MCA collaterals²⁰), alternatively the calcarine cortex may be supplied by a distinct branch of the PCA that fortuitously escapes compression.²¹

Reported causes of OLI include: post traumatic edema, tumor, abscess, SDH, unshunted hydrocephalus, and shunt malfunction.^{17,22,23}

The occipital poles are also particularly vulnerable to diffuse hypoxia²⁴; attested to by cases of cortical blindness after cardiac arrest.²⁵ Hypotension superimposed on compromised PCA circulation (from herniation or elevated ICP) may thus increase the risk of postgeniculate blindness.^{14,19}

Both coup and contrecoup trauma may produce OLI. Unlike a PCA occlusion infarct, macular sparing is not expected in traumatic occipital lobe injury.¹⁷

In patients with pregeniculate blindness

Elevated ICP transmits pressure to retina → bloodflow stasis, as well as mechanical trauma to optic chiasm from enlarging third ventricle (latter more commonly thought to be responsible for bitemporal hemianopia,¹⁰ but could, if unchecked, progress to complete visual loss). Also, if hypotension and anemia were present, consider the possibility of ischemic optic neuropathy^{26,27,28} which may be anterior, or posterior (the latter of which carries a poorer prognosis).

Presentation

These deficits are frequently unsuspected (altered mental state and the youth of many of these patients¹⁴ makes detection difficult); an examiner must persevere to detect homonymous hemianopsias in an obtunded patient.¹⁷

Pregenicate blindness is less often associated with depressed sensorium than is postgenicate (where direct compression and vascular compromise of midbrain are more likely¹⁴).

Prognosis

Cortical blindness after diffuse anoxia frequently improves (occasionally to normal); usually slowly (weeks to years quoted; several mos usually adequate).²⁵ Many reports of blindness after shunt malfunction are pre-CT era, thus the presence or extent of occipital lobe infarction not ascertained. Some optimistic outcomes reported,²⁹ however, permanent blindness or severe visual handicap are also described^{17,23}; no reliable predictor has been identified. As with infarcts elsewhere, younger patients fare better,²⁴ but extensive calcarine infarcts are probably incompatible with significant visual recovery.

24.5 CT/MRI criteria for hydrocephalus

24.5.1 General information

In general, hydrocephalus is best demonstrated on CT or MRI (□ Fig. 24.3). Occasionally, other means of determining the presence of hydrocephalus must be employed. Most experienced clinicians can recognize HCP by its appearance on CT or MRI. Numerous methods have been devised to attempt to quantitatively define radiographic criteria for hydrocephalus (HCP) (most date back to the early CT experience, and some are used definitionally for research purposes). Some are presented here for completeness. See also radiologic features of chronic HCP (p.400).

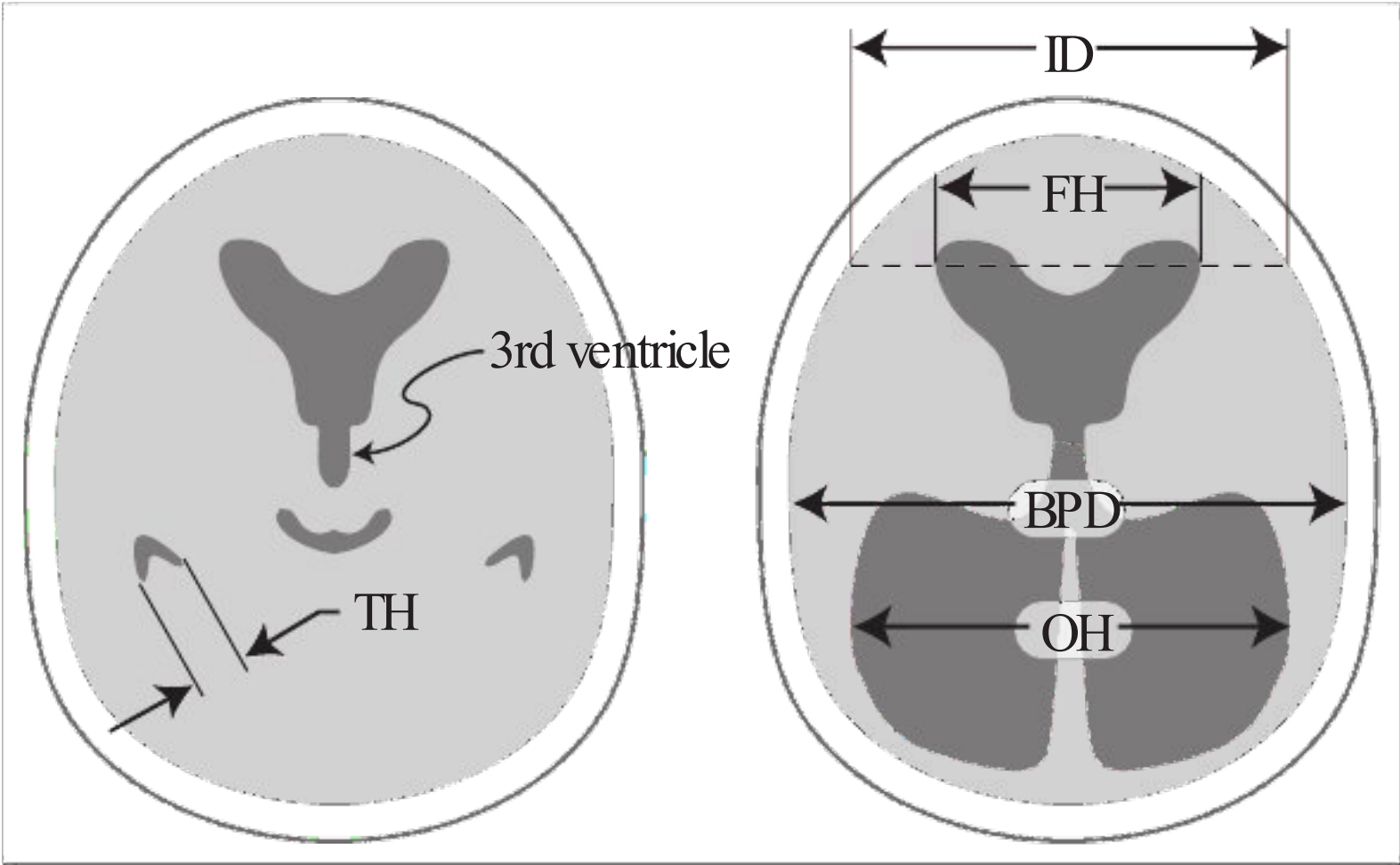


Fig. 24.3 Linear ventricular measurement for CT, MRI or U/S
Abbreviations: TH=temporal horns, FH=frontal horns, ID=internal diameter, BPD=biparietal diameter, OH=occipital horns

24.5.2 Specific imaging criteria for hydrocephalus

HCP is suggested when either³⁰:

1. the size of both temporal horns (TH) is ≥ 2 mm in width ($\square$ Fig. 24.3; in the absence of HCP, the temporal horns should be barely visible), and the sylvian & interhemispheric fissures and cerebral sulci are not visible
OR
2. both TH are ≥ 2 mm, and the ratio $\frac{FH}{ID} > 0.5$ (where FH is the largest width of the frontal horns, and ID is the internal diameter from inner-table to inner-table at this level, $\square$ Fig. 24.3)

Other features suggestive of hydrocephalus (see $\square$ Fig. 24.3 for measurements):

1. ballooning of frontal horns of lateral ventricles (“Mickey Mouse” ventricles) and/or 3rd ventricle (the 3rd ventricle should normally be slit-like)
2. periventricular low density on CT, or periventricular high intensity signal on T2WI on MRI suggesting transependymal absorption of CSF (note: a misnomer: CSF does not actually penetrate the ependymal lining, proven with CSF labeling studies; probably represents stasis of fluid in brain adjacent to ventricles)
3. used alone, the ratio
$$\frac{FH}{ID} : \begin{array}{ll} < 40\% & \text{normal} \\ 40 \text{ \AA } 50\% & \text{borderline} \\ > 50\% & \text{suggests hydrocephalus} \end{array}$$
4. Evans ratio (originally described for ventriculography³¹, or index; note: measurements that rely on the frontal horn diameter tend to underestimate hydrocephalus in pediatrics possibly because of disproportionate dilatation of the occipital horns in peds³²): ratio of FH to maximal biparietal diameter (BPD) measured in the same CT slice: > 0.3 suggests hydrocephalus
5. sagittal MRI may show thinning of the corpus callosum (generally present with chronic HCP) and/or upward bowing of the corpus callosum

24.6 Differential diagnosis of hydrocephalus

For etiologies of HCP, see above.

Conditions that may mimic HCP but are not due to inadequate CSF absorption are occasionally referred to as “Pseudohydrocephalus” and include:

- “hydrocephalus ex vacuo” enlargement of the ventricles due to loss of cerebral tissue (cerebral atrophy), usually as a function of normal aging, but accelerated or accentuated by certain disease processes (e.g. Alzheimer’s disease, Creutzfeldt-Jakob disease, traumatic brain injury). Does not represent altered CSF hydrodynamics, but is rather loss of brain tissue (p.400). See means of differentiating from true hydrocephalus (p.920)
- hydranencephaly (p.288)
- developmental anomalies where the ventricles or portions of the ventricles appear enlarged:
 - agenesis of the corpus callosum (p.259): may occasionally be associated with HCP, but more often merely represents expansion of the third ventricle and separation of the lateral ventricles
 - septo-optic dysplasia (p.260)
 - hydranencephaly (p.288): a post-neurulation defect. Total or near-total absence of the cerebrum most commonly due to bilateral ICA infarcts. It is critical to differentiate this from severe (“maximal”) hydrocephalus (HCP) since shunting for true HCP may produce some re-expansion of the cortical mantle; see means to differentiate (p.288)

Conditions that have been dubbed “hydrocephalus” but do not actually mimic the appearance of HCP

- otitic hydrocephalus: obsolete term used to describe the increased intracranial pressure seen in patients with otitis media; see Idiopathic intracranial hypertension (IIH) (p.766)
- external hydrocephalus (p.400): seen in infancy, enlarged subarachnoid space with increasing OFCs and normal or mildly dilated ventricles

24.7 Chronic HCP

Features indicative of chronic hydrocephalus (as opposed to acute hydrocephalus):

1. beaten copper cranium (some refer to beaten silver appearance) on plain skull x-ray.³³ By itself, does not correlate with increased ICP, however when associated with #3 and #4 below, does suggest ↑ ICP. May be seen in craniosynostosis; see description (p.252)
2. 3rd ventricle herniating into sella (seen on CT or MRI)
3. erosion of sella turcica (may be due to #2 above) which sometimes produces an empty sella, and erosion of the dorsum sellae
4. the temporal horns may be less prominent on imaging than in acute HCP
5. macrocrania: by convention, OFC greater than 98th percentile³⁴ (p.203)
6. atrophy of corpus callosum: best appreciated on sagittal MRI
7. in infants
 - a) sutural diastasis
 - b) delayed closure of fontanelles
 - c) failure to thrive or developmental delay

24.8 External hydrocephalus (AKA benign external hydrocephalus)

24.8.1 General information

Key concepts

- enlarged subarachnoid spaces over the frontal poles in the first year of life
- ventricles are normal or minimally enlarged
- may be distinguished from subdural hematoma by the “cortical vein sign”
- usually resolves spontaneously by 2 years of age

Enlarged subarachnoid space (usually over the cortical sulci of the frontal poles) seen in infancy (primarily in the first year of life) usually accompanied by abnormally increasing head circumference with normal or mildly dilated ventricles.³⁵ There are often enlarged basal cisterns and widening of the anterior interhemispheric fissure. No other symptoms or signs should be present (although there may be slight delay only in motor milestones due to the large head). Etiology is unclear, but a defect

in CSF resorption is postulated. External hydrocephalus (EH) may be a variant of communicating hydrocephalus.³⁶ No predisposing factor may be found in some cases, although EH may be associated with some craniosynostoses³⁷ (especially plagiocephaly) or it may follow intraventricular hemorrhage or superior vena cava obstruction.

24.8.2 Differential diagnosis

EH is probably distinct from benign subdural collections (or extra-axial fluid) of infancy (p.903). □ EH must be distinguished from symptomatic chronic extra-axial fluid collections (p.904) (or chronic subdural hematoma), which may be accompanied by seizures, vomiting, headache... and may be the result of child abuse. With EH, MRI or CT may demonstrate cortical veins extending from the surface of the brain to the inner table of the skull coursing through the fluid collection (“cortical vein sign”), whereas the collections in subdural hematomas compress the subarachnoid space which apposes the veins to the surface to the brain.^{38,39}

24.8.3 Treatment

EH usually compensates by 12–18 mos age without shunting.⁴⁰ Recommend: follow serial ultrasound and/or CT to rule out abnormal ventricular enlargement. Emphasize to parents that this does not represent cortical atrophy. Due to increased risk for positional molding, parents may need to periodically reposition the head while the child is sleeping.⁴¹

A shunt may rarely be indicated when the collections are bloody (consider the possibility of child abuse) or for cosmetic reasons for severe macrocrania or frontal bossing.

24.9 X-linked hydrocephalus

24.9.1 General information

Inherited hydrocephalus (HCP) with phenotypic expression in males passed on through carrier mothers who are phenotypically normal. Classical phenotypic expression will skip single generations.

Incidence: 1/25,000 to 1/60,000. Prevalence: $\approx$ 2 cases per 100 cases of hydrocephalus.

Gene located on Xq28.^{42,43,44}

24.9.2 Pathophysiology

L1CAM membrane bound receptor plays a significant role in CNS development for axonal migration to appropriate target locations through Integrin cell adhesion molecules and MAP Kinase signal cascade.^{42,43,44}

Abnormal gene expression results in poor differentiation and maturation of cortical neurons macroscopic anatomical abnormalities (bilateral absence of pyramidal tracts, see below).

Cytoplasmic domain loss of function mutations result in severe L1 syndrome, whereas mutations retaining expression of some functional protein (component imbedded in cell membrane) leads to mild L1 syndrome.

24.9.3 L1 syndromes

Classical syndromes include CRASH (corpus callosum hypoplasia, retardation, adducted thumbs (clasp thumbs), spastic paralysis, HCP), MSAS (mental handicap, aphasia, shuffling gate, adducted thumbs), HSAS (HCP with stenosis of the aqueduct of sylvius). Spectrum of disease also includes x-linked agenesis of the corpus callosum (ACC), and spastic paraparesis type 1.^{42,43}

Recent delineations⁴⁴:

- mild L1 syndrome: adducted thumbs, spastic paralysis, hypoplasia of CC
- severe L1 syndrome: as in mild L1 syndrome plus anterior cerebellar vermis hypoplasia, large massa intermedia, enlarged quadrigeminal plate, rippled ventricular wall following VP shunt placement (pathognomonic for X-linked HCP). Profound mental retardation in virtually all cases

Radiographic findings likely present if severe L1⁴⁵:

1. severe symmetric HCP with predominant posterior horn dilation
2. hypoplastic CC/ACC
3. hypoplastic anterior cerebellar vermis

4. large massa intermedia
5. large quadrigeminal plate
6. rippled ventricular wall following VP shunt placement (pathognomonic)

Treatment: no intervention demonstrates improvement in retardation status in observational papers.

1. VP shunt: main purpose is management of head size for improved care by caregiver. Does not improve neurologic outcome
2. there are no current genetic therapies for L1 CAM protein abnormalities
3. prenatal U/S: early (≈ 20 – 24 weeks gestational age) with frequent repeat scan in known carrier mothers. May allow for medically indicated termination early on
4. male infants with HCP and ≥ 2 clinical/radiographic signs should undergo genetic testing for L1 CAM mutation detection for future pregnancy counseling⁴²

24.10 “Arrested hydrocephalus”

24.10.1 General information

The exact definition of this term is not generally agreed upon, and some use the term compensated hydrocephalus interchangeably. Most clinicians use these terms to refer to a situation where there is no progression or deleterious sequelae due to hydrocephalus that would require the presence of a CSF shunt. Patients and families should be advised to seek medical attention if they develop symptoms of intracranial hypertension (decompensation) which may include: headaches, vomiting, ataxia or visual symptoms.⁴¹

Arrested hydrocephalus satisfies the following criteria in the absence of a CSF shunt:

1. near normal ventricular size
2. normal head growth curve
3. continued psychomotor development

24.10.2 Shunt independence

The concept of becoming independent of a shunt is not universally accepted.⁴⁶ Some feel that shunt independence occurs more commonly when the HCP is due to a block at the level of the arachnoid granulations (communicating hydrocephalus),⁴⁷ but others have shown that it can occur regardless of the etiology.⁴⁸ These patients must be followed closely as there are reports of death as late as 5 years after apparent shunt independence, sometimes without warning.⁴⁷

24.10.3 When to remove a disconnected or non-functioning shunt?

Note: a disconnected shunt may continue to function by CSF flow through an endothelialized subcutaneous tract. Recommendations on whether or not to repair vs. remove a disconnected or non-functioning shunt:

1. when in doubt, shunt
2. indications for shunt repair (vs. removal)
 - a) marginally functioning shunts
 - b) the presence of any signs or symptoms of increased ICP (vomiting, upgaze palsy, sometimes H/A alone⁴⁹...)
 - c) changes in cognitive function, $\downarrow$ attention span, or emotional changes
 - d) patients with aqueductal stenosis or spina bifida: most are shunt dependent
3. because of risks associated with shunt removal, surgery for this purpose alone should be performed only in the situation of a shunt infection⁵⁰
4. patients with a nonfunctioning shunt should be followed closely with serial CTs, and possibly with serial neuropsychological evaluations

24.11 Entrapped fourth ventricle

24.11.1 General information

AKA isolated fourth ventricle: 4th ventricle that neither communicates with the 3rd ventricle (through sylvian aqueduct) nor with the basal cisterns (through foramina of Luschka or Magendie). Usually seen with chronic shunting of the lateral ventricles, especially with post-infectious

hydrocephalus (fungal, in particular) or in those with repeated shunt infections. Possibly as a result of adhesions forming from prolonged apposition of the ependymal lining of the aqueduct due to the diversion of CSF through the shunt. Occurs in 2–3% of shunted patients.⁵¹ May also occur in Dandy Walker malformation (p.256) if the aqueduct is also obstructed. The choroid plexus of the 4th ventricle continues to produce CSF which enlarges the ventricle when there is 4th ventricular outlet obstruction or obstruction at the level of the arachnoid granulations.

24.11.2 Presentation

Presentation may include:

1. headache
2. lower cranial nerve palsies: swallowing difficulties
3. pressure on the floor of the 4th ventricle may compress the facial colliculus (p.576) → facial diplegia and bilateral abducens palsy
4. ataxia
5. reduced level of consciousness
6. nausea/vomiting
7. may also be an incidental finding (NB: some “atypical” findings, such as reduced attention span, may be related)

24.11.3 Treatment

Treatment of the entrapped 4th ventricle may alleviate associated slit ventricles.⁵² Most surgeons advocate shunting the ventricle either with a separate VP shunt, or linking into an existing shunt. Options:

1. usual first choice: insertion from below the tonsils under direct vision. The catheter may be brought out at the dural suture line, and may be anchored here by use of an angle adapter sutured to the dura
2. passage through a cerebellar hemisphere: potential complications include delayed injury to the brainstem by the catheter tip as the brainstem moves into its normal position with drainage of the 4th ventricle. This may be avoided by bringing the catheter into the 4th ventricle at a slight angle through the cerebellar hemisphere
3. Torkildsen shunt (ventriculocisternal shunt) is an option for obstructive hydrocephalus if it is certain that the arachnoid granulations are functional (usually not the case with hydrocephalus of infantile onset)
4. an LP shunt may be considered when the 4th ventricle outlets are patent

Cranial nerve palsies may occur with shunting of the 4th ventricle usually as a result of penetration of the brainstem by the catheter either at the time of catheter insertion, or in a delayed fashion as the 4th ventricle decreases in size,⁵³ but also possibly as a result of overshunting causing traction on the lower cranial nerves as the brainstem shifts posteriorly.⁵¹

24.12 Normal pressure hydrocephalus (NPH)

24.12.1 General information

Key concepts

- triad (not pathognomonic): dementia, gait disturbance, urinary incontinence
- communicating hydrocephalus on CT or MRI
- normal pressure on random LP
- symptoms may be remediable with CSF shunting

Normal pressure hydrocephalus (NPH), AKA Hakim-Adams syndrome, first described in 1965,⁵⁴ is clinically important because it may cause treatable symptoms, including one of the few forms of remediable dementia.

As originally described, the hydrocephalus of NPH was considered to be idiopathic (iNPH). However, in some cases of hydrocephalus with normal pressure, a predisposing condition may be

identified, suggesting that the ICP may have been elevated at some point in time. These patients may also respond to shunting.

Possible etiologies of “secondary NPH” include:

- 1. post-SAH
- 2. post-traumatic
- 3. post-meningitis
- 4. following posterior fossa surgery
- 5. tumors, including carcinomatous meningitis
- 6. also seen in ≈ 15% of patients with Alzheimer’s disease (AD)
- 7. deficiency of the arachnoid granulations
- 8. aqueductal stenosis may be an overlooked cause

To further complicate things, some patients considered to have NPH may actually have episodic elevations in ICP.

Throughout this text, discussion focuses on idiopathic NPH unless stated otherwise.

It is becoming increasingly acknowledged that the ventricular enlargement is likely not the primary underlying pathologic entity. Research continues in an effort to improve the understanding of this complicated condition.

24.12.2 Epidemiology

Incidence for idiopathic NPH (iNPH) may be as high as 5.5 per 100,000 per year.⁵⁵

The mean age for iNPH is older than that for secondary NPH.

24.12.3 Clinical

Clinical triad

See reference.⁵⁶

The triad is not pathognomonic, and similar features may also be seen e.g. in vascular dementia,⁵⁷ Alzheimer’s dementia and Parkinson’s disease.

- 1. gait disturbance: usually precedes other symptoms. Wide based with short, shuffling steps and unsteadiness on turning. Patients often feel like they are “glued to the floor” (so-called “magnetic gait”) and may have difficulty initiating steps or turns. Absence of appendicular ataxia
- 2. dementia: primarily memory impairment with bradyphrenia (slowness of thought) and bradykinesia □ Table 24.2 shows some differentiating features with Alzheimer’s disease)
- 3. urinary incontinence: usually unwitting (NB: a patient demented for any reason or with mobility impairment may have incontinence)

Table 24.2 Comparison of cognitive deficits in Alzheimer’s disease (AD) and NPH ^{a,b}		
Feature	AD	NPH
memory	↓	± auditory memory
executive function ^c	↓	±
attention concentration	↓	±
orientation	↓	
writing	↓	
learning	↓	
fine motor speed and accuracy	±	↓
psychomotor skills	±	slowed
language and reading	±	
behavioral or personality changes		±
^a modified ⁵⁸		
^b Key: ↓ =impaired; ±=borderline impaired		
^c see □ Table 24.3 for definition of executive function		

Table 24.3 Diagnostic guidelines for NPH ⁵⁸
Probable NPH
History ^a : must include:
1. insidious onset (vs. acute) 2. onset age ≥ 40 years 3. duration ≥ 3–6 months 4. no antecedent head trauma, ICH, meningitis or other known cause of secondary hydrocephalus 5. progression over time 6. no other neurological, psychiatric or general medical conditions that are sufficient to explain the presenting symptoms
Brain imaging: CT or MRI after onset of symptoms must show:
1. ventricular enlargement not attributable to cerebral atrophy or congenital enlargement ((Evan’s index^b > 0.3 or comparable measure) 2. no macroscopic obstruction to CSF flow 3. ≥ 1 of the following supportive features a) enlarged temporal horns not entirely attributable to hippocampal atrophy b) callosal angle ≥ 40° c) evidence of altered brain water content, including periventricular changes not attributable to microvascular ischemic changes or demyelination d) aqueductal or 4th ventricle flow void on MRI Other imaging findings that may support Probable designation but are not required: 1. pre-morbid study showing smaller or nonhydrocephalic ventricles 2. radionuclide cisternogram showing delayed clearance of radiotracer over the convexities after 48–72 hours 3. cine-MRI or other technique showing increased ventricular flow rate 4. SPECT showing decreased periventricular perfusion that is not altered by acetazolamide challenge
Physiological
CSF opening pressure (OP) on lateral decubitus LP: 5–18 mm Hg (70–245 mm H ₂ O)
Clinical: must show gait/balance disturbance, plus impairment in cognition and/or urinary function
1. gait/imbalance: ≥ 2 of the following (not entirely attributable to other conditions) a) decreased step height b) decreased step length c) decreased cadence (speed of walking) d) increased trunk sway while walking e) widened standing base f) toes turn outward while walking g) retropulsion (spontaneous or provoked) h) en bloc turning (≥ 3 steps to turn 180°) i) impaired walking balance: ≥ 2 corrections out of 8 tandem steps 2. cognition: documented impairment (adjusted for age & education) and/or decrease in performance on cognitive screening instrument (e.g. Monumental State examination), or evidence of ≥ 2 of the following not fully attributable to other conditions: a) psychomotor slowing (increased response latency) b) decreased fine motor speed c) decreased fine motor accuracy d) difficulty dividing or maintaining attention e) impaired recall, especially for recent events f) executive dysfunction: e.g. impairment in multistep procedures, working memory, formulation of abstractions/similarities, insight g) behavioral or personality changes 3. urinary dysfunction: a) any one of the following • episodic or persistent incontinence not attributable to primary urological disorder • persistent urinary incontinence • urinary and fecal incontinence b) or, any 2 of the following: • urinary urgency: frequent perception of a pressing need to void • urinary frequency (pollakiuria): voiding > 6 times in 12 hours with normal fluid intake • nocturia: needing to void > 2 times in an average night
Possible NPH
History: reported symptoms may:
1. have subacute or indeterminate mode of onset 2. onset at any age after childhood

Table 24.3 continued
3. duration: <3 months or indeterminate
4. may follow events such as mild head trauma, remote history of ICH, or childhood or adult meningitis or other conditions judged not likely to be causally related
5. coexist with other neurological, psychiatric, or general medical disorders but judged not to be entirely attributable to these conditions
6. be nonprogressive or not clearly progressive
Clinical: symptoms of either:
1. incontinence and/or cognitive impairment in the absence of observable gait/balance disturbance
2. gait disturbance or dementia alone
Brain imaging: ventricular enlargement consistent with hydrocephalus but associated with any of the following:
1. cerebral atrophy of sufficient severity to potentially explain ventricular enlargement
2. structural lesions that may influence ventricular size
Physiological
OP not available or outside of the range delineated for Probable NPH
Unlikely NPH
1. no ventriculomegaly
2. signs of increased ICP (e.g. papilledema)
3. no component of the clinical triad of NPH
4. symptoms explained by other causes (e.g. spinal stenosis)
^a history should be verified by individual familiar with premorbid and current condition
^b See definition and illustration of Evan’s index (p.399)

Other clinical features

Age usually >60 yrs. Slight male preponderance. Also below for other clinical information.

True aphasia is unusual, but speech output may be disturbed by impaired motivation or executive dysfunction.⁵⁸ As NPH progresses, cognitive impairment may become more generalized and less responsive to treatment.⁵⁸ Symptoms identical to those of idiopathic parkinsonism may occur in 11%⁵⁹

Case reports of a variety of psychiatric disturbances associated with NPH include: depression,⁶⁰ bipolar disorder,⁶¹ aggressiveness,⁶² paranoia.⁶³

Symptoms not expected with NPH

Although a variety of clinical features have been demonstrated to occur infrequently (e.g. SIADH,⁶⁴ syncope...), clinical features not expected solely as a result of NPH include: papilledema, seizures (prior to shunting), headaches.⁵⁸

24.12.4 Imaging in NPH

CT and MRI

- Features on CT⁶⁵ and MRI⁶⁶
- prerequisite: ventricular enlargement without block (i.e. communicating hydrocephalus)
 - features that correlate with favorable response to shunt. These features suggest that the hydrocephalus is not due to atrophy alone. (Note: atrophy / hydrocephalus ex vacuo, as in conditions such as Alzheimer’s disease, lessens the chance of, but does not preclude, responding to a shunt (cortical atrophy is a common finding in healthy individuals of advanced age⁶⁷)
 - periventricular low density on CT or high intensity on T2WI MRI: may represent transependymal absorption of CSF. May resolve with shunting
 - compression of convexity sulci (note: focal sulcal dilation may sometimes be seen and may represent atypical reservoirs of CSF which may diminish after shunting and should not be considered as atrophy⁶⁸)
 - rounding of the frontal horns

Although some patients improve with no change in ventricles,⁶⁹ clinical improvement most often accompanies reduction of ventricular size.

Radionuclide cisternography

Usefulness remains controversial. One study found that the cisternogram does not increase the diagnostic accuracy of clinical and CT criteria.⁷⁰ Use has been abandoned by most researchers.⁷¹

24.12.5 Ancillary tests for NPH

Lumbar puncture (LP)

Opening pressure

Normal LP opening pressure (OP) in the left lateral decubitus position averages 12.2 ± 3.4 cm H₂O (8.8 ± 0.9 mm Hg)⁷² and should be <180 mm H₂O (OP >24 cm H₂O suggests noncommunicating hydrocephalus rather than NPH^{58,73}). In NPH the average OP is 15 ± 4.5 cm H₂O (11 ± 3.3 mm Hg), slightly higher than, but overlapping with, normal. Based on expert opinion, an upper limit of 24 cm H₂O (17.6 mm Hg) is suggested for the definition of NPH. Patients with an initial OP >10 cm H₂O have a higher response rate to shunting.

CSF labs

Send CSF for routine labs (p.1506) to R/O infection, elevated protein (e.g. with tumor), SAH.

“Tap Test”

Consists of lumbar puncture with removal of a specific quantity of CSF and assessment of response.

The tap test has not undergone rigorous prospective evaluation. A positive response to withdrawal of 40–50 ml of CSF has a PPV in the range of 73–100%^{74,75,76} but sensitivity is low (26–61%). (Note: what constitutes a significant “response” has not been standardized, most experts prefer demonstrating objective improvement in gait, taking into account the fact that NPH patients can have day-to-day fluctuations in symptoms.)

Resistance testing

CSF Ro is considered to be the impedance of CSF absorptive mechanisms. 1/Ro is the conductance. Techniques and thresholds are center-specific. No clinical study has adequately addressed the fact that Ro normally increases with age.⁷⁷

Determination of CSF Ro may have a higher sensitivity (57–100%) but a similar PPV (75–92%) to the tap test.

Methodology

Numerous methods have been devised to measure Ro. Two illustrative methods:

- 1. bolus method⁷⁸: a known volume (usually ≈ 4 ml) is injected via LP at a rate of 1 ml/sec
- 2. Katzman test⁷⁹: infuse saline through LP at a known rate, Ro is given by Eq (24.1) (up to 19% of patients experience H/A after infusion studies⁸⁰)

$$Ro = \frac{\Delta \text{final steady state pressure} - \Delta \text{initial pressure}}{\text{infusion rate}}$$

Eq 24.1

Ambulatory lumbar drainage (ALD)

See reference.⁷⁴

A lumbar subarachnoid drain is placed with Tuohy needle, connected through a drip chamber to a closed drainage system. The drip chamber is placed at the level of the patient’s ear when they are recumbent, or at the level of the shoulder when sitting or ambulating.

A properly functioning drain should put out ≈ 300 ml of CSF per day.

If symptoms of nerve root irritation develop during the drainage, the catheter should be withdrawn several millimeters. Daily surveillance CSF cell counts and cultures should be performed (NB: a pleocytosis of ≈ 100 cells/mm³ can be seen normally just as a result of the presence of a drain).

A 5 day trial is recommended (mean time to improvement: 3 days).

Continuous CSF pressure monitoring

Some patients with a normal OP on LP demonstrate pressure peaks >270 mm H₂O or recurrent B-waves.⁸¹ These patients also, and may have a higher response rate to shunting than those without these findings.

Miscellaneous

Cerebral blood flow (CBF) measurements: Although some studies indicate otherwise, CBF measurements show no specific findings in NPH, and are not helpful in predicting who will respond to shunting. However, increased CBF after shunting correlates with clinical improvement.⁸²

EEG: There are no specific EEG findings in NPH.

24.12.6 Diagnostic criteria

24

Practice guideline: Diagnosis of NPH

Level II⁵⁸: Since strict diagnostic criteria cannot be formulated for NPH because of a lack of knowledge of the underlying pathophysiology at this time, it is recommended that the diagnosis be made in terms of Probable, Possible, and Unlikely NPH as described in □ Table 24.3.

Guidelines for diagnosis of NPH are shown in □ Table 24.3.

24.12.7 Differential diagnosis of NPH

□ Table 24.4 shows conditions with presentations similar to findings in NPH in the differential diagnosis.^{58,83} □ Table 24.5 compares some features of NPH, Alzheimer’s disease, and Parkinson’s disease.

24.12.8 Treatment

Management algorithm

1. based on history, physical exam, and imaging studies, categorize the patient as probable, possible, or unlikely NPH based on □ Table 24.3. For probable and possible NPH, without further testing, the degree of certainty of the diagnosis of NPH is $\approx 50\text{--}61\%$ ^{70,84,85} In an otherwise healthy patient in whom the diagnosis of NPH seems highly probable, it is not unreasonable to proceed to shunting⁷³
2. otherwise, to increase the certainty of response to shunting, one or more of the following tests is recommended⁷³
 - a) “tap test”: withdrawal of 40–50 ml of CSF via LP
 - positive response (p.407) increases likelihood of responding to a shunt (PPV) to the range of 73–100%
 - due to low sensitivity (26–61%), a negative response does not rule out the possibility of responding, and a subsequent supplemental test should be performed⁷³
 - if OP > 17.6 mm Hg (24 cm H₂O), consider further search for cause of secondary hydrocephalus (does not rule-out shunting as a treatment)
 - b) resistance testing: sensitivity (57–100%) > tap test, similar PPV (75–92%)
 - c) external lumbar drainage

CSF diversionary procedures

VP shunt is the procedure of choice. Lumbar-peritoneal shunts have been used, but disadvantages include: tendency to overshunt, difficult to tap, tendency to migrate. For most, use a medium pressure valve⁸⁶ (closing pressure 65–90 mm H₂O) to minimize the risk of subdural hematomas (see below), although response rate may be higher with a low-pressure valve.⁸⁷ Gradually sit patient up over a period of several days; proceed more slowly in patients who develop low-pressure headaches. Alternatively, the risk of developing SDH may be decreased with use of a programmable shunt valve, set initially at a high pressure (to reduce risk of subdural hematoma) and gradually decreasing the pressure setting over a number of weeks.

Follow patients clinically and with CT for $\approx 6\text{--}12$ months.

Patients who do not improve and whose ventricles do not change on imaging should be evaluated for shunt malfunction. If not obstructed, and if no subdural fluid collections have developed, a lower pressure valve may be tried (or a lower pressure selected on a programmable shunt).

Potential complications of shunting for NPH

Complication rates may be as high as $\approx 35\%$ (due in part to the frailty of the elderly brain).^{88,89}

Table 24.4 Conditions with similar presentation to NPH
Neurodegenerative disorders • Alzheimer’s disease • Parkinson’s disease • Lewy body disease • Huntington’s disease • frontotemporal dementia • corticobasal degeneration • progressive supranuclear palsy • amyotrophic lateral stenosis • multisystem atrophy • spongiform encephalopathy
Vascular dementia • cerebrovascular disease • multi-infarct dementia • Binswanger’s disease • CADASIL • vertebrobasilar insufficiency (VBI)
Other hydrocephalic disorders • aqueductal stenosis • arrested hydrocephalus • long-standing overt ventriculomegaly syndrome • noncommunicating hydrocephalus
Infectious disease • Lyme disease • HIV • syphilis
Urological disorders • urinary tract infection • bladder or prostate cancer • benign prostatic hypertrophy (BPH)
Miscellaneous • vitamin B12 deficiency • collagen vascular diseases • epilepsy • depression • traumatic brain injury • spinal stenosis • Chiari malformation • Wernicke’s encephalopathy • carcinomatous meningitis • spinal cord tumor

- Potential complications include⁹⁰:
1. subdural hematomas or hygroma (p.426): higher risk with low pressure valve and older patients who tend to have cerebral atrophy. Usually accompanied by headache, most resolve spontaneously or remain stable. Approximately one–third require evacuation and tying off of shunt (temporarily or permanently). Risk may be reduced by gradual mobilization post-op
 2. shunt infection
 3. intracerebral hemorrhage
 4. seizures (p.417)
 5. delayed complications include: above, plus shunt obstruction or disconnection

Endoscopic third ventriculostomy (ETV)

Initially reported for NPH in 1999.⁹¹ Mechanistically, it is difficult to explain why ETV would work for NPH, but it has been advocated by some⁹² in highly selected patients, using nonvalidated outcome measures, quoting post-op improvement in 69% of patients. At this time, ETV should not be considered a first line treatment for most cases of NPH.

Table 24.5 Comparison of NPH, Alzheimer’s & Parkinson’s disease ^a			
Feature	NPH	AD	IPA
gait disturbance ^b	+	±	±
postural instability	±		+
urinary disturbance	±	±	±
memory or cognitive impairment	±	+	±
difficulty performing familiar tasks	±	+	±
behavioral changes	±	+	±
limb rigidity			+
limb tremor			+
bradykinesia			+
^a abbreviations: AD+ Alzheimer’s disease; IPA=idiopathic paralysis agitans (Parkinson’s disease); += feature present; ±=feature partial or late			
^b in NPH, the gait is often wide based, in IPA often narrow stance			

24.12.9 Outcome

The most likely symptom to improve with shunting is incontinence, then gait disturbance, and lastly dementia. Black et al.⁸⁶ give the following markers for good candidates for improvement with shunting:

- clinical: presence of the classic triad (p.404).⁸⁸ Also 77%of patients with gait disturbance as the primary symptom improved with shunting. Patients with dementia and no gait disturbance rarely respond to shunting
- LP: OP>100 mm H₂O
- isotope cisternogram: typical NPH pattern. The mixed or normal pattern has no correlation with response to shunting
- continuous CSF pressure recording: pressure >180 mm H₂O or frequent Lundberg B waves (p.864)
- CTor MRI: large ventricles with flattened sulci (little atrophy)

Response is better when symptoms have been present for a shorter time.

NB: NPH patients with co-existing Alzheimer’s disease (AD) may still improve with VP shunts, thus AD should not exclude these patients from shunting.⁹³ However, patients with AD alone (with-out NPH) did not respond to shunting in a RPDB placebo-controlled trial.⁹⁴

In general, most responders eventually relapse, often after ≈ 5–7 years of good response. Shunt malfunction and subdural collections must be ruled out before ascribing this to the natural course of the underlying condition.

24.13 Hydrocephalus and pregnancy

24.13.1 General information

Patients with CSF shunts may become pregnant, and there are case reports of patients developing hydrocephalus during pregnancy requiring shunting.⁹⁵

Any of the shunt problems discussed in the following sections may occur in a pregnant patient with a shunt. With VP shunts, distal shunt problems may be higher in pregnancy. The following are management suggestions modified from Wisoff et al.⁹⁵

24.13.2 Preconception management of patients with shunts

1. evaluation, including:
 - a) evaluation of shunt function: preconception baseline MRI or CT. Further evaluation of shunt patency if any suspicion of malfunction. Patients with slit ventricles may have reduced compliance and may become symptomatic with very small changes in volume
 - b) assessment of medications, especially anticonvulsants

2. counselling, including:
 - a) genetic counselling: if the HCP is due to a neural tube defect (NTD), then there is a 2–3% chance that the baby will have a NTD
 - b) other recommendations include early administration of prenatal vitamins, and avoiding teratogenic drugs and excessive heat (e.g. hot-tubs): Neural tube defects, Risk factors (p.290).

24.13.3 Gravid management

1. close observation for signs of increased ICP: headache, N/V, lethargy, ataxia, seizures... Caution: these signs may mimic pre-eclampsia (which must also be ruled out). 58% of patients exhibit signs of increased ICP, which may be due to:
 - a) decompensation of partial shunt malfunction
 - b) shunt malfunction
 - c) some show signs of increased ICP in spite of adequate shunt function, may be due to increased cerebral hydration and venous engorgement
 - d) enlargement of tumor during pregnancy
 - e) cerebral venous thrombosis: including dural sinus thrombosis & cortical venous thrombosis
 - f) encephalopathy related to disordered autoregulation
2. patients developing symptoms of increased ICP should have CT or MRI to compare ventricle size to preconception baseline study
 - a) if no change from preconception study, puncture shunt to measure ICP and culture CSF. Consider radioisotope shunt-o-gram
 - b) if all studies are negative, then physiologic changes may be responsible. Treatment is bed rest, fluid restriction, and in severe cases steroids and/or diuretics. If symptoms do not abate, then early delivery is recommended as soon as fetal lung maturity can be documented (give prophylactic antibiotics for 48 hrs before delivery)
 - c) if ventricles have enlarged and/or shunt malfunction is demonstrated on testing, shunt revision is performed
 - in first two trimesters: VP shunt is preferred (do not use peritoneal trocar method after first trimester) and is tolerated well
 - in third trimester: VA or ventriculopleural shunt is used to avoid uterine trauma or induction of labor

24.13.4 Intrapartum management

1. prophylactic antibiotics are recommended during labor and delivery to reduce the incidence of shunt infection. Since coliforms are the most common pathogen in L&D, Wisoff et al. recommend ampicillin 2 gm IV q 6 hrs, and gentamicin 1.5 mg/kg IV q 8 hrs in labor and $\times$ 48 hrs post partum⁹⁵
2. in patients without symptoms: a vaginal delivery is performed if obstetrically feasible (lower risk of forming adhesions or infection of distal shunt). A shortened second stage is preferred since the increase in CSF pressure in this stage is probably greater than during other valsalva maneuvers⁹⁶
3. in the patient who becomes symptomatic near term or during labor, after stabilizing the patient a C-section under general anesthesia (epidurals are contraindicated with elevated ICP) is performed with careful fluid monitoring and, in severe cases, steroids and diuretics

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25 Treatment of Hydrocephalus

25.1 Medical treatment of hydrocephalus

HCP remains a surgically treated condition. Acetazolamide may be helpful for temporizing (see below).

25.1.1 Diuretic therapy

May be tried in premature infants with bloody CSF (as long as there is no evidence of active hydrocephalus) while waiting to see if there will be resumption of normal CSF absorption. However, at best this should only be considered as an adjunct to definitive treatment or as a temporizing measure.

Satisfactory control of HCP was reported in $\approx 50\%$ of patients of age < 1 year who had stable vital signs, normal renal function and no symptoms of elevated ICP (apnea, lethargy, vomiting) using the following¹ with the following regimen:

1. acetazolamide (a carbonic anhydrase inhibitor): 25 mg/kg/day PO divided TID $\times 1$ day, increase 25 mg/kg/day each day until 100 mg/kg/day is reached
2. simultaneously start furosemide: 1 mg/kg/day PO divided TID
3. to counteract acidosis, use tricitrate (Polycitra®):
 - a) start 4 ml/kg/day divided QID (each ml is equivalent to 2 mEq of bicarbonate, and contains 1 mEq K⁺ and 1 mEq Na⁺)
 - b) measure serial electrolytes, and adjust dosage to maintain serum HCO₃ > 18 mEq/L
 - c) change to Polycitra-K® (2 mEq K⁺ per ml, no Na⁺) if serum potassium becomes low, or to sodium bicarbonate if serum sodium becomes low
4. watch for electrolyte imbalance and acetazolamide side effects: lethargy, tachypnea, diarrhea, paresthesias (e.g. tingling in the fingertips)
5. perform weekly U/S or CT scan and insert ventricular shunt if progressive ventriculomegaly occurs. Otherwise, maintain therapy for a 6 month trial, then taper dosage over 2–4 weeks. Resume 3–4 mos of treatment if progressive HCP occurs

25.2 Spinal taps

HCP after intraventricular hemorrhage may be only transient. Serial taps (ventricular or LP²) may temporize until resorption resumes but LPs can only be performed for communicating HCP. If reabsorption does not resume when the protein content of the CSF is < 100 mg/dl, then it is unlikely that spontaneous resorption will occur in the near future (i.e. a shunt will usually be necessary).

25.3 Surgical

25.3.1 Goals of therapy

Normal sized ventricles is not the goal of therapy (some children have a paucity of brain tissue). Goals are optimum neurologic function (which usually requires normal intracranial pressure) and a good cosmetic result.

25.3.2 Surgical options

Options include

- third ventriculostomy: currently, endoscopic method is preferred (see below)
- shunting: various shunts are described below. The techniques of shunt placement are covered for VP shunts (p. 1515), for VA shunt (p. 1516), for ventriculopleural shunts (p. 1515), and for LP shunt (p. 1517)
- eliminating the obstruction: e.g. opening a stenosed sylvian aqueduct. Often higher morbidity and lower success rate than simple CSF diversion with shunts, except perhaps in the case of tumor
- choroid plexectomy: described by Dandy in 1918 for communicating hydrocephalus.³ May reduce the rate but does not totally halt CSF production (only a portion of CSF is secreted by the choroid plexus, other sources include the ependymal lining of the ventricles and the dural sleeves of spinal nerve roots). Open surgery was associated with a high mortality rate (possibly due to replacement